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FIBER-OPTIC COMMUNICATION

SYSTEMS AND COMPONENTS

Vivekanand Mishra
Sunita P. Ugale

WILEY

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Fiber-Optic Communication Systems and Components

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Preface

The main motivation of this book is to introduce the basic essentials of fiber-optic systems and their uses in both communication and non-communication areas, embracing various components. It is realized that in the current fast-changing technological scenario, new issues and concepts in the design of fiber-optic communication systems and networks should also be considered. However, a balanced approach is needed to optimize the efforts in the widening the scope of the topics covered and the depth to which each topic is addressed.

About the Book

This book is ideal for entry-level engineers, post-graduate students and other non-specialists as well. The introduction to fiber-optics offered in this book is thorough and accessible, and reflects the expertise of authors who have been teaching in this field for over 15 years. Using a mathematical approach with well-illustrated explanations, this book begins with the technical specifics of optical fibers, moves through the tools and techniques used to work with them and the devices used to connect fiber systems, and concludes with applications showing the various uses of fibers.

The book covers the fundamentals of fiber-optics, applications of fiber optics, types of fibers, cabling, light sources, transmitters, receivers, repeaters, regenerators and optical amplifiers, connectors, splicing, couplers and switches, other components, tools and accessories, measurements, system concepts, system design, and fiber-optic sensors.

This book is intended for a one-semester course in fiber-optics communication for juniors and seniors in science and engineering. Readers are assumed to have an understanding of optics and as well as a reasonable familiarity with the fundamentals of solid state physics. The text covers all the standard topics of a traditional optics course as well as few advanced topics which will be useful for PG students and researchers in this field. Design problems are included with solutions, which will be helpful for students from the examination point of view. Review questions, exercises and multiple-choice questions are provided to examine reader understanding and to assist the tutorial work. The core material of this book tries to get students to a stage where they can read more advance monographs, which provide an enhanced treatment of the practical areas, together with the incorporation of the relevant standardization issues from very recent research articles and papers. References at the end provide details of additional material for further in-depth studies.

Organization of the Book

The book is structured to introduce you to the basic concepts first, and then fiber system components and their applications in detail. The chapters are organized as follows:

- Chapter 1 gives the basics of a fiber-optic communication system, its advantages and evolution.
- Chapter 2 describes the wave properties of light and light propagation through optical fiber. It gives

the details of physical structures, constituent materials, and fabrication methods of optical fiber. Several generic fiber cable configurations are also illustrated. It also covers special-purpose fibers.

- Chapter 3 covers linear effects in optical fiber (such as attenuation and dispersion) as well as non-linear scattering and Kerr effects in optical fiber.
- Chapter 4 explains the structures, light-emitting principles and operating characteristics of light sources such as light-emitting diodes and lasers, in optical communication. It also includes line coding methods for signal transmission over a fiber, direct and external modulation techniques and the fiber connectors and splices.
- Chapter 5 covers the detection of the optical signal. The types of photo detectors, the noise associated with detection and optical receivers are discussed.
- Chapter 6 covers the design of point-to-point single-wavelength systems, with sample calculations. Multichannel concepts related to WDM and DWDM are explained. The optical couplers and other passive components used in fiber-optic systems are described and the system performance parameters are discussed.
- Chapter 7 covers the detailed discussion on various types of optical amplifier, followed by their system applications.
- Chapter 8 gives details of the fundamentals of fiber optic measurements and the related test equipments.
- Chapter 9 explains the principles and operation of fiber optic sensors.

We hope that by treating new aspects in addition to the “classical” topics, our book will give today’s students and researchers knowledge and tools that will prove useful for them in the future.

Acknowledgments

We express our gratitude to our parents who brought us up to this stage with their invaluable guidance and advice. We also express our gratitude to Dr. P. D. Porey, Director, SVNIT, Surat, Gujarat and Honorable Shri. Balasaheb Wagh, Chairman, K. K. Wagh Education Society, Nashik, Maharashtra for their constant encouragement and support.

Prof. Sunita P. Ugale would like to express her gratefulness to her husband Mr. Aniruddha Patil and son Kaustubh for their support, and understanding of the time spent while working on this project.

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Finally we sincerely express our thanks to Mr. Vikas Gupta, MD; Mr. Paras Bansal, Publisher; and the editorial team at Wiley India for bringing up this book in time.

Dr. Vivekanand Mishra
Prof. Sunita P. Ugale
March 2012

Contents

Preface	v
1 Overview of Optical Fiber Communication Systems	1
Learning Objectives	1
1.1 Introduction	1
1.2 Evolution of Fiber-Optic Systems	2
1.2.1 Evolution of WDM System Capacity	3
1.3 Advantages of Optical Fiber Communication	4
1.4 Disadvantages of Optical Fiber Communication	5
1.5 Elements of an Optical Fiber Transmission Link	5
1.5.1 Different Components Used in Optical Fiber Communication Systems	6
1.6 Applications of Fiber Optics in Day-to-Day Life	7
Summary	8
Multiple-Choice Questions	8
Review Questions	8
Answers	9
2 Optical Fibers: Structures, Wave Guiding, and Fabrication	11
Learning Objectives	11
2.1 Introduction	11
2.2 Electromagnetic Waves	11
2.3 Polarization	13
2.4 Diffraction	13
2.5 Reflection and Refraction	14
2.6 Total Internal Reflection	15
2.7 Waveguides	16
2.8 Light Propagation in Optical Fibers	16
2.8.1 Construction of a Fiber-Optic Cable	17
2.8.2 Light Propagation – Ray Theory	18
2.8.3 V Number	20
2.8.4 Fiber as a Dielectric Waveguide (Mode Theory)	21
2.9 Types of Optical Fibers	26
2.9.1 Multimode Step-Index Fibers	28
2.9.2 Multimode Graded-Index Fibers	28
2.10 Single-Mode Step-Index Fibers	31
2.10.1 Gaussian Beam	32

2.10.2	Mode-Field Diameter	33
2.10.3	Cutoff Wavelength	34
2.10.4	Single-Mode Graded-Index Fibers	35
2.11	Fiber Materials	35
2.11.1	Glass Fibers	36
2.11.2	Halide Glass Fibers	37
2.11.3	Chalcogenide Glass Fibers	37
2.11.4	Active Glass Fibers	37
2.11.5	Plastic Optical Fibers	37
2.12	Fabrication of Optical Fibers	37
2.12.1	Preform Generation Techniques	37
2.13	Fiber-Optic Cable Construction	41
2.14	Specialty Optical Fibers	45
2.14.1	Polarization-Maintaining Fibers	45
2.14.2	Photonic Crystal Fiber	46
2.14.3	Hollow Core Fibers	47
2.14.4	Dispersion-Compensated Fiber	47
2.14.5	Photosensitive Fiber	48
	Solved Problems	48
	Summary	56
	Multiple-Choice Questions	57
	Review Questions	58
	Exercises	58
	Answers	59
	References	59
3	Linear and Non-Linear Effects in Optical Fibers	61
	Learning Objectives	61
3.1	Introduction	61
3.2	Linear Effects in Optical Fiber	61
3.2.1	Attenuation	61
3.2.2	Absorption	63
3.2.3	Linear Scattering	64
3.2.4	Bending Losses	65
3.3	Dispersion	68
3.3.1	Intramodal Dispersion or Chromatic Dispersion	68
3.3.2	Intermodal Dispersion	72
3.3.3	Polarization-Mode Dispersion	72
3.3.4	Group Delay	74
3.3.5	Signal Distortion in Single-Mode Fibers	76
3.3.6	Dispersion and Maximum Bit Rate	77
3.4	Non-Linear Effects in Fiber	78
3.5	Scattering Effects	78
3.5.1	Stimulated Raman Scattering	78
3.5.2	Stimulated Brillouin Scattering	81

3.6	Kerr Effects	84
	3.6.1 <i>Four-Wave Mixing</i>	85
	3.6.2 <i>Self-Phase Modulation</i>	88
	3.6.3 <i>Cross-Phase Modulation (XPM or CPM)</i>	90
3.7	Comparison of Kerr Effect and Scattering Effect	91
3.8	The Formation of Solitons	92
	Solved Problems	93
	Summary	102
	Multiple-Choice Questions	103
	Review Questions	104
	Exercises	104
	Answers	105
	References	106
4	Optical Sources	109
	Learning Objectives	109
4.1	Semiconductor Fundamentals	110
	4.1.1 <i>Intrinsic and Extrinsic Materials</i>	110
	4.1.2 <i>pn Junction</i>	112
4.2	Light-Emitting Diode	114
	4.2.1 <i>Direct and Indirect Recombination</i>	116
	4.2.2 <i>LED Materials</i>	118
	4.2.3 <i>Quantum Efficiency of LED</i>	119
	4.2.4 <i>Double Heterojunction LED</i>	124
	4.2.5 <i>LED Structure</i>	125
	4.2.6 <i>LED Characteristics</i>	130
4.3	Laser	131
	4.3.1 <i>Laser Diode Rate Equations</i>	132
	4.3.2 <i>Construction of a Laser</i>	135
	4.3.3 <i>Laser Modes</i>	136
	4.3.4 <i>Semiconductor Lasers</i>	138
	4.3.5 <i>Laser Structures</i>	141
	4.3.6 <i>Laser Characteristics</i>	147
	4.3.7 <i>Laser Properties and Applications</i>	149
4.4	Line Coding	152
4.5	Modulation	154
4.6	Optical Transmitter	155
4.7	Optical Connectors	156
4.8	Optical Splices	157
	4.8.1 <i>Fusion Splicing</i>	158
	4.8.2 <i>Mechanical Splicing</i>	159
	4.8.3 <i>Losses in Fiber Joints</i>	160
	Solved Problems	161
	Summary	166
	Multiple-Choice Questions	167
	Review Questions	168

Exercises	168
Answers	169
References	169

5 Optical Receivers 171

Learning Objectives	171
5.1 Photodiode	171
5.1.1 Absorption Coefficient	173
5.1.2 Direct and Indirect Absorption	174
5.1.3 Photodetector Materials	174
5.1.4 Quantum Efficiency	176
5.1.5 Responsivity	176
5.1.6 Long-Wavelength Cutoff	178
5.2 Types of Photodiodes	178
5.2.1 <i>pn</i> Photodiode	178
5.2.2 PIN Photodiode	180
5.2.3 Avalanche Photodiode	183
5.3 Photodetector Noise	186
5.3.1 Minimum Detectable Power	186
5.3.2 Dark Current	186
5.3.3 Shunt Resistance (R_{sh})	186
5.4 Fundamental Photodiode Circuits	187
5.5 Phototransistor	188
5.5.1 Fundamental Phototransistor Circuits	189
5.6 Optical Receiver	190
5.6.1 Equivalent Circuit of Optical Fiber Receiver	192
5.6.2 Receiver Amplifier Configurations	192
Solved Problems	193
Summary	204
Multiple-Choice Questions	204
Review Questions	205
Exercises	206
Answers	207
References	207

6 Optical Transmission System: Concepts and Components 209

Learning Objectives	209
6.1 Point-to-Point Links	209
6.1.1 Digital System	211
6.1.2 Analog System	211
6.1.3 Local Area Networks	211
6.1.4 Distribution Networks	213
6.1.5 System Design Considerations	213
6.1.6 System Design Steps	215

6.2	Power Budgeting	215
6.3	Rise Time Budgeting	216
6.4	Multichannel Concept	217
	6.4.1 Wavelength Division Multiplexing (WDM)	218
	6.4.2 Advantages of WDM	219
	6.4.3 Special Case of WDM: Dense WDM	219
	6.4.4 WDM Standards	221
6.5	WDM Components	221
	6.5.1 Coupler/Splitter	221
	6.5.2 Optical Circulator	223
	6.5.3 Optical Add/Drop Multiplexer	226
	6.5.4 Michelson Interferometer	227
	6.5.5 Fabry–Perot Interferometer	230
	6.5.6 Mach–Zehnder Interferometer	231
	6.5.7 Arrayed Waveguide Grating	234
	6.5.8 Acousto-Optic Tunable Filter	237
	6.5.9 Ring Resonator-Based Wavelength Filters	239
	6.5.10 Fiber Grating	242
	6.5.11 Optical Switches	265
6.6	System Performance Parameters	274
	6.6.1 Bit Error Rate	274
	6.6.2 Eye Diagrams	275
	6.6.3 Optical Signal-to-Noise Ratio	276
	6.6.4 ASE Noise	276
	6.6.5 Crosstalk	278
	6.6.6 DWDM System Performance	279
	6.6.7 Non-Linear Effects	279
	Solved Problems	283
	Summary	291
	Multiple-Choice Questions	292
	Review Questions	293
	Exercises	294
	Answers	295
	References	295
7	Optical Amplifiers	299
	Learning Objectives	299
7.1	Basic Concept	299
7.2	Basic Applications of Optical Amplifiers	301
7.3	Types of Optical Amplifiers	302
7.4	Semiconductor Optical Amplifier (SOA)	302
	7.4.1 Types of SOA	304
	7.4.2 Cavity Gain of SOA	305
	7.4.3 Crosstalk in SOAs	306

7.5	Raman Amplifier	306
	7.5.1 Raman Scattering	306
	7.5.2 Raman Scattering in Optical Fibers	308
	7.5.3 Fiber Raman Amplifier	309
	7.5.4 Pumping Schemes for Raman Amplifier	311
	7.5.5 Broadbanding of Raman Amplifier	311
	7.5.6 Impact of Raman Scattering on WDM Systems	311
7.6	Rare-Earth-Doped Fiber Amplifiers	313
	7.6.1 Erbium-Doped Fiber Amplifier (EDFA)	314
	7.6.2 Thulium-Doped Fiber Amplifier (TDFA)	321
7.7	System Applications	322
	7.7.1 Wavelength Conversion	322
	7.7.2 Optical Gates	323
	7.7.3 Multiplexers	324
	7.7.4 Wavelength Selection and Spectral Inversion	324
	7.7.5 SOAs in CWDM Systems	324
	Solved Problems	324
	Summary	327
	Multiple-Choice Questions	327
	Review Questions	328
	Exercises	328
	Answers	329
	References	329
8	Basic Instrumentation for Optical Measurements	331
	Learning Objectives	331
8.1	Attenuation Measurement	331
	8.1.1 Cutback Method	331
	8.1.2 Insertion Loss Method	332
	8.1.3 Optical Time-Domain Reflectometer (OTDR) Method	333
8.2	Optical Power Measurement	337
8.3	Dispersion Measurement	338
	8.3.1 Pulse Dispersion Measurement in Time Domain	338
	8.3.2 Pulse Dispersion Measurement in Frequency Domain	338
8.4	Fiber Numerical Aperture Measurement	339
	8.4.1 Scanning Photodetector and Rotating Stage Method	340
	8.4.2 Trigonometric Method	340
8.5	Bandwidth and Data Rate	341
8.6	Optical Spectrum Analyzer	342
	8.6.1 Diffraction Grating-Based OSA	343
	8.6.2 Interferometer-Based Architectures	344
8.7	Wavelength Meter	345
8.8	Optical Tweezers	346
	8.8.1 Applications	346
	Solved Problems	347

	Summary	349
	Multiple-Choice Questions	349
	Review Questions	350
	Exercises	350
	Answers	350
	References	351
9	Optical Sensors	353
	Learning Objectives	353
9.1	Classification of the Fiber-Optic Sensors	353
9.2	Intrinsic Sensors	354
9.3	Extrinsic Sensors	354
9.4	Intensity-Modulated Sensors	355
	9.4.1 <i>Reflection-Type Sensors</i>	355
	9.4.2 <i>Transmission-Type Sensors</i>	355
	9.4.3 <i>Differential Intensity Sensors</i>	356
	9.4.4 <i>Microbending Sensor</i>	357
	9.4.5 <i>Evanescent Wave Fiber Sensors</i>	359
9.5	Phase-Modulated or Interferometric Sensors	359
	9.5.1 <i>Fiber Mach–Zehnder Interferometer Sensor</i>	359
	9.5.2 <i>Fiber Fabry–Perot Interferometer Sensors</i>	361
	9.5.3 <i>Fiber-Optic Gyroscope</i>	361
9.6	Wavelength-Modulated Sensors	363
	9.6.1 <i>Fiber Bragg Grating Sensors</i>	363
	9.6.2 <i>Fluorescent Sensors</i>	365
	9.6.3 <i>Blackbody Sensor</i>	366
9.7	Polarization Fiber Sensor	366
9.8	Fiber-Optic Biomedical Sensor	367
	9.8.1 <i>Blood Flow Measurement</i>	367
9.9	Fiber-Optic Liquid Level Sensor	368
9.10	Fiber-Optic Probe	368
9.11	Distributed Sensor	369
	9.11.1 <i>Wavelength Division Multiplexing Interrogation</i>	369
	9.11.2 <i>Time Division Multiplexing Interrogation</i>	370
	Summary	370
	Multiple-Choice Questions	371
	Review Questions	372
	Answers	372
	References	372
Index		375

1

Overview of Optical Fiber Communication Systems

Learning Objectives

After completing this chapter, you will be able to:

- Understand optical link evolution.
 - Understand the advantages of optical communication over copper wire communication.
 - Understand basic fiber-optic communication systems.
 - Gain familiarity with the recent trends in optical communication technology.
-

1.1 Introduction

Transmission of information from one place to another with the help of light is a very old technique. In 800 B.C., the Greeks used visible signals such as fire and smoke for sending information such as calls for help, victory in a war, enemy alerts, etc. However, only yes/no signals were conveyed using this type of technique. During the second century B.C., optical signals were encoded using signaling lamps, which facilitated the sending of any type of message. Till the end of the 18th century, there was negligible development in the field of optical communication. All the communication was of line of sight type with the human eye as the receiver. The speed of the optical communication link was limited due to the unreliable nature of the transmission paths, which were easily affected by atmospheric effects such as fog and rain.

During the middle of the 20th century, it was realized that if optical waves were used as the carrier, long-distance communication would be possible at high bit rates. In the old optical communication system, because of high transmission loss (10^5 – 10^7 dB/km), the bit rate-distance product is only about 1 (bit/s)-km. If the optical carrier is a coherent source such as a laser, more information can be sent to longer distances. A new era in optical communication started after laser was invented by Maiman in 1960. So from 1960 onward, communication engineers started using laser beams as optical carriers in an effective manner. Laser light is coherent, with high intensity, high monochromaticity, high directionality, and less divergence; hence it is capable of carrying a large amount of information compared to radio waves and microwaves.

Transmission of laser signals through air only works at very short distances or when atmospheric conditions are clear. Hence initially metallic and non-metallic waveguides were fabricated to guide light, but due to high losses associated with these waveguides, they were not suitable for telecommunication. Tyndall discovered that light could be transmitted through optical fibers by the phenomenon of total internal reflection. During the 1950s, optical fibers with large diameters (of about 1–2 mm) were used in endoscopes to see the inner parts of the human body. Kao and Hockham published a paper about an optical fiber communication system in 1966. Optical fibers can provide a much more reliable optical channel than the atmosphere, but give a very high loss of 1,000 dB/km as compared to a loss of few dB/km in atmosphere. Kao and his fellow workers realized that these high losses are due to impurities in the fiber material. Kapron, Keck, and Maurer (1970) reduced the fiber loss to 20 dB/km, which is

comparable to the losses in copper cable systems, by using pure silica as the fiber material. Thus, the optical fiber communication system became an engineering reality.

1.2 Evolution of Fiber-Optic Systems

High bit-rate optical transmission is limited by the electronic components and devices at transmitter and receiver ends and not due to the transmission channel fiber. During the period from 1973 to 1980, communication engineers focused on the development of multimode fibers. Due to their large cores, multimode fibers were easier to splice and couple with existing lasers. They were also suitable for coupling with the much more reliable and cheaper light-emitting diodes (LEDs). The cores of single-mode fibers were too small for efficient splicing and coupling into early lasers. At the same time, researchers became successful in developing reliable semiconductor lasers. These semiconductor lasers were small enough to work with single-mode fibers at the longer wavelengths at which attenuation in optical fibers was lowest.

By 1984, fibers with very small attenuations (of around 0.20 dB/km) were available. The lower attenuation was due to improved fabrication techniques that reduced impurities. Simultaneously there was development in the supporting technologies, such as lasers, detectors, and other components that operated at higher frequencies where intrinsic loss in silica fibers was lowest.

In 1976, field trials of fiber-optic telephone systems began. Corning and Western Electric opened full-scale production plants for optical fiber fabrication in 1980. In the same year, the first sea trial of a commercial optical fiber cable was also undertaken. By 1982, telephone companies began to switch from multimode to higher performance single-mode fiber production. AT&T's fiber-optic cable system between Washington, DC, and New York entered service in 1983. Moreover, lengths of commercial fiber increased from 2 km in 1982 to 25 km in 1987. On the other hand, the price per meter fell sixfold. AT-8, the first transatlantic optical fiber cable, was laid in 1988.

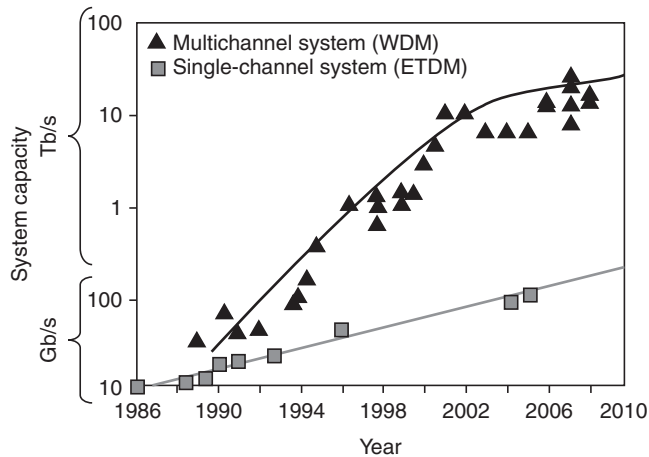
Tables 1.1 and 1.2 list key events in the evolution of optical communications. Figure 1.1 shows the historical evolution of fiber-optic systems capacity.

Table 1.1 | List of key events in the evolution of optical communication

1841	Daniel Colladon demonstrated light guiding in water jets in Geneva.
1854	British physicist John Tyndall shows that light is guided by a bending water jet and published the results in the <i>Proceedings of the Royal Society</i> .
1880	Alexander Graham Bell invents the photophone, using the optical effects of selenium.
1940s	Scientists at Corning develop the flame hydrolysis method/vapor deposition technique of making pure bulk silica (SiO_2).
1949–54	A Dutch group headed by Abraham C.S. van Heel, Technical University of Delft, develops the technique of cladding fibers to improve total internal reflection at the surface of the core fiber.
1960	Theodore Maiman, Hughes Aircraft Co., demonstrated the first laser, using a synthetic ruby.
1974	Bill French and William Tasker, Bell Labs, report attenuation of less than 2 dB/km, the lowest level yet reported.
1976	AT&T carried out a realistic optical fiber system experiment in Atlanta.
1979	Corning began production of optical waveguides in the world's first full-scale manufacturing facility in Wilmington, NC.
1988	First transatlantic optical-fiber cable was laid, with amplifiers about 65 km apart.
1997	Completion of FLAG (fiber-optic link around the globe), a 27,300 km fiber-optic cable system linking Great Britain and Japan, consisting of two 5.3 Gb/s optical-fiber pairs.

Table 1.2 | Different generations of optical-fiber communication systems

Generation	Type of Fiber and Wavelength of Optical Source (nm)	Bit Rate (Mb/s)	Repeater Spacing (km)	Loss (dB/km)	Existed up to
I	Multimode fiber (MMF), $\lambda = 800$ nm	4.5	10	1	1980
II	Single-mode fiber (SMF), $\lambda \approx 1,300$ nm	1.7×10^2	50	< 1	1987
III	SMF, $\lambda = 1,550$ nm	1.0×10^4	70	< 0.2	1990
IV	SMF, $\lambda = 1,550$ nm with coherent system	1.0×10^5	100	< 0.002	2000
V	SMF, $\lambda = 1,550$ nm with Erbium-doped fiber amplifier (EDFA)	$> 1.0 \times 10^9$	> 100	< 0.002	Still in use
VI	SMF, $\lambda = 1,550$ nm with EDFA and solitons	10 Gb/s	8,000,000 km	< 0.002	Still in use
Current System	SMF, $\lambda = 1,550$ nm with distributed Raman amplifier	40 Gb/s	100 km	< 0.002	2011

**Figure 1.1** | Historical evolution of fiber-optic systems capacity.

1.2.1 Evolution of WDM System Capacity

Repeater spacing for commercial systems is as follows:

1. *Long-haul systems* – 600 km repeater spacing.
2. *Ultra-long-haul systems* – 2,000 km repeater spacing (Raman + EDFA amplifiers, forward error correction coding, fast external modulators).
3. *Metro systems* – 100 km repeater spacing.
4. *State-of-the-art in DWDM* – channel spacing 50 GHz, 200 carriers, speed 10 GB/s, repeater spacing few thousand km.

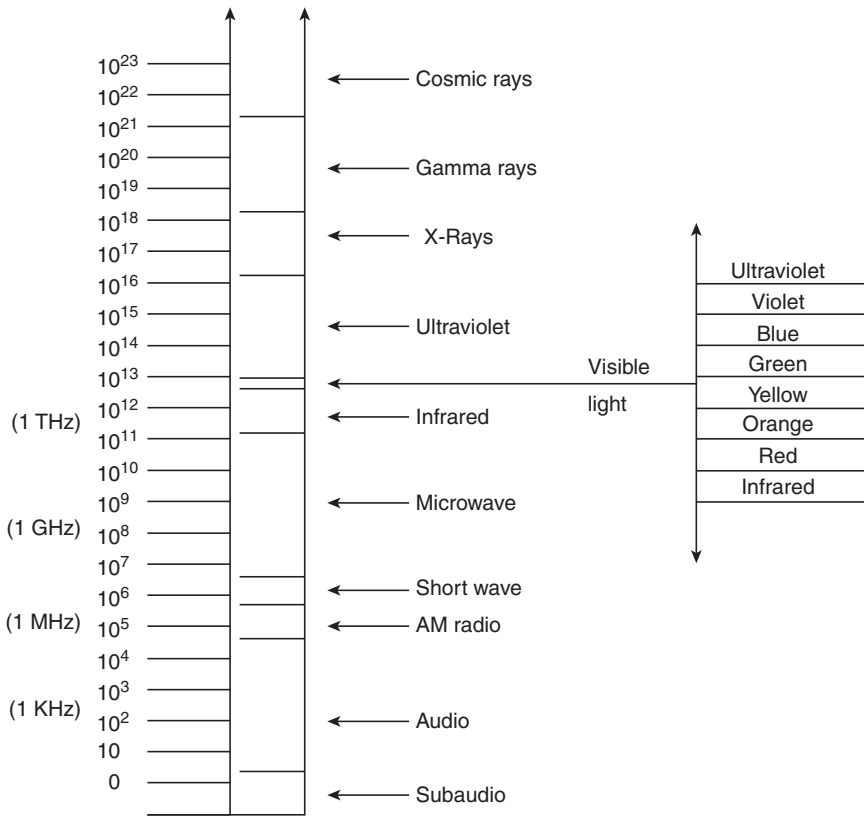


Figure 1.2 | The electromagnetic spectrum.

Figure 1.2 shows the electromagnetic spectrum. In optical fiber communication, a high frequency (around tens of Terahertz) light carrier is used for transmission. The higher the frequency of a carrier, the lower is its wavelength, the higher the available bandwidth and consequently the higher its capacity to transmit information; hence the cost/bps of transmitted information is very less.

1.3 Advantages of Optical Fiber Communication

Optical fiber communication has the following major advantages:

1. **Wider bandwidth:** The optical carrier frequency is in the range 10^{13} – 10^{15} Hz, whereas the microwave frequency is about 10^{10} Hz. Moreover, the radio-wave frequency is about 10^6 Hz and metallic cable system bandwidth is 500 MHz. The information-carrying capacity of a transmission system is directly proportional to the carrier frequency of the transmitted signals. Thus, the optical fiber yields a far greater transmission bandwidth than the conventional communication systems and metallic cable systems. Presently, the fiber bandwidth is not fully utilized. By sending several optical signals parallelly on the same fiber, bandwidth utilization can be greatly enhanced. This technique is called *wavelength division multiplexing* (WDM).
2. **Low transmission loss:** With the technical advancements in fabrication technology, fibers having a transmission loss as low as 0.002 dB/km are available. This transmission loss is extremely low as compared to

even the best copper conductors. Due to the usage of the ultra low-loss fibers and the erbium-doped silica fibers as optical amplifiers, almost lossless transmission can be achieved. As a result, communication links with wide repeater spacings of approximately more than 100 km can be implemented. In modern optical fiber telecommunication systems, optical amplification is achieved using erbium-doped silica fibers over a short length in the transmission path at selective points. Since the amplification is done in the optical domain itself, the distortion produced during the strengthening of the signal is almost negligible.

3. **Dielectric waveguide:** Optical fibers are made from silica or plastic polymer, which are electrical insulators. Therefore they do not pick up any electromagnetic wave or high-current lightning, unlike metallic conductors. This property is suitable in explosive and electrically hazardous environments. Optical fibers are not affected by any interference originating from power cables, railway power lines, and radio waves. There is no crosstalk between the fibers in a single cable as they are electrically isolated from one another.
4. **Signal security:** The signal transmitted through the fibers does not radiate out of the fiber significantly. Further the signal cannot be tapped from a fiber in a non-invasive manner. Therefore, optical-fiber communication provides a high degree of signal security. Hence it is suitable for military, banking, and data-transmission applications.
5. **Small size and weight:** Fiber-optic cables have small radii not greater than the diameter of a human hair, and they are flexible, compact, and lightweight. The fiber cables can be bent or twisted without damage. Further, optical-fiber cables are superior to the copper cables in terms of storage, handling, installation, and transportation, while maintaining comparable strength and durability. Hence these are most suited in aircrafts, satellites, and even ships.

1.4 Disadvantages of Optical Fiber Communication

Optical fiber communication has the following major disadvantages:

1. **Cost:** As compared to metallic cable-based communication systems, fiber-optic cables and the associated electronic equipments are more expensive. For two-way communication, two fibers are needed instead of just one. The connectors and the equipment needed to install them are more costly than those for copper cable.
2. **Signal distribution:** In metallic cable-based communications systems, many devices can share the same cable and communicate with each other. In fiber-optic cables, this is not practical, as signal transmission and reception are point-to-point. A central signal distribution device is necessary for interconnection.
3. **Difficulty in installation and maintenance:** Installation and maintenance of fiber-optic systems requires skilled technicians as compared to metallic cable-based communication systems.
4. **Sensitivity:** Workers dealing with fiber optics have to show extreme care not to damage or degrade the performance of the fiber-optic communications system since it is very susceptible to mishandling and dirt.

1.5 Elements of an Optical Fiber Transmission Link

Figure 1.3 shows the basic components in an optical fiber communication system. The function of the system is to convey the information signal from the source through the transmission medium to the destination. The input electrical signal modulates the intensity of light from the optical source. Then the signal is transmitted through the transmission medium (optical fiber cable) toward the receiver, where it is demodulated and transformed into original electrical information. In the transmission medium, the signal suffers loss, and its quality degrades due to noise. If the distance between transmitter and receiver is large, signal degradation will also be large. So there is a

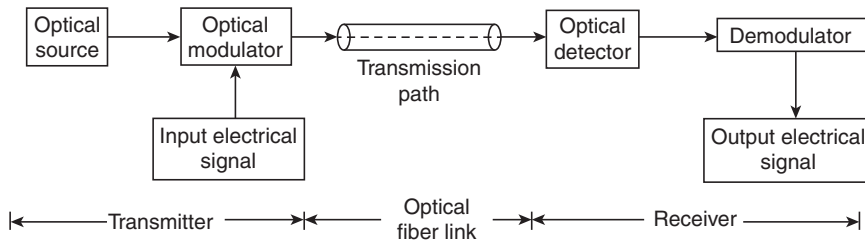


Figure 1.3 | Basic optical-fiber communication system.

limit on transmission distance beyond which signal recovery is not possible. Hence for long-haul communication, installation of repeaters at regular intervals is necessary.

In digital optical fiber communication systems, the input electrical signal is in the form of coded digital pulses generated by the encoder. These electric pulses modulate the intensity of the light from the laser diode or LED and convert them into optical pulses. In the receiver stage, a photodetector such as an avalanche photodiode (APD) or a p-i-n photodiode converts the optical pulses into electrical pulses. A decoder then converts the electrical pulses into the original electric signal.

1.5.1 Different Components Used in Optical Fiber Communication Systems

1. **Optical sources:** Heterojunction LEDs and lasers are the most commonly used optical sources in optical fiber communication. The important required characteristics of the light source are as follows:
 - (a) High output power.
 - (b) High linearity.
 - (c) Narrow spectral width.
 - (d) High modulation rate.
 - (e) Temperature stability.
 - (f) Long lifetime ($\sim 10^5$ hours).
 - (g) Low relative intensity noise (RIN).
2. **Optical fiber:** It is the transmission medium in an optical communication system. A signal transmitted through an optical fiber suffers from loss or degradation. Hence to recover the signal, electronic repeaters or optical amplifiers are used at regular intervals. Earlier systems used to employ multimode graded-index fiber (MMGI) for transmission of high bandwidth signals, that is, up to 1 Gbps data. Almost all systems that are employed today use single mode fibers. The important characteristics of the fiber transmission subsystem are the following:
 - (a) Low attenuation.
 - (b) Low dispersion.
 - (c) Lower fiber non-linearity.
 - (d) Higher optical signal-to-noise ratio (OSNR).
 - (e) Larger repeater span.
3. **Optical detectors:** Semiconductor photodiodes are used as optical detectors, which convert an optical signal to a proportional electrical signal. There are two types of most commonly used photodiodes: p-i-n photodiodes and APDs. For the photodetector, the requirements are as follows:
 - (a) Sensitive at operating λ .
 - (b) Low power consumption and operating voltage.

- (c) Fast response.
 - (d) Active area matched to fiber parameters.
 - (e) Insensitivity to variation in temperature and voltage.
 - (f) Small size and cost.
 - (g) Capability of internal gain.
 - (h) Low noise.
4. **Optical amplifiers:** In long-haul optical fiber communication systems, repeaters are installed at regular intervals of 100 km along the fiber length. A repeater converts the optical signal to an electrical signal and restores the signal (i.e., it compensates for signal loss and dispersion), after which it converts the signal back to an optical signal for further transmission. This method not only increases the cost and complexity of the optical communication system, but also reduces the operational bandwidth of the system. In modern communication systems, this amplification is achieved by erbium-doped fiber amplifiers. This gives improved signal-to-noise ratio due to all optical domain operation.
 5. **Fiber couplers:** A coupler is a device which distributes light from a main fiber into one or more branches of the fiber.
 6. **Fiber connectors:** In a fiber-optic communication link, one fiber can be connected with another fiber using fiber connectors. This connection may be permanent or demountable. Based on this, we have two types of joints – a permanent joint is made using *splice* and a demountable joint is done by *connector*.

1.6 Applications of Fiber Optics in Day-to-Day Life

Communication:

1. Cable television.
2. TV broadcast.
3. Videophones.
4. Voice communication (interoffice, intercity).
5. Local area networks (LAN).
6. Satellite ground stations.
7. Internet (e-mail, videoconferencing, web browsing).
8. Remote sensing and monitoring.
9. Power-generation station monitoring.
10. Aircraft monitoring.
11. Distance learning.
12. Access to remote libraries.

Healthcare:

1. Endoscopy.
2. Remote monitoring of patient.
3. Laser surgery.
4. Biomedical sensing.

Office:

1. Laser printer.
2. Photocopier.
3. Optical scanner.

4. Laser pointer.
5. Fax machine.
6. Overhead slide projector.

Others:

1. CD player.
2. Fluorescent lamp.
3. Bar-code reader.
4. Digital camera.

SUMMARY

1. Transmission of signals over a distance through copper cables, radio waves, and optical fiber cables is called telecommunication.
2. Optical fiber cables have proved to be the most advantageous transmission medium for long distance communication.
3. Fiber-optic communication has become common from 1980 onwards.
4. The basic fiber-optic communication system consists of an optical source, an optical modulator, a transmission medium (optical fiber), an optical detector, and an optical demodulator.
5. Due to advancement in fabrication technology, losses in the fiber are greatly reduced, so long distance communication with a minimum number of repeaters is possible.
6. Invention of optical amplifiers has greatly improved the performance of the optical fiber communication system.
7. To increase the transmission capacity and fully utilize the bandwidth capacity of an optical fiber, multichannel systems such as WDM, CWDM, and DWDM have come into existence.
8. There are numerous applications of fiber optics along with communication.

MULTIPLE-CHOICE QUESTIONS

1. The first practical application of fiber was
 - (a) In Graham Bell's photophone.
 - (b) In bundles to examine the insides of the stomach.
 - (c) To illuminate flowing jets of water.
 - (d) For communication via optical telegraph.
2. The major components of all optical communication system are
 - (a) Fiber and source
 - (b) Source, receiver, and fiber
 - (c) Fiber and receiver
 - (d) Fiber only
3. Which of the following is not an electromagnetic wave?
 - (a) Radio waves
 - (b) Light
 - (c) Infrared radiations
 - (d) Acoustic waves

REVIEW QUESTIONS

1. What are the advantages of optical fiber cables over coaxial cables?
2. State the frequency range of optical fiber communication.

3. Draw the basic elements of fiber-optic communication system.
4. List the optical sources used for optical fiber communication.
5. State the applications of optical fibers.

ANSWERS

Multiple-Choice Questions

1. (c)
2. (b)
3. (d)

2

Optical Fibers: Structures, Wave Guiding, and Fabrication

Learning Objectives

After completing this chapter, you will be able to:

- Describe how light is guided through optical fibers.
 - Differentiate between multimode and single-mode fibers.
 - Calculate numerical aperture (NA), intermodal dispersion, and material dispersion in optical fibers.
 - Understand zero-dispersion wavelength and dispersion-shifted fibers.
 - Gain familiarity with the different fabrication techniques of optical fiber.
 - Study the details of fiber-optic cables.
 - Study the specialty fibers used commercially.
-

2.1 Introduction

The great Scottish physicist James Clerk Maxwell (1831–1879) showed that electric and magnetic fields together can form a propagating wave, appropriately called an electromagnetic wave. It should be noted that light is the visible form of **electromagnetic radiation**. Moreover, it has properties in common with both waves and particles; but wave theory is most useful in understanding the interaction of polarized light with crystals. Each light wave consists of propagating electric and magnetic fields perpendicular to each other and to the direction of propagation. As the electrical interaction of light with crystals is much stronger than the magnetic interaction, hence the electric vector has been focused on in this chapter.

2.2 Electromagnetic Waves

Unlike mechanical waves, an electromagnetic wave does not require any medium to travel through. It can travel through vacuum also. The complete electromagnetic spectrum of light waves is shown in Fig. 2.1. It includes everything from the highest frequency and shortest wavelength gamma rays, followed by X-rays, ultraviolet, visible, infrared, microwaves, and the lowest frequency and the longest wavelength radio waves.

An electromagnetic wave in a vacuum consists of mutually perpendicular and oscillating electric and magnetic fields as shown in Fig. 2.2. It is a transverse wave, because the fields are perpendicular to the direction of wave travel. All electromagnetic waves, regardless of their frequency, travel through a vacuum at the same speed, that of light c ($= 3.00 \times 10^8$ m/s).

The frequency f and wavelength λ (lambda) of an electromagnetic wave in a vacuum are related to its speed through the following relation:

$$c = f\lambda \quad (2.1)$$

An electromagnetic wave propagating in the positive x -direction is shown in Fig. 2.2. Note that E and B are perpendicular to each other and in phase. The direction of propagation is given by the right-hand thumb rule. The amount of energy that flows per second across a unit area perpendicular to the direction of travel is called the **irradiance** or **flux density** of the wave.

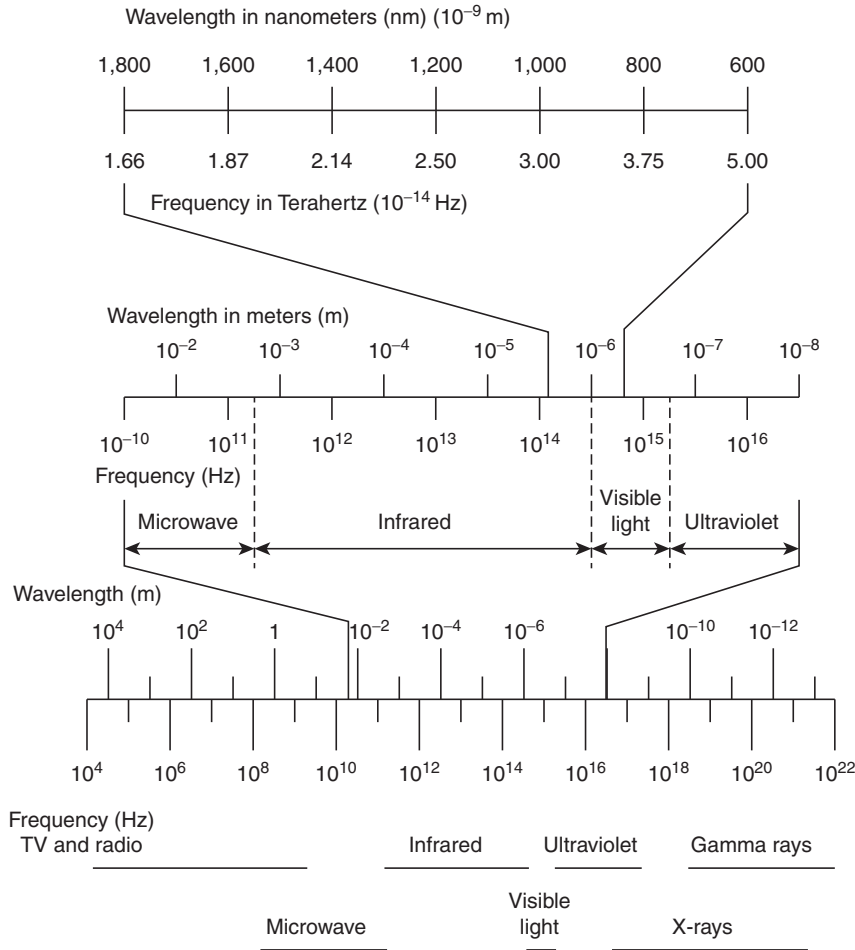


Figure 2.1 | Light wave spectrum.

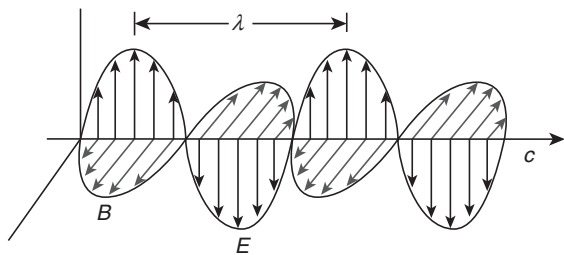


Figure 2.2 | An electromagnetic wave.

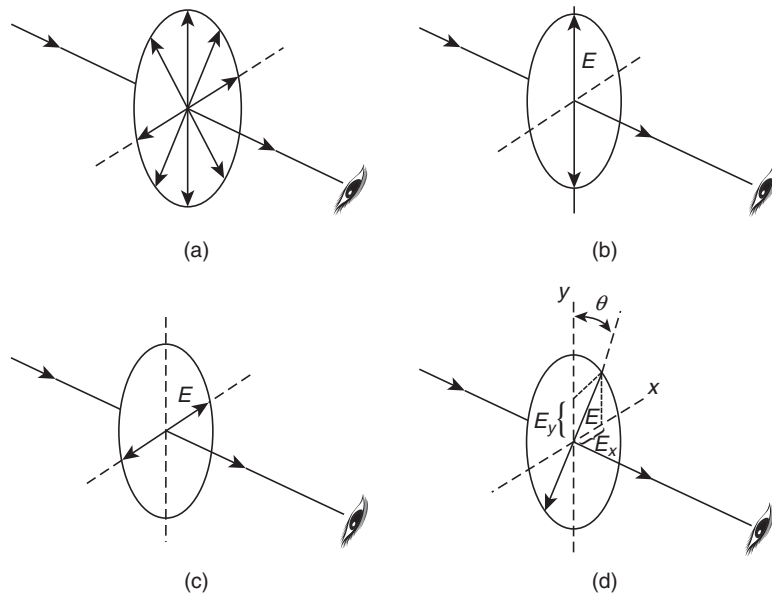


Figure 2.3 (a) Random vibrations of unpolarized light. (b) Linearly polarized light in a vertical direction. (c) Linearly polarized light in a horizontal direction. (d) Linearly polarized light in a direction making an angle θ with the vertical.

2.3 Polarization

Polarization arises from the direction of the E -field vector with respect to the direction of the light's propagation. A light wave's electric field vibrates in a direction perpendicular to its propagation, hence it is called a **transverse wave**, which is polarizable. If light is composed of vibrations in many different directions, with no preferred orientation, then it is **unpolarized** as shown in Fig. 2.3(a). Many light sources, for example arc lamps, incandescent bulbs and the sun produce unpolarized light. Vertically polarized light is shown in Fig. 2.3(b) and horizontally polarized light in Fig. 2.3(c). Each is an example of **linearly polarized light**. Figure 2.3(d) shows linearly polarized light making an angle of θ with the vertical. In this case, the tilted E vector can be described by its components, E_x and E_y .

If E_x and E_y are not in the same phase, that is, they do not reach their maxima and minima at the same time, then the E field does not remain oriented in a fixed, linear direction. Moreover, the light is *elliptically polarized*. *Circular polarization* is a special case of elliptical polarization. It occurs when E_x equals E_y and they are out of phase by 90° .

2.4 Diffraction

Diffraction is the phenomenon of bending of light rays at small apertures. It can be explained using Huygens' principle. In Fig. 2.4, a wave is incident on a barrier from the left. The barrier has a slit. Every point on the incident wave front can be considered as a new source of wave.

If the aperture is small compared to the wavelength of the light, the aperture behaves like a source and spherical waves result. As the slit width d increases, the diffracted wave becomes more and more like the incident plane wave, and a shadow is formed at the edges. The shadow is not precise and sharp as geometrical ray theory would predict, but rather diffracted a little into the dark region behind the obstacle, thus giving the shadow a fuzzy edge. This property of light that causes it to spread out as it travels by sharp edges or through tiny holes can be explained by light having wave-like properties.

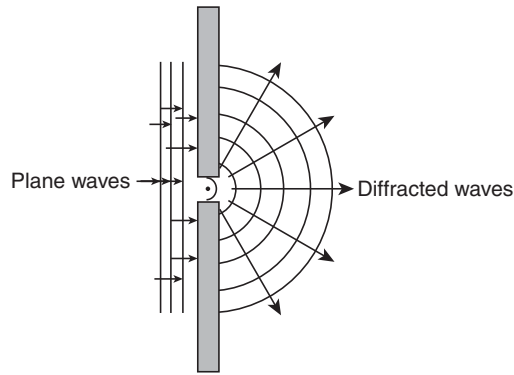


Figure 2.4 | Diffraction of light.

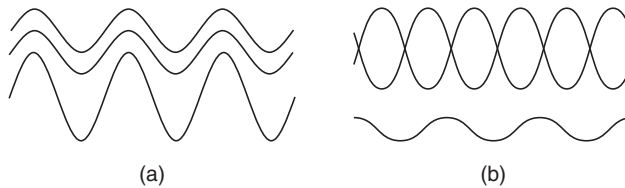


Figure 2.5 | (a) Two light waves in phase. (b) Two light waves out of phase.

Two light waves can interfere with each other to produce a resultant wave. If the two waves are in phase, the resultant wave will have a greater amplitude. However, if the two waves are out of phase by a value near $1/2$, the resultant wave will have a much reduced amplitude. Figure 2.5 shows two examples of interference of two light waves, with the resultant wave.

2.5 Reflection and Refraction

Light travels fastest in a vacuum. Its velocity in vacuum is $c = 3 \times 10^8$ m/s. In other substances the velocity of light is always less than 3×10^8 m/s. The ratio of the velocity of light in a vacuum to the velocity of light in another medium is called the **refractive index of the medium**. Refractive indices are always greater than 1.0. The refractive index of water (at 25°C) is 1.33. The refractive index of air (at 25°C and 1 bar) is 1.000293. In vacuum, light travels in a straight line at speed c , carrying energy from one place to another.

Figure 2.6 shows the refraction of light when it passes from air to glass. The frequency of the light wave does not change. But as it slows down in the glass, the wavelength becomes shorter, causing the light wave to bend.

Two key properties of light interacting with a medium are as follows:

1. It can be deflected upon passing from one medium to another (refraction).
2. It can be bounced off a surface (reflection).

Whenever a ray of light is incident on the boundary separating two different media, part of the ray is reflected back into the first medium and the remainder is refracted (bent in its path) as it enters the second medium. The degree of refraction or bending depends on the relative refractive indices of the two mediums according to Snell's Law. Mathematically it is given as

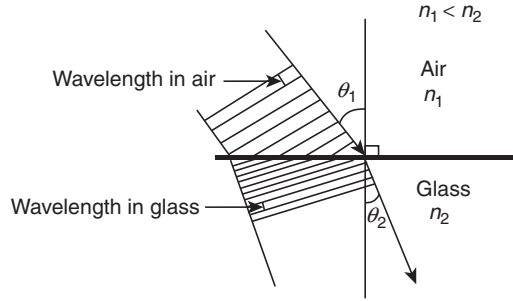


Figure 2.6 | Light refraction at the boundary of two media.

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1} \quad (2.2)$$

where n_1 and n_2 are the refractive indices of the two media, θ_1 is the angle of incidence, and θ_2 is the angle of refraction.

2.6 Total Internal Reflection

A light ray is propagating in a dielectric having a refractive index n_1 . It is incident at an angle θ_1 to the normal at the boundary of two mediums. The other medium is having refractive index n_2 , which is less than n_1 . The ray is

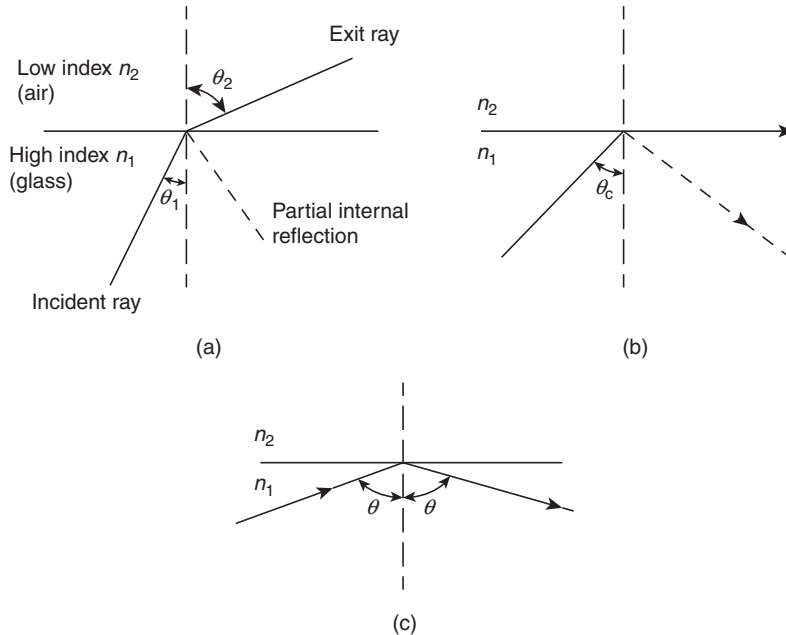


Figure 2.7 | (a) Refraction. (b) Critical angle. (c) Total internal reflection.

refracted at an angle θ_2 as shown in Fig. 2.7(a). A small amount of light is reflected back into the same medium and θ_2 is greater than θ_1 . Therefore, by Snell's law we can write

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (2.3)$$

As the value of θ_1 goes on increasing, the value of θ_2 also increases. At a particular stage, the value of θ_2 becomes 90° . This is called the **critical angle**. The refracted ray travels parallel to the interface between the dielectrics as shown in Fig. 2.7(b). This is the limiting case of refraction. The critical angle (θ_c) is given by

$$\sin \theta_c = \frac{n_2}{n_1} \quad (2.4)$$

If the angle of incidence is greater than critical angle, then the light is reflected back into the same medium. This is called **total internal reflection (TIR)** as shown in Fig. 2.7(c).

2.7 Waveguides

Waveguides are used to transfer electromagnetic power from one point in space to another. With the help of guided-wave optics technology, long-distance light transmission without the use of relay lenses is possible. A medium of one refractive index is embedded in a medium of lower refractive index to obtain optical confinement. The medium of high refractive index acts as a light trap. The light travels by multiple TIRs at the internal interfaces (boundaries). An optical waveguide is a light conductor consisting of a slab, strip, or cylinder of dielectric material surrounded by another dielectric material of a lower refractive index as shown in Fig. 2.8. Light is transported through the inner medium without penetrating into the surrounding medium. The most commonly used waveguide is the optical fiber, which is made of two concentric cylinders of low-loss dielectric material such as glass or plastic polymer.

In practice, the choice of the structure of waveguides is dictated by

1. The desired operating frequency.
2. The amount of power to be transferred.
3. The amount of transmission losses that can be tolerated.

2.8 Light Propagation in Optical Fibers

Optical fiber is a dielectric waveguide or medium in which information (voice, data, or video) is transmitted through a glass or plastic fiber, in the form of light. The basic structure of an optical fiber is shown in Fig. 2.9.

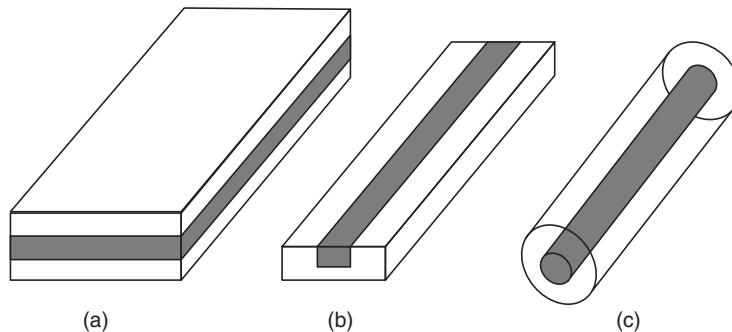


Figure 2.8 | Optical wave guides: (a) Slab; (b) strip; (c) fiber.

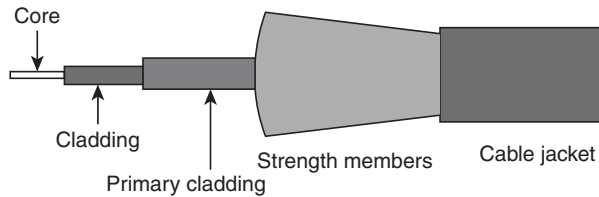


Figure 2.9 | Fiber-optic cable.

It consists of a transparent core with a refractive index n_1 surrounded by a transparent cladding of a slightly lower refractive index n_2 . The refractive index of the cladding is less than 1%, lower than that of the core. Typical values, for example, are a core refractive index of 1.47 and a cladding index of 1.46. The cladding supports the waveguide structure, protects the core from absorbing surface contaminants, and when adequately thick, substantially reduces the radiation loss to the surrounding air. Glass core fibers tend to have low loss in comparison with plastic core fibers. Additionally, most of the fibers are encapsulated in an elastic, abrasion-resistant plastic material that mechanically isolates the fibers from small geometrical irregularities and distortions. A set of guided electromagnetic waves, also called the modes of the waveguide, can describe the propagation of light along the waveguide. Only a certain number of modes are capable of propagating through the waveguide.

As light is launched into the fiber, it is confined to the fiber and travels by phenomenon of TIR. The refractive index determines the speed of the wave inside the material. The velocity of light v inside any material is given by the following expression:

$$v = \frac{c}{n} \quad (2.5)$$

where c is the *velocity* of light in free space $= 3 \times 10^8$ m/s and n is the refractive index. The light is guided in the core of an optical fiber by TIR at the boundary of the lower-index cladding.

2.8.1 Construction of a Fiber-Optic Cable

A fiber-optic cable consists of the following components:

1. **Core:** The central part of the fiber-optic cable that carries the light signal is called the fiber core. It is made up of silica glass. Fiber core is hair-thin in size. The diameter of the fiber core is typically $8 \mu\text{m}$ for single-mode fiber and $50 \mu\text{m}$ or $62.5 \mu\text{m}$ for multimode fiber.
2. **Cladding:** The layer that surrounds the fiber core is called the cladding. It is also made up of glass. The diameter of the cladding is typically $125 \mu\text{m}$. It helps in the light confinement inside the core. The refractive index of the cladding is always less than that of the core.
3. **Primary coating:** After the cladding, the primary coating made up of plastic is present. It is also known as the primary buffer. This layer does not participate in light propagation. It provides mechanical protection to the fiber core and cladding.
4. **Strength members:** The next layer is made up of aramid yarn or Kevlar strength members. This prevents the breakage of the fiber glass during installation. When fiber is pulled through a duct, the outer cover would stretch and the pulling load would be rested on the fiber. The strength members prevent this as their material is designed to take the strain.
5. **Cable jacket:** The last layer is the cable jacket. It provides mechanical protection to the fiber core and the cladding inside.

2.8.2 Light Propagation – Ray Theory

The transmission of light along an optical fiber follows two theories: ray theory and mode theory.

2.8.2.1 Ray Theory

This theory describes light as a simple ray. Ray theory is used to approximate the light acceptance and guiding properties of the optical fiber. The rays can be propagated by three paths along an optical fiber; these are the meridional, skew, and axial rays.

2.8.2.2 Different Rays

We have seen that rays approaching from within the cone of acceptance are successfully propagated along the fiber. The exact path taken by the ray depends on the position and the angle at which the ray strikes the core. There are three possible paths: *skew*, *meridional*, and *axial rays* as shown in Fig. 2.10.

The *skew ray* never passes through the center of the core. It reflects off the core–cladding interface and bounces around the outside of the core. It moves forward in a path similar to a spiral staircase as shown in Fig. 2.10(a).

The *meridional ray* passes through center of the core and is then reflected from the parallel surfaces of core and passes through the center. In the end view it appears as a horizontal line as shown in Fig. 2.10(b). It should be noted that the *axial ray* travels straight through the center of the core as shown in Fig. 2.10(c).

We can explain the basic transmission properties of optical fiber using these rays. Meridional rays are of two types: bound and unbound rays. Bound rays propagate through the fiber by TIR. They remain in the core and propagate along the axis of the fiber. Unbound rays are refracted out of the fiber core into the cladding as shown in Fig. 2.11.

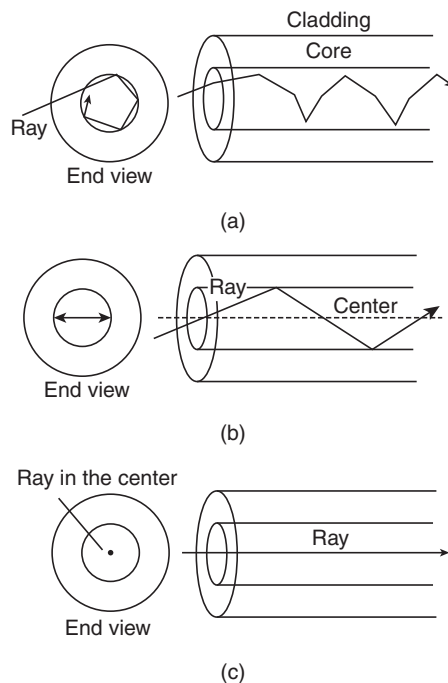


Figure 2.10 | (a) Skew ray. (b) Meridional ray. (c) Axial ray.

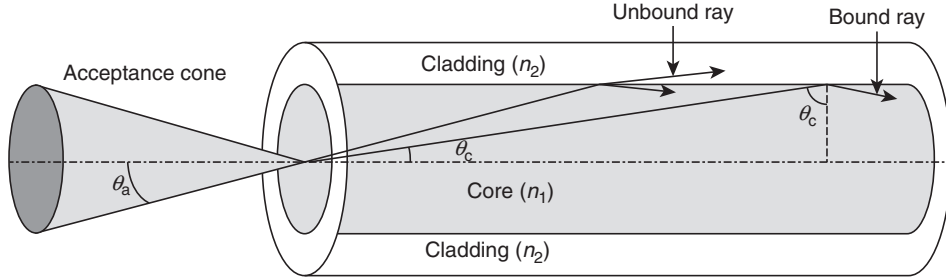


Figure 2.11 | Acceptance cone.

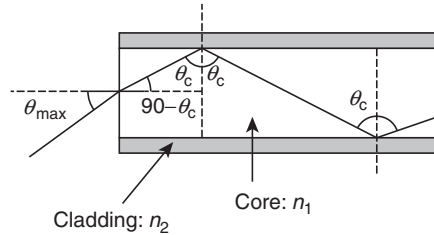


Figure 2.12 | Light propagation through the core by TIR.

In a perfect core–cladding interface, the bound rays remain in the core. A part of bound rays will be refracted out of the cladding due to imperfections at the core–cladding boundary. Light rays refracted into cladding will eventually escape from the fiber.

Meridional rays follow the laws of reflection and refraction. Bound rays propagate along the fiber due to TIR. Rays that enter the fiber must intersect the core–cladding interface at an angle greater than the critical angle θ_c . Only the rays that enter the fiber and strike the interface at angles greater than θ_c will be propagated along the fiber.

In order to be transmitted, the light ray incident on the fiber core must be within the acceptance cone defined by the acceptance angle. This is the angle over which light rays entering the fiber will be guided along its core. The acceptance angle is normally measured in terms of NA.

NA is a measurement of the ability of an optical fiber to capture light. It is a dimensionless quantity that specifies the angular range over which an optical fiber can accept light. In an optical system, NA is given by

$$NA = n_0 \sin(\theta_a) \quad (2.6)$$

where n_0 is the index of refraction of the surrounding medium and θ_a is half angle of the maximum cone of the light that can enter the optical system.

When a light ray is incident from a medium of refractive index n_0 to the core of index n_1 , Snell's law at a medium–core interface gives

$$n_0 \sin \theta_i = n_1 \sin \theta_r \quad (2.7)$$

where θ_i is the angle of incidence at the core and θ_r is the angle of refraction at medium core interface. From Fig. 2.12, using trigonometry, we get

$$\sin \theta_r = \sin(90 - \theta_c) = \cos \theta_c \quad (2.8)$$

where $\theta_c = \sin^{-1}(n_2/n_1)$ is the critical angle for TIR.

Substituting for $\sin \theta_r$ in Snell's law we get

$$\frac{n_0}{n_1} \sin \theta_i = \cos \theta_c \quad (2.9)$$

Squaring both sides, we get

$$\frac{n_0^2}{n_1^2} \sin^2 \theta_i = \cos^2 \theta_c = 1 - \sin^2 \theta_c = 1 - \frac{n_2^2}{n_1^2} \quad (2.10)$$

Thus,

$$n_0 \sin \theta_i = \sqrt{n_1^2 - n_2^2} \quad (2.11)$$

From Eqs. (2.6) and (2.11), we get

$$NA = \sqrt{n_1^2 - n_2^2} \quad (2.12)$$

where n_1 is the refractive index of the fiber core and n_2 is the refractive index of the cladding.

Core-cladding index difference Δ , also called fractional refractive index change, is given as

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \approx \frac{n_1 - n_2}{n_1} \quad (2.13)$$

Hence

$$NA = \sqrt{n_1^2 - n_2^2} = n_1 \sqrt{2\Delta} \quad (2.14)$$

The relationship between the acceptance angle, the indices of refraction – of the core (n_{core}), the cladding (n_{cladding}), the surrounding medium (n_0) – and NA is given by

$$NA = n_0 \sin(\theta_a) = \sqrt{n_{\text{core}}^2 - n_{\text{cladding}}^2} \quad (2.15)$$

Usually the surrounding medium is air, so $n_0 = 1$. In this case,

$$NA = \sin(\theta_a) \quad (2.16)$$

The acceptance angle of the fiber is given by

$$\theta_a = \sin^{-1}(NA) \quad (2.17)$$

NA is commonly used to specify a multimode fiber. It is a way of measuring the light-gathering ability of an optical fiber. It is used to measure source-to-fiber power-coupling efficiencies. A high NA indicates a high source-to-fiber coupling.

2.8.3 V Number

V number is a unit-less quantity; it is also called **normalized frequency**. The number of guided modes through a single-mode or multimode fiber depends upon the relative refractive index difference and the core radius of the fiber. V number also depends on these parameters. Therefore, V number and the number of guided modes are indirectly related to each other as follows:

$$\text{Number of guided modes in a step-index multimode fiber} = V^2/2$$

There is a cutoff value of V number, denoted by V_c , below which guided modes do not exist. The modes propagate as unguided modes. A step-index fiber becomes single-mode for $V < 2.405$. Now

$$V = \sqrt{U^2 + W^2} \quad (2.18)$$

where U is the radial propagation constant and W is the cladding decay parameter. U and W are given by

$$U = a \sqrt{n_1^2 k^2 - \beta^2} \quad (2.19)$$

and

$$W = a \sqrt{\beta^2 - k^2 n_2^2} \quad (2.20)$$

where a is the radius of the core, n_1 is the refractive index of core, n_2 is the refractive index of cladding, β is the propagation constant, and $k = 2\pi/\lambda$. Now

$$\begin{aligned} V^2 &= U^2 + W^2 \\ &= a^2 (n_1^2 k^2 - \beta^2) + a^2 (\beta^2 - k^2 n_2^2) \\ &= a^2 n_1^2 k^2 - a^2 \beta^2 + a^2 \beta^2 - a^2 k^2 n_2^2 \\ &= a^2 k^2 (n_1^2 - n_2^2) \end{aligned}$$

This implies

$$V = ak \sqrt{n_1^2 - n_2^2}$$

We know that

$$NA = \sqrt{n_1^2 - n_2^2}$$

Therefore

$$V = a \times k \times (NA) = \frac{2\pi}{\lambda} \times a \times (NA) = \frac{2\pi}{\lambda} \times a \times n_1 (2\Delta)^{1/2} \quad (2.21)$$

2.8.4 Fiber as a Dielectric Waveguide (Mode Theory)

In order to get a good understanding of the principal idea of the mode concept, without too much complex mathematics, the following simplified waveguide is considered: a plane layer which is infinite in extent in the lateral directions (y and z), but bounded by two parallel planes with respect to one direction (x). The refractive index between these two planar boundaries is n_1 , and that of the surrounding medium is n_2 , as shown in Fig. 2.13.

If the angular condition for total internal reflectance is met, then the light launched will be confined to the core because of its high refractive index as compared to cladding. The discrete set of guided electromagnetic waves is called the **modes of the waveguide**; it is determined by the fiber composition and geometry.

The general wave equation can be written as

$$\nabla^2 U = \frac{n^2}{c_v^2} \frac{\partial^2 U}{\partial t^2} \quad (2.22)$$

where U is the time-dependent wave function, n is the refractive index, and c_v is the speed of light in vacuum. The electric field can be written as

$$E(x, y, z, t) = E(x, y)e^{i(\omega t - \beta z)} \quad (2.23)$$

where $E(x, y)$ is the transverse electric field distribution which has to be solved, ω is the angular frequency, and β is a parameter called the propagation constant.

By combining Eqs. (2.22) and (2.23), we get

$$\frac{\partial^2 E(x, y)}{\partial x^2} + \frac{\partial^2 E(x, y)}{\partial y^2} + (k^2 n^2 - \beta^2) E(x, y) = 0 \quad (2.24)$$

in which

$$k = \frac{\omega}{c_v} = \frac{2\pi}{\lambda}$$

and

$$\omega = 2\pi\nu$$

where k is the wave number, λ is the wavelength, and ν is the frequency of the considered light.

Since the waveguide is assumed to be infinite in the y -direction, and because n is different inside and outside the waveguide, two separate equations for the different regions can be written.

Inside the waveguide, the wave equation is

$$\frac{\partial^2 E(x, y)}{\partial x^2} + (k^2 n_1^2 - \beta^2) E(x, y) = 0 \quad (2.25)$$

and outside the waveguide, the wave equation is

$$\frac{\partial^2 E(x, y)}{\partial x^2} + (k^2 n_2^2 - \beta^2) E(x, y) = 0 \quad (2.26)$$

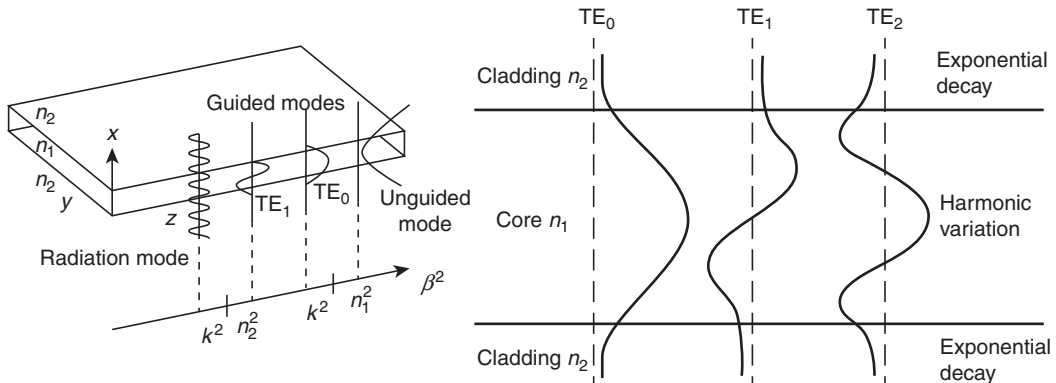


Figure 2.13 | Modes in a planar waveguide.

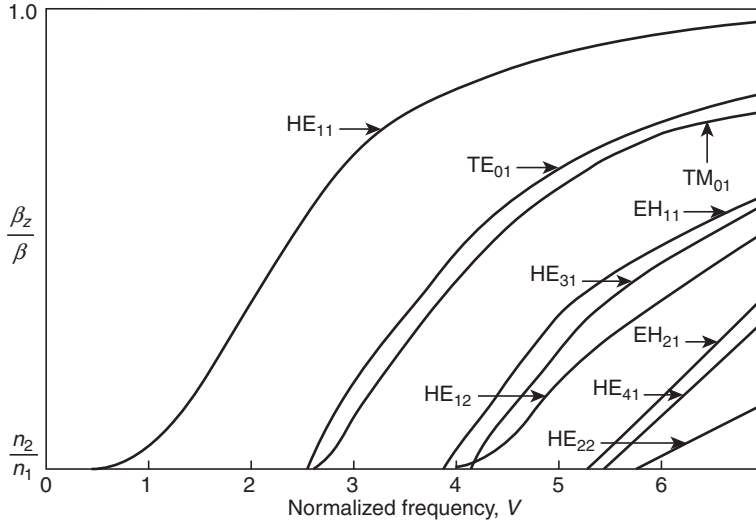


Figure 2.14 | Plots of the propagation constant as a function of normalized frequency for a few of the lowest-order modes.

Depending on the sign of the factor $(k^2 n_2^2 - \beta^2)$, the solutions of this simple differential equation are either sinusoidal or exponential functions. Therefore three different cases can be distinguished, which are graphically shown in Fig. 2.13.

1. $\beta^2 > k^2 n_1^2$: The factor $(k^2 n_2^2 - \beta^2)$ is negative in both regions ($n_1 > n_2$), and the solutions are exponential functions of x . These modes are called *unguided modes*.
2. $\beta^2 > k^2 n_2^2$: These modes are called *radiation modes*, and these are sinusoidal in both regions. Since the surroundings are not limited, all the energy will dissipate out of the waveguide.
3. $k^2 n_2^2 < \beta^2 < k^2 n_1^2$: These modes are called *guided modes*. Only in this case the solutions are sinusoidal functions inside the waveguide and exponentially decreasing functions outside. The electric field is now bounded to the waveguide. There are several guided modes in a waveguide, all with a different quantum number $m = 0, 1, 2$, etc. In other words, the propagation constant β is restricted to certain values, which represent the different modes. In Fig. 2.14, the zeroth ($m = 0$) and first-order ($m = 1$) transverse electric modes are drawn as TE_0 , TE_1 .

Because of the specific geometry of a fiber, it is easier to work with cylindrical coordinates. The wave equation is similar to Eq. (2.25). But instead of z , the radial parameter r will be used. Here, r is the radial distance from the centre of the fiber. Moreover, the angular dependency will lead to the introduction of another quantum number l ($l = 0, 1, 2$, etc.). The wave equation can therefore be written as

$$\frac{r^2 \partial^2 R}{\partial r^2} + \frac{r \partial R}{\partial r} + [(k^2 n^2 - \beta^2) r^2 - l^2] R = 0 \quad (2.27)$$

The solutions of this type of differential equations are **Bessel functions**. Because of the boundary condition, again a second quantum number m has to be introduced. The solutions of the differential equations are called TE_{lm}

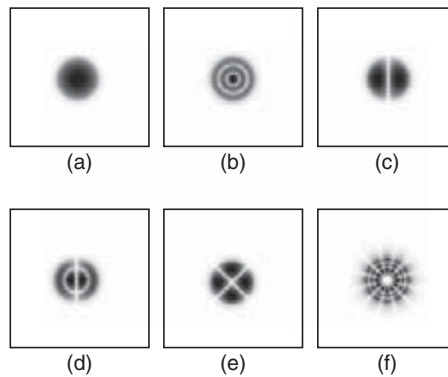


Figure 2.15 | Intensity patterns of some fiber mode examples: (a) TE₀₁, (b) TE₀₂, (c) TE₁₁, (d) TE₁₂, (e) TE₂₁, (f) TE₆₂.

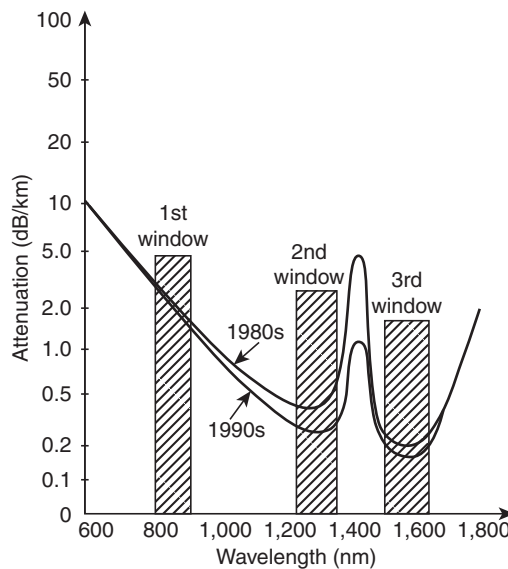


Figure 2.16 | Typical spectral attenuation in silica.

modes and some examples of the resulting intensity patterns are shown in Fig. 2.15. This figure clearly shows that whereas the zigzag paths of ray optics would lead to intensity distributions that change depending on the length of the fibers, the wave model provides a constant light–dark distribution that is independent of the length across the waveguide’s cross-section.

2.8.4.1 Attenuation

Due to absorption and scattering losses, optical power propagating in a fiber decays exponentially with length. Decay in the propagating power is nothing but attenuation. It is a wavelength-dependent phenomenon as shown

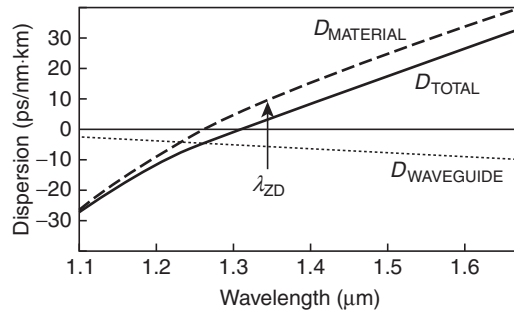


Figure 2.17 | Material and waveguide dispersion.

in Fig. 2.16. It is the most important parameter that determines spacing of repeaters needed to maintain acceptable signal levels and hence the cost of fiber-optic telecommunication systems.

In commercial fibers, the dominant cause of absorption is impurities in the form of hydroxyl ions. In the near-infrared and visible regions, the small absorption losses of pure silica are due to the tails of absorption bands in the far-infrared and ultraviolet. Due to recent advancements in fabrication techniques, fiber purity has greatly improved, leading to reduced attenuation losses. State-of-the-art systems can have attenuation to the tune of 0.1 dB/km.

Scattering can couple energy from guided to radiation modes, causing loss of energy from the fiber. Rayleigh scattering losses occur due to small-scale index fluctuations. This produces attenuation proportional to $1/\lambda^4$. Any irregularities in core diameter and geometry or changes in fiber axis direction also cause scattering.

2.8.4.2 Bandwidth Limitations

The amount of information that can be supported by an optical fiber determines the bandwidth, that is, it represents the maximum data rate. The maximum bandwidth of an optical fiber is limited due to the spreading of optical pulses. This spreading increases with distance and the pulses overlap each other. After some distance, the overlapping increases to such an extent that the pulses become indistinguishable. This pulse spreading phenomenon is known as **dispersion**. There are two main categories of dispersion: intermodal and intramodal. It is a wavelength-dependent phenomenon. Figure 2.17 shows the variation of material and waveguide dispersion with wavelength.

2.8.4.2.1 Intermodal Dispersion

Intermodal dispersion is applicable only to multimode fibers. It is a phenomenon between different modes in an optical fiber. In practice, light waves are composed of number of plane wave components of different frequencies. This group of waves with closely similar frequencies propagates together at *group velocity*. The different propagating modes have different group velocities; therefore each mode will reach at the end of a fixed distance at a different time. Therefore the transmitted pulse through a multimode fiber spreads.

2.8.4.2.2 Intramodal Dispersion

Intramodal dispersion is related to the material properties of an optical fiber. It occurs within the same mode; hence it is applicable to single-mode as well as multimode fiber. There are two distinct types of intramodal dispersion: material dispersion and waveguide dispersion.

Optical sources are not perfectly monochromatic. The source spectrum has a finite width ($\Delta\lambda$). This means that the output contains different wavelengths with a slight variation ($\lambda, \lambda \pm \Delta\lambda$). Therefore different

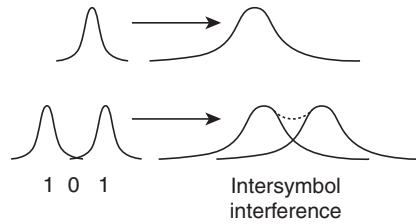


Figure 2.18 | Dispersion.

wavelengths will travel down an optical fiber at different velocities. This type of dispersion caused by a range of source wavelengths is called as **chromatic dispersion**. The pulse broadening due to dispersion is shown in figure 2.18. A parameter called full width at half maximum (FWHM) is used to measure dispersion. FWHM is the width of a pulse at the point where the signal power is at least half of maximum. A pulse with a wider FWHM will spread more than a pulse with a narrower FWHM. Therefore, a laser source with single longitudinal mode with very narrow line width is preferred for long-haul optical communication systems.

2.8.4.2.3 Polarization Mode Dispersion

Another type of material dispersion is polarization mode dispersion (PMD). If the fiber is not perfectly symmetric and homogeneous (i.e., if its refractive index is not uniform), then PMD arises. Ideally, the core of an optical fiber has an index of refraction that is uniform over the entire cross-section. Due to small changes in glass composition, geometry, and induced local strains during fiber drawing or cabling, there is slight change in refractive index. This can cause one of the orthogonal polarization modes to travel faster than the other. Due to this difference in travel speed, dispersion occurs.

2.9 Types of Optical Fibers

According to the number of modes that can propagate through optical fibers, they are classified as

1. Single-mode fibers.
2. Multimode fibers.

Single-mode fibers can propagate only the fundamental mode. Multimode fibers can propagate hundreds of modes.

According to an optical fiber's refractive index profile (the value of refractive index as a function of radial distance at any fiber diameter) and core size, further classification is as follows:

1. Multimode step-index fibers.
2. Multimode graded-index fibers.
3. Single-mode step-index fibers.
4. Single-mode graded-index fibers.

In a step-index fiber, the refractive index of the core is uniform, and there is a step change at the core-cladding interface. However, in graded-index fibers, the refractive index of the core is not constant throughout. It decreases gradually as a function of the radial distance from the fiber center.

Single-mode and multimode fibers can have a step-index or graded-index refractive index profile. Single-mode fibers have small diameters; therefore it is difficult to fabricate graded-index profile in single-mode fibers. The performance of single-mode graded-index fiber is also not very advantageous. Hence, single-mode fibers with

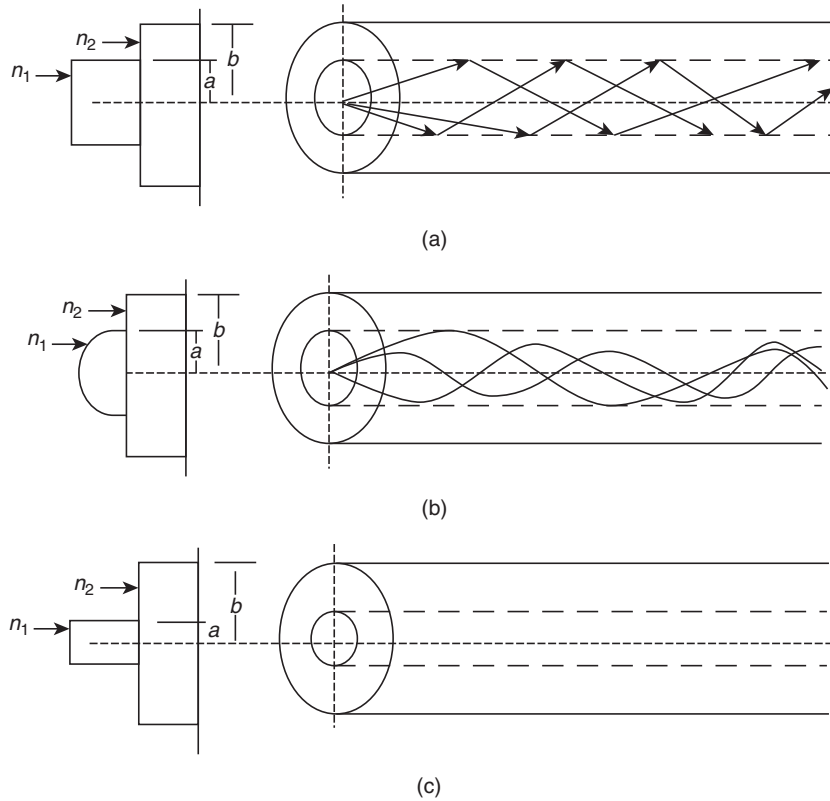


Figure 2.19 | The refractive index profiles and light propagation in (a) multimode step-index, (b) multimode graded-index, and (c) single-mode step-index fibers.

only step-index profile are fabricated. Depending upon the application, either multimode step-index profile or graded-index profile is chosen but the performance of multimode graded-index fibers is usually superior to that of multimode step-index fibers. Fiber transmission properties – such as attenuation and dispersion and hence system performance – may get affected due to a small change in core size and material composition. Figure 2.19 shows the refractive index profile for a multimode step-index fiber, a multimode graded-index fiber, and a single-mode step-index fiber. The figure also shows the propagation of light along each fiber.

The core size and the material of fiber are the two important selection criteria for fiber selection in optical system design.

Standard core sizes for various fibers are as follows:

1. Multimode step-index fibers: 50 μm and 100 μm .
2. Multimode graded-index fibers: 50 μm , 62.5 μm , 85 μm , and 100 μm .
3. Single-mode fibers: 8 μm and 10 μm .

The material from which optical fibers are fabricated also plays an important role in determining its properties. Optical fibers are fabricated from a high-quality glass (SiO_2) or plastic. With increase in purity of the fabrication material, the losses get reduced and the system performance improves. In order to get a slight variation in refractive index, a small amount of impurities or dopants like water or elements other than silica and oxygen are added to the glass material. Thus different refractive index profiles are obtained.

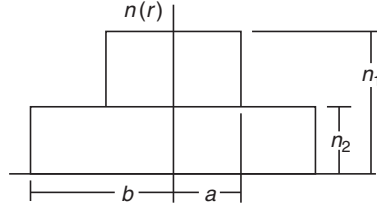


Figure 2.20 | Refractive index profile for multimode step-index fibers.

2.9.1 Multimode Step-Index Fibers

A multimode step-index fiber has a core of radius a and a constant refractive index n_1 . A cladding of slightly lower refractive index n_2 surrounds the core. Figure 2.20 shows the refractive index profile $n(r)$ for this type of fiber. Notice the step decrease in the value of refractive index at the core–cladding interface. Mathematically, we can write $n(r)$ as

$$n(r) = \begin{cases} n_1, & r < a \text{ (core)} \\ n_2, & r \geq a \text{ (cladding)} \end{cases}$$

The number of modes supported by a multimode step-index fiber depends on:

1. Transmission wavelength (λ).
2. Relative refractive index difference (Δ).
3. Core radius a of the fiber.

A finite number of guided modes can travel through a multimode step-index fiber, and there is a cut-off value of normalized frequency below which guided modes do not exist. Below the cut-off, only unguided or leaky modes can propagate. Unguided modes travel some distance and get lost in the cladding, while guided modes are confined to the fiber core over the full length of communication. Therefore, in multimode step-index fibers, the effect on mode propagation due to cladding properties, such as cladding diameter, is very limited.

The large core size (typically 50 μm or 100 μm) and large NA of multimode step-index fibers make it easier to couple light from a cheap light-emitting diode (LED) into the fiber, thus eliminating the need for a costly laser source. But the bandwidth capabilities and hence the information-carrying capabilities of multimode step-index fiber are limited due to the modal dispersion effect. Multimode step-index fiber selection depends on system application and design. Typically multimode step-index fibers are used in short-haul, limited bandwidth, low-cost applications.

2.9.2 Multimode Graded-Index Fibers

A multimode graded-index fiber has a core of radius a . The refractive index n_1 decreases as a function of the radial distance r . A cladding of slightly lower refractive index n_2 surrounds the core. Figure 2.21 shows the refractive index profile $n(r)$ for this type of fiber. Mathematically, we can write

$$n(r) = \begin{cases} n_1 \left(1 - 2\Delta \left(\frac{r}{a} \right)^\alpha \right)^{1/2}, & r < a \text{ (core)} \\ n_2 = n_1 (1 - 2\Delta)^{1/2}, & r \geq a \text{ (cladding)} \end{cases}$$

where Δ is the relative refractive index difference and α is the profile parameter.

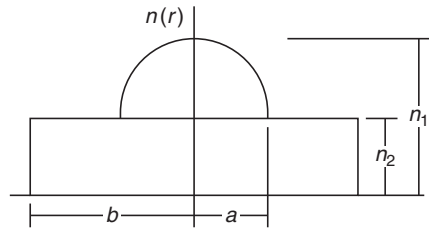


Figure 2.21 | The refractive index profile for multimode graded-index fibers.

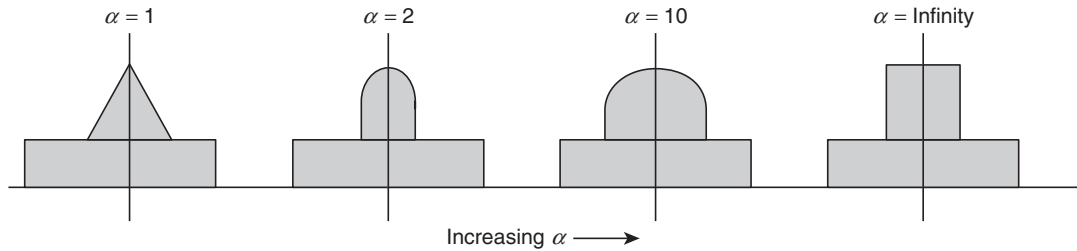


Figure 2.22 | Refractive index profiles for different values of alpha (α).

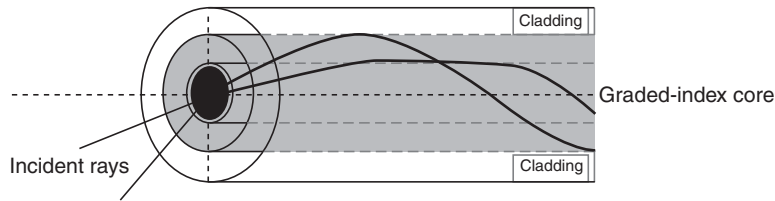


Figure 2.23 | Refractive index grading and light propagation in multimode graded-index fibers.

The *profile parameter* α determines the shape of the core's profile. As the value of α increases, the shape of the core's profile changes from a triangular shape to a step as shown in Fig. 2.22. Multimode fibers with parabolic graded-index profiles provide the best performance; hence most multimode graded-index fibers have a parabolic refractive index profile. Unless specified, assume that the core's refractive index profile is parabolic ($\alpha = 2$). The parabolic refractive index profile of the core causes multimode graded-index fibers to accept less light, but have less modal dispersion. Therefore, its performance is better than that of step-index fiber.

Light propagates in multimode graded-index fibers according to refraction and TIR. The gradual decrease in the core's refractive index from the center of the fiber creates much refraction of the rays because there are many high-to-low-index interfaces. The light rays become refracted or curved, which increases the angle of incidence at the next point of refraction to meet the condition of TIR. TIR occurs when the angle of incidence becomes larger than the critical angle of incidence. Figure 2.23 shows the process of refraction and TIR of light in multimode graded-index fibers.

In a graded-index fiber, NA is a function of radial distance from the fiber axis. Therefore its NA is also called *axial NA*. Its value is maximum at the fiber axis and gradually decreases. Hence, a graded-index fiber accepts less light as compared to step-index fiber.

A multimode graded-index fiber with the same normalized frequency as a multimode step-index fiber will have less propagating modes. The NA of graded-index (GI) fibers is a function of position across the core end face.

The local numerical aperture $\text{NA}(r)$ is given by

$$\text{NA}(r) = \begin{cases} \sqrt{n^2(r) - n_2^2} = \text{NA}(0) \sqrt{1 - \left(\frac{r}{a}\right)^\alpha} & \text{for } r \leq a \\ 0 & \text{for } r > a \end{cases} \quad (2.28)$$

where r is the radial distance from the fiber axis, a is the core radius, n_1 is the refractive index of the core, n_2 is the refractive index of cladding, α is the profile parameter, and $\text{NA}(0)$ is the axial NA, given by

$$\text{NA}(0) = \sqrt{n^2(0) - n_2^2} = \sqrt{n_1^2 - n_2^2} = n_1 \sqrt{2\Delta} \quad (2.29)$$

The number of modes in multimode graded-index fiber is given by

$$M_g = \left(\frac{\alpha}{\alpha + 2} \right) \left(\frac{V^2}{2} \right) \quad (2.30)$$

where α is the profile parameter and V is the normalized frequency.

2.9.2.1 Fiber Birefringence

Modes propagate with different phase velocities in an optical fiber, and the difference between their effective refractive indices is called *fiber birefringence*. Mathematically we can write the expression for fiber birefringence as

$$B_f = n_x - n_y \quad (2.31)$$

where n_x and n_y are the effective refractive indices of each mode (i.e., x and y mode).

2.9.2.2 Fiber Beat Length

When light is injected into the fiber such that both the modes are excited, one mode will be delayed in phase relative to the other mode as they propagate through fiber. When the phase difference is an integral multiple of 2π , the two modes will *beat* at that point, and the input polarization state is reproduced. The length over which this beat occurs is called the fiber beat length. It is given as

$$L_B = \frac{2\pi}{\beta_x - \beta_y} = \frac{\lambda}{n_x - n_y} = \frac{\lambda}{B_f} \quad (2.32)$$

where β_x and β_y are the propagation constants of the two orthogonal polarization modes, n_x and n_y are the effective refractive indices of two orthogonal modes, λ is the wavelength of injected light, and B_f is the birefringence.

A lower value of L_B means a higher value of $\beta_x - \beta_y$, which in turn corresponds to a fiber with high polarization-holding capacity. L_B gives the distance along the fiber after which the phase difference between the two fundamental modes becomes 2π . The state of polarization of the fundamental modes coupled in the fiber will be repeated after each distance L_B along the fiber.

Figure 2.23 shows possible paths that light may take when propagating in multimode graded-index fibers. The refractive index at the center is maximum. Moreover, the light ray has to travel less distance as compared to the ray away from the fiber axis, and speed of travel is less.

For the light ray traveling away from the fiber axis, the refractive index gradually decreases. It has to travel more distance and the speed is comparatively high. Light rays near the fiber axis travel less distance with less speed, and light rays away from the fiber axis travel more distance with faster speed; so they will arrive at each point along the fiber at nearly the same time. The decrease in time difference between the arrivals of light rays reduces modal dispersion and increases fiber bandwidth.

Multimode graded-index fiber is preferred in many systems because of following advantages:

1. Relatively high source to fiber-coupling efficiency.
2. Low loss.
3. Low sensitivity to microbending and macrobending.
4. High bandwidth.
5. Expansion capability.

Typically, the core diameter in a multimode fiber is 50, 62.5, or even 1,000 μm ; it has an operating wavelength range starting in the visible light region (650 nm) and a relative refractive index, Δ , that is a minimum of 1% and, typically, 2% and higher.

2.10 Single-Mode Step-Index Fibers

There are two types of single-mode step-index fiber: matched clad and depressed clad. In matched clad fiber the cladding has uniform refractive index distribution, while depressed clad fiber cladding has two regions, inner and outer, having different refractive indices n_2 and n_3 , respectively and $n_2 < n_3$ as shown in Fig. 2.24(b). A single-mode step-index matched clad fiber has a core of radius a and a constant refractive index n_1 . A cladding of slightly lower refractive index n_2 surrounds the core. Figure 2.24(a) shows the refractive index profile $n(r)$ for the step-index single-mode matched clad fiber.

A single-mode fiber supports only one mode of propagation in contrast to a multimode fiber, through which many modes of propagation are allowed. As already seen in Section 2.8.4.2.1, the multimode step-index fiber suffers from severe intermodal dispersion problem; and this single-mode step-index fiber is the solution to the problem. For a single-mode step-index fiber, NA ranges from 0.14 to 0.5.

The V number is given by

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda(\text{NA})} \quad (2.33)$$

$$= \frac{2\pi a}{\lambda} n_1 \sqrt{2\Delta} \leq 2.405 \quad (2.34)$$

where a is the core radius and λ is the cutoff wavelength.

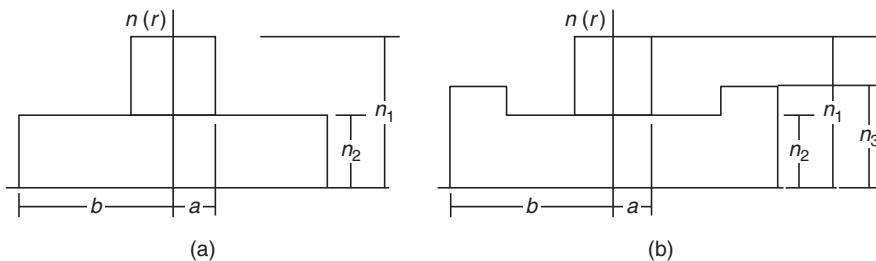


Figure 2.24 | Step-index single-mode fiber refractive index profile. (a) Matched clad fiber. (b) Depressed clad fiber.

The total number of modes is given by

$$M_s = \frac{V^2}{2} \quad (2.35)$$

The total average cladding power is given by

$$\left(\frac{P_{\text{clad}}}{P} \right)_{\text{total}} = \frac{4}{3} M_s^{-1/2} \quad (2.36)$$

Single-mode step-index fibers propagate only one mode, called the fundamental mode. Single-mode operation occurs when the value of the fiber's normalized frequency (V) is between 0 and 2.405 ($0 < V < 2.405$). The value of V should remain near the 2.405 level. This is because when the value of V is less than one, single-mode fibers carry a majority of the light power in the cladding material. The portion of light transmitted by the cladding material easily radiates out of the fiber. For example, light radiates out of the cladding material at fiber bends and splices. Therefore, to arrange the fiber so that it conducts only one mode, we can either decrease the core diameter d , increase the operating wavelength λ , or make n_2 as close to n_1 as possible.

Manufacturers use all these means to achieve their goal. The core diameter of a single-mode fiber is around 2–10 μm . Typically, the operating wavelength range starts at 1,300 nm, and the relative refractive index difference, $\Delta = (n_1 - n_2)/n_1$, is less than 0.4%. This fiber rejects all higher-order modes and conducts only one fundamental mode.

2.10.1 Gaussian Beam

If the cross-section of optical beam is observed, we can see that the beam is most intense in the center, with its intensity declining gradually from the center outward as shown in Fig. 2.25(b). This curve describes the change of a beam's intensity as a function of the beam's cross-sectional radius. The most popular model used in single-mode fibers is a Gaussian curve, given by

$$I(a) = I(0) \exp\left(-\frac{2a^2}{w_0^2}\right) \quad (2.37)$$

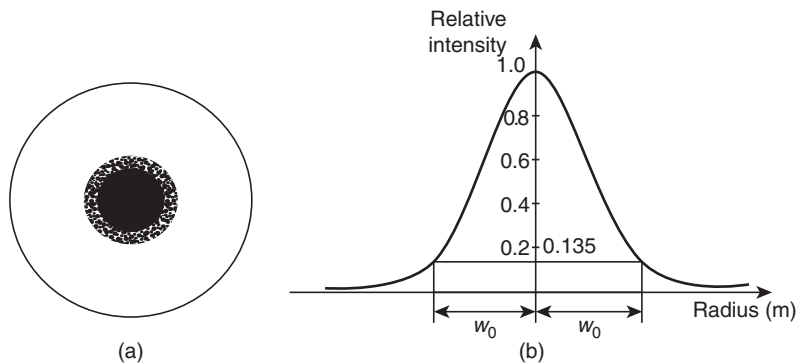


Figure 2.25 | Gaussian beam: (a) Physical appearance of the cross-sectional beam's structure; (b) Gaussian model of the beam's intensity.

where $I(a)$ is the current value of the beam's intensity at the radius r , $I(0)$ is the maximum beam intensity at radius $a = 0$, and w_0 is the mode-field radius.

There is no boundary to a beam's intensity and so, theoretically, a beam's light spreads to infinity. Hence, there is no natural measure of distribution of the beam's intensity. As usual in such a case, we have to introduce some measure by convention. This measure, known as the mode-field diameter (MFD), is equal to $2w_0$.

If we put $a = w_0$ into Eq. (2.37), we obtain

$$I(a) = 0.135 I(0) \quad (2.38)$$

Thus, MFD is the cross-sectional dimension $2w_0$, where the beam's intensity drops to $1/e^2$ ($= 0.135$) of its peak value.

The Gaussian model of distribution of a beam's intensity is the most popular because

1. It is close to the measured results.
2. It is easy to use in theoretical calculations.

2.10.2 Mode-Field Diameter

In multimode fibers, the NA and geometric sizes of both the core and the cladding are considered important performance parameters and given in the data sheets. But in single-mode fiber, manufacturers use the mode-field diameter (MFD) rather than the core diameter as a parameter that describes single-mode fibers. This is because the radial distribution of the optical power in the propagating fundamental mode is an important parameter, which is clear from Fig. 2.26. Moreover, it shows the wavelength-dependent field penetration into the fiber cladding.

In multimode fibers, we need to know the core diameter because all the light is confined within the core. In single-mode fibers, an essential portion of the light (typically, about 20%) is carried by the cladding, thereby making

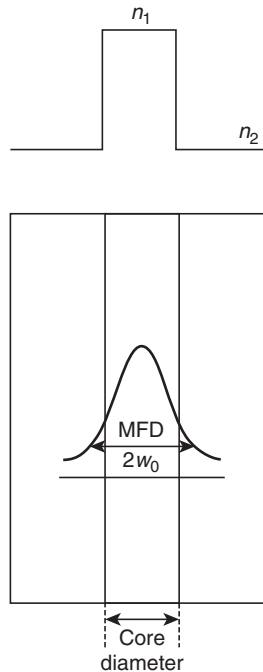


Figure 2.26 | Distribution of the beam's intensity in a single-mode fiber.

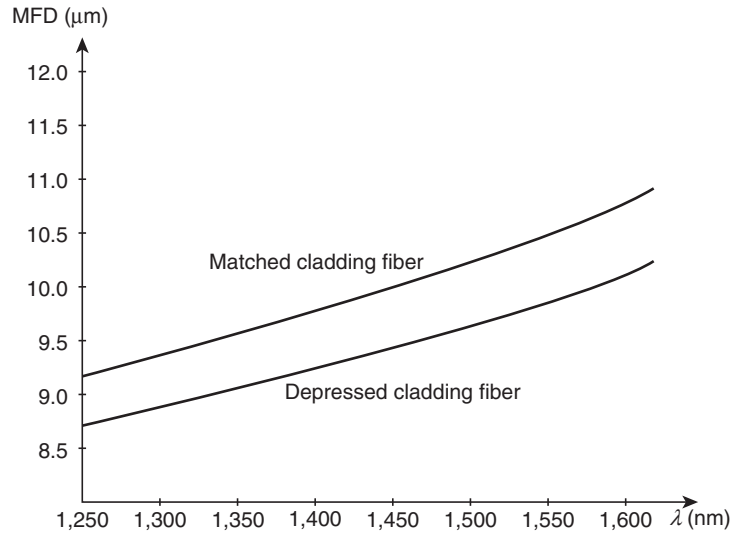


Figure 2.27 | Mode-field diameter as a function of wavelength.

the core diameter just an auxiliary parameter of a single-mode fiber. MFD is slightly larger than the single-mode fiber core diameter. Typically MFD is $9.3 \mu\text{m}$ for step-index single-mode fibers of $8.3 \mu\text{m}$ core diameters.

Thus, a light beam propagates in a single-mode fiber mostly within the core and partially within the cladding. Therefore the fiber's effective refractive index (n_{eff}) is the combination of the core and cladding refractive indices. The MFD depends on the operating wavelength. At shorter wavelengths, the beam is more focused and confined to the fiber's central axis. Therefore the MFD is small. The shorter the wavelength, the less the MFD. Typical graphs of MFD as a function of wavelength are shown in Fig. 2.27.

2.10.3 Cutoff Wavelength

The smallest operating wavelength when single-mode fibers propagate only the fundamental mode is called as **cutoff wavelength**. At this wavelength, the second-order mode becomes lossy and radiates into the fiber cladding. The fundamental mode becomes increasingly lossy as the operating wavelength becomes longer than the cutoff wavelength, and more power is transmitted through the fiber cladding. If the fundamental mode extends into the cladding material, it becomes increasingly sensitive to bending loss. To reduce losses caused by fiber bending and splicing, fiber manufacturers adjust the value of V by varying the core and cladding sizes and the relative refractive index difference (Δ).

A single-mode step-index fiber is suitable for long-haul, high-speed telecommunication systems because it has low attenuation and high bandwidth properties. However, the installation of a single system is expensive and difficult.

For a single-mode fiber with core diameter d , the cutoff wavelength (λ_c) when the V number is exactly 2.405 is given by

$$\lambda_c \leq \frac{\left\{ \pi d \sqrt{[n_1^2 - n_2^2]} \right\}}{2.405} \quad (2.39)$$

We have

$$\text{NA} = \sqrt{n_1^2 - n_2^2}$$

Therefore,

$$\lambda_c \leq 1.306d \text{ (NA)} \tag{2.40}$$

Cutoff wavelength is applicable to single-mode fibers only. This is the shortest wavelength at which a fiber can support single-mode operation. If we work with light at a wavelength shorter than λ_c , two, three, or more modes will propagate along the fiber. In other words, the same optical fiber can be single-mode or multimode, depending on the *operating wavelength*. The transition between single-mode and multimode operation occurs gradually. In practice, the fibers are designed specifically for either multimode or single-mode operation. For single-mode operation, the operating wavelength is always longer than λ_c .

2.10.4 Single-Mode Graded-Index Fibers

Single-mode graded-index fibers are not standard fibers. They are typically only used in specialty applications. The comparison between MMF and SMF is listed in Table 2.1.

Table 2.1 | Multimode versus single-mode fiber

Multimode Fibers	Single-Mode Fibers
They have a limit in terms of maximum bit rate of the order of 1 Gbit/1 km, due to multimode dispersion.	We will see that they are not affected by multimode dispersion, and their bandwidth limit is extremely higher.
They have a relatively large core, hence: 1. Splicing is easier. 2. Connectors are less expensive.	They have a small core, hence: 1. Splicing is more difficult. 2. Connectors are more expensive.
Installation is simpler.	Installation is more difficult.
They are intrinsically more resilient to mechanical and environmental stress. They are thus mostly used in LAN applications.	They are thus used in all applications where the distance to be covered is significantly higher than 1 km.

2.11 Fiber Materials

For the fabrication of optical fiber:

1. Two materials with slightly different refractive index for making of core and cladding are required.
2. These materials should be transparent to light in the operating wavelength range of around 800–1,600 nm.
3. The materials should have low attenuation.
4. These materials should have low intrinsic and scattering losses.
5. The materials must allow the making of long, thin, and flexible fibers.
6. The materials should be cheap and abundant.

Most of these requirements are met by glasses and plastic polymers.

Optical fibers require a high-index core and a low-index cladding, but the refractive index of pure silica is uniform. So fiber cannot be fabricated from pure silica. To change the refractive index of silica, dopant should be added. Thus in silica fibers, both core and cladding are made of silica, differentiated by different doping levels. Most dopants, when added, increase the refractive index of silica. So, doped silica can be used as core material. The most common core dopant material is germanium, with pure silica forming the cladding. A few materials like fluorine reduce the refractive index of silica, so they can form the cladding on the pure silica core.

Another material used for optical fiber fabrication is plastic. It is lightweight, inexpensive, flexible, has large NA, and is easy to handle, but has much higher attenuation. The best plastic fiber has a loss of approximately

Table 2.2 Various dopants used to make the core and cladding of various fibers

Core	Cladding
SiO_2	$\text{B}_2\text{O}_3\text{--SiO}_2$
$\text{GeO}_2\text{--SiO}_2$	SiO_2
$\text{P}_2\text{O}_5\text{--SiO}_2$	SiO_2

50 dB/km. At shorter wavelengths, the loss is less as compared to the near-infrared region. Therefore, these fibers have limited applications.

Different dopant materials are added in the fabrication of low-loss, long wavelength optical fibers. For example

1. Heavy-metal fluorides (e.g., zirconium and beryllium fluoride).
2. Chalcogenide glasses (e.g., arsenic/sulfur).
3. Crystalline materials (e.g., silver bromide and silver chloride).

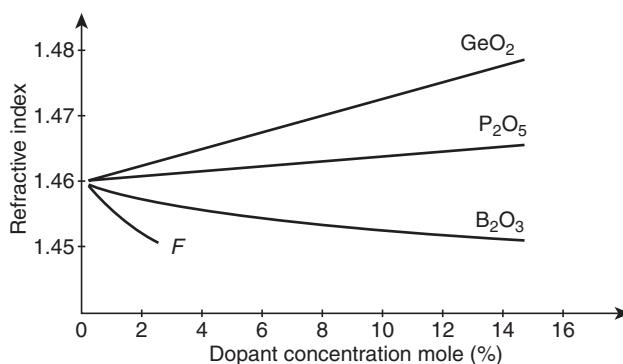
2.11.1 Glass Fibers

Optical fibers are made up of oxide glasses. The most common of which is silica (SiO_2). Glass composed of pure silica is referred to as *silica glass*, *fused silica*, or *vitreous silica*. The glass is made by fusing mixtures of metal oxides, sulfides, or selenides.

Some of its desirable properties are as follows:

1. A resistance to deformation at high temperatures.
2. A high resistance to breakage from thermal shock because of its low thermal expansion.
3. Good chemical durability.
4. A high transparency in both the visible and infrared regions of interest to many fiber-optic systems.

In order to produce two similar materials having slightly different indices of refraction for the core and the cladding, fluorine or various oxides are commonly added to the silica (see Table 2.2). These dopants can be classified into two basic groups: dopants which *increase* the refractive index and dopants which *decrease* the refractive index. For example, B_2O_3 and fluorine dopants decrease a material's refractive index, whereas GeO_2 and P_2O_5 will increase a material's refractive index as shown in Fig. 2.28.

**Figure 2.28** Refractive index as a function of dopant materials and their concentration.

2.11.2 Halide Glass Fibers

The absorption of silica rises rapidly at longer wavelengths. The materials that are transparent in this range are zirconium fluoride (ZrF_4) and barium fluoride (BaF_2) with some other components added to form the glass compound. It has been found that fluoride glasses have extremely low transmission losses at wavelengths in the range of 0.2–8 μm . Fluoride glasses belong to a general family of halide glasses in which the anions are from elements in Group VII of the periodic table, namely fluorine, chlorine, bromine, and iodine. These fibers are used in erbium-doped fiber amplifiers because of their desirable optical characteristics.

2.11.3 Chalcogenide Glass Fibers

Chalcogenide glass has high optical non-linearity, and so it is used in applications such as all-optical switches and fiber lasers. Chalcogenide glass fibers contain arsenic, germanium, phosphorus, sulfur, selenium, or tellurium. Theoretically, the minimum attenuation for these materials has been estimated at 1 dB/m.

2.11.4 Active Glass Fibers

When the rare-earth elements are incorporated into a normally passive glass, it results in the material having new optical and magnetic properties. These new properties allow the material to perform amplification, attenuation, and phase retardation on the light passing through it. Two commonly used materials for fiber lasers are erbium and neodymium.

2.11.5 Plastic Optical Fibers

Plastic fibers are generally used for short haul (up to 100 m) applications because they give rise to high attenuation of the optical signal. However, their mechanical strength is high. Plastic fibers are lighter and lower in cost than glass. Their operating temperature can go up to 125°C. For example, plastics can be used in medical applications and in some sensors where only shorter fiber lengths are needed. In addition, the mechanical flexibility of plastic allows these fibers to have large cores. These factors permit its use in inexpensive, economically attractive systems.

The following are examples of some of the compounds used in plastic fibers:

1. A polystyrene core/methyl methacrylate cladding.
2. A polymethyl methacrylate core/copolymer cladding.

2.12 Fabrication of Optical Fibers

There are different techniques of fiber fabrication. Most preferred techniques are as follows:

1. Preform generation techniques.
2. Double crucible technique.

2.12.1 Preform Generation Techniques

The various stages in the fabrication of all glass fibers are shown in Fig. 2.29. The first stage is producing a pure glass. It is then converted into the required glass composition for the core-cladding structure, that is, the *preform*. The preform has the same structure as the final optical fiber. The fiber is drawn from the preform at the correct diameter (125 μm), maintaining the same refractive index profile.

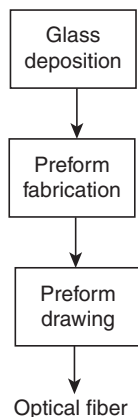


Figure 2.29 | Schematic of optical fiber fabrication.

A preform is a cylinder of silica material. Its diameter is in the range of 10–25 mm and its length varies from 60 to 120 cm. This preform consists of a core surrounded by a cladding with a desired refractive index profile. In other words, this is the desired optical fiber, but on a much larger scale.

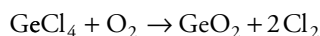
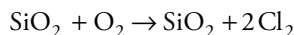
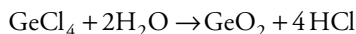
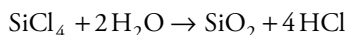
A preform is prepared to have a “drawable” material that is clean, and has low OH concentration, low metallic-ion contaminants, and is inexpensive. Many techniques have been developed to prepare these preforms. Some common commercially used methods are outside vapor deposition (OVD), modified chemical vapor deposition (MCVD), vapor phase axial deposition (VPAD), and plasma chemical vapor deposition (PCVD). They differ mainly by the way the soot is deposited.^[1–3]

The preform is made by vapor-phase oxidation, in which two gases, SiCl_4 and O_2 , are mixed at a high temperature to produce silicon dioxide (SiO_2) as follows:



Silicon dioxide, or pure silica, is usually obtained in the form of small particles (about $0.1 \mu\text{m}$ in diameter) called “soot.” A homogeneous transparent cladding material is formed by depositing the silica soot, layer upon layer, on the target rod or tube. To change the value of a cladding’s refractive index, some dopants are used. For example, fluorine (F_2) is used to decrease the cladding’s refractive index.

The soot for the core material is made by mixing three gases – SiCl_4 , GeCl_4 , and O_2 – through flame hydrolysis or chemical vapor deposition (CVD). This results in a mixture of SiO_2 and GeO_2 . Some typical reactions are given below:



The doping level is controlled by changing the amount of GeCl_4 gas added to the mixture.

Manufacturers deposit the soot on the surface of a glass substrate (mandrel) or inside a hollow tube by one of the following methods:

1. Modified chemical vapor deposition (MCVD).
2. Plasma-enhanced MCVD (PMCV).

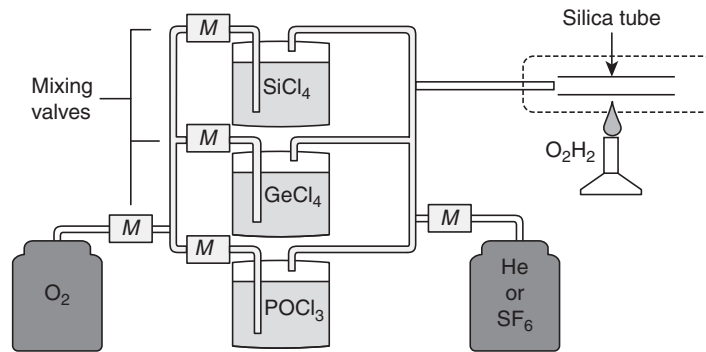


Figure 2.30 | MCVD complete system.

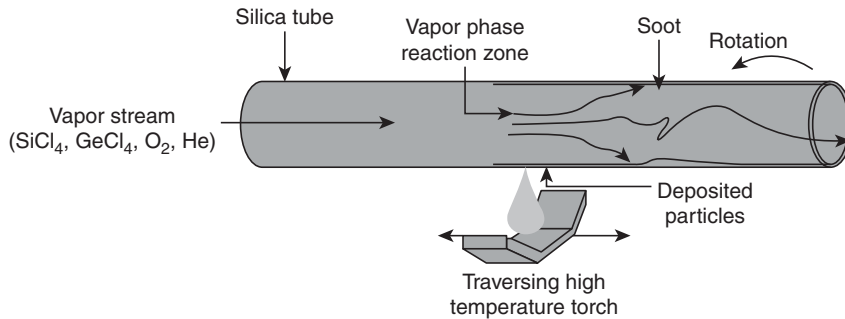


Figure 2.31 | MCVD – details of process inside the silica tube.

2.12.1.1 MCVD

The MCVD technique is widely used for the fabrication of optical fiber with the flexibility of producing the desired index profile in the core and optimizing performance over the desired spectral range (approximately $1.3\ \mu\text{m}$ – $1.6\ \mu\text{m}$). It offers a high level of control over the optical properties of the fiber.

The complete setup of MCVD process is shown in Fig. 2.30. Initially, the hollow fused silica tube is cleaned and etched to remove surface contaminants. The tube is then placed on a specialized lathe and gradually moved along its long axis. At the same time it is heated externally by a movable torch or furnace to temperatures of around $1,400$ – $1,600^\circ\text{C}$. This heat source is slowly moved in a continuous fashion along the length of the tube.^[4]

The vapor phase reactants halide and oxygen are passed through the hot zone as shown in Fig. 2.31. Glass particles formed during this reaction are deposited on the inner walls of the silica tube. The high-temperature traversing torch is moved back and forth, so that the glass particles will be deposited layer by layer on the inner wall of silica tube, which forms the cladding. When the layer becomes sufficiently thick, vaporized chlorides of germanium (GeCl_4) or phosphorus (POCl_3) are added to the gas flow to form the core of the fiber preform. Once the desired amount of material has accumulated, the torch's temperature is increased to $2,000^\circ\text{C}$, and the tube collapses, creating the preform. Careful control of rotation speed, temperature, and internal pressure of the tube as well as torch translation speed allow for good preservation of the structure of the core and cladding during collapse, minimizing the occurrences of bubbles and other defects. Different chemical mixtures can be used in successive layers of sintered glass to obtain different index profiles within the preform.^[5,6,7]

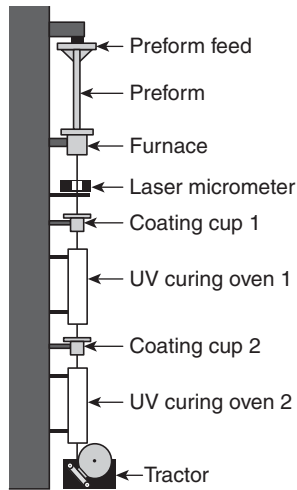


Figure 2.32 | Fiber drawing tower schematic.

2.12.1.2 PMCVD

Plasma enhanced MCVD (PMCVD) is nearly identical to MCVD. The main difference is the stimulation of oxide formation by isothermal plasma in a microwave cavity (3–5 MHz) at low pressure, which surrounds the silica tube. Volatile reactants supplied to the tube react within the microwave cavity. By controlling the microwave cavity, the reaction zone is moved backward and forward. With this technique, very thin layer deposition is possible. Hence in a single preform we can have up to 2,000 layers, and a good graded-index profile can be obtained. Deposition rates with PMCVD are achieved in excess of 3.5 g/min as compared to 1 g/min in MCVD.

2.12.1.3 Drawing and Coating

The preform obtained by any one method is placed vertically on a drawing tower as shown in Fig. 2.32. The lower end of the preform is heated to melting and the fiber is drawn by stretching. The thickness of fiber is controlled by the drawing rate. At faster rates, a narrower diameter fiber is drawn. The profile of the preform is preserved in the fiber. As the material leaves the preform, a mechanical feeder ensures a continuous flow of material. Any lag or overpressure would cause inconsistencies in the flow and therefore in the diameter of the fiber.

The thickness of fiber coming out of the furnace beneath the preform is measured by a laser micrometer. With the help of a feedback control system, the draw rate is controlled according to the error in the fiber diameter. If the fiber is measured to be too thin or too thick, it slows down or speeds up the draw rate accordingly. After this, the fiber is coated with a UV-sensitive epoxy or polymer followed by UV curing. There may be more than one coating phases depending on the fiber properties and application. Improper coating and curing may give rise to microbending losses in the fiber. The coating is necessary to protect the fiber from abrasive damage or environmental contaminants. The drawing and coating process has an effect on the performance properties of fibers. Transmission losses, bandwidth and fiber strength are all influenced by it.

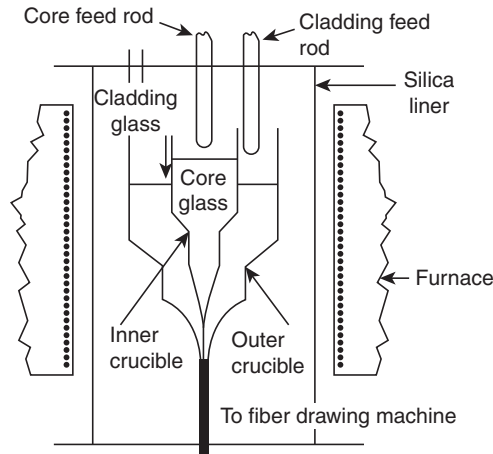


Figure 2.33 | Double-crucible method for fiber fabrication.

2.12.1.4 Double-Crucible Technique

This is a direct melt method of fabrication. The glass elements with the required refractive indices are made from the raw material of specially prepared constituents. Soda-lime-silicate and sodium-borosilicate glasses were made from materials purified to parts-per-billion (ppb) levels of transition metal impurities by ion exchange, electrolysis, recrystallization, or solvent extraction.

These starting glasses in the form of core feed rod and cladding feed rod are fed into an ingenious continuous casting system composed of concentric platinum crucibles, shown in Fig. 2.33. The crucibles are heated up to 800–1,100°C. The crucibles have nozzles in their bases. A thin stream of core glass poured from the upper crucible passes through the reservoir of cladding glass, and is concentrically surrounded by the cladding as it flows through the nozzle. Then the fiber coming out of the crucible is coated and immediately cured.

This process gives a high rate of production. Moreover, for large-scale production, it is very cheap and there is no restriction on dimensions, but the absorption losses are greater (for more details of absorption losses refer Section 3.1.1).

2.13 Fiber-Optic Cable Construction

Fiber-optic cables come in all sizes and shapes. Like coaxial cables, their actual construction depends on the intended application. Typical fiber construction is shown in Fig. 2.34. The basic optical fiber is provided with a buffer coating which is mainly used for protection during the manufacturing process. This buffer can be either a loose tube or a tight tube. Most data communication cables are made using either one of these two constructions. A third type, the ribbon cable, is frequently used in telecommunications as shown in Fig. 2.35. This buffer allows the fiber to flex and bend, particularly when going around corners or when being pulled through conduits.

Physical construction of optical cables is not governed by any agency. It is up to the designer of the system to make sure that the cable selected will meet the application requirements. However, there are five basic cable types as explained below.

1. **Simple cable:** The simple fiber-optic patch cable shown in Fig. 2.36 is ideal for use in office LAN connections, patchcords, pigtailed, and internal point-to-point links where frequent handling is likely

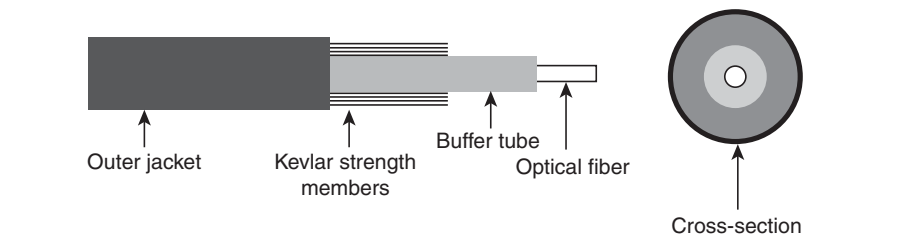


Figure 2.34 | Fiber construction.

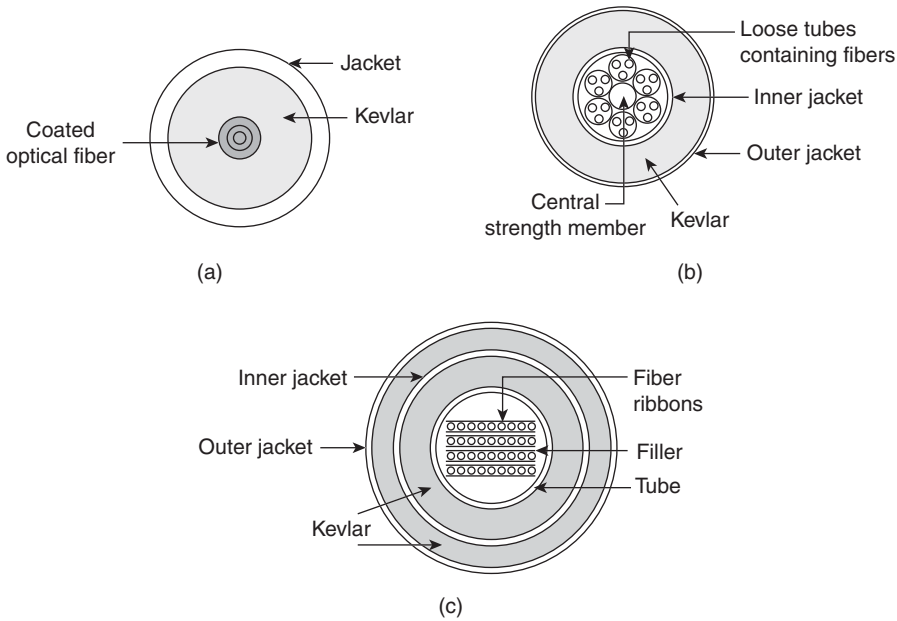


Figure 2.35 | (a) Tight-buffered fiber-optic cable. (b) Loose-tube fiber-optic cable. (c) Ribbon fiber-optic cable.

Table 2.3 | Comparison of loose-buffered and tight-buffered fiber cable

Loose-Buffered Cable	Tight-Buffered Cable
Each individual fiber bundle moves freely within the inner sheath	Fiber elements are held in place within the cable
More toxic when burned	Less toxic when burned
Less expensive	More flexible for manipulation and expensive
Higher fiber counts	Smaller in diameter with fewer fibers
More robust	More sensitive to outside forces
Optimized for long runs	Used in intrafacility applications
Used in applications requiring aerial, buried, or submerged installation of fiber cables	Cables are either distribution or breakout designs
	All fiber bundles are in a single jacket or each has a separate jacket

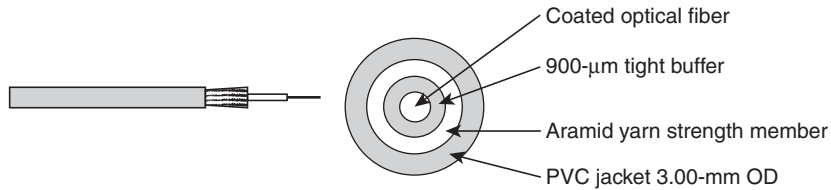


Figure 2.36 | Simple cable construction.

(e.g., data cabinets, equipment rooms, user outlets, etc.). Utilizing 600- or 900- μm buffered fiber, the cable is suitable for use with industry standard connectors and can be easily made into a patchcord. Its applications are as follows.

- (a) Patchcords.
 - (b) Pigtails.
 - (c) Internal interconnections.
2. **Zipcord cable:** Zip duplex fiber-optic patch cable shown in Fig. 2.37 is constructed with two simplex units joined together with a central web. This is also ideal for use in office LAN connections, patchcords, pigtails, and internal point-to-point links where frequent handling is likely (e.g., data cabinets, equipment rooms, user outlets, etc.). Like in the case of the simple cable, utilizing 600- or 900- μm buffered fiber, this cable is also suitable for use with industry standard connectors and can be easily made into a patchcord.

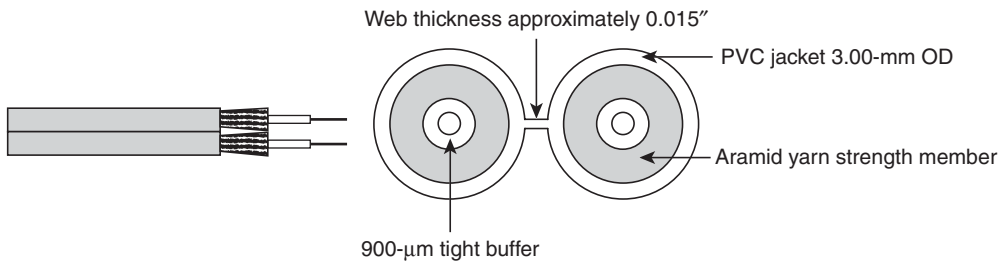


Figure 2.37 | Zipcord cable construction.

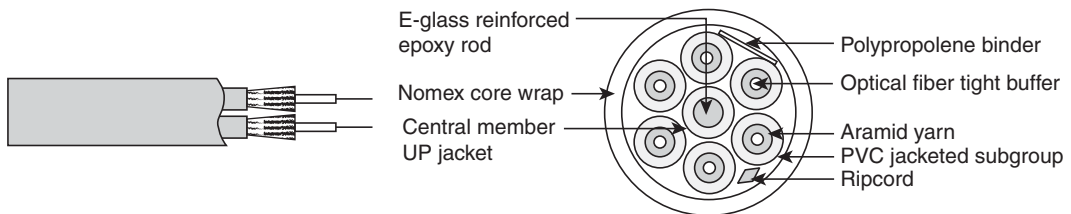


Figure 2.38 | Tight pack cable construction.

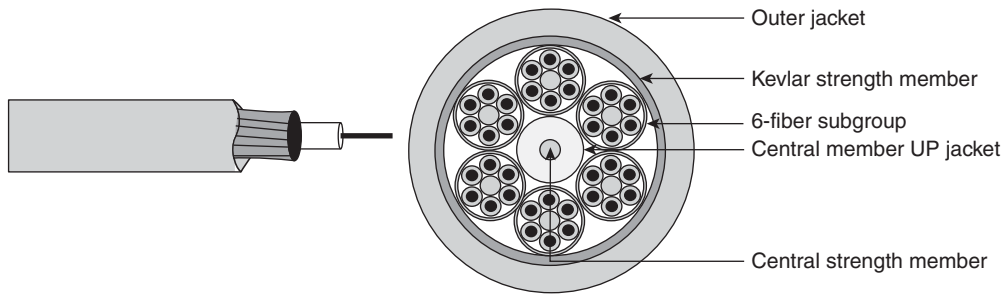


Figure 2.39 | Breakout cable construction.

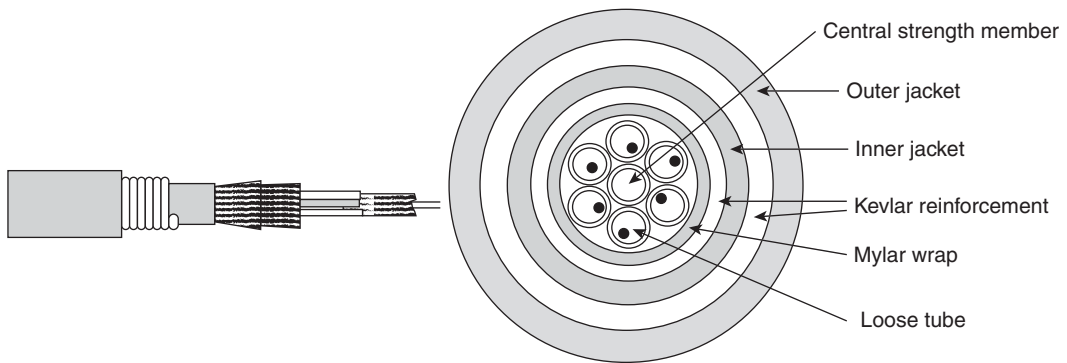


Figure 2.40 | Loose tube cable construction.

Its applications are as follows:

- (a) Patchcords.
 - (b) Pigtails.
 - (c) Internal interconnections.
3. **Tight-buffered cable:** Tight-buffered internal/external cables shown in Fig. 2.38 are constructed of 900- μm buffered fibers, surrounded by aramid yarn or jacketed e-glass strength members. Its applications are as follows:
 - (a) Internal cable for installation in trunking, under floor or ceiling spaces.
 - (b) Short-run external links between buildings.
 - (c) Fiber backbones in riser and horizontal configurations.
 4. **Breakout cable:** Breakout cables shown in Fig. 2.39 are constructed of up to 24, 2-mm simplex cables stranded around a central strength member. Each simplex cable has its own aramid yarn strength member and jacket. These sub-units are wrapped in a polyester tape and jacketed. Its applications are as follows:
 - (a) Internal cable for installation in trunking under floor or ceiling spaces.
 - (b) Short run external links between buildings.
 - (c) Fiber backbones in riser and horizontal configurations.
 5. **Loose tube cable:** The single loose tube cable shown in Fig. 2.40 is constructed of 250- μm fibers held in a gel-filled central loose tube, protected by high-strength e-glass members. This structure isolates the fiber mechanically from external forces as well as microbending losses. These cables are protected by high-strength aramid yarn or e-glass strength members, water blocking tape, and jacketed.

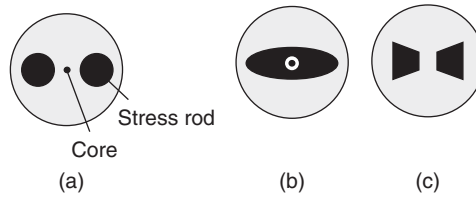


Figure 2.41 | Three types of PMFs: (a) Panda style, (b) elliptical-clad, and (c) bow-tie.

Loose tube cable is suitable for internal or external applications in ducts or intra-building links in campus environments.

2.14 Specialty Optical Fibers

Fibers used for special applications other than telecommunication have special properties and are called specialty optical fibers. There are many specialty fibers available commercially; some of them are discussed below.

2.14.1 Polarization-Maintaining Fibers

“Polarization-maintaining,” “Polarization-preserving,” “PM,” “HiBi,” or even occasionally “Polarization-retaining fiber” are all different names which describe an optical fiber that faithfully preserves and transmits the polarization state of the light launched into it, even when it is subjected to environmental perturbations.

PM fiber (PMF) is used in applications that require the transmission and delivery of polarized light at high bit rates. Currently, PMFs such as Panda and bow-tie fibers are used for polarization-maintaining applications or transmission lines.

In a standard single-mode fiber, guided light propagates as an ensemble of discrete waves, with their electric and magnetic fields randomly oriented and fluctuating in response to changes in environmental conditions that include temperature, pressure, and mechanical stresses. Fibers are never perfectly symmetric, and propagating light is thereby subject to a small but nonetheless significant modal birefringence.

Consequently, standard single-mode fiber can be thought of as a bi-mode fiber with each mode having orthogonal linear polarization. As the propagation characteristics of these two modes are quite similar, it is easy to transfer power between them, thereby altering the state of polarization of a guided wave.^[8]

Nowadays, three PM-fiber designs are in common use as shown in Fig. 2.41: bow-tie, Panda, and elliptical-clad. Because of its inherent uniformity and reproducibility, Panda fiber is widely used for all telecom applications.^[9]

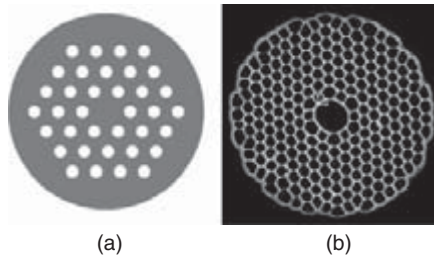


Figure 2.42 | Photonic crystal fibers: (a) Solid core structure and (b) hollow core structure.

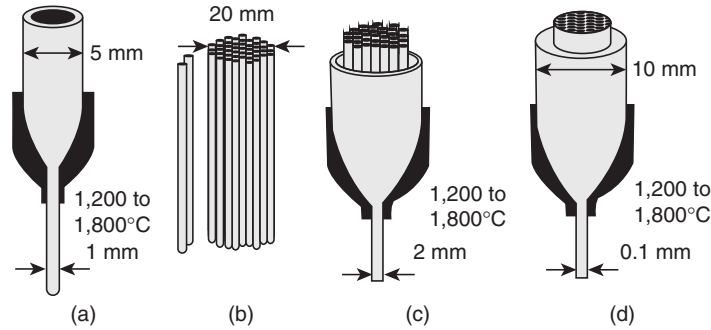


Figure 2.43 Photonic crystal fiber fabrication: (a) Individual capillary creation, (b) preform formation, (c) drawing of intermediate preform, (d) drawing of the final fiber.

If linearly polarized laser light is coupled into a PM fiber, the output beam is, in general, elliptically polarized. The degree of polarization depends on the relative angular position θ (azimuth) between the fiber birefringent axis and the polarization direction of the light at the input. Only when the polarization direction of the light at the input is perfectly aligned with the fiber birefringent axis (either the fast axis or the slow axis) can linear polarization be perfectly maintained at the output.

2.14.2 Photonic Crystal Fiber

A photonic crystal fiber (PCF) is made up of a periodic structure of capillaries filled with air and arranged in a hexagonal lattice as shown in Fig. 2.42. It has been shown that light can propagate in crystal structure defects. By removing one or more capillaries, the defect is created. PCF combines the properties of optical fibers and photonic crystals. The traditional fiber is proved as the best choice for many telecom and non-telecom applications. However, there are many constraints in the design of their structures (e.g., limited core diameter for single-mode fiber, limited choice for fiber material, mode cut-off wavelength, etc.). But, in the case of PCF we have high flexibility in the design. We can vary the parameters, such as type of lattice, lattice pitch, refractive index of glass, number of air holes, their shape and diameter, etc. By varying these parameters we can design a single-mode fiber, which can be operated in all wavelength ranges, where there is no cutoff wavelength.

By varying the structure, the desired dispersion characteristics can be obtained. It is possible to design a PCF with zero, low, or anomalous dispersion in the visible wavelength range. There are two guiding mechanisms in a PCF:

1. Index guiding (such as in a traditional optical fiber).
2. Photonic bandgap mechanism.

2.14.2.1 Fabrication of PCF

The step-by-step fabrication process for PCF is shown in Fig. 2.43.

1. Individual capillaries are fabricated. Due to design flexibility, individual capillaries with different diameter and wall thickness; cross-section (e.g. square, circular, hexagonal, etc.); glass material (e.g., silica, silicate, or different compositions of oxide) can be fabricated to obtain the desired characteristics.

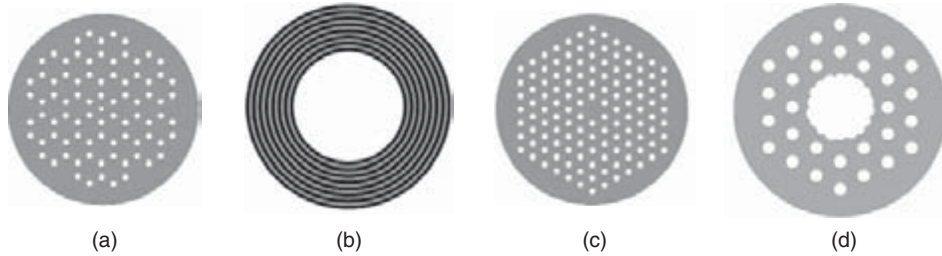


Figure 2.44 | Different PCF structures: (a) Honeycomb, (b) Bragg hollow core, (c) hexagonal lattice solid core, (d) large hollow core hexagonal.

2. A multicapillary preform is made by positioning the individual capillaries. The defect rod through which the light propagates is placed in the structure. This structure is then fused to form a preform.
3. The preform is drawn with a fiber drawing tower to a millimeter scale.
4. The final fiber is drawn.

It should be noted that polymer coating is done for providing mechanical protection.

2.14.3 Hollow Core Fibers

In hollow core fibers, light is guided through a hollow core surrounded by a microstructured cladding region with air holes. The two types of hollow core fiber are as follows:

1. A hollow core surrounded by microstructured cladding with a periodic arrangement of air holes in glass.
2. Photonic cladding made of circularly arranged air cells.

The light is guided by photonic bandgap-guiding mechanism because the TIR condition is not fulfilled, as the refractive index of the cladding is greater than that of the core (air). These fibers are almost bend-insensitive. A small fraction of light passes through the glass, but 98% is confined to the air core. Therefore all effects related to light–glass interaction, such as scattering and dispersion, are very less. This fiber allows high-power transmission without non-linear effects.

2.14.4 Dispersion-Compensated Fiber

The dispersion-compensated fiber (DCF) has negative dispersion. Such fibers can be inserted in a fiber-optic link to nullify the positive dispersion effect in a traditional optical fiber. For example, a 9-km DCF is required to compensate the dispersion in a 90-km G.652 fiber. But this fiber has higher attenuation, and the PMD coefficient is also double as compared to traditional fiber.

An optical network having small dispersion and high power density is more prone to non-linear effects, such as four-wave mixing. As the attenuation caused by a DCF is very high, it is generally placed between preamplifier and the booster amplifier of an erbium-doped fiber amplifier (EDFA), where power is high. If the optical link is overcompensated, non-linear effects will occur. Therefore to avoid this, the system is compensated very carefully.

Table 2.3 | Specifications of typical DCF

Characteristic	Unit	Value
Attenuation	dB/km	0.60
A_{eff}	μm^2	12.5
PMD coefficient	ps/ $\sqrt{\text{km}}$	0.1

2.14.5 Photosensitive Fiber

The photosensitivity of a glass fiber is an important phenomenon that allows the fabrication of special structures into the glass. By using lasers, the refractive index of such material can be varied. By this phenomenon, Bragg gratings and waveguides can be directly written into glass.

K.O. Hill and his co-workers discovered the photosensitivity in germanium-doped silica fiber for the first time in 1978. Photosensitivity of a fiber is its capability to change its refractive index locally when it is irradiated by UV light (this is explained in detail in Section 6.5.10.4.1).

Solved Problems

PROBLEM 1

Calculate the numerical aperture, of a single-mode fiber, where $n_1 = 1.4675$ and $n_2 = 1.4622$.

Solution: Numerical aperture is given by

$$\text{NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.4675)^2 - (1.4622)^2} = 0.125$$

PROBLEM 2

If the core layer of an optical fiber is made from silica with a refractive index of 1.45 and if the refractive index of the cladding layer is 1% less than that of the core, calculate (a) cladding layer refractive index, (b) critical angle, and (c) numerical aperture.

Solution: Core refractive index, $n_1 = 1.45$

(a) Cladding refractive index, $n_2 = 1\%$ smaller than n_1 . This implies $n_2 = 99\%$ of n_1 . Therefore

$$n_2 = \frac{99}{100} \times 1.45 = 1.4355$$

(b) Critical angle,

$$\begin{aligned} \theta_c &= \sin^{-1} (n_2/n_1) \\ &= \sin^{-1} (1.4355/1.45) = 81.89^\circ = 1.43 \text{ radians} \end{aligned}$$

- (c) Numerical aperture

$$NA = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.45)^2 - (1.4355)^2} = 0.2046$$

PROBLEM 3

An optical fiber has the following data: $n_1 = 1.5$, $n_2 = 1.45$. Calculate the following:

- (a) Critical angle.
 (b) Numerical aperture.
 (c) Acceptance angle.

Solution: We have $n_1 = 1.5$, $n_2 = 1.45$.

- (a) Critical angle is given by

$$\begin{aligned}\theta_c &= \sin^{-1}(n_2/n_1) \\ &= \sin^{-1}(1.45/1.5) = 75.17^\circ\end{aligned}$$

- (b) Numerical aperture is given by

$$NA = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.5)^2 - (1.45)^2} = 0.384$$

- (c) Acceptance angle is given by

$$\theta_a = \sin^{-1}(NA) = \sin^{-1}(0.384) = 22.58^\circ$$

PROBLEM 4

Calculate the refractive indices of the core and the cladding materials of an optical fiber whose $NA = 0.35$ and $\Delta = 0.01$.

Solution: It is given that $NA = 0.35$ and $\Delta = 0.01$. Now

$$\Delta = \frac{n_1 - n_2}{n_1}$$

and

$$NA = n_1 \sqrt{2\Delta}$$

$$\Rightarrow n_1 = \frac{NA}{\sqrt{2\Delta}} = \frac{0.35}{\sqrt{2 \times 0.01}} = 2.48$$

So

$$\Delta = \frac{n_1 - n_2}{n_2} = 1 - \frac{n_2}{n_1}$$

$$n_2 = n_1 (1 - \Delta) = 2.48 (1 - 0.01)$$

$$n_2 = 2.45$$

PROBLEM 5

A step-index fiber has a core radius of 8 microns. The core and cladding indices are 1.458 and 1.44, respectively. What is the V number of the fiber if the operating wavelength is 1,300 nm? How many modes will be supported in the fiber? If a fiber must be operated as a single-mode fiber at 1,500 nm with the indices as given, what must be the radius of the fiber?

Solution: V number is given by

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi \times 8 \times 10^{-6}}{1,300 \times 10^{-9}} \sqrt{(1.458)^2 - (1.44)^2} = 8.83$$

Total number of modes is given by

$$M_s = \frac{V^2}{2} = 39$$

For single-mode operation, $V = 2.405$. Therefore

$$a = \frac{2.405\lambda}{2\pi \sqrt{n_1^2 - n_2^2}} = \frac{2.405 \times 1,500}{2\pi \sqrt{(1.458)^2 - (1.44)^2}} = 2.5 \mu\text{m}$$

The fiber should not have radius larger than 2.5 μm for single-mode operation at 1,500 nm.

PROBLEM 6

Calculate the V number and the number of modes supported by the step-index fiber having $n_1 = 1.53$, $n_2 = 1.5$ and with a core radius of 50 μm operating at 1,500 nm.

Solution: V number is given by

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi \times 50 \times 10^{-6}}{1,500 \times 10^{-9}} \sqrt{(1.53)^2 - (1.5)^2} = 62.8$$

Total number of modes is given by

$$M_s = \frac{V^2}{2} = \frac{(62.8)^2}{2} \approx 1972$$

PROBLEM 7

Determine the normalized frequency at 820 nm for a step-index fiber having a core radius of 30 μm , $n_1 = 1.48$, $n_2 = 1.46$. Calculate the number of modes propagating in the fiber at 820 nm, 1,300 nm, and 1,550 nm.

Solution:

(a) At $\lambda = 820$ nm, V number is given by

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi \times 30 \times 10^{-6}}{820 \times 10^{-9}} \sqrt{(1.48)^2 - (1.46)^2} = 55.74$$

Total number of modes is given by

$$M_s = \frac{V^2}{2} = \frac{(55.74)^2}{2} \approx 1,553$$

(b) At $\lambda = 1,300$ nm, V number is given by

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi \times 30 \times 10^{-6}}{1,300 \times 10^{-9}} \sqrt{(1.48)^2 - (1.46)^2} = 35.15$$

Total number of modes is given by

$$M_s = \frac{V^2}{2} = \frac{(35.15)^2}{2} \approx 618$$

(c) At $\lambda = 1,550$ nm, V number is given by

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi \times 30 \times 10^{-6}}{1,550 \times 10^{-9}} \sqrt{(1.48)^2 - (1.46)^2} = 29.48$$

Total number of modes is given by

$$M_s = \frac{V^2}{2} = \frac{(29.48)^2}{2} \approx 435$$

PROBLEM 8

A multimode step-index fiber has a relative refractive index difference of 1% and a core refractive index of 1.5. The number of modes operating at a wavelength of 1,300 nm is 1,100. Calculate the fiber core diameter.

Solution: Given

$$\Delta = 1\% = 0.01; n_1 = 1.5; \lambda = 1,300 \text{ nm}; M_s = 1,100$$

Total number of modes is given by

$$M_s = \frac{V^2}{2}$$

$$\Rightarrow V = \sqrt{2M_s} = \sqrt{2 \times 1,100} = 46.9$$

We have

$$V = \frac{2\pi a}{\lambda} n_1 \sqrt{2\Delta}$$

Therefore core diameter,

$$2a = \frac{V\lambda}{\pi n_1 \sqrt{2\Delta}} = \frac{46.9 \times 1,300 \times 10^{-9}}{\pi \times 1.5 \times \sqrt{2 \times 0.01}} = 91.49 \mu\text{m}$$

Hence core diameter = 91.49 μm .

PROBLEM 9

Calculate the core radius and cladding refractive index of a step-index silicon fiber having $\text{NA} = 0.3$, $V = 75$, $n_1 = 1.5$ and it is to be operated at 850 nm.

Solution: V number is given by

$$V = \frac{2\pi a}{\lambda}(\text{NA})$$

$$\Rightarrow a = \frac{\lambda V}{2\pi(\text{NA})} = \frac{850 \times 10^{-9} \times 75}{2\pi \times 0.3} = 33.8 \mu\text{m}$$

Core radius $a = 33.8 \mu\text{m}$. We have

$$\text{NA} = \sqrt{n_1^2 - n_2^2}$$

$$\Rightarrow n_2 = \sqrt{n_1^2 - (\text{NA})^2} = \sqrt{1.5^2 - 0.3^2} = 1.47$$

So cladding refractive index, $n_2 = 1.47$.

PROBLEM 10

Find the cutoff wavelength for a step-index fiber to be operated as single-mode fiber when the core refractive index is 1.46, relative refractive index difference is 0.25%, and the core diameter is 9 μm .

Solution: $\Delta = 0.25\% = 0.0025$, $n_1 = 1.46$, diameter = 9 μm , therefore $a = 4.5 \mu\text{m}$. We have

$$\text{NA} = n_1 \sqrt{2\Delta} = 1.46 \times \sqrt{2 \times 0.0025}$$

$$\text{NA} = 0.103$$

For single-mode fiber, cutoff number, $V = 2.405$. Therefore

$$V = \frac{2\pi a}{\lambda}(\text{NA}) = 2.405$$

$$\Rightarrow \lambda = \frac{2\pi a}{V}(\text{NA})$$

$$= \frac{2\pi \times 4.5 \times 10^{-6} \times 0.103}{2.405}$$

$$= 1.21 \mu\text{m}$$

The cutoff wavelength $\lambda = 1.21 \mu\text{m}$.

PROBLEM 11

A graded-index fiber having parabolic profile of refractive index has $NA = 0.3$ in air and a core diameter of $60 \mu\text{m}$ and supports 742 guided modes. Calculate the wavelength of light propagating in the fiber.

Solution: The number of modes in multimode graded-index fiber is given by

$$M_g = \left(\frac{\alpha}{\alpha + 2} \right) \left(\frac{V^2}{2} \right)$$

where α is the profile parameter and V is the normalized frequency. Parabolic refractive index profile $\alpha = 2$. So

$$\begin{aligned} M_g &= \frac{V^2}{4} \\ \Rightarrow 742 &= \frac{V^2}{4} \\ \Rightarrow V &= 54.48 \end{aligned}$$

We have

$$\begin{aligned} V &= \frac{2\pi a}{\lambda} (NA) \\ \Rightarrow \lambda &= \frac{2\pi a}{V} (NA) = \frac{2\pi \times 30 \times 10^{-6} \times 0.3}{54.48} = 1.03 \mu\text{m} \end{aligned}$$

Wavelength of light propagation is $\lambda = 1.03 \mu\text{m}$.

PROBLEM 12

A graded-index fiber with parabolic refractive index profile has a diameter of $50 \mu\text{m}$ and $NA = 0.2$. Calculate the total number of guided modes propagating in the fiber operating at a wavelength of $1 \mu\text{m}$.

Solution: We have

$$V = \frac{2\pi a}{\lambda} (NA) = \frac{2\pi \times 25 \times 10^{-6} \times 0.2}{1 \times 10^{-6}} = 31.4$$

The number of modes in multimode graded-index fiber is given by

$$M_g = \left(\frac{\alpha}{\alpha + 2} \right) \left(\frac{V^2}{2} \right)$$

where α is the profile parameter and V is the normalized frequency. Parabolic refractive index profile $\alpha = 2$. So

$$M_g = \frac{V^2}{4} = \frac{(31.4)^2}{4} \approx 247$$

PROBLEM 13

The relative refractive index difference between of a fiber is 0.7% and its core refractive index is 1.45. Calculate the values for NA of the fiber when

- (a) Index profile is not taken into consideration.
 (b) Index profile is triangular.

Solution: $n_1=1.45$ and $\Delta = 0.7\% = 0.07$. Now

$$\Delta = \frac{n_1 - n_2}{n_1}$$

$$\Delta = \frac{n_1 - n_2}{n_1} = 1 - \frac{n_2}{n_1}$$

$$n_2 = n_1 (1 - \Delta) = 1.45(1 - 0.07)$$

$$n_2 = 1.3485$$

- (a) When index profile is not taken into account:

$$\text{NA}(0) = \text{axial NA}$$

$$\text{NA}(0) = \sqrt{n_1^2 - n_2^2} = \sqrt{1.45^2 - 1.3485^2} = 0.533$$

- (b) When the index profile is triangular:

$$\alpha = 1$$

The local numerical aperture $\text{NA}(r)$ is given by

$$\text{NA}(r) = \text{NA}(0) \sqrt{1 - \left(\frac{r}{a}\right)^\alpha}$$

If $r = a$, then $\text{NA}(r) = 0.533$.

PROBLEM 14

For a step-index fiber $n_1 = 1.465$ and $n_2 = 1.46$, normalized frequency is 2.4. Calculate the core radius and numerical aperture at 0.8 μm .

Solution: For a step-index fiber $n_1 = 1.465$, $n_2 = 1.46$, $V = 2.4$, $\lambda = 0.8 \mu\text{m}$. Now

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$$

Therefore core radius

$$a = \frac{V\lambda}{2\pi\sqrt{n_1^2 - n_2^2}} = \frac{2.4 \times 0.8 \times 10^{-6}}{2\pi\sqrt{(1.465)^2 - (1.46)^2}} = 2.53 \mu\text{m}$$

Numerical aperture

$$\text{NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.465)^2 - (1.46)^2} = 0.121$$

PROBLEM 15

A symmetrical step-index planar waveguide is made of glass with $n_1 = 1.49$ and $n_2 = 1.48$. The thickness of the guide layer is $10\text{ }\mu\text{m}$ and the guide is excited by a source of wavelength $\lambda = 850\text{ nm}$. What is the range of propagation constant? What is the maximum number of modes supported by the guide?

Solution: The phase propagation constant β lies between β_1 and β_2 . Now

$$\beta_1 = kn_1 = \frac{2\pi n_1}{\lambda} = \frac{2\pi \times 1.49}{850 \times 10^{-9}} = 11.0140 \times 10^6\text{ m}^{-1}$$

$$\beta_2 = kn_2 = \frac{2\pi n_2}{\lambda} = \frac{2\pi \times 1.48}{850 \times 10^{-9}} = 10.941 \times 10^6\text{ m}^{-1}$$

The normalized frequency, cutoff parameter, V parameter or V number is given by

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi \times 10 \times 10^{-6}}{850 \times 10^{-9}} \sqrt{1.49^2 - 1.48^2} = 73.91 \times 0.1723 = 12.73$$

where a is the core radius and λ is the cutoff wavelength. The total number of modes is given by

$$M_s = \frac{V^2}{2} = \frac{12.73^2}{2} \approx 81$$

The guide can support maximum 81 modes.

PROBLEM 16

What should be the maximum thickness of the guide slab of a symmetrical step-index planar waveguide so that it supports only the fundamental TE mode if $n_1 = 2.6$ and $n_2 = 2.46$ and operating wavelength is 850 nm .

Solution: Single-mode step-index fibers propagate only one mode, called the fundamental mode. Single-mode operation occurs when the value of the fiber's normalized frequency (V) is between 0 and 2.405 ($0 < V < 2.405$). The value of V should remain near the 2.405 level. Now

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$$

$$2.4 > \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$$

$$\frac{2.4 \times 850 \times 10^{-9}}{\pi \sqrt{2.6^2 - 2.46^2}} > 2a$$

$$2a < 0.7715 \times 10^{-6}\text{ m}$$

Therefore, the thickness of the guide slab should be more than $0.7715\text{ }\mu\text{m}$.

PROBLEM 17

Calculate the maximum thickness of the guide slab of a symmetrical planar waveguide so that it supports the first 10 modes, if $n_1 = 2.6$ and $n_2 = 2.46$ and operating wavelength is 850 nm . Also calculate the maximum and minimum values of the propagation constant β .

Solution: We have

$$\beta_1 = kn_1 = \frac{2\pi n_1}{\lambda} = \frac{2\pi \times 2.6}{850 \times 10^{-9}} = 19.219 \times 10^6 \text{ m}^{-1}$$

$$\beta_2 = kn_2 = \frac{2\pi n_2}{\lambda} = \frac{2\pi \times 2.46}{850 \times 10^{-9}} = 18.184 \times 10^6 \text{ m}^{-1}$$

The maximum thickness of the guide slab of a symmetrical planar waveguide so that it supports the first 10 modes is

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$$

$$10 > \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$$

$$\frac{10 \times 850 \times 10^{-9}}{\pi \sqrt{2.6^2 - 2.46^2}} > 2a$$

$$2a < 3.21 \times 10^{-6} \text{ m}$$

PROBLEM 18

The beat length of 12 cm is observed in a typical single-mode fiber when light of wavelength 1,100 nm is launched into it. Calculate the difference between the propagation constant for the two orthogonally polarized modes and the modal birefringence.

Solution: The length over which this beat occurs is called the fiber beat length. It is given as

$$L_B = \frac{2\pi}{\beta_x - \beta_y} = \frac{\lambda}{n_x - n_y} = \frac{\lambda}{B_f}$$

Difference between propagation constant

$$\beta_x - \beta_y = \frac{2\pi}{L_B} = \frac{2\pi}{12 \times 10^{-2}} = 52.35 \text{ m}^{-1}$$

We have

$$L_B = \frac{2\pi}{\beta_x - \beta_y} = \frac{\lambda}{n_x - n_y}$$

Therefore, the modal birefringence is

$$n_x - n_y = \beta_x - \beta_y \times \frac{\lambda}{2\pi} = 52.35 \text{ m}^{-1} \times \frac{1,100 \times 10^{-9}}{2\pi} = 9.164 \times 10^{-6}$$

SUMMARY

1. Light is the visible form of **electromagnetic radiation**.
2. Light travels by the principle of **total internal reflection** through optical fiber.

3. There are different types of fiber. Depending upon the number of modes that can propagate through the fiber, they are classified as single-mode and multimode fibers. Depending upon the refractive index profile there are two types of fibers: step-index fiber and graded-index fiber.
4. Fiber properties depend on core-cladding material and core-cladding refractive index profile.
5. The number of modes transmitted through a fiber depends on its core diameter and refractive index profile.
6. The fabrication of glass fiber is a two-stage process. The first stage is producing a pure glass, and then converting it into the preform. In second stage preform is drawn into a fiber.
7. Optical fibers can be enclosed in a loose tube or tight buffer cable configuration.
8. Hollow core fibers allow high-power transmission without non-linear effects.
9. Dispersion compensated fibers have negative dispersion and are hence used to nullify the positive dispersion effect in a fiber-optic communication system link.

MULTIPLE-CHOICE QUESTIONS

1. What is the necessity of cladding for an optical fiber?
 - (a) To provide proper light guidance inside the core.
 - (b) To avoid leakage of light from the fiber.
 - (c) To provide mechanical strength to the fiber.
 - (d) To protect the core from scratches and other mechanical damages.
2. What are the uses of optical fibers?
 - (a) To transmit information which is in the form of coded signals for telephone communication, computer data, etc.
 - (b) To transmit optical images (e.g., endoscopy).
 - (c) To act as a light source at inaccessible places.
 - (d) To act as sensors for mechanical, electrical, and magnetic measurements.
3. Why do we prefer step-index single-mode fiber for long distance communication? Step-index single-mode fiber has
 - (a) Low attenuation due to smaller core diameter.
 - (b) Higher bandwidth.
 - (c) Very low dispersion.
4. Plastic optic fibers
 - (a) Have lower losses than glass fibers.
 - (b) Are used in the automobile industry.
 - (c) Are suitable for long-distance communications.
 - (d) None of these.
5. A ray of light in a transparent material of refractive index 1.5 is approaching a material with a refractive index of 1.48. At the boundary, the critical angle is
 - (a) 90°
 - (b) 9.4°
 - (c) 75.2°
 - (d) 80.6°
6. The common windows used in fiber-optic communications are centered on wavelengths of
 - (a) 1,300 nm, 1,550 nm, and 850 nm
 - (b) 850 nm, 1,500 nm, and 1,300 nm
 - (c) 1,350 nm, 1,500 nm, and 850 nm
 - (d) 800 nm, 1,300 nm, and 1,550 nm
7. As the meridional ray is propagated along the optic fiber, it
 - (a) Travels in a sort of spiral shape.
 - (b) Stays in the center of the fiber.
 - (c) Passes repeatedly through the center of the core.
 - (d) Is reflected off the inside surface of the primary buffer. This is called TIR.
8. Modal dispersion is largest in which type of fiber
 - (a) Step-index multimode.
 - (b) Graded-index multimode.
 - (c) Step-index single-mode.
 - (d) Dispersion-shifted single-mode.
 - (e) Polarization-maintaining fibers.
9. What is the cutoff wavelength of a single-mode step-index fiber with core diameter of 8 μm , core index of 1.5, and cladding index of 1.496?

- (a) $0.89\ \mu\text{m}$
 - (b) $1.15\ \mu\text{m}$
 - (c) $1.28\ \mu\text{m}$
 - (d) $1.31\ \mu\text{m}$
 - (e) $1.495\ \mu\text{m}$
10. What is the acceptance angle for a large-core step-index fiber with core index of 1.5 and cladding index of 1.495?
- (a) 4.7°
 - (b) 7.0°
 - (c) 9.4°
 - (d) 11°
 - (e) 14°
11. Increase in the concentration of GeO_2 in SiO_2 will
- (a) Decrease the refractive index.
 - (b) Increase the refractive index.
 - (c) Change refractive index randomly.
 - (d) Not change refractive index.
12. In a multifiber cable the strength member is
- (a) Placed along the central axis of the cable.
 - (b) Placed in a coaxial cylindrical configuration.
 - (c) Placed anywhere within the cable.
 - (d) Not required.

REVIEW QUESTIONS

1. Write the expression for the refractive index in graded-index fibers.
2. Define numerical aperture of a step-index fiber.
3. Define mode-field diameter.
4. What is Snell's law?
5. What is the principle used in the working of fibers as light guides?
6. What are step-index and graded-index fibers?
7. Define acceptance angle.
8. Define relative refractive index difference.
9. What is V number of fiber or normalized frequency of fiber?
10. What are the conditions for total internal reflection?
11. Explain why the index of the core must be larger than that of the cladding in a step-index fiber.
12. What is meant by step-index fiber and graded-index fiber?
13. What is single-mode fiber and multimode fiber?
14. What is fiber birefringence?
15. Give the expression for numerical aperture in graded-index fibers.
16. Define cutoff wavelength of the fiber.
17. Define and explain the following terms of fibers:
 - (a) Index profile
 - (b) Modes
18. Define MFD. How is it related to the V parameter?
19. Describe the construction of optical fiber cable.
20. With neat sketch describe double crucible method for producing optical fibers.

EXERCISES

1. A step-index fiber has a core index of 1.5 and cladding index of 1.456. What is the acceptance angle of the fiber?
2. A multimode step-index fiber has a relative refractive index difference of 1% and a core refractive index of 1.45. The number of modes propagating at a wavelength of $1.3\ \mu\text{m}$ is 1,100. Estimate the diameter of the fiber core.
3. What is the maximum core diameter for a fiber to operate in single mode at a wavelength of $1,310\ \text{nm}$ if the NA is 0.12?
4. Determine the normalized frequency at $850\ \text{nm}$ for a step-index fiber, which has a core radius of $25\ \mu\text{m}$, a core refractive index of 1.48, and a cladding refractive index of 1.46. How many modes propagate in this fiber at $1,320\ \text{nm}$ and $1,550\ \text{nm}$?
5. If a multimode step-index fiber having a core refractive index of 1.5, a cladding refractive index of 1.38, and a core radius of $25\ \mu\text{m}$ operates at a wavelength of $1,300\ \text{nm}$, calculate:

- (a) Numerical aperture.
 - (b) Normalized frequency.
 - (c) Solid acceptance angle.
 - (d) Total number of modes entering the fiber.
6. A multimode step-index fiber with a core diameter of $80\text{ }\mu\text{m}$ and a relative refractive index difference of 1.5% is operating at a wavelength of $0.85\text{ }\mu\text{m}$. If the core refractive index is 1.48 , calculate normalized frequency for the fiber and the number of guided modes.
7. Calculate the difference between the propagation constant for the two orthogonally polarized modes and the modal birefringence, when the beat length of 10 cm is observed in a typical single-mode fiber when light of wavelength $1,300\text{ nm}$ is launched into it.
8. Calculate the core radius and numerical aperture at $0.85\text{ }\mu\text{m}$, for a step-index fiber having $n_1 = 1.47$ and $n_2 = 1.465$, normalized frequency is 2.4 .

ANSWERS

Multiple-Choice Questions

- 1. (a)
- 2. (a)
- 3. (c)
- 4. (d)
- 5. (d)
- 6. (a)
- 7. (d)
- 8. (a)
- 9. (b)
- 10. (b)
- 11. (b)
- 12. (c)

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3 | Linear and Non-Linear Effects in Optical Fibers

Learning Objectives

After completing this chapter, you will be able to:

- Understand the origin of loss and causes of various dispersions in optical fibers.
 - Study various linear effects, such as attenuation and dispersion in optical fibers.
 - Study various non-linear effects, such as scattering and Kerr effects.
 - Study how physical effects inside optical fiber can limit communication speeds and distances.
 - Study the applications of non-linear effects.
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3.1 Introduction

Optical fibers are not a lossless or distortionless medium. Thirty years ago, optical fibers typically had a loss of 20 dB/km. Today, the loss can be as small as 0.2 dB/km or even less than that, which is a significant improvement because of technology advancement. This unit discusses the various attenuation sources and other signal-degrading phenomena.

The demand for bandwidth is increasing day by day. For long-haul networks, the installation of new communication link infrastructure involves huge investments. Therefore, it is preferable to use existing infrastructure with increased data rate capacity, using techniques such as dense wavelength division multiplexing (DWDM). The available communication network infrastructure uses optical fibers with poor physical characteristics, which limits their transmission capability at high speeds.

The quality of the signal and the maximum data rate are mostly affected by the pulse broadening effect, called **dispersion**. This is caused by the physical properties of fiber. For long-haul communication, single-mode fibers are used, which suffer from chromatic dispersion and polarization mode dispersion, that is, the spreading of pulses causes overlapping of adjacent pulses. This overlapping increases with distance and after some time, it becomes difficult to interpret the adjacent pulses. This increases the bit error rate. The broadening of pulses is restricted to 10% of the pulse duration of single bit of information and is done to preserve the quality of signal.

3.2 Linear Effects in Optical Fiber

3.2.1 Attenuation

Loss of signal strength while traveling through the optical fiber from the transmitting end to the receiving end is called **attenuation**. As the distance traveled by an optical signal along the fiber increases, the reduction in signal also increases. Therefore, attenuation is always expressed in decibel per kilometer (dB/km). Loss of optical signal is shown as reduction in the intensity of optical signal in Fig. 3.1.



Figure 3.1 | Attenuation in optical fiber.

The two power levels are compared with the unit decibel. If P_s is the input source optical power launched into the optical fiber and P_o is the received output power from the fiber, then

$$\text{Number of decibels (dB)} = 10 \log_{10} \frac{P_s}{P_o} \quad (3.1)$$

$$\Rightarrow \frac{P_s}{P_o} = 10^{\frac{\text{dB}}{10}} \quad (3.2)$$

In optical communication, attenuation is usually expressed in decibels per unit length. Therefore

$$\alpha_{\text{dB}} L = 10 \log_{10} \frac{P_s}{P_o} = 4.343 \alpha_p (\text{km}^{-1}) \quad (3.3)$$

where α_{dB} is the signal attenuation per unit length in decibels and L is the fiber length.

Attenuation is a wavelength-dependent phenomenon as shown in Fig. 3.2. Typically, attenuation decreases with increase in wavelength in the 800–1,600 nm range.^[1] The first low-loss window is around 850 nm.

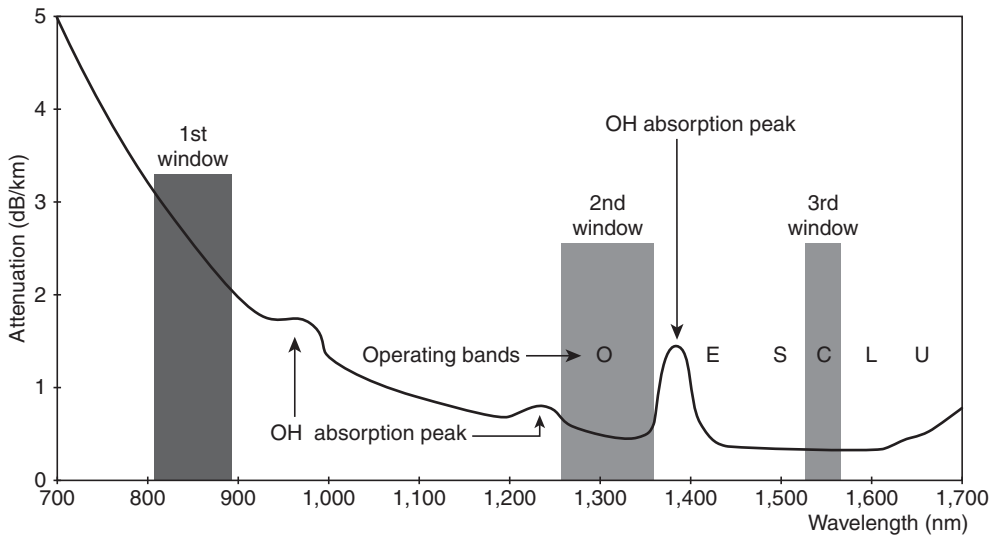


Figure 3.2 | Fiber attenuation.

This window exhibits a relatively high loss, of the order of 3 dB/km. The optical systems working in this window typically employ a light-emitting diode (LED) source and a silicon-based photodiode. The second low-loss window is around 1300 nm with a lower attenuation of around 0.5 dB/km. This window has theoretically zero dispersion. The third window, around 1550 nm, offers minimum attenuation of around 0.2 dB/km (though dispersion is non-zero) for silica fibers. The attenuation also depends on the type of fiber; multimode silica fibers have more attenuation as compared to single-mode silica fibers.

Single-mode fibers possess lower attenuation in the second (1310 nm) and third (1550 nm) windows of operation, hence they are most suited for long-haul telecommunication. On the other hand, multimode fibers are operated in the first (850 nm) and sometimes second (1310 nm) window. Optical fibers possess more attenuation at 850 nm, but this is compensated using a less expensive optical source, and hence they are preferred for short-distance communication.

Scattering, absorption, and bending are the main causes of fiber attenuation. These are mainly dependent on the fiber material properties and the type of fiber. Another cause of signal loss is fiber connections. Signal loss occurs in fiber components such as connectors, splices, and couplers used to interconnect the fiber.

3.2.2 Absorption

The main cause of absorption is the presence of impurities such as metal particles or moisture in the fiber. Due to these impurities, light of a particular wavelength is absorbed and dissipated as heat. Figure 3.2 shows that the OH absorption peaks around 1385 nm.

Absorption is defined as the portion of attenuation resulting from the conversion of optical power into another energy form, such as heat. The absorption in optical fiber is influenced by the following factors:

1. The atomic structure of fiber material.
2. Impurities in fiber material.
3. Basic fiber material.

Defects in atomic structure, such as missing molecules or oxygen defects, cause absorption. Another cause is the diffusion of hydrogen molecules into the glass. Thus, the main cause of absorption is intrinsic and extrinsic material properties.

3.2.2.1 Intrinsic Absorption

The silica fiber is the preferred channel for most communication applications because of its low absorption property in the required wavelength range – from 700 to 1600 nm. Pure silica has low absorption due to its basic material structure. The attenuation in this wavelength range for pure silica is shown in Fig. 3.3; the low attenuation range is between the ultraviolet and infrared regions. Intrinsic absorption in this region is due to the stimulation of electron transitions within the glass by higher energy excitation. The tail of the ultraviolet absorption band is shown in Fig. 3.3. It may extend into the short wavelength region and also into the infrared and far infrared regions. Intrinsic absorption is mainly caused by the interaction of photons with the molecular vibrations within the glass. The vibrations of Si–O, Ge–O, B–O, and P–O give rise to absorption.

3.2.2.2 Extrinsic Absorption

Extrinsic absorption is caused by metallic impurities such as iron, nickel, and chromium introduced into the fiber material during fabrication. Another major cause is absorption due to water dissolved in glass. Water in silica glass forms silicon-hydroxyl (Si–OH) bonds, which are bonded into the glass structure and have stretching vibrations

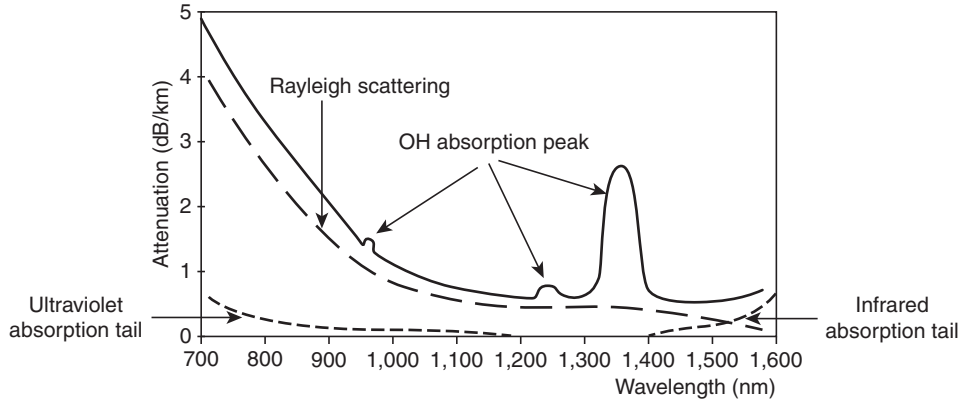


Figure 3.3 | Fiber losses.

between 2700 and 4200 nm. However, the harmonics or overtones of the fundamental absorption occur at 1.38, 1.25, and 0.95 μm . Figure 3.3 shows the presence of the three OH harmonics. The level of the OH harmonic absorption is also indicated.

3.2.3 Linear Scattering

The scattering phenomenon transfers optical power of one mode to another mode. During this process, the signal may get attenuated because the transfer of power may be to a leaky mode, which does not continue to propagate through fiber. There are two categories of linear scattering:

1. Rayleigh scattering.
2. Mie scattering.

3.2.3.1 Rayleigh Scattering

Rayleigh scattering occurs due to material inhomogeneties. The density and compositional variations cause refractive index fluctuations. This effect can be reduced by improving the fabrication process. Attenuation caused by Rayleigh scattering is proportional to the fourth power of the wavelength ($1/\lambda^4$).^[2, 3]

Mathematically, Rayleigh scattering is given as

$$\gamma_R = \frac{8\pi^3}{3\lambda^4} n^8 p^2 \beta_C K T_F \quad (3.4)$$

where γ_R is the Rayleigh scattering coefficient, λ is the optical wavelength, n is the refractive index of the medium, p is the average photo elastic coefficient, β_C is the isothermal compressibility at a fictive temperature T_F , and K is Boltzmann's constant. Now

$$\text{Transmissivity or transmission factor of the fiber} = e^{-\gamma_R L} \quad (3.5)$$

where L is the length of fiber. From the equation of γ_R , we can say that Rayleigh scattering can be strongly reduced by operating at the longest possible wavelength.

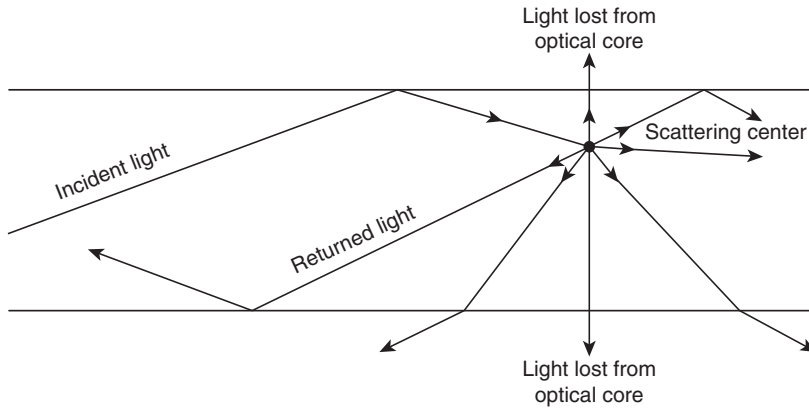


Figure 3.4 | Scattering in optical fiber.

The attenuation due to Rayleigh scattering in dB/km is given as

$$\text{Attenuation} = 10 \log \left(\frac{1}{\text{Transmissivity or transmission factor of the fiber}} \right) \quad (3.6)$$

3.2.3.2 Mie Scattering

Mie scattering occurs when the size of the inhomogenities is comparable to the guided wavelength, for example, non-perfect cylindrical structure, core–cladding refractive index difference, irregularities in core–cladding interface, change in fiber diameter with length, presence of air bubbles, etc.

The scattering created by such inhomogenities is mainly in the forward direction. Inhomogenities can be reduced by the following ways:

1. Removing imperfections such as non-uniform fiber diameter, presence of bubbles, etc.
2. Increasing the relative refractive index difference to make the fiber more guiding.
3. Carefully controlled extrusion and coating of the fiber.

3.2.4 Bending Losses

In an optical fiber, radiation losses take place at the bends, because the light traveling inside the fiber is slower when compared to the evanescent field outside the fiber. (The light wave propagating through the fiber decays exponentially away from the core–cladding boundary. This part of light wave in cladding is called **evanescent field**. As the evanescent wave is the integral part of a guided wave, any modification of the evanescent field modifies the guided wave.) This velocity difference inhibits the guiding of light.

Depending upon the radius of the bend, there are two types of bends:

1. Microbends.
2. Macrobends.

If the radius of curvature is a few micrometers, the bend is called a **microbend** [refer to Fig. 3.5(a)]. Microbends can occur due to any of the following reasons:

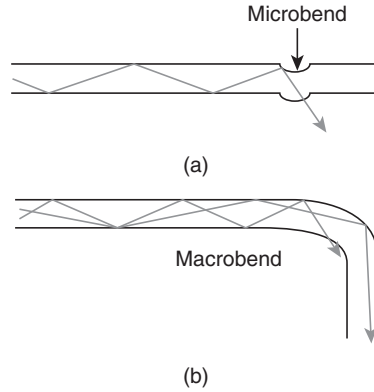


Figure 3.5 | Bending losses: (a) Microbend and (b) macrobend.

1. Non-uniformities in the manufacturing of fiber.
2. Non-uniform mechanical tensile forces by which the fiber is pressed against a rough surface.
3. Non-uniform lateral pressure created during cabling of fiber.

If the radius is large compared to the fiber diameter, the bend is called a **macrobend** [refer to Fig. 3.5(b)]. Bending losses can be reduced by using

1. The shortest possible operating wavelength.
2. A fiber with a large relative refractive index difference.

3.2.4.1 Microbending Losses

Microbends are small repetitive variations in the radius of curvature along the length of a fiber. Microbends can arise when the fibers are incorporated into cables or can be caused by manufacturing defects. Therefore, the losses due to these are also referred to as **cabling losses** or **packaging losses**. The core of the fiber gets deformed as shown in Fig. 3.6. Microbending causes repetitive coupling of energy between guided and non-guided modes, thus causing light to escape outside the fiber core.

Microbending losses can be minimized by extruding a compressible jacket over fiber. Another way to reduce the losses in single-mode fiber is to choose the V number near cutoff.^[4, 5]

In multimode fibers, the microbending loss is expressed mathematically as

$$\text{Loss}_{M_{\text{microbend}}} = Nb^2 \left(\frac{a^4}{b^6 \Delta^3} \right) \left(\frac{E}{E_F} \right)^{3/2} \quad (3.7)$$

where N is the number of bumps, b is the height of the bump per unit length, b is the fiber diameter, a is the core radius, Δ is the relative refractive index difference, E is the elastic modulus of the surrounding medium, and E_F is the elastic modulus of the fiber.

In single-mode fiber, the microbending loss is expressed mathematically as

$$\text{Loss}_{S_{\text{microbend}}} = 0.05 \alpha_m \frac{K^4 (Fd)^6 (\text{NA})^4}{a^2} \quad (3.8)$$

where α_m is the attenuation constant, K is the wave vector $= 2\pi/\lambda$, a is the core radius, and Fd is the half of mode field diameter in single-mode fiber.

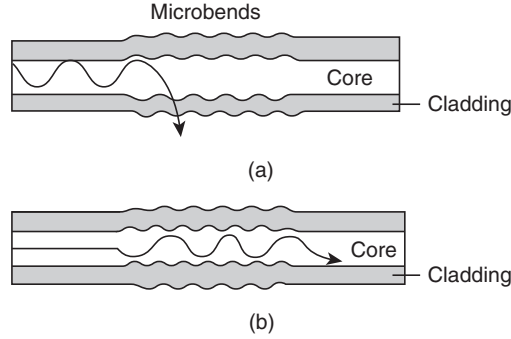


Figure 3.6 | Microbending: (a) Power loss from higher order mode; (b) power coupling to higher order modes.

3.2.4.2 Macrobending Losses

These losses are also called “large-curvature radiation losses,” because these are present in fibers whose radius of curvature is larger than the fiber diameter. The modal electric field distribution has a small portion extending into the cladding, which is called the **evanescent field**. As we go away from the core, the evanescent field decays exponentially. At the macrobend, the evanescent field, which is at the far end from the center of core, must move faster to keep up with the speed of the field in the core. If the fiber bend radius is gradually reduced, at some value of radius the loss becomes maximum, and this radius is called the **critical radius**. Here, a very high mode phase velocity (nearly equal to speed of light) is required, which is impossible. So, some modes are converted to higher order modes, which are then radiated outside as shown in Fig. 3.7.

The macrobend losses occur when optical fibers are packed for transportation during the installation process. Mathematically, the macrobend loss is expressed as

$$\text{Loss}_{\text{macrobend}} = 10 \log \left[\frac{(\alpha + 2)}{(2\alpha) \left(\frac{a}{R\Delta} \right)} \right] \quad (3.9)$$

where α is the profile parameter, a is the core radius, R is the bend radius, and Δ is the index difference.

Large bending losses tend to occur in multimode fibers at a critical radius of curvature R_{CM} , which is given as^[7]

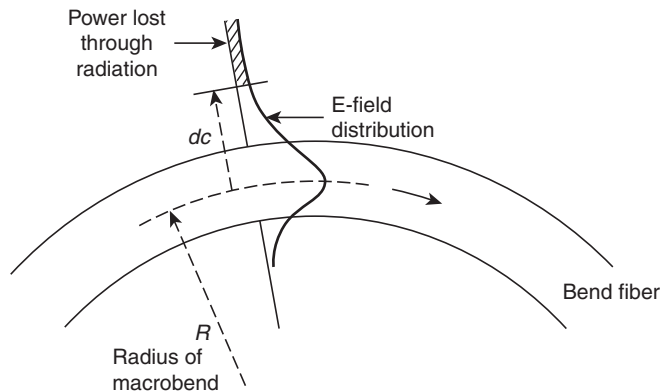


Figure 3.7 | Radiation loss at macrobending.

$$R_{CM} = \frac{3n_1^2 \lambda}{4\pi(n_1^2 - n_2^2)^{3/2}} \quad (3.10)$$

The critical radius of curvature R_{CS} for single-mode fiber is given as^[4]

$$R_{CS} = \frac{20\lambda}{(n_1 - n_2)^{3/2}} \left(2.748 - 0.996 \frac{\lambda}{\lambda_C} \right)^{-3}$$

3.3 Dispersion

In digital communication, information is transmitted in the form of coded pulses, which become unrecognizable after traveling a long distance. Because of pulse broadening, adjacent pulses get overlapped and hence limit the maximum number of pulses sent per second. This phenomenon of pulse broadening is referred to as **pulse dispersion**, and media that possess this property are called **dispersive media**. An example of dispersion seen in nature is a rainbow, in which the white light is split into different color components. Alternatively, using the prism arrangement, dispersion causes separation of white color into different wavelength components.

There are two different types of dispersion in optical fibers:

1. Intramodal dispersion.
2. Intermodal dispersion.

Intramodal or chromatic dispersion occurs in all types of fibers. Intermodal or modal dispersion occurs only in multimode fibers. Each type of dispersion mechanism leads to pulse spreading. The spreading of the optical pulse limits the information-carrying capacity of the fiber.

3.3.1 Intramodal Dispersion or Chromatic Dispersion

Intramodal or **chromatic** dispersion depends primarily on fiber materials. There are two types of intramodal dispersion:

1. Material dispersion.
2. Waveguide dispersion.

Light is a type of electromagnetic wave sent through an optical fiber. It is a combination of electric and magnetic fields orthogonal to each other. This wave travels through free space at a constant speed of 3×10^8 m/s. When the same electromagnetic wave travels through a medium of refractive index n , the speed is reduced to

$$c' = \frac{3 \times 10^8 \text{ m/s}}{n}$$

The above equation indicates that if the refractive index is higher, that is, if the material is denser, the speed of light in the material is lower than the speed of light in vacuum.

The refractive index of a material is a frequency-dependent parameter, therefore the speed of light in a particular material is a frequency-dependent phenomenon, and is called **chromatic dispersion**.

The optical pulse used in telecommunication applications has a special shape, and is called a *Gaussian pulse* as shown in Fig. 3.8. It contains a single bit of information and consists of thousands of photons. This pulse is very intense because the photons are more concentrated in the center of pulse.

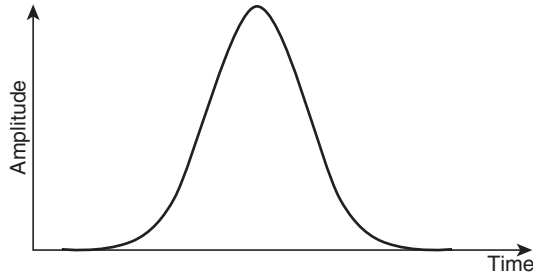


Figure 3.8 | A Gaussian pulse.

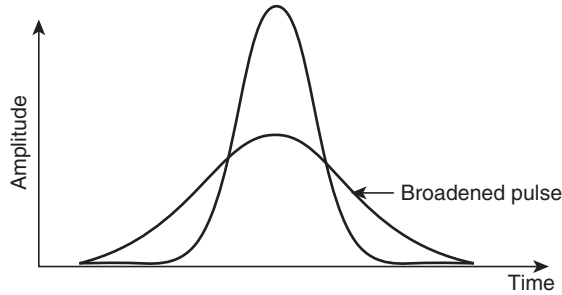


Figure 3.9 | A pulse broadens due to chromatic dispersion.

The optical pulses generated by an ideal monochromatic source should contain photons of a single frequency, and all photons will then travel at same speed. But practically such a source is not available. In practical systems an LED or laser source is used, which is not truly monochromatic. Thus an optical pulse contains photons with different frequencies. So the traveling speed of these photons will be slightly different and this will cause the pulse broadening, as shown in Fig. 3.9.

The broadened pulse has reduced peak intensity as compared to transmitted pulse (as indicated in Fig. 3.10); hence, if the pulses are overlapped, it becomes very difficult to detect. Thus, chromatic dispersion is the main limiting factor for high-data-rate transmission.

This phenomenon can be useful if it is combined with self-phase modulation to obtain optical solutions.

3.3.1.1 Material Dispersion

Material dispersion is caused by refractive-index variation as a function of the optical wavelength. The various spectral components of a given mode will propagate with different speeds depending on the wavelength. Therefore, if the spectral width $\Delta\lambda$ of the source is larger (e.g., when an LED is used), the pulse broadening due to this effect may be significant. Thus, material dispersion depends on the spectral width of the source. All the wavelengths ($\lambda \pm \Delta\lambda$) in the spectral width of the optical source can propagate through the optical fiber with different speeds, causing more dispersion. Material dispersion is an intramodal effect and of particular importance for single-mode and LED systems. Let the refractive index variation be $n(\lambda)$. Then the propagation constant is given as

$$\beta = \frac{2\pi n(\lambda)}{\lambda} = kn(\lambda)$$

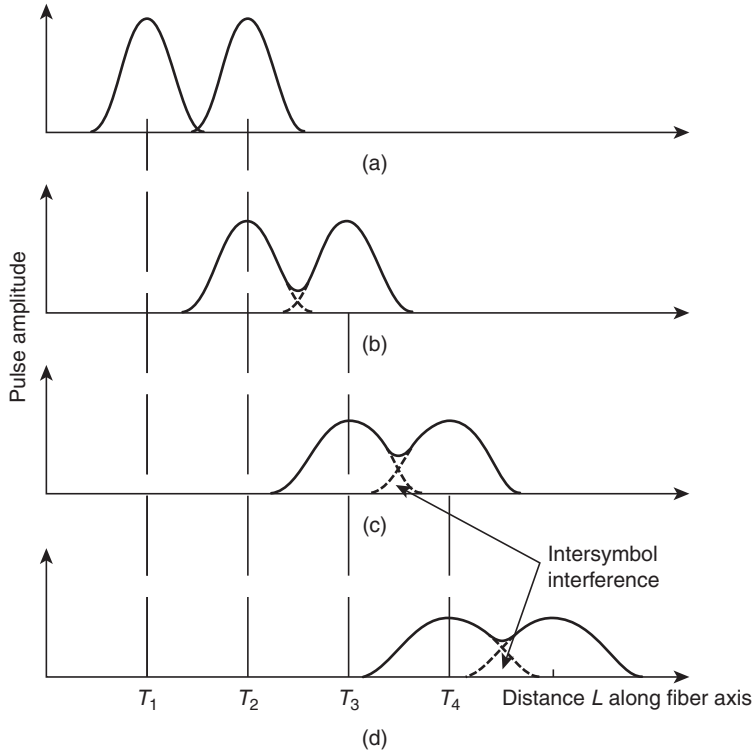


Figure 3.10 | Pulse overlap: (a) Input pulses at T_1 ; (b) distinguishable pulses at time $T_2 > T_1$; (c) barely distinguishable pulses at $T_3 > T_2$; and (d) indistinguishable pulses at $T_4 > T_3$.

The group delay due to material dispersion is then

$$\frac{\tau_{\text{MATERIAL}}}{L} = \frac{1}{V_g} = (-) \frac{\lambda^2}{2\pi c} \left[\frac{d\beta}{d\lambda} \right] = (-) \frac{\lambda^2}{2\pi c} \left\{ \frac{\partial}{\partial \lambda} \left[\frac{2\pi n(\lambda)}{\lambda} \right] \right\} \quad (3.11)$$

$$\frac{\tau_{\text{MATERIAL}}}{L} = (-) \frac{\lambda^2}{2\pi c} \left\{ 2\pi \left[\frac{-n(\lambda)}{\lambda^2} + \frac{1}{\lambda} \frac{dn}{d\lambda} \right] \right\} = \frac{1}{c} \left[n(\lambda) - \lambda \frac{dn}{d\lambda} \right] \quad (3.12)$$

The total dispersion of a length L is then

$$\tau_{\text{MATERIAL}} = \frac{L}{c} [n(\lambda) - \lambda \frac{dn(\lambda)}{d\lambda}] \quad (3.13)$$

The pulse spread σ_{MATERIAL} for a source of spectral width σ_λ is found by differentiating group delay with respect to wavelength and multiplying by σ_λ :

$$\sigma_{\text{MATERIAL}} \approx \left| \frac{d\tau_{\text{MATERIAL}}}{d\lambda} \right| \sigma_\lambda = \frac{\sigma_\lambda L}{c} \left| \lambda \frac{d^2 n}{d\lambda^2} \right| = \sigma_\lambda L |D_{\text{MATERIAL}}(\lambda)| \quad (3.14)$$

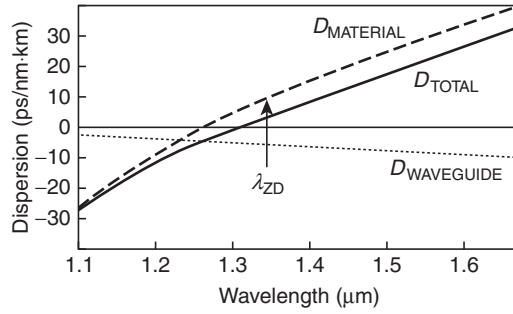


Figure 3.11 | Variation of material and waveguide dispersion as a function of wavelength.

Therefore, the material dispersion of optical fiber D_{MATERIAL} is

$$D_{\text{MATERIAL}} = \frac{\lambda}{c} \left| \frac{d^2 n}{d\lambda^2} \right| \quad (3.15)$$

D_{MATERIAL} is zero at $\lambda = 1.276 \mu\text{m}$ for pure silica, and hence this wavelength is referred to as the *zero dispersion wavelength* (λ_{ZD}) as shown in Fig. 3.11.

3.3.1.2 Waveguide Dispersion

Waveguide dispersion is an intramodal effect. It occurs because a single-mode fiber confines only about 80% of the optical power to the core of the fiber. It occurs within one mode for a single-mode fiber, while in a multimode fiber it occurs in each mode. Every mode in a multimode system will have its own waveguide dispersion. In terms of the modal propagation constant, β varies as a/λ . The optical fiber radial dimension (radius a) relative to the light wavelength λ is now the important parameter.

The group delay in terms of the normalized propagation constant b is

$$b = 1 - \left(\frac{ua}{V} \right)^2 = \frac{\frac{\beta^2}{k^2} - n_2^2}{n_1^2 - n_2^2} \quad (3.16)$$

For a small value of refractive index difference $\Delta = n_1 - n_2/n_1 \ll 1$, that is, for the weakly guided condition,

$$b = \frac{\frac{\beta}{k} - n_2}{n_1 - n_2} \quad (3.17)$$

Solving for β , we get

$$\frac{\beta}{k} = n_2 + b(n_1 - n_2) \quad (3.18)$$

We have

$$n_2 = n_1(1 - \Delta) \Rightarrow n_1 = \frac{n_2}{1 - \Delta} \quad (3.19)$$

Therefore

$$n_1 - n_2 = \frac{n_2}{1 - \Delta} - n_2 = \frac{n_2 - n_2(1 - \Delta)}{1 - \Delta} = n_2 \frac{\Delta}{1 - \Delta} \quad (3.20)$$

Putting these values in Eq. (3.18) yields

$$\frac{\beta}{k} = n_2 + bn_2 \frac{\Delta}{1 - \Delta} \quad (3.21)$$

$$\Rightarrow \beta = kn_2 + kbn_2 \frac{\Delta}{1 - \Delta} \quad (3.22)$$

Since $\Delta = n_1 - n_2/n_1 \ll 1$, we have

$$\beta \approx kn_2 + kbn_2 \frac{\Delta}{1} = kn_2(1 + b\Delta) \quad (3.23)$$

The group delay $\tau_{\text{WAVEGUIDE}}$ arising from waveguide dispersion is

$$\tau_{\text{WAVEGUIDE}} = \frac{L}{c} \frac{d\beta}{dk} = \frac{L}{c} \left[n_2 + n_2 \Delta \frac{d(kb)}{dk} \right] \quad (3.24)$$

We have normalized frequency

$$V = ka(n_1^2 - n_2^2)^{1/2} \cong kan_1 \sqrt{2\Delta}$$

for small values of Δ . Therefore

$$\tau_{\text{WAVEGUIDE}} = \frac{L}{c} \left[n_2 + n_2 \Delta \frac{d(Vb)}{dV} \right] \quad (3.25)$$

The first term in the equation is a constant. It is the time delay for a light pulse traveling in a waveguide where $n_2 = \text{constant}$. The second term above is the group delay arising from waveguide dispersion. The group delay is different for every guided mode. For small-radius waveguides, waveguide dispersion can be significant. For large radius waveguides (multimode), waveguide dispersion is very small and can be neglected.^[8]

3.3.2 Intermodal Dispersion

Intermodal or modal dispersion occurs only in multimode fiber. The input light pulse is made up of a number of modes. All these different modes travel through the optical fiber with different speeds and in different directions. So the distance traveled by each mode is also different. Thus, the time taken by each mode to cover the same distance is different, as shown in Fig. 3.12. This causes pulse broadening. The pulse broadening effect increases with increase in distance traveled by a light pulse through optical fiber. Modal dispersion is the main source of dispersion in multimode fibers. As single-mode fibers propagate only the fundamental mode, modal dispersion does not exist in single-mode fibers. Therefore, single-mode fibers exhibit the lowest amount of total dispersion. Single-mode fibers also exhibit the highest possible bandwidth, and are therefore preferred for long-haul communication.

3.3.3 Polarization-Mode Dispersion

Another source of pulse broadening is the fiber birefringence in the polarization states of an optical signal. Birefringence can result from intrinsic factors such as geometric irregularities of the fiber core or internal stress on it. Other causes of birefringence are bending, twisting, or pinching of the fiber.

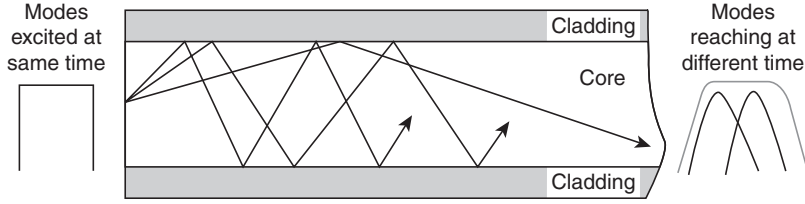


Figure 3.12 | Distance traveled by each mode over the same time span (intermodal dispersion).

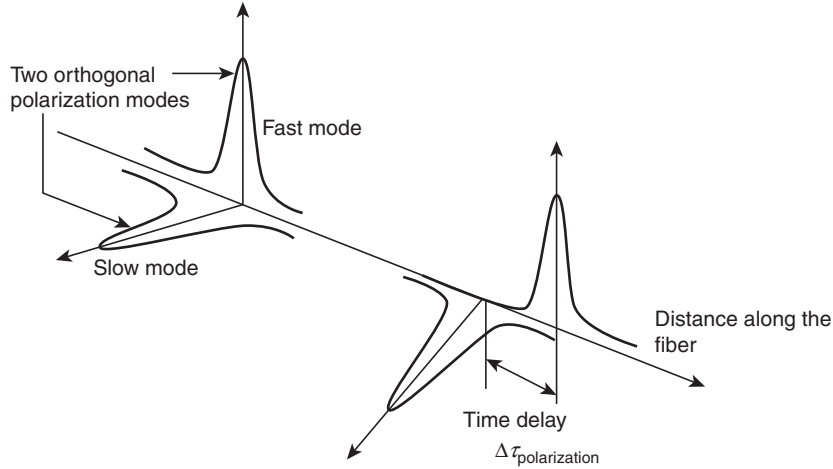


Figure 3.13 | Polarization mode dispersion.

Polarization state is a fundamental property of an optical signal. Polarization refers to the electric field orientation of the light signal. If the polarization state varies with distance traveled by light through the fiber, as shown in Fig. 3.13, and the fiber is birefringent, it causes each polarization mode to travel at a slightly different velocity. The resulting difference in propagation times $\Delta\tau_{\text{polarization}}$ between the two orthogonal polarizing modes will result in pulse broadening. This is the polarization mode dispersion (PMD).

The arrival time difference between two orthogonal polarizations is given by

$$\Delta\tau_{\text{polarization}} = \left[\frac{L}{v_{gx}} - \frac{L}{v_{gy}} \right] \quad (3.26)$$

Temperature and movement are big factors in causing birefringence in the fiber. Thus, PMD varies randomly along a fiber, particularly in aerial cables. This can be the limiting factor, particularly in long-haul fiber-optic communication systems operating at high bit rates.^[9]

3.3.3.1 Information Capacity Determination

A light pulse broadens as it travels along the fiber. The pulse broadening causes neighboring pulses to overlap. After a certain amount of overlap, neighboring pulses will become indistinguishable. Hence, this dispersive mechanism limits the information capacity of a fiber. One measure of the information capacity of an optical waveguide is called the *bandwidth-distance product*. This value is expressed in Hz-km. For example, if a fiber is having a

bandwidth-distance product of 250 MHz-km and the length of the link is 5 km, then the usable bandwidth over this link is

$$\text{Usable bandwidth} = \frac{250 \text{ MHz-km}}{5 \text{ km}} = 50 \text{ MHz}$$

3.3.4 Group Delay

Consider a linear system in which an optical source launches light power into a fiber where all the modes carry equal power. The spectral components will be assumed to travel independently. Each component will undergo a time delay called the *group delay* per unit length in the direction of travel.

$$\text{Velocity}_{\text{group}} = V_g = \frac{c}{n(\lambda)} \quad (3.27)$$

$$\frac{\text{Time delays (sec)}}{\text{Length (m)}} = \frac{\tau_g}{L} = \frac{1}{\text{Velocity}_{\text{group}}} = \frac{1}{V_g} \quad (3.28)$$

$$\tau_g = \frac{L}{V_g} \quad (3.29)$$

As a monochromatic light wave propagates along a waveguide in the z -direction, all points of constant phase travel at a phase velocity V_p given by

$$V_p = \frac{\omega}{\beta} \quad (3.30)$$

The group velocity is then

$$V_g = \frac{d\omega}{d\beta} \quad (3.31)$$

where

$$\omega = 2\pi\nu = \frac{2\pi c}{\lambda}$$

Therefore

$$V_g = \frac{d\left(\frac{2\pi c}{\lambda}\right)}{d\beta} = \frac{d(kc)}{d\beta} = c \frac{dk}{d\beta} \quad (3.32)$$

$$\Rightarrow V_g = 2\pi c \frac{d(\lambda^{-1})}{d\beta} \quad (3.33)$$

By use of the chain rule, we have

$$V_g = \frac{(-)2\pi c}{\lambda^2} \frac{d\lambda}{d\beta} \quad (3.34)$$

$$\Rightarrow \frac{1}{V_g} = \frac{(-)\lambda^2}{2\pi c} \frac{d\beta}{d\lambda} \quad (3.35)$$

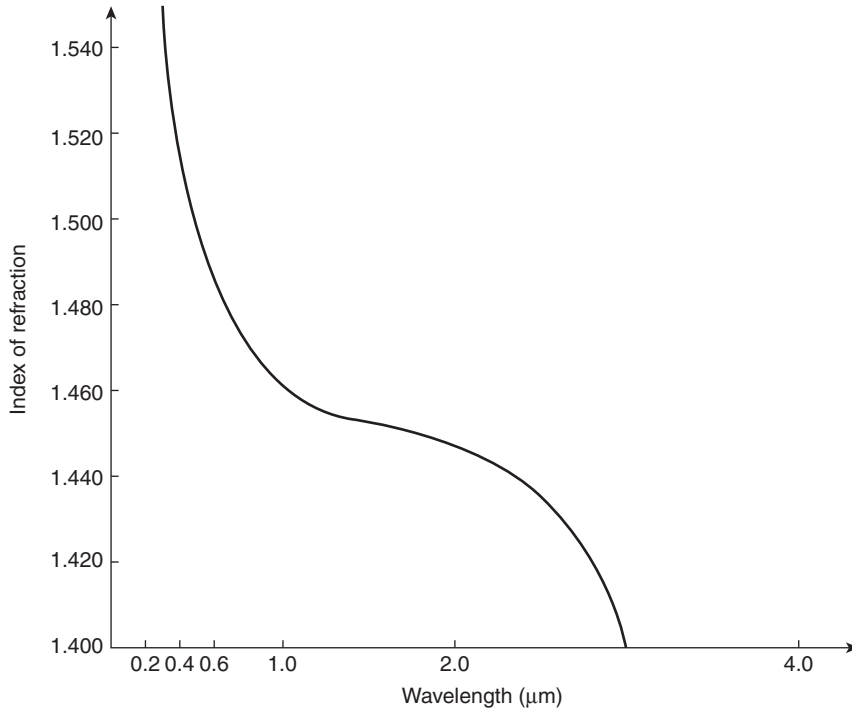


Figure 3.14 | Refractive index variation as a function of optical wavelength.

Since the index of refraction depends on the wavelength as shown in Fig. 3.14 and the velocity of light in a medium depends on the index of refraction, then the group velocity depends on the wavelength of the light.

$$\beta = k \cdot n(\lambda) = \frac{2\pi}{\lambda} n(\lambda) \quad (3.36)$$

Under chromatic dispersion condition, each spectral component will undergo a different time delay and the pulse will spread with transmission distance.

$$\frac{\tau_g}{L} = \frac{1}{V_g} = (-) \frac{\lambda^2}{2\pi c} \frac{d\beta}{d\lambda} \quad (3.37)$$

$$\Rightarrow \tau_g = (-) \frac{L\lambda^2}{2\pi c} \frac{d\beta}{d\lambda} \quad (3.38)$$

Putting the group time delay τ_g into the time spread equation gives

$$\delta\tau = \frac{d}{d\lambda} \left[(-) \frac{L\lambda^2}{2\pi c} \frac{d\beta}{d\lambda} \right] \delta\lambda \quad (3.39)$$

$$\Rightarrow \delta\tau = (-) \frac{L}{2\pi c} \left[2\lambda \frac{d\beta}{d\lambda} + \lambda^2 \frac{d^2\beta}{d\lambda^2} \right] \delta\lambda \quad (3.40)$$

β_2 is the group velocity dispersion (GVD). It gives the magnitude of pulse broadening as the pulse propagates inside the fiber:

$$\beta_2 = \frac{d^2\beta}{d\omega^2} \quad (3.41)$$

If the spectral width of the optical source is moderate, the delay difference per unit wavelength along the propagation path is approximately $d\tau_g/d\lambda$. Therefore

$$\delta\tau = \frac{d\tau_g}{d\lambda} \delta\lambda \quad (3.42)$$

and

$$D = \frac{1}{L} \frac{d\tau_g}{d\lambda} = (-) \frac{L}{2\pi c} \left[2\lambda \frac{d\beta}{d\lambda} + \lambda^2 \frac{d^2\beta}{d\lambda^2} \right] \quad (3.43)$$

where D is the dispersion. It is defined as the pulse spread as a function of wavelength and is measured in picoseconds per kilometer per nanometer [ps/(nm-km)]. It is a combined result of material dispersion (D_{MATERIAL}) and waveguide dispersion ($D_{\text{WAVEGUIDE}}$):

$$\text{Total dispersion} = D_{\text{MATERIAL}} + D_{\text{WAVEGUIDE}}$$

3.3.5 Signal Distortion in Single-Mode Fibers

For single-mode fibers, material τ_{MATERIAL} and waveguide $\tau_{\text{WAVEGUIDE}}$ dispersion can be nearly equal. The pulse spread $\sigma_{\text{WAVEGUIDE}}$ occurring over a distribution of wavelengths σ_λ can be found by taking the derivative of the group delay with respect to wavelength λ :

$$\sigma_{\text{WAVEGUIDE}} \approx \left| \frac{d\tau_{\text{WAVEGUIDE}}}{d\lambda} \right| \sigma_\lambda = \sigma_\lambda L |D_{\text{WAVEGUIDE}}(\lambda)| \quad (3.44)$$

where $D_{\text{WAVEGUIDE}}(\lambda)$ is the waveguide dispersion. Using chain rule, we get

$$\frac{d\tau_{\text{WAVEGUIDE}}}{d\lambda} = \frac{d\tau_{\text{WAVEGUIDE}}}{dV} \frac{dV}{d\lambda} = \left(\frac{d\tau_{\text{WAVEGUIDE}}}{dV} \right) \left(\frac{-V}{\lambda} \right) \quad (3.45)$$

We know that $V = 2\pi a / \lambda(\text{NA})$. Differentiating this, we get

$$dV = \frac{-2\pi a}{\lambda^2} (\text{NA}) d\lambda \quad (3.46)$$

$$\frac{dV}{d\lambda} = -\frac{2\pi a}{\lambda^2} (\text{NA}) = \frac{-V}{\lambda} \quad (3.47)$$

We know that

$$\frac{d\tau_{\text{WAVEGUIDE}}}{d\lambda} = \left(\frac{d\tau_{\text{WAVEGUIDE}}}{dV} \right) \left(\frac{-V}{\lambda} \right) \quad (3.48)$$

$$\Rightarrow \tau_{\text{WAVEGUIDE}} = \frac{L}{c} \left[n_2 + n_2 \Delta \frac{d(bV)}{dV} \right] \quad (3.49)$$

$$\Rightarrow \frac{d\tau_{\text{WAVEGUIDE}}}{dV} = \frac{L}{c} \frac{d}{dV} \left[n_2 + n_2 \Delta \frac{d(bV)}{dV} \right] = \frac{L}{c} \left[n_2 \Delta \frac{d^2(bV)}{dV^2} \right] \quad (3.50)$$

Thus

$$\frac{d\tau_{\text{WAVEGUIDE}}}{d\lambda} = \frac{L}{c} \left[n_2 \Delta \frac{d^2(bV)}{dV^2} \right] \left(-\frac{V}{\lambda} \right) \quad (3.51)$$

Total waveguide dispersion over a finite wavelength spread is

$$\sigma_{\text{WAVEGUIDE}} = \frac{L}{c} \left[n_2 \Delta \frac{d^2(bV)}{dV^2} \right] \left(-\frac{V}{\lambda} \right) \sigma_\lambda = \frac{n_2 L \Delta \sigma_\lambda}{c \lambda} \left[V \frac{d^2(bV)}{dV^2} \right] = \sigma_\lambda L |D_{\text{WAVEGUIDE}}(\lambda)| \quad (3.52)$$

Therefore

$$D_{\text{WAVEGUIDE}}(\lambda) = -\frac{n_2 \Delta}{c \lambda} \left[V \frac{d^2(bV)}{dV^2} \right] \quad (3.53)$$

Calculation of actual values of the term within square bracket is not a simple task, and hence the following easy-to-use empirical formula may be used:

$$V \frac{d^2(bV)}{dV^2} \approx 0.080 + 0.549(2.834 - V)^2 \quad (3.54)$$

Refer to Fig. 3.11 for the behavior of the dispersion by wavelength, which can be summarized as

At longer wavelength, $\tau_{\text{MATERIAL}} > \tau_{\text{WAVEGUIDE}}$

At shorter wavelength, $\tau_{\text{MATERIAL}} < \tau_{\text{WAVEGUIDE}}$

3.3.6 Dispersion and Maximum Bit Rate

In a digital communication system, pulse broadening results in an overlap of pulses. The optical power associated with a broadened pulse gets reduced and hence it becomes difficult to detect such pulses at receiver, resulting in greater BER. Thus along with attenuation, pulse broadening is one of the phenomena which limits the distance between two repeaters in a fiber-optic link. If the pulse broadening is more, less number of pulses per second can be transmitted in a digital link. Thus it limits the maximum permissible bit rate (B_{max}). NRZ is the most common technique of coding used in digital communication; if τ is the pulse dispersion then we have

$$B_{\text{max}} = \frac{0.7}{\tau} \quad (3.55)$$

The above formula takes into account only the limitation imposed by pulse dispersion in the fiber. In an actual link, the source and detector characteristics are also taken into account while calculating the maximum bit rate. In an optical fiber, the pulse dispersion is caused by

1. Intermodal dispersion.
2. Material dispersion.
3. Waveguide dispersion.

However, waveguide dispersion is important only in single-mode fibers, so it can be neglected in the analysis for multimode fibers. Thus (considering multimode fibers), if τ_{MODAL} and τ_{MATERIAL} are the dispersion due to inter-modal and material dispersions, respectively, the total dispersion is given by

$$\tau = \sqrt{\tau_{\text{MODAL}}^2 + \tau_{\text{MATERIAL}}^2} \quad (3.56)$$

3.4 Non-Linear Effects in Fiber

The refractive index of a material is a function of the frequency as well as intensity of light traveling through it. To overcome the losses in optical power, the input optical power can be increased. However, after a certain threshold power level, that is, at high intensity, if there are a number of signal wavelengths present, then the non-linear effects in fiber are significant. Non-linear effects can cause reduction in power or gain in power at different wavelengths, crosstalk between adjacent channels, or wavelength conversion. Thus the optical system performance may degrade. In some applications, non-linear effects can be useful also. Non-linear effects can be divided into two cases based on their origins:

1. Stimulated scattering.
2. Optical Kerr effects.

Optical Kerr effect is the result of intensity dependence of the refractive index of an optical fiber, leading to a phase constant that is a function of the optical intensity, whereas stimulated scattering is a result of scattering leading to an intensity-dependent attenuation constant. There are two stimulated scattering phenomena in an optical fiber: Raman scattering and Brillouin scattering. The intensity dependence of the refractive index results in self-phase modulation (SPM), cross-phase modulation (XPM or CPM), and four-wave mixing (FWM). Another difference between stimulated scatterings and the effects of a non-linear refractive index is that the former is associated with threshold powers at which their effects become significant.^[10–13]

3.5 Scattering Effects

In scattering effects, energy gets transferred from one light wave to another wave at a longer wavelength or lower energy. The lost energy is absorbed by the molecular vibrations or photons in the medium. This second wave is called the Stokes wave. The first wave can be thought of as being a “pump” wave that causes amplification of Stokes wave. As the pump propagates in fiber, it loses power and the Stokes wave gains power.

3.5.1 Stimulated Raman Scattering

The interaction between the incident optical signal (photon) and molecular vibrations gives rise to stimulated Raman scattering (SRS). Due to the interaction, some part of the energy of the incident photon is absorbed by the molecule. This results in the scattering of the photon. The frequency as well as energy of the incident photon reduces in this process, and this modified photon is called the Stokes photon. The reduction in frequency is equal to the molecular vibration frequency, which is called the Stokes frequency and the incident optical signal is called a pump wave. In general, the criterion used to determine the level of scattering effects is the threshold power P_{RamanTh} defined as the input power level that can induce the scattering effect so that half of the power (3-dB power reduction) is lost at the output of an optical fiber of length L . The Raman threshold power P_{RamanTh} for a single-channel optical system is given as

$$P_{\text{RamanTh}} = \frac{32\alpha A_{\text{eff}}}{g_R} \quad (3.57)$$

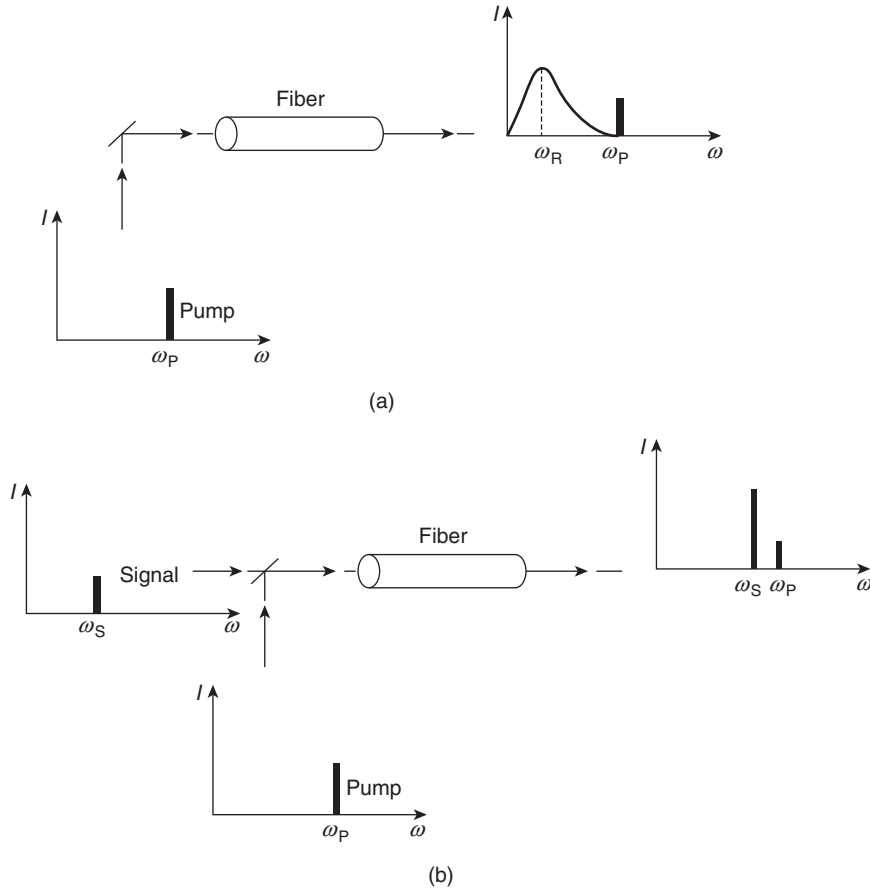


Figure 3.15 | (a) Spontaneous Raman scattering phenomenon. (b) SRS phenomenon.

where α is the attenuation coefficient, A_{eff} is the effective core area of an optical fiber, and g_R is the Raman gain coefficient.

When a monochromatic light beam propagates in an optical fiber, spontaneous Raman scattering [refer to Fig. 3.15(a)] occurs. In this process, some of the photons are transferred to new frequencies. The scattered photons may lose energy corresponding to Stokes shift or gain energy corresponding to anti-Stokes shift. If the pump beam is linearly polarized, the polarization of the scattered photons may be the same (parallel scattering) or orthogonal (perpendicular scattering). The probability of scattering at other frequencies increases if the photons at other frequencies are already present. This process is known as stimulated Raman scattering [refer to Fig. 3.15(b)]. In SRS, a coincident photon at the downshifted frequency will receive a gain. This feature of Raman scattering is exploited in Raman amplifiers for signal amplification.^[14–17]

SRS occurs when light waves interact with molecular vibrations in a solid lattice. If two or more signals at different wavelengths are injected into a fiber, SRS causes power to be transferred from the lower wavelength channel to the higher wavelength channel as shown in Fig. 3.16.

The energy of a photon at wavelength λ is hc/λ , where h is Planck's constant (6.63×10^{-34} Js). Thus a photon of lower λ has higher energy. Transfer of energy from a signal of lower wavelength to signal of higher wavelength corresponds to emission of photons of lower energy caused by photons of higher energy.

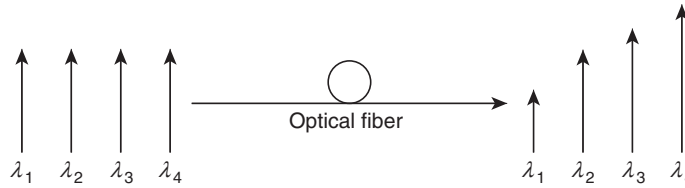


Figure 3.16 | Power from lower wavelength channel is transferred to the higher wavelength channel.

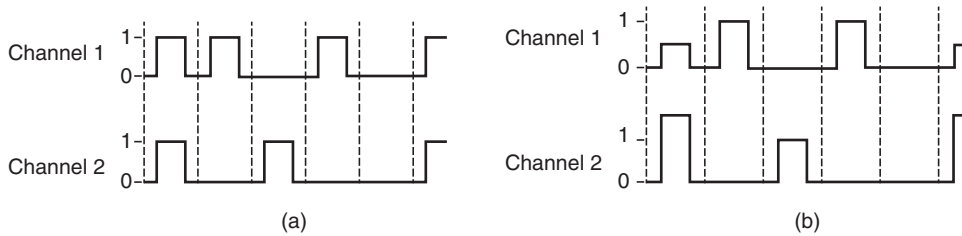


Figure 3.17 | SRS for 2-channel system.

In WDM systems, channels at shorter wavelengths will act as pump signals and suffer from excess loss. On the other hand, the channels at longer wavelengths acting as probe signals are amplified via SRS. In the two-channel system shown in Fig. 3.17, the pump channel (channel 1) is located at λ_1 whereas the probe channel (channel 2) is at λ_2 ($\lambda_2 > \lambda_1$). Here the effect of dispersion is neglected. Under the presence of SRS, the signal in channel 1 is depleted and transferred to channel 2 whenever both channels carry bit 1, as shown in Fig. 3.17(b). However, there is no SRS when both channels carry bit 0.

In WDM systems, with the increase in the number of channels, the chance that all channels transmit bit 1 simultaneously decreases. This decreases the effect of SRS. The effect of SRS is also reduced under the presence of dispersion. Fiber dispersion causes the signals at different wavelengths to travel at different speeds, causing walk-off between bit sequences at different channels. The walk-offs among channels decrease the effect of SRS, hence increasing the threshold power. In general, SRS is not the limiting factor in optical communication systems compared with the other non-linear effects due to its high threshold power.

It is a broadband effect and causes coupling in both directions – forward and reverse. Raman shift between two wavelengths is about 13 THz (100 nm in 1550 nm). Due to the relatively large core diameter of multi-mode fibers, the threshold optical power level is extremely high, hence this effect is not usually observed in multimode fibers.

3.5.1.1 Reduction in SRS Penalty

Different schemes used for reduction of power penalty in SRS process are as follows:

1. Dispersion phenomenon in a channel reduces the SRS penalty. The signals in different channels travel at different velocities and hence reduce the chances of overlap between pulses propagating at different wavelengths.
2. By reducing channel spacing, SRS can be reduced.
3. By keeping the power level of each channel below the threshold level, SRS can be reduced. To achieve this, distance between amplifiers has to be reduced.

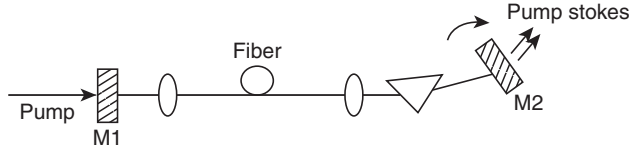


Figure 3.18 | Tunable Raman laser.

3.5.1.2 Applications of SRS Phenomenon

1. **Raman Fiber Laser:** A fiber-based Raman laser employing the SRS phenomenon is shown in Fig. 3.18. A Fabry–Perot cavity is formed using the partially reflecting mirrors M1 and M2. A piece of single-mode fiber is placed inside a cavity in which the SRS process occurs due to wavelength-selective feedback for the Stokes light, which gives an intense output. An intra-cavity prism is used for tuning the laser. The Raman amplification during a round trip should be large enough to compensate the cavity losses, and this determines the Raman threshold power.
2. **Raman Fiber Amplifier:** The SRS phenomenon may be used to provide optical amplification within optical fibers. The SRS process in a fiber causes energy transfer from the pump to the signal. The Raman amplification may occur at any wavelength as long as the appropriate pump laser is available. (For details refer to Section 7.5.)

3.5.2 Stimulated Brillouin Scattering

The interaction between a strong optical signal and an acoustic wave gives rise to the non-linear effect called **stimulated Brillouin scattering (SBS)**. It causes refractive index variations, that in turn cause the scattering of the optical signal in the backward direction. Due to back-scattering there is a loss of signal power. The frequency of scattered light also gets down-shifted. The shift in frequency is equal to frequency of an acoustic wave. The Brillouin threshold power is given as

$$P_{\text{BrillouinTh}} = 42 \frac{\alpha A_{\text{eff}}}{g_B} \left(1 + \frac{v_p}{v_B} \right) \quad (3.58)$$

where g_B is the Brillouin gain coefficient, v_p is the signal line width, and v_B is the Brillouin gain bandwidth.

For a $1.55 \mu\text{m}$ system employing a standard single-mode optical fiber, $P_{\text{BrillouinTh}}$ can be as low as 2 mW when the signal is continuous wave and has ideal zero line width $v_p \approx 0$.

The electrostriction process generates a macroscopic acoustic wave at a frequency ω_B in an oscillating electric field at the pump frequency ω_p . The Brillouin scattering may be spontaneous or stimulated as shown in Figs. 3.19(a) and (b).

In spontaneous Brillouin scattering, there is absence of a pump photon, which results in the creation of a Stokes photon and an acoustic phonon simultaneously. For energy conservation, the Stokes shift ω_B must be equal to $(\omega_p - \omega_s)$, where ω_p is frequency of pump and ω_s is frequency of Stokes waves.

For momentum conservation

$$k_A = (k_p - k_s)$$

where k_A are momentum vectors of acoustic waves, k_p are momentum vectors of pump waves, and k_s are momentum vectors of Stokes waves.

If v_A is acoustic velocity then dispersion relation can be written as

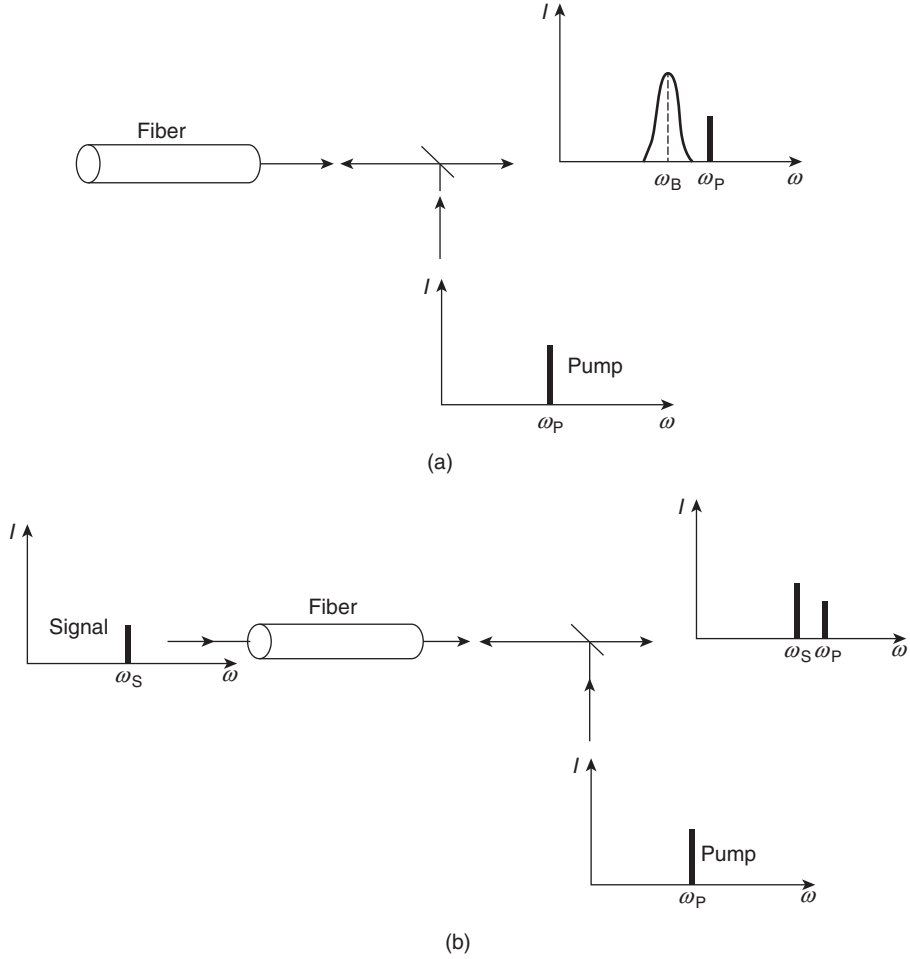


Figure 3.19 | (a) Spontaneous Brillouin scattering phenomenon. (b) SBS phenomenon.

$$\omega_B = v_A |\vec{k}_A| = v_A |\vec{k}_P - \vec{k}_S| \quad (3.59)$$

or

$$\omega_B = 2v_A |\vec{k}_P| \sin(\theta/2) \quad (3.60)$$

where θ is the angle between the pump and Stokes momentum vectors and modulus of k_P and k_S is taken as nearly equal. From the above expression, it is clear that the frequency shift depends on the angle θ . For $\theta = 0^\circ$, shift is zero, that is, there is no frequency shift in forward direction (no Brillouin scattering). $\theta = \pi$ represents the backward direction and in this situation the shift is maximum. The maximum backward frequency shift ($\nu_B = \omega_B/2\pi$) is calculated from Eq. (3.58) and the relationship

$$|\vec{k}_P| = \frac{2\pi n}{\lambda_P}$$

as

$$v_B = \frac{2nv_A}{\lambda_p} \quad (3.61)$$

where n is the mode index.

If the wavelength spacing is much greater than 20 MHz, SBS does not cause any interaction between different wavelengths. The interference of the spontaneously produced scattered wave and the pump wave generates spatial modulation in intensity, which results in amplification of the acoustic wave by the *electrostriction effect* (elasto-optic effect). The amplified acoustic wave raises the spatial modulation of intensity and hence the amplitude of scattered wave. Again there is an increment in the amplitude of the acoustic wave. This positive feedback is responsible for the SBS, which can transfer all power from the pump to the scattered wave.^[18–22]

3.5.2.1 Brillouin Gain Spectrum

The Brillouin gain spectrum describes the dependence of gain on frequency. The finite lifetime T_B of acoustic photons is the root cause of frequency dependence of the gain (g_B). This is also a reason for the small spectral width of the gain spectrum. The nature of decay of acoustic waves is exponential and is of the form $\exp[-t/T_B]$. The Brillouin gain may be written as

$$g_B(\omega) = \frac{g_B(\omega_B)}{1 + (\omega - \omega_B)^2 T_B^2} \quad (3.62)$$

At $\omega = \omega_B$, Brillouin gain is maximum and the gain $g_B(\omega)$ is affected by different fiber parameters such as doping concentration, non-uniform distribution of dopants, etc. The Brillouin gain spectrum for different types of fiber is shown in Fig. 3.20.

3.5.2.2 Reduction in Power Penalty

When any non-linear effect contributes to signal impairment, an additional amount of power is needed at the receiver to maintain the same BER as in the absence of non-linear effects.

The different schemes used to reduce the power penalty due to SBS are as follows:

1. Power level per WDM channel should be kept much below the SBS threshold. So the spacing between amplifier stages should be reduced in long-haul systems.
2. The linewidth can be increased by direct modulation of the source laser. But this may result in significant dispersion, which can be reduced by suitable dispersion management.

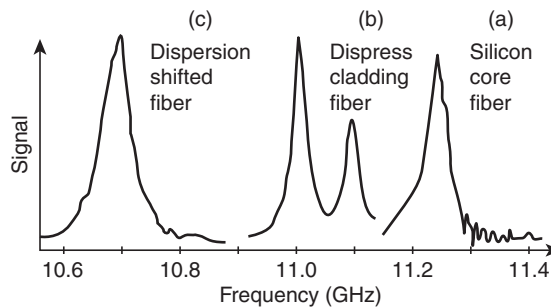


Figure 3.20 | Brillouin-gain spectra at pump wavelength 1525 nm.

3. Instead of amplitude modulation, phase modulation method reduces the power present in an optical fiber, which in turn reduces the SBS penalty.

3.5.2.3 Applications of the SBS Phenomenon

Some applications of SBS in real life are as follows:

1. **Fiber Sensors**
Fiber sensors can sense the temperature and strain over long distances. The refractive index of silica changes with change in temperature or strain. This change produces change in Brillouin shift. By monitoring the change in Brillouin shift, the distribution of temperature and strain over long distances can be obtained. Sometimes such sensors are also known as distributed fiber sensors.^[23]
2. **Brillouin Fiber Amplifiers**
If the frequency shift of a weak signal from the pump frequency is equal to the Brillouin shift, the optical gain in an SBS process can be utilized for amplification of the weak signal. In a Brillouin fiber amplifier, a part of pump power is transmitted to signal through the SBS process and hence amplification in signal power occurs.^[24]
3. **Pulse Delaying and Advancement**
The SBS process is helpful in controlling the group velocity of an optical pulse. This can enable the delaying or advancement of a pulse in the range of tens of nanoseconds. These group delay changes can be obtained in conventionally used optical fibers.^[25]

3.5.2.4 Comparison of Raman and Brillouin Process

In spite of many similarities between SBS and SRS, the SBS differs from the SRS in several ways.

1. Brillouin shift originates from the photon–acoustic photon interaction while Raman shift is due to photon–optical photon interaction.
2. SBS occurs only in the backward direction whereas SRS can occur in both forward and backward directions.
3. Brillouin scattering occurs due to Bragg-type scattering from propagating acoustic waves, and a large number of molecules are involved, while Raman scattering is the result of individual molecular motion.
4. The SBS Stokes shift is smaller (0.09 nm shift in 1550 nm) as compared to SRS Stokes shift (100 nm shift in 1550 nm).
5. The Brillouin gain bandwidth is extremely narrow in comparison to Raman gain bandwidth.
6. The threshold power level for SBS is quite low compared to that of SRS.
7. SBS does not produce crosstalk between adjacent channels, while SRS produces crosstalk.

3.6 Kerr Effects

Electro-optic effects refer to changes in the refractive index of a material induced by the application of an external electric field. This field modulates the optical properties of the device.

Electro-optic effects are classified as first- and second-order effects. Let the refractive index n be a function of the applied electric field $n = n(E)$. Expanding this by a Taylor series in E , we get

$$n' = n + a_1 E + a_2 E^2 + \cdots \quad (3.63)$$

The coefficient a_1 is called the linear electro-optic effect, the coefficient a_2 is called the second order electro-optic effect, and the higher order terms have been found to be negligible for the highest practical electric fields and are ignored.

The change in n due to the linear term is called *Pockels effect*:

$$\Delta n = a_1 E \quad (3.64)$$

The change in n due to the second-order term is called *Kerr effect*:

$$\Delta n = a_2 E^2 = (\lambda K) E^2 \quad (3.65)$$

Here K is the Kerr coefficient measured in m/V^2 . It is noteworthy that K depends on the wavelength. A typical value for the Kerr coefficient of glass is $3 \times 10^{-15} \text{ m/V}^2$.

All materials exhibit the Kerr effect. However, only **non-centrosymmetric materials** exhibit the Pockels effect. GaAs is an example of a material that exhibits Kerr effect.

The *Kerr effect* or the *quadratic electro-optic effect* (QEO effect) is a change in the refractive index of a material in response to an electric field. It is distinct from the Pockels effect in that the induced index change is directly proportional to the *square* of the electric field instead of the magnitude of the field.

Optical Kerr effect was discovered by the Scottish physicist John Kerr in 1875. In this effect, the electric field is due to the light itself. This causes a variation in the index of refraction which is proportional to the local irradiance of the light. This refractive index variation is responsible for the non-linear optical effects of self-focusing and self-phase modulation, and is the basis for Kerr lens mode locking. This effect only becomes significant with very intense beams such as those from lasers.

3.6.1 Four-Wave Mixing

Consider three optical frequencies, f_1 , f_2 , and f_3 , closely spaced (in terms of wavelength). Then, from the interaction of the three, a fourth optical wave frequency f_{fwm} is generated, such that $f_{\text{fwm}} = f_1 + f_2 - f_3$. This is known as *four-wave mixing* (FWM) or *four-photon mixing*. If the order of lightwave frequencies is f_1 , f_{fwm} , f_3 , and f_2 , as shown in Fig. 3.21, then the generated FWM component is $f_{\text{fwm}} = f_1 \pm f_2 \pm f_3$.

The FWM component, E_{FWM} , at the output of a fiber segment, L , generated by the three components E_1 , E_2 , and E_3 , with angular frequency ω , refractive index n , non-linear refractive index χ , and loss α is described by

$$E_{\text{FWM}} = j \left[\frac{2\pi\omega}{nc} \right] d\chi E_1 E_2 E_3 e^{-\alpha(L/2)} F(\alpha, L, \Delta\beta) \quad (3.66)$$

where $F(\alpha, L, \Delta\beta)$ is a function of fiber loss, fiber length, and propagation variation (phase mismatch) related to channel spacing and dispersion.

The output power of the f_{fwm} and the efficiency of four-wave mixing depend on the following factors:

1. Refractive index.
2. Fiber length.
3. Chromatic dispersion of the fiber.

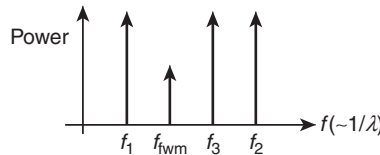


Figure 3.21 | Four-wave mixing [three optical frequencies, f_1 , f_2 , and f_3 , may interact to produce a fourth frequency, f_{fwm} ($f_{\text{fwm}} = f_1 + f_2 - f_3$)].

4. Channel spacing.
5. Power intensity of the contributing frequencies f_1 , f_2 , and f_3 .
6. Higher order polarization properties of the material (non-linear Kerr coefficient).

Four-wave mixing is independent of bit rate.

When N signals are involved in the FWM process, the number of FWM generated signals is given by

$$M = \frac{N^2(N-1)}{2} \quad (3.67)$$

For example, when 12 signals are four-wave mixed together, 792 signals are generated.

If the intensity and wavelengths are in a specific relationship for two signals, FWM may occur. In this case, the fiber refractive index is modulated at the beat frequency of the two wavelengths. The phase modulation creates two sidebands (at frequencies given by this difference), the intensity of which is weak as compared with the intensity of the mixing products from three signals.

The effects of FWM on optical transmission in single-mode fiber are as follows:

1. *Reduction in signal power* due to signal power sharing among the contributing channels to the FWM generated channel.
2. *Crosstalk* due to superposition of uncorrelated data from the contributing channels.
3. *Signal-to-noise degradation* due to superposition of noise and random data from the contributing frequencies f_1 , f_2 , and f_3 .

With increase in the signal input power of f_1 , f_2 , and f_3 and reduction in the channel spacing (due to more wavelength-channels in the same fiber), the FWM output term f_{fwm} increases. ITU-T (G.663) recommends that the critical optical power for FWM should be greater than 10 mW, although this depends on channel spacing. At a channel spacing of 200 GHz, the FWM effect is drastically reduced as compared with 100 GHz or less spacing.

Due to the bandwidth demand, the current trend is to accommodate more and more channels into the available spectrum. As a result of this, the channel spacing decreases to less than 100 GHz (50 GHz for 80 channels in the C-band and going down to 25 GHz). Therefore, FWM crosstalk becomes more significant. A widely used formula for FWM-induced crosstalk is

$$P_{ijk}(L) = (\eta/9) D^2 \gamma^2 P_i P_j P_k e^{-\alpha L} \{[1 - \exp^{-\alpha L}]^2 / \alpha^2\} \quad (3.68)$$

where P_i , P_j and P_k are the input powers of the three input signals f_1 , f_2 , and f_3 ; L is the optical traveled length (i.e., fiber length); α is the attenuation coefficient; η is the FWM efficiency; D is a degenerative factor (equal to 3 for degenerative FWM or 6 for non-degenerative FWM); and γ is a non-linear coefficient (for the fiber medium) given by

$$\gamma = \frac{2\pi n_2}{\lambda A_{\text{eff}}} \quad (3.69)$$

where λ is the wavelength in free space, and A_{eff} and n_2 are the effective area and the non-linear refractive index of the fiber, respectively.

The non-linear refractive index n_2 is related to non-linear susceptibility χ_{1111} of the optical medium and to the refractive index of the core n of the fiber as follows:

$$n_2 = \left[\frac{48\pi^2}{cn^2} \right] \chi_{1111} \quad (3.70)$$

The FWM efficiency depends, among other parameters, on a phase-matching factor, which depends on fiber dispersion and channel spacing. In DWDM communications, the zero-dispersion wavelength is avoided and the

wavelength operating point is preferably positive and slightly above the zero-dispersion wavelength because the FWM phenomenon has the highest efficiency at that condition. Thus, four-wave mixing requires strong phase matching of coincident energy from all three wavelengths. Moreover, both chromatic dispersion and the length of fiber reduce the intensity of the FWM product.

Thus the FWM efficiency depends on the following parameters:

1. Material dispersion.
2. Channel separation.
3. Fiber length.
4. Optical power level of each contributing channel.

The resultant FWM component affects the quality of the signal in terms of crosstalk and bit error rate and thus FWM limits the channel capacity of a fiber system. As the density of wavelengths (channels) in DWDM systems increases, crosstalk increases. The following points are to be noted:

1. FWM becomes more intense if the optical power of each channel is increased.
2. FWM becomes more intense if the channel (wavelength) density increases or if the channels are spaced very close to each other.
3. If the launched optical power of each channel is reduced, then the actual fiber length is decreased to ensure that the arriving signal can be detected reliably. This may necessitate optical amplification to extend the fiber path, thus increasing the system cost.
4. If the channels are spaced farther apart, then fewer wavelength channels can fit into the fiber.

3.6.1.1 Reduction in FWM

Four-wave mixing is a phenomenon that cannot be entirely eliminated. However, it can be greatly minimized so that its effect is not destructive. Different methods to reduce FWM are as follows:

1. Uneven spacing between channels.
2. Increasing channel spacing.
3. Power to be launched into the fiber is reduced.
4. Near-zero net chromatic dispersion is maintained by using segments of fibers with opposing non-zero dispersion characteristics after long spans of standard fiber cable.

3.6.1.2 Applications of FWM Process

Two important applications of FWM are squeezing and wavelength conversion. These are described below.

1. **Squeezing:** It is the phenomenon of reducing quantum noise using the FWM process. It generates the special state of an electromagnetic field for which noise fluctuations, in some frequency range, are reduced below the quantum-noise level. FWM can be used for squeezing as noise components at the signal and idler frequencies are coupled through the fiber non-linearity.

Practically, squeezing is similar to deamplification of a signal and idler waves for certain values of relative phase between the two waves. Photons of random phases are generated due to spontaneous emission at the signal and idler frequencies. Depending on the relative phases of signal–idler photon pairs, the four-wave mixing process increases or decreases its number. Noise is reduced below the quantum-noise level when the phase of the local oscillator is adjusted to match the relative phase corresponding to the photon pair, whose number was reduced as a result of FWM process.

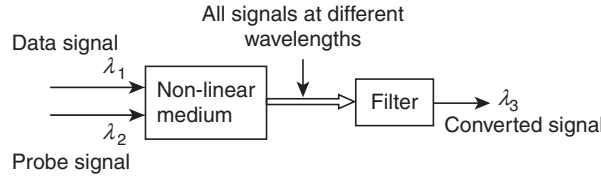


Figure 3.22 | Wavelength conversion through FWM process.

2. **Wavelength conversion:** The four-wave-mixing phenomenon can be used for wavelength conversion. The wavelength converter transforms information from one wavelength to another. The conversion phenomenon is shown in Fig. 3.22. When a data input (λ_1) and a probe signal (λ_2) are injected into a non-linear medium, due to the mixing process a new signal (λ_3) is generated in association with other signal wavelengths such that

$$\frac{1}{\lambda_3} = \frac{2}{\lambda_1} - \frac{1}{\lambda_2} \quad (3.71)$$

$$\Rightarrow \omega_3 = 2\omega_1 - \omega_2 \quad (3.72)$$

where ω is an angular frequency. Wavelength conversion is an important component in all optical networks. The wavelength of the incoming signal may already be in use by another information channel. Conversion of the incoming signal to a new wavelength allows both information channels to traverse the same fiber simultaneously.

3.6.2 Self-Phase Modulation

In self-phase modulation, the optical pulse exhibits a phase shift induced by refractive index and the refractive index varies with intensity of optical signal. The most intense regions of the pulse are slowed down the most, so they exhibit the greatest phase shift. A phase shift changes the distances between the peaks of an oscillating function and the oscillation frequency along the horizontal axis.

A phase shift is equivalent to stretching out or squishing part of an oscillating function along its horizontal axis. Figure 3.23(a) shows a sine wave having constant frequency throughout the length. Figure 3.23(b) shows the same sine wave after undergoing a phase shift such that the right-hand side of the wave has a lower frequency than the left-hand side. As a result, the wave is chirped. There is an ordered variation in frequency along the horizontal axis.

The same concepts of phase shift and chirp may be applied to an optical pulse. Figure 3.24(a) shows an unchirped Gaussian pulse and Fig. 3.24(b) shows the same pulse after being chirped by a phase shift. Notice also that the chirped pulse has the same envelope as the unchirped pulse. This is because self-phase modulation only broadens the pulse in the frequency domain, not the time domain.

As in Fig. 3.24, self-phase modulation leads to chirping with higher frequencies on the trailing (left-hand) side of the pulse and lower frequencies on the leading (right-hand) side. Self-phase modulation may cause errors at the receiving end of a fiber-optic communication system. Particularly in wavelength-division multiplexed systems, the frequencies of individual signals need to stay within strict upper and lower bounds to avoid encroaching on the other signals.

For a silica optical fiber, the refractive index n is given by

$$n = n_0 + n_2 I(\tau) \quad (3.73)$$

where n_0 is the linear refractive index of the material, its value is approximately 1.5; n_2 is the non-linear refractive index, its value is around $3 \times 10^{-20} \text{ m}^2/\text{W}$; and $I(\tau)$ is the optical intensity in units of W/m^2 .

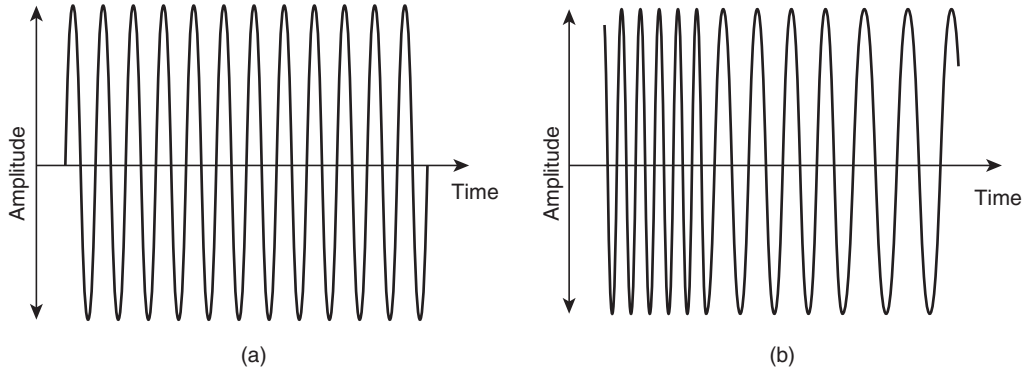


Figure 3.23 | (a) An unchirped sine wave. (b) A chirped sine wave.

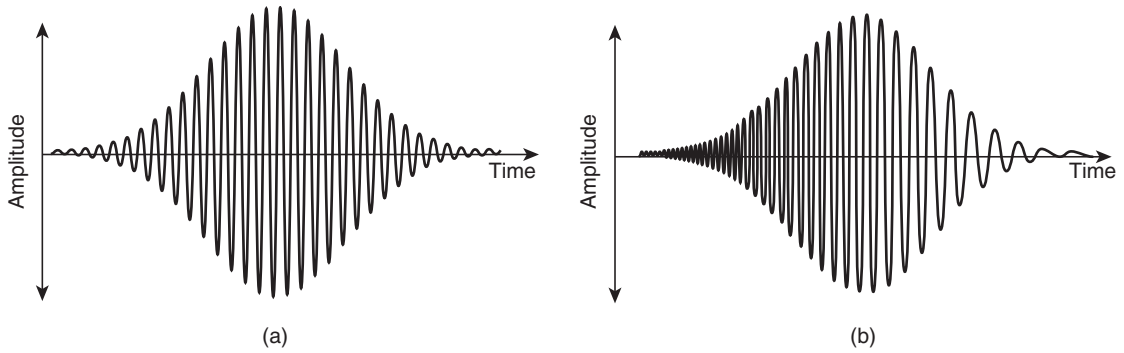


Figure 3.24 | (a) An unchirped Gaussian pulse. (b) A chirped Gaussian pulse.

Although the value of n_2 seems very small, the effect of non-linear refractive index is not negligible at higher signal intensity and long transmission distance. For a propagation distance of L , the phase of the signal is given by

$$\phi(L, \tau) = \frac{2\pi n_0 L}{\lambda} + \frac{2\pi n_2 I(\tau) L_{\text{eff}}}{\lambda} \quad (3.74)$$

where L_{eff} is the effective transmission distance taking into account of the fiber attenuation, and it is given by

$$L_{\text{eff}} = \frac{1 - e^{-\alpha L}}{\alpha} \quad (3.75)$$

Note that Eq. (3.74) is derived under the assumption that a plane wave propagates in an infinite uniform medium. The first term in Eq. (3.74) corresponds to a linear phase shift and depends only on the transmission distance L . On the other hand, the second term in Eq. (3.74) depends on the transmission distance L by way of L_{eff} , as well as the intensity variation of the signal $I(\tau)$. The intensity dependence of refractive index causes a non-linear phase shift, which is proportional to the intensity of the signal $I(\tau)$; hence this effect is called **self-phase modulation**. L_{eff} is less than L and indicates that fiber attenuation reduces the effect of non-linear phase shift. SPM broadens the signal spectrum, but does not affect the intensity profile of the signal. The spectral broadening effect can be understood from the fact that the time-dependent phase variation causes instantaneous frequency deviation $\delta\omega(\tau)$, which is given by

$$\delta\omega(\tau) = -\frac{\partial\phi}{\partial\tau} = -\frac{2\pi n_2 L_{\text{eff}}}{\lambda} \frac{\partial I(\tau)}{\partial\tau} \quad (3.76)$$

The above equation suggests that the magnitude of instantaneous frequency deviation increases with distance and the intensity variation of the signal. The spectrum broadening effect caused by SPM may result in interchannel crosstalk among channels in WDM systems.

SPM alone does not affect the intensity profile of the signal but this is no longer true in the presence of dispersion. The combined effect of dispersion and SPM strongly depends on the sign of dispersion.

3.6.2.1 Applications of SPM Phenomenon

Two important applications of SPM are in solitons and in-pulse compression.^[26] The various applications are as follows:

1. **Solitons:** SPM causes chirping of optical signal with higher frequencies in the trailing edge and lower frequencies in the leading edge. On the other hand, chirping caused by linear dispersion is associated with lower frequencies in the trailing edge and higher frequencies in leading edge. Both these effects are opposite to each other. By proper choice of pulse shape such as a hyperbolic secant-shape and the pulse power, one effect can be compensated with the other. In such a situation, the pulse would propagate undistorted by mutual compensation of dispersion and SPM. An optical pulse having no broadening effect in time domain or frequency domain is called a **soliton**. The high-bandwidth applications in optical communication are mainly limited due to pulse broadening effect, so in such applications solitons can be used efficiently.
2. **Pulse compression:** The SPM phenomenon can be used in pulse compression. In the wavelength region where chromatic dispersion is positive, the blue-shifted trailing edge travels faster, and also moves toward the center of the pulse. Similarly, the red-shifted leading edge of the pulse travels slower and moves toward the center of pulse. In this situation, SPM causes pulses to narrow.
3. **Optically tunable delays:** In ultra-high speed optical communications, the transmission data rate is limited due to optical-to-electronic conversion of information. Therefore, it is desirable to have all-optical components for buffering and tunable delaying signal pulses for application in telecommunication, optical coherence tomography, and optical sampling. There is a novel technique for all-optical delays which involves spectral broadening via self-phase modulation and wavelength filtering.
4. **Optical 40 Gb/s 3R regenerator:** Combined effect of self-phase modulation and cross-absorption modulation is utilized in all optical 3R regenerators.^[26] The performance of such regenerators is experimentally verified for 40 Gb/s data rate. The introduction of a pre-distortion block configuration including a highly non-linear fiber enhances the chromatic dispersion tolerance.

3.6.3 Cross-Phase Modulation (XPM or CPM)

Cross-phase modulation (XPM) is due to the non-linear effect of the refractive index on the optical intensity. The total non-linear phase shift on a given channel is due to the combined intensities of all transmitted channels, which can result in crosstalk among WDM channels. When N channels are transmitted in a single optical fiber, the non-linear phase shift on the j th channel is given by

$$\phi(L, \tau) = \frac{2\pi n_0 L}{\lambda} + \frac{2\pi n_2 L_{\text{eff}}}{\lambda} \left(I_j(\tau) + 2 \sum_{m \neq j}^N I_m(\tau) \right) \quad (3.77)$$

where $I_m(\tau)$ is the optical intensity of the m^{th} channel. The first term in the parentheses on the right-hand side of Eq. (3.77) corresponds to SPM discussed in the previous section, whereas the second term is responsible for CPM. The factor of 2 in Eq. (3.77) indicates that the effect of CPM from a neighboring channel is two times stronger than that caused by SPM itself.

In continuous-wave signals, CPM would dominate over SPM according to Eq. (3.77). For pulses at different wavelengths, the effect of CPM depends on the relative temporal locations of those pulses. CPM is strongest when pulses completely overlap one another. Additionally, the probability that all channels transmit bit 1 has to be taken into consideration, similar to the case of SRS. The low probability of all channels simultaneously transmitting bit 1 reduces the effect of CPM on average. Under the presence of dispersion, pulses at different wavelengths travel at different group velocities, which cause pulses to walk off from one another, reduce the effect of CPM. If the dispersion discrepancy among channels is more, the pulses walk off from one another more rapidly. In other words, the effect of CPM is inversely proportional to dispersion discrepancies among channels in WDM systems. Thus, in order to minimize the impairment caused by CPM, the channel separation and/or local dispersion have to be properly chosen in WDM systems.

3.6.3.1 Applications of the CPM Phenomenon

Optical switching and pulse compression can be done using the CPM phenomenon.

1. **Optical switching:** It can be obtained by phase shift in an optical pulse due to CPM phenomenon. For ultra-fast optical switching, the interferometric method employing CPM-induced phase shift is used. An interferometer divides a weak signal pulse equally between its two arms. This signal experiences identical phase shifts in each arm and is transmitted through constructive interference. When a pump pulse at different wavelengths is injected into one of the arms, the CPM phenomenon in that arm changes the signal phase. If the CPM-induced phase shift is large (close to π), it will produce destructive interference and hence no transmission of signal pulse. Thus an intense pump pulse can switch the signal pulse.
2. **Pulse compression:** Like the SPM-induced frequency chirp, the CPM-induced frequency chirp can also be used for pulse compression. Unlike the SPM techniques, CPM is able to compress even weak input pulses because the co-propagating intense pump pulse produces the frequency chirp. The CPM-induced chirp is affected by pulse walk-off and depends on the initial relative pump signal delay. The use of CPM-induced pulse compression requires a careful control of the pump pulse parameters such as its wavelength, peak power, width and initial delay relative to the signal pulse.
3. **Pulse re-timing:** In an anomalous-dispersion polarization-maintained fiber, ultra-fast optical pulses can be re-timed by utilizing cross-phase modulation phenomenon. Spectral, temporal, and spatial properties of ultra-short pulses can be controlled with help of this phenomenon.

3.7 Comparison of Kerr Effect and Scattering Effect

Kerr and scattering non-linear effects are compared as below:

1. Population inversion is not required in Kerr non-linear effects, while it is needed for non-linear scattering effects (SRS and SBS).
2. Kerr non-linear effects are due to intensity dependence of refractive index, whereas this does not happen in the case of non-linear scattering effects.
3. Kerr non-linearities are elastic, that is, they involve no energy transference. Non-linear scattering effects are inelastic, that is, energy transfer occurs from pump wave to the Stokes wave.

3.8 The Formation of Solitons

We have now learned the origins of chromatic dispersion and self-phase modulation in previous sections and how they separately affect the propagation of an optical pulse. Both these phenomena are undesirable in fiber-optic communication if considered separately. Since the effects are opposite to each other, so we can design a system such that the effects caused by the two phenomena will cancel each other and there will not be any kind of pulse broadening.

Self-phase modulation leads to higher frequencies at the trailing side of the pulse and lower frequencies at the leading side of the pulse, while dispersion causes higher frequencies to travel faster than lower frequencies. Therefore, dispersion causes the trailing side of the pulse to travel faster than the leading side, effectively compressing the pulse and undoing the frequency chirp induced by self-phase modulation. If the properties of the pulse are just right, the instantaneous effects of self-phase modulation and dispersion cancel each other completely, and the pulse remains unchirped and retains its initial width along the entire length of the fiber. In other words, we have formed a soliton!

The formation of solitons requires a careful balance among the properties of the pulse and the fiber. Specifically, the following relationship must be satisfied:

$$P = -\frac{\beta_2}{\gamma T_0^2} \quad (3.78)$$

where P is the peak power of the pulse, β_2 is the dispersion parameter, γ is the non-linear coefficient, and T_0 is the half-width of the pulse at its $1/e$ (36.79%) of the intensity point.

The dispersion parameter and the non-linear coefficient depend on the properties of both the pulse and the fiber. They are given by

$$\beta_2 = \frac{\lambda^3}{2\pi c^2} \frac{\partial^2 n}{\partial \lambda^2} \quad \text{and} \quad \gamma = \frac{2\pi n_2}{\lambda A_{\text{eff}}} \quad (3.79)$$

where λ is the pulse wavelength, c is the speed of light in a vacuum (3.0×10^8 m/s), n is the refractive index, n_2 is the non-linear index coefficient (a measure of the degree of fiber non-linearity), and A_{eff} is the effective core area of the fiber. In addition, the pulse must be unchirped and have a hyperbolic-secant shape, as shown in Fig. 3.25. If all of these qualifications are met, solitons may be successfully generated and propagated through a fiber.

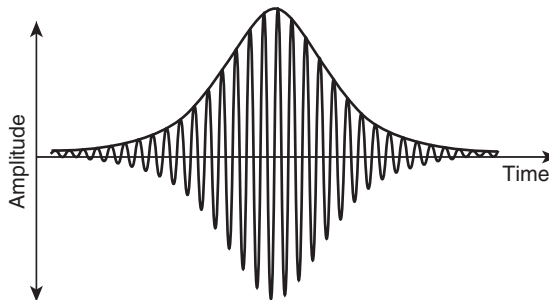


Figure 3.25 | An optical soliton with a hyperbolic-secant envelope.

Solved Problems

PROBLEM 1

Determine the overall signal attenuation in decibels through an optical fiber of length 8 km, assuming no splices and connectors, when the mean optical power launched is $120 \mu\text{W}$ and the output power is $5 \mu\text{W}$.

Solution: We know that

$$\text{Number of decibels (dB)} = 10 \log_{10} \frac{P_s}{P_o}$$

$$\Rightarrow \text{Number of decibels (dB)} = 10 \log_{10} \frac{120 \times 10^{-6}}{5 \times 10^{-6}} = 24.0 \text{ dB}$$

PROBLEM 2

Determine the signal attenuation per kilometer for the optical fiber of length 10 km, assuming no splices and connectors, when the mean optical power launched is $100 \mu\text{W}$ and the output power is $5 \mu\text{W}$.

Solution:

$$\alpha_{\text{dB}} L = 10 \log_{10} \frac{P_s}{P_o} = 10 \log_{10} \frac{100 \times 10^{-6}}{5 \times 10^{-6}} = 20.0 \text{ dB}$$

Therefore

$$\alpha_{\text{dB}} = \frac{20.0}{10 \text{ km}} = 2 \text{ dB/km}$$

PROBLEM 3

Determine the overall signal attenuation for the optical fiber of length 20 km with splices at 2 km intervals, each giving an attenuation of 1.5 dB, when the mean optical power launched is $200 \mu\text{W}$ and the output power is $15 \mu\text{W}$.

Solution:

$$\text{Number of decibels (dB)} = 10 \log_{10} \frac{P_s}{P_o} = 10 \log_{10} \frac{200 \times 10^{-6}}{15 \times 10^{-6}} = 13.33 \text{ dB}$$

The link has 9 splices (at 2 km intervals), each with an attenuation of 1.5 dB.

Therefore, total loss due to splices is $9 \times 1.5 \text{ dB} = 13.5 \text{ dB}$.

Hence the overall signal attenuation for the link is

$$\text{Total signal attenuation} = 13.33 \text{ dB} + 13.5 \text{ dB} = 26.83 \text{ dB}$$

PROBLEM 4

Determine the attenuation (in dB/km) in silica fiber due to Rayleigh scattering at an optical wavelength of 630 nm. The different parameters of silica fiber are as follows:

n is the refractive index of medium = 1.46

p is the average photo elastic coefficient = 0.286

β_C is the isothermal compressibility = $7 \times 10^{-11} \text{ m}^2 \text{N}^{-1}$

Fictive temperature $T_F = 1400 \text{ K}$

k is the Boltzmann's constant = $1.381 \times 10^{-23} \text{ JK}^{-1}$

Solution: Mathematically, Rayleigh scattering is given as

$$\begin{aligned}\gamma_R &= \frac{8\pi^3}{3\lambda^4} n^8 p^2 \beta_C K T_F \\ &= \frac{8\pi^3}{3(630 \times 10^{-9})^4} \times (1.46)^8 \times (0.286)^2 \times 7 \times 10^{-11} \times 1.381 \times 10^{-23} \times 1400 \\ &= 1.199 \times 10^{-2} \text{ m}^{-1}\end{aligned}$$

Transmissivity or transmission factor of the fiber = $\exp(-\gamma_R L)$. Therefore

$$\text{Transmissivity} = \exp(-1.199 \times 10^{-2} \times 10^{-3}) = 0.301$$

The attenuation due to Rayleigh scattering in dB/km is given as

$$\begin{aligned}\text{Attenuation} &= 10 \log \left(\frac{1}{\text{Transmissivity or transmission factor of the fiber}} \right) \\ &= 10 \log \left(\frac{1}{0.301} \right) = 5.2 \text{ dB/km}\end{aligned}$$

PROBLEM 5

Calculate the macrobend loss of a GI fiber with index profile $\alpha = 2$, core diameter of $50 \mu\text{m}$, and wavelength 850 nm , bent curve radius $R = 2.0 \text{ cm}$. The core refractive index is 1.45, $\text{NA} = 0.21$, $\Delta = 0.02$.

Solution: Mathematically, the macrobend loss is expressed as

$$\text{Loss}_{\text{macrobend}} = 10 \log \frac{(\alpha + 2)}{(2\alpha) \left(\frac{a}{R\Delta} \right)}$$

Substituting the values we get

$$\text{Loss}_{\text{macrobend}} = 10 \log \frac{(2 + 2)}{(2 \times 2) \left(\frac{25 \times 10^{-6}}{2 \times 10^{-2} \times 0.02} \right)} = 12 \text{ dB}$$

PROBLEM 6

Calculate attenuation in dB/km in an optical fiber of length 600 m , in which 80% of optical signal is lost while propagating through fiber.

Solution: Using the relevant formulae, we can calculate

$$\frac{P_s}{P_o} = 80\% = 0.8$$

Therefore

$$\alpha_{\text{dB}} L = 10 \log_{10} \frac{P_s}{P_o} = 10 \log_{10}(0.8)$$

$$\alpha_{\text{dB}} = -1.615 \text{ dB/km}$$

PROBLEM 7

For an optical fiber communication system of 2 km length, the 3-dB pulse widths at the input and output are 0.5 ns and 10 ns, respectively. Find the pulse broadening for the fiber and the maximum bit rate.

Solution:

$$\text{Pulse broadening, } \tau = 10 \text{ ns} - 0.5 \text{ ns} = 9.5 \text{ ns}$$

So

$$B_{\text{max}} = \frac{0.7}{\tau} = \frac{0.7}{9.5 \times 10^{-9}} = 73.68 \times 10^6 \text{ bits per second}$$

PROBLEM 8

An LED operating at 850 nm has a spectral width of 55 nm and launches the power in 1 km fiber. What is the pulse spreading in ns/km due to material dispersion? Given:

$$\frac{d^2 n}{d\lambda^2} = 4 \times 10^{-2} \mu\text{m}^{-2}$$

Solution: The pulse spread σ_{MATERIAL} for a source of spectral width σ_λ is

$$\sigma_{\text{MATERIAL}} \approx \left| \frac{d\tau_{\text{MATERIAL}}}{d\lambda} \right| \sigma_\lambda = \frac{\sigma_\lambda L}{c} \left| \lambda \frac{d^2 n}{d\lambda^2} \right| = \sigma_\lambda L |D_{\text{MATERIAL}}(\lambda)|$$

$$\sigma_{\text{MATERIAL}} \approx \frac{1}{3 \times 10^8} \times 850 \times 4 \times \frac{10^{-2}}{10^{-6}} \times 55$$

$$\sigma_{\text{MATERIAL}} = 6.23 \text{ ns/km}$$

PROBLEM 9

An optical power of 10 μW is fed into fiber of 10 km length. The output power is found to be 5 μW . Calculate

- Overall loss in dB assuming no connectors or splices.
- Overall loss in dB/km.
- Overall signal attenuation for a 20 km optical link using the same fiber with splices at 2 km interval, each of 1 dB attenuation.
- The numerical input–output power ratio in part (c).

Solution:

- (a) Number of decibels(dB) = $10 \log_{10} \frac{P_s}{P_o}$. So

$$\text{Number of decibels(dB)} = 10 \log_{10} \frac{10 \times 10^{-6}}{5 \times 10^{-6}} = 3.0 \text{ dB}$$

- (b) We have

$$\alpha_{\text{dB}} L = 10 \log_{10} \frac{P_s}{P_o} = 3.0 \text{ dB}$$

$$\alpha_{\text{dB}} = \frac{3.0}{10 \text{ km}} = 3.0 \text{ dB/km}$$

- (c) Here $L = 20 \text{ km}$

$$\text{Loss along 20 km fiber} = \frac{0.3 \text{ dB}}{\text{km}} \times 20 \text{ km} = 6 \text{ dB}$$

The link has 9 splices (at 2-km intervals) each with an attenuation of 1.0 dB.

Therefore total loss due to splices is $9 \times 1.0 \text{ dB} = 9.0 \text{ dB}$.

Hence the overall signal attenuation for the link is

$$\text{Total signal attenuation} = 6.0 \text{ dB} + 9.0 \text{ dB} = 15.0 \text{ dB}$$

- (d) Input–output power ratio is

$$\frac{P_s}{P_o} = 10^{\text{dB}/10} = 10^{15/10} = 31.6$$

PROBLEM 10

An LED with a spectral width of 17 nm is used as an optical source and the length of the fiber is 30 km. Due to material dispersion, the pulse spreads at the output at 1.76 ns/km. Calculate the amount of material dispersion.

Solution: We know that

$$\sigma_{\text{MATERIAL}} \approx \left| \frac{d\tau_{\text{MATERIAL}}}{d\lambda} \right| \sigma_{\lambda} = \frac{\sigma_{\lambda} L}{c} \left| \lambda \frac{d^2 n}{d\lambda^2} \right| = \sigma_{\lambda} L |D_{\text{MATERIAL}}(\lambda)|$$

Therefore, material dispersion is

$$D_{\text{MATERIAL}} = \frac{\sigma_{\text{MATERIAL}}}{\sigma_{\lambda} L} = \frac{1.76}{17 \times 30} = 3.45 \times 10^{-3} \text{ ns/nm-km}$$

PROBLEM 11

An optical fiber has losses of 0.6 dB/km at 1300 nm. If 100 μW of power is injected into the fiber, how much power will reach at a distance of 20 km?

Solution:

$$\begin{aligned}\alpha_{\text{dB}} L &= 10 \log_{10} \frac{P_s}{P_o} = 10 \log_{10} \frac{100 \times 10^{-6}}{P_o} \\ \frac{0.6 \text{ dB}}{\text{km}} \times 20 \text{ km} &= 10 \log_{10} \frac{100 \times 10^{-6}}{P_o} \\ 1.2 &= \log_{10} \frac{100 \times 10^{-6}}{P_o} \\ \frac{100 \times 10^{-6}}{P_o} &= 10^{1.2} \\ P_o &= 6.3 \mu\text{W}\end{aligned}$$

PROBLEM 12

Determine the attenuation in dB/km due to Rayleigh scattering in silica at an operating wavelength of 700 nm. Silica has fictive temperature of 1400 K with isothermal compressibility of $7 \times 10^{-11} \text{ m}^2 \text{N}^{-1}$. Refractive index is 1.46 and the photo-elastic coefficient is 0.29.

Solution: Mathematically, Rayleigh scattering is given as

$$\begin{aligned}\gamma_R &= \frac{8\pi^3}{3\lambda^4} n^8 p^2 \beta_C K T_F \\ &= \frac{8\pi^3}{3(700 \times 10^{-9})^4} \times (1.46)^8 \times (0.29)^2 \times 7 \times 10^{-11} \times 1.381 \times 10^{-23} \times 1400 \\ &= 0.809 \times 10^{-3} \text{ m}^{-1}\end{aligned}$$

Transmissivity or transmission factor of the fiber = $\exp(-\gamma_R L)$. Hence

$$\text{Transmissivity} = \exp(-0.809 \times 10^{-3} \times 10^{-3}) = 0.445$$

The attenuation due to Rayleigh scattering in dB/km is given as

$$\begin{aligned}\text{Attenuation} &= 10 \log \left(\frac{1}{\text{Transmissivity or transmission factor of the fiber}} \right) \\ \text{Attenuation} &= 10 \log \left(\frac{1}{0.445} \right) = 3.5 \text{ dB/km}\end{aligned}$$

PROBLEM 13

A step-index single-mode fiber has a core refractive index of 1.45, relative refractive index difference of 0.3%, and a core diameter of 8.2 μm . Calculate the waveguide dispersion parameter for the given fiber at $\lambda = 1300 \text{ nm}$.

Solution: We have

$$n_2 = n_1(1 - \Delta) = 1.45(1 - 0.003) = 1.4456$$

Since

$$V = \frac{2\pi a}{\lambda} \times n_1 \sqrt{2\Delta} = \frac{\pi \times 8.2}{\lambda} \times 1.45 \sqrt{2 \times 0.003} = 2.2256$$

we have

$$V \frac{d^2(bV)}{dV^2} \approx 0.080 + 0.549(2.834 - V)^2$$

$$V \frac{d^2(bV)}{dV^2} \approx 0.080 + 0.549(2.834 - 2.2256)^2 = 0.2832$$

The waveguide dispersion parameter is given as

$$D_{\text{WAVEGUIDE}}(\lambda = 1300 \text{ nm}) = -\frac{n_2 \Delta}{c\lambda} \left[V \frac{d^2(bV)}{dV^2} \right]$$

$$D_{\text{WAVEGUIDE}}(\lambda = 1300 \text{ nm}) = -\frac{1.4456 \times 0.003}{3 \times 10^5 \text{ km/s} \times 1300} [0.2832]$$

$$D_{\text{WAVEGUIDE}}(\lambda = 1300 \text{ nm}) = -3.149 \text{ ps/nm-km}$$

PROBLEM 14

A step-index single-mode fiber exhibits material dispersion of 6 ps/nm-km at an operating wavelength of 1550 nm. If $n_1 = 1.45$ and $\Delta = 0.5\%$, calculate the diameter of the core of the fiber needed to make the total dispersion zero at operating wavelength.

Solution:

$$\text{Total dispersion} = D_{\text{MATERIAL}} + D_{\text{WAVEGUIDE}}$$

Since

$$D_{\text{MATERIAL}} = 6 \text{ ps/nm-km}, \quad D_{\text{WAVEGUIDE}} = -6 \text{ ps/nm-km}$$

and

$$n_2 = n_1(1 - \Delta) = 1.45(1 - 0.005) = 1.4427$$

we have

$$D_{\text{WAVEGUIDE}}(\lambda = 1550 \text{ nm}) = -\frac{n_2 \Delta}{c\lambda} [0.080 + 0.549(2.834 - V)^2]$$

$$D_{\text{WAVEGUIDE}}(\lambda = 1550 \text{ nm}) = \frac{1.4427 \times 0.005}{3 \times 10^8 \times 1550 \times 10^{-9}} [0.080 + 0.549(2.834 - V)^2]$$

As the required $D_{\text{WAVEGUIDE}} = -6 \text{ ps/nm-km}$, we have

$$-6 \text{ ps/nm-km} = 15.51 \text{ ps/nm-km} \times [0.080 + 0.549(2.834 - V)^2]$$

$$\Rightarrow V = 2.086$$

Since

$$V = \frac{2\pi a}{\lambda} \times n_1 \sqrt{2\Delta}$$

we have

$$2a = \frac{V\lambda}{\pi n_1 \sqrt{2\Delta}} = \frac{2.086 \times (1550 \text{ nm})}{\pi \times 1.45 \times \sqrt{2 \times 0.005}} = 7.1 \text{ } \mu\text{m}$$

PROBLEM 15

Find the radius of curvature of macrobend in optical fiber at which the power decreases by 50%. $\alpha = 2$, $n_2 = 1.45$, $\Delta = 0.02$, and $a = 25 \text{ } \mu\text{m}$ launched in fiber = 24 dB.

Solution: Mathematically, the macrobend loss is expressed as

$$\text{Loss}_{\text{macrobend}} = 10 \log \frac{(\alpha + 2)}{(2\alpha) \left(\frac{a}{R\Delta} \right)}$$

where α is the profile parameter, a is the core radius, R is the bend radius, and Δ is the index difference. Now substituting the values we get

$$12 \text{ dB} = 10 \log \frac{(2 + 2)}{(2 \times 2) \left(\frac{25 \times 10^{-6}}{R \times 0.02} \right)}$$

$$1.2 = \log \frac{(2 + 2)}{(2 \times 2) \left(\frac{25 \times 10^{-6}}{R \times 0.02} \right)}$$

$$15.85 = \frac{1}{\left(\frac{25 \times 10^{-6}}{R \times 0.02} \right)}$$

$$15.85 = \frac{R \times 0.02}{(25 \times 10^{-6})}$$

$$R = 2.0 \times 10^{-2} \text{ m or } 2.0 \text{ cm}$$

PROBLEM 16

Calculate the material loss in GI fiber with profile parameter = 2, at 850 nm wavelength. It is bent with radius of 1 cm. The core refractive index is 1.45, $\Delta = 0.02$, and core diameter is 60 μm .

Solution: The macrobend loss is expressed as

$$\text{Loss}_{\text{macrobend}} = 10 \log \frac{(\alpha + 2)}{(2\alpha) \left(\frac{a}{R\Delta} \right)}$$

Substituting the values we get

$$\text{Loss}_{\text{macro bend}} = 10 \log \frac{(2+2)}{(2 \times 2) \left(\frac{30 \times 10^{-6}}{1 \times 10^{-2} \times 0.02} \right)} = 10 \log 6.66 = 8.23 \text{ dB}$$

PROBLEM 17

Calculate the waveguide dispersion in a 1-km long link having 9/125 single mode fiber with $n_1 = 1.48$, $\Delta = 0.22\%$ operating at 1300 nm.

Solution: We have

$$n_2 = n_1(1 - \Delta) = 1.48(1 - 0.0022) = 1.477$$

Now

$$V = \frac{2\pi a}{\lambda} \times n_1 \sqrt{2\Delta}$$

$$V = \frac{\pi \times 9}{1300 \times 10^{-9}} \times 1.48 \sqrt{2 \times 0.0022} = 2.14$$

We have

$$V = \frac{d^2(bV)}{dV^2} \approx 0.080 + 0.549(2.834 - V)^2$$

$$\approx 0.080 + 0.549(2.834 - 2.14)^2 = 0.48$$

The waveguide dispersion parameter is given as

$$D_{\text{WAVEGUIDE}}(\lambda = 1300 \text{ nm}) = -\frac{n_2 \Delta}{c \lambda} \left[V \frac{d^2(bV)}{dV^2} \right]$$

$$= -\frac{1.477 \times 0.0022}{3 \times 10^5 \text{ km/s} \times 1300} [0.48]$$

$$= -4 \text{ ps/nm-km}$$

PROBLEM 18

For a fiber-optic link of length 1 km, the material dispersion parameter is 25 ps/nm-km. The relative spectral width σ_λ / λ of the source is 0.0012 at 850 nm wavelength. Calculate the pulse broadening per km.

Solution: We have

$$\sigma_\lambda = \left(\frac{\sigma_\lambda}{\lambda} \right) \lambda$$

$$= 0.0012 \times 850 \times 10^{-9} = 1.02 \text{ nm}$$

Pulse broadening per km due to material dispersion: The pulse spread σ_{MATERIAL} for a source of spectral width σ_λ is found by differentiating group delay with respect to wavelength and multiplying by σ_λ :

$$\sigma_{\text{MATERIAL}} \approx \left| \frac{d\tau_{\text{MATERIAL}}}{d\tau} \right| \sigma_\lambda = \frac{\sigma_\lambda L}{c} \left| \lambda \frac{d^2 n}{d\lambda^2} \right| = \sigma_\lambda L |D_{\text{MATERIAL}}(\lambda)|$$

The pulse spread σ_{MATERIAL} for a source of spectral width σ_λ is

$$\begin{aligned} \sigma_{\text{MATERIAL}} &= \sigma_\lambda L |D_{\text{MATERIAL}}(\lambda)| \\ &= 1.02 \times 1 \times 25 \\ &= 25.5 \text{ ps/km} \end{aligned}$$

PROBLEM 19

For a fiber optic link of 1 km, the waveguide dispersion parameter $\left[V \frac{d^2(bV)}{dV^2} \right] = 0.2$, $n_2 = 1.48$, $\Delta = 0.02$, $\sigma_\lambda = 1 \text{ nm}$ at 900 nm wavelength. Calculate the pulse broadening per km due to waveguide dispersion.

Solution: Total pulse broadening due to waveguide dispersion over a finite wavelength spread is

$$\begin{aligned} \sigma_{\text{WAVEGUIDE}} &= -\frac{n_2 L \Delta \sigma_\lambda}{c \lambda} \left[V \frac{d^2(bV)}{dV^2} \right] \\ &= -\frac{1.48 \times 1 \times 0.02 \times 1 \times 10^{-9}}{3 \times 10^8 \times 900 \times 10^{-9}} [0.2] \\ &= 21.9 \text{ ps/km} \end{aligned}$$

PROBLEM 20

Calculate the critical radius of curvature at which large bending losses occur in a multimode fiber, with core refractive index of 1.5, relative refractive index difference of 3%, and an operating wavelength of 0.82 μm .

Solution: We have

$$n_2 = n_1(1 - \Delta) = 1.5(1 - 0.03) = 1.455$$

Large bending losses tend to occur in multimode fiber at critical radius of curvature R_c , which is given as

$$R_c = \frac{3n_1^2 \lambda}{4\pi(n_1^2 - n_2^2)^{3/2}}$$

$$\begin{aligned}
 &= \frac{3 \times (1.5)^2 \times 0.82 \times 10^{-6}}{4\pi((1.5)^2 - (1.455)^2)^{3/2}} \\
 &= 9 \text{ } \mu\text{m}
 \end{aligned}$$

PROBLEM 21

Calculate the critical radius of curvature at which large bending losses occur in a single-mode fiber having 8 μm core diameter, with core refractive index of 1.5, relative refractive index difference of 0.3%, and an operating wavelength of 1550 nm.

Solution: We have

$$n_2 = n_1(1 - \Delta) = 1.5(1 - 0.003) = 1.4955$$

Cutoff wavelength for a single mode fiber is given as

$$\begin{aligned}
 \lambda_c &= \frac{2\pi a n_1 (2\Delta)^{1/2}}{2.405} \\
 &= \frac{2\pi \times 4 \times 10^{-6} \times 1.5(2 \times 0.003)^{1/2}}{2.405} \\
 &= 1.214 \text{ } \mu\text{m}
 \end{aligned}$$

The critical radius of curvature R_{CS} for single-mode fiber is given as

$$\begin{aligned}
 R_{CS} &= \frac{20\lambda}{(n_1 - n_2)^{3/2}} \left(2.784 - 0.996 \frac{\lambda}{\lambda_c} \right)^{-3} \\
 &= \frac{20 \times 1550 \times 10^{-9}}{(1.5 - 1.4955)^{3/2}} \left(2.748 - 0.996 \frac{1550 \times 10^{-9}}{1.214 \times 10^{-6}} \right)^{-3} \\
 &= 34 \text{ mm}
 \end{aligned}$$

SUMMARY

1. Attenuation and dispersion are the linear effects observed in optical fibers.
2. Loss of signal strength while traveling through the optical fiber from the transmitting end to the receiving end is called attenuation. Scattering, absorption, and bending are the main causes of fiber attenuation.
3. The main cause of absorption is the presence of impurities such as metal particles or moisture in the fiber. There are two types of absorption, namely, intrinsic absorption and extrinsic absorption.
4. Scattering phenomenon transfers the optical power of one mode to another mode. There are two categories of linear scattering as follows: Rayleigh scattering and Mie scattering.
5. In an optical fiber, radiation losses take place at the bends. Depending upon the radius of bend,

there are two types of bends, namely, microbend and macrobend.

6. Dispersion causes the spreading of pulses that can limit the transmission speed in analog or digital systems. There are two types of dispersion in optical fibers: intermodal and intramodal dispersion.
7. Material and waveguide dispersion add to give intramodal dispersion. The pulse spreading due to intramodal dispersion is proportional to the spectral width of transmitter as well as length of fiber.
8. Intermodal or modal dispersion occurs only in multimode fiber.
9. The non-linear effects in optical fiber are divided into two categories as follows: stimulated scattering and optical Kerr effects.
10. The interaction between incident optical signal and molecular vibrations gives rise to stimulated Raman scattering (SRS).
11. The interaction between strong optical signal and acoustic wave gives rise to stimulated Brillouin scattering (SBS).
12. The intensity dependence of refractive index results in self-phase modulation (SPM), cross-phase modulation (XPM or CPM), and four-wave mixing (FWM).

MULTIPLE-CHOICE QUESTIONS

1. A fiber optic link of length 50 km length has an attenuation of 0.2 dB/km. The receiver sensitivity is 20 nW. How much power must be supplied by the source?
 - (a) 20 nW
 - (b) 0.20 μ W
 - (c) 2.0 μ W
 - (d) 1 W
2. Which type of fiber cable offers lowest dispersion?
 - (a) Single-mode step index
 - (b) Multimode step index
 - (c) Multimode graded index
 - (d) None of these
3. A graded index profile in optical fiber cable provides
 - (a) Less waveguide dispersion than step index profile
 - (b) Less material dispersion than step index profile
 - (c) Less attenuation than step index profile
 - (d) Less modal dispersion than step index profile
4. Attenuation in optical fiber is usually measured in
 - (a) dB/km
 - (b) dB/m
 - (c) KdB/m
 - (d) dBm/m
5. Dispersion in silica fiber is minimum at $\lambda = ?$
 - (a) 850 nm
 - (b) 1300 nm
 - (c) 1550 nm
 - (d) 1670 nm
6. Scattering loss in optical fiber varies with wavelength as
 - (a) $\frac{1}{\lambda}$
 - (b) $\frac{1}{\lambda^2}$
 - (c) $\frac{1}{\lambda^3}$
 - (d) $\frac{1}{\lambda^4}$
7. Which index profile value of α results in minimum dispersion for Graded index fiber?
 - (a) 1.0
 - (b) 2.0
 - (c) 3.0
 - (d) 2.1
8. A fiber has attenuation of 0.00435 dB/m. What is the total attenuation of a 10 km length?
 - (a) 0.0435 dB
 - (b) 4.35 dB
 - (c) 43.5 dB
 - (d) Cannot tell without knowing wavelength
9. A single-mode fiber has material dispersion of 20 ps/nm-km and waveguide dispersion of -15 ps/nm-km at the signal wavelength. What is the total chromatic dispersion?
 - (a) 35 ps/nm-km
 - (b) -35 ps/nm-km
 - (c) 5 ps/nm-km
 - (d) -5 ps/nm-km
10. The threshold power level for SBS is
 - (a) Quite lower than SRS
 - (b) Quite higher than SRS
 - (c) Equal in SBS and SRS
 - (d) Cannot say

REVIEW QUESTIONS

1. What are the causes of attenuation in optical fiber?
2. Explain in brief the curve for wavelength versus attenuation for different ranges of the signal.
3. Briefly describe the absorption losses in optical fiber comparing the intrinsic and extrinsic absorption.
4. With neat sketches explain the microbending and macrobending effects in optical fiber. How to minimize bending losses?
5. Explain the following linear scattering losses in optical fiber:
 - (a) Rayleigh scattering
 - (b) Mie scattering
6. Describe the intermodal and intramodal dispersion in optical fiber.
7. How is the data rate of the fiber link related to total dispersion?
8. Explain Raman scattering and Brillouin scattering in optical fiber.
9. Give the applications of stimulated Raman scattering.
10. What is optical Kerr effect?
11. Differentiate between Kerr effect and scattering effect in optical fiber.
12. Explain self-phase modulation and cross-phase modulation.
13. Write a note on applications of self-phase modulation and cross-phase modulation.
14. Compare Raman scattering and Brillouin scattering in optical fiber.
15. Describe four-wave mixing in optical fiber.
16. Give the applications of self-phase modulation and cross-phase modulation.
17. What is soliton? Describe the soliton formation in brief.
18. How can one reduce four-wave mixing phenomenon in optical fiber?

EXERCISES

1. A 1 km length fiber transmits 99.9% of the light entering it. How much light will remain after 20 km of fiber?
2. If an instantaneous pulse is transmitted through 20 km multimode fiber with total dispersion of 10 ns/km at the signal wavelength, what will be the pulse length at the end?
3. A step index single-mode fiber has a core refractive index of 1.45, relative refractive index difference of 0.25%, and a core diameter of 8.2 μm . Calculate the waveguide dispersion parameter for the given fiber at $\lambda = 1550$ nm.
4. A step index single-mode fiber exhibits material dispersion of 7 ps/nm-km at an operating wavelength of 1550 nm. $n_1 = 1.48$, $\Delta = 1\%$. Calculate the diameter of the core of the fiber needed to make the total dispersion zero at operating wavelength.
5. Calculate the amount of pulse spreading in pure silica fiber of length 10 km, if LED source

operating at 820 nm and having spectral width of 20 nm is used to supply power. Given:

$$\frac{d^2 n}{d\lambda^2} = 4 \times 10^{-2} \mu\text{m}^{-2}$$

6. Calculate the waveguide dispersion at 1550 nm for a single-mode fiber with core and cladding diameters of 9 μm and 125 μm , respectively. Core refractive index $n_1 = 1.45$ and relative refractive index difference is 0.4%.
7. Calculate the critical radius of curvature at which large bending losses occurs in a single-mode fiber having 8 μm core diameter, with core refractive index of 1.5, relative refractive index difference of 0.4%, and an operating wavelength of 1550 nm.
8. Find the radius of curvature of macrobend in optical fiber at which the power decreases by 50%. $\alpha = 2$, $n_2 = 1.45$, $\Delta = 0.02$, and $a = 25 \mu\text{m}$, power launched in fiber = 35 dB.

9. An LED with a spectral width of 20 nm is used as an optical source and the length of the fiber is 20 km. Due to material dispersion, the pulse spreads at the output at 1.76 ns/km. Calculate the amount of material dispersion.
10. An optical power of 20 μW is fed into fiber of 15 km length. The output power is found to be 5 μW . Calculate
 - (a) Overall loss in dB assuming no connectors or splices.
 - (b) Overall loss in dB/km.
 - (c) Overall signal attenuation for a 20 km optical link using the same fiber with splices at 2 km interval, each of 1 dB attenuation.
 - (d) The numerical input output ratio of (c).
11. Determine the overall signal attenuation in decibels through the optical fiber of length 8 km, assuming no splices and connectors, when the mean optical power launched is 250 μW and the output power is 20 μW .
12. Calculate the macrobend loss of a GI fiber with index profile $\alpha = 1$, core diameter of 50 μm , wavelength 850 nm, bent curve radius $R = 1.5$ cm. The core refractive index is 1.45, $\text{NA} = 0.4$, $\Delta = 0.02$.
13. Calculate attenuation in dB/km in an optical fiber of length 900 m, in which 75% of optical signal is lost while propagating through fiber.
14. For an optical fiber communication system of 2 km length, the 3 dB pulse widths at the input and output are 1.5 ns and 18 ns, respectively. Find the pulse broadening for the fiber and the maximum bit rate.
15. Find the pulse spreading due to material dispersion in an optical system having an LED source operating at 830 nm with a spectral width of 60 nm. The source launches the power in a 1 km fiber. Given:

$$\frac{d^2 n}{d\lambda^2} = 4 \times 10^{-2} \mu\text{m}^{-2}$$
16. Estimate the amount of material dispersion in an optical fiber system. An LED with a spectral width of 25 nm is used as an optical source and the length of the fiber is 40 km. Due to material dispersion, the pulse spreads at the output at 1.8 ns/km.
17. An optical fiber losses 0.7 dB/km at 1350 nm. If 120 μW of power is injected into the fiber, how much power will reach at a distance of 15 km?
18. Calculate the diameter of the core of the fiber needed to make the total dispersion zero at 1550 nm for a step index single-mode fiber having material dispersion of 6.5 ps/nm-km, $n_1 = 1.45$, $\Delta = 0.5\%$.
19. Find the radius of curvature of macrobend in optical fiber at which the power decreases by 40%. $\alpha = 2$, $n_2 = 1.45$, $\Delta = 0.02$, $a = 30 \mu\text{m}$, power launched in fiber = 32 dB.
20. Calculate the critical radius of curvature at which large bending losses occurs in a multimode fiber, with core refractive index of 1.45, relative refractive index difference of 4%, and an operating wavelength of 0.85 μm .

ANSWERS

Multiple-Choice Questions

1. (b)
2. (a)
3. (c)
4. (a)
5. (b)
6. (d)
7. (b)
8. (c)
9. (c)
10. (a)

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4 | Optical Sources

Learning Objectives

After completing this chapter, you will be able to understand:

- Fundamentals of semiconductor physics.
 - Construction and operation of LED.
 - Internal and external quantum efficiency of LED.
 - Basics of laser.
 - Construction, operation, and applications of laser diodes (LDs).
 - Construction, operation, and applications of distributed-feedback (DFB) lasers.
 - Differences between LEDs and laser diodes and discuss them with respect to performance characteristics.
-

The optical transmitter converts an electrical input signal into the corresponding optical signal and then launches it into the optical fiber channel in the communication system. The main component of optical transmitters is an optical source. Three types of optical sources are available: incandescent lamp, LEDs, and lasers. The most preferred sources are LEDs and lasers because of several advantages such as compact size, high efficiency, good reliability, right wavelength range, small emissive area compatible with fiber core dimensions, and possibility of direct modulation at relatively high frequencies.

The selection criteria for light sources in optical communication are as follows:

1. **Single longitudinal mode:** Multimode sources are not good for communication; they may cause pulse broadening due to dispersion. If the power distribution among the longitudinal modes is random, it causes relative intensity noise (RIN) and mode partitioning noise (MPN). Therefore, many single longitudinal mode lasers have been introduced such as distributed feedback (DFB) laser, distributed Bragg reflection (DBR) laser, etc. They work on the principle of Bragg reflection to generate only a single mode.
2. **Low noise:** A low noise is important to achieve a low bit error rate (BER) in digital communication and a sufficiently large signal-to-noise ratio (SNR) in analog communication.
3. **Small linewidth:** In wavelength domain, the width of output spectrum is called the *linewidth*. In general, the more coherent the output light, the smaller is the linewidth. For incoherent communication, wider linewidth will result in pulse dispersion similar to a multimode source.
4. **Low threshold current:** For laser diodes, lasing cannot start until the bias current is higher than a minimum value, called the **threshold current** (I_{th}). The output power is proportional to $(I - I_{th})$, where I is the bias current. Therefore, a lower threshold current allows a smaller bias current for the same output power. This minimizes the power dissipation problem in driving the laser diode.
5. **Wavelength:** Light waves at different wavelengths have different propagation characteristics. First-generation laser diodes were operated at a wavelength of around 870 nm using a GaAs laser diode. To have a lower attenuation and dispersion, the quaternary alloy technology of III–V compounds has been successfully developed. The wavelength is now increased to 1300 nm and 1550 nm band.

6. **Linearity:** For analog communication, signal distortion due to light source non-linearity should be minimized. Non-linearity will introduce higher harmonic terms and crosstalk.
7. **Tunability:** For applications such as wavelength division multiplexing, the capability to tune the wavelength of source is essential.
8. **Large modulation bandwidth:** The information in pass-band communication is transmitted via carrier modulation. Two techniques are followed: direct modulation and external modulation. Out of the two, direct modulation is simple and is frequently used as compared to external modulation. Direct modulation uses the transmitted signal to drive the light source directly. The light source should respond fast enough to the time-varying input signal; therefore, the modulation bandwidth should be as large as possible. On the other hand, in external modulation, an external device modulates the continuous wave output from light source, where a large modulation bandwidth is not necessary.
9. **Size and configuration:** It should be compatible with optical fibers to launch the light efficiently.
10. **Cost:** The source should be comparatively cheap and highly reliable in order to compete with conventional transmission techniques.

4.1 Semiconductor Fundamentals

The properties of semiconductor materials fall between those of insulators and conductors. The semiconductor elements silicon (Si) and germanium (Ge) are suitable for solid-state electronic applications due to their unique properties. These materials have four valence electrons and form crystal lattices, in which substituted atoms or dopants can dramatically change the electrical properties. In solid-state electronics, either pure silicon or germanium may be used as the intrinsic semiconductor, which forms the starting point for fabrication. Though both silicon and germanium have four valence electrons, germanium has more free electrons at a given temperature and a higher conductivity. Silicon is the most widely used semiconductor for electronics, because it can be used at much higher temperatures than germanium.

4.1.1 Intrinsic and Extrinsic Materials

Semiconductor materials are classified into two types:

1. **Intrinsic semiconductors:** Intrinsic means “belonging naturally.” Pure semiconductor crystals are insulators or very poor conductors at room temperature, since only those electrons that are naturally present in the material participate in conduction.
2. **Extrinsic semiconductors:** Extrinsic means “added externally.” Extrinsic semiconductors can be obtained by adding impurities into intrinsic semiconductors. The addition of impurity is known as *doping*. Here, addition of impurity means addition of other materials so as to increase the conductivity of the pure semiconductor. Extrinsic semiconductors are good conductors at room temperature. Depending on the type of impurity, there are two types of extrinsic semiconductors:
 - (a) p-type extrinsic semiconductors.
 - (b) n-type extrinsic semiconductors.

In the following sub-sections we will be discussing these semiconductors in detail.

4.1.1.1 Intrinsic Semiconductors

4.1.1.1.1 Generation of Electron–Hole Pairs

A silicon crystal is different from an insulator because at any temperature above the absolute zero temperature, some of the electrons absorb the thermal energy and move to higher orbit, breaking the covalent bond. When an

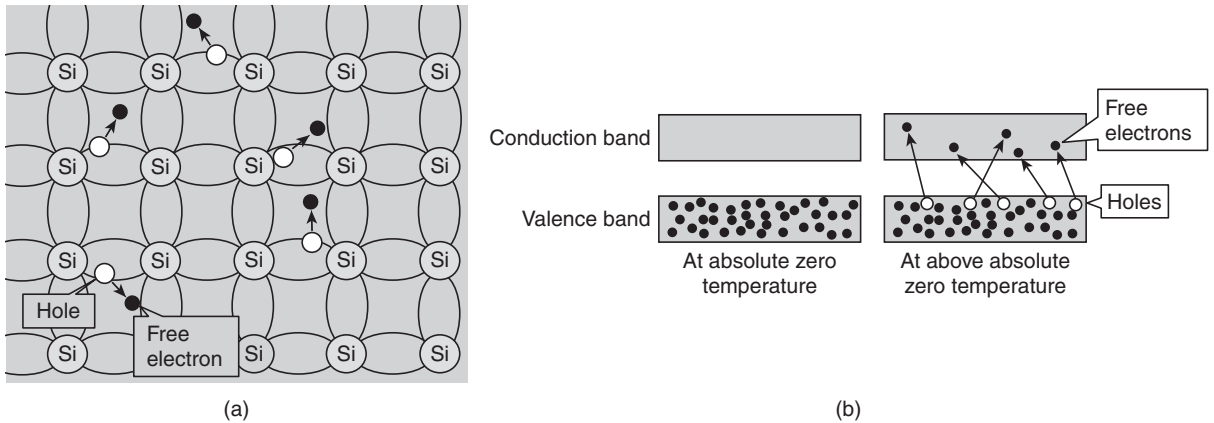


Figure 4.1 | A hole is created when an electron breaks its covalent bond. (a) Silicon crystal structure. (b) Energy band diagram of silicon.

electron breaks a covalent bond and becomes free, a vacancy is created in the broken covalent bond. This resulting vacancy is referred to as a **hole**.

The electron has a negative charge; its absence represents the loss of a negative charge. Therefore, a hole has a characteristic of a positively charged particle. As an electron jumps randomly from one valence shell to other valence shell, it leaves a hole behind it. Each corresponding electron and hole is referred to as an electron–hole pair. Therefore, in intrinsic semiconductor, the number of electrons generated due to increased temperature is always equal to the number of holes. With increase in temperature, the number of electron–hole pair increases. If a voltage is applied, then more electron–hole pairs are generated. Figure 4.1 shows the generation of electron–hole pair due to thermal energy using band theory.

4.1.1.1.2 Current Flow in Intrinsic Semiconductor

When pure semiconductor material is subjected to a voltage as shown in Fig. 4.2, the free electrons are attracted to the positive terminal of the source. The holes move toward the negative terminal. Both electrons and holes contribute to current flow in an intrinsic semiconductor.

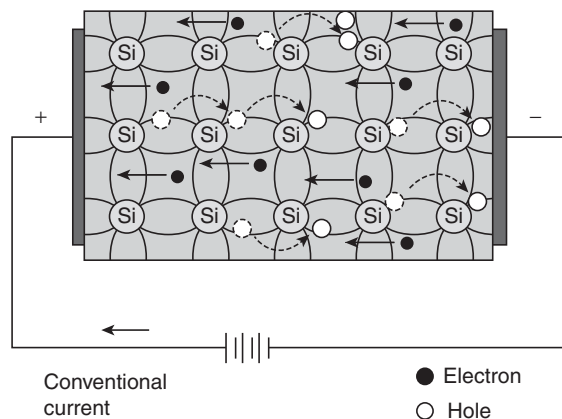


Figure 4.2 | Current flow in pure semiconductor material.

At room temperature, very little current conduction is allowed. The amount of current flow is determined by the number of hole–electron pairs in a material. This current is highly temperature-dependent.

As the temperature increases, its resistance decreases, which allows the crystal to support current flow. In silicon for every 6°C rise in temperature, resistance reduces to half value. While in germanium, for every 10°C rise in temperature, resistance reduces to half value. Thus, germanium is more stable than silicon for temperature changes. This implies that the current flow in an intrinsic semiconductor is influenced by the temperature or application of increased voltage. However, the conductivity does not increase significantly; hence, practically they are not used for manufacturing electronics devices.

4.1.1.2 Extrinsic Semiconductors

We know that to increase the conductivity of pure semiconductor materials, a process called doping is used. *Doping* is the process by which small amounts of selected additives, called impurities, are added to semiconductors. Semiconductors that undergo this treatment are referred to as *extrinsic semiconductors*.

There are two types of impurities:

1. **Donor impurity:** When pure semiconductor material is doped with impurity like pentavalent material (having five valence electrons) then the impurity is referred as donor impurity because it loses its extra electron to the semiconductor causing it to have an excess number of free electrons. Some examples are arsenic (As) and antimony.
2. **Acceptor impurity:** When pure semiconductor material is doped with impurity like trivalent material (having three valence electrons) then the impurity is referred to as acceptor impurity because it reduces the number of free electrons causing it to have more holes. Some examples are indium (In) and gallium.

4.1.1.2.1 n-type Semiconductor

An n-type semiconductor is one that is doped with an n-type or donor impurity. Current flow in an n-type material is similar to conduction in a copper wire. That is, with voltage applied across the material, electrons will move through the crystal toward the positive terminal just like current flows in a copper wire. Additional free electron–hole pairs are generated due to the breaking of covalent bond. The corresponding electrons move toward the positive terminal while holes move toward negative terminal.

4.1.1.2.2 p-type Semiconductor

A p-type semiconductor is one that is doped with a p-type or acceptor impurity. Current flow in a p-type material is by positive holes, instead of negative electrons. Unlike the electrons, the holes move from the positive terminal of the p-type material to the negative terminal. In addition to the holes provided by the acceptor atom, additional electrons and holes are generated due to breaking of the covalent bond. The corresponding electrons move toward the positive terminal while holes move toward the negative terminal.

4.1.2 pn Junction

When a section of n-type material bonds to a section of p-type material, a pn junction is formed. This arrangement conducts electricity in only one direction. When no external voltage is applied to the terminals of the pn junction, the junction is referred to as unbiased. Under this unbiased condition at the pn junction, the electrons from n-type material diffuse into the p-type material and recombine with hole. Each electron leaving the n-side leaves behind positive ions, as shown in Fig. 2.1(a). When an electron falls into a hole on the p-side, it creates a negative ion out of the trivalent atom that accepts this electron, as shown in Fig. 2.1(a). The ions are fixed in the crystal structure due to covalent bonding and they cannot move around like free electrons and holes. Hence, ions are known as immobile charges.

Due to recombination process of electrons and holes, large numbers of negative ions accumulate near the junction of the p-side and large numbers of positive ions accumulate near the junction of the n-side, as shown in Fig. 2.1(b). The negatively charged ions on the p-side repel the electrons, which attempt to diffuse into the p-side. Hence, after some time, accumulation of ions on either side stops diffusion completely. This state is referred as state of equilibrium of the pn junction.

Thus, in the state of equilibrium, only ions are present at the junction and there are no mobile free charge carriers such as electrons and holes. This region is known as *depletion region*. In other words, a region near the junction where the electrons and holes are depleted is called as the *depletion region* or *space charge region*.

4.1.2.1 Barrier Potential or Cut-In Voltage

Due to the presence immobile ions on either side of the pn junction, an electric field is developed across the junction. This field is known as the *barrier potential* or *cut-in voltage*. The polarity of this potential is shown in Fig. 4.3. This barrier potential is fixed, having value of 0.7 V (typical) for silicon and of 0.3 V (typical) for germanium.

4.1.2.2 Biasing

Biasing is the process of application of external DC voltage. When no external voltage is applied, the diode remains in equilibrium state and there is no current flowing through it. In order to make the current to flow through the diode, biasing is applied. There are two types of biasing:

1. Forward bias.
2. Reverse bias.

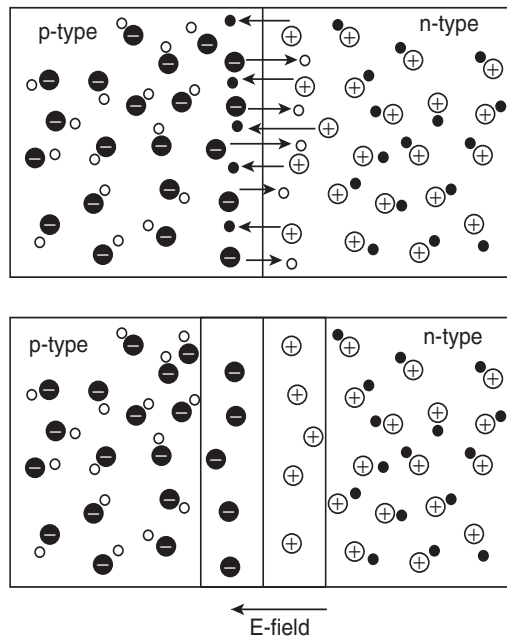


Figure 4.3 | Formation of pn junction and formation of depletion region. At the junction, free electrons from the n-type material fill holes from the p-type material. This creates an insulating layer in the middle of the diode called the depletion region.

4.1.2.2.1 Forward Bias

To avoid the depletion region, the electrons from the n-type area should be forced toward the p-type area and holes in the reverse direction. This can be achieved by connecting the n-type side of the diode to the negative end of the battery voltage and the p-type side to the positive end, that is, **forward-biased condition**. The free electrons in the n-type material are repelled by the negative electrode and drawn to the positive electrode. The holes in the p-type material move the other way.

If applied DC voltage is less than the barrier potential of the junction, the free electrons do not have enough energy to cross the depletion region. When electrons enter the depletion region, ions will push them back into the n-side. Hence, there is no current through the diode.

When applied DC voltage is greater than the barrier potential of the junction, the free electrons and holes get enough energy to pass through the depletion region. As a result of increased number of holes and electrons crossing the junction, depletion region reduces and so barrier potential reduces. Hence, large number of electrons and holes cross the junction under the influence of applied voltage and constitute the current flow. This current is known as *forward current*.

4.1.2.2.2 Reverse Bias

Current will not flow if the polarity of battery voltage is reversed, that is, the p-type side is connected to the negative end of the battery voltage and the n-type side is connected to the positive end. This is the **reverse-biased condition**. The negative electrons in the n-type material are attracted to the positive electrode. The positive holes in the p-type material are attracted to the negative electrode. Due to the movement of electrons and holes are away from the junction, depletion region increases. Hence, no current flows across the junction.

4.2 Light-Emitting Diode

A light-emitting diode (LED) is essentially a pn junction diode. When carriers are injected across a forward-biased junction, it emits incoherent light. Most of the commercial LEDs are realized using a highly doped n- and a p-junction. To understand the principle, let us consider an unbiased pn+ junction (Fig. 4.4 shows the pn+ energy band diagram).

The depletion region extends mainly into the p-side. There is a potential barrier from E_c on the n-side to E_c on the p-side, called the *built-in voltage*, V_0 . This potential barrier prevents the excess free electrons on the n+ side from diffusing into the p-side.

When a voltage V is applied across the junction as shown in Fig. 4.5, the built-in potential is reduced from V_0 to $(V_0 - V)$. This allows the electrons from the n+ side to get injected into the p-side. Since electrons are the minority carriers in the p-side, this process is called *minority carrier injection*. However, the hole injection from the p-side to n+ side is very less and so the current is primarily due to the flow of electrons into the p-side.

The recombination of the holes and minority carriers injected in the p-side results in spontaneous emission of a light photon of energy $E_g = h\nu$ as shown in Fig. 4.6. This spontaneous emission process is called *injection electroluminescence*, because the optical emission takes place due to injection of electrons. The reabsorption of emitted photons is avoided by properly designing the LED structure. The wavelength of the light emitted, and hence the color, depends on the band gap energy of the materials forming the pn junction.

The emitted photon energy is approximately equal to the bandgap energy of the semiconductor. The following equation relates the wavelength and the bandgap energy:

$$h\nu = E_g \quad (4.1)$$

$$\Rightarrow h \frac{c}{\lambda} = E_g \quad (4.2)$$

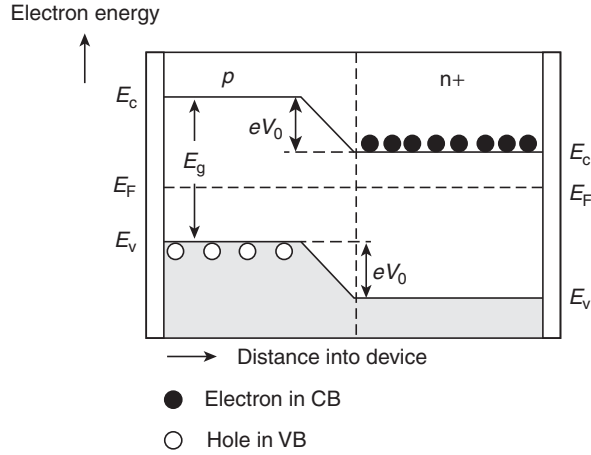


Figure 4.4 | pn+ junction under unbiased conditions.

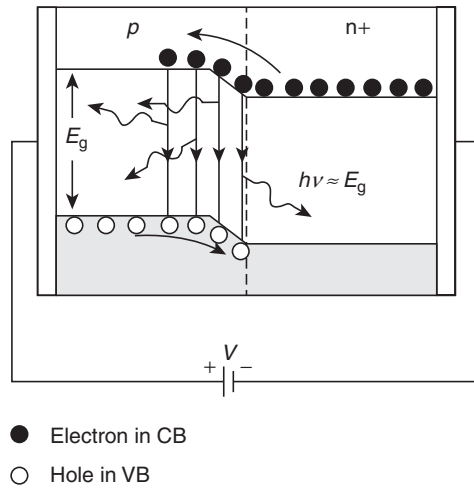


Figure 4.5 | pn+ junction under biased conditions.

$$\Rightarrow \lambda = h \frac{c}{E_g} \quad (4.3)$$

where h is Planck's constant, c is the speed of the light, and E_g is the bandgap energy. Substituting the values of h and c gives

$$\lambda = \frac{1.24}{E_g} \quad (4.4)$$

where λ is written in μm and E_g in eV. Thus, a semiconductor with a 2 eV bandgap emits light at about 620 nm in the red. A 3 eV bandgap material would emit at 414 nm in the violet.

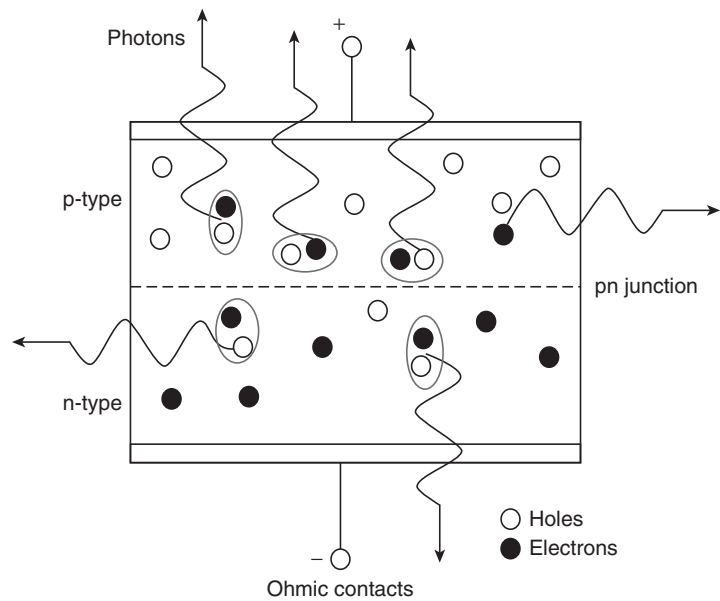


Figure 4.6 | Carrier recombination resulting into spontaneous emission of light in a pn junction.

4.2.1 Direct and Indirect Recombination

The recombination can be classified into the following two kinds:

1. Direct recombination.
2. Indirect recombination.

4.2.1.1 Direct Recombination

In direct bandgap materials, the minimum energy of the conduction band lies directly above the maximum energy of the valence band in momentum space energy. Figure 4.7 shows the electron energy (E)–momentum (k) plot of a direct bandgap material. In this material, free electrons at the bottom of the conduction band can recombine directly with free holes at the top of the valence band, as the momentum of the two particles is the same. This direct transition of electron from the conduction band to the valence band provides an efficient mechanism for photon emission and the average time the minority carrier remains in a free state before recombination, that is, the minority carrier lifetime is short (10^{-8} to 10^{-10} s). This is known as *direct recombination*. Direct recombination occurs spontaneously. Commonly used direct bandgap semiconductor materials are listed in Table 4.1.

Table 4.1 | Direct bandgap semiconductor materials

Semiconductor material	Energy bandgap (eV)	Recombination coefficient B_r ($\text{cm}^3 \text{s}^{-1}$)
GaAs	1.43	7.21×10^{-10}
GaSb	0.73	2.39×10^{-10}
InAs	0.35	8.5×10^{-11}
InSb	0.18	4.58×10^{-11}

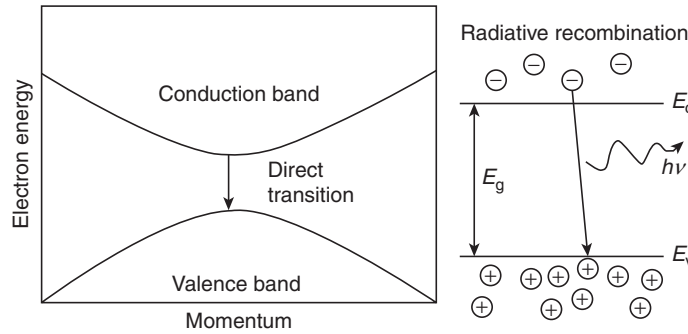


Figure 4.7 | Direct band gap and direct recombination.

4.2.1.2 Indirect Recombination

In the indirect bandgap materials, the minimum energy in the conduction band is shifted by a k -vector relative to the valence band. The k -vector difference represents a difference in momentum. Due to this difference in momentum, the probability of direct electron–hole recombination is less and relatively slow (10^{-2} to 10^{-4}).

For electron–hole recombination to take place, it is essential that the electron loses momentum such that it has a momentum value corresponding to the maximum energy of the valence band. In these materials, additional dopants (impurities) are added which form very shallow donor states. These donor states capture the free electrons locally, providing the necessary momentum shift for recombination. These donor states serve as the recombination centers. This is called *indirect (non-radiative) recombination*. Figure 4.8 shows the electron energy (E)–momentum (k) plot of an indirect bandgap material and an example of how nitrogen serves as a recombination center in GaAsP. In this case, it creates a donor state when SiC (silicon carbide) is doped with Al. Its recombination takes place through an acceptor level. The indirect recombination should satisfy both conservation of energy and momentum. The minority carrier lifetime is much longer and the probability of non-radiative transitions is greater. Thus, the indirect bandgap materials, listed in Table 4.2, such as silicon and germanium, give insignificant levels of electroluminescence.

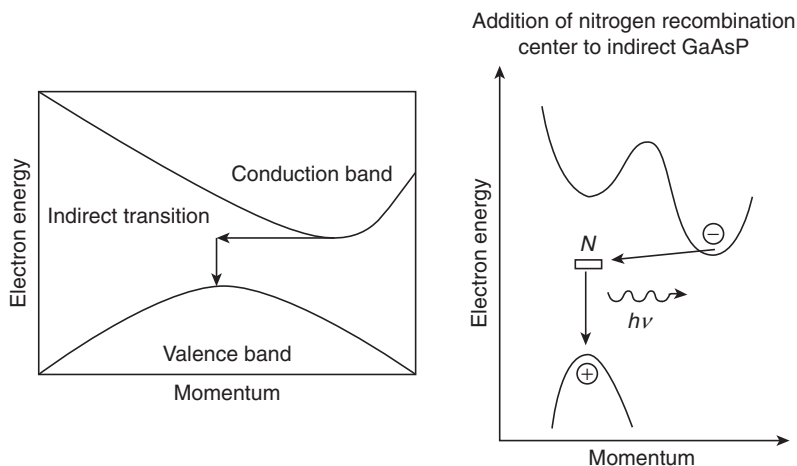


Figure 4.8 | Indirect bandgap and non-radiative recombination.

Table 4.2 | Indirect bandgap materials

Semiconductor material	Energy bandgap (eV)	Recombination coefficient B_r (cm^3s^{-1})
Si	1.12	1.79×10^{-15}
Ge	0.67	5.25×10^{-14}
GaP	2.26	5.37×10^{-14}

The radiative minority carrier lifetime is given by

$$\tau_r = [B_r(N + P)]^{-1} \quad (4.5)$$

where N is majority charge carrier concentration in the n-type region and P is majority charge carrier concentration in the p-type region.

4.2.2 LED Materials

LEDs are fabricated by epitaxially growing doped semiconductor layers on a suitable substrate such as GaAs or GaP. An important class of commercial LEDs that cover the visible spectrum is constructed from III–V compounds corresponding to IIIa and Va columns of the periodic table (see Table 4.3).

Ternary alloys are based on alloying GaAs and GaP. These are denoted by $\text{GaAs}_{1-y}\text{P}_y$. InGaAlP is an example of quarternary (four elements) III–V alloys with a direct bandgap. Various electroluminescent semiconductors of commercial LEDs with their emission wavelengths and colors are listed in Table 4.4.

Table 4.3 | Compounds corresponding to IIIa and Va columns of the periodic table

IIIa	Va
Aluminum (Al)	Nitrogen (N)
Gallium (Ga)	Phosphorus (P)
Indium (In)	Arsenic (As)
	Antimony (Sb)

Table 4.4 | Electroluminescent semiconductor with emission wavelength and color

Semiconductor material	Wavelength	Color
Gallium arsenide (GaAs)	>760 nm	Infrared
Aluminum gallium arsenide (AlGaAs)		
Aluminum gallium arsenide (AlGaAs)	$610 < \lambda < 760$	Red
Gallium arsenide phosphide (GaAsP)		
Aluminum gallium indium phosphide (AlGaInP)		
Gallium arsenide phosphide (GaAsP)	$590 < \lambda < 610$	Orange
Aluminum gallium indium phosphide (AlGaInP)		
Gallium phosphide (GaP)		
Gallium arsenide phosphide (GaAsP)	$570 < \lambda < 590$	Yellow
Aluminum gallium indium phosphide (AlGaInP)		
Gallium phosphide (GaP)		

(continued)

Table 4.4 | (continued)

Semiconductor material	Wavelength	Color
Indium gallium nitride (InGaN)/Gallium nitride (GaN)	$500 < \lambda < 570$	Green
Gallium phosphide (GaP)		
Aluminum gallium indium phosphide (AlGaInP)		
Aluminum gallium phosphide (AlGaP)		
Zinc selenide (ZnSe)	$450 < \lambda < 500$	Blue
Indium gallium nitride (InGaN)		
Silicon carbide (SiC) as substrate		
Indium gallium nitride (InGaN)	$400 < \lambda < 450$	Violet
Aluminum gallium nitride (AlGaIn)		
Aluminum nitride (AlN)		
Aluminum gallium indium nitride (AlGaInN)		
	$\lambda < 400$	Ultraviolet

A new technology is developing around semiconducting polymers. These are plastics that have the electrical characteristics of semiconductors. Conventional plastics are insulators, because they bond electrons tightly. Semiconducting plastics can be doped to make p- and n-materials, and electrical devices including LEDs can be made from them.

4.2.3 Quantum Efficiency of LED

The internal quantum efficiency of LED is the ratio of photons generated to injected electrons. The internal quantum efficiency of simple homojunction LED is only about 50% due to absence of stimulated emission. It totally relies on spontaneous emission, which allows non-radiative recombination within the structure due to crystalline imperfections and impurities. This reduces the efficiency greatly. With double heterojunction structure, internal efficiencies of about 60%–80% can be obtained.

The power generated internally by an LED can be determined by considering the minority carriers, when it is forward-biased and carriers are injected. The excess density of electrons (Δn) and holes (Δp) is equal since injected carriers are created and recombined in pairs in order to maintain charge neutrality.^[1]

In extrinsic materials one carrier type has a much higher concentration than the other. For example, in the p-type region, the hole concentration is much greater than the electron concentration. This excess minority carrier density decays exponentially with time t as

$$\Delta n = \Delta n(0)e^{-t/\tau} \quad (4.6)$$

where $\Delta n(0)$ is the initial injected excess electron density and τ is the total carrier recombination lifetime. When there is a constant current flow into an LED, an equilibrium condition is established. The total rate at which the carriers are generated is the sum of externally supplied and thermally generated rates. Externally supplied rate is $Jled$, where J is current density in A/cm^2 , e is the electron charge, and d is the thickness of recombination region. Thermal generation rate is $\Delta n/\tau$. The rate equation for carrier recombination in the LED can be written as

$$\frac{d(\Delta n)}{dt} = \frac{J}{ed} - \frac{\Delta n}{\tau} \text{ (m}^{-3}\text{s}^{-1}\text{)} \quad (4.7)$$

The equilibrium condition is found by setting

$$\frac{d(\Delta n)}{dt} = 0$$

Therefore

$$\Delta n = \frac{J\tau}{ed} \text{ m}^{-3} \quad (4.8)$$

This relationship gives the steady-state electron density in the active region when a constant current is flowing through it.

In steady state the recombination rate r_t is

$$r_t = \frac{J}{ed} = r_r + r_{nr} \quad (4.9)$$

where r_r is the radiative recombination rate per unit volume and r_{nr} is the non-radiative recombination rate per unit volume.

If the forward-bias LED current is i , then the total number of recombinations per second (R_t) is given as

$$R_t = \frac{i}{e} \quad (4.10)$$

The internal quantum efficiency (η_{int}) of LED is the ratio of the radiative recombination rate to the total recombination rate. Mathematically, we can write

$$\eta_{int} = \frac{r_r}{r_t} = \frac{r_r}{r_t + r_{nr}} \quad (4.11)$$

$$\Rightarrow \eta_{int} = \frac{R_r}{R_t} \quad (4.12)$$

This is because by definition of efficiency. Internal quantum efficiency is the ratio of radiative recombination per second to total recombinations per second:

$$R_t = \frac{R_r}{\eta_{int}} \quad (4.13)$$

where R_r is the total number of radiative recombinations per second and R_t is the total recombinations per second. Then from Eqs. (4.10) and (4.13) we can write

$$R_t = \frac{i}{e} = \frac{R_r}{\eta_{int}} \quad (4.14)$$

Therefore

$$R_r = \eta_{int} \frac{i}{e} \quad (4.15)$$

R_r is equivalent to total number of photons generated per second. The optical power generated internally by the LED (P_{int}) is

$$P_{int} = \text{Total no. of photons generated per second} \times \text{Energy of each photon}$$

$$P_{int} = R_r \times hf \quad (4.16)$$

$$P_{int} = \eta_{int} \frac{i}{e} \times hf = \eta_{int} \frac{hci}{e\lambda} \quad (4.17)$$

For exponential decay of excess carriers, the radiative recombination lifetime is

$$\tau_r = \frac{\Delta n}{r_r}$$

The non-radiative recombination lifetime is

$$\tau_{nr} = \frac{\Delta n}{r_{nr}}$$

and the total recombination lifetime

$$\tau = \frac{\Delta n}{r_t}$$

From Eq. (4.9) we can write

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} \quad (4.18)$$

Therefore, the internal quantum efficiency is

$$\begin{aligned} \eta_{int} &= \frac{1}{1 + (r_{nr}/r_r)} = \frac{1}{1 + (\tau_r/\tau_{nr})} \\ \Rightarrow \eta_{int} &= \frac{\tau}{\tau_r} \end{aligned} \quad (4.19)$$

All internally generated photons will not exit the device. The actual emitted power can be obtained from the external quantum efficiency η_{ext} . The **external quantum efficiency** of LED is defined as the ratio of photons emitted from the LED to the number of internally generated photons. To find the external quantum efficiency, we need to consider the radiation geometry for an LED and reflection effects at the surface of LED. Most of the light generated within the device is trapped by total internal reflection as indicated in Fig. 4.9, when it is radiated at an angle greater than critical angle for crystal–air interface. Only light falling within a cone defined by the critical angle will be emitted from an optical source.

The cone defined by critical angle is

$$\phi_c = (\pi/2) - \theta_c$$

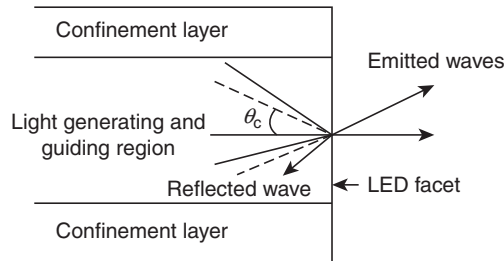


Figure 4.9 | Light emitted by LED.

Only fraction of light falling within the cone defined by critical angle will cross the interface. Consider the equation below [explained in Chapter 2, Eq. (2.4)]:

$$\sin \phi_c = \frac{n_2}{n_1}$$

Here n_1 is the refractive index of semiconductor material and n_2 is the refractive index of outside material (= 1 for air). Therefore external quantum efficiency is given as

$$\eta_{\text{ext}} = \frac{1}{4\pi} \int_0^{\phi_c} T(\phi) (2 \sin \phi) d\phi \quad (4.20)$$

For normal incidence

$$T(\phi) = \text{Fresnel transmissivity} = T(0)$$

where

$$\begin{aligned} T(0) &= \frac{4n_1 n_2}{(n_1 + n_2)^2} \\ &= \frac{4n}{(n+1)^2} \quad [\text{putting } n_1 = n \text{ and } n_2 = 1 \text{ (for air)}] \end{aligned} \quad (4.21)$$

Therefore

$$\eta_{\text{ext}} \approx \frac{1}{n(n+1)^2} \quad (4.22)$$

Optical power emitted from LED is

$$P_{\text{ext}} = \eta_{\text{ext}} P_{\text{int}} \approx \frac{P_{\text{int}}}{n(n+1)^2} \quad (4.23)$$

Further loss in optical power is encountered when light is coupled from source to fiber, as shown in Fig. 4.10.^[2]

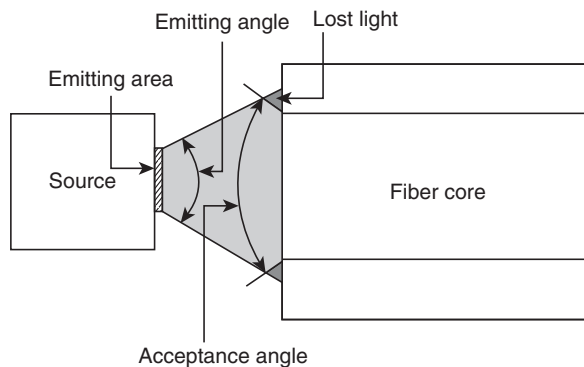


Figure 4.10 | Light transfer from source to fiber.

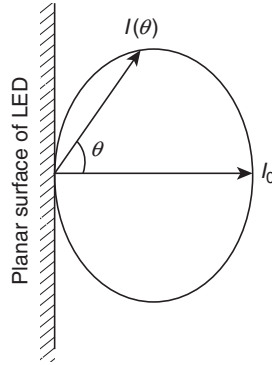


Figure 4.11 | Lambertian intensity distribution of planar LED.

The nature of radiation geometry for an LED which emits through planar surface is Lambertian, as shown in Fig. 4.11. The intensity distribution is maximum along the perpendicular to the planar surface and gradually decreases on both sides; it is proportional to cosine of viewing angle θ .

To calculate the coupling efficiency, consider a step-index fiber in which all the light incident on the fiber core within the acceptance angle θ_a is coupled, while incident light at angle greater than θ_a will not couple. For a Lambertian source,

$$I(\theta) = I_0 \cos \theta \quad (4.24)$$

The coupling efficiency η_c is given by

$$\eta_c = \frac{\int_0^{\theta_a} I(\theta) \sin \theta d\theta}{\int_0^{\pi/2} I(\theta) \sin \theta d\theta} \quad (4.25)$$

Putting value of $I(\theta)$, we get

$$\begin{aligned} \eta_c &= \frac{\int_0^{\theta_a} I_0 \cos \theta \sin \theta d\theta}{\int_0^{\pi/2} I_0 \cos \theta \sin \theta d\theta} \\ &= \frac{\int_0^{\theta_a} I_0 \sin 2\theta d\theta}{\int_0^{\pi/2} I_0 \sin 2\theta d\theta} \\ &\Rightarrow \eta_c = \frac{[-I_0 \cos 2\theta/2]_0^{\theta_a}}{[-I_0 \cos 2\theta/2]_0^{\pi/2}} \\ &\Rightarrow \eta_c = \sin^2 \theta_a = (\text{NA})^2 \end{aligned} \quad (4.26)$$

4.2.4 Double Heterojunction LED

The LED realized using two differently doped semiconductors that are of the same material is called a *homojunction*. When the LEDs are realized using different bandgap materials, they are called a *heterostructure device*. A heterostructure LED is brighter than a homojunction LED.

Heterojunctions are classified as an isotype (n-n or p-p) or an anisotype (p-n). This sandwich structure confines the carriers and the optical field in the central active layer. It effectively reduces the carrier diffusion length and hence the area of region where radiative recombination may take place.

A heterojunction is formed between two semiconductors which have different bandgaps but same lattice parameters so that they can be grown together as a single crystal. For example, a heterojunction may be formed between GaAs and its ternary alloy $\text{Ga}_{1-x}\text{Al}_x\text{As}$. The narrow bandgap material such as n- or p-type GaAs is sandwiched

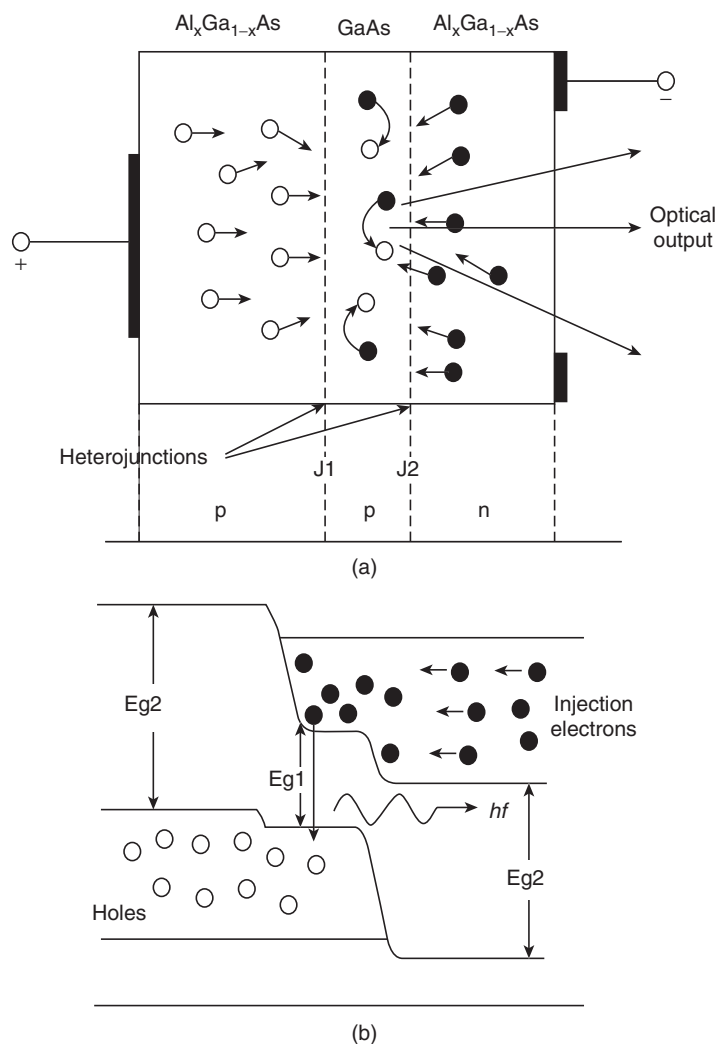


Figure 4.12 | Double heterojunction LED: (a) Layer structure and (b) energy band diagram.

between layers of wider bandgap materials such as p- or n-type GaAlAs to form a double heterostructure (DH) as shown in Fig. 4.12.

When the DH-LED is forward-biased, the holes from p-GaAlAs are injected into n-GaAs, but are prevented from going into n-GaAlAs by a potential barrier at J2. Similarly, the electrons from n-GaAlAs are injected into n-GaAs but prevented from going further by potential barrier at junction J1. Thus, a large number of carriers are confined in the central layer of n-GaAs, where they recombine to produce optical radiation of wavelength corresponding to the bandgap of n-GaAs. As most of the activity takes place in the central layer, it is called an *active layer*. This structure gives more radiative recombinations, hence a brighter LED. The radiations generated by band to band transitions in the active layer cannot excite the carriers in the adjoining layers because E_{g1} is less than E_{g2} . Thus, the confining layers of wider bandgap material are transparent to this radiation. The radiation may be collected through the surface or edge.

The limitation of GaAs/Ga_{1-x}Al_xAs based LED is that the range of wavelengths (800–900 nm) emitted is outside the wavelength limits of lowest attenuation and zero total dispersion of optical fibers. Therefore, such emitters cannot be used in long-haul communication systems.

Highly efficient DH-LED emitting longer wavelengths can be fabricated using quaternary alloy indium–gallium–arsenide–phosphide/indium–phosphide (In_xGa_{1-x}As_yP_{1-y}/InP). It operates in 930–1650 nm wavelength range and hence is a better choice for fiber-optic system.^[3]

4.2.5 LED Structure

The major types of LED structures are as follows:

1. Planar LED.
2. Dome LED.
3. Surface-emitting LED.
4. Edge-emitting LED.
5. Super luminescent LED.
6. Polymer LED.

4.2.5.1 Planar LED

The geometry and internal structure of LED play an important role in emitting light from its surface. Simple planar LEDs emit light in all directions, as shown in Fig. 4.13(a) and most of the emissions come from their surfaces. The light is emitted in a broad cone, with intensity falling off roughly with the cosine of the angle from the normal to the semiconductor junction, that is, Lambertian distribution.

The LED structures are organized such that most of the recombinations take place on the surface in the following two ways^[4]:

1. The doping concentration of the substrate is increased, so that additional free minority charge carrier electrons move to the top, recombine, and emit light at the surface.
2. The diffusion length L is increased. Now

$$L = \sqrt{D\tau}$$

where D is the diffusion coefficient and τ is the carrier lifetime.

The LED structure should be such that the photons generated from the device are emitted without being reabsorbed. This is achieved by making p-layer on the top thin enough to create a depletion layer. There are different dome structures for efficient emitting as shown in Figs. 4.13(b) and 4.14.

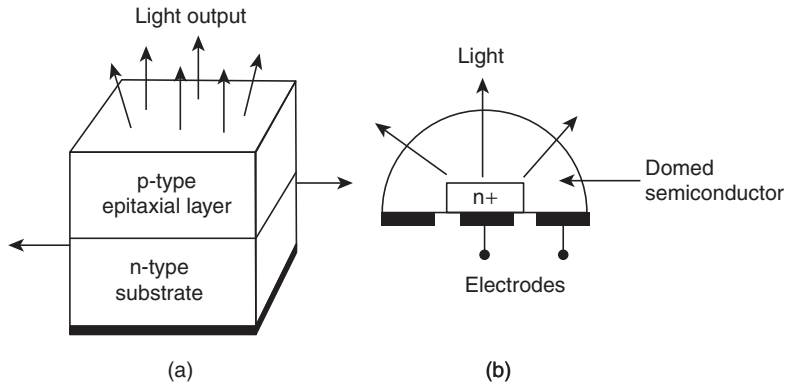


Figure 4.13 | LED structure: (a) Planar structure and (b) dome structure.

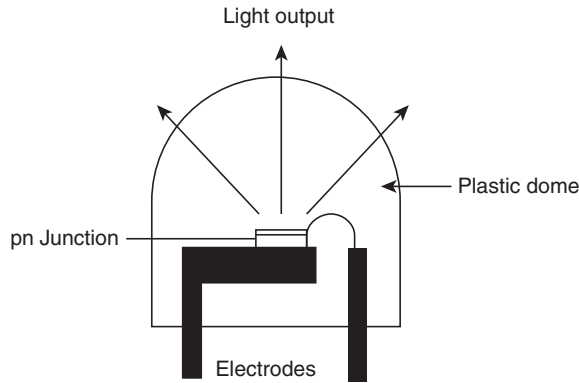


Figure 4.14 | LED encapsulated into dome-shaped plastic medium.

4.2.5.2 Dome LED

Another widely used structure is dome LED as shown in Fig. 4.13(b) where a hemisphere of n-type GaAs is formed around a diffused p-type region. The diameter of the dome is chosen to maximize the amount of internal emission reaching the surface within the critical angle of the GaAs–air interface.

Let us calculate the critical angle for GaAs–air interface: For GaAs, the refractive index $n_1 = 3.6$ and for air, the refractive index $n_2 = 1$. Therefore

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1}{3.6}\right) = 16.12^\circ$$

The optical rays making angle of incidence greater than 16° will suffer total internal reflection. To reduce the problem, dome or hemisphere structure is preferred so that the light rays will strike the surface at an angle $< \theta_c$ and the surface does not experience the total internal reflection. However, it is practically expensive to make pn junction of the dome shape. Therefore, it is encapsulated into dome-shaped plastic medium having higher refractive index than that of air as shown in Fig. 4.14.

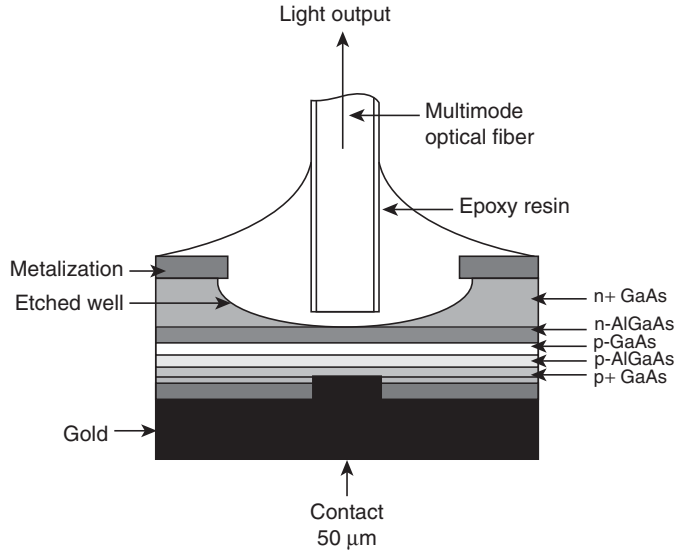


Figure 4.15 | AlGaAs DH surface-emitting LED.

4.2.5.3 Surface-Emitting LED

In the case of surface-emitting light diodes (SELED), the light beam is emitted from the top of the diode as shown in Fig. 4.15 and not from the side. The beam's emission cone is typically 120° and the guiding has a circular symmetry. The large beam emission angle is caused by a difference in refractive index between the semiconductor (high index) and the fiber or the glass material used for coupling with the fiber (low index). The total output power is between $500 \mu\text{W}$ and 1 mW .

The bottom p+ GaAs and top n+ GaAs layers are included for the realization of low-resistance Ohmic contacts. A deep well is etched to reach the top n-AlGaAs layer to avoid the reabsorption of the emitted radiation in the top n-GaAs layer. This well also gives support to the fiber, which is butt-coupled to the device and held in place with an epoxy resin of appropriate refractive index to enhance the external power efficiency of the device. Photons are generated in the thin p-GaAs region and emitted from the top surface. The heterostructure and reflection from the back crystal face ensures surface emission. Thus, the forward radiance of these devices is very high. The top n-GaAs contact layer ensures low contact resistance and thermal resistance and allows high current densities and high radiation intensity. The fiber is properly aligned to optimize coupling of the emitted radiation. The power coupled (P_c) into a multimode step index fiber can be calculated from

$$P_c = \pi(1 - r)AR_D(\text{NA})^2 \quad (4.27)$$

where r is the Fresnel reflection coefficient at the fiber surface, A is the fiber cross-section, and R_D is the radiance of source. Several other types of lens coupling are used with a surface-emitting heterostructure to improve the coupling efficiency.

4.2.5.4 Edge-Emitting LED

An edge-emitting LED (EELED) is a high-radiance structure. The light is produced in a very thin active layer and it is emitted from the side of a structure. Due to carrier confinement layers, the beam divergence is narrowed to 30° in a plane perpendicular to the junction, while in the plane of junction due to absence of confinement, a

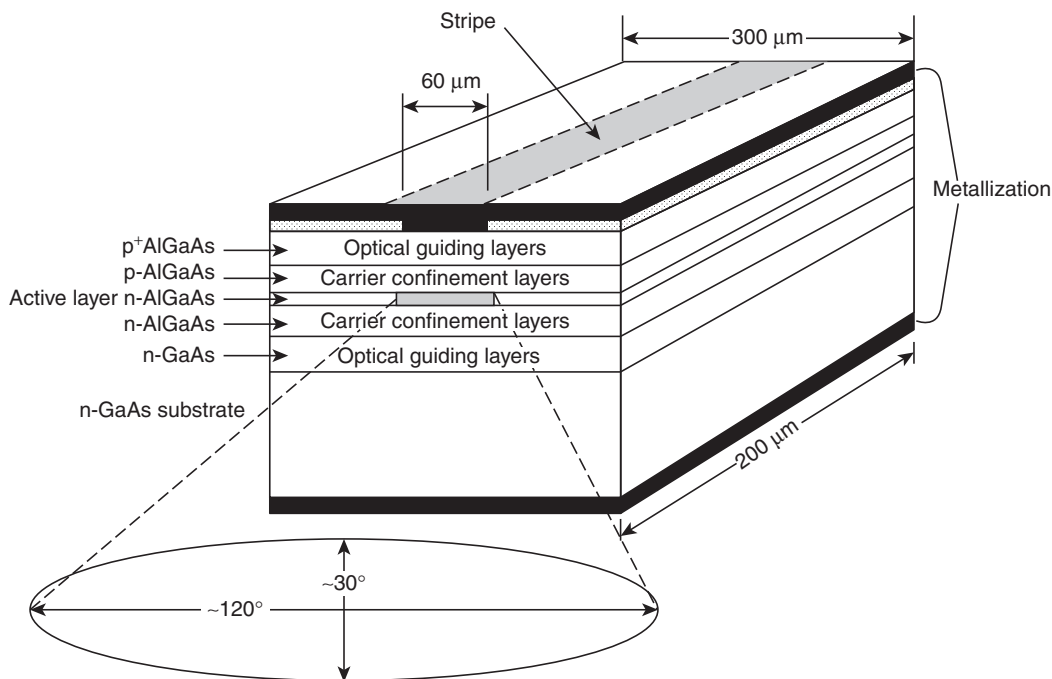


Figure 4.16 | Strip geometry DH AlGaAs edge-emitting LED.

Lambertian pattern output gives 120° divergence. Thus, an elliptical pattern is obtained. At low injection current, the efficiency of EELED is less, hence preferably it is operated at high intensity current. This type of LED gives the output between $500 \mu\text{W}$ and 1 mW .

In optical communication applications where tight coupling of emitted light to a fiber or waveguide is required, more collimated strip geometry is incorporated as shown in Fig. 4.16.

The device consists of an active junction region, which is the source of the incoherent light. A very large population of carriers for recombination is created in this region by forward-bias injection, and two InGaAsP layers on both sides serve as carrier confinement layers, on the outer sides of which are doped InP layers. These serve as the cladding layers and the region in between forms an optical waveguide. The photons are generated in the very thin active region and spread into the guiding layers, without reabsorption, because of their larger bandgaps. The stripe geometry is made by selective metallization on the top surface through a window opened in a SiO_2 layer, which allows higher carrier injection densities for the same drive current. To match the typical fiber core diameter ($50\text{--}100 \mu\text{m}$), the contact stripes for the edge emitter are made $50\text{--}70 \mu\text{m}$ wide. The emission pattern of the edge emitter is more directional than that of the surface emitter.

4.2.5.5 Super Luminescent LED

Super luminescent LEDs (SLEDs) are different from EELEDs and surface-emitting LEDs in several ways. In an SLED, there is stimulated emission with amplification but insufficient feedback for laser oscillations to occur.

It has the following advantages over EELEDs and SELEDs:

1. The spectral width is narrower.
2. Light coming from an SLED is more coherent and its degree of polarization is higher.

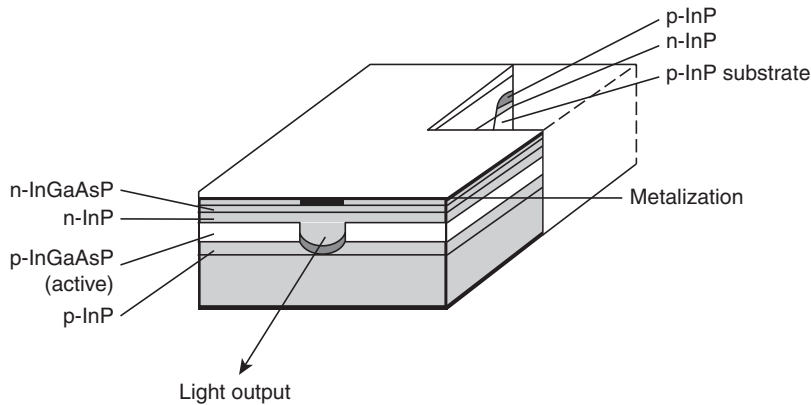


Figure 4.17 | InGaAsP super luminescent LED.

3. The output beam is more directional, like a laser, which allows for better coupling in the fibers.
4. A U-shaped cut in the active layer shown in Fig. 4.17 increases the density of the carriers, which improves the power efficiency, reaching 18 or 20 mW.

GaAsP/InP SLED emits at 1300 nm. It comprises a buried active layer within U-shaped groove on the p-type InP substrate. One end of the structure is made optically lossy to prevent reflections and thus suppress lasing; the output is taken from the opposite end. For operation, the injected current is increased until stimulated emission and hence amplification occur, but because there is high loss at one end of the device, no optical feedback takes place, and hence no laser oscillations build up. Because of stimulated emissions, single-pass amplification takes place, which results into high optical output power with narrowing of spectral width to approximately 30–40 nm.^[5]

Most of the light emitted from LED is coupled into the optical fiber. The coupling efficiency can be improved by using lenses, specifically when the fiber core diameter is significantly larger than width of the emission region. There are several lens coupling configurations such as

1. Spherically polished structures.
2. Spherical-ended or tapered fiber coupling.
3. Truncated spherical micro lenses.
4. Integral lens structure.

4.2.5.6 Polymer LED (PLED)

PLEDs are the polymer light-emitting diodes. The polymer emits light when exposed to electricity. PLEDs are thin film displays that are created by sandwiching conjugated polymer between two proper electrodes at a short distance. The prototype device was first assembled in 1990 by researchers based at Cavendish Laboratory at Cambridge University. It functioned and emitted a green/yellow glow. Figure 4.18 shows the schematic structure of the first single-layer PPV polymer LED device.^[6]

PLEDs consist of a thin layer of PPV between two electrodes, deposited on a glass substrate. In the structure shown in Fig. 4.18, layer of aluminum (or magnesium or calcium) is the cathode, acting as the electron injection layer and layer of indium/tin oxide is the anode, acting as the hole injection layer. The light-emitting layer shown in Fig. 4.18 is poly(p-phenylenevinylene) (PPV). It is the most widely used light-emitting polymer. The cathode is made up of aluminum or calcium. These metals have low work function, that is, the work function of metal

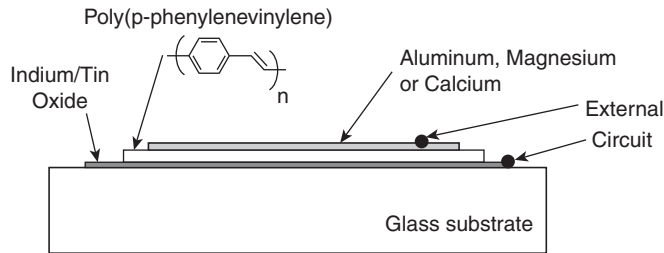


Figure 4.18 | Schematic structure of polymer LED.

indicates the bonding tightness of metal electron with metal surface. The material with low work function is very suitable for cathode because electron injection from cathode into PPV layer can be done efficiently. The anode is made up of relatively high work function material such as indium or tin oxide, which facilitates the easy injection of holes into the PPV layer. The recombination takes place at the layer interface giving rise to photon emission. The anode material (indium/tin oxide) is transparent to light; it provides the route for the emitted light to exit through the device.

The following are some advantages of polymer LED:

1. High brightness and long lifetime.
2. Low power consumption.
3. View angle can be large as 160° .
4. Permits flexible lighting and displays.
5. Much cheaper.
6. Useful to large area lighting.

4.2.6 LED Characteristics

LED is a very linear device and hence it tends to be more suitable for analog transmission. Output power versus forward current characteristic of LED is shown in Fig. 4.19.

The surface emitter radiates significantly more optical power into the air than the edge emitter, and both devices are reasonably linear at moderate drive currents. SLED gives high output power for more forward current. Practically, LEDs do exhibit non-linearity; therefore, it is required to use some linearizing circuit to ensure linear performance.

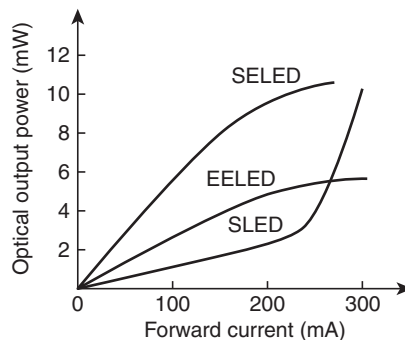


Figure 4.19 | Output power versus forward current characteristic of LED.

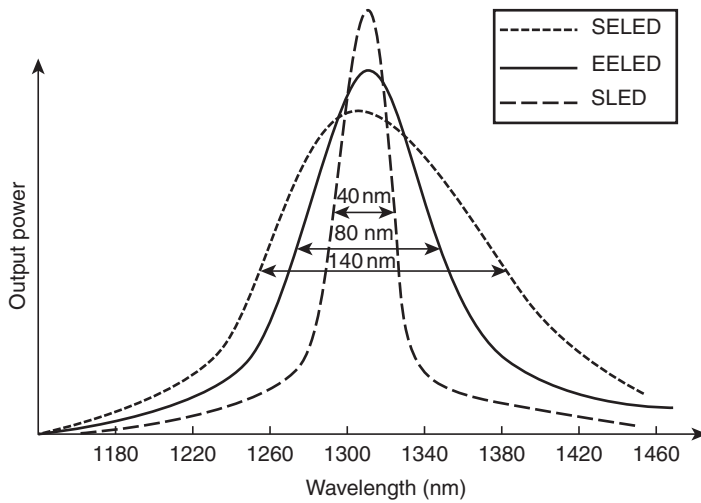


Figure 4.20 | Spectral response of LED.

The spectral width of the source is important as it determines the contribution to material dispersion. Low spectral width allows increased data rate. Spectral responses of SELED, EELED, and SLED are shown in Fig. 4.20. For super luminescent LED, spectral width as narrow as 30–40 nm is obtained.

The internal quantum efficiency of LEDs decreases exponentially with increasing temperature; hence light output decreases as the device temperature increases as shown in the figure. Edge-emitting device has greater temperature-dependence than the surface emitter. The SLED output power also strongly depends on junction temperature.

4.3 Laser

Another important light source used in optical communication is the laser diode (LD). A basic LD structure is similar to that of edge-emitting LED. By adding additional structure for photon confinement, coherent light can be generated. The principle of semiconductor laser is based on external pumping and internal light amplification. When a laser has several energy states, external pumping excites carrier to a higher energy state. When they return to the ground state, they release energy and generate photons. Photon generation from external pumping is not sufficient for coherent light generation. An additional amplification mechanism is needed to multiply photons of the same frequency and phase. In a laser, this is made possible by a quantum phenomenon called *stimulated emission*.

Several photon emission and absorption processes exist in two-level atomic system. When a carrier is pumped to the upper energy level, it can come back to the ground state either spontaneously or by stimulation.

An electron in an atom is excited from an energy level E_1 to a higher energy level E_2 by the absorption of a photon of energy $h\nu = E_2 - E_1$. The electron in higher energy level can come to lower energy level spontaneously or can be forced or stimulated by another photon for the downward transition. In this transition, a photon of energy E_g is released. In spontaneous emission, the electron falls down from energy level E_2 to E_1 and emits a photon of energy $h\nu = E_2 - E_1$ in a random direction as shown in Fig. 4.21(b). The transition is spontaneous and a random photon is emitted.

In stimulated emission, an incoming photon of energy $h\nu = E_2 - E_1$ stimulates the whole emission process by forcing the electron at E_2 to transit down to E_1 as shown in Fig. 4.21(c). The emitted photon is in-phase with the incoming photon; it is in the same direction; it has the same polarization; and it has the same energy.

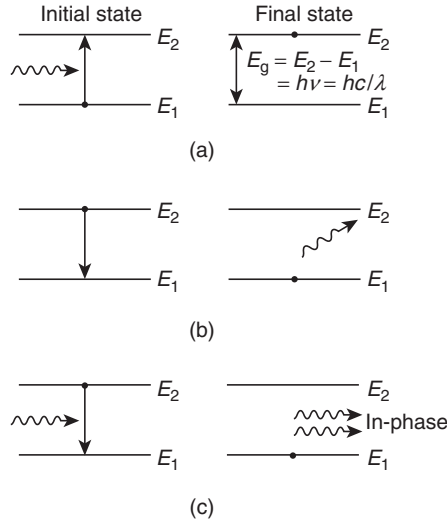


Figure 4.21 Energy state diagram: (a) Absorption, (b) spontaneous emission, and (c) stimulated emission.

4.3.1 Laser Diode Rate Equations

An optical medium has density of atoms N_1 in lower energy state E_1 and density of atoms N_2 in higher energy state E_2 . Due to photon absorption, there is an upward transition from energy level E_1 to E_2 . This transition rate is proportional to number of atoms N_1 and number of photons having energy $E_2 - E_1 = h\nu$. The upward transition rate is

$$R_{12} = B_{12}N_1\rho(h\nu) \quad (4.28)$$

where B_{12} is a constant of proportionality, called *Einstein coefficient*. The downward transition from E_2 to E_1 involves spontaneous and stimulated emission. The spontaneous emission depends on concentration N_2 of atoms at E_2 and the stimulated emission depends on both N_2 and the photon concentration $\rho(h\nu)$ with energy $h\nu = E_2 - E_1$.

The downward transition rate is

$$R_{21} = A_{21}N_2 + B_{21}N_2\rho(h\nu) \quad (4.29)$$

where A_{21} is a constant of proportionality. It is known as Einstein coefficient for spontaneous emission. B_{21} is a constant of proportionality as Einstein coefficient for stimulated emission. In order to find the constants of proportionality, consider the thermal equilibrium condition. There is no net change with time in the populations at E_1 and E_2 , that is,

$$R_{12} = R_{21} \quad (4.30)$$

$$\Rightarrow B_{12}N_1\rho(h\nu) = A_{21}N_2 + B_{21}N_2\rho(h\nu)$$

$$\Rightarrow \rho(h\nu) = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2} \quad (4.31)$$

$$\Rightarrow \rho(h\nu) = \frac{A_{21}/B_{21}}{B_{12}N_1/B_{21}N_2 - 1} \quad (4.32)$$

And, furthermore, in thermal equilibrium Boltzmann statistics demands that

$$\frac{N_2}{N_1} = \exp\left[-\frac{E_2 - E_1}{kT}\right] \quad (4.33)$$

where k is the Boltzmann constant and T is the absolute temperature. In thermal equilibrium condition, the atomic system produces a radiation density identical to blackbody radiation. It is given by Planck's blackbody radiation distribution law:

$$\rho_{\text{equilibrium}}(h\nu) = \frac{8\pi h\nu^3}{c^3 \left[\exp\left(\frac{h\nu}{kT}\right) - 1 \right]} \quad (4.34)$$

$$\rho(h\nu) = \frac{A_{21}/B_{21}}{\frac{B_{12}}{B_{21}} \left(\exp\left[-\frac{E_2 - E_1}{kT}\right] \right) - 1} \quad (4.35)$$

Comparing Eqs. (4.34) and (4.35), we can write

$$B_{12} = B_{21}$$

and

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad (4.36)$$

The ratio of stimulated emission to spontaneous emission is

$$\frac{R_{21}(\text{stimulated})}{R_{21}(\text{spontaneous})} = \frac{B_{21}N_2\rho(h\nu)}{A_{21}N_2} = \frac{B_{21}\rho(h\nu)}{A_{21}} \quad (4.37)$$

Substituting the value of A_{21}/B_{21} from Eq. (4.36) we get

$$\frac{R_{21}(\text{stimulated})}{R_{21}(\text{spontaneous})} = \frac{c^3}{8\pi h\nu^3} \rho(h\nu)$$

The ratio of stimulated emission to absorption is

$$\frac{R_{21}(\text{stimulated})}{R_{12}(\text{absorption})} = \frac{N_2}{N_1} \quad (4.38)$$

From the above equations we can conclude that for stimulated photon emission to exceed photon absorption, we need to achieve population inversion, that is, $N_2 > N_1$. For stimulated emission to far exceed spontaneous emission, we should have a large photon concentration which is achieved by building an optical cavity to contain the photons.

Stimulated emission is the basis for obtaining photon amplification since one incoming photon results in two outgoing photons. To obtain stimulated emission, the incoming photon should not be absorbed by another atom at E_1 . When there are more atoms at E_2 than at E_1 , population inversion is achieved. With only two energy levels, population of E_2 cannot be made greater than E_1 , because in steady state the incoming photon causes as many upward excitations as downward stimulated emissions.

Consider a three-energy-level system. Step-by-step process involved in light amplification by stimulated emission of radiation (LASER) is shown in Figs. 4.22(a)–(d).

The four-level system shown in Fig. 4.23 has much lower pumping requirements. The pumping excites the atoms from ground state into energy level E_4 and they decay rapidly to the metastable level E_3 . Since the

population of E_4 and E_2 remains essentially unchanged, a small increase in the number of atoms in energy level E_3 creates population inversion, and lasing takes place between this level and level E_2 .

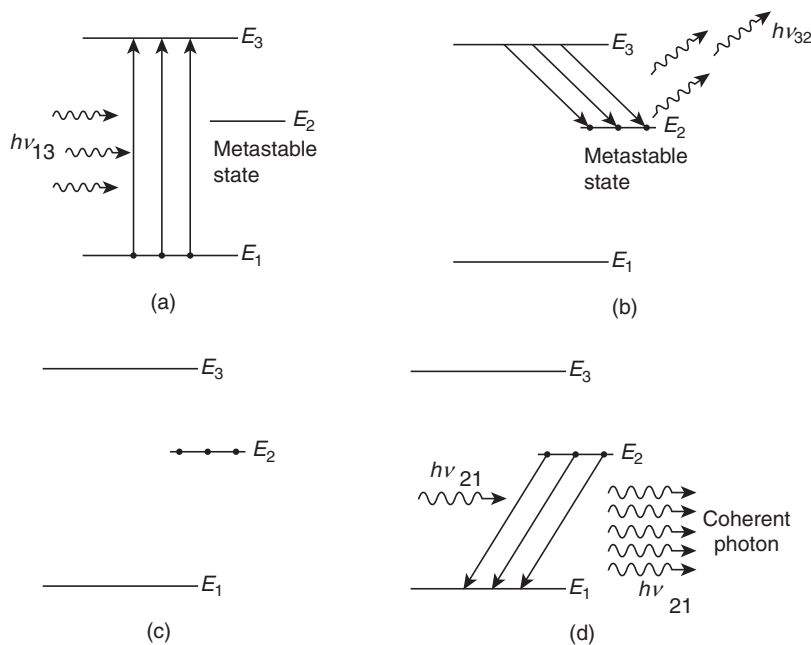


Figure 4.22 The principle of laser. (a) Atoms in the ground state E_1 are pumped to the energy level E_3 by a photon of energy $h\nu_{13} = E_3 - E_1$. (b) Atoms at E_3 rapidly decay to the metastable state at energy level E_2 by releasing a photon or lattice vibrations of energy $h\nu_{32} = E_3 - E_2$. (c) Population inversion between E_2 and E_1 . (d) A photon of energy $h\nu_{21} = E_2 - E_1$ initiates the process of stimulated emission, and avalanche action takes place and coherent photons are emitted.

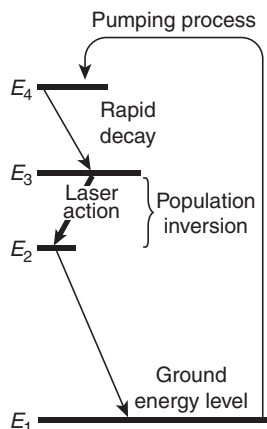


Figure 4.23 Level diagram for four-level system (He-Ne laser).

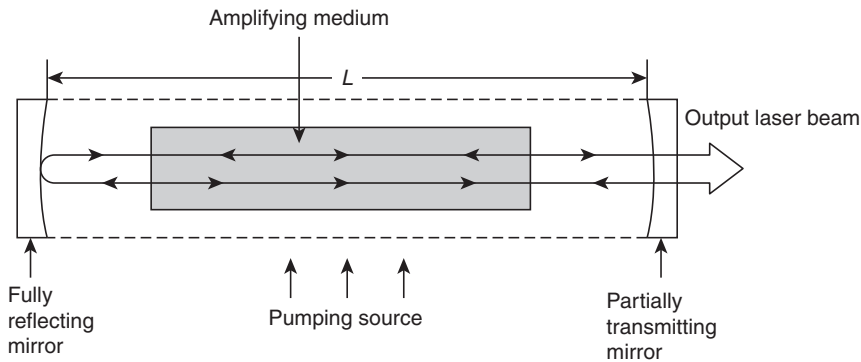


Figure 4.24 | Basic laser structure showing optical feedback.

4.3.2 Construction of a Laser

A laser consists of an active laser material, a source of excitation energy, and a resonator or feedback mechanism to perform the three stages of laser action. The general construction of a laser is shown in Fig. 4.24.

4.3.2.1 Laser Material

Depending upon the type of laser, different laser materials are available as listed below:

1. **Solid laser:** For example, Ruby, glass lasers.
2. **Liquid:** For example, dye lasers.
3. **Gas:** For example, He–Ne, CO₂.

The material is in the normal state if the number of atoms in higher energy level is less than the number of atoms in lower energy level. The material is in excited state when the number of atoms in higher energy level is more than the number of atoms in lower energy level; this state is called as *population inversion*. The more populated atoms in higher energy level are forced to lower energy level, and stimulated emission occurs. The number of energy levels involved in this process classifies the laser as two-level, three-level, or four-level system. A two-level system is not used in commercial laser, as it is very difficult to achieve population inversion in this system.

4.3.2.2 Excitation Source

To excite or pump more atoms to higher energy level, an external pumping energy source is used. There are two types of pumping, electrical and optical. If an AC or DC electrical discharge is used, it is called as *electrical pumping*. If an optical source is used, it is optical pumping. Electrical pumping is used in gas and semiconductor lasers. Optical pumping is used for most of the solid-state and dye lasers.

4.3.2.3 Resonator

In most lasers, a Fabry–Perot cavity having a pair of mirrors, one at each end of the laser, is used as a resonator. One of the mirrors is completely reflective while the other mirror is partially transparent. The reflection of the laser beam between the two mirrors results in increased power. The beam is reflected back for amplification, until a specific threshold power is reached. The portion of the laser beam with the necessary power is coupled as output through the partially transparent mirror.

4.3.3 Laser Modes

4.3.3.1 Longitudinal Modes

Oscillations occur in the laser cavity over a small range of frequencies where the cavity gain is sufficient to overcome the absorption, scattering, and diffraction losses at the mirrors. Therefore, the device is not a perfectly monochromatic source but emits over a narrow spectral band. The central frequency of this spectral band is determined by the mean energy level difference of the stimulated emission transition. The spectral emission from the device contains other oscillation frequencies, which result due to thermal motion of atoms and atomic collisions.

Inside a resonant cavity of length L with parallel end mirrors, when sufficient population inversion is achieved, the radiation builds up and the reflections from end mirrors give rise to traveling waves in opposite direction within the cavity. These oppositely traveling waves interfere constructively to set up standing waves that are stationary electromagnetic (EM) oscillations between mirrors as shown in Fig. 4.25.

These standing waves exist only at frequencies for which the distance between mirrors is an integral number of half-wavelengths. Within the optical cavity, several thousand peaks for each standing wave occur, but the complete half-wavelengths that can fit within the distance between two mirrors can only exist and others will cancel each other. Each standing wave corresponds to one longitudinal mode. If more than one standing wave exists, it means there are more longitudinal modes. The separation between frequency of longitudinal modes depends upon cavity length L . The resonance condition along the axis of cavity is given as

$$L = \frac{\lambda q}{2n} \quad (4.39)$$

where L is the optical spacing between the mirrors, λ is the emission wavelength, n is the refractive index of amplifying medium, and q is an integer.

The discrete emission frequencies ν are given as

$$\nu = \frac{qc}{2nL} \quad (4.40)$$

where c is the velocity of light. Each discrete standing wave is referred to as a longitudinal mode associated with the laser cavity. Figure 4.26 shows two longitudinal modes within a cavity. There will always be one or more longitudinal modes depending on the frequency or wavelength of the laser gain medium. The longitudinal modes generated are indistinguishable unless a spectrum analyzer is used to analyze the beam. They all travel in the same direction, and their color is indistinguishable because their wavelengths (frequencies) are so similar, as indicated above.

Different modes are separated by a frequency interval $\delta\nu$ as shown in Fig. 4.27:

$$\delta\nu = \frac{c}{2nL} \quad (4.41)$$

Assuming $\delta\nu \ll \nu$ and $\nu = c/\lambda$, we can write

$$\delta\lambda = \frac{\lambda\delta\nu}{\nu} = \frac{\lambda^2}{c} \delta\nu \quad (4.42)$$

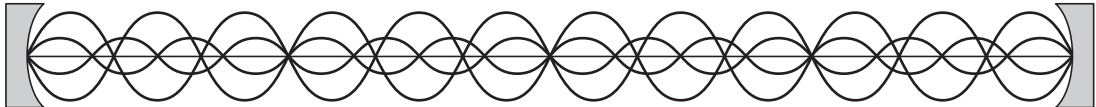


Figure 4.25 | Standing waves between mirrors.

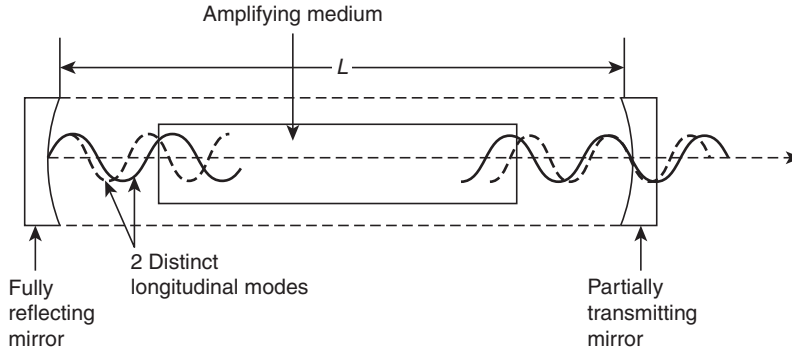


Figure 4.26 | Longitudinal modes in laser cavity.

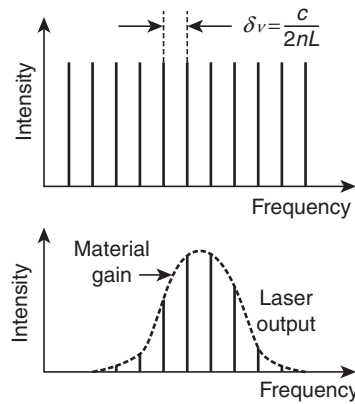


Figure 4.27 | Longitudinal mode frequency domain representation.

Substituting the value of $\delta\nu$ gives

$$\delta\lambda = \frac{\lambda^2}{2nL} \quad (4.43)$$

4.3.3.2 Transverse Modes

The different longitudinal modes having different wavelengths existing in the optical cavity travel the same path while the different transverse modes travel slightly different paths, in slightly different directions. Hence the path length traveled is also different and their frequencies are also different. Each of these modes evolves because after the one round trip in the cavity, each mode travels the same path again, which results into steady beam. Each transverse mode may consist of several longitudinal modes. The frequency separation between longitudinal modes is given by

$$\Delta\nu = \frac{c}{2L}$$

Unlike the longitudinal modes, which contribute only a single spot of light to the laser output, transverse modes give rise to a pattern of spots at the output as shown in Fig. 4.29. For TEM_{00} mode, all parts of the propagating

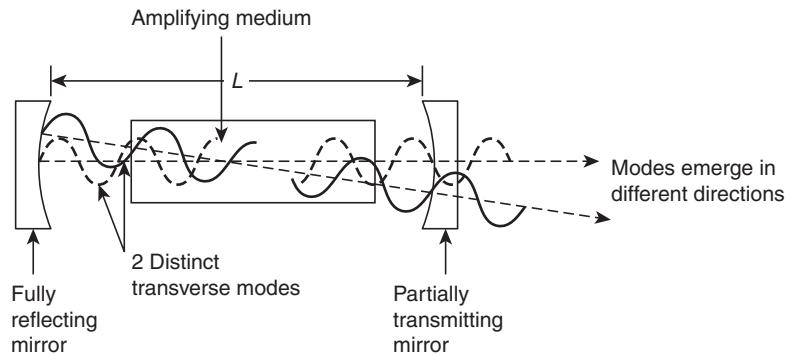


Figure 4.28 | Transverse modes in laser cavity.

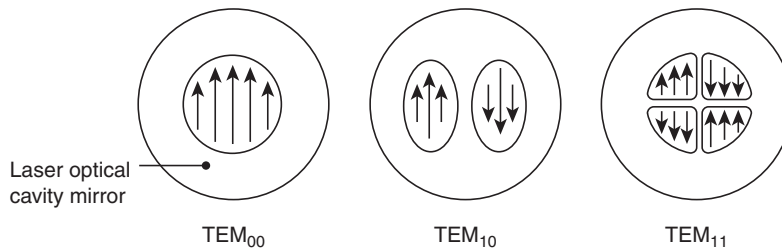


Figure 4.29 | Transverse cavity modes of laser and their fields.

wavefront are in phase. For higher order modes, phase reversal produces various patterns. Higher order transverse modes occur only when the width of cavity is sufficient for them to oscillate.

4.3.4 Semiconductor Lasers

Semiconductor lasers are very similar to light-emitting diodes (LEDs) in structure. Laser diodes and LEDs are pn junction semiconductors that convert electrical energy applied across the junction into optical radiation. In both laser diodes and LEDs, the wavelength of the output radiation depends on the energy gap across the pn junction. However, the output from a laser diode is highly coherent and collimated, while the output from an LED has many phases and is dispersed in different directions.

A laser diode consists of a junction between a p-type and an n-type material. The construction of a laser diode is shown in Fig. 4.30. When a positive voltage is applied to the p-type material and a negative voltage is applied to the n-type material, a local excess of minority carriers (electrons for the p-type and holes for the n-type) is formed on each side of the junction. This condition is known as forward-biasing the diode. It causes the carriers to cross the junction resulting in the electron–hole recombination. The electron–hole recombination generates photons with a wavelength proportional to the bandgap energy at the junction. Population inversion and optical feedback is necessary for laser action. By passing a high current through the diode, majority of charge carriers are excited to higher energy level and population inversion is achieved. Using the electrical contacts attached to the flat surface of diode, voltage is applied across the diode. By cleaving the ends of semiconductor crystal, flat parallel surfaces are obtained, and then the structure works as optical cavity. The cleaved ends act as mirror and enable feedback and output coupling. Direct bandgap semiconductors such as GaAs and InP are used in practice in the

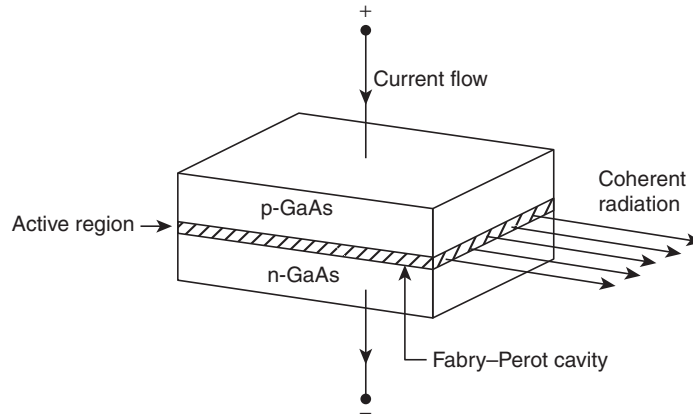


Figure 4.30 | Semiconductor laser diode structure.

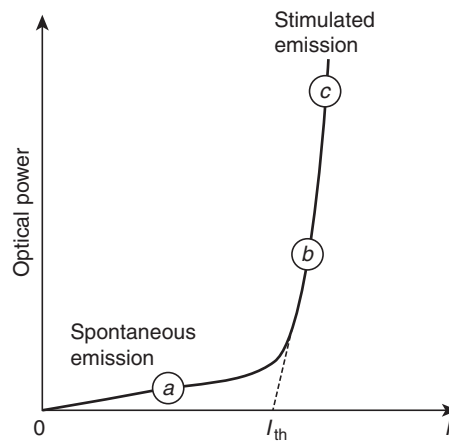


Figure 4.31 | Typical optical output power versus diode current characteristics.

manufacture of laser diodes. A commercial laser consists of a number of layers of different materials in addition to the semiconductor material. The advantages of having several layers include ability to vary the wavelength of output, better confinement of laser light, and facilitate lasing action.^[7]

Lasing radiation is only obtained when the optical gain in the medium can overcome the photon losses from cavity, which requires the diode current I to exceed a threshold value I_{th} . Below I_{th} , the light from the device is due to spontaneous emission and not stimulated emission as shown in Fig. 4.31. The light output is then composed of incoherent photons that are emitted randomly and the device behaves like an LED. The characteristic are shown in Fig. 4.32(a). Lasing oscillations occur only when the optical gain in the medium can overcome the photon losses from the cavity, that is when the optical gain reaches the threshold gain. This occurs at threshold current I_{th} . Above I_{th} , the light intensity becomes coherent radiation consisting of cavity wavelengths and increases steeply with the current. The number of modes in the output spectrum and their relative strengths depend on the diode current as shown in Figs. 4.32(b) and (c).

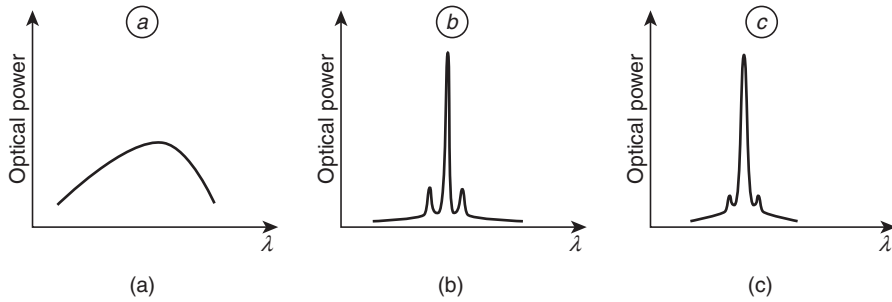


Figure 4.32 Output spectrum of laser at the corresponding points on the output characteristics curve in Fig. 4.31. (a) Point (a) LED operation, (b) Point (b) laser operation, (c) Point (c) laser operation.

There are several laser diode efficiency definitions. The external differential efficiency η_{EDQE} of laser diode is defined as

$$\eta_{\text{EDQE}} = \frac{\text{Increase in number of output photons from the diode (per unit second)}}{\text{Increase in number of injected electrons into diode (per unit second)}}$$

If P_o is the optical power emitted, I is the current, e is charge on an electron, and hf is photon energy, then

$$\eta_{\text{EDQE}} = \frac{dP_o / hf}{dI / e} = \frac{e \cdot dP_o}{dI (E_g)} \quad (4.44)$$

where

$$E_g = hf = \frac{hc}{\lambda}$$

is the bandgap energy expressed in electron volts. The external quantum efficiency η_{EQE} of laser diode is defined as

$$\begin{aligned} \eta_{\text{EQE}} &= \frac{\text{Number of output photons from the diode (per unit second)}}{\text{Number of injected electrons into diode (per unit second)}} \\ \Rightarrow \eta_{\text{EQE}} &= \frac{P_o / hf}{I / e} = \frac{e \cdot P_o}{I (E_g)} \end{aligned} \quad (4.45)$$

Since emitted power changes linearly when the injection current I is greater than the threshold current I_{th} , we have

$$\eta_{\text{EQE}} = \eta_{\text{EDQE}} \left(1 - \frac{I_{\text{th}}}{I} \right) \quad (4.46)$$

External power efficiency η_{EPE} of the device in converting electrical input to optical output is given by

$$\begin{aligned} \eta_{\text{EPE}} &= \frac{\text{Optical output power}}{\text{Electrical input power}} \\ \eta_{\text{EPE}} &= \frac{P_o}{P} \times 100 = \frac{P_o}{I \times V} = \frac{P_o e}{I E_g} \times \frac{1}{V} \times \frac{E_g}{e} \end{aligned}$$

This implies

$$\eta_{\text{EFE}} = \frac{\eta_{\text{EQE}} E_g}{eV} \quad (4.47)$$

4.3.5 Laser Structures

The broad classification of laser is given in Fig. 4.33. We discuss them below.

1. **Edge-emitting laser:** The light emerges from the edge of the device, where the junction intersects the surface. The configuration is simple and easy to fabricate. Most laser diodes are edge emitters. They suffer from the drawbacks that the volume of material that can contribute to the laser emission is limited and they are difficult to package as 2-D arrays.
 - (a) *Homojunction laser:* One type of semiconductor material with different dopants is used to fabricate the pn junction shown in Fig. 4.34(a). The index of refraction depends on the impurity added and the doping level. The lightly doped material has the highest index of refraction and the more heavily doped p-type material has a lower refractive index. This produces the light pipe effect, which helps to confine the laser light to the active junction region. In this structure the index difference is low and much light is lost.
 - (b) *Single heterojunction:* The fraction of Ga in the p-type layer is replaced by Al to reduce the index of refraction, as shown in Fig. 4.34(b), which results in better confinement of laser light to the optical cavity. This leads to lower losses, lower current, reduced damage, and longer lifetime for the diodes.

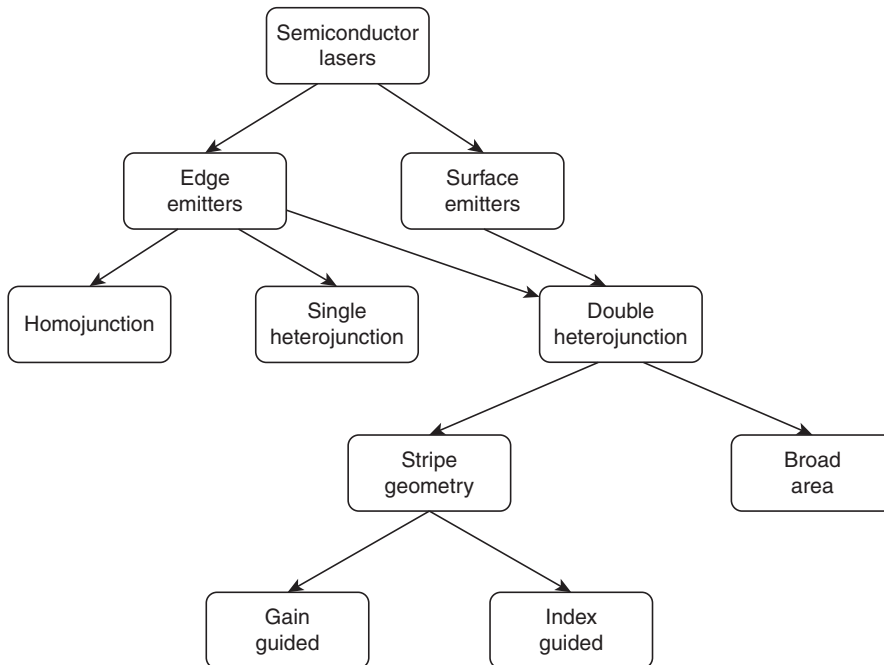


Figure 4.33 | Classification of laser structures.

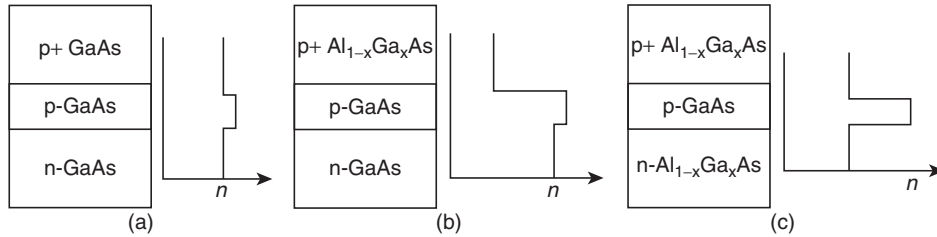


Figure 4.34 Structure and refractive index profile of laser: (a) Homostructure, (b) single heterojunction, and (c) double heterojunction.

2. **Surface-emitting laser:** The light emerges from the surface of the chip rather than from the edge. The devices can be packed densely on a semiconductor wafer and it is possible to fabricate 2-D arrays easily.
 - (a) **Double heterojunction:** Only the junction region is composed of GaAs; both p- and n-regions are of AlGaAs as shown in Fig. 4.34(c). Much better confinement is obtained in this structure. It leads to reduced loss but there are two additional difficulties. The optical radiation is so well confined that the output power may easily reach the damage threshold. The tight confinement also reduces the effective beam width of the output aperture of the laser. This increases the divergence angle in the direction perpendicular to the junction.

In most modern semiconductor lasers, the current is injected only within a narrow region beneath a stripe contact several μm wide, in order to keep the threshold current low and to control the optical field distribution in the lateral direction.

Compared with broad-area lasers, where the entire laser chip is excited, the threshold current of lasers with stripe geometry is reduced roughly proportional to the area of contact.

The two types of stripe geometry structures are as follows:

1. **Gain-guided laser:** The current injection is restricted to a small region along the junction plane.
2. **Index-guided laser:** A built-in refractive index variation in the lateral direction is incorporated.

4.3.5.1 Gain-Guided Lasers

The current injection is restricted to a narrow region beneath a stripe as shown in Fig. 4.35. The active region is planar and continuous. Lasing occurs only in a limited region of the active layer beneath the stripe contact where high density of current flows. This horizontal confinement of light wave propagating through the active region is thereby accomplished by the small refractive index variation produced by the current-generated population inversion. If the light wave spreads in the horizontal plane outside of the horizontal dimensions of the stripe, it will be absorbed by the unexcited region of the active layer. In the vertical directions, the lower refractive indices of the surrounding layers reflect the optical wave back into the active region.

The current restriction serves several purposes:

1. It allows continuous-wave (CW) operation with reasonable low threshold currents (10–100 mA).
2. It can allow fundamental-mode operation along the junction plane, which is necessary for applications where the optical wave is coupled into a single-mode optical fiber.
3. The requirements for heat sinking are low.

Such lasers are termed as *gain-guided lasers* because the optical intensity distribution in the lateral direction is determined by the gain profile produced by carrier density distribution.

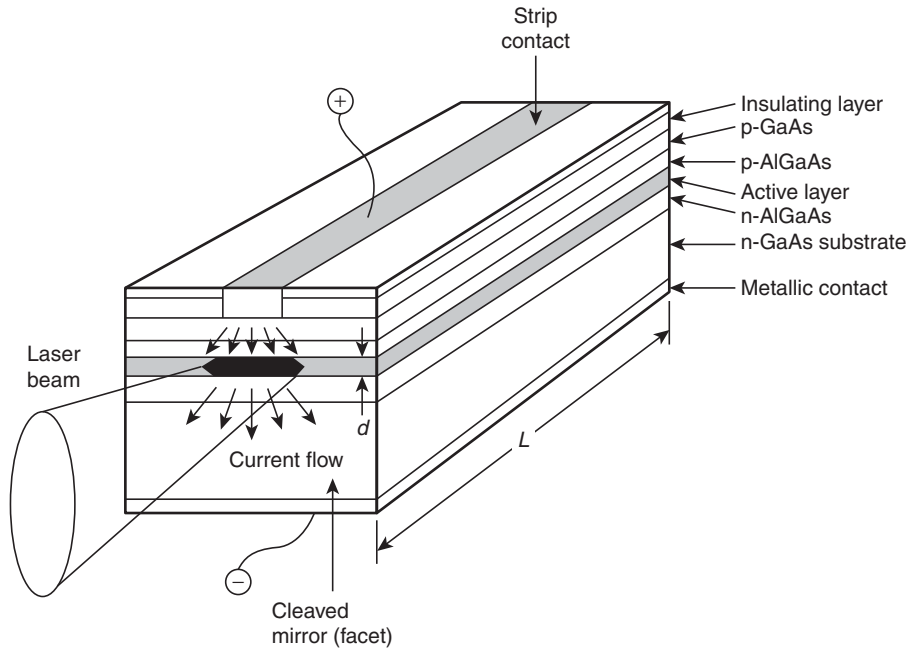


Figure 4.35 | Gain-guided laser.

4.3.5.2 Index-Guided Lasers

The transversal mode control in laser diodes can be achieved using index guiding along the junction plane. The mode control is necessary for improving the optical wave current linearity and the modulation response of lasers.

The active region is surrounded by materials with lower refractive indices in both the vertical (y) and lateral (x) transverse directions. The active region is buried in lower refractive indices layers (e.g., InP) on all sides as shown in Fig. 4.36. For this reason, these lasers are called *buried-heterostructure lasers*. The lateral index step along the junction plane is about two magnitudes larger than the carrier induced effects. As a result, the lasing characteristics of buried-heterostructure lasers are primarily determined by the rectangular waveguide that confines the mode inside the buried region. The transverse dimensions of the active region and the index discontinuities are chosen so that only the lowest order transverse modes can propagate in the waveguide. These index-guided devices produce beams with much higher beam quality, but are typically limited in power to only a few hundred milliwatts. Another important feature of this laser is the confinement of the injected carriers to the active region.

4.3.5.3 Single Frequency Solid-State Lasers

Ideally the output spectrum from a laser diode should be as narrow as possible. This implies that it should have only single-mode output. For single-mode operation, the optical output from laser should contain single transverse mode and single longitudinal mode.

Single transverse mode operation can be obtained by reducing the aperture of the resonant cavity to < 0.4 mm, such that only TEM_{00} mode is supported. Single longitudinal mode operation can be obtained by reducing the length of cavity until the frequency separation of the adjacent modes given by $\delta\nu = c/2nL$ is larger than the laser transition linewidth. Only single mode that falls within the transition linewidth can oscillate within the cavity.

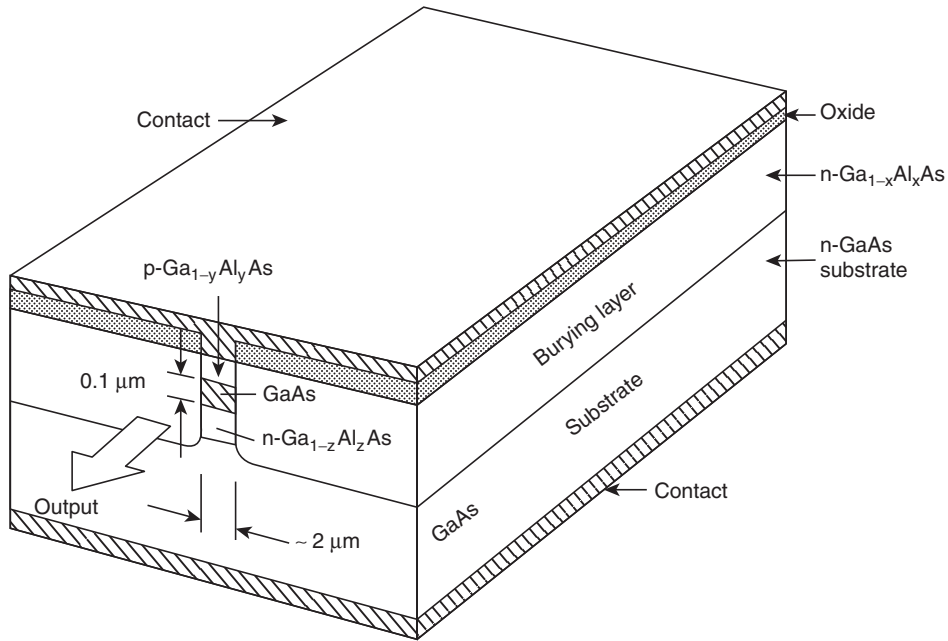


Figure 4.36 | Index-guided laser.

The conventional cleaved mirror structures are difficult to fabricate with cavity lengths below 50 mm. Another possible method is to insert an optical feedback in the device to eliminate other frequencies. Periodic gratings incorporated within the lasers waveguide can be utilized as a means of optical feedback. In distributed feedback (DFB) laser, the optical grating is incorporated in the active region which is pumped. On the other hand, in distributed Bragg reflector (DBR) laser, the grating is incorporated near the cavity end. Unlike Fabry–Perot laser, DFB and DBR lasers give rise to a single longitudinal mode even if modulated at high speed.

4.3.5.3.1 Distributed Bragg Reflector (DBR)

The DBR is a mirror that has been designed like a reflection-type diffraction grating. It has a periodic corrugated structure [refer Fig. 4.37(a)]. The partial reflections of waves from the corrugations interfere constructively to give a reflected wave only when the wavelength corresponds to twice the corrugation periodicity. Two partially reflected waves A and B, shown in Fig. 4.37(b), have optical path difference of 2Λ , where Λ is the corrugation period. These waves can interfere constructively only if 2Λ is a multiple of wavelength within the medium. These wavelengths are called Bragg wavelengths. For more details please refer Section 6.5.10.

The condition for in-phase interference is

$$2\Lambda = q \frac{\lambda_B}{n} \quad (4.48)$$

where n is the refractive index of corrugated material and $q = 1, 2, \dots$ is an integer called diffraction order. The DBR has high reflectance around Bragg wavelength λ_B and low reflectance away from λ_B . Due to this only that particular Fabry–Perot cavity mode, within the optical gain curve, which is close to λ_B , can lase and exist in the output.

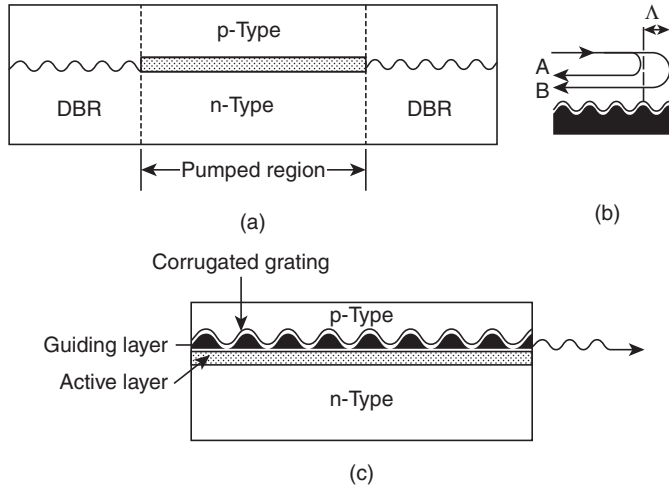


Figure 4.37 | Distributed feedback laser: (a) DBR laser, (b) details of Bragg reflection, and (c) DFB laser.

4.3.5.3.2 Distributed Feedback (DFB) Laser

In DFB laser, there is a corrugated layer, called the guiding layer (grating), next to the active layer as shown in Fig. 4.37(c). These corrugations in the refractive index act as optical feedback over the length of the cavity by producing partial reflections. Thus, the optical feedback is distributed over the cavity length.^[8,9]

The radiation is fed from active layer to guiding layer along the whole cavity length. A left-traveling wave in the guiding layer experiences partial reflections and these reflected waves are optically amplified by the medium to constitute a right going wave.

In DFB structure, traveling waves are reflected partially and periodically as they propagate. The left and right traveling waves couple coherently to setup mode if their frequency is related to the corrugation periodicity Λ . The DFB modes are at wavelengths symmetrically placed about Bragg wavelength λ_B . If λ_q is an allowed DFB lasing mode then

$$\lambda_q = \lambda_B \pm \frac{(\lambda_B)^2}{2nL}(q+1) \quad (4.49)$$

where q is a mode integer, $q = 0, 1, 2, \dots$, and L is the effective length of the diffraction grating. The relative threshold gain for higher modes is so large that only the $q = 0$ mode effectively lases.

4.3.5.3.3 Cleaved Coupled Cavity (CCC) Laser

Two different cavities of different lengths L_1 and L_2 are coupled as shown in Fig. 4.38(a). As the lasers are of different lengths and pumped with different currents, the mode spacing for the two cavities is different, as shown in Fig. 4.38(b). Modes in cavity L_1 are spaced more closely than modes in L_2 cavity. The modes which are common in both cavities are allowed because the cavities are coupled. The modes in L_1 and L_2 coincide only at far-spaced interval as shown in Fig. 4.38(b), so there is a wide separation between modes in coupled cavity which results into single-mode operation.^[10]

4.3.5.4 Quantum Well Devices

The double heterojunction (DH) laser fabricated with very thin active layer of around 10 nm thickness of narrow bandgap semiconductor – such as GaAs – sandwiched between two wider bandgap semiconductors – such as

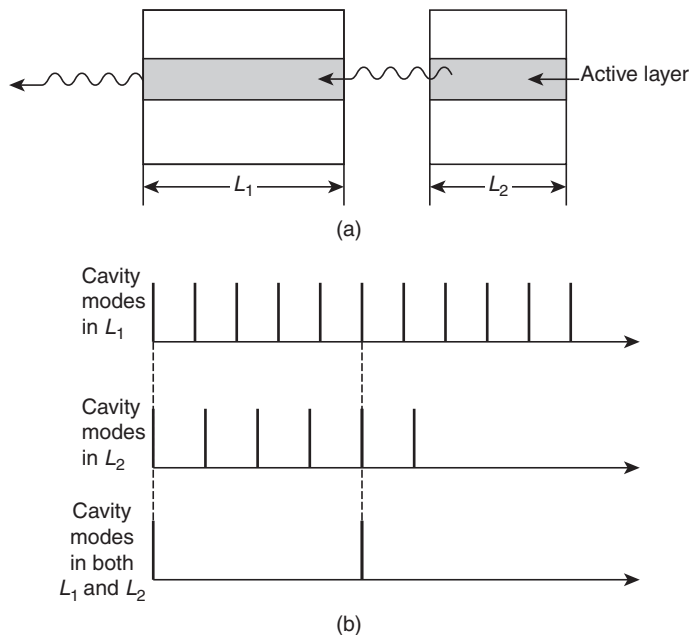


Figure 4.38 | Cleaved coupled cavity laser: (a) structure and (b) cavity modes.

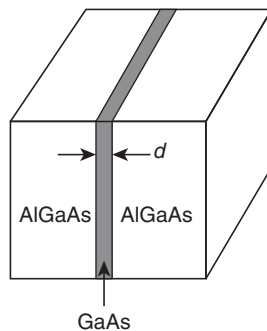


Figure 4.39 | A quantum well laser structure.

AlGaAs – is shown in Fig. 4.39. In this structure, the carrier motion normal to the active layer is restricted, resulting in a quantization of kinetic energy into discrete energy levels for the carriers moving in that direction. The thin active layer causes drastic changes to the electronic and optical properties in comparison with conventional DH laser having active layer of $0.1\text{--}0.3\text{ }\mu\text{m}$ thickness.

Under a forward-bias condition, electrons are injected into the conduction band of the GaAs layer, which serves as an active layer. The injected electrons populate ample number of states at E_1 . Hence population inversion occurs quickly without the need for a large current. The stimulated transition from E_1 to E_1' results into lasing action as shown in Fig. 4.40(a).

The advantages of quantum well structure are as follows:

1. The threshold current for lasing action in single quantum well (SQW) laser is typically $0.5\text{--}1\text{ mA}$, whereas in DH laser the threshold current is in the range of $10\text{--}50\text{ mA}$.

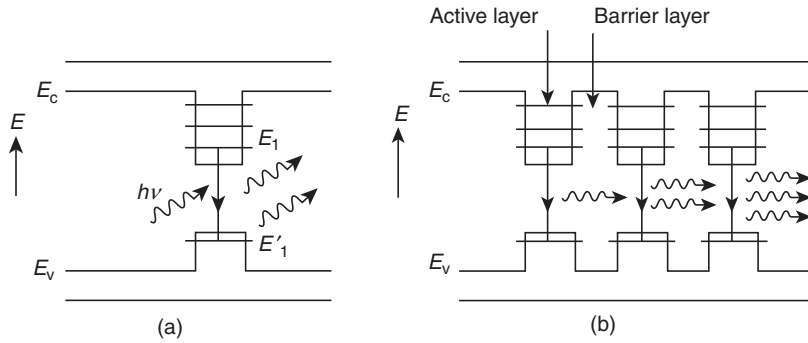


Figure 4.40 | (a) Single quantum well laser structure and (b) multiple quantum well structure.

2. Since the majority of the electrons are at and near E_1 and holes are at E'_1 , the range of emitted photon energies is close to $E_1 - E'_1$ and hence the wavelength and linewidth in output spectrum are narrower than those of the bulk semiconductor laser.

The advantages of single quantum well laser structure can be extended to a larger volume of crystal by using multiple quantum wells (MQW) as indicated in Fig. 4.40(b). In MQW laser, the structure has alternating ultrathin layers of wide and narrow bandgap semiconductor as shown in Fig. 4.40(b). The smaller bandgap layers are active layers, where the lasing action takes place, and the wider bandgap layers are the barrier layers. Better confinement of the optical mode is obtained in MQW laser as compared to SQW laser.

4.3.5.5 Vertical Cavity Surface Emitting Lasers (VCSELs)

In vertical cavity surface-emitting laser (VCSEL), multilayer mirrors are fabricated on the top and bottom of the quantum well gain region to give feedback. Consequently, the laser output is perpendicular to the active quantum well plane. Dielectric mirrors made from alternating high and low refractive indices form the reflectors at the ends of the cavity. This is quarter wave thick multilayer structure having 20–30 layers, which works as DBR. The active layer is generally very thin (less than $0.1 \mu\text{m}$) and is likely to be MQW for improved threshold current.

GaAs is used as a substrate, while InGaAs is the active layer and provides emission at 980 nm. The dielectric mirrors are then alternating layers of AlGaAs with different compositions and hence different bandgaps and refractive indices. The vertical cavity is circular in cross-section, as shown in Fig. 4.41, so that the emitted beam has circular cross-section. The height of vertical cavity is in the range of microns, therefore such a laser is also called as *microlaser*.^[11]

4.3.6 Laser Characteristics

1. **Temperature dependence:** Generally, the threshold current tends to increase with temperature as shown in Fig. 4.42. The temperature dependence of threshold current density J_{th} is approximately exponential and it is given by

$$J_{\text{th}} \propto e^{T/T_0} \quad (4.50)$$

where T is the device absolute temperature and T_0 is the threshold temperature coefficient.

2. **Mode hopping:** In single-mode laser diode, the peak emission wavelength exhibits “jumps” at certain temperatures as shown in Fig. 4.43. A jump corresponds to a mode hop in the output. That is, at a

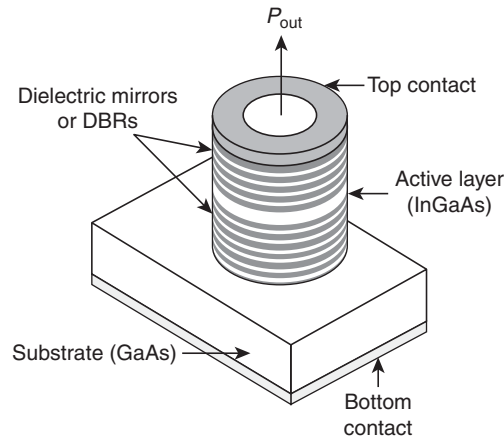


Figure 4.41 | Vertical cavity surface-emitting lasers (VCSELs).

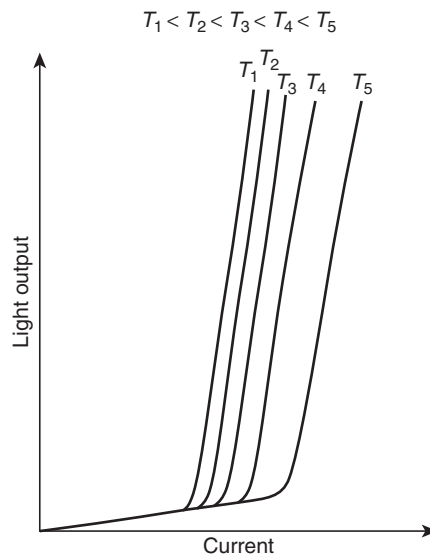


Figure 4.42 | Variation in threshold current with temperature for laser.

new operating temperature, another mode fulfills the laser oscillation condition which means a discrete change in the laser oscillation wavelength. The peak emission wavelength increases slowly with temperature due to slight increase in refractive index and cavity length with temperature.

3. **Dynamic response:** When a current pulse is applied to the device, it results into switch on delay (t_d) followed by high frequency damped oscillations known as relaxation oscillations as shown in Fig. 4.44. Consequently, the laser output can comprise several pulses as the electron density is repetitively built up and quickly reduced, thus causing relaxation oscillations. At higher data rates above 100 Mbps, this behavior can deteriorate the pulse shape since $t_d = 0.5$ ns and relaxation oscillation can last twice t_d .
4. **Noise:** The random intensity fluctuations caused by temperature variation create a noise called *relative intensity noise* (RIN). Typically, RIN for single-mode semiconductor laser lies in the range of 130 to 160 dB/Hz.

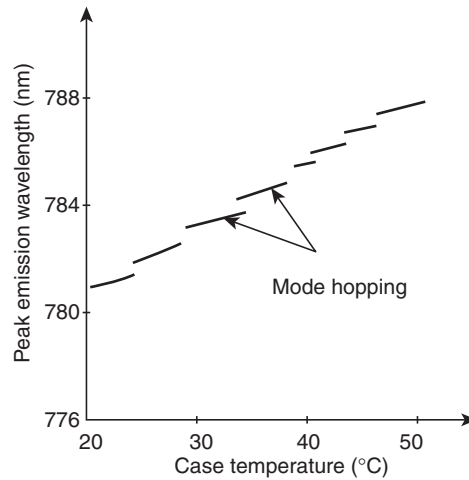


Figure 4.43 | Peak wavelength versus case temperature characteristics showing mode hopping in output spectrum of single-mode laser diode.

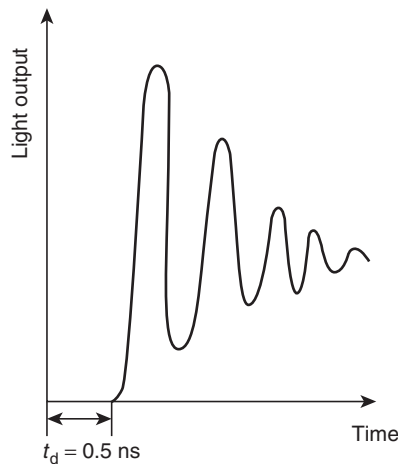


Figure 4.44 | Dynamic behavior of laser.

It decreases as the injection current level I increases. Mode partition noise is a phenomenon associated with multimode semiconductor lasers when the modes are not stabilized. Temperature changes can cause the relative intensities of the various longitudinal modes in laser's output spectrum to vary from one pulse to next, even though the total output power from a laser is maintained nearly constant.

4.3.7 Laser Properties and Applications

1. **Collimation:** In collimated light, all the light rays or waves are traveling in a specific direction and are all parallel to each other. Lasers produce the most collimated light as compared to other light sources. There are numerous applications of collimated light such as a reference beam in construction, leveling, and

grading land; alignment of pipe such as sewer pipe, and sending light over long distances without suffering significant divergence, laser pointers, etc. The cavity mirror properties, like the radii of curvature of the mirrors and the separation between mirrors, decide the collimation of light.

2. **Monochromaticity:** It defines the purity of a signal in terms of frequency or wavelength or indicates how narrow the laser beam frequency bandwidth is. If the laser is operating in a single longitudinal mode, as most solid-state and semiconductor lasers do, the actual laser linewidth can be significantly narrower. Most lasers normally provide a sufficiently narrow frequency output bandwidth, of the order of 10^9 – 10^{11} Hz suitable for applications requiring a single narrow wavelength. In some applications a much narrower linewidth is required. By operating with a single longitudinal and single transverse mode (TEM_{00}), linewidths of the order of 1 MHz (10^6 Hz) or less can be obtained. The narrowing is enhanced by choosing highly reflecting mirrors, constructing a very stable mirror cavity in conjunction with the amplifier by eliminating vibrations of the mirrors and other cavity elements, and providing temperature stability.
3. **Coherence:** Coherent light means the light waves that are “in-phase” with one another or in other words how similar they are in-phase. Temporal or longitudinal coherence means the closeness in-phase of various portions of the laser frequency bandwidth. Spatial or transverse coherence means the closeness in-phase of different spatial portions of the beam after the beam has propagated a certain distance. This phased relationship determines how readily the various portions of the beam can interfere with each other, after the beam has propagated a specific distance, to produce such effects as diffraction (bending) of light and related applications such as holography.
4. **Intensity and radiance:** Intensity or irradiance is the power of the laser beam divided by the cross-sectional area of the beam. It is expressed in watts per square centimeter (W/cm^2). It is a measure of the amount of energy that can be applied to a specific region within a given amount of time. Intensity and radiance is the most important specification of laser in material processing applications such as welding, cutting, heat treating, ablating, and drilling, as well as laser surgery.
5. **Focusability:** Lasers can be focused to a very small spot size as compared to other light sources. In applications like a compact disk player, it is desirable to have properly focused laser beam. The information storage capacity is decided by diameter of the spot size of laser beam. The information is stored in smallest grooves possible on the disk and the width of the grooves is determined by the ability of a laser beam to access a single groove without overlapping adjacent grooves.

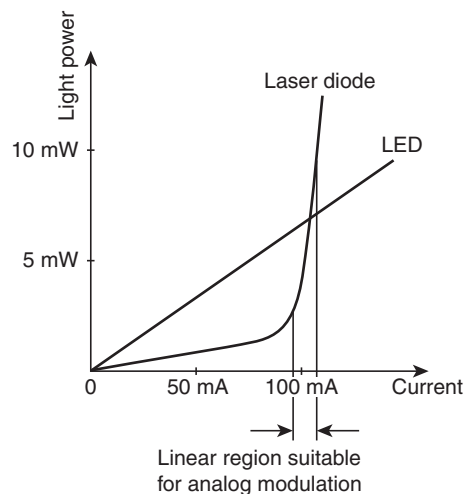


Figure 4.45 | Typical optical power output versus forward current characteristics for LED and laser.

Table 4.1 | Comparison of LED and laser source

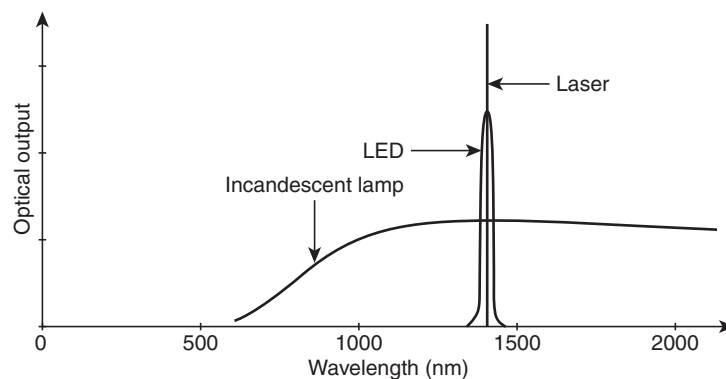
S. No.	Light-emitting diode	Laser
1	Light is mostly monochromatic (narrow energy spread comparable to the distribution of electrons/ hole populations in the band edges)	Light is essentially single wavelength (highly monochromatic)
2	Light is from spontaneous emission (random events in time and thus phase)	Light is from “stimulated emission” (timed to be in-phase with other photons)
3	Light diverges significantly	Light has significantly lower divergence

The electrical power required to operate both LED and laser is generally similar, but the optical output power against current characteristic for both varies considerably as shown in Fig. 4.45. Laser must be operated in the region of stimulated emission, that is, above threshold to obtain linear operation, where the optical output power levels are in the range of 1 to 10 mW. The comparison of LED and laser is listed in Table 4.1.

Fiber-optic sources must operate in the low-loss transmission windows of glass fiber. LEDs are typically used at the 850 nm and 1310 nm transmission wavelengths, whereas lasers are primarily used at 1310 nm and 1550 nm.

LEDs are used in lower data-rate (few hundreds of megahertz), shorter distance multimode systems because of their bandwidth limitations and lower output power. Due to the fiber's small NA and core diameter, the coupling of LED into single-mode fiber is difficult. Therefore, LEDs are most often used with multimode fiber. The output spectrum of a typical LED is about 40 nm, which limits its performance because of severe chromatic dispersion. LEDs operate in a more linearly fashion than laser diodes, hence they are more suitable for analog modulation.

Laser diodes (LDs) are used in those applications in long-haul communication where higher data rates (GHz or even THz) are required. LD has a much higher output power than an LED. Therefore, it is capable of transmitting information over longer distances. LD has a much narrower spectral width (refer Fig. 4.46). It can provide high-bandwidth communication over long distances. The LD's smaller NA also allows it to be more effectively coupled with single-mode fiber. The disadvantage of LDs is that they are inherently non-linear, which makes analog transmission more difficult. They are also very sensitive to fluctuations in temperature and drive current, which causes their output wavelength to drift.

**Figure 4.46** | Spectral linewidths of different optical sources.

4.4 Line Coding

For digital data transmission the timing information of incoming signal must be extracted precisely by the receiver circuit and the error in received signal should be detected by adding extra bits into the raw data stream. So the raw data are arranged in a particular format. This process is called *channel* or *line coding*.

Line coding is important in digital communications for the following reasons:

1. Easy timing recovery at the receiver.
2. To shape the spectrum of the signal as required by various elements in the communication systems.

The following three basic coding schemes are used (refer Fig. 4.47):

1. Non-return-to-zero (NRZ).
2. Return-to-zero (RZ).
3. Phase encoded (PE).

The details of these three coding schemes and the different sub coding techniques are as follows:

1. **Non-return-to-zero (NRZ):** The NRZ method occupies the entire bit width (bit period). In a serial data stream, ON–OFF signal is send, “1” is send as a pulse and “0” as no pulse. This code is simple to generate and decode, but it has no error monitoring or correcting capabilities as well as no self-clocking features. As the long string of ones and zeros contain no timing information and the clock is not very stable, N identical bits may be misinterpreted as either $(N + 1)$ or $(N - 1)$ bits.
 - (a) **On–Off (NRZ):** In this form of line codes, bit 1 is represented by some positive voltage (+5 V, for example) and bit 0 by 0 V. The pulses corresponding to binary 1 remain at the positive voltage for the whole duration of the bit period. It does not return to zero at any time during the bit period. Therefore, it is called as non-return-to-zero.
 - (b) **Polar (NRZ):** In this form of line codes, bit 1 is represented by some positive voltage (+5 V, for example) and bit 0 is represented by negative of that voltage (so it would be -5 V). The pulses corresponding to binary 1 and binary 0 remain at the positive and negative voltages, respectively, for the whole duration of the bit period (they do not return to zero). The advantage of this line code over the ON–OFF (NRZ) is that it has zero DC value when the number of binary 1s is equal to the number of binary 0s. A line code with zero DC is desired in some applications that require that the transmitted signal to have no DC.
2. **Return-to-zero (RZ):** In applications having larger available bandwidth, each bit is coded as two optical line code bits. Such type of coding is known as RZ code. The RZ method occupies only a portion of the bit period. There are different types of RZ code. In unipolar RZ (Fig. 4.47), a “1” bit is represented by a half-period optical pulse that can occur in either the first or second half of the bit-period. A “0” is represented by no signal during the bit period.
 - (a) **On–Off (RZ):** In this line code, bit 1 is represented by some positive voltage (+5 V, for example) for half of the bit period and zero in the other half of the bit period and bit 0 is represented by zero for the whole bit period. This is why this line code is a return-to-zero line code (because any pulse corresponding to binary 1 always returns back to zero). The advantage of this line code over the previous line codes is that a long sequence of 1s always has transitions at the center of each bit and therefore bit synchronization becomes easy for long sequences of 1s. Long sequence of 0s is still difficult to be synchronized.

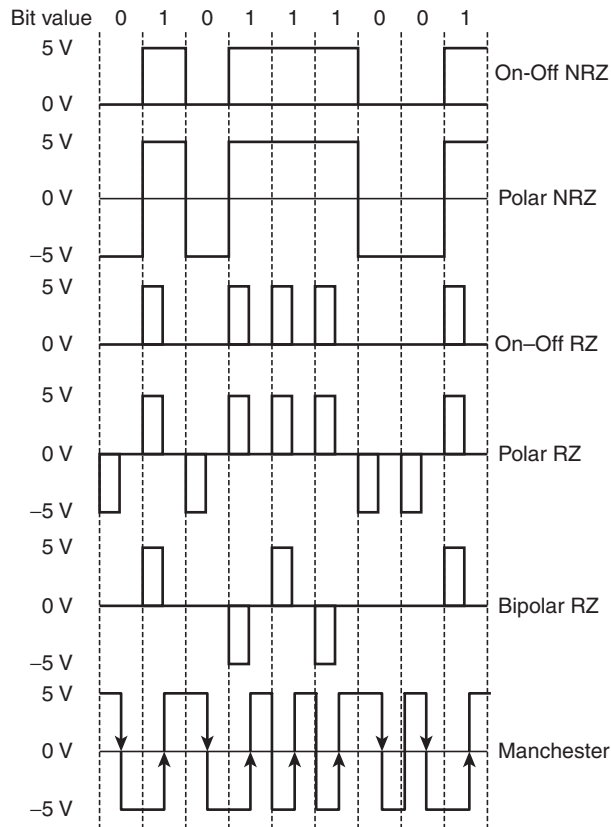


Figure 4.47 | Digital line coding schemes.

- (b) **Polar (RZ):** In this line code, bit 1 is represented by some positive voltage (+5 V, for example) for half of the bit period and zero in the other half of the bit period and bit 0 is represented by the negative of that voltage for half of the period and zero for the other half. The advantage of this line code over the previous ones is that long sequences of 1s or 0s have transitions at the center of each bit and therefore bit synchronization becomes easy for long sequences of 1s or 0s. Also, this line code has zero DC when the number of 1s and 0s is the same.
- (c) **Bipolar (RZ):** In this line code, bit 0 is represented by 0 V for the whole bit period. A bit of 1 is represented by some positive voltage (+5 V, for example) for half of the bit period and 0 in the other half of the bit period. However, the next bit of 1 (whether it is the next bit or 1000 bits later) is represented by the negative of the voltage for half of the bit period and 0 for the second half. So, the bits of 1s are represented by alternating positive and negative pulses. This ensures that the DC value of the signal is always 0 even if we have non-equal number of 1s and 0s.
3. **Phase encoded:** Manchester coding is a bi-phase coding. In this line code, bit 1 is represented by -5 V for the first half of the bit and +5 V for the second half of the bit, and bit 0 is represented by +5 V for the first half of the bit period and -5 V for the second half. Thus, an up transition in the middle of the bit represents bit 1 and a down transition in the middle of the bit represents bit 0.

4.5 Modulation

Lasers and LEDs used in telecommunication applications are modulated using one of the following two methods:

1. **Direct modulation:** The output power of the device varies directly with the input drive current as shown in Fig. 4.48. Both LEDs and lasers can be directly modulated using analog and digital signals. The benefit of direct modulation is that it is simple and cheap. The disadvantage is that it is slower than indirect modulation with limits of less than approximately 3 GHz.
2. **External modulation:** The external modulation setup is as shown in Fig. 4.49. The continuous output of laser source is intensity-modulated or phase-modulated using external modulator. The electrical modulating signal applied to modulator acts like a controller. For high-speed operation in long-haul communication systems, external modulation is preferred rather than direct modulation. It can be used with high power lasers but it requires complex circuit.

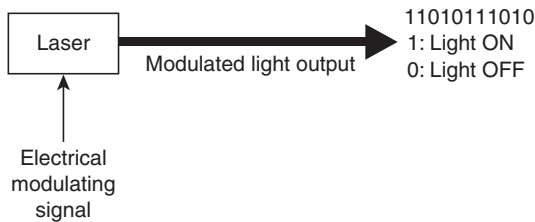


Figure 4.48 | Direct modulation.

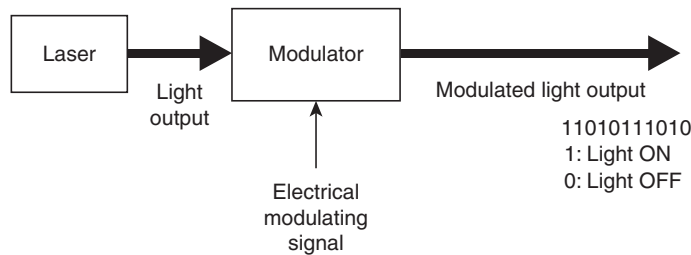


Figure 4.49 | External modulation.

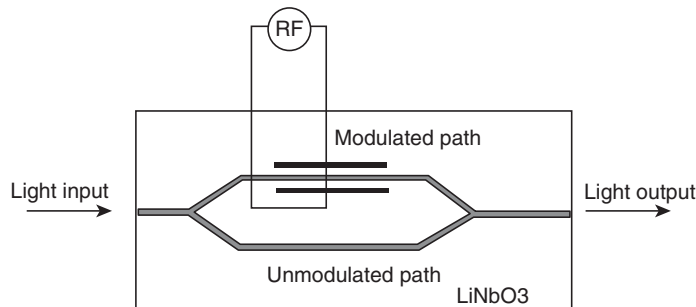


Figure 4.50 | External modulation using Mach-Zehnder waveguide interferometer.

An integrated optical modulator incorporating waveguide Mach–Zehnder interferometer fabricated on lithium niobate (LiNbO_3) slab, as shown in Fig. 4.50, is used as an external modulator. For guiding the light through the device, the waveguide region is slightly doped with impurities to increase the refractive index. The light entering the modulator gets split into two paths. One path is unmodulated. Across the other path electrodes are placed. Lithium niobate (LiNbO_3) is an electro-optic material. When a voltage is applied to the electrodes placed across the arm of waveguide, the refractive index of material of that arm changes. This causes phase delay proportional to applied voltage. After passing through the two arms again when the light recombines, the waves interfere.

- (a) Output will be ON if the two waves are in phase and interfere constructively.
- (b) Output will be OFF if the two waves are out of phase and cancel each other.

The induced phase shift is given as

$$\text{Phase shift} = \Delta\theta = 180^\circ \times \frac{V_{\text{in}}}{V_\pi}$$

where V_{in} is the modulating voltage and V_π is the input voltage corresponding to 180° phase shift.

4.6 Optical Transmitter

In most cases LEDs are operated from a DC supply, with a series resistor (Fig. 4.51). The series resistor limits the forward current to a suitable value. The forward current requirement changes according to the application. Simple pilot lamp or status indicator may require around 5–6 mA, while other applications where more light output is needed may require up to 20 mA or more. If forward current is I_F , the supply voltage is V_S , and the expected forward voltage drop of the LED at this current level is V_F , then

$$R_S = \frac{V_S - V_F}{I_F} \quad (4.52)$$

If an LED is operated from an AC supply, or from an unreliable source, it can be protected by one of the methods shown in Fig. 4.52. The simplest approach is shown in Fig. 4.52(a) where a reverse-biased diode is connected across the LED to limit any reverse voltage to 0.7 V. The LED is protected from reverse voltage but efficiency of circuit is very less, during negative half cycle of AC input, supply current is drawn but LED does not emit light. So the light output and efficiency are both effectively halved.

The second method shown in Fig. 4.52(b) is more efficient and also maintains the LED light output. Here a bridge of four diodes is used to ensure that the current always flows through the LED in the forward direction, regardless of supply polarity.

For obtaining higher light output, usually LED is operated from a pulsed current of short duty cycle (ON–OFF ratio) instead of DC supply. This scheme reduces the power dissipation and gives more light output during on time of pulse (Fig. 4.53).

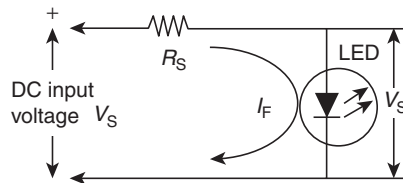


Figure 4.51 | LED operated from DC supply.

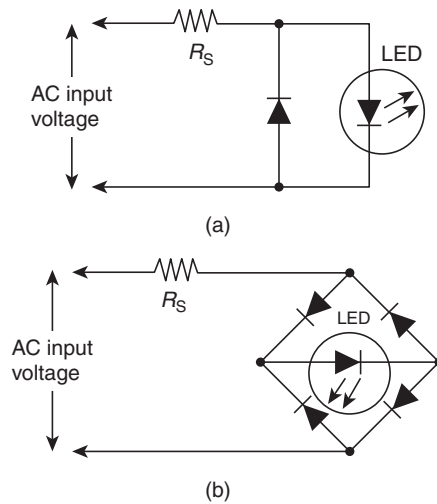


Figure 4.52 | LED operated from AC supply.

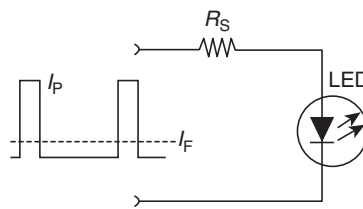


Figure 4.53 | LED operated from higher but pulsed current.

4.7 Optical Connectors

Optical connectors are demountable and are used to connect fiber-optic cable to peripheral equipment and to other fibers. These connectors are very high precision devices since the fiber dimensions are few microns. To couple the light between fibers efficiently, the tolerance requirements are stringent, because it is required to have repeated connection and disconnection without any fiber alignment problem. To maintain the optimum performance, the fiber ends should be protected from any damage; thus the optical connector should be insensitive to environmental factors; it should be low cost component and should be easily fitted.

There are many optical connectors in use today. The SMA connector is the most popular type of connector. Figure 4.54 shows an exploded view of the parts of this connector.

The most popular type of multimode connector in use today is the ST connector, shown in Fig. 4.55. This connector provides lower overall losses approximately less than 1 dB (20%) of loss. It uses a twist lock type of design and does not require alignment sleeves. It has anti-rotation tab, which assures that every time the connectors are mated, the fibers always return to the same rotational position giving constant and uniform performance. ST connectors are available for both multimode and single-mode fibers, the primary difference being the overall tolerances. Multimode ST connectors will only perform properly with multimode fibers. More expensive single-mode ST connectors will perform properly with both single-mode and multimode fibers.

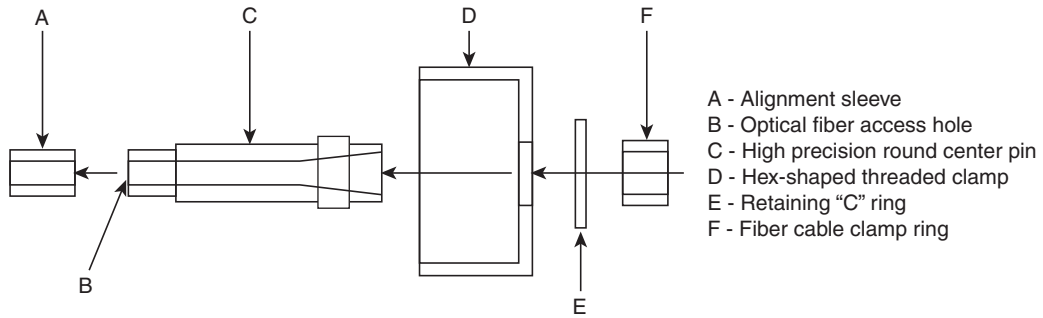


Figure 4.54 | Construction of SMA connector.

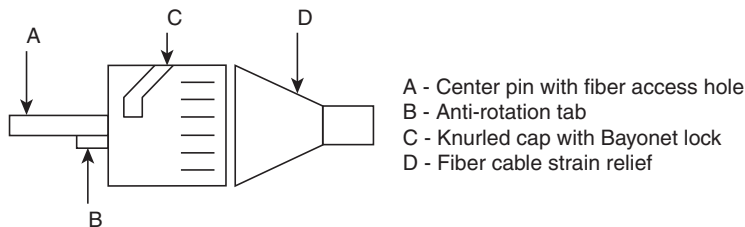


Figure 4.55 | Major features of the "Industry Standard" ST connector.

4.8 Optical Splices

While interconnecting the fiber with demountable connectors, significant loss is observed. If a permanent connection is required then another method with much less loss is splicing. Before splicing the two fibers, to obtain a joint for efficient transfer of optical energy, both ends of fiber must be cut precisely square and polished flat because diameter of the core in an optical fiber is very small and any irregularity (such as a join) can result in significant loss of power. They must then be joined together so that there is minimum air (or water) between the two ends and these ends must match up nearly exactly.

Joining fibers together is not an easy task because the dimensions of a fiber are tiny and accuracy of alignment is critical. Light traveling in a fiber is a guided wave and the fiber is a waveguide. Any imperfection or irregularity (such as a join) will result in loss.^[13]

There are three general ways of joining fibers:

1. Fusion splicing.
2. Use of index matching epoxy glues.
3. Mechanical connectors of different types.

In all types of joints, the alignment of fiber core is the initial requirement. Even if the fiber is aligned, it is not confirmed that the core will also be aligned, because the core is not always at the center of fiber. Irrespective of great care taken by the manufacturers, there is variation always. Difference between the axis of the core and the axis of the cladding is expressed as the "concentricity" of the fiber. So if the cores are not aligned there is random misalignment and hence losses will be there. The major improvement in connector losses observed between the middle 1980s and the 1990s was due generally to better fiber manufacture as well as due to better connectors and connection techniques.

4.8.1 Fusion Splicing

In fusion splicing, the ends of two pieces of fiber are actually melted (fused) together. The splicing process requires special expensive splicing equipment but can be performed very quickly, so the cost becomes reasonable if done in good quantity. The fiber ends are cut, polished, butted up to one another, and fused by heat. Very high temperature is required, which is obtained by electric arc method. The joint thus produced is very fragile and mechanical means are used to protect it. Practically the loss of around 0.1 dB is obtained in fusion splicing, which is much better than connector loss of around 1 dB.

A device setup for fusion splicing is illustrated in Fig. 4.56. The stepwise splicing procedure is as follows:

1. The primary coating of each fiber end is stripped/cleaved such that it is square.
2. The fiber ends are positioned a few mm from one another and clamped to positioning blocks.
3. The fiber ends are then aligned with one another and brought closer together.
4. When alignment is satisfactory, an electric arc is started between the two electrodes and the fibers are brought into contact. Heat from the arc melts the glass and the joint is made.

The two important things to be considered are – alignment of the fiber and precise control of the heating arc. Automatic fusion splicers have the facility of visual display for positioning. The fiber end faces and core can be clearly seen. A microcomputer in the splicer examines the images of the fiber ends and aligns them automatically. Using this method very low-loss splices can be obtained.

4.8.1.1 Control of Heating

To make a good mechanical joint, precise heat control is very essential. Both the ends of fiber need to be melted completely and glass from the two fibers mixed. If more heat is applied, the overlap of fiber ends will be more and the joint will have a strong perturbation of the refractive index and hence high loss. If precise heat is applied and very thin layer on the end face of each fiber is heated, then best optical properties can be obtained. But the mechanical characteristics of this joint are not good. To provide the mechanical strength, the newly formed splice should be protected by covering it with a sleeve of heat-shrink material and then applying gentle heat. The material contracts around the fiber and protects the splice. A metal strengthening pin is often integrated into the side of the wrapping to provide additional mechanical strength.

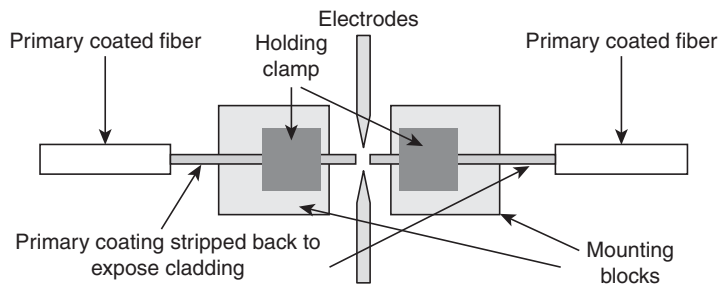


Figure 4.56 | Fusion splicing schematic.

4.8.1.2 Cleaving the Fiber

Before splicing the two ends of fiber, it is important to square cut the ends of fiber. The further refinement to get the desired shape of cut can be done by polishing. The technique to get a good square fiber end is called “cleaving.”

1. To destroy the local surface tension and to make a crack on one point of fiber surface, a scratch or nick is made in the side of the fiber.
2. A stress is applied such that the fiber will crack across its diameter. This can be done by bending the fiber around a rod. But this may create a “lip” on the side of the fiber opposite to the nick. Like automatic splicing machines, cleaving machines are also available in the market, which deliver a short sharp blow to the fiber in exactly the right place and produce a good square cleave.

4.8.2 Mechanical Splicing

The stepwise procedure of a mechanical splice is as follows:

1. The ends of two fibers are cleaned and stripped.
2. The fibers with prepared ends are carefully butted together and aligned using a mechanical assembly.
3. A gel is used at the point of contact to reduce light reflection and keep the splice loss to a minimum.
4. The ends of the fiber are held together by friction or compression so that the fibers remain aligned.
5. The gap between the two fibers is filled with an epoxy resin having the same RI as the fiber core.

For mechanical splicing, there are various ways of aligning the fiber ends without reference to alignment of the cores:

1. A glass tube is used into each end of the fiber which is pushed. Before insertion of fiber, a small amount of the epoxy resin is placed on the end of one of the fibers. A small hole is made in the tube at the point of joint so that excess epoxy can escape.
2. The other methods of obtaining mechanical alignment of the outside of the fibers are as follows:
 - (a) V-grooves.
 - (b) Slots.
 - (c) Alignment rods.

This splicing technique has the lowest cost but it is not very good. The quality of the joint depends on the following:

1. The concentricity of the fiber.
2. The tolerances and precision of the alignment device.
3. The accuracy of the outside diameter of the fiber.
4. The circularity of the outside of the fiber.

However, this makes a solid, permanent connection and is used for fiber-to-fiber joints in many situations.

4.8.2.1 Mechanical Splicing with Alignment and Bonding

This process is similar to mechanical splicing but the fibers are precisely positioned in the same way as with fusion splicing. The cleaved fibers are inserted into silica sleeves and bonded in place. The fiber ends are polished to get a very accurate surface. Then the sleeves are actively aligned so that the maximum optical power is transferred. They are then bonded with epoxy and covered with another protective sleeve. This technique provides very high quality splices but it is very time consuming.

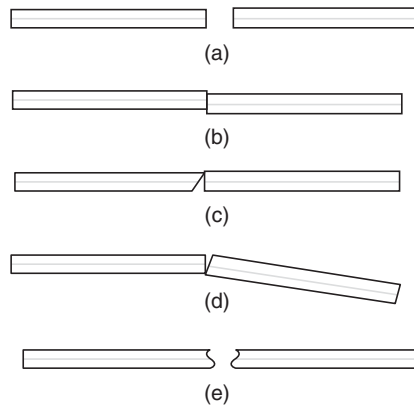


Figure 4.57 Sources of extrinsic losses: (a) Longitudinal misalignment, (b) lateral misalignment, (c) angular misalignment, (d) fiber end not cut square, and (e) fiber end irregular or rough.

4.8.3 Losses in Fiber Joints

Losses in fiber joints are commonly classified into two kinds:

1. **Extrinsic losses:** These losses are those caused by factors concerned in joining the fiber but are unrelated to the properties of the fiber itself (refer Fig. 4.57). The sources of extrinsic losses are listed below.
 - (a) Longitudinal misalignment.
 - (b) Lateral misalignment.
 - (c) Angular misalignment.
 - (d) Fiber end not cut square.
 - (e) Fiber end irregular or rough.
2. **Intrinsic losses:** These are losses caused by some property inherent in the construction of the fiber (refer Fig. 4.58).
 - (a) Core concentricity.
 - (b) Core shape (Ellipticity).
 - (c) Core diameter.
 - (d) Cladding diameter.
 - (e) Mode field diameter.
 - (f) Numerical aperture.
 - (g) Refractive index profile.

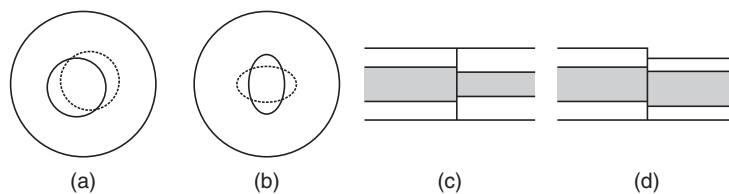


Figure 4.58 Sources of intrinsic losses: (a) Core concentricity, (b) core shape, (c) core diameter, and (d) cladding diameter

Solved Problems

PROBLEM 1

Calculate the radiative minority carrier lifetime in gallium arsenide and germanium when the minority carriers are injected into p-type region, which has hole concentration of 10^{18} cm^{-3} .

Solution: The radiative minority carrier lifetime is given by

$$\tau_r = [B_r(N + P)]^{-1}$$

where N is majority charge carrier concentration in n-type region and P is majority charge carrier concentration in p-type region. In the p-region, the hole concentration determines the radiative carrier lifetime as $P \gg N$. Hence

$$\tau_r = [B_r N]^{-1}$$

For gallium arsenide:

Recombination coefficient of gallium arsenide is $B_r = 7.21 \times 10^{-10} \text{ cm}^3/\text{s}$. So

$$\begin{aligned}\tau_r &= [7.21 \times 10^{-10} \times 10^{18}]^{-1} \\ \Rightarrow \tau_r &= 1.39 \times 10^{-9} \text{ s or } 1.39 \text{ ns}\end{aligned}$$

For germanium:

Recombination coefficient of germanium is $B_r = 5.25 \times 10^{-14} \text{ cm}^3/\text{s}$. So

$$\begin{aligned}\tau_r &= [5.25 \times 10^{-14} \times 10^{18}]^{-1} \\ \Rightarrow \tau_r &= 19 \mu\text{s}\end{aligned}$$

Note: Gallium arsenide is a direct bandgap material and has the radiative minority carrier lifetime much less than germanium (indirect bandgap material).

PROBLEM 2

For a double heterojunction LED having peak emission wavelength of 870 nm and drive current of 50 mA, determine the total carrier recombination lifetime and power internally generated within the device. The radiative and non-radiative recombination lifetimes of the minority carriers are 50 ns and 90 ns, respectively.

Solution: Total carrier recombination lifetime is given by

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}}$$

Therefore

$$\tau = \frac{\tau_r \cdot \tau_{nr}}{\tau_r + \tau_{nr}} = \frac{50 \text{ ns} \times 90 \text{ ns}}{(50 + 90) \text{ ns}} = 32.14 \text{ ns}$$

The internal quantum efficiency is

$$\eta_{\text{int}} = \frac{\tau}{\tau_r} = \frac{32.14}{50} = 0.642$$

Optical power generated internally by the LED, P_{int} , is

$$P_{\text{int}} = \eta_{\text{int}} \frac{h\nu}{e\lambda} = \frac{0.642 \times 6.626 \times 10^{-34} \times 3 \times 10^8 \times 50 \times 10^{-3}}{1.602 \times 10^{-19} \times 870 \times 10^{-9}} = 45.78 \text{ mW}$$

PROBLEM 3

The refractive index of a material used for fabricating an LED is 3.5. Calculate its external quantum efficiency.

Solution: The external quantum efficiency is

$$\eta_{\text{ext}} \approx \frac{1}{n(n+1)^2} \approx \frac{1}{3.5(3.5+1)^2}$$

$$\Rightarrow \eta_{\text{ext}} \approx 1.41\%$$

PROBLEM 4

For a GaAs planar LED having refractive index of 3.4, calculate optical power emitted by the LED when the optical power generated internally is 30 mW.

Solution: Given: refractive index $n = 3.4$, power generated internally $P_{\text{int}} = 30 \text{ mW}$. The external quantum efficiency is

$$\eta_{\text{ext}} \approx \frac{1}{n(n+1)^2}$$

Optical power emitted from LED is

$$P_{\text{ext}} = \eta_{\text{ext}} P_{\text{int}} \approx \frac{P_{\text{int}}}{n(n+1)^2} = \frac{30 \times 10^{-3}}{3.4(3.4+1)^2} = 0.455 \text{ mW}$$

PROBLEM 5

The light output from GaAs LED is coupled into a step index fiber having numerical aperture of 0.22 and core refractive index of 1.42. Calculate coupling efficiency and optical power loss in dB, relative to the power emitted from LED, when coupling the light output into the fiber.

Solution: Coupling efficiency is given as

$$\eta_c = \sin^2 \theta_a = (NA)^2 = (0.22)^2 = 0.0484$$

$$\Rightarrow \eta_c = 4.84\%$$

Thus, 4.84% of externally emitted power is coupled into the fiber. The power coupled with fiber = P_f and power externally emitted = P_{ext} . Therefore

$$\text{Loss} = -10 \log \frac{P_f}{P_{\text{ext}}} = 10 \log \eta_c$$

$$\Rightarrow \text{Loss} = -10 \log 0.0484 = 13.15 \text{ dB}$$

PROBLEM 6

In an InGaAsP–InP laser diode which has an optical cavity of length 250 μm , the peak radiation is at 1550 nm and the refractive index of InGaAsP is 4.0. The optical gain bandwidth is 2 nm.

- (a) What is the mode integer q of the peak radiation?
- (b) What is the separation between the modes of the cavity?
- (c) How many modes are there in the cavity?

Solution: Given, $L = 250 \mu\text{m}$, $\lambda = 1550 \text{ nm}$, optical gain bandwidth = 2 nm.

- (a) *Mode integer of peak radiation:* We have

$$L = \frac{\lambda q}{2n}$$

Therefore

$$q = \frac{2Ln}{\lambda} = \frac{2 \times 4 \times 250 \times 10^{-6}}{1550 \times 10^{-9}} = 1290.3$$

- (b) Different modes are separated by $\delta\lambda$, given by

$$\delta\lambda = \frac{\lambda^2}{2nL} = \frac{(1550 \times 10^{-9})^2}{2 \times 4 \times 250 \times 10^{-6}} = 1.201 \text{ nm}$$

- (c) The number of modes are

$$\text{No. of modes} = \frac{\text{Optical gain bandwidth}}{\delta\lambda} = \frac{2 \text{ nm}}{1.2 \text{ nm}}$$

Therefore maximum two modes are there in the cavity.

PROBLEM 7

Determine the number of modes and their frequency of separation in a ruby laser having peak emission wavelength of 550 nm, if the crystal length is 5 cm and refractive index is 1.78.

Solution: The number of modes is

$$q = \frac{2Ln}{\lambda} = \frac{2 \times 1.78 \times 0.05}{550 \times 10^{-9}} = 3.2 \times 10^5$$

Different modes are separated by a frequency interval $\delta\nu$, given by

$$\delta\nu = \frac{c}{2nL} = \frac{3 \times 10^8}{2 \times 1.78 \times 0.05} = 1.68 \text{ GHz}$$

PROBLEM 8

A commercial laser diode with an emission wavelength of 670 nm has threshold current at 25°C equal to 76 mA. At $I = 80 \text{ mA}$, the output optical power is 2 mW and the voltage across the diode is 2.3 V. If the diode current is increased to 82 mA, the output power increases to 3 mW. Calculate the external quantum efficiency and external power efficiency of the laser diode.

Solution: Emitted power changes linearly when the injection current I is greater than the threshold current I_{th} . In this case,

$$\eta_{EQE} = \eta_{EDQE} \left(1 - \frac{I_{th}}{I} \right)$$

We have

$$\begin{aligned} \eta_{EDQE} &= \frac{dP_0/hf}{dI/e} = \frac{P_{02} - P_{01}}{I_2 - I_1} \times \frac{e}{E_g} \\ \Rightarrow \eta_{EDQE} &= \frac{(3 - 2) \text{ mW}}{(82 - 80) \text{ mA}} \times \frac{1.602 \times 10^{-19}}{E_g} \end{aligned}$$

Now

$$E_g = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{670 \text{ nm}} = 2.966 \times 10^{-19}$$

Therefore

$$\begin{aligned} \eta_{EDQE} &= \frac{(3 - 2) \text{ mW}}{(82 - 80) \text{ mA}} \times \frac{1.602 \times 10^{-19}}{2.966 \times 10^{-19}} \\ \Rightarrow \eta_{EDQE} &= 27\% \end{aligned}$$

So

$$\eta_{EQE} = 27 \left(1 - \frac{76 \text{ mA}}{80 \text{ mA}} \right) = 1.35\%$$

External power efficiency is

$$\eta_{EPE} = \frac{P_0}{P} \times 100 = \frac{P_0}{I \times V} = \frac{2 \text{ mW}}{80 \text{ mA} \times 2.3} = 1.086\%$$

PROBLEM 9

For an InGaAsP laser diode operating at $\lambda = 1310 \text{ nm}$, $I = 40 \text{ mA}$, the output power is 3 mW and the voltage across the diode is 1.4 V . If the diode current is increased to 45 mA , the optical output power increases to 4 mW . Calculate external quantum efficiency, external differential quantum efficiency, and external power efficiency of the laser diode.

Solution: We have

$$E_g = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1310 \text{ nm}} = 1.517 \times 10^{-19}$$

External differential quantum efficiency is

$$\eta_{EDQE} = \frac{dP_0/hf}{dI/e} = \frac{P_{02} - P_{01}}{I_2 - I_1} \times \frac{e}{E_g}$$

$$\Rightarrow \eta_{\text{EDQE}} = \frac{(4-3)\text{mW}}{(45-40)\text{mA}} \times \frac{1.602 \times 10^{-19}}{1.517 \times 10^{-19}}$$

$$\Rightarrow \eta_{\text{EDQE}} = 21.12\%$$

External power efficiency is

$$\eta_{\text{EPE}} = \frac{P_0}{P} \times 100 = \frac{P_0}{I \times V} = \frac{3\text{mW}}{40\text{mA} \times 1.4} = 5.3\%$$

External quantum efficiency is

$$\eta_{\text{EQE}} = \frac{P_0/hf}{Ie} = \frac{e \cdot P_0}{I(E_g)} = \frac{1.602 \times 10^{-19} \times 3 \times 10^{-3}}{1.57 \times 10^{-19} \times 40 \times 10^{-3}} = 7.9\%$$

PROBLEM 10

The total efficiency of an injection laser with a GaAs active region is 22%. The voltage applied to the device is 2.7 V and the bandgap energy for GaAs is 1.43 eV. Calculate the external power efficiency of the device.

Solution: We have

$$\eta_{\text{EPE}} = \frac{\eta_{\text{EQE}} E_g}{eV}$$

$$\Rightarrow \eta_{\text{EPE}} = \frac{0.22 \times 1.43}{2.7} \times 100 = 11.65\%$$

PROBLEM 11

For an AlGaAs-based heterostructure laser diode having cavity length of 200 μm , radiation wavelength of 870 nm, and refractive index of GaAs is about 3.7, calculate the mode integer q of the peak radiation and the separation between the modes of cavity. If the optical gain bandwidth is 6 nm, how many modes are there within this bandwidth?

Solution: Given, $L = 200 \mu\text{m}$, $\lambda = 870 \text{ nm}$, optical gain BW = 6 nm. For mode integer of peak radiation we have

$$L = \frac{\lambda q}{2n}$$

Therefore

$$q = \frac{2Ln}{\lambda} = \frac{2 \times 3.7 \times 200 \times 10^{-6}}{870 \times 10^{-9}} = 1701.1$$

Different modes are separated by $\delta\lambda$, given by

$$\delta\lambda = \frac{\lambda^2}{2nL} = \frac{(870 \times 10^{-9})^2}{2 \times 3.7 \times 200 \times 10^{-6}} = 0.511 \text{ nm}$$

The number of modes is

$$\text{No. of modes} = \frac{\text{Optical gain BW}}{\delta\lambda} = \frac{6 \text{ nm}}{0.511 \text{ nm}} \approx 11$$

Therefore, maximum 11 modes are there in the cavity.

PROBLEM 12

For a DFB laser having corrugation period Λ of 220 nm, a grating length of 400 μm , and the effective refractive index of medium 3.5, calculate the Bragg wavelength and mode wavelength assuming first-order grating.

Solution: Given: corrugation period $\Lambda = 220 \text{ nm}$, grating length $L = 400 \mu\text{m}$, effective refractive index of medium $n = 3.5$, Bragg wavelength $\lambda_B = ?$

We have

$$2\Lambda = q \frac{\lambda_B}{n}$$

Therefore

$$\lambda_B = \frac{2\Lambda n}{q} = \frac{2 \times 220 \times 10^{-9} \times 3.5}{1} = 1540 \text{ nm}$$

The symmetric mode wavelengths about λ_B are

$$\begin{aligned} \lambda_q &= \lambda_B \pm \frac{\lambda_B^2}{2nL}(q+1) \\ &= 1540 \pm \frac{(1540 \text{ nm})^2}{2 \times 3.5 \times 400 \times 10^{-6}}(0+1) \end{aligned}$$

So $q = 0$ mode wavelengths are

$$\lambda_0 = 1539 \text{ nm or } 1541 \text{ nm}$$

SUMMARY

1. LEDs are fabricated by epitaxially growing doped semiconductor layers on a suitable substrate such as GaAs or GaP.
2. The internal quantum efficiency of an LED is the ratio of photons generated to injected electrons.
3. The external quantum efficiency of an LED is defined as the ratio of photons emitted from the LED to the number of internally generated photons.
4. An LED that is realized using two differently doped semiconductors that are of the same material is called a homojunction. When LEDs are realized using different bandgap materials, they are called heterostructure devices. A heterostructure LED is brighter than a homojunction LED.
5. The geometry and internal structure of an LED play an important role in emitting light from its surface.

6. A laser consists of an active laser material, a source of excitation energy, and a resonator or feedback mechanism.
7. The principle of semiconductor laser is based on external pumping and internal light amplification.
8. To excite or pump more atoms to higher energy level, an external pumping energy source is used. There are two types of pumping, electrical and optical. If an AC or DC electrical discharge is used, it is called electrical pumping. If an optical source is used, it is optical pumping.
9. In most lasers, a Fabry–Perot cavity having a pair of mirrors, one at each end of the laser, is used as a resonator.
10. LEDs are used in lower data-rate (few hundreds of megahertz), shorter distance multimode systems because of their bandwidth limitations and lower output power.
11. Laser diodes (LD) are used in long-haul communication where higher data rates (GHz or even THz) are required.
12. Optical connectors are demountable and are used to connect fiber-optic cable to peripheral equipment and to other fibers. If a permanent connection is required then another method with much less loss is splicing.

MULTIPLE-CHOICE QUESTIONS

1. Operating wavelength of GaAlAs LEDs and lasers are
 - (a) 820 and 850 nm
 - (b) 665 nm
 - (c) 1300 nm
 - (d) 1550 nm
 - (e) None of the above
2. Light emission from an LED is modulated by
 - (a) Applied voltage
 - (b) Drive current
 - (c) Illumination
 - (d) All of the above
3. Which of the following material is not suitable for making LED?
 - (a) GaAs
 - (b) Silicon
 - (c) InGaAsP
 - (d) GaAlAs
4. In LED, which of the following factors affects the efficiency of the diode most severely and cannot be eliminated even in principle?
 - (a) Total internal reflection
 - (b) Absorption
 - (c) Back emission
 - (d) Fresnel reflection
5. The material for making an efficient LED should be
 - (a) A metal
 - (b) An insulator
 - (c) A direct bandgap type semiconductor
 - (d) An indirect bandgap type semiconductor
6. The color of an LED can be changed by
 - (a) Changing the doping level of the semiconductor
 - (b) Changing applied voltage
 - (c) Using different bandgap semiconductor
 - (d) None of the above
7. The difference between laser and LEDs is
 - (a) Semiconductor material
 - (b) Structure of device that confine light and electrical current to produce lase action
 - (c) Higher drive current used for laser
 - (d) Both (b) and (c)
8. Which of the following material pairs are suitable for making a heterojunction?
 - (a) GaAs and AlAs
 - (b) GaAs and GaP
 - (c) GaAs and Ge
 - (d) GaAs and GaAlAs
9. Spectral width of LED is typically
 - (a) 1–20 nm
 - (b) 30–50 nm
 - (c) 100–200 nm
 - (d) Greater than 200 nm
10. Spectral width of laser is typically
 - (a) 1–3 nm
 - (b) 0.1–0.5 nm
 - (c) 3–10 nm
 - (d) 10–30 nm

REVIEW QUESTIONS

1. Enlist the requirements of LED materials.
2. What are direct and indirect bandgap semiconductor materials? Which of these are more suitable for fabricating LEDs? Why?
3. With neat sketches explain LED characteristics and derive the mathematical expression for internal and external quantum efficiencies.
4. Explain the basic principle of operation of a LASER.
5. Explain the terms absorption, spontaneous emission, and stimulated emission of radiation.
6. What do you mean by population inversion in connection with "LASER"?
7. What are the main components of a LASER?
8. Write two important characteristics of a laser beam.
9. What do mean by optical cavity?
10. What do you mean by the efficiency of a laser?
11. What is an optical resonator?
12. Draw schematic diagram of a Fabry–Perot resonator.
13. How do spontaneous and stimulated emissions occur?
14. Explain the principle of working of a semiconductor LASER using suitable diagram.
15. Explain the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from device.
16. What is meant by optical confinement in a laser? How is it achieved? Explain.
17. What are the major drawbacks of homostructure LED and what are the major advantages of a heterostructure LED over a homostructure one?
18. Briefly describe what is meant by the following terms when they are used in relation to injection LASER.
 - (a) Relaxation oscillation.
 - (b) Mode hopping.
19. What do you understand by splicing? Explain types of splices and steps involved in splicing fiber.
20. Explain the losses caused by longitudinal lateral and angular displacement in splicing of fiber?

EXERCISES

1. For a double heterojunction InGaAsP LED emitting peak wavelength of 1310 nm and drive current of 50 mA, determine the total carrier recombination lifetime and power internally generated within the device. The radiative and nonradiative recombination lifetimes of the minority carriers are 30 ns and 100 ns, respectively.
2. Calculate its external quantum efficiency of LED fabricated from material having a refractive index of 3.4.
3. A GaAs LED is forward-biased with a current of 130 mA and a voltage of 1.5 V. Each emitted photon possesses energy of 1.43 eV, and the refractive index of GaAs is 3.7. Assuming the internal quantum efficiency of the LED to be 60%, calculate the external power efficiency.
4. For a GaAs planar LED having refractive index of 3.3, calculate optical power emitted by the LED when the optical power generated internally is 50 mW.
5. The light output from GaAs LED is coupled into a step index fiber having numerical aperture of 0.25 and core refractive index of 1.44. Calculate coupling efficiency and optical power loss in dB, relative to the power emitted from LED, when coupling the light output into the fiber.
6. For an InGaAsP-InP based laser diode having cavity length of 270 μm , radiation wavelength of 1550 nm and refractive index of InGaAsP is 4, calculate the mode integer q of the peak radiation and the separation between the modes of cavity. If the optical gain bandwidth is 2 nm how many modes are there within this bandwidth? Also calculate the number of modes in the cavity.
7. A GaAs laser operating at 850 nm has a 400 μm length and a refractive index $n = 3.7$. What are the frequency and wavelength spacings?

8. A double heterostructure edge-emitting Fabry–Perot AlGaAs laser is $300\text{ }\mu\text{m}$ long, which emits at 900 nm and has a refractive index of 4.3. Find the number of lasing modes and spacing between them.
9. A double-heterojunction InGaAsP LED emitting at a peak wavelength of 1550 nm has radiative and non-radiative recombination times of 30 and 90 ns, respectively. The drive current is 40 mA.
 - (a) Find the internal quantum efficiency and the internal power level.
 - (b) If the refractive index of the light source material is $n = 4$, find the power emitted from the device.
10. Calculate the mode integer q of the peak radiation, the separation between the modes of the cavity, and number of modes in the cavity for an InGaAsP–InP laser diode. It has an optical cavity of length $200\text{ }\mu\text{m}$, the peak radiation is at 1550 nm , and the refractive index of InGaAsP is 4.0. The optical gain bandwidth is 3 nm .
11. If the crystal length is 7 cm and refractive index is 1.78, determine the number of modes and their frequency of separation in a ruby laser having peak emission wavelength of 570 nm .
12. Calculate external quantum efficiency, external differential quantum efficiency, and external power efficiency of an InGaAsP laser diode operating at $\lambda = 1350\text{ nm}$, $I = 40\text{ mA}$, the output power is 4 mW , and the voltage across the diode is 1.5 V . If the diode current is increased to 50 mA , the optical output power increases to 6 mW .
13. Calculate the external power efficiency of an injection laser. The total efficiency of the device with a GaAs active region is 30%. The voltage applied to the device is 4.2 V and the bandgap energy for GaAs is 1.43 eV .
14. Calculate the mode integer q of the peak radiation and the separation between the modes of cavity for an AlGaAs-based heterostructure laser diode having cavity length of $300\text{ }\mu\text{m}$, radiation wavelength of 820 nm , and refractive index of GaAs is about 3.7. If the optical gain bandwidth is 5 nm , how many modes are there within this bandwidth?
15. Calculate the Bragg wavelength and mode wavelength for a DFB laser having corrugation period Λ of 230 nm , a grating length of $420\text{ }\mu\text{m}$, and the effective refractive index of medium 3.35, assuming first-order grating.

ANSWERS

Multiple-Choice Questions

- | | |
|--------|---------|
| 1. (a) | 6. (c) |
| 2. (b) | 7. (d) |
| 3. (b) | 8. (d) |
| 4. (a) | 9. (b) |
| 5. (c) | 10. (a) |

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5 | Optical Receivers

Learning Objectives

After completing this chapter, you will be able to:

- Study mechanism of electron–hole pair (EHP) generation in semiconductors by photon absorption.
 - Gain familiarity with performance parameters such as quantum efficiency, responsivity, and long-wavelength cutoff.
 - Study the different types of photodiodes.
 - Overview avalanche photodiodes (APDs) for internal current gain.
 - Understand noise properties of the photodetection process in photodiodes.
 - Analyze optical receivers and study their blocks.
 - Study different receiver amplifier configurations.
-

A photodetector converts electromagnetic radiation into an electronic signal whose amplitude is proportional to the incident light intensity. Photodiodes are fabricated from semiconductor materials. The most commonly used materials are silicon (Si), gallium arsenide (GaAs), indium antimonide (InSb), indium arsenide (InAs), lead selenide (PbSe), and lead sulfide (PbS). These materials absorb light over a wavelength range of 250–1,100 nm for silicon and 800–2000 nm for GaAs. The improvement in optical receiver and transmitter sides is more desirable because fewer repeaters will be needed and the spacing between repeaters can be increased.

The following performance criteria of optical detectors are important for applications in optical communication systems:

1. Sensitivity has to be matched to the emission spectra of the optical transmitter.
2. There should be a linear relationship between the intensity of the incident signal and the electrical output signal of the photodetector.
3. High quantum efficiency/high spectral sensitivity.
4. Fast response time to obtain high data rate operation.
5. Stability of performance irrespective of ambient conditions, such as temperature.
6. Reliability and robustness.
7. Long life.
8. Low noise.
9. Low cost.

5.1 Photodiode

Figure 5.1 shows the cross-section of a photodiode. A pn junction is formed by the p-layer material at the active surface and the n-layer material at the substrate. This structure operates as a photodetector. When a photon of light is absorbed, it excites an electron and produces a single pair of charge carriers, an electron and a hole, where

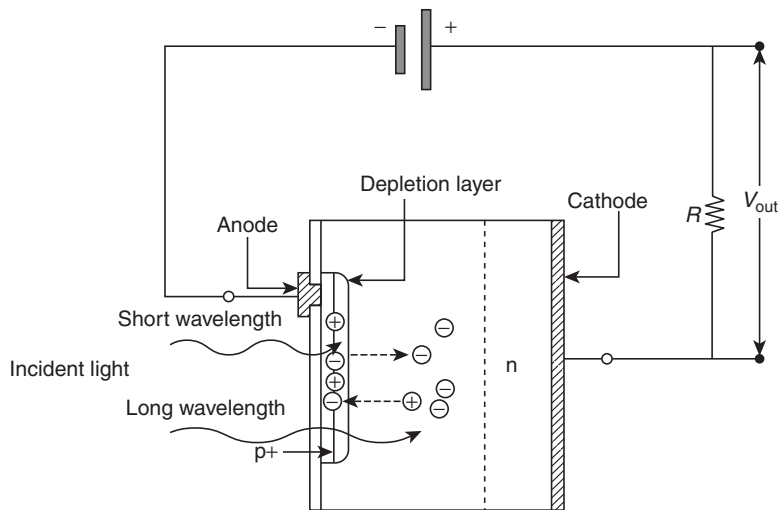


Figure 5.1 | Reverse-biased pn junction photodiode.

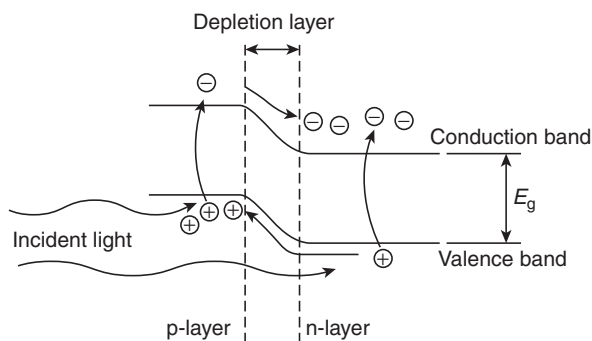


Figure 5.2 | Energy band diagram of the reverse-biased pn junction.

a hole is simply the absence of an electron in the semiconductor lattice. If the light energy is greater than the bandgap energy E_g , the electrons are pulled up into the conduction band, leaving holes in their place in valence band. These electron–hole pairs (EHPs) occur throughout the p-layer, depletion layer, and n-layer materials. In the depletion region, the electric field accelerates these electrons toward the n-layer and holes toward the p-layer. In this manner, EHPs in n- and p-layers, which are generated in proportion to the amount of incident light, are collected. If the external circuit is connected, electrons will flow away from the p-layer and holes will flow away from n-layer toward the respective electrodes as shown in Fig. 5.2. These electrons and holes generate current flow in the semiconductor. The trick in a photodiode is to collect the photon-induced charge carriers as current or voltage at the electrodes, before they have a chance to recombine.

The depletion region must be sufficiently thick to allow a large fraction of incident light to be absorbed in order to achieve maximum carrier–pair generation. Since long carrier drift times in the depletion region restrict the speed of operation of the photodiode, it is necessary to limit its width. Thus, there is a tradeoff between the number of photons absorbed and the speed of response.

5.1.1 Absorption Coefficient

The absorption of photons of a specific wavelength in a photodiode produces the EHPs, and the photocurrent produced depends on the absorption coefficient α of the semiconductor for that particular wavelength. For this process, the energy of the incident photon should be at least equal to the bandgap energy E_g of the semiconductor material to excite an electron from the valence band to the conduction band. Therefore, the upper or long cutoff wavelength λ_c for photon-generative absorption depends on bandgap energy. It is given as

$$\lambda_c (\mu\text{m}) = \frac{1.24}{E_g (\text{eV})} \quad (5.1)$$

At a specific wavelength the photocurrent I_{photo} produced by incident optical power P_{in} is given by

$$I_{\text{photo}} = \frac{P_{\text{in}} e(1-r)}{hf} [1 - e^{-\alpha_0 d}] \quad (5.2)$$

where e is the charge on electron, r is the Fresnel reflection coefficient at semiconductor–air interface, and d is the width of absorption region. Absorption coefficient is a wavelength-dependent parameter as shown in Fig. 5.3. The incident photon having wavelength shorter than λ_g gets absorbed as it travels in the semiconductor and the light intensity decays exponentially with distance. The light intensity at distance x from the semiconductor surface is given by

$$I(x) = I_0 e^{-\alpha x} \quad (5.3)$$

where I_0 is the intensity of incident radiation and α is the absorption coefficient.

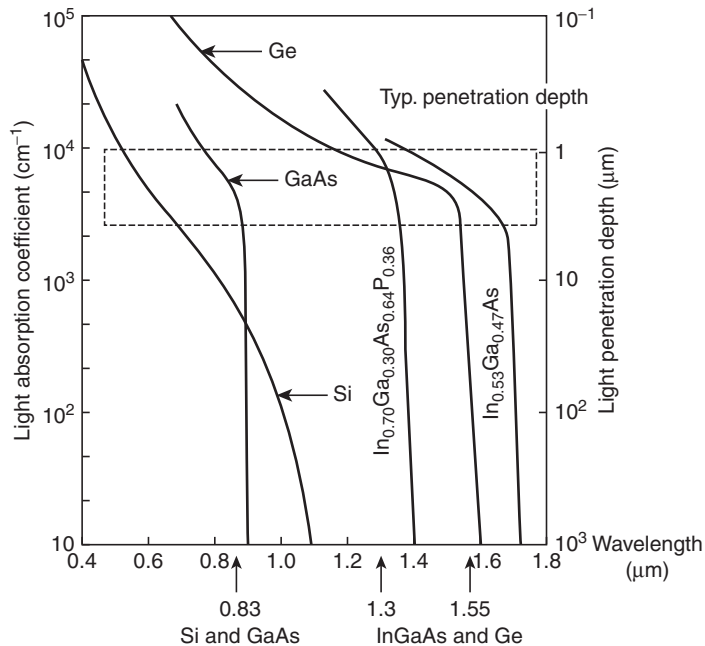


Figure 5.3 | Optical absorption curve for different semiconductor materials.^[1,2]

5.1.2 Direct and Indirect Absorption

In direct bandgap semiconductors such as III–V semiconductors (e.g., GaAs, InAs, InP, GaSb and in their alloys, e.g., InGaAs, GaAsSb), the photon absorption process is a direct process that requires no assistance from lattice vibrations. The photon is absorbed and the electron is directly moved from the valence band to the conduction band without any change in crystal momentum [see Fig. 5.4(a)].

In indirect bandgap semiconductors such as Si and Ge, the assistance of phonon is required to keep the energy and momentum conserved as shown in Fig. 5.4(b). This reduces the transition probability as compared to direct absorption where no phonon is involved. The cutoff wavelength is not as sharp as direct bandgap semiconductor.^[3]

Ideally the bandgap energy of photodiode material should be slightly less than the photon energy corresponding to longest operating wavelength of the system. This ensures high absorption coefficient and good response. The bandgap energy for direct absorption in silicon is 4.10 eV (Table 5.1), corresponding to a threshold of 0.30 μm in ultraviolet range, which is well outside the wavelength range of interest. For germanium, the threshold for direct absorption is 0.81 eV, corresponding to 1.53 μm , that is, it strongly absorbs below 1.53 μm . Thus, germanium can be used in the fabrication of a photodetector which is to be operated in the required wavelength range. However, the major disadvantage of germanium photodiodes is large dark current due to narrow bandgap as compared to other semiconductor materials.

5.1.3 Photodetector Materials

Sensitivity of photodiodes is wavelength-dependent as well as material-dependent (as shown in Fig. 5.5).

1. **Short-wave band (800–900 nm):** Silicon PIN diodes are used for the short-wave band. They are very inexpensive, reliable, and easy to handle.
2. **Medium-wave band (1,250–1,350 nm):** In this band, germanium and different compound semiconductors are used. Germanium can be used up to 1,600 nm, because it has a lower bandgap energy of 0.67 eV. Indium gallium arsenide phosphide (InGaAsP) has an optical bandgap of 0.89 eV, therefore it is suitable for the medium wavelength band.

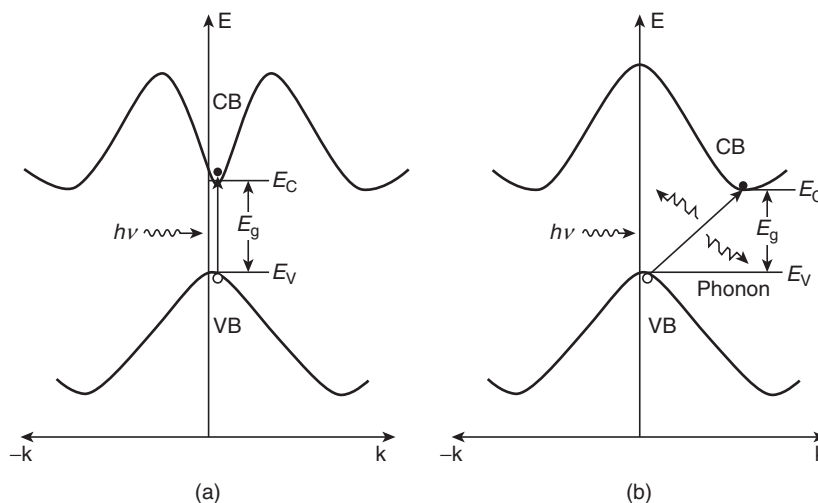


Figure 5.4 | Photon absorption: (a) GaAs (direct bandgap); (b) Si (indirect bandgap).

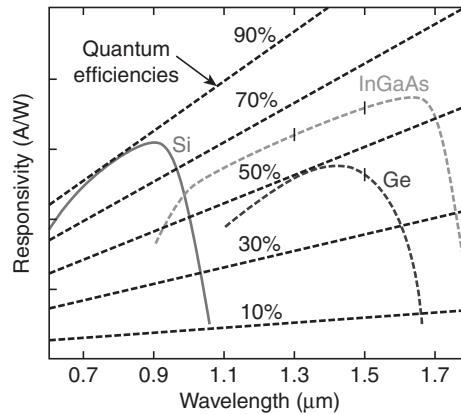


Figure 5.5 | Sensitivity of PIN diodes for various materials.^[4]

3. **Long-wave band (1,500–1,600 nm):** For long-wave band operation, the optical bandgap of the material should be very small. InGaAs has bandgap energy of 0.77 eV. However, at room temperature, a large number of carriers are excited and so a heterostructure of InGaAs is used to solve this problem.

The direct bandgap III–V alloys are superior as compared to germanium, because their bandgaps can be tailored to the desired wavelength by changing the relative concentrations of their constituents. Higher speeds of operation can be obtained by using their heterojunction structures. The ternary alloys listed in Table 5.2 are used to obtain longer wavelength operation.

Table 5.1 | Photodiode materials and their bandgaps^[2]

Photodiode Semiconductor Material	Indirect Bandgap (eV)	Direct Bandgap (eV)
Si	1.14	4.10
Ge	0.67	0.81
GaAs	–	1.43
InAs	–	0.35
InP	–	1.35
GaSb	–	0.73
InGaAs	–	0.75

Table 5.2 | Photodiode materials with their application area

Photodiode Semiconductor Material	Application
<i>Indirect bandgap materials</i>	
Si	Used for avalanche photodiodes for LAN applications
Ge	Used for avalanche photodiodes for LAN applications and long-distance communications

(continued)

Table 5.2 | (continued)

Photodiode Semiconductor Material	Application
<i>Direct bandgap materials</i>	
InGaAs	Excellent material for long-haul communications, 1.55 μm
AlGaSb	Excellent optical properties, used for long-haul communications
InGaAsP	Suitable for both 1.55- and 1.3- μm applications for long-haul applications
HgCdTe	Excellent material for long-wavelength applications in night vision and thermal imaging ^[5]
AlGaIn	UV photon counting, chemical and biological agent spectroscopy and threat detection, non-line-of-sight communications, flame detection and monitoring, environmental monitoring, astronomy, early missile threat warning and covert space-to-space communications ^[6]

5.1.4 Quantum Efficiency

To quantify the photon absorption ability, a parameter called the *quantum efficiency* (η) is used. It is the ratio of the number of excited EHPs to the total number of incident photons:

$$\eta = \frac{\text{Number of excited EHPs}}{\text{Total number of incident photons}} \quad (5.4)$$

Therefore,

$$\eta = \frac{\text{Number of EHPs per second}}{\text{Total number of incident photons per second}} = \frac{r_{\text{EHP}}}{r_{\text{photon}}} \quad (5.5)$$

Including the reflection at the device interface, the quantum efficiency is given by

$$\eta = (1 - R)(1 - e^{-\alpha d}) \quad (5.6)$$

where d is the thickness of absorption thickness, α is the absorption coefficient, and R is the reflectivity of front interface. Equation (5.6) shows that quantum efficiency can be increased by having

1. Smaller reflectivity R .
2. A larger absorption coefficient α .
3. A larger thickness d .

5.1.5 Responsivity

The responsivity R of a photodetector is defined as the output photocurrent per unit incident optical power:

$$R = \frac{I_{\text{photo}}}{P_{\text{in}}} \text{ A/W} \quad (5.7)$$

Responsivity gives the transfer characteristic of the detector. The relation between responsivity and quantum efficiency can be obtained as follows: The energy of each photon is $E = hf$. Thus, incident photon rate is

$$r_{\text{photon}} = \frac{P_{\text{in}}}{hf} \quad (5.8)$$

From Eq. (5.5) we can write

$$r_{\text{EHP}} = \eta \times r_{\text{photon}} \quad (5.9)$$

$$\Rightarrow r_{\text{EHP}} = \frac{\eta P_{\text{in}}}{hf} \quad (5.10)$$

Therefore, the output photocurrent is

$$I_{\text{photo}} = \frac{\eta P_{\text{in}} e}{hf} \quad (5.11)$$

where e is the charge on an electron. Thus, responsivity R is

$$R = \frac{\eta e}{hf} = \frac{\eta \lambda}{1.24} \text{ A/W} \quad (5.12)$$

Frequency f of the incident photon is

$$f = \frac{c}{\lambda}$$

Therefore,

$$R = \frac{\eta e \lambda}{hc} \quad (5.13)$$

In general, responsivity increases with increase in wavelength, but because of wavelength cutoff condition, responsivity has a sudden drop as shown in Fig. 5.6.

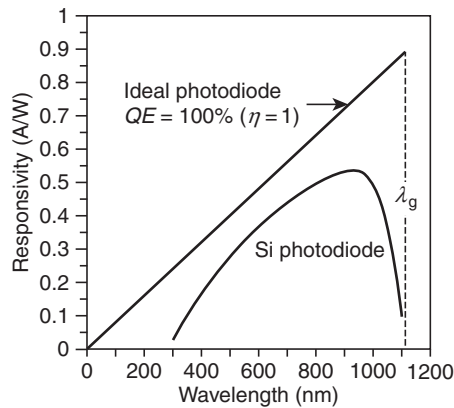


Figure 5.6 | Responsivity against wavelength curve for ideal and typical silicon photodiodes.

5.1.6 Long-Wavelength Cutoff

In an intrinsic absorption process, the energy of incident photons must be greater than or equal to the bandgap energy (E_g) of photodetector fabrication material. Thus, the photon energy

$$\frac{hc}{\lambda} \geq E_g$$

We can write

$$\lambda \leq \frac{hc}{E_g}$$

Thus, there is a long wavelength cutoff λ_c , above which photons are not absorbed by the semiconductor. It is given as

$$\lambda_c = \frac{hc}{E_g}$$

where $h = 6.626 \times 10^{-34}$ Js, $c = 3 \times 10^8$ m/s, $1 \text{ eV} = 1.6 \times 10^{-19}$ J. Therefore,

$$\lambda_c (\mu\text{m}) = \frac{1.24}{E_g (\text{eV})}$$

5.2 Types of Photodiodes

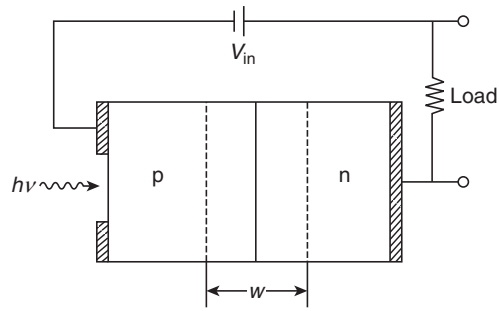
5.2.1 pn Photodiode

It is the simplest structure and is always operated in reverse-bias condition. The depletion region is formed as shown in Fig. 5.7. The width of the depletion region depends on the doping concentration. Wider depletion regions can be obtained by having lower doping concentration. Incident photons of energy $h\nu$ are absorbed inside the depletion region as well as the diffusion region. The photons absorbed within the depletion region generate EHPs and drift under the influence of the electric field. The resulting flow of photocurrent constitutes the response of the photodiode to the incident optical power. The response time is governed by the transit time τ_{drift} . It is given as

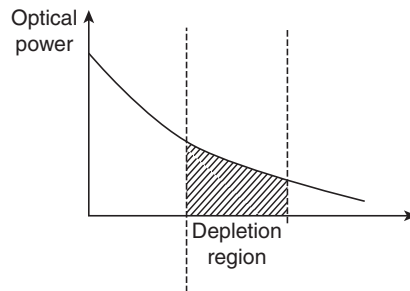
$$\tau_{\text{drift}} = \frac{w}{v_{\text{drift}}} \quad (5.14)$$

where w is the depletion region width and v_{drift} is the average drift velocity.

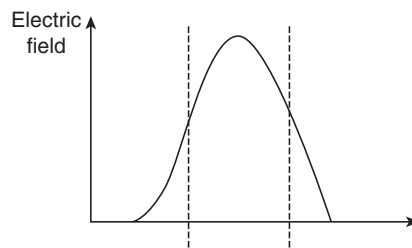
The structure of a pn photodiode is shown in Fig. 5.8. The incident photons are also absorbed outside the depletion region. The electrons generated in the p-region have to diffuse to the depletion region before they can drift to the n-region under the built-in electric field. Similarly, the holes generated in the n-region have to diffuse to the depletion region for their drift toward the p-region. The diffusion process is slow as compared to drift process and hence limits the response of the photodiode. Therefore, most of the photons should be absorbed in the depletion region; hence, it is made as long as possible by decreasing the doping in the n-region. This width is normally 1–3 μm . The detection wavelength range of silicon photodiode is 400–700 nm and for germanium it is 700–900 nm.



(a)



(b)



(c)

Figure 5.7 (a) Structure of pn photodiode in reverse-bias condition. (b) Optical power variation. (c) Electric field variation.

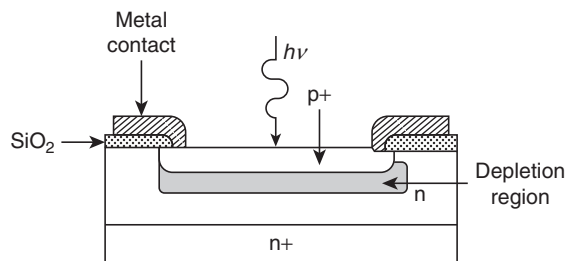


Figure 5.8 A pn photodiode.

5.2.2 PIN Photodiode

The diffusion component of a pn photodiode can be reduced by decreasing the widths of the p-region and the n-region and increasing the width of the depletion region. By this, most of the incident photons are absorbed inside it as shown in Fig. 5.9. It also allows longer wavelength operation, where the light penetrates more deeply into the wider depletion region, typically 5–50 μm . The n-material is very lightly doped, so that it can be considered as intrinsic region and inserted between p- and n-regions. This creates the PIN structure (Fig. 5.10). The middle layer offers very high resistance, hence most of the voltage drop occurs across it. Most of the incident photons are absorbed inside the intrinsic region. Therefore, the drift component of photocurrent dominates when compared to the diffusion component.

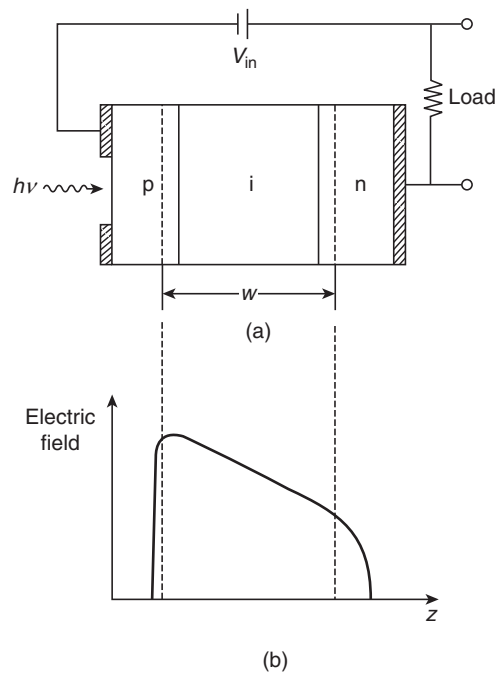


Figure 5.9 | (a) Structure of PIN photodiode in reverse-bias condition. (b) Electric field variation.

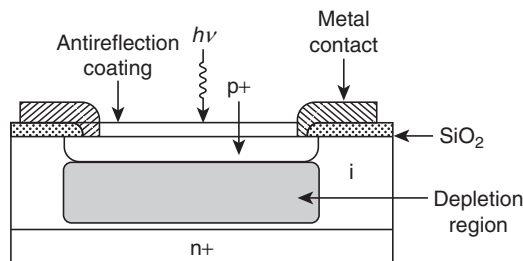


Figure 5.10 | A PIN photodiode.

The two very thin layers of negative and positive charges – separated by intrinsic Si of width w – form a structure similar to parallel-plate capacitor. The junction or depletion layer capacitance of a PIN diode is given by

$$C_{\text{depletion}} = \frac{\epsilon_s A}{w} \quad (5.15)$$

where A is the cross-sectional area, ϵ_s is the permittivity of the semiconductor, and w is the width of intrinsic semiconductor.

The typical value of this depletion capacitance is of the order of picofarads. The PIN structure is designed so that the photon absorption occurs within intrinsic Si-layer. The photo-generated EHPs are separated by the field E and drifted toward the n+ and p+ regions. This gives rise to external photocurrent. The photocurrent is detected as a voltage across a small load resistor.

The response time of a PIN photodiode is determined by three main factors. These are explained below.

1. Drift time of carriers through the depletion region: The response time of a PIN photodiode is determined by the transit times of photo-generated carriers across the width w of the intrinsic Si layer. Increasing w allows more photons to be absorbed and thus increases the quantum efficiency, but it slows down the speed of response as carriers transit time becomes longer. The transit time or drift time for charge carriers generated at the edge of the intrinsic Si layer is

$$\tau_{\text{drift}} = \frac{w}{v_{\text{drift}}} \quad (5.16)$$

where w is the depletion region width and v_{drift} is the average drift velocity. To increase the speed, drift time has to be reduced. This is achieved by increasing v_{drift} . Hence we have to apply more field E . When the field in the depletion region exceeds a saturation value, then the carriers may be assumed to travel at a constant drift velocity.

2. Diffusion time of carriers generated outside the depletion region: Carrier diffusion is a slow process. It is given as

$$\tau_{\text{diffusion}} = \frac{d^2}{2D_C} \quad (5.17)$$

where D_C is the minority carrier diffusion coefficient and d is the diffusion distance traveled by carriers.

3. Time constant due to depletion capacitance and load resistor: The reverse-biased photodiode exhibits a voltage-dependent capacitance. The junction or depletion layer capacitance of a PIN diode is given by

$$C_{\text{depletion}} = \frac{\epsilon_s A}{w} \quad (5.18)$$

where A is the cross-sectional area, ϵ_s is the permittivity of the semiconductor, and w is the width of intrinsic semiconductor.

All the above factors affect the response time of the photodiode. The bandwidth of device is mainly limited by the drift time of carriers through the depletion region. If it is assumed that no carriers are generated outside the depletion region and depletion capacitance is negligible, then the maximum 3-dB bandwidth is given by^[7]

$$B_{\text{max}} = \frac{1}{2\pi\tau_{\text{drift}}} = \frac{v_{\text{drift}}}{2\pi w} \quad (5.19)$$

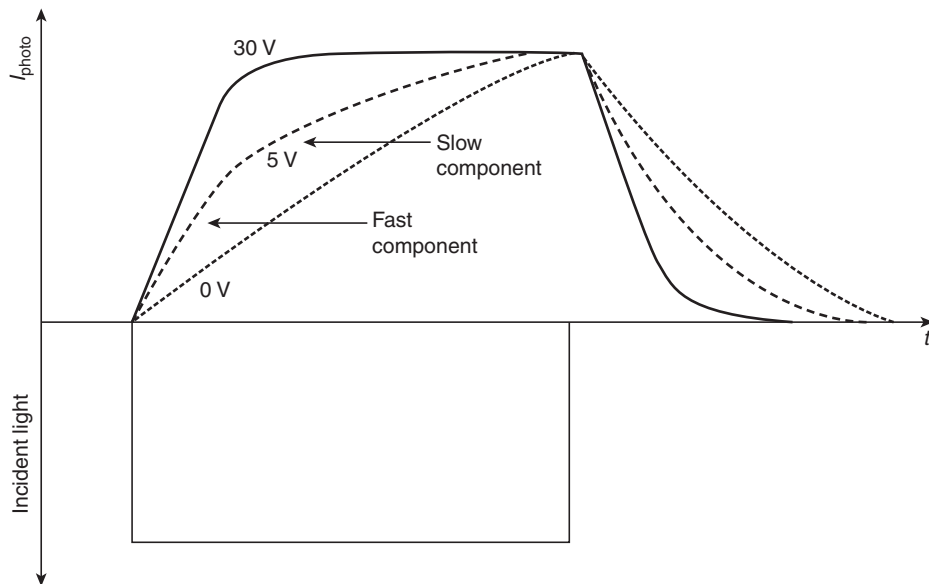


Figure 5.11 | Transient response of a photodiode to a square pulse of radiation.

Therefore,

$$\text{Maximum response time} = \frac{1}{B_{\max}} \quad (5.20)$$

In many applications, the dynamic performance of the device is very important. The photodiode response time is the quadrature sum of the charge collection time and the time constant of RC circuit $\tau = R \times C$, where R is the circuit resistance and C is the circuit capacitance.

Charge collection time is made up of a fast and a slow component and is voltage-dependent. The fast component is produced by travel of electrons and holes through the depletion region under the influence of the electric field. Photons absorbed outside the depletion region produce carriers that are collected by diffusion. The response time of these carriers is relatively slow. The RC time constant effect is due to series resistance plus load resistances and the junction plus stray capacitances. Figure 5.11 illustrates the transient response of a photodiode to a square pulse of radiation.

In the unbiased condition, the diffusion component of photodiode is significant and the rise time is much larger, approximately $0.5 \mu\text{s}$ depending upon the operating wavelength. For a fast response time, majority of the carriers should be generated within depletion layer, which can be achieved by proper selection of silicon resistivity and operating voltage. Transit time depends on both electron and hole drift velocities. Figure 5.12 illustrates the way drift velocity increases with electric field. Except at very high fields, the electron velocity is 2.5 times that of holes. Hence, hole transit time tends to dominate.

As the operating wavelength increases, the depletion depth necessary for full absorption increases. Therefore transit time also increases. Hence it becomes difficult to achieve rise times faster than 15–20 ns at 1,064 nm, whereas below 900 nm, rise time as small as 2 ns are also available.

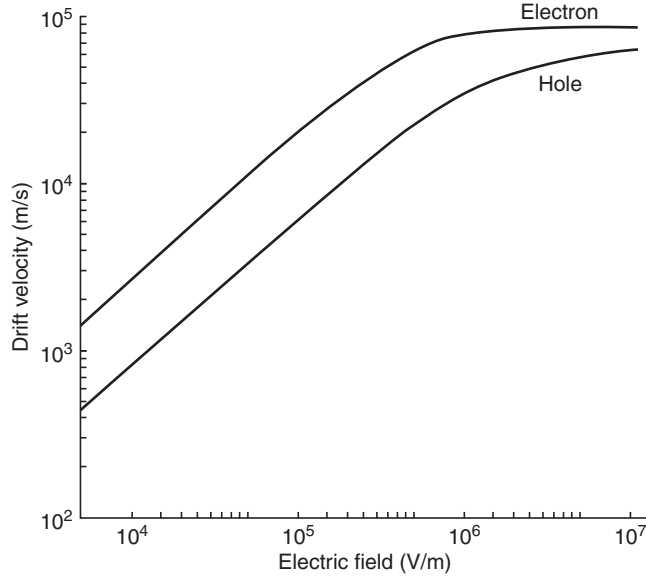


Figure 5.12 | Drift velocity versus electric field for holes and electrons.

5.2.2.1 Rise Time

It is measured by applying a step light input to the photodiode and is defined as the time required for the output to change from 10% to 90% of the steady output level. The rise time depends on the incident light wavelength and load resistance as follows:

$$t_r = \frac{0.35}{f_c} \quad (5.21)$$

where f_c is the cutoff frequency.

5.2.3 Avalanche Photodiode

One limitation of the PIN photodiode is the lack of internal gain. Incident photon produces only one EHP. For many years, the solution of optical amplification was the photomultiplier tube (PMT). This gives high gain, but there are few shortcomings, such as, it is a bulky vacuum tube and it generates heat. As compared to a photodiode, it offers limited linearity, a narrow spectral response range, and a low quantum efficiency of less than 25%. The avalanche photodiode (APD) is a solid-state efficient alternative for PMT.

In an APD, similar to photodiodes, the incident photons produce EHPs. However, the APD is operated with a large reverse bias, which creates the high-field region within the device. This accelerates the photon-generated electrons, which acquire so much kinetic energy that they collide on bound electrons in the valence band and create secondary hole–electron pairs. This process is called as **impact ionization**. If the field created is high enough, these secondary carrier pairs create new EHPs. This is called the **avalanche effect**. Thus, the carriers get multiplied, which results in a large photocurrent.

A simplified schematic of silicon reach-through APD is shown in Fig. 5.13. The n+ layer is a thin layer; the photons to be detected fall on this layer through the window. There are three p-type layers of different doping levels next to the n+ layer. The first is a thin p-type layer and second layer is a thick, lightly doped p-type layer, which is almost intrinsic. The third layer is a heavily doped p-layer. Under no-bias condition, the depletion layer

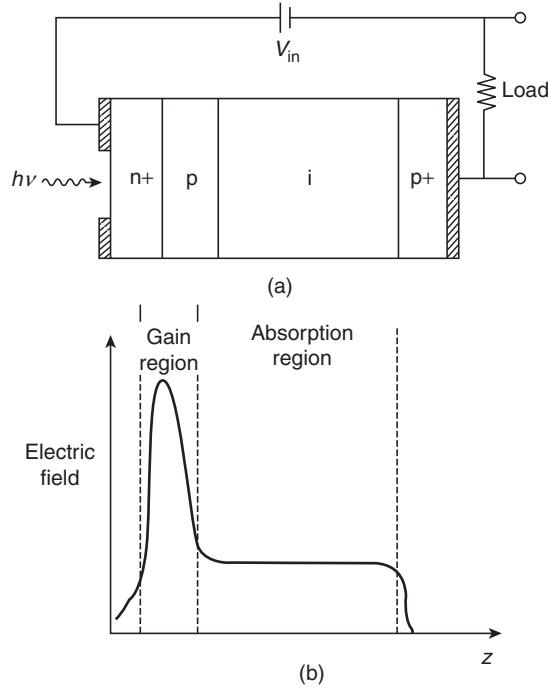


Figure 5.13 (a) Structure of silicon reach-through avalanche photodiode (RAPD) in reverse-bias condition. (b) Electric field variation.

does not extend across the intrinsic layer. The diode is reverse-biased to increase the field in the depletion region. If a sufficient reverse-bias is applied, the depletion in the p-layer widens to reach through to the intrinsic layer; hence it is named as **reach-through APD**.

The electric field distribution across the junction is shown in Fig. 5.13(b). It is maximum at n+p junction, then decreases slowly through the p-layer and the field vanishes at the end of narrow depletion layer in the p+ layer. The nearly uniform field in intrinsic region separates the hole–electron pairs and drifts them toward the n+ and p+ layers. When these electrons reach the p-layer, they experience greater electric field. They acquire sufficient kinetic energy, greater than E_g , to impact ionize some of the Si covalent bonds and release another EHP. These electrons further cause another impact ionization and release more EHPs. This chain multiplication process leads to an avalanche of impact ionization process. Thus, one electron entering the p-layer can generate more number of EHPs, which contribute to the multiplied photocurrent. Hence we can say that this photodiode possesses internal gain mechanism.

Thus, the photo-generation takes place in the intrinsic region, whereas avalanche multiplication takes place in the p-region. The avalanche multiplication is a statistical process and leads to carrier-generation fluctuation that leads to excess noise in the avalanche multiplied photocurrent.

The avalanche multiplication factor M of an APD is defined as

$$M = \frac{\text{Multiplied output photocurrent}}{\text{Initial or primary unmultiplied photocurrent}} = \frac{I_{\text{output}}}{I_{\text{photo}}} \quad (5.22)$$

5.3 Photodetector Noise

The different figures of merit used to assess the noise performance of optical detectors are as follows:

1. Noise equivalent power (NEP).
2. Detectivity (D).
3. Specific detectivity (D^*).

5.3.1 Minimum Detectable Power

In many design applications, the designer is concerned with the minimum detectable power of a photodiode. The minimum incident power on a photodiode required to generate a photocurrent equal to the total photodiode noise current is defined as the noise equivalent power, or NEP. As a mathematical expression, this may be written as

$$\text{NEP} = \frac{\text{RMS noise current (A)}}{\text{Photodiode sensitivity (A/W)}} \quad (5.24)$$

The NEP is dependent on the bandwidth of the measuring system, frequency of modulated signal, detector area, and operating temperature. In order to remove the bandwidth dependence, the figure is divided by the square root of the bandwidth. This gives an NEP density in $\text{W/Hz}^{1/2}$. Since the power-to-current conversion of a diode depends on radiation wavelength, the NEP figure is always quoted at a particular wavelength.

There are several sources of noise in photodetectors. These include *shot noise* from the detector dark current, *shot noise* from the photocurrent, *Johnson noise* from thermal fluctuations in the detector impedance, and *flicker noise* that is inversely proportional to the measurement frequency (also referred to as “ $1/f$ ” noise). APDs also have “*excess noise*” that depends on the ratio of the ionization coefficients for electrons and holes and the avalanche gain.

The shot noise current produced by the reverse leakage current of a device is given by the formula

$$I_S = (2ei_D B)^{1/2} \quad (5.25)$$

where I_S is the shot noise current, e is the electronic charge (1.6×10^{-19} coulomb), B is the bandwidth of system, and i_D is the dark leakage current (amps).

5.3.2 Dark Current

Dark current is a small current which flows when a reverse voltage is applied to a photodiode in dark condition (no light). This is a major source of noise in applications where reverse voltage is applied to photodiode. In contrast to this, the applications where no reverse voltage is applied, noise resulting from shunt resistance is predominant.

5.3.3 Shunt Resistance (R_{sh})

It is the voltage-to-current ratio in the vicinity of 0 V:

$$R_{sh} = \frac{10 \text{ mV}}{i_D} \quad (5.26)$$

where i_D is the dark current at $V_{in} = 10 \text{ mV}$. The Johnson or thermal noise contribution is provided principally by the shunt resistance of the device. The Johnson noise current is given by

$$I_J = \left[\frac{4kTB}{R_{sh}} \right]^{1/2} \quad (5.27)$$

where I_J is the Johnson noise current (A, RMS), k is the Boltzmann constant (1.33×10^{-23} J/K), T is the absolute temperature (K), and R_{sh} is the shunt resistance giving rise to noise.

The total noise current is the quadrature sum of the individual noise current contributions:

$$I_N = (I_S^2 + I_J^2)^{1/2} \quad (5.28)$$

and

$$NEP = \frac{I_N}{R} \quad (5.29)$$

Shot noise is the dominant component for a reverse-biased photodiode especially for a large area device operated at a high voltage. If a device is operated with zero bias, then the Johnson noise dominates, since the dark current tends to zero. It is usually the case when operating in this mode that noise current is reduced to such a degree that the NEP, and hence the minimum detectable signal, is reduced in spite of some loss of absolute sensitivity.

The detectivity D is defined as the inverse of NEP

$$D = \frac{1}{NEP} \quad (5.30)$$

The specific detectivity D^* is a parameter which takes into account the area of the photodetector. It is given as

$$D^* = DA^{1/2} \quad (5.31)$$

5.4 Fundamental Photodiode Circuits

The photodiode shown in Fig. 5.14(a) is unbiased. The photocurrent I_{photo} produced by incident photons is converted into output voltage by connecting load resistor R_L . The output voltage (V_{OUT}) is given as $V_{OUT} = I_{photo} \times R_L$. Output voltage is proportional to the amount of incident light when $V_{OUT} < V_{OC}$; V_{OC} is the open terminal voltage, that is, when $R_L = 0$. Figure 5.14(b) shows the operating point for a load resistor (R_L) for unbiased photodiode.

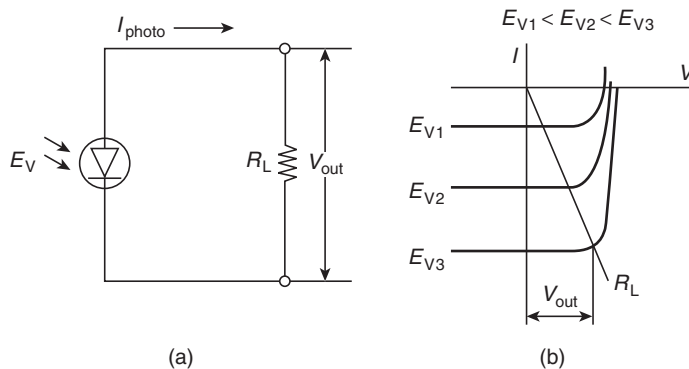


Figure 5.14 | (a) Photodiode circuit without bias. (b) Operating point for a load resistor (R_L).

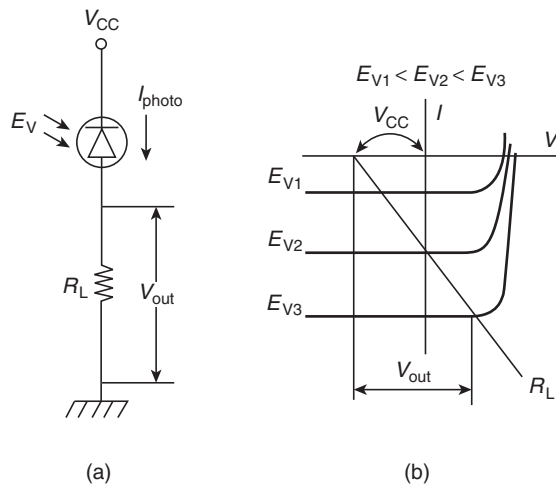


Figure 5.15 | (a) Photodiode circuit with bias. (b) Operating point for a load resistor (R_L).

Figure 5.15(a) shows a circuit in which the photodiode is reverse-biased by V_{CC} . The photocurrent (I_{photo}) is converted into an output voltage by the load resistor R_L . The output voltage V_{OUT} is given as $V_{OUT} = I_{photo} \times R_L$. Output voltage is proportional to the amount of incident light when $V_{OUT} < (V_{OC} + V_{CC})$. Figure 5.15(b) shows the operating point for a load resistor R_L with reverse-bias applied to the photodiode.

Features of a circuit used with a reverse-biased photodiode are as follows:

1. High-speed response.
2. Wide-proportional range of output.

Therefore, this circuit is generally used.

5.5 Phototransistor

Phototransistors are solid-state light detectors having internal gain. These are much more sensitive than photodiodes. These devices can be used to provide either an analog or a digital output signal. The advantageous features of phototransistors are as follows:

1. Low-cost visible and near-IR photodetection.
2. Available with gains from 100 to over 1,500.
3. Moderately fast response times.
4. Available in a wide range of packages.
5. Can be used with any visible or near infrared light source such as infrared emitting diodes (IREDs); neon, fluorescent, and incandescent bulbs; lasers; flame sources; sunlight; etc.
6. In many applications a greater output signal is required than the output generated by photodiode alone. The output signal of photodiode can be amplified externally by using op-amp transistor circuit. However, this approach is often not as practical or as cost effective as the use of phototransistors.

The equivalent circuit of a phototransistor is shown in Fig. 5.16. It shows that it is the same as a photodiode whose output photocurrent is fed into the base of a conventional small signal transistor. Base connection is not required for the operation of phototransistor, but it is always provided so that the designer has the option of using base current to bias the transistor. The typical gain of a phototransistor can range from 100 to over 1,500.

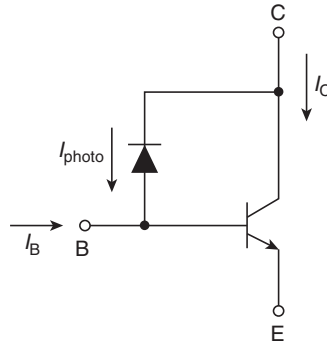


Figure 5.16 | Equivalent circuit of phototransistor.

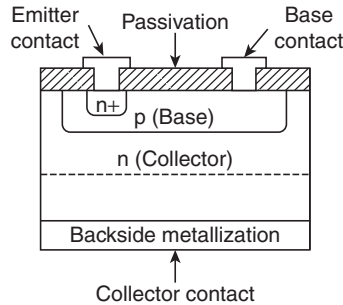


Figure 5.17 | Phototransistor chip cut away view.

The simplified structure of a phototransistor is shown in Fig. 5.17. Radiations incident on the device pass through the wide bandgap emitter and are absorbed in the base, the base–collector depletion region, and the collector. A large secondary photocurrent is obtained when the photo-generated holes are swept into base.

5.5.1 Fundamental Phototransistor Circuits

The circuit shown in Fig. 5.18(a) is a common-emitter amplifier. Light input at the base causes the output voltage (V_{OUT}) to decrease from high to low. The circuit shown in Fig. 5.18(b) is a common-collector amplifier with an output voltage (V_{OUT}) increasing from low to high in response to the light input. For the circuits in Fig. 5.18 to operate in the switching mode, the load resistor (R_L) should be set in relation with the collector current (I_C) as $V_{CC} < R_L \times I_C$.

5.5.1.1 Applications

Phototransistors can be used in various applications. Few of them are indicated in Fig. 5.19.

1. **Optoisolator:** It is similar to a transformer in that the output is electronically isolated from the input.
2. **Optical switch:** An object is detected when it enters the gap of the optical switch and blocks the light path between the emitter and the detector.
3. **Retro sensor:** It detects the presence of an object by generating light and then looking for its reflectance off the object to be sensed.

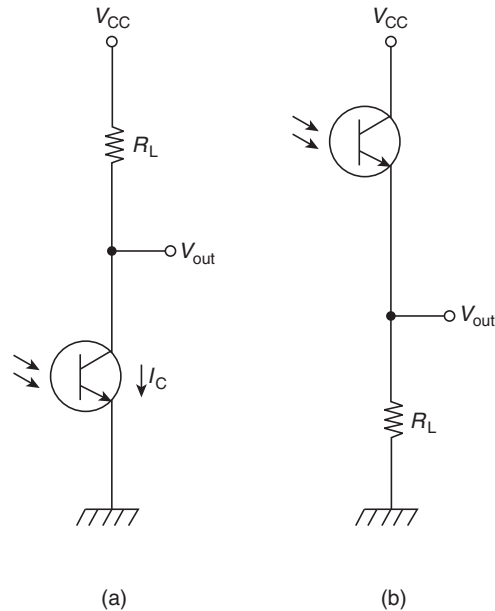


Figure 5.18 | Fundamental phototransistor circuit.

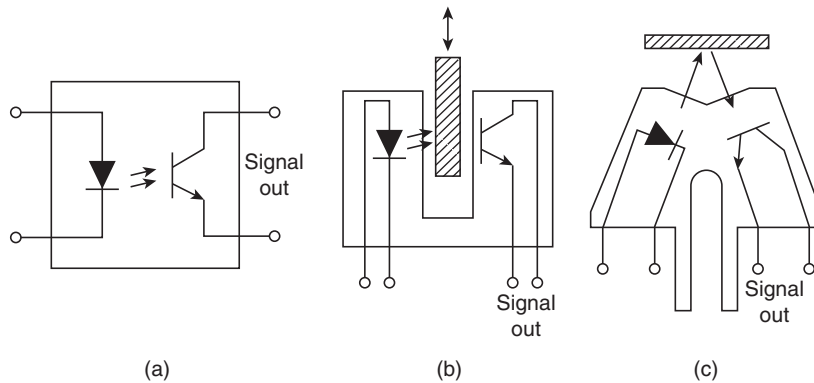


Figure 5.19 | (a) Optoisolator. (b) Optical switch. (c) Retro sensor.

5.6 Optical Receiver

In the previous section, we have discussed various photodetector structures that convert the incident optical power to the photocurrent. This is the very first stage of the optical receiver. This raw electrical signal needs further processing to extract the original information. The weak photocurrent requires amplification and conversion to voltage. The converted signal may be analog or digital depending upon the type of system (see Fig. 5.20). The next stages are as follows:

1. **Amplification:** The power of the received optical signal is less than $10 \mu\text{W}$. The detected signal is in microampere range, which is to be amplified for further signal processing. Before that, it needs the

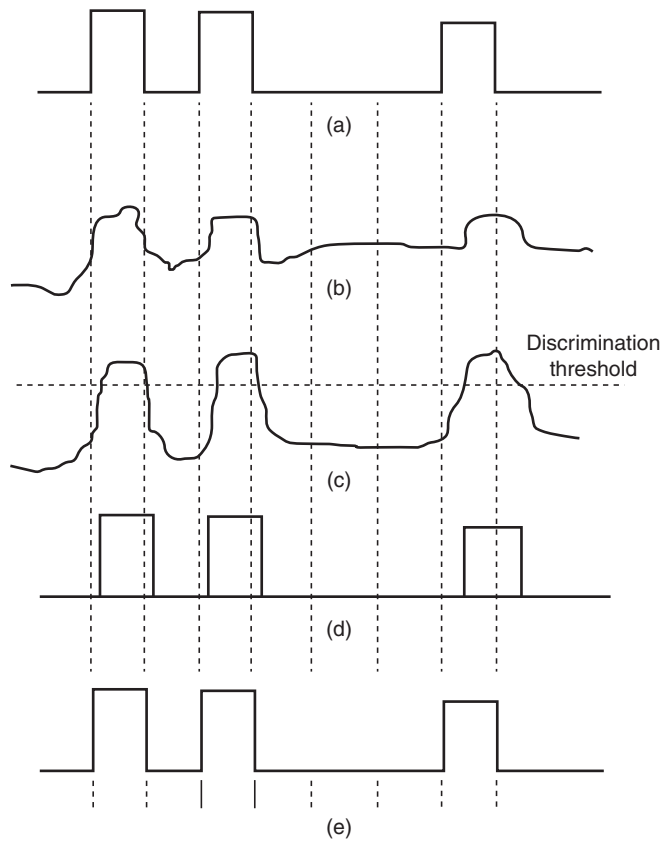


Figure 5.20 (a) Original digital signal. (b) Photodetected signal. (c) Amplified signal. (d) Discriminated signal. (e) Re-synchronized signal.

conversion from current to voltage. The receiver may have one or more amplification stages. The special low-noise amplifier for weak input is used for pre-amplification. Further amplifier may be added to amplify the signal to high level.

2. **Equalization:** Detection and amplification may distort the received signal. All the frequencies are not amplified by the same factor. Therefore, equalization circuit takes care of this and equalizes this difference or unevenness in amplification.
3. **Filtering:** This stage removes the undesired frequencies (e.g., noise or harmonics) and improves the signal-to-noise ratio.
4. **Discrimination:** This circuit generates digital pulses by comparing the signal with discrimination level as shown in Figs. 5.20(c) and (d). The decision circuit produces ON pulse if signal level is above discrimination threshold and OFF pulse if signal level is below discrimination threshold.
5. **Re-synchronization:** This circuit recreates the clock signal and puts the discriminated pulses in correct time slot.

The sources of noise at photodetector and amplification stage are as follows:

1. Quantum noise.
2. Dark current: When no light is falling on the photodetector, a small current flows through the circuit.

This current under no light or dark condition is called the dark current. Ideally no current should flow. Practically, a good photodetector should have dark current as less as possible.

3. Excess noise due to random fluctuations (only for APD).
4. Thermal noise due to bias resistor.
5. Amplifier noise.

5.6.1 Equivalent Circuit of Optical Fiber Receiver

In the equivalent circuit of optical fiber receiver shown in Fig. 5.21, i_{det} represents the optical detector as a current source, C_d is the detector capacitance, R_L is the detector load resistance, i_t is the thermal noise current, i_{TS} is the total shot noise current, R_a is the amplifier input resistance, C_a is the amplifier input capacitance, and i_{amp} is the amplifier noise current.

5.6.2 Receiver Amplifier Configurations

There are three commonly used amplifier configurations, as follows:

1. **Low impedance front end:** In Fig. 5.22(a), R_a is the amplifier input resistance and R_b is the detector bias resistor. The total load resistance of photodiode is

$$R_{\text{TL}} = \frac{R_a R_b}{R_a + R_b}$$

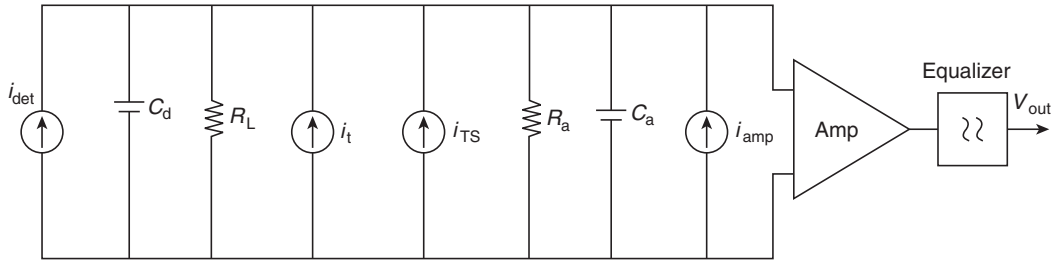


Figure 5.21 | Equivalent circuit of optical fiber receiver.

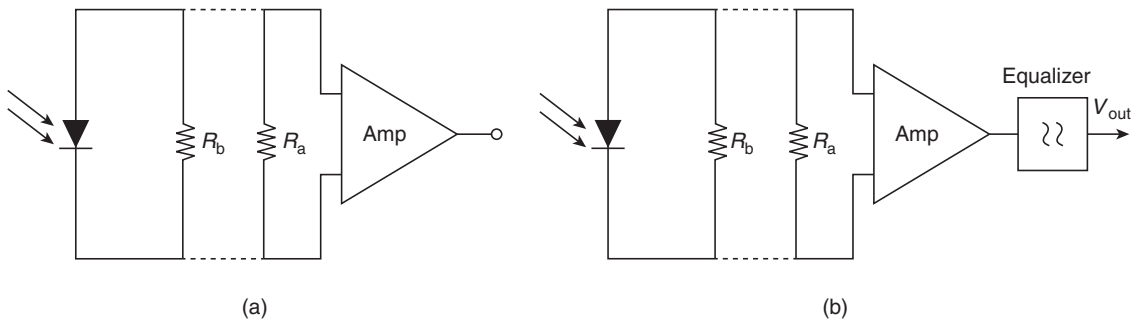


Figure 5.22 | (a) Low impedance front end. (b) High impedance front end.

Bandwidth and noise are the important parameters in the design of receiver. From the equivalent circuit of receiver, the front-end total capacitance is

$$C_T = C_d + C_a$$

and if the load resistance is R_L , then the post-detection bandwidth B is given as

$$B \geq \frac{1}{2\pi R_{TL} C_T}$$

From the above equation, it is clear that to increase the post-detection bandwidth B , R_L should be reduced. However, increase in B and decrease in R_L increases the thermal noise, which is undesirable. It severely limits the sensitivity of receiver. Thus, the circuit is impractical for long-distance communication.

2. **High impedance front end:** In this circuit, the effect of thermal noise is reduced by increasing the detector bias resistor and input impedance of amplifier. However, the frequency response is degraded. The detector output is integrated over long time constant and then restored by differentiation with the help of equalizer as shown in Fig. 5.22(b). Thus, this structure gives much better sensitivity as compared to low impedance front-end configuration, but needs equalization and has limited dynamic range.
3. **Transimpedance front end:** In transimpedance front-end configuration (Fig. 5.23), a low noise, high input impedance amplifier with negative feedback is used to overcome the drawbacks of the high impedance front-end configuration. It operates in a current mode amplifier, and the high input impedance is reduced by negative feedback. This configuration provides much greater bandwidth as well as reduced thermal noise.

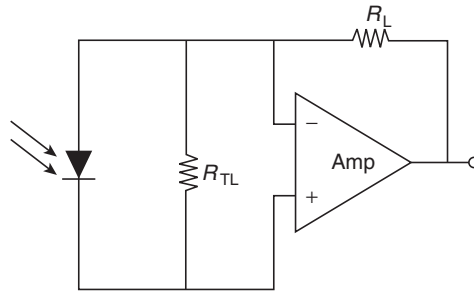


Figure 5.23 | Transimpedance front-end.

Solved Problems

PROBLEM 1

In a photodetection device having absorption layer of thickness $10\text{ }\mu\text{m}$ with refractive index $n = 3.5$ and absorption coefficient $\alpha = 10^5\text{ m}^{-1}$, calculate quantum efficiency if reflectivity is 0.31.

Solution: The quantum efficiency is given by

$$\eta = (1 - R)(1 - e^{-\alpha d})$$

$$\Rightarrow \eta = (1 - 0.31)(1 - e^{-10^5 \times 10 \times 10^{-6}}) = 0.43$$

PROBLEM 2

Determine the cutoff wavelength for GaAs photodetector having bandgap energy of 1.43 eV.

Solution: We have

$$\lambda_c (\mu\text{m}) = \frac{1.24}{E_g (\text{eV})} = \frac{1.24}{1.43} = 0.876$$

The GaAs photodetector will cease to operate above 876 nm.

PROBLEM 3

A pn photodiode has a quantum efficiency of 65% for photons of energy 1.52×10^{-19} J. Calculate (a) wavelength at which the diode is operating and (b) the optical power required to achieve a photocurrent of 3 μA , when the wavelength of incident photon is that calculated in part (a).

Solution: The energy of each photon is

$$E = hf = \frac{hc}{\lambda}$$

Therefore,

$$\lambda = \frac{hc}{E} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.52 \times 10^{-19}} = 1,300 \text{ nm}$$

Now

$$R = \frac{\eta e}{hf} = \frac{0.65 \times 1.6 \times 10^{-19}}{1.52 \times 10^{-19}} = 0.684 \text{ A/W}$$

Also we know that

$$R = \frac{I_{\text{photo}}}{P_{\text{in}}} \text{ in A/W}$$

So

$$P_{\text{in}} = \frac{3 \times 10^{-6}}{0.684} \text{ A/W} = 4.38 \mu\text{W}$$

PROBLEM 4

A PIN photodiode generates one electron–hole pair per three incident photons at a wavelength of 850 nm. Calculate the quantum efficiency of diode and mean output photocurrent when the incident optical power is 15 μW .

Solution: We have

$$\eta = \frac{1}{3} = 0.33 = 33\%$$

Now

$$R = \frac{I_{\text{photo}}}{P_{\text{in}}} \text{ A/W}$$

Also

$$\begin{aligned}\lambda_c (\mu\text{m}) &= \frac{1.24}{E_g (\text{eV})} \\ \Rightarrow E_g (\text{eV}) &= \frac{1.24}{\lambda_c (\mu\text{m})} = \frac{1.24}{0.85} = 1.4588 \\ \Rightarrow E_g &= 2.33 \times 10^{-19} \text{ J}\end{aligned}$$

Therefore

$$I_{\text{photo}} = R \times P_{\text{in}} = \frac{\eta e}{hf} = \frac{0.33 \times 1.6 \times 10^{-19}}{2.33 \times 10^{-19}} \times 15 \times 10^{-6} = 3.399 \mu\text{A}$$

PROBLEM 5

Photons of 1,300 nm wavelength are incident on a pn photodiode at a rate of $5 \times 10^{10} \text{ s}^{-1}$ and the electrons are collected at a rate of $2 \times 10^{10} \text{ s}^{-1}$. Calculate the quantum efficiency and responsivity of the diode.

Solution: We have

$$\eta = \frac{2 \times 10^{10}}{5 \times 10^{10}} = 0.4$$

Now

$$\begin{aligned}R &= \frac{\eta e \lambda}{hc} \\ &= \frac{0.4 \times 1.6 \times 10^{-19} \times 1,300 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^8} \\ &= 0.418 \text{ A/W}\end{aligned}$$

PROBLEM 6

Calculate the maximum response time for silicon PIN photodiode having depletion layer width of 30 μm and carrier velocity of $3 \times 10^4 \text{ m/s}$, assuming no carriers are generated outside the depletion region and depletion capacitance is negligible.

Solution: The maximum 3-dB bandwidth is given by

$$B_{\text{max}} = \frac{1}{2\pi\tau_{\text{drift}}} = \frac{v_{\text{drift}}}{2\pi w} = \frac{3 \times 10^4}{2\pi \times 30 \times 10^{-6}} = 1.59 \times 10^8 \text{ Hz}$$

Therefore,

$$\text{Maximum response time} = \frac{1}{B_{\max}} = \frac{1}{1.59 \times 10^8} = 6.2 \text{ ns}$$

PROBLEM 7

The quantum efficiency of APD is 35% at 1,300 nm. Calculate the multiplication factor if it produces the output photocurrent of 6 μA for incident optical power of 0.35 μW .

Solution: The avalanche multiplication factor M of an APD is defined as

$$M = \frac{\text{Multiplied output photocurrent}}{\text{Initial or primary unmultiplied photocurrent}} = \frac{I_{\text{output}}}{I_{\text{photo}}}$$

We have

$$I_{\text{photo}} = R \times P_{\text{in}} = \frac{\eta e \lambda}{hc} P_{\text{in}}$$

Therefore,

$$M = \frac{I_{\text{output}}}{I_{\text{photo}}} = \frac{I_{\text{output}}}{\frac{\eta e \lambda}{hc} P_{\text{in}}} = \frac{I_{\text{output}} hc}{\eta e \lambda P_{\text{in}}}$$

Substituting the values, we get

$$M = \frac{6 \times 10^{-6} \times 6.626 \times 10^{-34} \times 3 \times 10^8}{0.35 \times 10^{-6} \times 0.35 \times 1.6 \times 10^{-19} \times 1,300 \times 10^{-9}} = 46.6$$

PROBLEM 8

Determine the multiplication factor for silicon RAPD, if its quantum efficiency is 70%, incident optical power is 0.5 μW , and output current is 10 μA at 900 nm.

Solution: We have

$$\begin{aligned} R &= \frac{\eta e \lambda}{hc} \\ &= \frac{0.7 \times 1.6 \times 10^{-19} \times 900 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^8} \\ &= 0.507 \text{ A/W} \end{aligned}$$

Now

$$I_{\text{photo}} = R \times P_{\text{in}} = 0.507 \times 0.5 \times 10^{-6} = 0.2535 \mu\text{A}$$

So

$$M = \frac{I_{\text{output}}}{I_{\text{photo}}} = \frac{10 \times 10^{-6}}{0.2535 \times 10^{-6}} = 39.44$$

The multiplication factor of the photodiode is approximately 40.

PROBLEM 9

Find the multiplication factor for silicon APD having quantum efficiency of 72% at 890 nm, and 0.75 μW optical power produces a multiplied photocurrent of 15 μA .

Solution: We have

$$\begin{aligned} R &= \frac{\eta e \lambda}{hc} \\ &= \frac{0.72 \times 1.6 \times 10^{-19} \times 890 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^8} \\ &= 0.5157 \text{ A/W} \end{aligned}$$

Therefore

$$I_{\text{photo}} = R \times P_{\text{in}} = 0.5157 \times 0.75 \times 10^{-6} = 0.3867 \mu\text{A}$$

So

$$M = \frac{I_{\text{output}}}{I_{\text{photo}}} = \frac{15 \times 10^{-6}}{0.3867 \times 10^{-6}} = 38.78$$

The primary photocurrent is multiplied by factor of 39.

PROBLEM 10

If a diode has $i_D = 2 \text{ nA}$, a shunt resistance R_{sh} of $5 \times 10^8 \Omega$, and responsivity $R = 0.5 \text{ A/W}$, then for $B = 1 \text{ Hz}$, calculate NEP.

Solution: Shot noise current I_S is given by

$$\begin{aligned} I_S &= (2ei_D B)^{1/2} \\ &= (2 \times 1.6 \times 10^{-19} \times 2 \times 10^{-9} \times 1)^{1/2} \\ &= 2.5 \times 10^{-14} \text{ A} \end{aligned}$$

Johnson noise current I_J is

$$\begin{aligned} I_J &= \left[\frac{4kTB}{R_{\text{sh}}} \right]^{1/2} \\ &= \left[\frac{4 \times 1.33 \times 10^{-23} \times 300 \times 1}{5 \times 10^8} \right]^{1/2} \end{aligned}$$

$$= 5.6 \times 10^{-15} \text{ A}$$

Total noise current is

$$\begin{aligned} I_N &= (I_S^2 + I_J^2)^{1/2} \\ &= [(2.5 \times 10^{-14})^2 + (5.6 \times 10^{-15})^2]^{1/2} \\ &= 2.56 \times 10^{-14} \text{ A} \end{aligned}$$

Hence

$$\text{NEP} = \frac{I_N}{R} = \frac{2.56 \times 10^{-14}}{0.5} = 5.12 \times 10^{-14} \text{ W}$$

PROBLEM 11

Determine the quantum efficiency and responsivity of InGaAs photodetector if a 80 ns pulse, having 6.2×10^6 photons at a wavelength of 1,300 nm, falls on it and average EHPs generated are 5.5×10^6 .

Solution: The quantum efficiency is

$$\begin{aligned} \eta &= \frac{\text{Number of excited EHPs per second}}{\text{Total number of incident pairs per second}} \\ &= \frac{r_{\text{EHP}}}{r_{\text{photon}}} \\ &= \frac{5.5 \times 10^6}{6.2 \times 10^6} \\ &= 0.887 \end{aligned}$$

Hence

$$\eta = 88.7\%$$

Responsivity R is

$$\begin{aligned} R &= \frac{\eta e}{hf} \\ &= \frac{\eta \lambda (\mu\text{m})}{1.24} \text{ A/W} \\ &= \frac{0.887 \times 1.3}{1.24} \text{ A/W} \\ &= 0.93 \text{ A/W} \end{aligned}$$

PROBLEM 12

If the responsivity of PIN silicon photodetector is 0.55 A/W at 800 nm wavelength, calculate its efficiency.

Solution: Responsivity R is

$$R = \frac{\eta \lambda \text{ (}\mu\text{m)}}{1.24} \text{ A/W}$$

Therefore efficiency is

$$\begin{aligned} \eta &= \frac{1.24R}{\lambda \text{ (}\mu\text{m)}} \\ &= \frac{1.24 \times 0.55}{0.8} = 0.8525 \\ &= 85.25\% \end{aligned}$$

PROBLEM 13

Calculate the photocurrent generated in the output of a photodiode when photons of energy 1.4×10^{-19} are incident, the optical power level is $12 \mu\text{W}$, and its responsivity is 0.7 A/W .

Solution: We have

$$I_{\text{photo}} = R \times P_{\text{in}} = 0.7 \times 12 \times 10^{-6} = 8.4 \mu\text{A}$$

PROBLEM 14

If the responsivity of a PIN diode is 0.6 A/W and its dark current is 1.5 nA , calculate the minimum power detectable by it.

Solution: We can recognize the presence of optical power that produces a signal current equal to dark current. Therefore, we have

$$R = \frac{I_{\text{photo}}}{P_{\text{in}}} = \frac{i_{\text{D}}}{P_{\text{in}}} \text{ A/W}$$

Therefore,

$$P_{\text{in}} = \frac{i_{\text{D}}}{R} = \frac{1.5 \times 10^{-9}}{0.6} = 2.5 \text{ nW}$$

Thus, the minimum detectable optical power is 2.5 nW .

PROBLEM 15

An InGaAs PIN photodiode operating at $1,300 \text{ nm}$ has dark current of 4 nA , efficiency is 72% , load resistance is $1 \text{ k}\Omega$, incident optical power is 400 nW , and receiver bandwidth is 20 MHz . Calculate the signal-to-noise ratio.

Solution: Responsivity R is

$$\begin{aligned} R &= \frac{\eta \lambda \text{ (}\mu\text{m)}}{1.24} \\ &= \frac{0.72 \times 1.3 \text{ (}\mu\text{m)}}{1.24} \text{ A/W} \\ &= 0.7548 \text{ A/W} \end{aligned}$$

But

$$R = \frac{I_{\text{photo}}}{P_{\text{in}}} \text{ A/W}$$

Therefore

$$I_{\text{photo}} = R \times P_{\text{in}} = 0.72 \times 400 \times 10^{-9} = 288 \text{ nA}$$

Now

$$I_{\text{signal}} = I_{\text{photo}} = 288 \text{ nA}$$

Shot noise current

$$\begin{aligned} I_s &= (2ei_D B)^{1/2} \\ &= (2 \times 1.6 \times 10^{-19} \times 4 \times 10^{-9} \times 20 \times 10^6)^{1/2} \\ &= 0.16 \times 10^{-9} \text{ A} \end{aligned}$$

Johnson noise current is

$$\begin{aligned} I_J &= \left[\frac{4kTB}{R_{\text{sh}}} \right]^{1/2} \\ &= \left[\frac{4 \times 1.33 \times 10^{-23} \times 300 \times 20 \times 10^6}{1 \times 10^3} \right]^{1/2} \\ &= 18.19 \times 10^{-9} \text{ A} \end{aligned}$$

Total noise current

$$\begin{aligned} I_N &= (I_s^2 + I_J^2)^{1/2} \\ &= [(0.16 \times 10^{-9})^2 + (18.19 \times 10^{-9})^2]^{1/2} \\ &= 18.19 \times 10^{-9} \text{ A} \end{aligned}$$

Hence signal-to-noise ratio is

$$\begin{aligned} \text{SNR} &= \frac{I_{\text{signal}}}{I_N} = \frac{288 \text{ nA}}{18.19 \text{ nA}} = 15.83 \\ \text{SNR (in dB)} &= 20 \log(15.83) = 23.99 \text{ dB} \end{aligned}$$

PROBLEM 16

The photodetector operating at 800 nm has a quantum efficiency of 15%. Load resistance is 100 Ω and the optical power absorbed by it is 1.7 μW . Calculate the voltage across load.

Solution: Responsivity R is

$$\begin{aligned} R &= \frac{\eta \lambda \text{ (}\mu\text{m)}}{1.24} \\ &= \frac{0.15 \times 0.8 \text{ (}\mu\text{m)}}{1.24} \text{ A/W} \\ &= 0.0967 \end{aligned}$$

Now

$$R = \frac{I_{\text{photo}}}{P_{\text{in}}} \text{ A/W}$$

Therefore,

$$I_{\text{photo}} = R \times P_{\text{in}} = 0.0967 \times 1.7 \times 10^{-6} = 164.39 \text{ nA}$$

The voltage across the load is

$$\begin{aligned} V_L &= I_{\text{photo}} \times R_L \\ &= 164.39 \times 10^{-9} \times 100 \\ &= 16.43 \text{ }\mu\text{V} \end{aligned}$$

PROBLEM 17

A PIN photodiode is having a depletion layer width of 30 μm and a carrier velocity of 3×10^4 m/s. Calculate the maximum 3-dB bandwidth for it.

Solution: The transit time or drift time for charge carriers generated at the edge of intrinsic Si-layer is

$$\tau_{\text{drift}} = \frac{W}{v_{\text{drift}}} = \frac{30 \times 10^{-6}}{3 \times 10^4} = 10 \times 10^{-10} \text{ s}$$

We have

$$t_r = \frac{0.35}{f_c}$$

Therefore, maximum 3-dB bandwidth is

$$f_{3\text{-dB}} = \frac{0.35}{t_r} = \frac{0.35}{10 \times 10^{-10}} = 3.5 \times 10^8 \text{ Hz}$$

So

$$f_{3\text{dB}} = 350 \text{ MHz}$$

PROBLEM 18

A silicon PIN photodiode has depletion region width = 12 μm , cross-sectional area 0.4 mm^2 , load resistance of 10 $\text{M}\Omega$, $\epsilon_s = 10 \times 10^{-13} \text{ F/m}$, and saturation velocity of 10^5 m/s . Calculate transit time, junction capacitance, and time constant of photodiode.

Solution: The transit time or drift time for charge carriers generated at the edge of intrinsic Si-layer is

$$\tau_{\text{drift}} = \frac{w}{v_{\text{drift}}} = \frac{12 \times 10^{-6}}{10^5} = 12 \times 10^{-11} \text{ s} = 0.12 \text{ ns}$$

The junction or depletion layer capacitance of PIN diode is given by

$$C_{\text{depletion}} = \frac{\epsilon_s A}{w} = \frac{10 \times 10^{-13} \times 0.4 \times 10^{-6}}{12 \times 10^{-6}} = 0.33333 \times 10^{-13}$$

This implies

$$C_{\text{depletion}} = 3.3 \text{ pF}$$

Time constant is

$$\begin{aligned} t &= R_L C_{\text{depletion}} \\ &= 10 \times 10^6 \times 3.3 \times 10^{-12} \\ &= 33 \times 10^{-6} \end{aligned}$$

Hence

$$t = 33 \text{ } \mu\text{s}$$

PROBLEM 19

The commercial Ge pn junction photodiode has a photosensitive area of 0.006 mm^2 (its responsivity curve is shown in Fig. 5.24). The dark current is 0.35 μA , junction capacitance is 4 pF, rise time is 0.54 ns when the reverse bias voltage is 12 V. Calculate its quantum efficiency at 1,300 nm and 1,550 nm. Also find the intensity of light at 1,550 nm that gives a photocurrent equal to the dark current.

Solution: We know that

$$R = \frac{\eta e \lambda}{hc}$$

The responsivity curve is given in Fig. 5.24, and we have at $\lambda = 1,550 \text{ nm}$, $R = 0.7$. So

$$\eta = \frac{0.7 \times 6.626 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 1,550 \times 10^{-9}} = 0.561$$

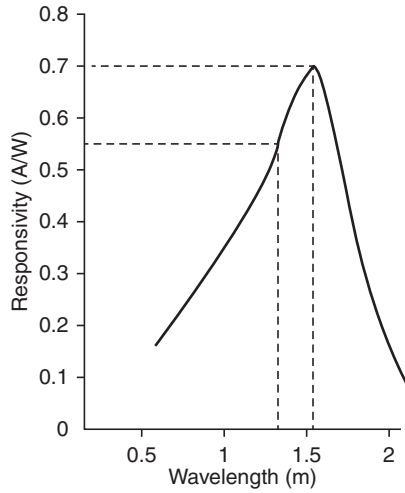


Figure 5.24 | Responsivity curve of Ge pn junction photodiode.

This implies $\eta = 56.10\%$. Again from Fig. 5.24, we have at $\lambda = 1,300 \text{ nm}$, $R = 0.6$. So

$$\eta = \frac{0.55 \times 6.626 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 1,300 \times 10^{-9}} = 0.5256$$

This means $\eta = 52.56\%$. Now

$$R = \frac{I_{\text{photo}}}{P_{\text{in}}} \text{ A/W}$$

Dark current is $0.3 \mu\text{A}$, at $1,550 \text{ nm}$, $I_{\text{photo}} = i_D$. Therefore, incident optical power is

$$P_{\text{in}} = \frac{I_{\text{photo}}}{R} = \frac{0.35 \times 10^{-6}}{0.7} = 0.5 \times 10^{-6} \text{ W}$$

Intensity of light is

$$\frac{P_{\text{in}}}{\text{Photosensitive area}} = \frac{0.5 \times 10^{-6}}{0.006 \times 10^{-6}} = 83.33 \text{ W/m}^2$$

PROBLEM 20

An Si APD has a quantum efficiency of 75% at 850 nm without multiplication. The APD is biased to operate with a multiplication factor of 100 . If the incident optical power is 12 nW , calculate the photocurrent.

Solution: We have

$$R = \frac{\eta e \lambda}{hc} = \frac{0.75 \times 1.6 \times 10^{-19} \times 850 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^8} = 0.5131 \text{ A/W}$$

The photocurrent without multiplication is given by

$$I_{\text{photo}} = R \times P_{\text{in}} = 0.5131 \times 12 \times 10^{-9} = 6.1 \text{ nA}$$

The avalanche multiplication factor M of an APD is defined as

$$M = \frac{\text{Multiplied output photocurrent}}{\text{Initial or primary unmultiplied photocurrent}} = \frac{I_{\text{output}}}{I_{\text{photo}}}$$

Multiplied photocurrent is

$$I_{\text{output}} = M \times I_{\text{photo}} = 100 \times 6.1 \text{ nA} = 610 \text{ nA}$$

SUMMARY

1. A photodetector converts electromagnetic radiation into an electronic signal, which is proportional to the incident light intensity.
2. Photodiodes are fabricated from semiconductor materials. The most commonly used materials are silicon (Si), gallium arsenide (GaAs), indium antimonide (InSb), indium arsenide (InAs), lead selenide (PbSe), and lead sulfide (PbS).
3. A photodetector is a reverse-biased pn junction. When a photon of light is absorbed, it excites an electron and produces a single pair of charge carriers, an electron and a hole. These two charge carriers are swept in opposite directions by the externally applied bias voltage, and the photocurrent flows in the external circuit.
4. To quantify the photon absorption ability, a parameter called the *quantum efficiency* (η) is used. It is the ratio of the number of excited EHPs to the total number of incident photons.
5. The responsivity R of a photodetector is defined as the output photocurrent per unit incident optical power.
6. The wavelength at which the detectors respond depends on their composition. Silicon is used at 400–1100 nm, InGaAs is used at 1000–1700 nm, and Germanium is used at 800–1800 nm.
7. There are three types of photodiodes, namely (a) pn photodiode, (b) PIN photodiode, (c) Avalanche photodiode (APD). The pn and PIN photodiodes produce output current without gain, while APD produces output current with gain. Thus, the APD is a solid-state efficient alternative for a photomultiplier tube (PMT).
8. PIN photodiodes have high speed and improved sensitivity.
9. The minimum incident power on a photodiode required to generate a photocurrent equal to the total photodiode noise current is defined as the noise equivalent power, or NEP.
10. A photodetector is the very first stage of the optical receiver, the further stages are amplification, equalization, filtering, discrimination, and resynchronization.

MULTIPLE-CHOICE QUESTIONS

1. Fiber optic photodiode detectors are normally
 - (a) Forward-biased
 - (b) Reverse-biased
 - (c) Unbiased to generate voltage similar to solar cell
 - (d) Thermoelectrically cooled
2. To create a hole–electron pair in a photodiode, the energy of incident photon should be
 - (a) Equal to E_g
 - (b) Less than E_g

- (c) Greater than E_g
 - (d) Much greater than E_g
 - (e) Both (a) and (c)
3. Silicon photodetectors are useful at which wavelength?
 - (a) 800–900 nm
 - (b) 1,300 nm
 - (c) 1,550 nm
 - (d) All of the above
 4. Which detector material is most often used in 1,550 nm window?
 - (a) Silicon
 - (b) GaAs
 - (c) Germanium
 - (d) InGaAs
 5. What is a PIN photodiode?
 - (a) Circuit element used in amplification stage of receiver
 - (b) Photovoltaic detector
 - (c) Semiconductor detector with an undoped intrinsic region between p and n materials
 - (d) Point contact diode detector
 6. Which type of detector has internal gain mechanism to give amplified output?
 - (a) pn photodiode
 - (b) PIN photodiode
 - (c) Avalanche photodiode
 - (d) Photovoltaic detector
 7. The bandgap energy of silicon is 1.1 eV. What is the maximum wavelength that can be absorbed by it?
 - (a) 780 nm
 - (b) 950 nm
 - (c) 1,120 nm
 - (d) 1,550 nm
 8. The highest wavelength absorbed by Germanium is 1,850 nm. What is its bandgap energy?
 - (a) 0.5 eV
 - (b) 0.67 eV
 - (c) 0.8 eV
 - (d) 0.9 eV
 9. A pn photodiode generates three electron–hole pairs per ten incident photons at a wavelength of 1,100 nm. What is the quantum efficiency of the diode, if all the photo-generated electrons are collected?
 - (a) 20%
 - (b) 30%
 - (c) 40%
 - (d) 50%
 10. A pn photodiode has a quantum efficiency of 70% for photons of energy 1.52×10^{-19} J. What is its operating wavelength?
 - (a) 980 nm
 - (b) 1,100 nm
 - (c) 1,300 nm
 - (d) 1,550 nm

REVIEW QUESTIONS

1. Explain detection process in PIN photodiode. Compare this device with APD photodiode and give the drawbacks of APD.
2. Define the quantum efficiency and responsivity of a photodetector.
3. Briefly explain the following terms related to a photodetector.
 - (a) Dark current
 - (b) Detectivity
 - (c) Spectral response
4. List the requirements of a photodetector.
5. Derive an expression for the responsivity of an intrinsic photodiode in term of the quantum efficiency of the device and the wavelength of the incident radiation. Also find the wavelength at which quantum efficiency and the responsivity are equal.
6. Give the definition and explain the meaning of the noise equivalent power.
7. Derive the formula for SNR involving the quantum efficiency of a photodiode.
8. Give the definition of the bandwidth of a photodiode. What are the factors that restrict the bandwidth of a pn photodiode?
9. Explain the avalanche multiplication noise and the temperature effects on avalanche gain.
10. Draw the equivalent circuit of optical receiver and explain the related terms.
11. With neat sketches explain different receiver amplifier configurations.

EXERCISES

1. Calculate the quantum efficiency of an InGaAs Photodetector operating at 1300 nm, when a 100 ns pulse having 6.2×10^6 photons falls on it and 5.5×10^6 EHPs are generated.
2. Photons of 850 nm wavelength are incident on a pn photodiode at a rate of $3 \times 10^{11} \text{ s}^{-1}$ and the electrons are collected at a rate of $1.2 \times 10^{11} \text{ s}^{-1}$. Calculate quantum efficiency and responsivity of diode.
3. Calculate the responsivity of a PIN photodiode at 1300 nm and 1550 nm if the quantum efficiency is 70%. At what wavelength is the photodiode more responsive? Why?
4. A photodiode has a quantum efficiency of 70%, when photons of energy $1.3 \times 10^{-19} \text{ J}$ are incident upon it. Find the operating wavelength of photodiode. Also calculate the incident optical power required to obtain photocurrent of 2.5 μA .
5. Find the photocurrent generated, if the incident photon's energy is $1.6 \times 10^{-19} \text{ J}$, optical power level is 12 μW , and the responsivity of photodiode is 0.7 A/W.
6. Calculate the responsivity of InGaAs photodiode in the wavelength range of 1300–1600 nm, if the quantum efficiency is 92%.
7. A silicon photodiode operating at 800 nm, 27°C has dark current of 0.1 A, responsivity is 0.5 A/W, load resistance is 10 M Ω , source resistance = 100 Ω , incident optical power is 2 μW , active area is $3 \times 10^{-3} \text{ cm}^2$ and receiver bandwidth is 500Hz. Calculate signal-to-noise ratio and NEP.
8. An APD has a quantum efficiency of 45% at 1300 nm. Calculate the multiplication factor of the diode if it produces an output photocurrent of 6 μA after avalanche multiplication for the incident power of 0.3 μW .
9. For a InGaAs PIN photodiode is having responsivity of 0.85 A/W, dark current is 5 nA, what is the optical power at a wavelength of 1550 nm that will give a photocurrent that is twice the dark current?
10. Calculate quantum efficiency in a photo-detection device having absorption layer of thickness 10 μm with refractive index $n = 3.5$ and absorption coefficient $\alpha = 10^5 \text{ m}^{-1}$, if reflectivity is 0.35.
11. Determine the cutoff wavelength for InP photodetector having bandgap energy of 1.35 eV.
12. A pn photodiode has a quantum efficiency of 75% for photons of energy $1.52 \times 10^{-19} \text{ J}$. Calculate (a) wavelength at which the diode is operating and (b) the optical power required to achieve a photocurrent of 4 μA , when the wavelength of incident photon is that calculated in part (a).
13. Calculate the quantum efficiency of a PIN photodiode and mean output photocurrent when the incident optical power is 15 μW and the photodiode generates one electron-hole pair per two incident photons at a wavelength of 850 nm.
14. Calculate the quantum efficiency and responsivity of the diode when photons of 1,350 nm wavelength are incident on a pn photodiode at a rate of $5.5 \times 10^{10} \text{ s}^{-1}$ and the electrons are collected at a rate of $2.5 \times 10^{10} \text{ s}^{-1}$.
15. Calculate the maximum response time for silicon PIN photodiode having a depletion layer width of 40 μm and carrier velocity of $3.5 \times 10^4 \text{ m/s}$, assuming no carriers are generated outside the depletion region and depletion capacitance is negligible.
16. Calculate the multiplication factor of APD having the quantum efficiency of 45% at 1,350 nm and it produces the output photocurrent of 7 μA for incident optical power of 0.40 μW .
17. Determine the multiplication factor for silicon RAPD, if its quantum efficiency is 65%, incident optical power is 0.55 μW , and output current is 12 μA at 830 nm.
18. Calculate NEP of a diode having $i_D = 3 \text{ nA}$, a shunt resistance R_{sh} of $7 \times 10^8 \Omega$, and responsivity (R) = 0.6 A/W, then for $B = 1 \text{ Hz}$.
19. Determine the quantum efficiency and responsivity of InGaAs photodetector if a 100 ns pulse, having 7.0×10^6 photons at a wavelength of 1,350 nm, falls on it and average EHPs generated are 6.0×10^6 .
20. If the responsivity of PIN silicon photodetector is 0.65 A/W at 850 nm wavelength, calculate its efficiency.

21. Calculate the photocurrent generated in the output of a photodiode when photons of energy 1.4×10^{-19} are incident, the optical power level is $15 \mu\text{W}$, and its responsivity is 0.75 A/W .
22. If the responsivity of a PIN diode is 0.75 A/W and its dark current is 1.9 nA , calculate the minimum power detectable by it.
23. The photodetector operating at 820 nm has a quantum efficiency of 25% . Load resistance is 120Ω and the optical power absorbed by it is $2 \mu\text{W}$. Calculate the voltage across load.
24. Calculate the maximum 3-dB bandwidth of a PIN photodiode is having a depletion layer width of $35 \mu\text{m}$ and a carrier velocity of $3.1 \times 10^4 \text{ m/s}$.
25. Calculate transit time, junction capacitance, and time constant of a silicon PIN photodiode having depletion region width = $15 \mu\text{m}$, cross-sectional area 0.5 mm^2 , load resistance of $10 \text{ M}\Omega$, $\epsilon_s = 10 \times 10^{-13} \text{ F/m}$, and saturation velocity of 10^5 m/s .

ANSWERS

Multiple-Choice Questions

- | | |
|--------|---------|
| 1. (b) | 6. (c) |
| 2. (e) | 7. (c) |
| 3. (a) | 8. (b) |
| 4. (d) | 9. (b) |
| 5. (c) | 10. (c) |

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6

Optical Transmission System: Concepts and Components

Learning Objectives

After completing this chapter, you will be able to:

- Analyze signal distortion between transmitter and receiver in a simple fiber-optic point-to-point link with respect to attenuation and pulse dispersion.
 - Do a qualitative estimation of the maximum transmission distance L at a given signal data rate.
 - Do a quantitative characterization of the signal detection process and definition of the minimal detectable optical power for a particular type of signal distortion with respect to a detection error criterion.
 - Undertake a detailed study of multichannel optical systems and the ITU standards for the same.
 - Understand optical system components with typical specifications and applications.
-

In the preceding chapters, we studied the three main components of fiber-optic communication systems, namely, optical fibers, optical transmitters, and optical receivers. This chapter covers issues related to system design and performance parameters for a practical light-wave system.

Optical accessories such as connectors, couplers, switches, splices, and multiplexing devices are essential to the successful operation of fiber links and networks. From a selection set of device components and a given set of system requirements, analysis is required to determine if the system being designed will meet the requirements.

From their architectural point of view, fiber-optic systems are classified into the following three broad categories:

1. Point-to-point links.
2. Local area networks.
3. Distribution networks.

6.1 Point-to-Point Links

A typical point-to-point fiber-optic communication network consists of a transmitter [light-emitting diode (LED)/laser source], a transport medium (optical fiber), and a receiver (PIN/APD photodetector) as shown in Fig. 6.1. The evolution of a fiber optic point-to-point link is given in Table 6.1.

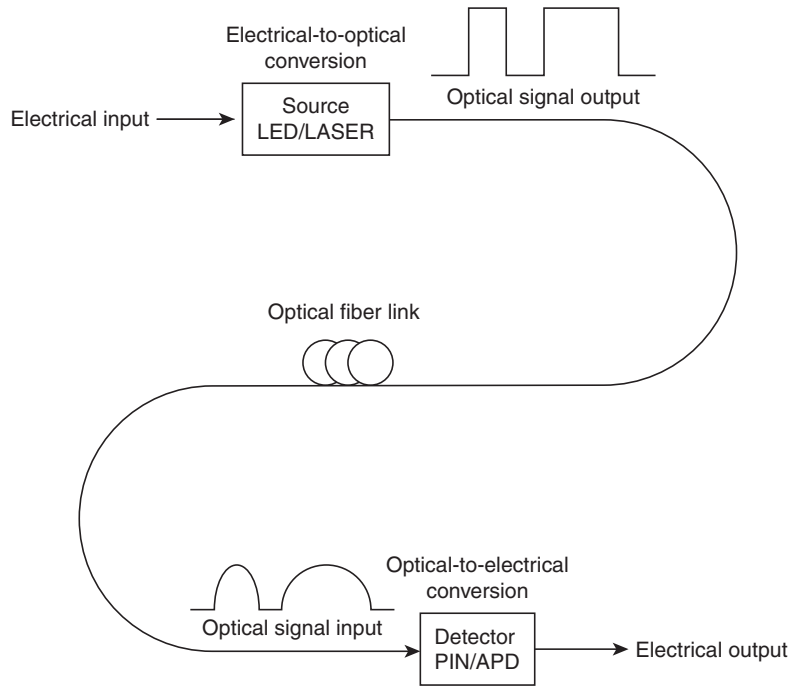


Figure 6.1 | Simplified block diagram of point-to-point optical communication link.

Table 6.1 | Evolution of fiber-optic point-to-point link

Decade	Generation	System features
Early 1970s	1 st Generation	- Multimode fiber-optic waveguides > 5 dB/km attenuation - Room temperature GaAs LEDs and multimode FP Lasers @ 830 nm
Early 1980s	2 nd Generation	- Low-loss single-mode optical fibers 1 dB/km @ 1,310 nm - Multimode Fabry–Perot 1,310 nm lasers
Mid-1980s to late 1980s	3 rd Generation	- Operation in the low loss window of 0.2 dB/km @ 1,550 nm but high dispersion @ 1,550 nm - Development of single-frequency distributed feedback (DFB) 1,310 nm and 1,550 nm lasers
Late 1980s to mid-1990s	4 th Generation	- Multichannel erbium-doped fiber amplifiers (EDFAs) @ 1,550 nm deployed - New dispersion-shifted fiber yields zero dispersion @ 1,550 nm and 0.5 dB/km loss @ 1,310 nm - Multichannel (WDM) @ 1,550 nm. Number of channels and channel spacing limited by fiber four-wave mixing (FWM)
Mid-1990s	5 th Generation	- AT&T true wave fiber and Corning large optical core fiber reduce fiber FWM - Optical solitons, dispersion compensation

6.1.1 Digital System

Digital data transmission is more advantageous than analog transmission. The analog signal is converted into digital pulses by pulse code modulation (PCM) after passing through the stages of sampling, quantizing, and encoding. The digital data in the form of pulses is then line coded with any one of the following techniques (for details refer to Section 4.4):

1. Return-to-zero (RZ).
2. Non-return-to-zero (NRZ).
3. Phase encoded (PE).

Line coding is important for data security and the precise extraction of timing signal. The line-coded data is then modulated. The output of the modulator operates the optical source and the optical signal is launched into optical fiber cable. At the receiver end, the optical detector converts the received optical signal to an electrical signal, which is then amplified, decoded, and demodulated to get back the original transmitted data.

For digital system, the total system rise time should not exceed 70% of a bit interval (data interval unit) for non-return-to-zero (NRZ) data or 35% of a bit interval for return-to-zero (RZ) data. In other words, for good reception of signal

$$T_{\text{sys}} \leq 0.7 \times \text{Pulse width} \quad (6.1)$$

where for NRZ data format

$$\text{Pulse width} = \frac{1}{\text{Bit rate}} \quad (6.2)$$

and for RZ data format

$$\text{Pulse width} = \frac{1}{2 \times \text{Bit rate}} \quad (6.3)$$

6.1.2 Analog System

Analog transmission avoids the cost and complexity of digital equipments. Irrespective of some disadvantages, analog systems are used for short-distance communication, such as Cable TV (CATV) and direct cable television. The analog baseband signal is modulated by the following systems:

1. Direct modulation.
2. Subcarrier modulation.

Direct intensity modulation (D-IM) is used for a single baseband analog signal. However, when a number of baseband channels are to be transmitted on a single optical fiber link, subcarrier modulation is used. The baseband signals are modulated with radio frequency (RF) subcarriers, then multiplexed and intensity-modulated. This way, a two-level modulation [Amplitude modulation intensity modulation (AM-IM), frequency modulation intensity modulation (FM-IM), or phase modulation intensity modulation (PM-IM)] is done. This scheme gives improved signal-to-noise ratio. If instead of RF, a microwave carrier is used, then it is called *subcarrier multiplexing* (SCM).

6.1.3 Local Area Networks

Local area network or LAN interconnects large number of individual computers, terminals, and devices, such as printers, fax modems within a local area (e.g., a work group, department, university campus, small company,

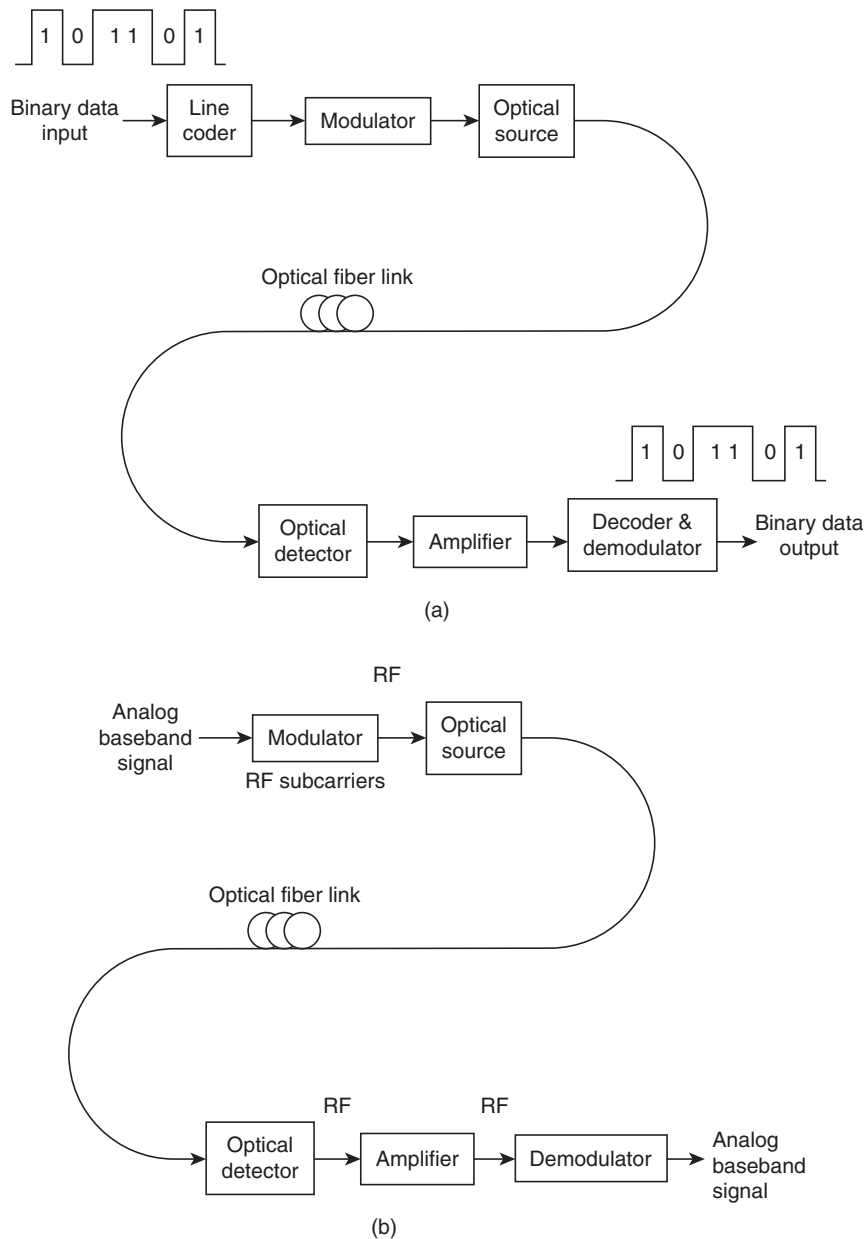


Figure 6.2 | Point-to-point fiber-optic link: (a) Digital; (b) analog.

office, etc.). Different LANs may interface with each other and create corporate network. Any user can randomly transmit data to any other user. The data transmission through optical fiber cable takes place in the form of data packets. These data packets carry header information. It is then routed to a particular device depending upon header information.

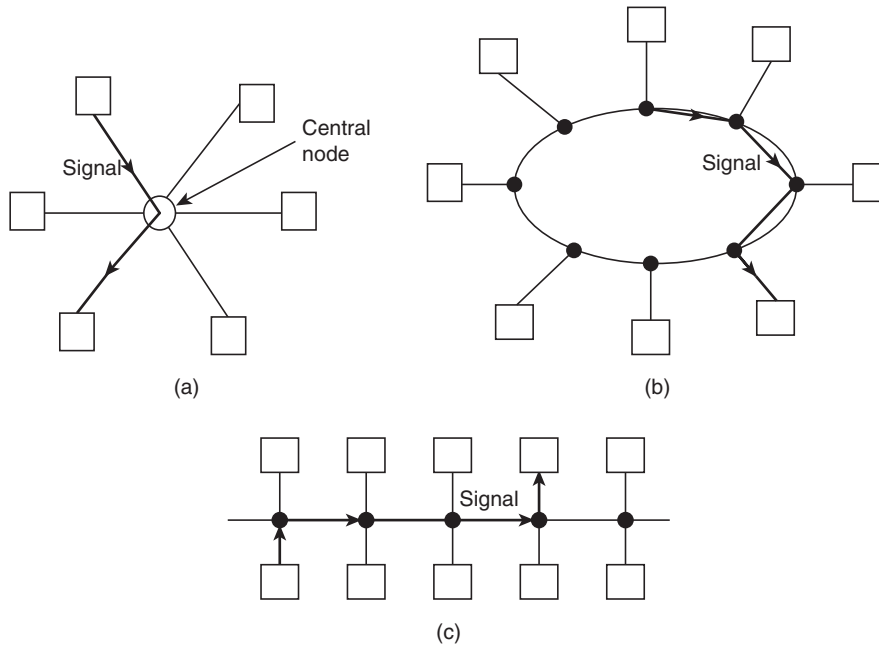


Figure 6.3 | LAN topologies: (a) Star; (b) ring; (c) data bus.

The three basic LAN approaches are: star, ring, and data-bus topology. In star topology, all signals pass through a central node as shown in Fig. 6.3(a). In ring topology [Fig. 6.3(b)], nodes are connected by point-to-point links to form a closed ring. Data packets are sent from one node to next node through optical fiber cable, and the device accepts the data if its address is found in the header. In data-bus topology [Fig. 6.3(c)], all nodes are connected through common transmission medium, that is, a single fiber cable carries multichannel data throughout the area of service and power is distributed through the optical taps, which divert small fraction of optical power to each device.

6.1.4 Distribution Networks

Optical fiber system can be used to distribute the information to a group of subscribers, for example, CATV, distribution of telephone signals from central switching office, etc. The common distribution network configurations are hub topology and bus topology. Several hubs can share the single optical fiber cable because of its large bandwidth for connection with main hub as shown in Fig. 6.4. Bus topology is same as shown in Fig. 6.3(c).

6.1.5 System Design Considerations

The goal of any optical system is to deliver virtually error-free long-distance communication at low cost. While designing an optical communication network systems, following criteria are considered^[1,2]:

1. **Signals to be transmitted:** This will determine the types of equipment, that is, source and detector, and the bit rate on the fiber-optic cable.
2. **Type of fiber:** The two types of fiber are:
 - (a) Multimode.
 - (b) Single mode.

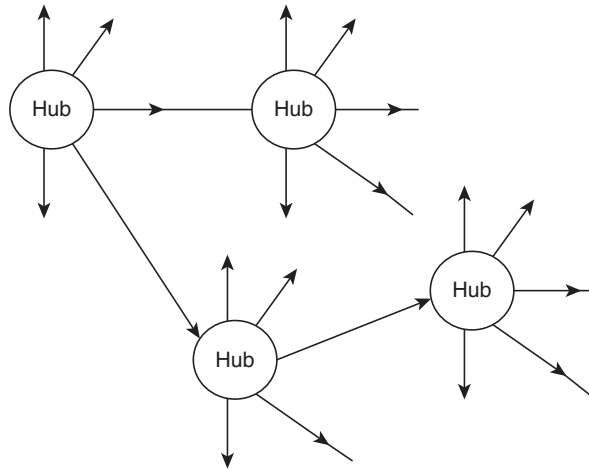


Figure 6.4 | Distribution network hub topology.

Due to high susceptibility to dispersion effects, the bandwidth of multimode fibers is less as compared to single-mode fiber (SMF). Multimode fibers are not suitable for coarse and dense wavelength division multiplexing (WDM). For long-haul applications, SMF is preferred.

3. **Number of fibers available:** The optical components and the electronics associated with wavelength division-multiplexed optical fiber system and the cost of system are very high. This can radically increase the price of an optical system. If a new infrastructure of fiber network is to be established, it is better to lay multiple fibers to avoid multiplexing.
4. **Power budget of the system:** The power budget of optical fiber system includes
 - (a) Loss or attenuation along the fiber.
 - (b) Losses due to fiber connectors, splices, patch panels.

As previously discussed, attenuation is a wavelength-dependent parameter. Around 1,300 nm approximately, the loss is 0.4 dB, whereas near 1,550 nm, it is 0.3 dB/km. A fiber link is established by connecting several pieces of fibers through fiber connectors and splices. At every joint, the signal gets attenuated. Approximately loss at each joint is 0.5 dB. All these losses are added with system margin of around 6–7 dB.
5. **Dispersion limit of the system:** The broadening of pulse limits the maximum data rate of the system and decides the maximum bandwidth limit of fiber-optic system. This effect is more pronounced in multimode and high data-rate systems.

The choice of various components depends on:

1. The distance between transmitter and receiver.
2. Information rate.
3. Cost of system.
4. Reliability of components.
5. Possibility of upgrading.

The parameters to be considered in the selection of different components of optical communication link are listed in Table 6.2.

Table 6.2 Parameters to be considered in the selection of components of communication link

Single-mode/multimode fiber	LED/LASER source	PIN/APD photodetector
Core size	Output power	Operating wavelength
Core refractive index profile	Emission wavelength	Responsivity
Numerical aperture	Emission pattern	Speed
Attenuation	Spectral line width	Sensitivity
Dispersion	Number of modes	

The power level reaching the detector determines the performance of the link in terms of *bit error rate* (BER). Moreover, the rise times of the source, detector, and fiber determine the bandwidth available for transmission. Therefore, the design of a fiber-optic system is usually carried out using power budgeting and rise time budgeting.

6.1.6 System Design Steps

- Operating wavelength selection:**
Short distance: 800–900 nm
Long distance: around 1,300 nm or 1,550 nm for low attenuation
- Photodetector selection:** Compared to APDs, PINs are less expensive and more stable with temperature. However, they have less sensitivity.
- Optical source selection:**
 - LEDs: 150 (Mb/s)-km with wavelength 800–900 nm and larger than 1.5 (Gb/s)-km with wavelength 1,330 nm.
 - InGaAsP lasers: 25 (Gb/s)-km with wavelength 1,330 nm and ideally around 500 (Gb/s)-km with wavelength 1,550 nm. 10–15 dB more power. However, these are more costly and have complex circuitry.
For data rate 2.5 (Gb/s)-km or higher, a laser must be used.
- Optical fiber selection:** SMFs are used with lasers or edge-emitting LEDs. Multimode fibers are used with LEDs. NA and core-cladding index difference should be optimized for any particular application.

6.2 Power Budgeting

A typical point-to-point link model with various power losses is shown in Fig. 6.5. The source emitting an input power P_i couples light into the optical fiber with the help of coupler. At this point, there is a coupling loss l_c . The light propagates through the fiber, suffering a transmission loss αL over a fiber of length L . In the fiber link wherever splices are there, the loss due to splicing is l_s . Finally, the light signal reaches the detector. The light falling on the detector is P_o .

If P_i and P_o are the optical powers entering the fiber and leaving the fiber, respectively, then the total loss in dB is given as

$$\begin{aligned} \text{Total loss} &= P_t = 10 \log \frac{P_o}{P_i} \\ \Rightarrow P_t &= P_i - P_o \end{aligned} \quad (6.4)$$

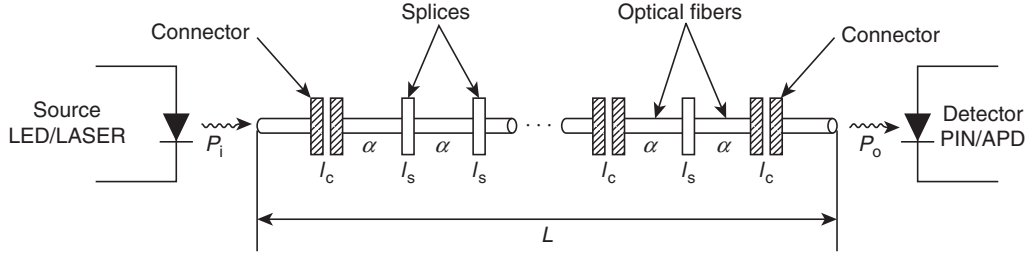


Figure 6.5 | Point-to-point link model with various power losses.

Now

$$P_t = \text{Connector loss} + \text{Splice loss} + \text{Fiber attenuation} + \text{System margin} \quad (6.5)$$

$$P_t = (N_c \times l_c) + (N_s \times l_s) + (\alpha \times L) + \text{System margin}$$

where P_i is the link input power, P_o is the link output power, N_c is the number of connectors, l_c is the connector coupling loss, N_s is the number of splices, l_s is the splicing loss, L is the length of the fiber, and α is the attenuation constant of fiber.

System Margin: Apart from the actual losses, while designing a link, usually a margin of 6–8 dB is incorporated for the splices or components that may have to be added in future and also to allow for any deterioration of system components due to aging.

6.3 Rise Time Budgeting

Rise time budget is used to check the maximum bit rate at which the system can work properly. In other words, we can say that it determines the dispersion limitation of an optical fiber link. Dispersion is the broadening of pulse, which causes overlapping of pulses, that is, intersymbol interference in addition to limiting the maximum data rate.

Manufacturers' data sheet provides various information on bandwidth of the fiber, namely multimode pulse delay (ns/km), material dispersion (ns/nm·km), and/or 3-dB optical bandwidth (MHz km). For digital systems, a complete rise time analysis – including the source, fiber, and detector rise time – is required.

System rise time is given as

$$T_{\text{SYS}} = 1.1 \sqrt{T_{\text{TX}}^2 + T_{\text{RX}}^2 + T_{\text{FIBER}}^2} \quad (6.6)$$

where T_{SYS} is the system rise time, T_{TX} is the transmitter rise time, T_{RX} is the receiver rise time, and T_{FIBER} is the fiber rise time.

The rise time for different types of fibers can be written as follows:

1. Modal distortion:

(a) Multimode step index fiber:

$$T_{\text{MODAL}} = \frac{n_1 \Delta}{c} L \quad (6.7)$$

(b) Multimode graded index fiber:

i. Parabolic index

$$T_{\text{MODAL}} = \frac{n_1}{2c} \Delta^2 L \quad (6.8)$$

ii. Optimum profile

$$T_{\text{MODAL}} = \frac{n_1}{8c} \Delta^2 L \quad (6.9)$$

2. Material dispersion:

$$T_{\text{MATERIAL}} \cong 85 L \Delta \lambda \text{ ps} (\lambda_0 \sim 850 \text{ nm}) \quad (6.10a)$$

$$T_{\text{MATERIAL}} \cong 0.5 L \Delta \lambda \text{ ps} (\lambda_0 \sim 1,300 \text{ nm}) \quad (6.10b)$$

$$T_{\text{MATERIAL}} \cong 20 L \Delta \lambda \text{ ps} (\lambda_0 \sim 1,500 \text{ nm}) \quad (6.10c)$$

Here L is in kilometers and $\Delta \lambda$ in nanometers. The total fiber rise time is given as

$$T_{\text{FIBER}} = \sqrt{T_{\text{MODAL}}^2 + T_{\text{MATERIAL}}^2} \quad (6.11)$$

where T_{MODAL} is the rise time resulting from intermodal distortion and T_{MATERIAL} is the rise time resulting from material dispersion. If spectral width of fiber is given, then

$$T_{\text{FIBER}} = D \times \Delta \lambda \times L \quad (6.12)$$

where D is the fiber dispersion in ps/km nm, $\Delta \lambda$ is the spectral width in nm, and L is the length of fiber. Now

$$T_{\text{FIBER}} \cong 2L\Delta\lambda (\lambda_o \sim 1,300 \text{ nm}, \lambda_z \sim 1,300 \text{ nm}) \quad (6.13a)$$

$$T_{\text{FIBER}} \cong 16L\Delta\lambda (\lambda_o \sim 1,500 \text{ nm}, \lambda_z \sim 1,300 \text{ nm}) \quad (6.13b)$$

$$T_{\text{FIBER}} \cong 2L\Delta\lambda (\lambda_o \sim 1,550 \text{ nm}, \lambda_z \sim 1,550 \text{ nm}) \quad (6.13c)$$

where λ_0 is the operating wavelength and λ_z is the zero dispersion wavelength.

6.4 Multichannel Concept

The demand for bandwidth has increased by manifolds over the past few years and is likely to continue in the near future. To be able to meet customer's needs, telecommunication companies have to find solutions for increasing their channel capacity at the lowest possible cost. Wavelength division multiplexing (WDM) seems to be a promising solution. In WDM, several different wavelengths are transmitted over one SMF at the same time. The main advantages of WDM systems are their capability of being compatible to existing hardware, being modular, and having the ability of saving a lot of costly equipment if designed properly.^[3]

The following are the different ways to increase channel capacity:

1. **Installing new fiber:** By installing more fibers, more channels can be added to increase the bandwidth of a system. But because of high cost associated with laying new fiber, this approach is impractical. The approximate estimated cost for adding a cable is around INR 35,00,000 per kilometer.
2. **Increasing the bit rate:** Another approach is increasing the bit rate. But the required faster electronic circuitry for high data-rate application is more complex and expensive. Moreover, the available electronic circuits have a speed limit of approximately 100 GHz. The other technical issues are chromatic dispersion and non-linear effects. With increase in data rate and transmitted power, both the effects increase almost linearly.
3. **Using TDM:** Time division multiplexing (TDM) is a good option to increase the data rate and hence throughput of fiber-optic system. The existing low-speed channels get divided into smaller bits and then

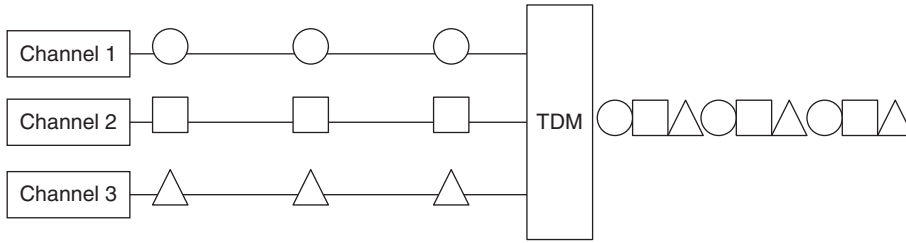


Figure 6.6 | Time division multiplexing system.

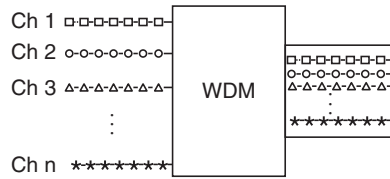


Figure 6.7 | WDM system with n channels.

transmitted piece after piece over the fast channel as shown in Fig. 6.6. On the input side, there are three channels (channels 1–3) that work with normal speed. One channel can carry these four signals by splitting each channel into small section of bits and then transmitting a piece of each channel in a round robin manner.

4. **Using multiple wavelengths:** Presently in a practical single-channel system, data rate of 100 Gbps is utilized. Around 1,550 nm, an SMF can carry the data at the rate of around 25,000 Gbps. But the devices in transmitter and receiver systems, such as lasers, modulators, switches, etc., cannot work at such a faster rate. Thus, we can say that the high data rate is not limited due to fiber, but due to the optical components in transmitter and receiver sections. The available optical communication band is 1,450–1,650 nm. So using a single wavelength over a fiber is wastage of its capacity.

To fully utilize the bandwidth capacity of optical fiber, it is logical to use several wavelengths. If n different wavelengths are used, as shown in Fig. 6.7, the capacity of the channel can be increased by a factor of n , without changing the modulation speed of the single laser.

6.4.1 Wavelength Division Multiplexing (WDM)

Figure 6.8 illustrates the basic concept of a WDM. Monochromatic optical signals of wavelengths $\lambda_1, \lambda_2, \dots, \lambda_N$ are generated by laser diodes (LDs) and sent through N fibers to a multiplexer (MUX). The MUX combines these input signals into a polychromatic output signal through the process of multiplexing. The large bandwidth capacity of optical fiber is exploited through the multiplexing process. This multiplexed, polychromatic signal is launched into a single optical fiber for transmission. At the destination, a demultiplexer (DEMUX) separates the polychromatic signal into original monochromatic wavelengths $\lambda_1, \lambda_2, \dots, \lambda_N$.

The DEMUX must be designed such that the center wavelengths of channels should be same as the original wavelengths and channel spectral widths, $\Delta\lambda_N$. Therefore, to avoid overlapping of the channels, the pass bands should be chosen to accommodate system tolerances. The MUX and DEMUX are not tunable devices; therefore, to obtain good performance, the design and fabrication should be perfect.

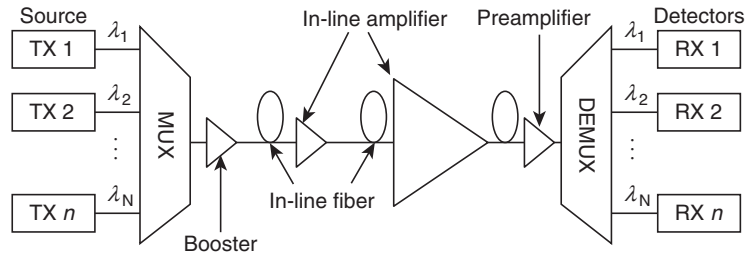


Figure 6.8 | Typical WDM system.

Depending on application needs, different types of WDM systems are developed, such as point-to-point long-distance transmission, local access network, reconfigurable network, etc. Each of these systems needs different WDM components.

6.4.2 Advantages of WDM

WDM has the following advantages over the other presented approaches to increase the capacity of a link:

1. With the existing available infrastructure, high-capacity link can be established with WDM because it works with single-mode communication fiber and the already installed low-speed equipment.
2. *It is scalable:* The number of channels can be increased easily to increase the transmission capacity instead of changing the technology itself. Changing the technology is a costly affair. In WDM, to get high-capacity operation, telecom companies can pay for the extra bandwidth needed.
3. *It is transparent:* Each optical channel can have its own data format for transmission purpose. It is independent of transmission protocol. The data can be synchronous or asynchronous, analog or digital, and so on.
4. Depending upon the customer need, the capacity of channel can be easily increased. The fiber channel can be distributed part-by-part among number of customers on a lease basis.
5. *Wavelength switching:* It allows for reconfigurable optical layer. To enable wavelength multiplexing, there must be an array of components that are wavelength-dependent and perform various functions.

Key components of WDM systems are as follows:

1. Optical add/drop MUXs.
2. Optical cross connects (OXC)s.
3. Wavelength converters.
4. Couplers.
5. Switchers.
6. Fused fiber directional couplers.
7. Mach–Zender interferometer-based MUXs.
8. Fiber grating couplers – reflective and transmissive.

The most expensive component is the fiber. The cost involves cost of fiber and installation cost.

6.4.3 Special Case of WDM: Dense WDM

Rapid and reliable communication networks are important in every sector, such as business, entertainment, government, academia, etc. Therefore, it is difficult for telecommunication industry to meet the increasing demand

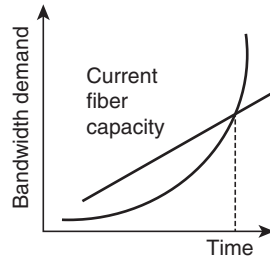


Figure 6.9 | Fiber capacity and increasing bandwidth demand.

of bandwidth with time as shown in Fig. 6.9. The predictions related to future requirement of fiber capacity have proven wrong. The advent of the Internet alone has put much burden on fiber-optic system. Moreover, there are millions of users today and the number is increasing drastically day by day.

To meet the higher channel capacity requirement, there are three possible solutions as follows:

1. Installation of new fiber.
2. Adoption of new technology such as TDM to achieve faster bit rates.
3. Deploy dense WDM.

Instead of using different wavelengths (e.g., the three main communication windows at 850 nm, 1,300 nm, and 1,550 nm), different wavelengths can be chosen nearer together. In the late 1990s, the minimum spacing achieved was 25 GHz, which is equivalent to a difference in channel wavelength of approximately 0.6 nm. Generally, if the spacing between the channels is below 200 GHz, then it is called as dense WDM (DWDM).

6.4.3.1 Advantages of DWDM Over General WDM

1. More closely spaced channels gives rise to higher capacity operation. In DWDM, the channel capacity of single fiber can be increased 16 times, and a throughput of 40 Gbps can be obtained. This new technology along with network management systems and add/drop multiplexers enable the system designer to meet the increasing demand of channel capacity at much lower cost without installation of new fiber.

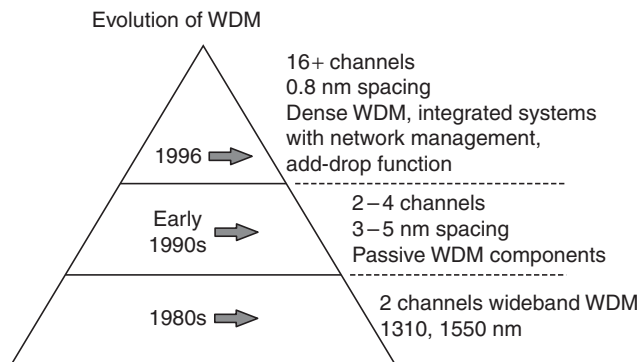


Figure 6.10 | Evolution of WDM technology.

2. The erbium-doped fiber amplifier (EDFA) can amplify all the channels in DWDM system simultaneously. If the channels lay around the 1,550 nm communication window, all channels can be amplified simultaneously using optical amplifiers, such as EDFAs. This saves additional equipment and therefore money.

As shown in Fig. 6.10, the early application of WDM was restricted to two widely spaced wideband channels. With the evolution of technology, the number of channels are increasing and the spacing between channels is reducing. Closely spaced more number of WDM channels is nothing but DWDM.

In DWDM technology, a composite optical signal carries multiple information streams and each is transmitted on a separate optical wavelength. A number of wavelengths can be densely packed and integrated into a transmission system, with extremely high-frequency signals in the 192–200 THz range. This concept is being used in recent technologies.

6.4.4 WDM Standards

The International Telecommunication Union (ITU)^[4] drafted standard G.692 entitled “Optical Interfaces for Multichannel Systems with Optical Amplifiers.” This covers the long-distance point-to-point WDM systems using STM-4, STM-16, and/or STM-64 on 4, 8, 16, or 32 channels. The maximum link distance for a system without amplifiers is 160 km or up to 640 km with optical amplification. A wavelength grid is based on 100 GHz spacing and a reference frequency of 193.1 THz (i.e., 1,553.5 nm). Users are free to use any wavelength on the grid in an arbitrary way. Also the users are free to select any part of the spectrum and unequally spaced channels are allowed provided the channel wavelengths are situated on the grid.

6.5 WDM Components

6.5.1 Coupler/Splitter

A coupler combines light into or splits light out of a fiber. A splitter is a type of coupler that divides the optical signal on one fiber to two or more fibers. The most common splitter is a 1×2 splitter, as shown in Fig. 6.11(a). The *splitting ratio* α is the amount of power that goes to each output. For a two-port splitter, the most common splitting ratio is 50:50, though splitters with any ratio can be manufactured. Combiners [see Fig. 6.11(b)] are the reverse of splitters, and when turned around, a combiner can be used as a splitter. An input signal to the combiner suffers a power loss of about 3 dB. A 2×2 coupler [see Fig. 6.11(c)], in general, is a 2×1 combiner followed immediately by a 1×2 splitter, which has the effect of broadcasting the signals from two input fibers onto two output fibers.

The $N \times M$ coupler is a simple fundamental device that is used in switching optical signals. A common construction is the *fused biconical tapered* fiber coupler. This is fabricated by twisting together, melting, and pulling two SMFs so they get fused together over a uniform section of length W , as shown in Fig. 6.12.

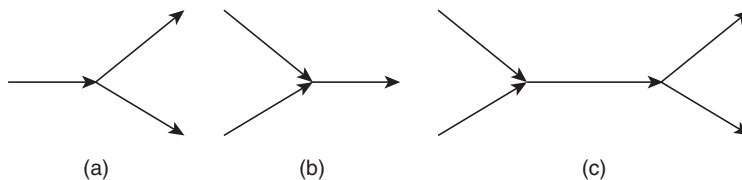


Figure 6.11 | (a) Splitter, (b) combiner, and (c) coupler.

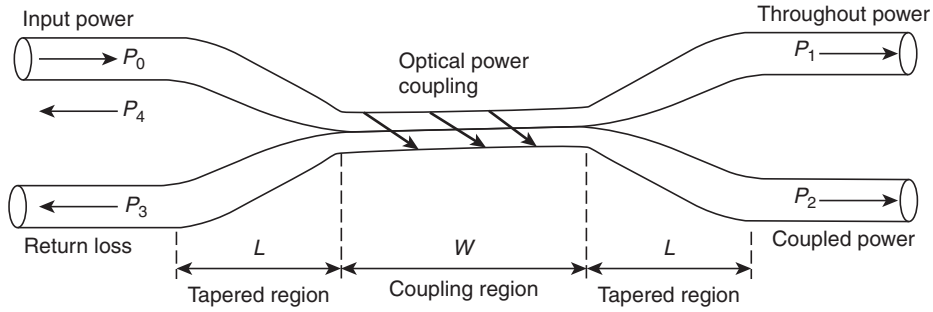


Figure 6.12 | Fused-fiber coupler.

Since the transverse dimensions are gradually reduced down to that of the coupling region when the fibers are pulled during the fusion process, each input and output fiber has a long tapered section of length L . The total draw length is $2L + W$. In Fig. 6.12, P_0 is the input power, P_1 is the throughput power, and P_2 is the power coupled into the second fiber. The parameters P_3 and P_4 are extremely low signal levels resulting from backward reflections and scattering due to bending in and packing of the device.

The optical power is coupled from one fiber to another through the following three parameters:

1. Axial length of coupling region.
2. Size of reduced radius r .
3. The difference in the radii (Δr) of the two fibers in the coupling region.

The performance of the optical coupler is indicated by splitting ratio or coupling ratio, which indicates the percentage of division of optical power between the output ports. If P_0 is the input power and P_1 and P_2 are the output powers, then

$$\text{Splitting ratio} = \left(\frac{P_2}{P_1 + P_2} \right) \times 100\% \quad (6.14)$$

In addition to the 50:50 power split incurred in a coupler, the signal also experiences *return loss*. If the signal enters an input of the coupler, roughly half of the signal's power goes to each output of the coupler. However, a small amount of power is reflected in the opposite direction and is directed back to the inputs of the coupler. This measure of input isolation is also called *crosstalk*. Typically, the amount of power returned by a coupler is 40–50 dB below the input power:

$$\text{Return loss or crosstalk} = 10 \log \left(\frac{P_3}{P_0} \right) \quad (6.15)$$

Another type of loss is *insertion loss*. One source of insertion loss is the loss incurred when directing the light from a fiber into the coupler device. Ideally, the axes of the fiber core and the coupler input port must be perfectly aligned, but full perfection may not be achievable due to the very small dimensions. Insertion loss is a loss for a particular port-to-port path. For the path from input port i to the output port j , it is given as

$$\text{Insertion loss} = 10 \log \left(\frac{P_i}{P_j} \right) \quad (6.16)$$

The Passive Star Coupler (PSC) is a multiport device in which light coming into any input port is broadcast to every output port. The PSC is attractive because the optical power gets equally divided amongst the output ports. Mathematically it is given as ^[9]

$$P_{\text{out}} = \frac{P_{\text{in}}}{N} \quad (6.17)$$

where P_{in} is the optical power introduced into the star by a single node, N is the number of output ports of the star, and P_{out} is the optical power received at each output port. Note that this expression ignores the *excess loss* caused by flaws introduced in the manufacturing process that the signal experiences when passing through each coupling element. The excess loss is defined as the ratio of the input power to the total output power. For a 2×2 coupler, it is given as

$$\text{Excess loss} = 10 \log \left(\frac{P_0}{P_1 + P_2} \right) \quad (6.18)$$

6.5.2 Optical Circulator

6.5.2.1 Isolators

An optical isolator is an optical component that allows light transmission in only one direction. It is used to block the unwanted back-flow of optical signal. The operation of device is based on the Faraday effect. When a beam of plane-polarized light is passed through any transparent solid or liquid placed in a uniform magnetic field, in a direction parallel to magnetic lines of force then the transmitted light remains plane-polarized but the plane of polarization gets rotated by an angle proportional to the magnetic field intensity. This optical rotation is called Faraday effect or Faraday rotation.

Figure 6.13 shows the principle of operation of an isolator.^[3,5] The incoming light has only vertical polarization state. It is passed through a polarizer, which passes only light energy in the vertical polarization state and blocks light energy in the horizontal polarization state. The Faraday rotator is used to rotate the polarization state, clockwise by 45° , irrespective of the direction of propagation. This is again followed by polarizer that passes only polarization states with 45° orientation. Thus, the light signal from left to right is passed through the device without any loss. On the other hand, light entering the device from the right due to the reflections, with the same 45° orientation, is rotated another 45° by the Faraday rotator, and thus is blocked by the first polarizer. Practically, polarization state of input light cannot be controlled; so an isolator should work irrespective of input polarization state.

Figure 6.14 shows the working principle of a polarization-independent device. The isolator consists of special walk-off polarizer (SWP) and half-wave plate. Incoming light with arbitrary polarization state is sent through SWP. The SWP is used to split the light into two components, horizontal and vertical. Each component goes through a Faraday rotator, which rotates the polarization state by 45° . The Faraday rotator is followed by a half-wave plate. The half-wave plate is similar to the Faraday rotator except that it rotates the light coming from left to right by 45° clockwise and the light coming from right to left by 45° . The combination of Faraday rotator and

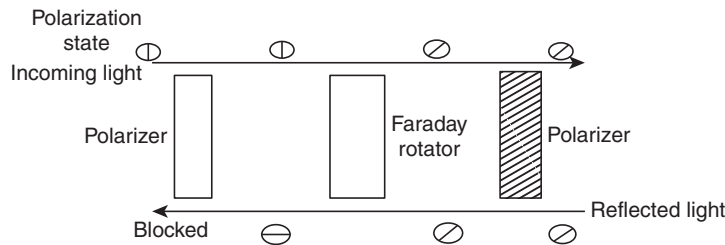


Figure 6.13 | Principle of operation of an isolator.

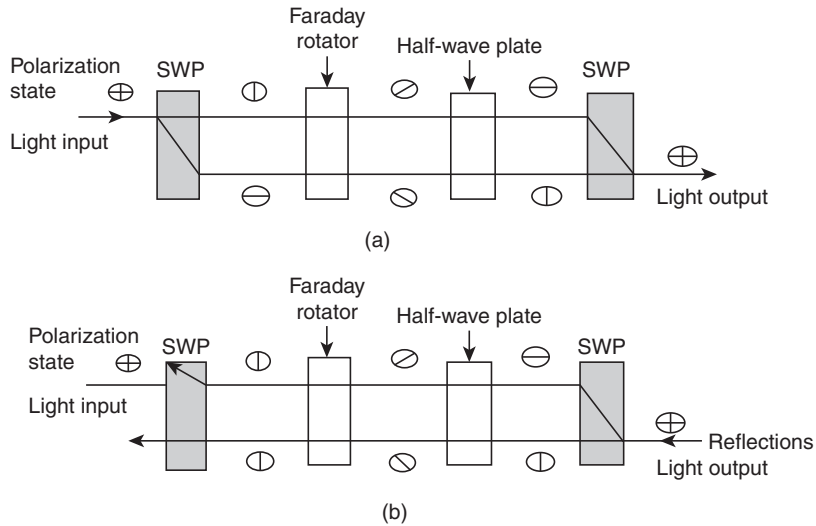


Figure 6.14 | Propagation of light in a polarization-independent optical isolator (a) from left to right and (b) from right to left.

half-wave plate changes the polarization state from vertical to horizontal and vice versa, and then these signals are recombined by another SWP at the output.

For a back-reflected signal in reverse direction, Faraday rotator and half-wave plate cancel each other's effect and the polarization state remains unchanged. Hence it is not recombined by the SWP at the input.

6.5.2.2 Circulators

An optical circulator is an isolator having three or more ports. In contrast to an isolator, which causes loss in the isolation direction, a circulator collects the light and directs it to a non-reciprocal output port. Several possible circulator configurations are shown in Fig. 6.15.

A four-port circulator operation is illustrated in Fig. 6.15(a). The optical input is sequentially passed from one port to other, for example in the sequence port 1 → port 2 → port 3 → port 4 → port 1. This configuration is a strict-sense circulator, since every input port has a specific non-reciprocal output port.

A non-strict-sense circulator having any number of ports greater than two is illustrated in Fig. 6.15(b). In this case, each input port has a specific non-reciprocal output port. The last port is exceptional and the light input to the last port is lost. There is no connection between the first and the last ports.

The three-port non-strict-sense circulator is illustrated in Fig. 6.15(c). In some telecommunication applications, return of light from port 3 to port 1 is not required. An example of this is light reflected from fiber Bragg grating (FBG) in dispersion compensation or optical wavelength drop-and-add demultiplexer/multiplexer (OADM). (Refer to Section 6.6.10.4 for more details.) Thus, this circulator is useful in such applications.

Optical circulators are non-reciprocal optical devices. Light from one fiber is routed to another fiber depending upon the direction of light propagation by these devices. The light never goes to other ports.

There are number of applications of optical circulators as listed below:

1. To convert an existing unidirectional fiber-optic communication link to a full duplex communication link by installing an optical circulator at each end of the link.

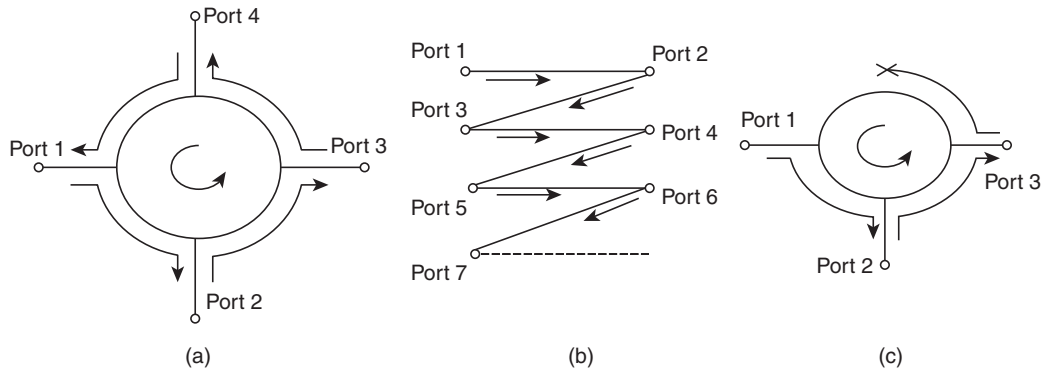


Figure 6.15 | Optical circulator connections: (a) Strict-sense circulator with four ports. (b) Non-strict-sense circulator in ladder topology. (c) Non-strict-sense three-port circulator.

2. Optical circulators are also used in:
 - (a) Fiber amplification systems.
 - (b) WDM networks.
 - (c) Optical time-domain reflectometers (OTDRs) and for test instruments.

6.5.2.2.1 Working

A signal entering at port 1 exits only at port 2, and a signal entering at port 2 exits only at port 3. A signal entering at port 3 should exit only at port 1 in a “perfect” circulator. But in many applications, this last condition is unnecessary. Thus, most commercial circulators are designed to be “imperfect” so as to absorb any signal directed into port 3. The typical specifications of a circulator are listed in Table 6.3.

Table 6.3 | Typical specifications of circulator

Parameter	Value
No. of ports	3
Central wavelength (nm)	1,310 or 1,550
Typ. insertion loss ± 20 nm, 0~70°C, all SOP(dB)	0.8
Max. WDL(dB)	0.20
Directivity (dB)	65
Maximum polarization-dependent loss (PDL) (dB)	0.10
Maximum PMD (ps)	0.10
Return loss (dB)	60
Fiber length (m)	1.0
Minimum tensile load (N)	5
Power handling (mW)	320
Operating temperature (°C)	0~75
Storage temperature (°C)	-30~85
Dimension (mm)	Ø6.0 × L60.0

6.5.3 Optical Add/Drop Multiplexer

Optical multiplexers are components specifically designed for WDM systems. The main function of an optical multiplexer is to couple two or more wavelengths into the same fiber. On the other hand, the demultiplexer separates a multiplicity of wavelengths in a fiber and directs them to many fibers (Fig. 6.16).

On a point-to-point optical fiber link, individual wavelengths could be removed, inserted, or routed to add more flexibility in the optical system, using OADM.^[6,7] They can be used at different points along the optical link to insert or remove route-selected channels, increasing the network flexibility.

An add/drop multiplexer consists of three ports: input port, output port, and add/drop port. Assume that there are $\lambda_1, \dots, \lambda_{N-1}, \lambda_N$ wavelengths at the input and λ_i is dropped, and then at add/drop port, λ_i is diverted. At the output port, all the other wavelengths ($\lambda_1, \dots, \lambda_{i-1}, \lambda_{i+1}, \dots, \lambda_{N-1}, \lambda_N$) will be present except λ_i as shown in Fig. 6.17(a). When wavelength λ_i is added, then at the input port, λ_i should not be there (i.e., input has $\lambda_1, \dots, \lambda_{i-1}, \lambda_{i+1}, \dots, \lambda_{N-1}, \lambda_N$). At the output port, all the wavelengths ($\lambda_1, \dots, \lambda_{N-1}, \lambda_N$) will be present as shown in Fig. 6.17(b).

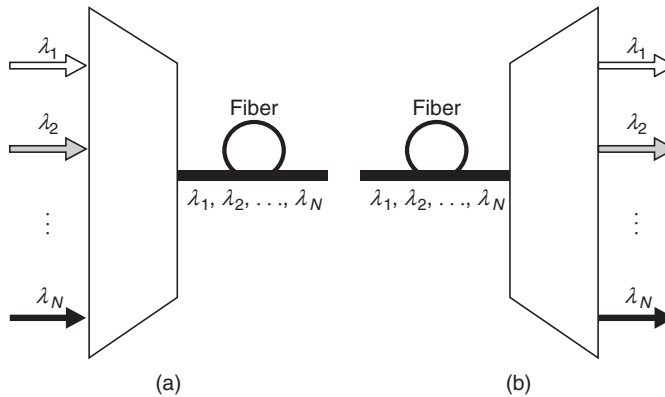


Figure 6.16 | (a) Optical multiplexer and (b) optical demultiplexer.

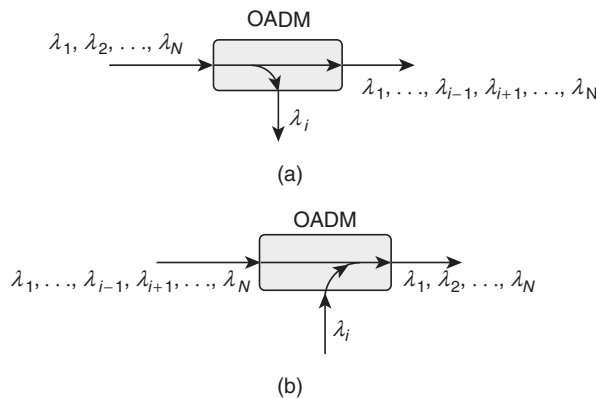


Figure 6.17 | OADM: (a) Drop function in OADM; (b) add function in OADM.

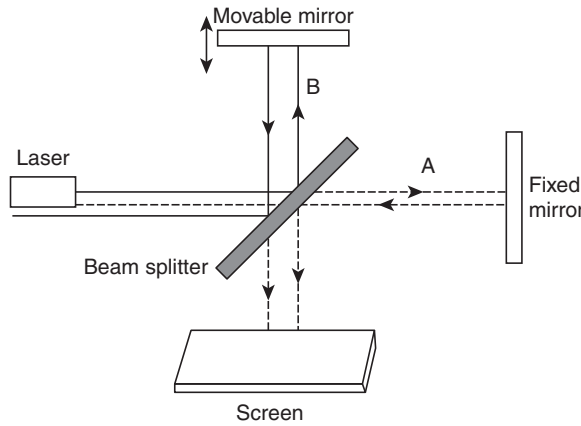


Figure 6.19 | Michelson interferometer.

therefore their recombination may result in an interference pattern. The commonly used arrangement is shown in Fig. 6.19, which is referred to as a *Michelson interferometer*.

This device can be used to calculate the wavelength of a laser source, as well as the refractive index of CO₂ and air. As per Fig. 6.19, when light from the laser strikes the beam splitter, it is divided in two half parts. Half of the light is sent along path *B* toward the movable mirror and the other half is sent down path *A* toward a fixed mirror. The light then returns from each mirror and is recombined at the output and viewed on a screen. If the path lengths of *A* and *B* are different, the light waves will be at different phases when they are recombined. Depending upon the path-length difference, the two waves will combine and give rise to interference pattern.

As the position of mirror along path *A* is fixed, path *A* cannot be altered. However, by moving the mirror in horizontal direction, path *B* can be changed. With the help of a screw and lever assembly attached to the rear of the mirror at the end of path *B*, path length can be increased or decreased.

Alternatively, by changing the medium through which the light travels, path length *B* can be changed instead of moving the mirror.

6.5.4.1 Theory: Translation

In the relationship between the wavelength of the input light and the translation of the movable mirror, the amplitude *A* and intensity *I* of a wave are generally described as

$$A(x, t) = E \cos(\omega t - kx) \quad (6.19)$$

$$I \propto A^2 \quad (6.20)$$

The amplitude is split at the beam splitter and travels the distance between the beam splitter and the mirrors twice ($2kL_1$) and then to the projection screen (L_p), yielding

$$A_1(x, t) = \frac{E}{\sqrt{2}} \cos(\omega t - 2kL_1 - kL_p) \quad (6.21)$$

$$A_2(x, t) = \frac{E}{\sqrt{2}} \cos(\omega t - 2kL_2 - kL_p) \quad (6.22)$$

where half of the energy travels along each path and their phases are described by the distances to the mirrors L_{1i} and the projection screen L_p . By solving the above equations, the output of the interferometer is

$$I_p = I_0 \cos^2 [k(L_1 - L_2)] \quad (6.23)$$

By substituting the standard definition of the wave number k into Eq. (6.23) we get

$$k = \frac{2\pi}{\lambda}$$

We can see that maxima and minima (fringes and extinctions) will occur when the following conditions are met:

$$\text{max: } \Delta L = |L_1 - L_2| = \frac{m\lambda}{2}, \quad m = 0, 1, 2, \dots \quad (6.24)$$

$$\text{min: } \Delta L = |L_1 - L_2| = \frac{[m + (1/2)]\lambda}{2}, \quad m = 0, 1, 2, \dots \quad (6.25)$$

When the path length B is varied, due to path-length difference the phase of two beams relative to each other is also varied. When this phase shift is half of the wavelength or integer multiple of wavelength, we can have maximum output. If the two beams are completely out of phase (i.e., phase difference is 180°), then output will be minimum. Therefore, when path length B is increased or decreased by a distance equal to the wavelength of the light or half of the wavelength, then the entire fringe pattern can be seen. We have

$$2d = \lambda \quad (6.26)$$

and, therefore, a train of N fringes will behave as

$$\begin{aligned} 2d &= N\lambda \\ \Rightarrow \lambda &= \frac{2d}{N} \end{aligned} \quad (6.27)$$

where N is the total number of fringes in the train seen during a translation of length d .

6.5.4.2 Theory: Index of Refraction

Instead of physically moving the mirror along path B, a medium with higher refractive index may also be introduced in order to increase optical path length. An evacuated cell has optical path length equal to its physical length t (light travels at speed c in a vacuum). A cell filled with a medium of refractive index n has path length L_i defined as

$$L_i = nt \quad (6.28)$$

Therefore, the increase ΔL_i in optical path length will be

$$\Delta L_i = 2(n-1)t \quad (6.29)$$

The “2” is present because the beam passes through the chamber twice during its trip. This increase then corresponds to the $2d$ from Eq. (6.27). Substitution of optical path length from Eq. (6.29) in Eq. (6.27) yields

$$\lambda = \frac{2(n-1)t}{N} \quad (6.30)$$

Thus, by counting the fringes and knowing the wavelength of the input light, the index of refraction n of any transparent material may be determined as follows:

$$n = \left(\frac{N\lambda}{2t} \right) + 1 \quad (6.31)$$

6.5.5 Fabry–Perot Interferometer

Consider a slab of thickness h of a dielectric material of refractive index n_2 , surrounded by a material of refractive index n_1 , as illustrated in Fig. 6.20. A plane wave of amplitude A_i is incident on the plate at point U with an angle θ to its normal. Part of the incident wave energy will be reflected at the boundary between the two media with the angle θ_{refl} to the normal, while the remaining will be refracted into the slab with the angle θ_{refr} . The refraction and reflection angles are related to the angle of incidence θ according to Snell's law, which is given by

$$n_1 \sin \theta = n_2 \sin \theta_{\text{refr}} \quad (6.32)$$

$$\theta = -\theta_{\text{refr}} \quad (6.33)$$

The refracted beam will then reach the second interface of the slab at point W , where part of its energy will be reflected back into the slab, the remaining part being refracted again in the medium of refractive index n_1 . The process will carry on with multiple reflections and refractions at the interfaces between the two dielectric media. This will ultimately lead to the creation of a number of secondary reflected and transmitted beams of decreasing intensity. Depending on their phase relation, the reflected or transmitted beams will interfere constructively or destructively. This structure represented in Fig. 6.20 is known as a Fabry–Perot interferometer and belongs to the category of “multiple-beam interferometers” or “interferometers by division of amplitude.” In practice, such a cavity, also known as etalon, can be realized by a slab of glass in air (in which case $n_2 > n_1 \cong 1$) or by an air gap between two glass plates ($n_1 > n_2 \cong 1$).

These general principles can be applied to the design of optical filters to be used in WDM systems as follows: A Fabry–Perot interferometer can be used as an optical band-pass filter if a high reflectivity, resulting in a high finesse, can be achieved. In practice, Fabry–Perot filters can be realized by inserting a slab of dielectric material between the tips of lensed optical fibers or by creating an air gap between two reflection-coated facets of optical fiber. The two configurations are represented in Figs. 6.21(a) and (b), respectively.

In the first case, wavelength tunability can be obtained by varying the angle of refraction θ_{refr} by rotating the dielectric slab. A transmission peak will be obtained at wavelength λ_p , provided

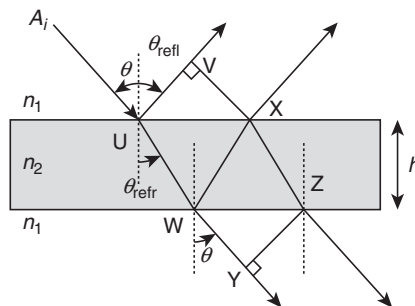


Figure 6.20 | Fabry–Perot interference principle.

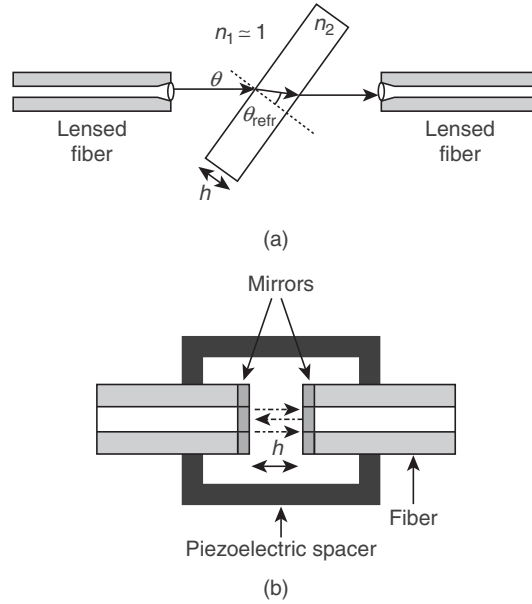


Figure 6.21 Fiber Fabry-Perot design where tuning is achieved (a) by rotating the slab and (b) by setting the cavity length as h .

$$\cos \theta_{\text{refr}} = k \frac{\lambda_p}{2n_2 h} \quad (6.34)$$

where k is an integer. In the second case, tuning is achieved by controlling the thickness of the cavity h using a piezoelectric element. A transmission peak is obtained at wavelength λ_p when the thickness is set to the value

$$h = k \frac{\lambda_p}{2} \quad (6.35)$$

where $k \in \mathbb{N}$.

In WDM systems, Fabry-Perot filters can be used to select a single channel out of N channels spaced at $\Delta\lambda$. It can also be used to select a group of regularly spaced channels from the WDM spectrum. The free spectral range (FSR) should be made equal to the spacing of the channels to be selected. Design tradeoffs for Fabry-Perot filters are their tuning range, the required reflectivity and bandwidth, and finesse can be achieved by proper choice of the cavity length h , angle of incidence θ , and materials and coatings.

Fabry-Perot filters are attractive due to their wavelength tunability. But the level of accuracy required in the adjustment of the cavity length or angle of incidence is high. Fiber Fabry-Perot filters are prone to instabilities and require wavelength stabilization schemes to be implemented.

6.5.6 Mach-Zehnder Interferometer

Devices such as optical filters, wavelength multiplexers, and demultiplexers, which manipulate the optical properties in wavelength domain, are essential in WDM optical communication system. One of the most

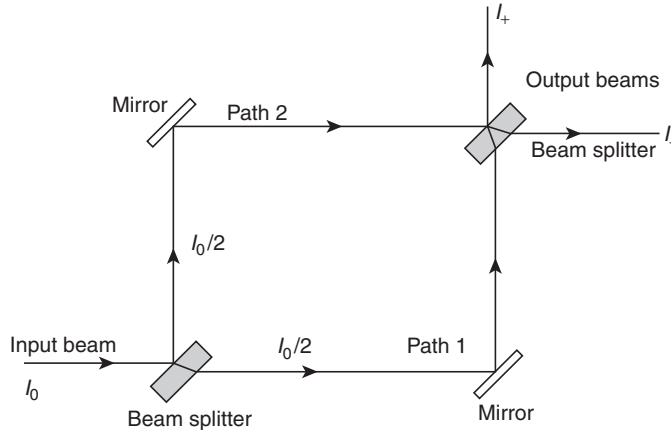


Figure 6.22 | Mach-Zehnder interferometer.

useful devices is Mach-Zehnder interferometer (MZI). A light beam is first split into two parts by a beam splitter and then recombined by a second beam splitter. Depending on the relative phase acquired by the beam along the two paths, the second beam splitter will reflect the beam with efficiency between 0 and 100%. The phase, on reaching the second beam splitter, is simply the path length divided by the wavelength, multiplied by 2π . On recombination at the beam splitter, if the two paths are of equal length, then the phases are equal.^[8,9]

The MZI consists of two 100% reflective mirrors and two 50% reflective beam splitters as shown in Fig. 6.22. It is an amplitude-splitting interferometer. The input beam is split into two parts; these two beams travel different paths (path 1 and path 2 shown in Fig. 6.22) through the interferometer. Beam in one path is phase shifted with respect to the beam in the other path. When these beams recombine, they cause a constructive or destructive interference. The phase shift to one of the beam paths serves as an amplitude modulator and controls the output signal from the device.

MZIs may be constructed in a variety of ways depending on the task they are designed to perform. The key concept in the working of MZI is the interference caused by the phase difference between the waves in the two arms. The steps involved are as follows:

1. The input beam is launched into the MZI.
2. By using beam splitter, it is split into two beams in a 50:50 ratio.
3. The two beams travel through different paths (path 1 and path 2).
4. The phase shift is introduced in one of the two paths (path 2 in Fig. 6.23).
5. Then the two beams are recombined. They produce an interference pattern that decides the output of MZI.

The typical modes of operation of MZI are ON and OFF modes:

1. If the two beams are in-phase (i.e., phase difference is 2π or multiple of 2π), then they interfere constructively and add (ON mode).
2. If the two beams are out-of-phase by an odd factor of π , then they interfere destructively and cancel each other (OFF mode).
3. If the two beams are not completely in-phase or out-of-phase, then the interference of two beams will result into some non-zero output and the amplitude depends on phase shift.

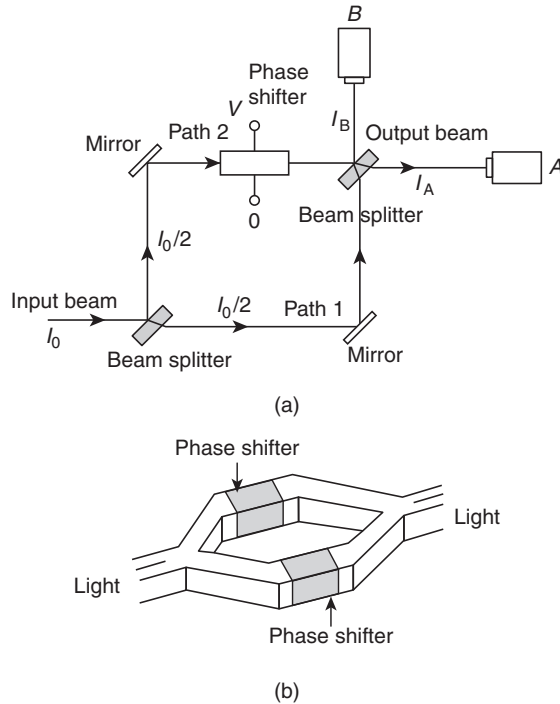


Figure 6.23 | Mach-Zehnder interferometer designs.

The phase modulation in path 2 is obtained by an electro-optically active material. Externally, the voltage V is applied to change the refractive index of the material that will effectively change the path length. Other methods of phase modulation are thermal, magneto-optical, as well as stress-optical phase modulation.

Generally, the phase modulation is done in only one arm of MZI as shown in Fig. 6.23(a). Sometimes the phase modulation can be introduced in both the arms as shown in Fig. 6.23(b). In electro-optically active material such as lithium niobate (LiNbO_3), by application of electric field, the refractive index of material can be changed. This will change the path length which results into phase shift. Thus, by changing the bias voltage, the phase shift can be changed.

6.5.6.1 MZ Intensity Modulator

An integrated MZ intensity modulator device is illustrated in Fig. 6.23(a). The substrate is usually silicon, and the waveguide and cladding regions are silica (SiO_2). The bias voltage V is applied to the phase shifter to create the phase difference between the two optical beams traveling through the two arms of MZI. By changing the voltage, phase difference can be changed that will change the intensity of output. If the applied voltage gives the phase shift of 2π or multiple of 2π , then we can get 100% transmittance because of constructive interference. If the two beams are out-of-phase completely, that is, the phase shift is π , output optical power is 0% (destructive interference). Also if the phase shift is in between, then output intensity varies accordingly. Thus, the device acts as intensity modulator. Intensity is modulated by changing the bias voltage of electro-optic device connected in one or both arms of MZI.

6.5.6.2 MZI Working as Demultiplexer

Refer to Fig. 6.24 for the operation of MZI as a demultiplexer.^[3] Input 1 has a signal, and after the first directional coupler, input 1 signal power is divided equally between the two arms of the MZI: the signal in lower arm lags the upper arm by $\pi/2$ phase. Because of the path difference of ΔL between the two arms, there is further phase lag of $\beta\Delta L$ in lower arm signal. When the signal passes through the second directional coupler, it goes through phase delay of $\pi/2$ to the first output relative to the signal from the upper arm. Therefore the total relative phase difference at the output 1 is

$$\pi/2 + \beta\Delta L + \pi/2$$

The signal in output 2 is having a relative phase difference of

$$\pi/2 + \beta\Delta L - \pi/2 = \beta\Delta L$$

If $\beta\Delta L = k\pi$ and if k is odd, then signals at output 1 add in-phase, while signals at output 2 add with opposite phase and cancel each other. This indicates that

1. The wavelengths passed from input 1 to output 1 are those wavelengths for which $\beta\Delta L = k\pi$ and k is odd.
2. The wavelengths passed from input 1 to output 2 are those wavelengths for which $\beta\Delta L = k\pi$ and k is even.

In Fig. 6.25, a four-channel wavelength multiplexer is constructed using three 2×2 MZIs. All four input wavelengths can be switched or multiplexed into the same output port using combination of three MZIs. One arm of each MZI is made longer than the other to provide a wavelength-dependent phase shift between the two arms. The path-length difference is chosen such that the two different wavelengths at different input ports appear at only one output port. Both λ_1 and λ_3 are combined using MZI₁; λ_2 and λ_4 are combined using MZI₂; and MZI₃ combines the outputs of MZI₁ and MZI₂, that is, λ_1 , λ_2 , λ_3 , and λ_4 .

6.5.7 Arrayed Waveguide Grating

Arrayed waveguide gratings (AWGs) are essential components for the realization of DWDM and optical networks. They are optical wavelength (de)multiplexers used in DWDM. While performing basic (de)multiplexing functions, they can be combined with other components to create add/drop multiplexers.^[10–14]

An A WG is a generalization of MZI. The device consists of two multiport couplers interconnected by an array of waveguide as shown in Fig. 6.26. In MZI, two copies of the same signal, shifted in-phase by different amounts, are added together, whereas in A WG, several copies of same signal, shifted in-phase by different amounts, are

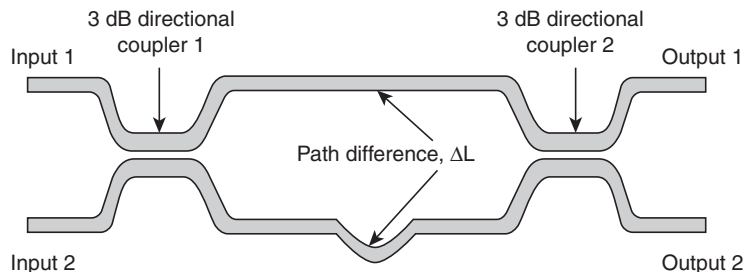


Figure 6.24 | MZI as a demultiplexer.

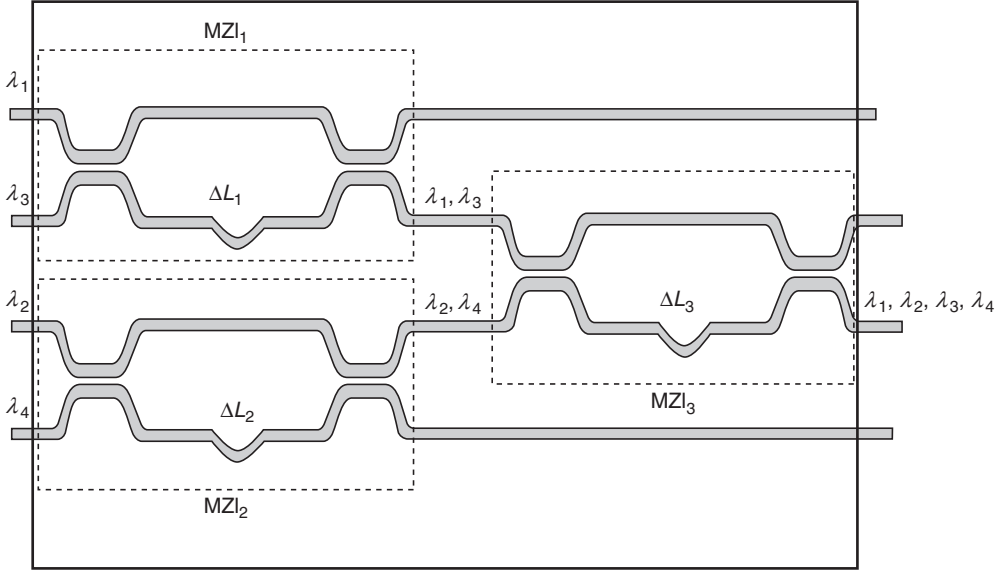


Figure 6.25 | Four-channel wavelength multiplexer using three 2×2 MZIs.

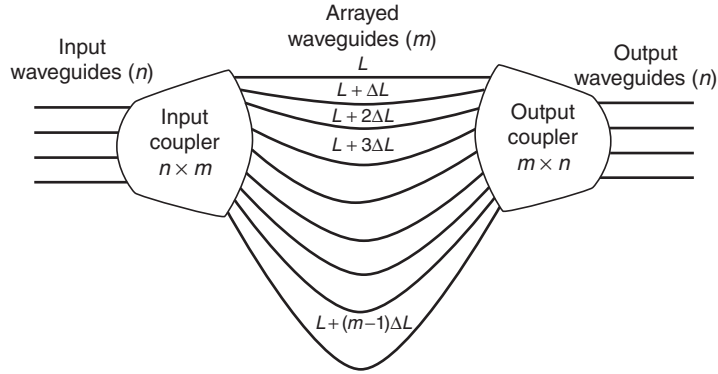


Figure 6.26 | An arrayed waveguide grating.

added together. AWG consists of $n \times m$ input and output couplers, interconnected by m waveguides. The lengths of these waveguides are chosen such that the difference in length between consecutive waveguides is ΔL .

The first coupler splits the signal into m parts. The relative phases of these parts are determined by the distance traveled in the coupler from the input waveguide to the arrayed waveguide.

The distance traveled between input waveguide x and arrayed waveguide y is denoted by d_{xy}^{in} . The arrayed waveguide y has a path length larger than arrayed waveguide $(y-1)$ by ΔL . The distance traveled between arrayed waveguide y and output waveguide z is given by d_{yz}^{out} . Then the relative phases of the signals from input x to output z traversing m different paths between them are given by

$$\phi_{xyz} = \frac{2\pi}{\lambda} (n_1 d_{xy}^{\text{in}} + n_2 k \Delta L + n_1 d_{yz}^{\text{out}}), \quad k=1, 2, \dots, m \quad (6.36)$$

where n_1 is the refractive index in the input and output directional couplers, and n_2 is the refractive index in the arrayed waveguide. If the input and output couplers are designed such that

$$d_{xy}^{\text{in}} = d_x^{\text{in}} + k\delta_x^{\text{in}} \quad \text{and} \quad d_{yz}^{\text{out}} = d_z^{\text{out}} + k\delta_z^{\text{out}} \quad (6.37)$$

then Eq. (6.36) can be written as

$$\phi_{xyz} = \frac{2\pi}{\lambda} (n_1 d_x^{\text{in}} + n_1 d_z^{\text{out}}) + \frac{2\pi k}{\lambda} (n_1 \delta_x^{\text{in}} + n_2 \Delta L + n_1 \delta_z^{\text{out}}), \quad k = 1, 2, \dots, m \quad (6.38)$$

From Eq. (6.38), it is clear that the wavelengths λ that are present at input x and satisfy

$$(n_1 \delta_x^{\text{in}} + n_2 \Delta L + n_1 \delta_z^{\text{out}}) = p\lambda$$

for some integer p , add in phase at output z .

6.5.7.1 AWG as Demultiplexer

At the input of AWG, several channels are applied ($\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$) as shown in Fig. 6.27. In commercial system, the number of channels ranges from 8 to 40, and the channel spacing is typically 50 GHz or 100 GHz with operating wavelength of 1,550 nm. This operating wavelength is most suitable since at this wavelength the attenuation is lowest in optical fiber.

From the input waveguide, light gets coupled into the free propagation region (FPR). This light then disperses and enters into arrayed waveguides (AWs). The length of each AW increases by ΔL successively. Mathematically, this increase in length is given as

$$\Delta L = \frac{m\lambda_w}{n_{\text{eff}}} \quad (6.39)$$

where m is an integer number, λ_w is the central operational wavelength, and n_{eff} the effective refractive index.

The light is focused at the center of the plane at the central frequency as shown in Fig. 6.28. The output waveguide is positioned to capture the focused light. Due to different path lengths ($L + \Delta L, L + 2\Delta L, L + 3\Delta L, \dots$), different wavelengths have different phase shifts. The phases will change along the AW output plane in incremental

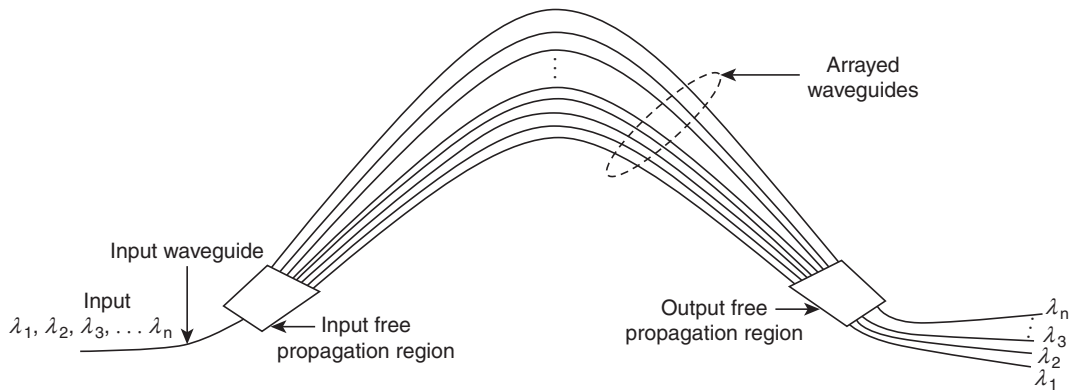


Figure 6.27 | An arrayed waveguide grating as demultiplexer.

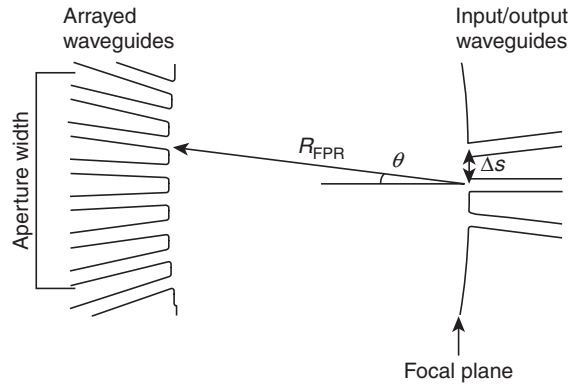


Figure 6.28 | Output FPR.

fashion. Due to this, the focal point will move along the focal plane at the end of FPR. An output waveguide picks up each wavelength.

6.5.7.2 AWGs Performance Issues

1. **Crosstalk:** Inter-channel crosstalk may be caused by imperfect design or fabrication of AWG. It may be caused by an overlap of focused spot in the output FPR with adjacent output waveguide. The imperfect design or flaws in fabrication may result into spreading of spot, which gives rise to increased crosstalk.
2. **Insertion loss:** This loss is mainly due to imperfect coupling at input FPR with AWs, and coupling of AWs with output FPR. The other causes of loss are material losses, scattering due to fabrication errors and waveguide roughness, de-focusing of the spot on the output plane due to phase errors, decreasing coupling efficiency into the output waveguide, etc.
3. **Polarization-dependent dispersion:** In strongly confining waveguide, the boundary conditions may give rise to the birefringence phenomenon. Birefringence may be introduced due to fabrication process. It causes the quasi-TE and quasi-TM polarized modes to propagate at different speeds. Due to this, a second shadow spot appears on the output plane of the FPR. The TE and TM mode with same polarization experience different phase shifts and may get coupled with wrong waveguide. This can cause inter-channel crosstalk.

6.5.8 Acousto-Optic Tunable Filter

The angular relationship between the laser beam and the acoustic wave is shown in Fig. 6.29. When the light beam strikes the acousto-optic material, it gets diffracted. The intensity of diffracted light is proportional to the acoustic power, geometric factors (L/H), and material figure of merit, and it is inversely proportional to the square of the wavelength.^[15]

Depending on the laser parameters, such as laser wavelength, power density, and polarization, variety of different acousto-optic materials are used for constructing the modulators.

For visible and near-infrared regions, gallium phosphide, tellurium dioxide, indium phosphide or fused quartz are used. For infrared region, germanium with relatively high figure of merit is used. For high-frequency (GHz) signal-processing devices, lithium niobate, indium phosphide, and gallium phosphide materials are used as listed in Table 6.5.

Table 6.5 | Acousto-optic materials

Material	Optical Range (Micron)
Fused quartz	0.2–4.5
Flint glass SF6	0.45–2.0
Chalcogenide glass	1.0–2.2
Gallium phosphide	0.59–10.0
Germanium	2.0–12.0
Indium phosphide	1.0–1.6
Lithium niobate	0.6–4.5
Tellurium oxide	0.4–5.0

6.5.8.1 Device Construction

After selection of proper acousto-optic material, it is optically polished. Then the lithium niobate transducer is bonded to the modulator medium using metal pressure bonding. This bonding is superior to epoxy bonding. The transducer is then lapped to the fundamental resonant frequency.

The important performance parameter for acousto-optic modulator is the modulation speed. This depends on the transit time, t . The transit time, t , and the rise time, t_r , are given by

$$t = \frac{V}{d} \quad (6.40)$$

and

$$t_r = 0.85 \tau \quad (6.41)$$

To obtain a high modulation speed, t should be as small as possible.

6.5.8.2 Tunable Filter

An acousto-optic tunable filter is a solid-state electro-optical device. It is an electronically tunable spectral band-pass filter. It has no moving parts. It consists of a TeO_2 crystal. The acoustic waves, at radio frequencies (RF), are used to separate a single wavelength of light from a broadband source. By changing the frequency of the RF source, different wavelengths can be filtered. This wavelength does not depend on the device geometry.

To obtain the near-infrared region operation, tellurium dioxide (TeO_2) crystal-based acousto-optic tunable filter is used. The optical waves and acoustic waves travel with different angles through crystal. The acoustic transducer is attached to one side of the TeO_2 crystal as shown in Fig. 6.29.

The RF signal is applied to the acoustic transducer. The transducer emits acoustic waves or vibrations. Due to these acoustic waves passing through the TeO_2 crystal, alternate compression and relaxation of crystal lattice takes place. This results into periodic refractive index variation like Bragg diffracter. This structure diffracts a single wavelength, thus the filtering operation is obtained.

The wavelength of diffracted light is determined by the “phase matching” condition as given below:

$$\lambda = \Delta n \alpha \frac{V_a}{f_a} \quad (6.42)$$

where Δn is the birefringence of the TeO_2 crystal, f_a is the acoustic frequency, V_a is the acoustic velocity, and α is a complex parameter depending on the design of the acousto-optic tunable filter (AOTF). By changing the frequency of the applied RF, the diffracted wavelength of the light can be varied.

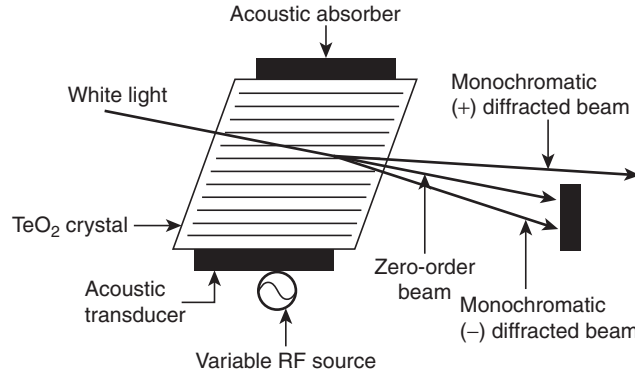


Figure 6.29 | Schematic representation of acousto-optic tunable filter.

As shown in Fig. 6.29, after diffraction the light travels along two first-order orthogonally polarized beams [monochromatic (+) and (–) beams]. The undiffracted light is blocked by beam stop. The device material and wavelength of operation decides the bandwidth of filtered light. The bandwidth may be as narrow as 1 nm [full width at half maximum (FWHM)]. By varying the RF power, the intensity of filtered light can be precisely controlled. Thus, transmission frequency as high as 98% can be obtained.

6.5.9 Ring Resonator-Based Wavelength Filters

The characteristics of microring resonators are very similar to Fabry–Perot filters. In these filters, the injected and reflected signals are separated in individual waveguides. The design of filter does not require gratings or facets and therefore it is much simple. Due to their miniature size (few tens of micrometers), they are most suitable for integrated optics and VLSI photonics applications.

These filters can be used in following applications^[16,17]:

1. Laser cavities.
2. Optical amplifiers.
3. Logic elements.
4. Add/drop multiplexers/demultiplexers.
5. Sensors, etc.

Ring resonators consist of a waveguide in a closed loop along with two coupled port waveguides as shown in Fig. 6.30. Few selected wavelengths resonate in the ring. To couple light into the ring, a bus waveguide is placed close to it. Similarly, another waveguide is used to extract the resonant light from the ring resonator.

The details of ring resonator as two sets of coupled waveguides are shown in Fig. 6.31. Here E_{in} is the input field, E_{out} is the output/throughput field, E_d is the dropped field, $E_{radd} \dots E_{rdrop}$ are the fields in the ring as indicated, κ_1 is the field coupling coefficient between the input bus and the ring, and κ_2 is the field coupling coefficient between the input port and the through port. The port waveguides and the ring are evanescently coupled, and a fraction k_1 of the incoming signal is transferred to the ring. Constructive interference occurs when the optical path length of a roundtrip is a multiple of the effective wavelength. The light signal accumulates and the microring resonator (MR) is on resonance. This results into periodic fringes in the wavelength response at the output ports as shown in Fig. 6.32. Thus, the drop port shows maximum transmission at resonance because a fraction k_2 of the built-up field inside the ring is coupled to this port. Ideally all the power is directed toward drop port.

After a round trip, the light coupled back to the output port is phase shifted by 180° with respect to incoming light, so no light exists in output port at resonance.

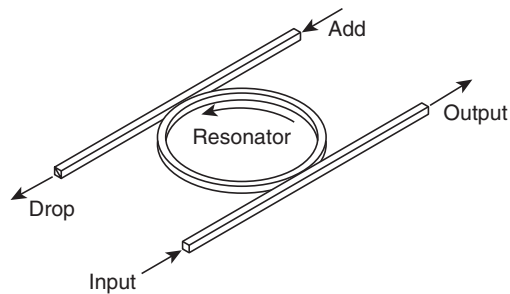


Figure 6.30 | Schematic drawing of 4-port ring resonator.

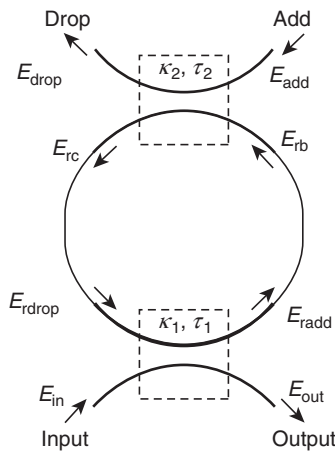


Figure 6.31 | Ring resonator as two sets of coupled waveguides.

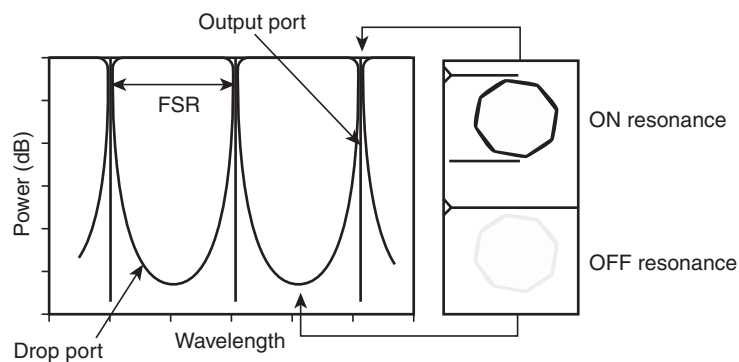


Figure 6.32 | Response of microring resonator for output port and drop port.

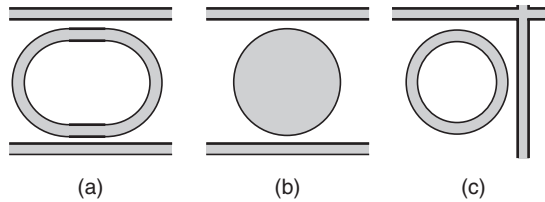


Figure 6.33 | Other microring resonators shapes: (a) Racetrack, (b) disk, and (c) Manhattan configuration.

6.5.9.1 Micro-Resonator Geometries

Any geometrical path that provides optical feedback will act like the microring. Thus, the shape of an MR is not restricted to a circle.

1. **Racetrack:** It is a frequently used geometry and is shown in Fig. 6.33(a). In this geometry, the couplers of the resonator are straight waveguides.
Advantage: Accurate control of the coupling constants.
Disadvantage: Somewhat larger structure and hence have a reduced FSR.
2. **Disk:** It is another common shape and is shown in Fig. 6.31(b). Because of larger cross-section and lateral width, it is difficult to make the disk single mode. The response of device is influenced by total roundtrip length, wherein the place of adjacent waveguide is not important.
3. **Manhattan:** It is an additional geometry as shown in Fig. 6.33(c). This structure is scalable and allows for optimum use of area.

6.5.9.2 Tuning of Microring Resonator

In the previous section, the geometrical and material parameters of MR are considered fixed; so it acts like a passive device. Operational flexibility can be obtained and the wavelength response can be changed by varying some parameters.

By changing the refractive index of ring, the resonant wavelength of MR can be tuned. There are various methods to change the refractive index as follows:

1. **Electro-optic effect:** By varying an electrical field, the change in refractive index is obtained.
2. **Thermo-optic effect:** By applying heat to the ring, the refractive index of the material is changed.
3. **Carrier injection:** Optical pumping creates free carriers, which change the loss parameter and the refractive index of the material.
4. **Opto-optical effect:** The light itself causes a change in index via non-linear effects.
5. **Changing the material:** The total wavelength response gets shifted by $\Delta\lambda$ due to change in refractive index. Thus, the band-pass filter can be tuned to the desired wavelength by changing the refractive index of ring.

In optical fiber communication systems, resonators are combined in different architectures (instead of using a single device) to achieve the improved performance number of microrings. A few such applications are as follows:

1. **Higher-order filters:** Higher-order MR filters are devices where several feedback paths contribute to the response of the filter. Figure 6.34 shows two different setups, normally designated as serial cascade and parallel cascade in literature. In both the cases, the additional feedback paths result in an improved filter response, that is, we get a more flat-top response in the pass band and a steeper roll-off at the pass-band edges.

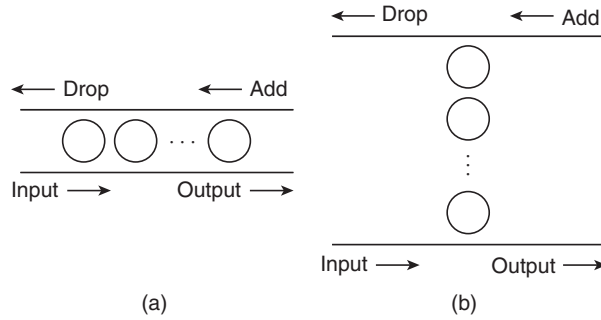


Figure 6.34 | Cascaded ring filters: (a) Serially arranged and (b) parallel.

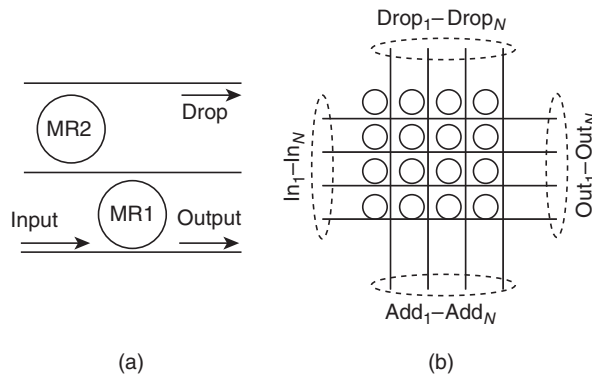


Figure 6.35 | (a) Wavelength selective switch based on two MRs; (b) reconfigurable optical add/drop multiplexer.

2. **Wavelength-selective MR switch:** A schematic drawing of an MR-based switch is shown in Fig. 6.35(a). This structure is based on two MRs connected through a straight waveguide and acts as a wavelength-selective switch. The first MR selects a certain wavelength. The second tunable ring switches this wavelength to the drop port or to a non-used channel. It acts like an absorber.
3. **Reconfigurable optical add/drop multiplexer:** Another example of complex structures based on MRs is the $N \times M$ matrix switch based on the Manhattan configuration as shown schematically in Fig. 6.35(b). This compact configuration allows adding and dropping any of the M wavelengths to any of the N input/output ports. In order to improve the performance, each building block consisting of a single MR can be replaced by a higher-order filter.

6.5.10 Fiber Grating

Fiber grating is one of the key elements in the established and emerging fields of optical communication systems. In optical telecommunication system, a major drawback is on the component side for controlling the light like coupling, in and out filtering, etc., which mainly relies on bulk optics. Bulk optics has high losses, stringent tolerance for alignment, and huge size. These disadvantages are overcome using fiber gratings, which have low losses, high stability, small size compatible with fiber sizes, and low cost. Their applications also spread into the area of optical fiber sensing.

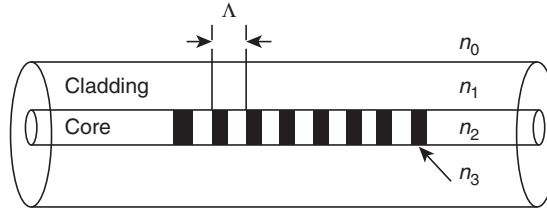


Figure 6.36 | Fiber Bragg grating structure.

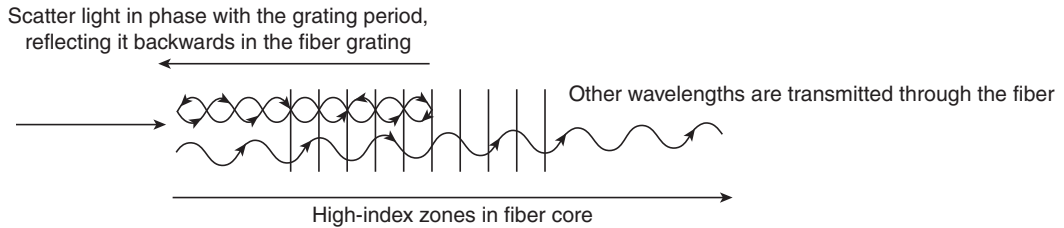


Figure 6.37 | Fiber Bragg grating reflects light at wavelengths that match the grating period and transmits other wavelength.

Fiber gratings are passive optical components, which selectively reflect and transmit light of certain wavelengths. FBGs were discovered in 1978 when Hill et al. sent 488 nm light from an argon-ion laser into the core of a germanium-doped optical fiber. The fundamental principle behind the operation of an FBG is Fresnel reflection.^[18] Here light traveling between media of different refractive indices may both reflect and refract at the interface. In fiber grating, a periodic perturbation of refractive index along the fiber length is formed. These perturbations scatter light. It selectively reflects a narrow range of wavelength. Each time the light hits a region of higher refractive index, a bit is scattered backward. If the wavelength matches the spacing of the high-index zone in the fiber, the waves scattered from each high-index zone interfere constructively, producing strong reflections. The wavelength selected is twice the distance between the lines written into the fiber because the light wave has to go through the region between them twice, once into the fiber grating and once when reflected back out as shown in Fig. 6.37. If Λ is the grating period and n_{eff} is the effective refractive index of grating in the fiber core as shown in Fig. 6.36, then the reflected wavelength is called Bragg wavelength and is defined by the relationship

$$\lambda_B = 2n_{\text{eff}}\Lambda \quad (6.43)$$

where

$$n_{\text{eff}} = \frac{n_2 + n_3}{2}$$

Here n_2 and n_3 are the refractive indices of core as shown in Fig. 6.36.

The high-index regions also scatter light at other wavelengths, but the scattered waves differ in phase and so they cancel each other by destructive interference. Thus, these non-resonant wavelengths are transmitted through the grating with low loss.

6.5.10.1 Core and Cladding Mode Coupling

The propagation of light along a waveguide can be described in terms of a set of guided electromagnetic waves called the *modes of waveguide*. In optical fibers, the core-cladding boundary conditions lead to coupling between the electric and magnetic field components.

Each mode has its specific propagation constants. If the periodic perturbation is introduced alongside the fiber, the mode will exchange its power. This phenomenon is known as *mode coupling*. Fiber gratings can be broadly classified into two types: (a) Bragg gratings (also called reflection and short-period gratings), in which coupling occurs between modes traveling in opposite directions; and (b) transmission gratings (also called long-period gratings – LPG), in which the coupling is between modes traveling in the same direction.^[19,20]

A fiber grating is simply an optical diffraction grating, and thus its effect upon a light wave incident on the grating at an angle can be described by the familiar grating equation:

$$n \sin \theta_2 = n \sin \theta_1 + m (\lambda / \Lambda) \quad (6.44)$$

where n is the refractive index of medium (core), θ_1 is the angle of incidence, θ_2 is the angle of the diffracted wave, and the integer m determines the diffraction order.

Figure 6.38 illustrates reflection by a Bragg grating of a mode with a bounce angle of θ_1 into the same mode traveling in the opposite direction with a bounce angle of $\theta_2 = -\theta_1$. β is the z -component of wave propagation constant k and is the main parameter in describing fiber modes, is simply

$$\beta = (2\pi / \lambda) n_{\text{eff}} \quad (6.45)$$

where $n_{\text{eff}} = n_{\text{co}} \sin \theta$. The mode remains guided as long as β satisfies the condition $n_{\text{cl}} k < \beta < n_{\text{co}} k$, where n_{co} and n_{cl} are core and cladding refractive indices and $k = 2\pi / \lambda$.

The boundary between truly guided modes and leaky modes is defined by the cutoff condition $\beta = n_{\text{cl}} k$. As soon as β becomes smaller than $n_{\text{cl}} k$, power leaks out of the core into the cladding region.

We find that the resonant wavelength for reflection of a mode of index $n_{\text{eff}1}$ into a mode of index $n_{\text{eff}2}$ is

$$\lambda = (n_{\text{eff}1} + n_{\text{eff}2}) \Lambda \quad (6.46)$$

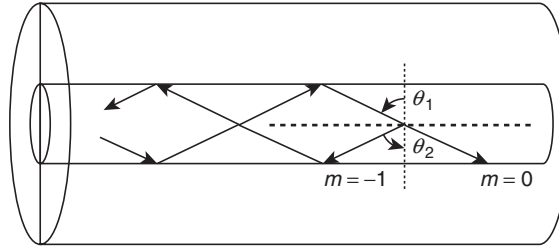


Figure 6.38 | Core mode Bragg reflection by an FBG.

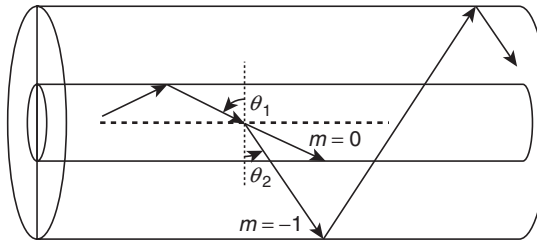


Figure 6.39 | Cladding mode coupling by long-period grating.

If the two modes are identical, we get the familiar result for Bragg reflection:

$$\lambda = 2n_{\text{eff}}\Lambda \quad (6.47)$$

Diffraction by a transmission grating of a mode with a bounce angle of θ_1 into a co-propagating mode with a bounce angle of θ_2 is illustrated in Fig. 6.39. In this illustration, the first mode is a core mode, whereas the second is a cladding mode. Since here $\beta_2 > 0$, the resonant wavelength for a transmission grating is

$$\lambda = (n_{\text{eff1}} - n_{\text{eff2}}) \Lambda \quad (6.48)$$

6.5.10.2 Types of Gratings

Depending upon refractive index profile and grating period variation, gratings are of different types as shown in Fig. 6.40. The grating period can be uniform or graded and either localized or distributed in structure.

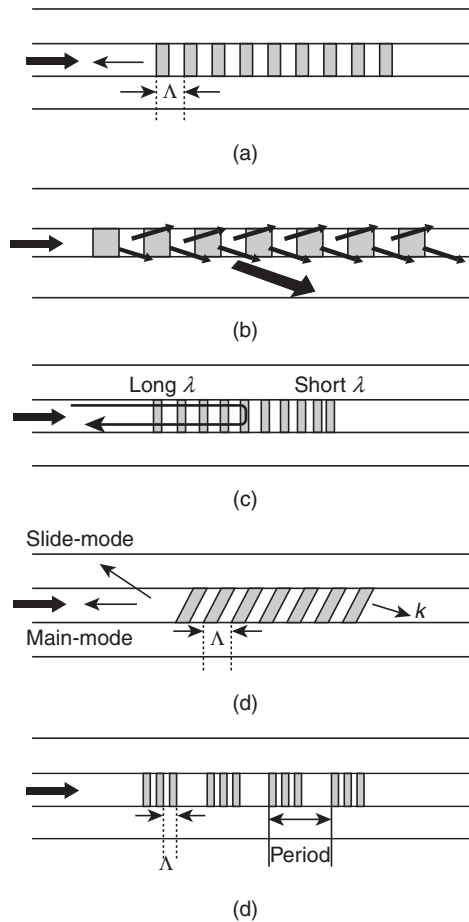


Figure 6.40 Types of fiber gratings: (a) Fiber Bragg grating, (b) long-period fiber grating, (c) chirped fiber grating, (d) tilted fiber grating, (e) sampled fiber grating.

The various types of grating are as follows:

1. **Fiber Bragg grating (reflection grating):** For a uniform short-period FBG, the refractive index perturbation is given by

$$n(z) = n_{co} + \Delta n_0 A(z) n_d(z) \quad (6.49)$$

where n_{co} is the core refractive index, $n_d(z)$ is the index variation function, Δn_0 is the maximum index variation, and $A(z)$ is the Apodization function.

For uniform FBG with no apodization index, the variation function $n_d(z)$ is given as

$$n_d(z) = \frac{\cos 2\pi z}{\Lambda} \quad (6.50)$$

where Λ is the constant grating period, $A = 1$. Typical reflectivity spectrum for linear FBG is as shown in Fig. 6.41.

2. **Long-period fiber grating (transmission grating):** Long-period fiber gratings (LPFGs) typically have grating periods of the order of 100 μm to 1 mm, and are therefore much easier to manufacture.^[21] An LPFG can couple the forward propagating core mode to one or a few of the forward propagating cladding modes. The transmission characteristics of an LPFG, whether formed in an SMF or waveguide, can be analyzed by the coupled-mode theory. There exists a specific wavelength (resonance wavelength) at which the coupling between the guided mode and a specific cladding mode is the strongest. The transmission spectrum of an LPFG usually consists of a number of rejection bands centered at resonance wavelengths that correspond to different cladding modes. The center wavelength of a rejection band is, in general, sensitive to a number of physical parameters (temperature, strain, external refractive index, fiber/waveguide dimensions, grating pitch, etc.). Moreover, its sensitivity depends critically on the

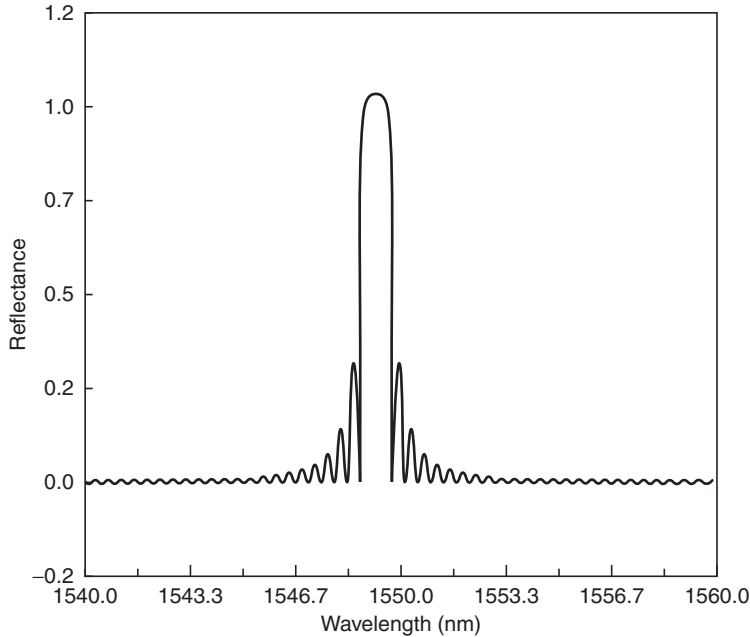


Figure 6.41 | Typical reflectivity spectrum for linear FBG.

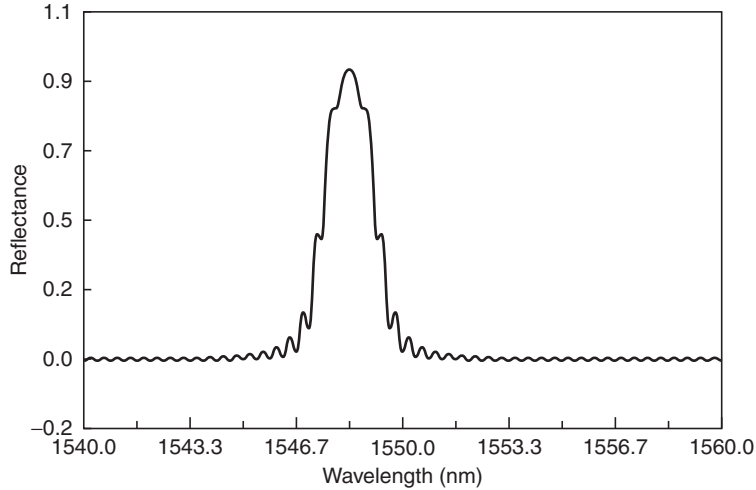


Figure 6.42 | Typical reflectivity spectrum for linear chirped (-0.006 nm/cm) FBG.

dispersion properties of the modes. This property of the LPFG has been explored for the realization of tunable devices and sensors. The bandwidth of a typical centimeters-long LPFG is of the order of 10 nm. The bandwidth can be increased significantly by manipulating the dispersion properties of the modes. To obtain more sophisticated transmission characteristics, one can vary the coupling coefficient (apodization) and the pitch (chirping) along the grating, or introduce phase shifts along the grating.

3. **Chirped fiber grating:** The refractive index profile of the grating is modified as a linear variation in the grating period, and the resulting grating is said to be *chirped*. The reflected wavelength changes with the grating period, broadening the reflected spectrum. A grating possessing a chirp has the property of adding dispersion since each wavelength component is reflected at different positions, which results in a delay time difference for different reflected wavelengths. Typical reflectivity spectrum for linear chirped FBG is as shown in Fig. 6.42.
4. **Tilted fiber grating:** The variation of refractive index occurs at an angle to the optical axis. The angle of tilt has an effect on the reflected wavelength and the bandwidth. A tilted fiber grating can couple the forward propagating core mode to the backward propagating core mode and a backward propagating cladding mode. The resonant wavelength for each mode depends differentially on external perturbations and thus can be used to make the fiber grating into a multimodal sensor.
5. **Sampled fiber grating:** A sampled fiber grating can reflect several wavelength components with equal wavelength spacing.
6. **Apodized grating:** A uniform FBG yields highly undesirable side lobes due to the sharp boundaries of the grating. These side lobes cause high crosstalk between the adjacent channels in a DWDM communications system. A well-discussed method to reduce these side lobes is to apodize the grating couple strength along the grating by gradually tapering the refractive index modulation amplitude to zero at both ends of the grating as shown in Fig. 6.43. For sinc apodization,

$$A(z) = \text{sinc} \left(\frac{z - (L/2)}{\Lambda_T} \right) \quad (6.51)$$

where Λ_T is the sinc function parameter.

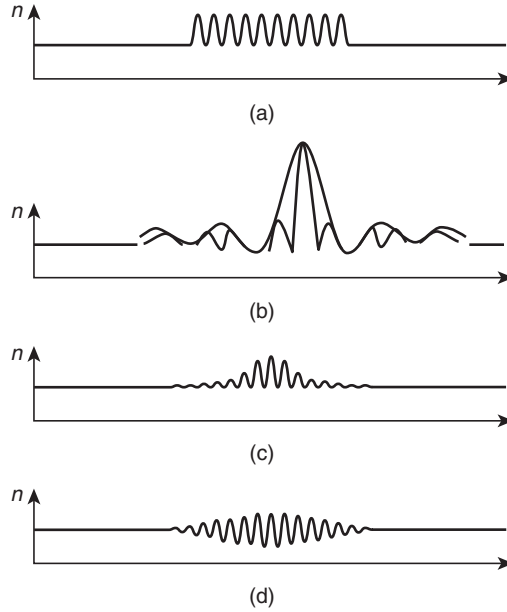


Figure 6.43 | Refractive index variation function. (a) No apodization; (b) sync apodization; (c) Gaussian apodization; (d) raised cosine apodization.

For Gaussian apodization,

$$A(z) = \exp\left(-\alpha_g \left(\frac{z - (L/2)}{L}\right)^2\right) \quad (6.52)$$

where α_g is the Gaussian parameter.

For raised cosine apodization,

$$A(z) = \alpha_r \left(1 + \cos\left[\frac{\pi[z - (L/2)]}{L}\right]\right) \quad (6.53)$$

where α_r is raised cosine parameter. The apodization function $A(z)$ in Eq. (6.49) can be changed to reduce the side lobes.

6.5.10.3 FBG Characteristics

The characteristics response from Bragg grating can be fully described by

1. The center wavelength of grating λ_B .
2. Peak reflectivity $R_{\max}$ of grating which occurs at λ_B .
3. Physical length of grating L .
4. Refractive index of core of optical fiber n_{co} .
5. Amplitude of induced core index perturbation Δn .

For a grating with uniform index modulation and period the reflectivity is given by

$$R(L, \lambda) = \frac{k^2 \sinh^2(SL)}{\Delta\beta^2 \sinh^2(SL) + k^2 \cosh^2(SL)} \quad (6.54)$$

where R is the grating reflectivity as a function of both grating length and wavelength, L is the total length of grating, κ is the coupling constant given by

$$\kappa = \pi \Delta n / \lambda$$

$\Delta\beta$ is the wave vector detuning given by

$$\Delta\beta = \beta - (\pi/\Lambda)$$

β is the fiber core propagation constant given by

$$\beta = 2\pi n_0 / \lambda$$

and

$$S = \sqrt{k^2 - \Delta\beta^2}$$

For light at the Bragg grating center wavelength, λ_B , there is no wave vector detuning. Hence, $\Delta\beta = 0$. The reflectivity function then becomes

$$R(L, \lambda) = \tanh^2(SL) \quad (6.55)$$

The index of refraction change is assumed uniform over the grating length; however, the value of the change is reduced accordingly with increasing grating length such that the maximum grating reflectivity remains constant. Theoretically, Bragg gratings may be constructed with extremely small bandwidths by simply increasing the grating length. However, in practice, such devices are not easy to manufacture. The error associated with the spacing between the periods of a grating during manufacturing is cumulative; therefore, with increasing grating length, the total error will increase, resulting in out-of-phase periods leading to broadening of the Bragg grating. Furthermore, if a long perfect Bragg grating is constructed, the effects of the environment have to be considered very carefully. For example, any strain or temperature fluctuations on any part of the grating will cause the periods to move out-of-phase, resulting in broadening the spectral responses of the Bragg grating.

1. **Grating length dependence:** The spectral response of fiber grating is greatly affected by length of the grating. Here, as shown in Fig. 6.44, the index of refraction change is assumed uniform over the grating length. However, the value of the change is reduced with increasing grating length in such a way that the maximum grating reflectivity remains constant. The bandwidth of the gratings decreases with increasing length. Theoretically, Bragg gratings may be constructed with extremely small bandwidths by simply increasing the grating length. However, in practice such devices are not easy to manufacture. The error associated with the spacing between the periods of a grating (during manufacturing) is cumulative, therefore, with increasing grating length the total error increases, resulting in out-of-phase periods and leading to broadening of the Bragg grating reflection. Furthermore, if a long perfect Bragg grating is constructed, the effects of the environment have to be considered very carefully. For example, any strain or temperature fluctuations on any part of the grating will cause the periods to move out-of-phase resulting in broadening of the Bragg grating reflection.
2. **Index of refraction dependence:** As shown in Fig. 6.45, the reflectivity and bandwidth of FBG decreases with decrease in change of refractive index. The bandwidth approaches a minimum value and remains constant for further reductions in the index of refraction change.

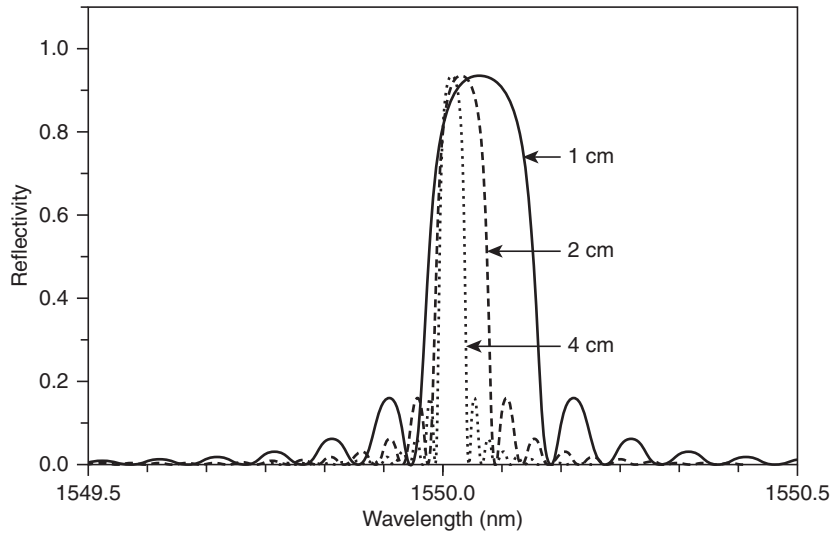


Figure 6.44 | Spectral reflectivity response of uniform Bragg gratings for different grating lengths.

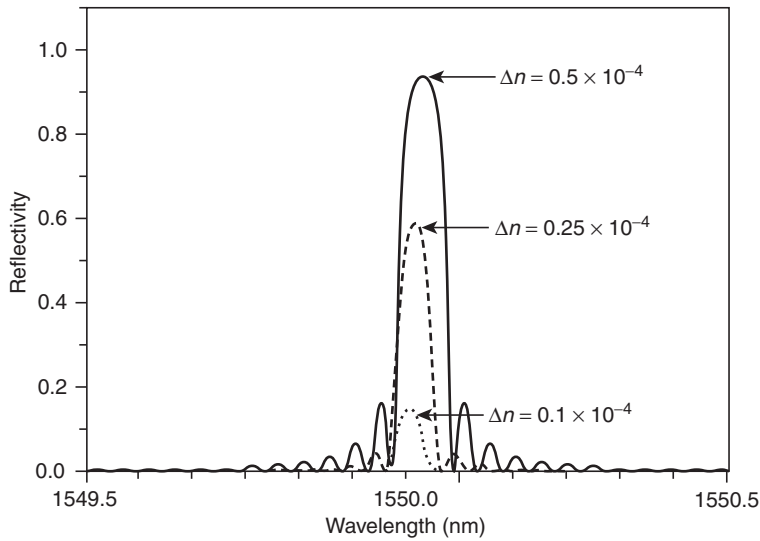


Figure 6.45 | Spectral reflectivity response of uniform Bragg grating 2 cm in length for different refractive indices.

3. **Strain and temperature sensitivity of Bragg gratings:** The Bragg grating resonance, which is the center wavelength of light back-reflected from a Bragg grating, depends on the effective index of refraction of the core and the periodicity of the grating. The effective index of refraction, as well as the periodic spacing between the grating planes, will be affected by changes in strain and temperature. The shift in the Bragg grating center wavelength $\Delta\lambda_B$ due to strain and temperature changes is given by

$$\Delta\lambda_B = 2 \left(\Lambda \frac{\partial n_{\text{eff}}}{\partial L} + n_{\text{eff}} \frac{\partial \Lambda}{\partial T} \right) \Delta L + 2 \left(\Lambda \frac{\partial n_{\text{eff}}}{\partial T} + n_{\text{eff}} \frac{\partial \Lambda}{\partial T} \right) \Delta T \quad (6.56)$$

The first term in Eq. (6.56) represents the strain effect on an optical fiber. This corresponds to a change in the grating period and the strain-induced change in the refractive index due to the photoelastic effect. To make the calculation easier, the strain affected may be expressed as

$$\Delta\lambda_B = \lambda_B (1 - P_c) \varepsilon \quad (6.57)$$

where P_c is an effective strain-optic constant defined as

$$P_c = \frac{n_{\text{eff}}^2}{2[P_{12} - \nu(P_{11} + P_{12})]} \quad (6.58)$$

P_{11} and P_{12} are components of the strain-optic sensor, ν is Poisson's ratio, and ε is the applied strain. The second term in Eq. (6.56) represents the effect of temperature on an optical fiber. A shift in the Bragg wavelength due to the thermal expansion changes the grating period and the refractive index. The wavelength shift due to the temperature change may be expressed as

$$\Delta\lambda_B = \lambda_B (\alpha_\Lambda + \alpha_n) \Delta T \quad (6.59)$$

where

$$\alpha_\Lambda = \left(\frac{1}{\Lambda} \right) \left(\frac{\partial \Lambda}{\partial T} \right)$$

is the thermal expansion coefficient and

$$\alpha_n = \left(\frac{1}{n_{\text{eff}}} \right) \left(\frac{\partial n_{\text{eff}}}{\partial T} \right)$$

represents the thermo-optic coefficient.

4. **Long-period fiber grating characteristics:** The transmission characteristics of LPFG can be analyzed by coupled mode theory. For an SMF with the propagation constant of the fundamental mode, LP_{01} , denoted by β_{01} and the propagation constants of the cladding modes denoted by $\beta_{cl}^{(n)}$, the phase-matching condition between the guided mode and the forward propagating cladding modes is given by

$$\beta_{01} - \beta_{cl}^{(n)} = \frac{2\pi}{\Lambda} \quad (6.60)$$

where Λ is the grating periodicity required to couple the fundamental mode to the n th cladding mode and the superscript n denotes the order of the mode. In this case, the phase matching vector is short resulting in a long Λ , typically of the order of hundreds of microns

There exists a specific wavelength (resonance wavelength or Bragg wavelength λ_{Br}) at which the coupling between the guided mode and a specific cladding mode is strongest:

$$\lambda_{Br} = 2n_{\text{eff}}^{\text{core}} \Lambda_{Br} \quad (6.61)$$

where $n_{\text{eff}}^{\text{core}}$ is the effective refractive index of the fundamental mode and Λ_{Br} is the grating period. The transmission spectrum of an LPFG, as shown in Fig. 6.46, usually consists of a number of rejection bands centered at resonance wavelengths that correspond to different cladding modes. For a grating with uniform index, variation across the core LP_{01} mode is coupled with cladding modes having same symmetry, namely

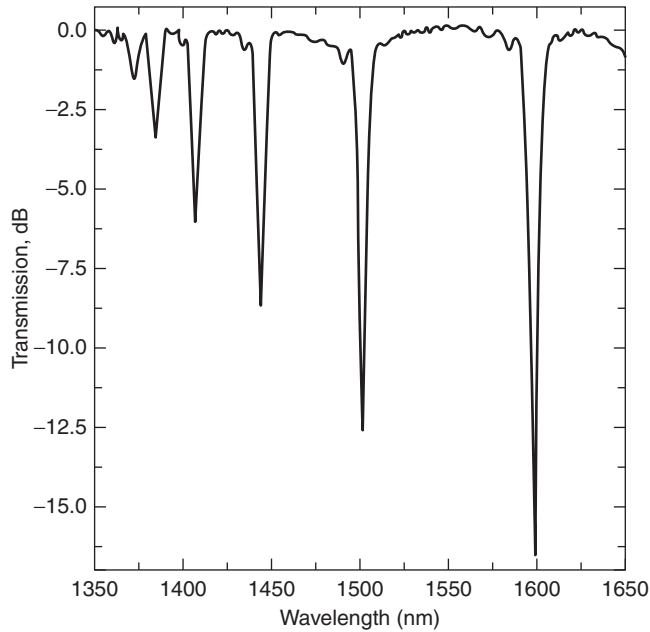


Figure 6.46 | A typical transmission spectrum of an LPFG.

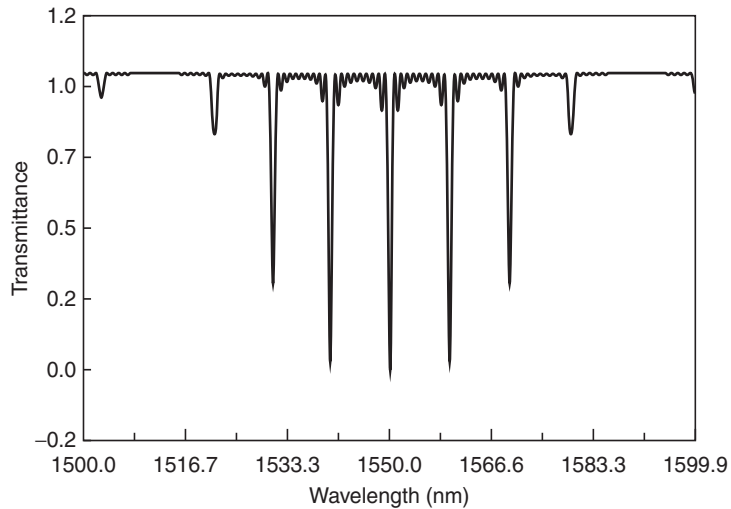


Figure 6.47 | Transmission spectrum of cascaded LPFG (length: 20 mm, period: 500 μm , no. of cascaded LPG: 10, length of grating-free region: 60 mm).

LP_{0m} ($m \geq 2$) modes. For a grating with asymmetric index distribution across the fiber cross-section, LP_{01} mode can be coupled to a large number of modes, including those that do not possess circular symmetry.

5. **Cascading of LPFG:** Cascaded LPFGs can be used as a multichannel filter for multiwavelength signal generation in WDM systems. The transmission spectrum of the filter, as shown in Fig. 6.47, has a sinusoidal

fringe pattern with an envelope governed by the shape of the rejection band of the individual grating. The channel spacing can be controlled by changing the physical separation of the two gratings.

6.5.10.4 Fabrication Techniques

Fiber Bragg grating can be internally written or externally written. Internally inscribed Bragg gratings were first demonstrated in 1978 by Hill and co-workers. They discovered the photosensitivity in optical fiber for the first time in the experimental setup as shown in Fig. 6.48. Continuous wave argon ion laser was launched into an SMF and the intensity of back reflected light was monitored. Initially, the reflected light intensity was low, but after a period of few minutes they observed growth in back-reflected power and almost all the light launched into the fiber were back reflected. The growth in back-reflected light was explained in terms of a new non-linear effect called “*photosensitivity*.”^[21] The photosensitivity of a fiber is its capability to change its refractive index locally when it is irradiated by a UV light. This enables an index grating to be written in the fiber. The reasoning is as follows. Coherent light propagating in the fiber interferes with a small amount of light reflected back from the end of the fiber to setup a standing wave pattern which through photosensitivity writes an index grating in the fiber core. Magnitude of refractive index change obtained depends on composition of glassy material, wavelength, intensity, and total dosage of irradiating light. As the strength of grating increases, the intensity of the back-reflected light increases until it saturates near 100%. But the Hill gratings function only at light wavelengths in visible range close to wavelength of writing light (generally 488 nm). To overcome this limitation, external writing has been developed.

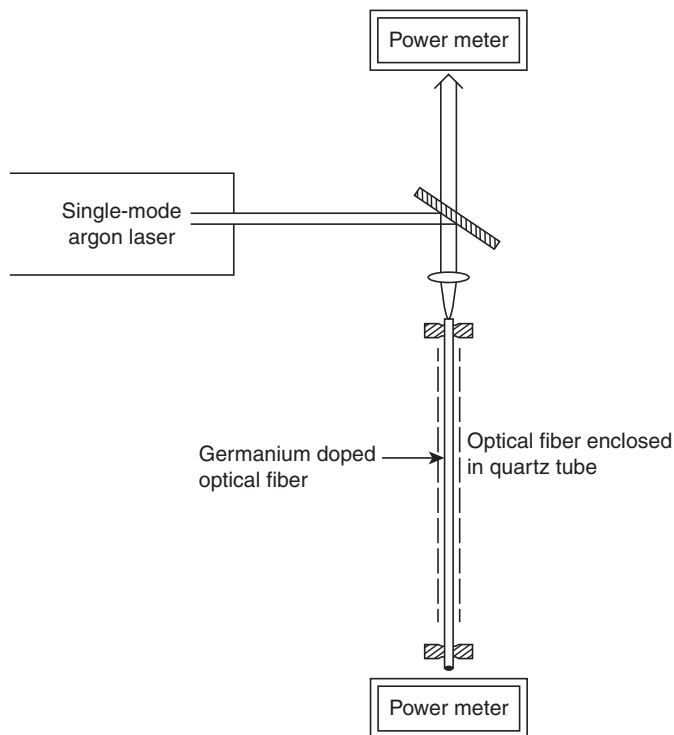


Figure 6.48 | Schematic for recording Hill gratings in optical fiber.

6.5.10.4.1 Photosensitivity

Bragg gratings have been written in many types of optical fibers using various methods. However, the mechanism of index change is not fully understood. Several models have been proposed for these photo-induced refractive-index changes. Photosensitivity of a fiber is its capability to locally change its refractive index when it is irradiated by a UV light. The only common element in these theories is that the germanium–oxygen vacancy defects.

Sometimes a weaker bond is formed between Ge and the germanium–oxygen vacancy defects, Si atoms, or between two Ge atoms, which lacks the intermediate oxygen atom as shown in Fig. 6.49. This weaker bond requires 5.06 eV of energy to break, which lies within ultraviolet range. Single-photon absorption of 244 nm radiation from a frequency-doubled argon-ion laser breaks this defect bond, and the released electrons are trapped at hole-defect sites to form color centers, such as Ge(1) and Ge(2). As products of photochemical reaction, GeE , SiO_3^+ or GeO_3^+ and e^- are created and the absorption of light near 5.06 eV is reduced. Resulting changes in the absorption spectrum are accompanied by a corresponding index change through the Kramers–Kronig relation.

Fluorine and trivalent boron (as B_2O_3) are other dopants commonly used in germania-doped silica fiber. A major difference between germanium and fluorine/boron is that while the refractive index increases with increasing concentration of germanium, it decreases with boron/fluorine. With fluorine, only modest reductions in the refractive index are possible (-0.1%), whereas with boron, large index reductions (>0.02) are possible.

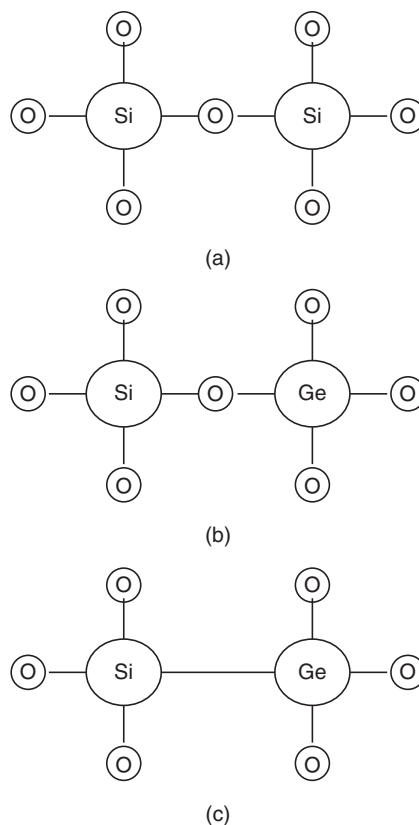


Figure 6.49 (a) Standard SiO_2 glass structure. (b) Ge–Si structure after Ge doping. (c) Ge–Si bond lacking the intermediate oxygen atom (germanium oxygen defect).

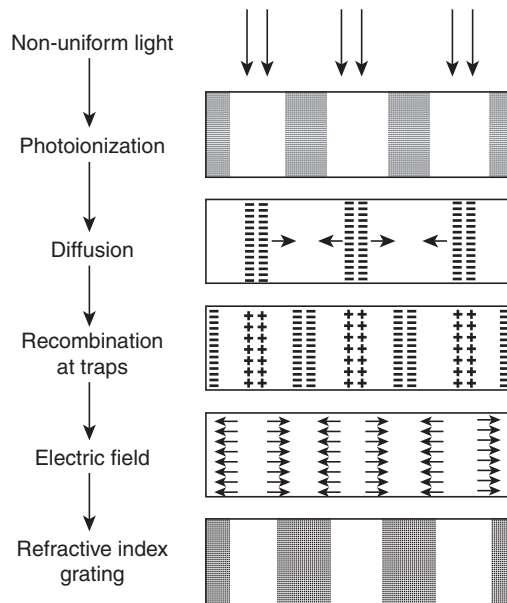


Figure 6.50 | Response of a photorefractive material to a sinusoidal spatial light pattern.

Boron, being trivalent, also changes the topology of the glass. Boron and germanium together allow a low refractive index difference between the core and cladding to be maintained with a large concentration of both elements. On the other hand, a depressed cladding fiber can be fabricated by incorporating boron in the cladding to substantially reduce the refractive index.

Boron-doped silica glass is generally free of defects, with a much-reduced melting temperature. Being a lighter atom, the vibrational contribution to the absorption loss extends deeper into the short wavelength region and increases the absorption loss in the 1,500 nm window. Boron with germanium doping is excellent for photosensitivity. In addition to this, aluminosilicate, phosphosilicate, chalcogenide, and fluoride fibers are also photosensitive.

During the writing process of a Bragg grating, when the fiber is exposed to a UV interference pattern, free electrons in the high intensity regions diffuse until they are trapped by defects in the low intensity regions as shown in Fig. 6.50. This redistribution of charges within the fiber will create periodic space-charge electric fields. The periodic refractive-index change is proportional to $\chi(3)E^2$, where $\chi(3)$ is the third-order non-linear coefficient and E is the electric field of the dipole source.

Recently, hydrogen loading (hydrogenation), flame brushing, and boron co-doping have been used for enhancing the photosensitivity of germanosilicate fibers. Hydrogenation or hydrogen loading of optical fibers is a simple technique for achieving very high UV photosensitivity in germanosilicate optical fibers. Hydrogen loading is carried out by diffusing hydrogen molecules into fiber cores at high pressures and temperatures. It should be noted that the increased fiber/waveguide photosensitivity, as a result of hydrogen loading, is not a permanent effect, and as the hydrogen diffuses out, the photosensitivity decreases.

6.5.10.4.2 Inscribing Bragg Grating in Optical Fiber

Depending on the fabrication technique, Bragg gratings may be labeled as internally or externally written. Externally written Bragg gratings – gratings inscribed using techniques such as interferometric, point-by-point, and

phase-mask – overcome the limitations of internally written gratings and are considered far more useful. Although most of these inscription techniques were initially considered difficult due to the requirements of sub-micron resolution and thus stability, they are well controlled today and the inscription of Bragg gratings using these techniques is considered routine.

Transverse Holographic Method

The interferometric fabrication technique, which is also called as transverse holographic method, is an external writing approach for inscribing Bragg gratings in photosensitive fibers. It was first demonstrated by Meltz and co-workers, who used an amplitude splitting interferometer to fabricate FBGs in an experimental arrangement similar to the one shown in Fig. 6.51. An excimer-pumped dye laser operating at a wavelength in the range of 486–500 nm was frequency-doubled using a non-linear crystal. This provided a UV source in the 244 nm band with adequate coherence length (a critical parameter in this inscription technique). The UV radiation was split into two beams of equal intensity that were recombined to produce an interference pattern, normal to the fiber axis. A pair of cylindrical lenses focused the light onto the fiber, and the resulting focal line was approximately 4 mm long by 124 μm wide. A broadband source was also used in conjunction with a high-resolution monochromator to monitor the reflection and transmission spectra of the grating. The grating was formed in a 2.6 μm diameter core, 6.6-mol% GeO_2 -doped fiber after 5 min exposure to a 244 nm interference pattern with an average power of 18.5 mW. The length of the exposed region was estimated to be between 4.2 and 4.6 mm.

In interferometer system, the interference fringe pattern period Λ depends on both the irradiation wavelength λ_w and the half-angle between the intersecting UV beams. Since the Bragg grating period is identical to the period of the interference fringe pattern, the fiber grating period is given by

$$\Lambda = \frac{\lambda_w}{2 \sin \phi} \quad (6.62)$$

Bragg condition is given as

$$\lambda_B = 2n_{\text{eff}} \Lambda$$

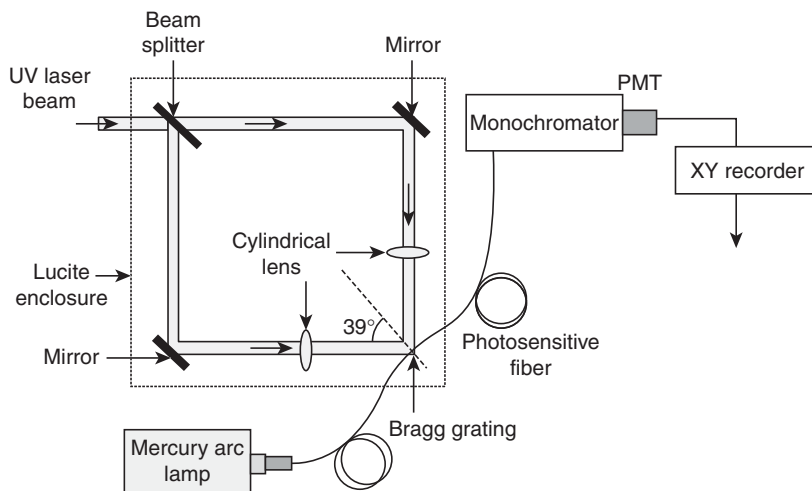


Figure 6.51 | An amplitude-splitting interferometer used by Meltz et al., which demonstrated the first externally fabricated Bragg grating in optical fiber.

$$\Rightarrow \lambda_B = \frac{n_{\text{eff}} \lambda_w}{\sin \phi} \quad (6.63)$$

where n_{eff} is the effective core index. Bragg grating wavelength can be changed by changing λ_w and/or ϕ . This technique is advantageous as compared to internal writing technique as Bragg gratings can be photo-imprinted in the fiber core without removing the glass cladding. The period of photo-induced grating depends on the angle between the two interfering coherent UV light beams. Bragg gratings could be made to function at much longer wavelengths in a spectral region of interest for devices which have applications in fiber-optic communications and optical sensors.

Phase-Mask Method

One of the most effective methods for inscribing Bragg gratings in photosensitive fiber is the phase-mask technique. This method employs a diffractive optical element (phase mask) to spatially modulate the UV writing beam (Fig. 6.52). KrF excimer lasers are the most common UV sources used to fabricate Bragg gratings with a phase mask. The UV laser sources typically have low spatial and temporal coherence. The low spatial coherence requires the fiber to be placed in close contact with the grating corrugations on the phase mask in order to induce maximum modulation in the index of refraction. The further the fiber is placed from the phase mask, the lower is the induced index modulation, which results in lower reflectivity Bragg gratings. Clearly, the separation of the fiber from the phase mask is a critical parameter in producing high-quality gratings.

Optical fiber is placed almost in contact with the corrugations of the phase. Ultraviolet light which is incident normal to the phase mask passes through and is diffracted by the periodic corrugations of phase mask. Normally, most of the diffracted light is contained in the 0, +1, and -1 diffracted orders. However, the phase mask is designed to suppress the diffraction into the zero order by controlling the depth of the corrugations in the phase mask.

The two ± 1 diffracted order beams interfere to produce a periodic pattern that photo-imprints a corresponding grating in the optical fiber. If the period of the phase mask grating is Λ_{mask} and the period of photo-imprinted index grating is $\Lambda_{\text{mask}}/2$, then this period is independent of the wavelength of the ultraviolet light irradiating the phase mask.

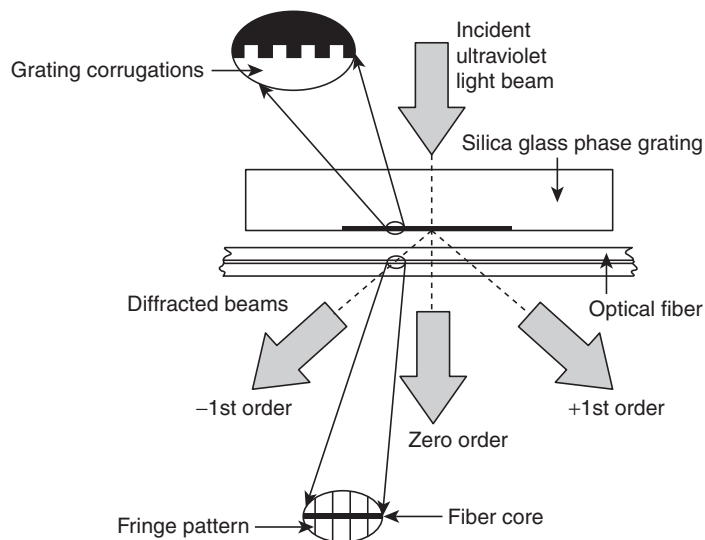


Figure 6.52 | Phase-mask geometry for inscribing Bragg gratings in optical fibers.

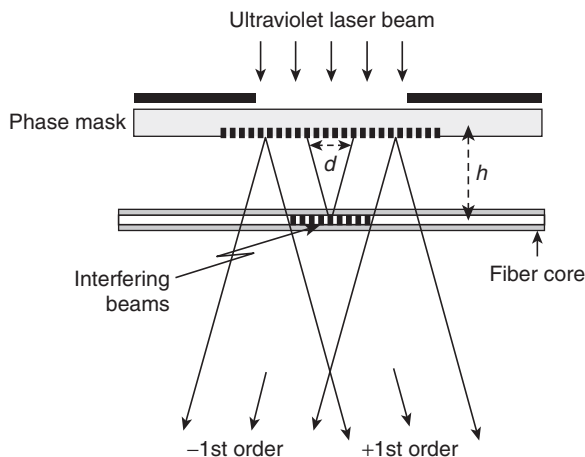


Figure 6.53 | Simple schematic of phase-mask geometry for inscribing Bragg gratings in optical fibers.

To understand the details refer to Fig. 6.53. The fiber core is at a distance h from the phase mask. The transmitted *plus* and *minus* first orders that interfere to form the fringe pattern on the fiber emanate from different parts of the mask (referred to as distance d in Fig. 6.53). Since the distance of the fiber from the phase mask is identical for the two interfering beams, the requirement for temporal coherence is not critical for the formation of a high-contrast fringe pattern. On the other hand, as the distance h increases, the separation d between the two interfering beams emerging from the mask increases as well. In this case, the requirement for good spatial coherence is critical for the formation of a high-contrast fringe pattern. As the distance h extends beyond the spatial coherence of the incident UV beam, the interference fringe contrast will deteriorate, eventually resulting in no interference at all. It should also be noted that if the zeroth-order beam is not significantly suppressed, interference will occur between zeroth- and first-order diffracted beams; in this case, the interference pattern will change as a function of the fiber-phase-mask separation resulting in fringes that vary from half the phase-mask period to one period of the mask.

As compared to holographic technique, phase-mask technique is very simple and offers easier alignment of fiber for photo-imprinting. Cheaper UV excimer laser source can be used since the stability and coherence requirements are reduced. The technique is flexible to fabricate gratings with controlled spectral response characteristics. Also it is possible to manufacture several gratings at a time in a single exposure by irradiating parallel fibers through the phase mask.

The shortcoming of this technique is that a separate phase mask is required for each different Bragg wavelength.

Point-by-Point Fabrication of Bragg Gratings

The point-by-point technique for fabricating Bragg gratings is accomplished by inducing a change in the index of refraction a step at a time. A focused single pulse from an excimer laser produces each grating plane separately. A single pulse of UV light from an excimer laser passes through a mask containing a slit. A focusing lens images the slit onto the core of the optical fiber from the side as shown in Fig. 6.54, and the refractive index of the core increases locally in the irradiated fiber section. The details are shown in Fig. 6.55. The fiber is then translated through a distance Λ corresponding to the grating pitch in a direction parallel to the fiber axis. This process is repeated to form the grating structure in the fiber core. Essential to the point-by-point fabrication technique is a very stable and precise submicron translational system.

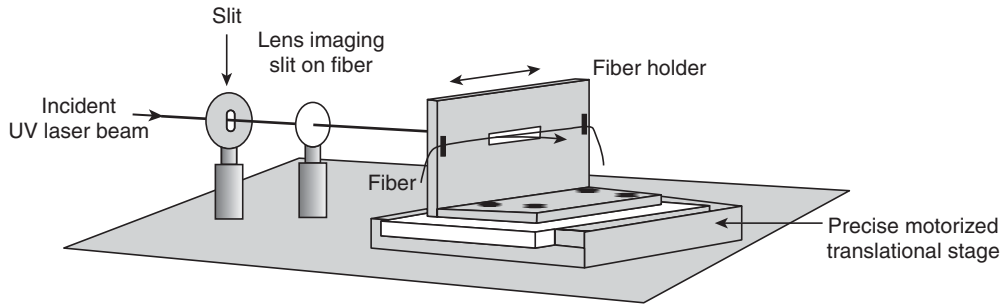


Figure 6.54 | Schematic of setup for fabricating Bragg gratings using the point-by-point technique.

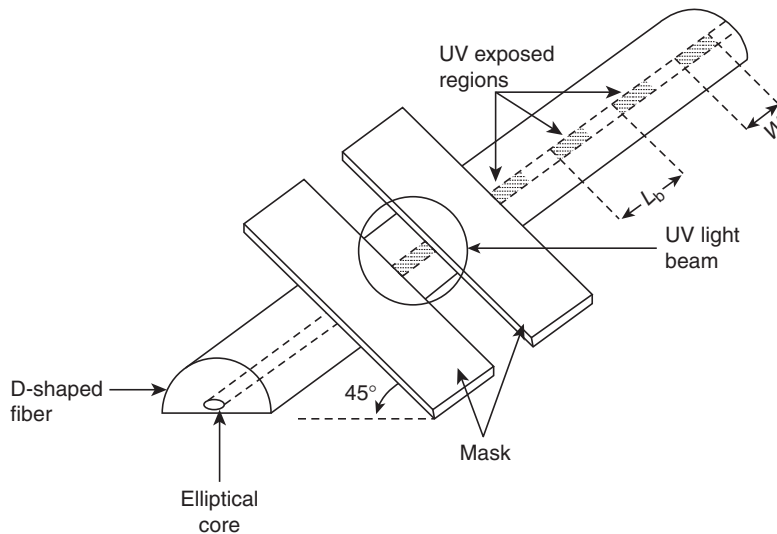


Figure 6.55 | Details of point-by-point writing.

One disadvantage of the point-by-point technique is that it is a tedious process. Because it is a step-by-step procedure, this method requires a relatively long process time. Errors in the grating spacing due to thermal effects and/or small variations in the fiber's strain can occur. This limits the gratings to very short lengths. Typically, the grating period required for first-order reflection at 1,550 nm is approximately 530 nm. Because of the sub-micron translation and tight focusing required, first-order 1,550 nm Bragg gratings are yet to be demonstrated using the point-by-point technique. Malo et al. have only been able to fabricate Bragg gratings, which reflect light in the second and third order, that have a grating pitch of approximately 1 μm and 1.5 μm , respectively.

6.5.10.4.3 Erasing of Grating

The grating can be erased by monotonic UV illumination. New gratings can also be formed at the positions where the previous gratings had been erased. Thermal erasing of grating is also possible. The grating is erased by heating up to 450°C for 15 min.

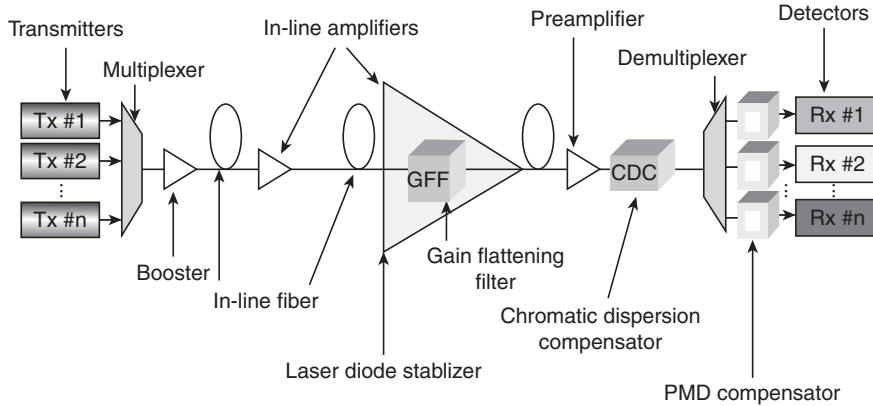


Figure 6.56 | Typical WDM system.

6.5.10.5 Applications of FBGs

Fiber Bragg gratings (FBGs) have emerged as important components in a variety of light-wave applications. Sharp filtering characteristics, direct connectivity to fibers, and ease of manufacturing are very attractive features of FBGs when they are applied to WDM systems as shown in Fig. 6.56. In addition to band-pass filters and band-rejection filters, FBGs have many other applications such as wavelength lockers for stabilizing laser spectra, gain equalizers for EDFAs, optical add/drop multiplexers, and dispersion compensators. This section deals with these applications of FBGs.

6.5.10.5.1 Gain Equalizers for EDFAs

With the development of high-capacity WDM optical communication systems, the gain flattening of EDFAs has been a research issue in the recent years. For single-channel systems, the gain variation is not a problem. However, as the number of channels increases, the transmission problem arises because a conventional EDFA has intrinsic non-uniform gain. They typically present gain peaking at about 1,530 nm, and the useful gain bandwidth may be reduced to less than 10 nm. The gain of EDFAs depends on a large number of device parameters, such as erbium-ion concentration, amplifier length, core radius, and pump power. To increase the gain bandwidth of an amplified light-wave system, several methods can be used. However, equalizing optical filters operating as spectrally selective loss elements appear to be the best solution.

Chirped FBG meets requirements for practical gain equalizers.^[22] The gain flattening filter (GFF) is fabricated by changing the grating period so that the refractive-index modulation realizes the required transmission loss spectrum, which should have an inverse shape of the EDFA gain spectrum (see Fig. 6.57). This device can cover a large optical bandwidth over >35 nm.

6.5.10.5.2 Chirped FBGs for Dispersion Compensation

One of the most useful applications of FBGs is the dispersion compensator that can compensate for group-velocity dispersion (GVD) of fibers for transmission. The standard single-mode fiber (SMF) has a zero-GVD wavelength at 1.3 μm ; therefore, it has an anomalous GVD as large as 16 ps/nm/km at 1.5 μm . Chirped FBGs can be designed to have a GVD that compensates for fiber dispersion precisely.

Due to anomalous GVD of SMF, the longer wavelength component of the signal travels through the fiber slower than the shorter wavelength component, resulting in the pulse-width broadening. The grating period of

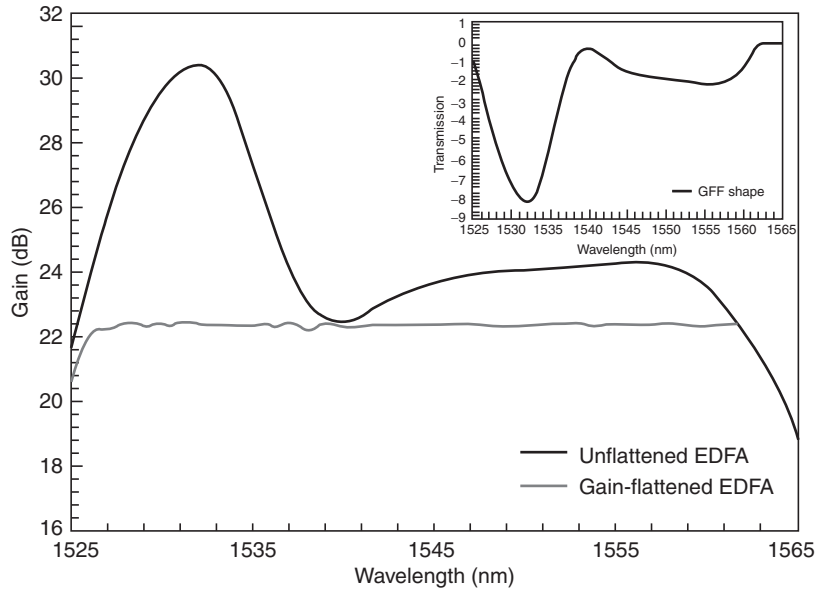


Figure 6.57 | Gain spectrum of a gain-flattened EDFA by using an FBG-based gain equalizer.

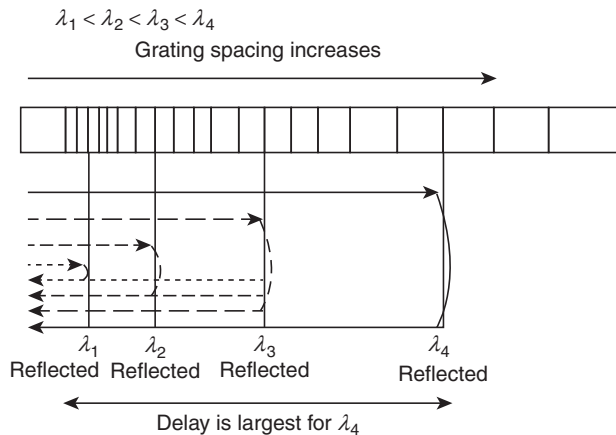


Figure 6.58 | Fiber grating works as a delay line.

the chirped FBG is linearly chirped, and becomes shorter from the input end toward the far end. Figure 6.58 illustrates the working principle of the chirped FBG as a dispersion compensator.

The longer wavelengths in a pulse arrive first, and the shorter wavelengths arrive last. As shown in Fig. 6.59, the grating is made with different spacings, which reflect different wavelengths at different positions along the length of the grating. In this case, longest wavelengths λ_4 arrive first and are transmitted through the grating toward the last part. The first part of grating reflects the shortest wavelengths, which arrive last. The longer wavelengths have to travel a longer distance, so they are delayed, allowing the shorter wavelengths to catch up.

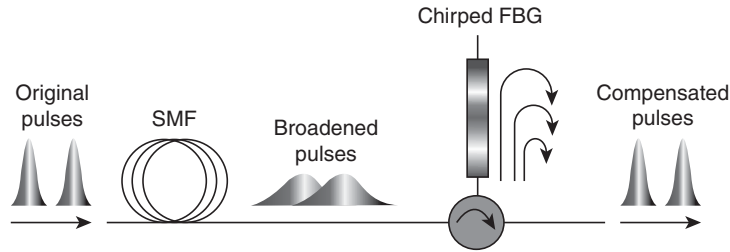


Figure 6.59 | Dispersion compensation by chirped FBG.

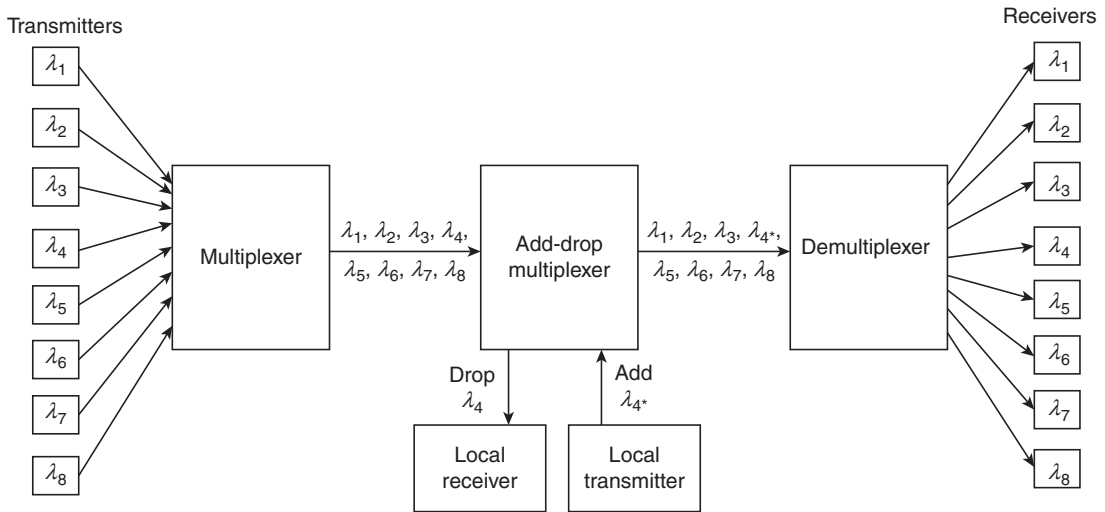


Figure 6.60 | A WDM system.

6.5.10.5.3 Optical Add/Drop Multiplexers

The ability of fiber gratings to select one or more wavelengths is important in WDM, or where pump and signal wavelengths must be combined or separated as shown in Fig. 6.60. Other optical devices can do the same thing, but fiber gratings select a narrow range of wavelengths and fit naturally into fiber-optic systems.

The FBG can be used to construct an optical add/drop multiplexer (OADM) in WDM networks. The configuration of the OADM is shown in Fig. 6.61, where two circulators are connected with an FBG. Wavelength channels $\lambda_1, \lambda_2, \dots, \lambda_N$ are coming from the network and incident on the OADM from port 1. Only the wavelength that coincides with the Bragg wavelength λ_k of the FBG is dropped from port 2. Other non-resonant wavelengths pass through the FBG and come back to the network from port 4. On the other hand, the wavelength λ_k can be added to the network from port 3.

In *MZI* add/drop multiplexer, as shown in Fig. 6.62, a signal is launched into the input port. It is equally split through the first 3 dB coupler. Bragg wavelength is back-reflected on each arm of the *MZI*, whereas all other wavelengths pass through the FBG ($\lambda_{B,drop}$). All other phase-matched wavelengths ($\lambda_i \neq \lambda_B$) leave the device via the output port. A signal at the channel wavelength (λ_B) launched into the add port emerges at the output port as well.

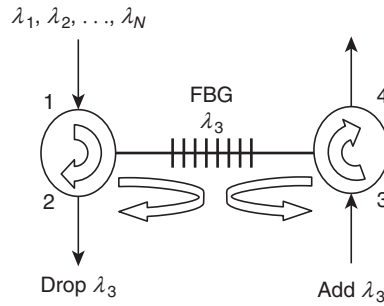


Figure 6.61 | FBG-based optical add/drop multiplexer.

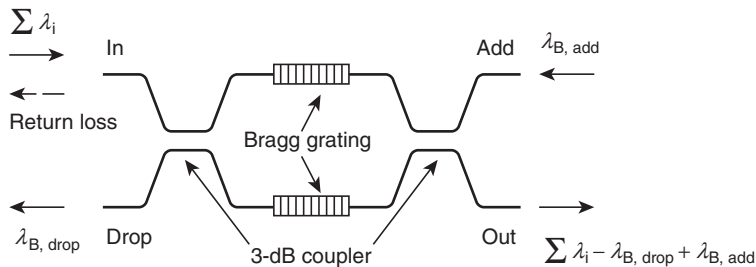


Figure 6.62 | Mach-Zehnder interferometer add/drop multiplexer.

6.5.10.5.4 Wavelength-Stabilized Lasers

When an FBG is coupled to a semiconductor optical amplifier, the oscillation wavelength of the laser is defined by the Bragg wavelength of the FBG. Since the temperature sensitivity of the Bragg wavelength is small, the oscillation wavelength can be stabilized precisely to the International Telecommunication Union (ITU) grid. The FBG is also applied to the wavelength stabilization of 980 nm semiconductor lasers for EDFA pump sources as shown by Fig. 6.63. The FBG acts as wavelength locker to produce a narrow spectrum near the Bragg wavelength.^[23] The narrow and stabilized spectrum of 980 nm lasers is of importance for pumping erbium–ytterbium fiber amplifiers. The grating can be directly written inside the rare-earth-doped fiber-laser cavity with a $\pi/2$ -phase shift in the middle of the cavity length.

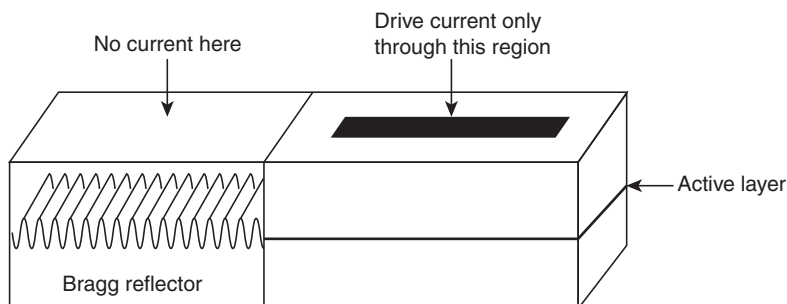


Figure 6.63 | Single-frequency laser.

Such phase-shifted FBG enables the fiber distributed feedback (DFB) laser to oscillate at the Bragg wavelength. The fiber DFB laser has shown superior low noise, high output power, and narrow line width. Another advantage of this type of laser is that its oscillation wavelength can be tuned without mode hopping just by changing the Bragg wavelength.

6.5.10.5.5 Sensor Technology

Fiber and integrated optics technologies were primarily developed for telecommunication applications. However, the advances in the development of high-quality and competitive price opto-electronic components and fibers have largely contributed to the expansion of guided wave technology for sensing as well.^[24,25] The main reasons that make guided wave optics attractive for sensing can be summarized as follows:

1. Non-electrical method of operation, which is explosion-proof and offers intrinsic immunity to radio frequency and, more generally, to any kind of electromagnetic interference.
2. Small size/weight and great flexibility that allow access to otherwise restricted areas.
3. Capability of resisting to chemically aggressive and ionizing environments.
4. Easy interface with optical data communication systems and secure data transmission.

Many types of fiber gratings can be used in sensing applications including Bragg gratings, LPGs, and chirped gratings. The basic principle of FBG sensors is the measurement of an induced shift in the wavelength of an optical source due to a measurand, such as strain or temperature. A basic reflective FBG sensor system is shown in Fig. 6.64. A broadband light source is used to interrogate the grating, from which a narrowband slice is reflected. The peak wavelength of the reflected spectrum can be compared to λ_B .

It should be noted that LPFGs are attracting much interest for use in sensing applications. In SMFs, an LPFG couples the fundamental HE_{11} core mode and multiple forward-propagating HE_{1m} cladding modes. Thus, they have a higher sensitivity to external perturbations. They are more sensitive to measurands than FBGs and are easier to manufacture. A typical LPFG has a length of tens of mm with a grating period of hundreds of μm . The high attenuation of the cladding modes results in a series of minima occurring in the transmission spectrum of the fiber. This means that the spectral response is strongly influenced by the optical properties of the cladding and surrounding medium. This can be exploited for chemical sensing as shown in Fig. 6.65, where a broadband source is used to interrogate an LPFG wavelength shift of the output spectrum.

Refractive index sensing is the backbone of various chemical and biological processes. In LPFGs, the fundamental guided mode gets coupled to different order cladding modes that attenuate quickly resulting in multiple loss wavelength resonance bands in transmission. The loss resonance wavelength at which coupling takes place satisfies the phase matching condition, that is,

$$\lambda_i = [n_{\text{eff}}^{\text{co}} - n_{\text{eff}}^{\text{cl}}] \Lambda$$

where $n_{\text{eff}}^{\text{co}}$ and $n_{\text{eff}}^{\text{cl}}$ are the effective refractive indices of the fundamental core mode and i^{th} cladding mode, respectively, and Λ is the period of the LPFG.

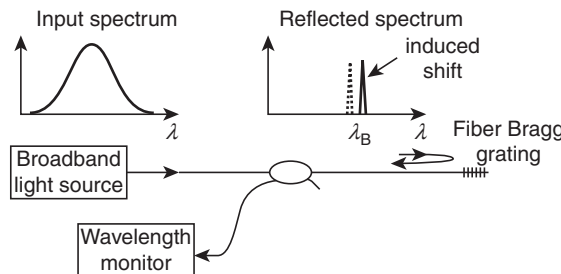


Figure 6.64 | Basic reflective FBG sensor system.

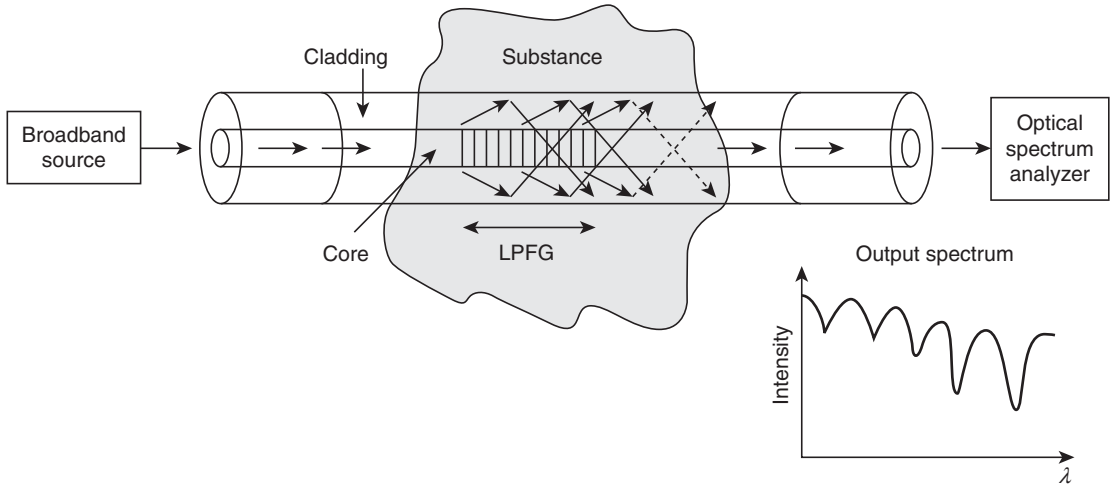


Figure 6.65 | Detail of LPFG chemical sensor.

Since effective index of a cladding mode is dependent upon the refractive index of the surrounding medium, any change in the latter alters the loss resonance wavelength. This is the basic principle of an LPFG-based refractive index sensor.

The effect of refractive index of the surrounding medium on the loss resonance wavelength is given as

$$\frac{d\lambda_i}{dn_{\text{sur}}} = \frac{d\lambda_i}{dn_{\text{icff}}^{\text{cl}}} \cdot \frac{dn_{\text{icff}}^{\text{cl}}}{dn_{\text{sur}}} \quad (6.64)$$

where n_{sur} is the refractive index of the surrounding medium.

Variation in n_{sur} is reflected as shift in loss resonance wavelength as well as variation in intensity of the loss resonance peak. As the order of cladding mode increases, it becomes more sensitive to ambient index change.

In applications where fiber sensors offer new capabilities, however, such as distributed sensing, fiber sensors appear to have a distinct edge over the competition. Here fibers with sensor arrays can be embedded into the materials to allow measurement of parameters such as load, strain, temperature, and vibration, from which the health of the structure can be assessed and tracked on a real-time basis. Gratings may also prove to be useful as the optical-sensing element in a range of other fiber-sensor configurations, such as grating-based chemical sensors, pressure sensors, and accelerometers.

Bragg gratings are extremely susceptible to fluctuations in ambient temperatures. By using special packaging methods, shift in central wavelength of FBG due to temperature drift can be minimized. The temperature compensating package normally comprises two materials having different thermal expansion coefficients or alternatively only one material with a negative thermal expansion coefficient. The extraordinarily negative thermal expansion coefficient material is zirconium tungstate (ZrW_2O_8). It is a ceramic having a negative thermal expansion coefficient over a wide temperature range (0–1,050 K).

6.5.11 Optical Switches

In WDM systems, tens to hundreds of wavelengths, each modulated at 10 Gbps or more, are transmitted on a single fiber at a time. In older systems, the switching of signals from one channel to other was done by electronic switching. In this technique at every switching node, optical signals are converted to electrical form (O/E conversion), which

are then buffered electronically. Then again these are converted to optical form (E/O conversion) and subsequently are forwarded to their next hop.

Electronic switching is a well-studied, sophisticated, and extensively used technology. However, with increasing data-rate capacity and channel capacity, this technology is unable to keep up as it is highly data-rate- and protocol-dependent. Whenever the system upgradation is required, the electronic equipments need to be replaced and/or added.

These drawbacks can be eliminated if the signal switching is performed in only optical domain. Optical switching is done without electrical-to-optical and optical-to-electrical conversion. Thus, it is data rate as well as protocol-independent. This results in reduction in the network equipment, an increase in the switching speed, and thus network throughput, and a decrease in the operating power.^[26,27]

6.5.11.1 Optical-Switching Technologies

Some prominent optical-switching technologies and their features are as follows:

6.5.11.1.1 Optomechanical Switches

It is the old, well-established, and commercially available optical-switching technology. The switching function is performed by some mechanical means in these switches. These mechanical means are as follows:

1. Prisms.
2. Mirrors.
3. Directional couplers.

Advantages of optomechanical switches are as follows:

1. Exhibit low insertion losses.
2. Low PDL.
3. Low crosstalk.
4. Low fabrication cost.

Disadvantages of optomechanical switches are as follows:

1. Low switching speed (few milliseconds).
2. Lack of scalability.
3. Long-term reliability problem.

Optomechanical switch configurations are limited to 1×2 and 2×2 port sizes. Larger port counts can be obtained by combining several 1×2 and 2×2 switches. However, this increases cost and degrades performance. Optomechanical switches are mainly used in fiber protection and very low-port count wavelengths add/drop applications.

6.5.11.1.2 Microelectromechanical System Devices

Microelectromechanical systems (MEMS) use tiny reflective surfaces to redirect the beams of light to a desired port. There are two possible methods for this:

1. By steering the light beam directly to a port.
2. By ricocheting off the light of neighboring reflective surfaces to a port.

Two MEMS approaches for optical switching are as follows:

1. Two-dimensional (2-D) or digital.
2. Three-dimensional (3-D) or analog.

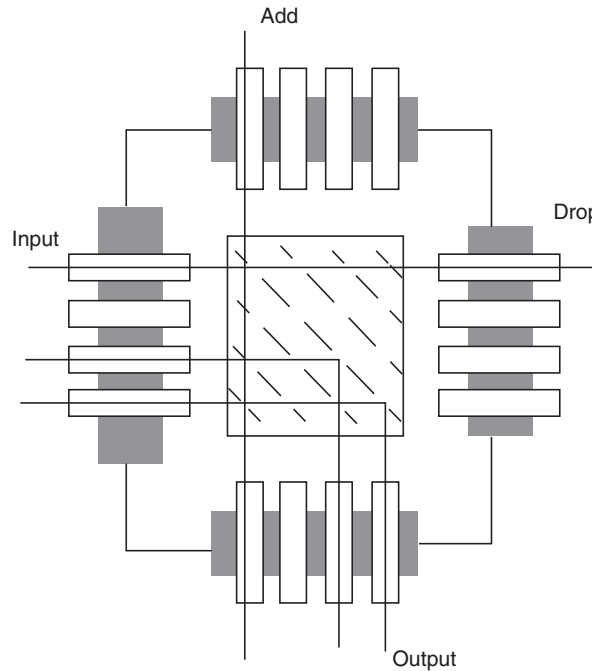


Figure 6.66 | 2-D MEMS switching technology.

Figure 6.66 shows a top view of a 2-D MEMS device. In this device, the microscopic mirrors are arranged in a crossbar configuration. The mirror position is either ON or OFF (bistable). Due to this, the operation of switch is digital and switch driving is also very straightforward. This type of switch is used for adding or dropping optical wavelength channels. These are not preferred for applications requiring more than 32-input, 32-output port configuration because of high insertion loss.

In 3-D MEMS, for each input and output port, there is a dedicated movable mirror. A connection path from an input port to a selected output port is established by tilting two mirrors independently. The light mirrors can be tilted freely about two axes, thus giving analog mode operation. Very large-port count switches having 1,000 input and output ports can be established by using this technology. A complex and expensive feedback system is required to maintain the position of mirrors against any external disturbances as drift is the main drawback of this technology.

6.5.11.1.3 Electro-Optic Switches

A directional coupler having changeable coupling ratio can be used as 2×2 switch. The coupling ratio is changed by varying the refractive index of the material in the coupling region. The switch is constructed from the waveguide of commonly used electro-optic material, lithium niobate (LiNbO_3). The refractive index of the substrate is changed by applying an electrical voltage to the electrodes. The light path through the waveguide toward the desired port is manipulated by changing the refractive index. The switching time is limited due to capacitance of electrode configuration and its state can be changed very rapidly in less than few nanoseconds. Thus, these switches are fast and reliable. But it is polarization-dependent, and the insertion loss is very high. By applying higher driving voltage, the problem of polarization-dependence can be solved. However, this limits the switching speed of device.

Several 2×2 switches are integrated on a single substrate to realize large switches. But they have higher insertion loss and polarization-dependence loss. They are more expensive than mechanical switches.

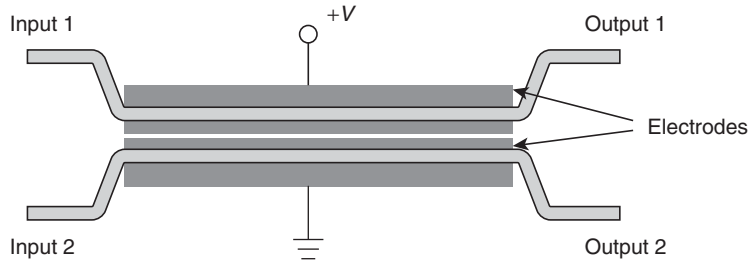


Figure 6.67 | An electro-optic directional coupler switch.

6.5.11.1.4 Thermo-Optic Switches

Thermo-optic switches work on the principle of *thermo-optic effect*. The variation of the refractive index of a dielectric material due to temperature variation of the material itself is called thermo-optic effect. There are two categories of thermo-optic switches:

1. **Interferometric switches** are based on MZIs. They consist of two 3 dB couplers. The coupler splits the signal into two beams. The signal traveling through the first coupler gets equally divided into two parts, travels through the two arms of equal length. It is combined in second coupler and finally splits again.
The application of heat to one arm of the interferometer causes change in the refractive index. This changes the optical path of light through that arm of the interferometer. Change in path length causes phase shift in the signal of the two arms. Thus, by changing the amount of heat applied to one arm, phase difference between the light beams through the two arms of interferometer can be changed. Depending upon the phase difference, the two signals interfere constructively or destructively. Thus, the two signals get added or subtracted, respectively. The output port is thus selected.
2. **Digital optical switches** are integrated optical devices. Two interacting waveguide arms constitute the switch. The light propagates through the waveguide arms. The output port is decided by the phase error between the two beams. One waveguide arm is heated using an electrode controlled through electronics circuit. The heating causes change in refractive index of waveguide material, and the light is transmitted down one path rather than the other.

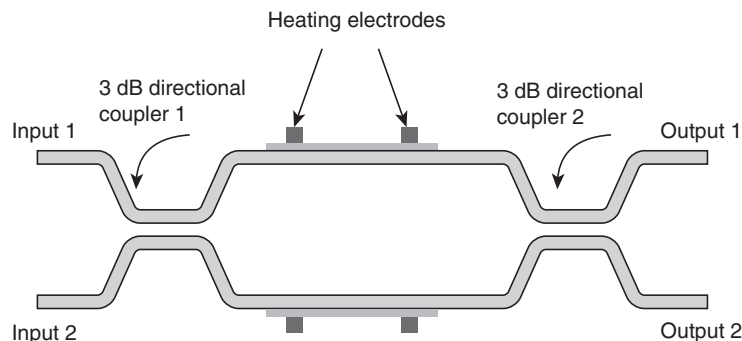


Figure 6.68 | A 2×2 interferometric switch.

Advantages:

1. Miniature size.
2. This technology allows the integration of variable optical attenuators and wavelength selective elements (AWGs) on the same chip with the same technology.

Disadvantages:

1. High-driving-power characteristics.
2. Limited integration density (large die area).
3. High-power dissipation.
4. Requires forced air cooling for reliable operation.
5. Optical performance parameters, such as crosstalk and insertion loss, may be unacceptable for some applications.

6.5.11.1.5 Liquid-Crystal Switches

Many organic materials exhibit liquid-crystal state over certain temperature ranges. Due to permanent electrical dipole moment, the molecules can take up certain mean relative orientation in liquid-crystal state. The orientation of these molecules can be controlled by applying a suitable voltage across a cell filled with liquid-crystal material. Thus, optical properties of the material can be altered.

By applying the electric field over the liquid crystal, the polarization state of incident light over liquid crystal can be controlled. Liquid-crystal optical switches work on this principle. Optical space switching is obtained by the change of polarization in combination with polarization selective beam splitters. Polarization diversity scheme is implemented to make devices polarization-insensitive by treating each polarization mode differently. The input signal is decomposed into its TE and TM components.

Figure 6.69 gives the structure of a 1×2 liquid-crystal optical switch. The birefringent plate at the input port is used to change the polarization of incoming light beam into the desired state. The birefringent material has different refractive index along the different axes. Without bias voltage, the input signal passes through the

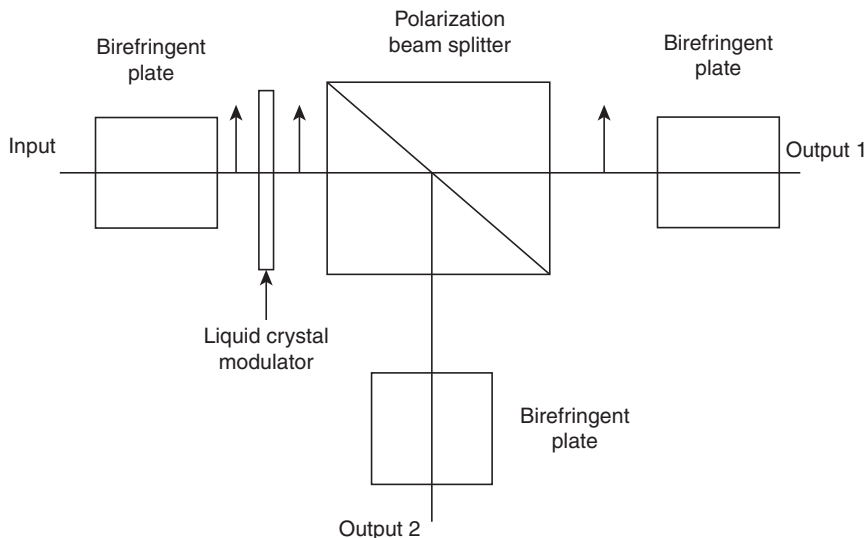


Figure 6.69 | A 1×2 liquid-crystal optical switch.

liquid-crystal cell, and polarization beam splitter with the same polarization and signal is available at output 1. When the bias voltage is applied on the liquid crystal modulator, the molecules rotate the polarizations of the signal passing through them. If a sufficiently large bias is applied, then the molecules rotate the polarizations of the signal passing through them by 90° (orthogonally) and the polarization beam splitter reflects the signal to the other output port (output 2).

Advantages:

1. No moving parts.
2. Very reliable.
3. Their optical performance is satisfactory.

Disadvantage:

1. Gets affected by extreme temperatures if not properly designed.

6.5.11.1.6 Acousto-Optic Switches

The operation of acousto-optic switches is based on the interaction of sound waves and light waves. The input polarization beam splitter splits the input signal into two orthogonally polarized components. These two orthogonal components are directed to propagate through two separate parallel waveguides. Subsequently, a surface-acoustic wave is created and directed in the direction of light waves. Due to an acousto-optic effect in the waveguide material, the structure formed is equivalent to a moving grating. At selected wavelength, the two waves are phase-matched and the phase-matched signal is flipped from TM to TE mode and vice versa. The output polarization beam splitter directs it to the lower output (selected output). A signal that is not phase-matched is directed toward the upper output (unselected output).

It is possible to have several acoustic waves in the material with different frequencies at the same time. Therefore, it is possible to switch several different wavelengths simultaneously if the incoming signal is multiwavelength.

The switching speed of acousto-optic switches is limited by the speed of sound, and typically it is in microseconds.

6.5.11.1.7 Semiconductor Optical Amplifier Switches

Semiconductor optical amplifiers (SOAs) are used in optical network to serve various purposes. A SOA can be used as an ON–OFF switch. If a small bias voltage is applied, population inversion condition will not occur and the input signal will be absorbed by the device (OFF condition of switch). On the other hand, if sufficient bias voltage is applied, population inversion will take place and the input signal will get amplified (ON condition of switch), thus the switching operation can be achieved.

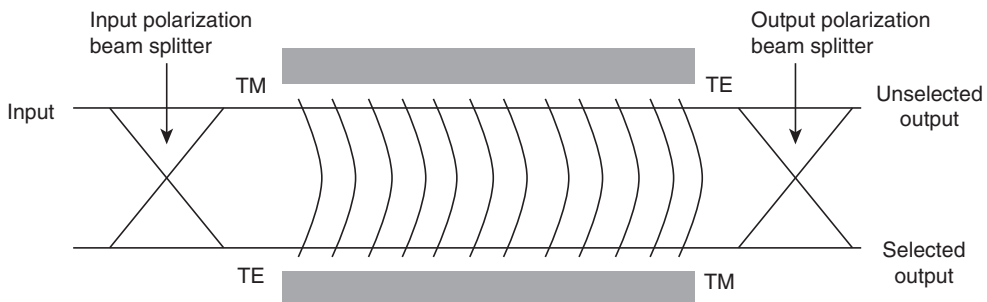


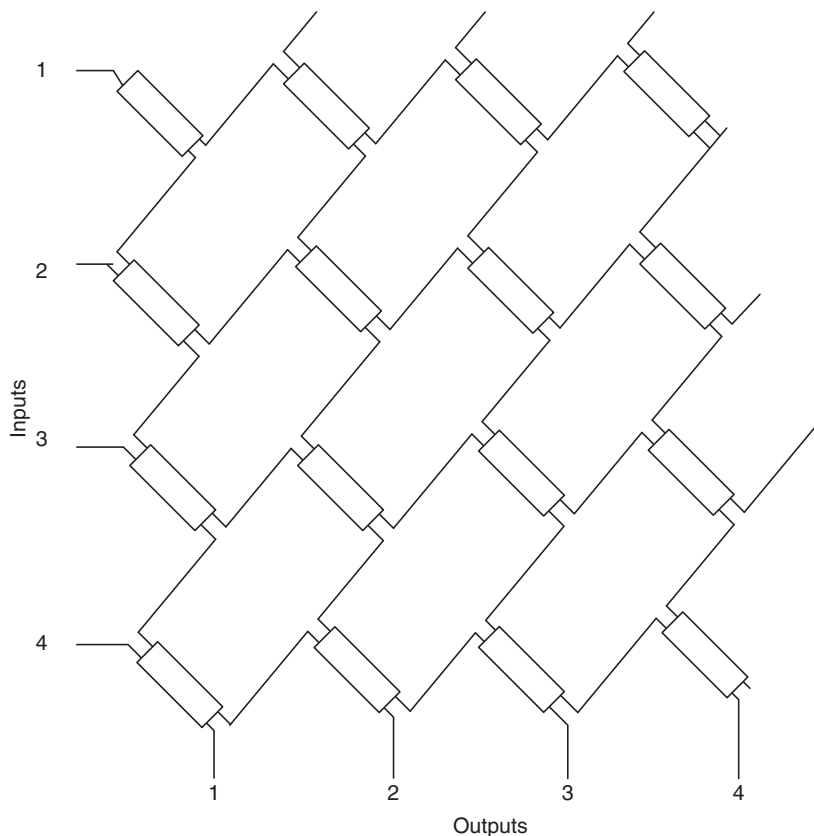
Figure 6.70 | A polarization-independent acousto-optic switch.

Table 6.6 | Matrix switch characteristics

Electrical-to-Optical-to-Electrical (OEO) Switch	All-Optical (OOO) Switch
Converts signal to electrical for switching	Keeps signal in optical domain
Lossless switching – Retimes, reclocks and regenerates signals at specific data rates	Lossy switching – Optical insertion loss from input to output
Unidirectional switching	Bidirectional switching
Point-to-point and broadcast modes	Point-to-point and broadcast mode using photonic multi-casting architecture
Specific digital data rate formats	Independent of data protocol and data rate
Available up to $1,024 \times 1,024$ (and higher)	Maximum input/output typically 256×256

Table 6.6 enlists the various comparisons between all-optical switch and electrical-to-optical-to-electrical switch.

6.5.11.1.8 Large Switches

**Figure 6.71** | A 4×4 crossbar switch realized using 16, 2×2 switches.

Switch sizes larger than 2×2 can be realized by appropriately cascading small switches as shown in Fig. 6.71. The most popular architectures for building large switches are the crossbar. The interconnection between the inputs and the outputs is achieved by appropriately setting the states of the 2×2 switches. The connection rule that is used states that to connect input i to output j , the path taken traverses the 2×2 switches in row i until it reaches column j and then traverses the switches in column j until it reaches output j .

6.5.11.2 Parameters for Evaluating an Optical Switch

In WDM systems, all-optical switching fabrics are important in transmission and switching. In these devices, optical-to-electrical and electrical-to-optical (OEO) conversion is not required, which results in allowing switching directly in the optical domain.

Switching time is the most important parameter of a switch. Different applications have different switching time requirements. The various performance parameters for optical switch are as follows:

1. **Insertion loss:** The fraction of signal power that is lost because of the switch is called *insertion loss*. This loss must be as small as possible. The insertion loss of a switch should be about the same for all input–output connections. This loss is usually measured in decibels.
2. **Extinction ratio (ON–OFF switches):** This is the ratio of the output power in the ON-state to the output power in the OFF-state. This ratio should be as large as possible.
3. **Crosstalk:** It is the ratio of the power at a specific output from the desired input to the power from all other inputs.
4. **Polarization-dependent loss (PDL):** If the loss of the switch is not equal for both states of polarization of the optical signal, the switch is said to have PDL. It is desirable that optical switches have low PDL.

Other important parameters are reliability, energy usage, scalability, and temperature resistance.

6.5.11.3 Applications of Switches

6.5.11.3.1 Optical Cross-Connects (OXC)

Optical switches are used to establish an interconnect between two network nodes. A dedicated wavelength is assigned to each light path or optical link. Optical switches are used in OXCs. Optical switches reconfigure the OXCs to support new light paths as shown in Fig. 6.72. In optical networks, OXCs are basic elements for routing optical signals.

Optical switches in OXCs should fulfill the following requirements:

1. The ability to switch to a specific optical path without disrupting the other optical paths.
2. The ability to switch with low loss, high reliability, good uniformity of optical signals independent of path length.
3. Scalability.
4. High port count switches.

In the current optical networks, cross connects cannot operate fully in the optical domain. The optical signal is first converted to electrical. Then the signal is switched using electrical means, and again it is converted to optical signal. This switching scheme is called the OEO switching.

There are a number of disadvantages of OEO switching listed as follows:

1. The switching speed of electronics cannot compete with the capacity of optics.
2. OEO switching is protocol- and data-rate-dependent. Therefore, if system upgradation is required, the expensive transceivers and electrical switch core have to be replaced.

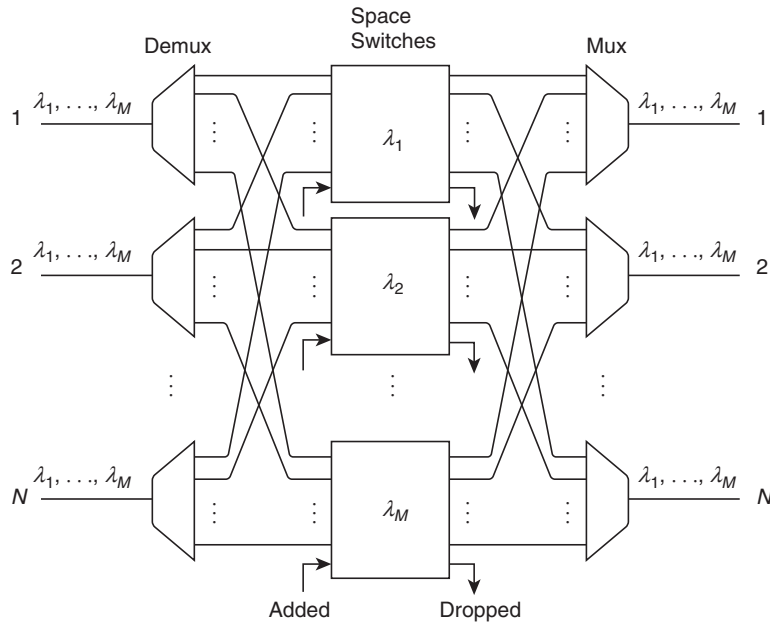


Figure 6.72 | An optical cross-connect based on optical switches.

All these disadvantages are overcome in OOO cross-connects. The electrical-to-optical and optical-to-electrical conversion is not required; switching is done in optical domain. Thus, OOO cross-connects are protocol- and data-rate-independent, and system upgradation is easily possible without replacement of costly equipments. Along with these advantages, the OOO cross-connects have reduced cost, size, and complexity.

6.5.11.3.2 Optical Add/Drop Multiplexing

Optical add/drop multiplexers (OADMs) located in network nodes extract (drop) an optical signal at particular wavelength or insert (add) another optical signal at the same wavelength. Multiwavelength signals add/drop functioning can also be obtained using an OADM without any electronic processing. OADM switches are wavelength-selective.

6.5.11.3.3 Protection Switching

During the system- or network-level errors, traffic transmission is completed by protection switching. Generally more reliable and smaller port count switches like 1×2 or 2×2 are used in optical protection switching.

6.5.11.3.4 Optical Signal Monitoring

For network management, optical signal monitoring (OSM) is extremely important. By using optical tap or coupler, 1% of signal is diverted for testing without interrupting the traffic. The tapped aggregated WDM signal is processed to separate out the different channel wavelengths. The individual wavelengths are then monitored for each channel's optical spectra for wavelength accuracy, optical power levels, and optical crosstalk.

6.5.11.3.5 Network Provisioning

When new data routes have to be established or existing routes need to be modified, network provisioning occurs. Presently, a slow manual process is there that takes several weeks or more. By using high-capacity

reconfigurable switches, automatic and quick service can be obtained. Thus, network flexibility, bandwidth, and profitability can be increased.

6.6 System Performance Parameters

6.6.1 Bit Error Rate

Bit error rate is the most common figure of merit for digital links. It is defined as the number of bit errors (N_{error}) occurring over a specific time interval, divided by the total number of bits (N_{total}) sent during that interval^[28] and is commonly abbreviated as BER. Figure 6.73 shows the plot of BER versus quality factor Q . Mathematically, it is expressed as

$$\text{BER} = \frac{N_{\text{error}}}{N_{\text{total}}} \quad (6.65)$$

If the error rate is 10^{-9} , it indicates that on an average one error occurs for every one billion pulses sent. The typical value of error rate ranges from 10^{-9} to 10^{-15} for optical fiber telecommunication systems.

The BER is also known as the *error probability* and commonly abbreviated as P_{error} . The mathematical relation between signal-to-noise ratio (SNR) and BER is

$$\text{BER} = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} \exp\left(-\frac{x^2}{2}\right) dx \approx \frac{1}{\sqrt{2\pi}} \frac{e^{-Q^2/2}}{Q} \quad (6.66)$$

where Q value is given as

$$Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0} \quad (6.67)$$

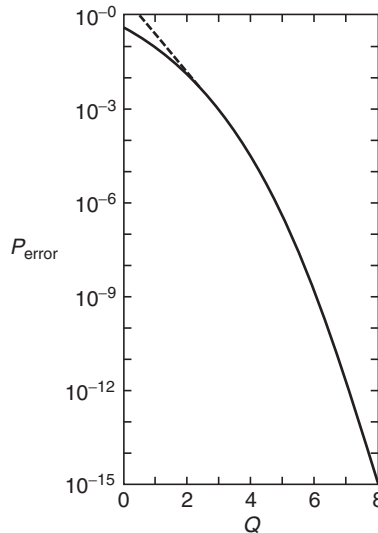


Figure 6.73 | Plot of BER versus factor Q .

where I_1 is the average detected signal currents for bit 1, I_0 is the average detected signal currents for bit 0, σ_1 and σ_0 are the detected *root mean square* (rms) noise values for bit 1 and 0, respectively, assuming a non-return-to-zero (NRZ) code and an equal number of 1 and 0 pulses.

6.6.2 Eye Diagrams

The eye diagram is a simple and powerful measurement technique for assessing the data-handling ability of a digital transmission system. Figure 6.74 shows a typical eye pattern or an eye diagram. It gives the useful system performance information.

The eye opening is an important parameter. The larger size of eye opening is indication of goodness of signal. For this case, signal detection is easier and error rate is low. On the other hand, if the eye is nearly closed, signal detection is very difficult or impossible.

The width of eye opening gives the time interval during which the received signal can be sampled without causing the intersymbol interference, that is, interference from adjacent pulses.

When the height of eye opening is the largest, it is the best time to sample the received signal. Due to amplitude distortion in data signal, the height of eye opening reduces. The degree of distortion is given by the vertical distance between the top of the eye opening and the maximum signal level. It becomes more and more difficult to distinguish between 1s and 0s in the signal as the eye closes more. Thus, the height of the eye opening gives the noise margin or immunity to noise at the specified sampling time.

Noise margin is the ratio of the peak signal voltage V_1 for an alternating bit sequence to the maximum signal voltage V_2 measured from the threshold level. It is always expressed in percentage. Mathematically, we can write

$$\text{Noise margin (\%)} = \frac{V_1}{V_2} \times 100 \quad (6.68)$$

The noise in the receiver and the pulse distortion in optical fiber give rise to *timing jitter*. It is also called as eye jitter or phase distortion. The amount of jitter in the signal is indicated by the horizontal eye opening. Wider horizontal eye opening indicated less amount of jitter. Timing jitter is also indicated by the thickness of the band of signals at the zero-crossing point.

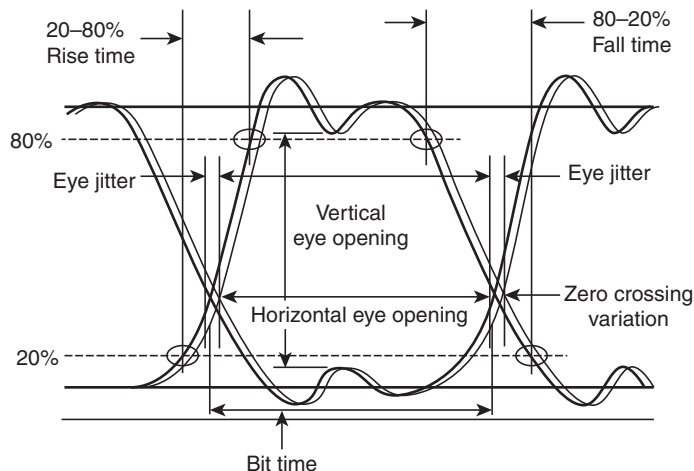


Figure 6.74 | Eye diagram.

The rate of eye closure is indicated by the slope of eye diagram. This determines the sensitivity of the system to timing errors. The possibility of timing errors increases as the slope becomes more horizontal.

Traditionally, the *rise time* is defined as the time required by a signal to reach from 10% of its maximum value to 90% of its maximum value. But in the case of optical signals, the 10% and 90% points on rising edge of signal are often obscured by noise and jitter effects. Hence, the measurement of time interval is taken between 20% and 80% of its maximum value.

The approximate relationship to convert from 20–80% rise time to a 10–90% rise time is given as

$$T_{10-90} = 1.25 \times T_{20-80} \quad (6.69)$$

For a linear system, all the eye openings will be identical and symmetrical. The eye pattern will become asymmetric due to any non-linearity of optical channel transfer characteristic.

6.6.3 Optical Signal-to-Noise Ratio

In the installation and performance evaluation of WDM link in operation, optical signal-to-noise ratio (OSNR) is the important parameter. Principally BER is determined from OSNR.

The OSNR is independent of data format, pulse shape, or optical filter bandwidth. It is the ratio of the net signal power P_{signal} to the net noise power P_{noise} :

$$\text{OSNR} = \frac{\text{Net signal power}}{\text{Net noise power}}$$

OSNR gives the information of quality of signal after amplification also, because it is the ratio of signal and noise power, and both will get amplified by same factor. Thus, OSNR is given by the following equation:

$$\text{OSNR (dB)} = 10 \log \frac{P_{\text{signal}}}{P_{\text{noise}}} + 10 \log \frac{B_{\text{meas}}}{B_{\text{ref}}} \quad (6.70)$$

where B_{meas} is the noise equivalent measurement bandwidth of the instrument and B_{ref} is the reference spectral bandwidth. The various sources of noise are transmitter noise, crosstalk, or amplified spontaneous emission (ASE) from an EDFA. OSNR is useful in the design and installation of optical networks. It is also used to check the health and status of individual optical channels in WDM system.

In long-haul optical communication system, there is a chain of optical amplifiers installed at regular intervals. The ASE noise from amplifiers dominates the other sources of noise such as receiver thermal noise, shot noise, etc. Thus, neglecting this noise we can write the relation between the parameter Q and the OSNR as follows:

$$Q = \frac{2\sqrt{B_{\text{opt}}/B_{\text{elect}}} \text{ OSNR}}{1 + \sqrt{1 + 4(\text{OSNR})}} \quad (6.71)$$

where B_{opt} is the bandwidth of an optical band-pass filter in front of receiver and B_{elect} is the electrical noise-equivalent bandwidth in the receiver.

6.6.4 ASE Noise

Electrical repeaters have the advantage that they can completely relaunch the signal by regenerating and further retransmitting. But it requires demultiplexing of WDM signal, which is not efficient and cost-effective. This

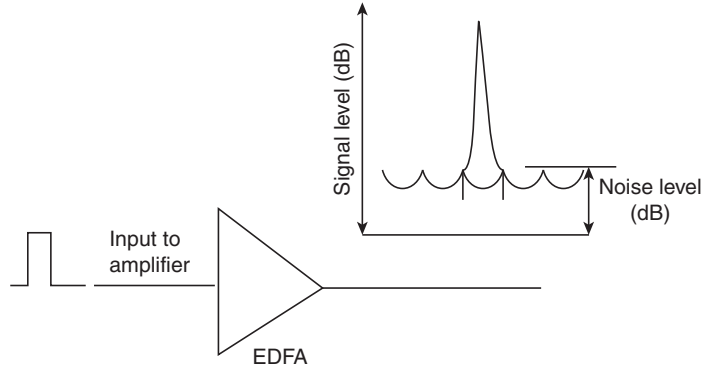


Figure 6.75 | ASE noise associated with single stage optical amplifier.

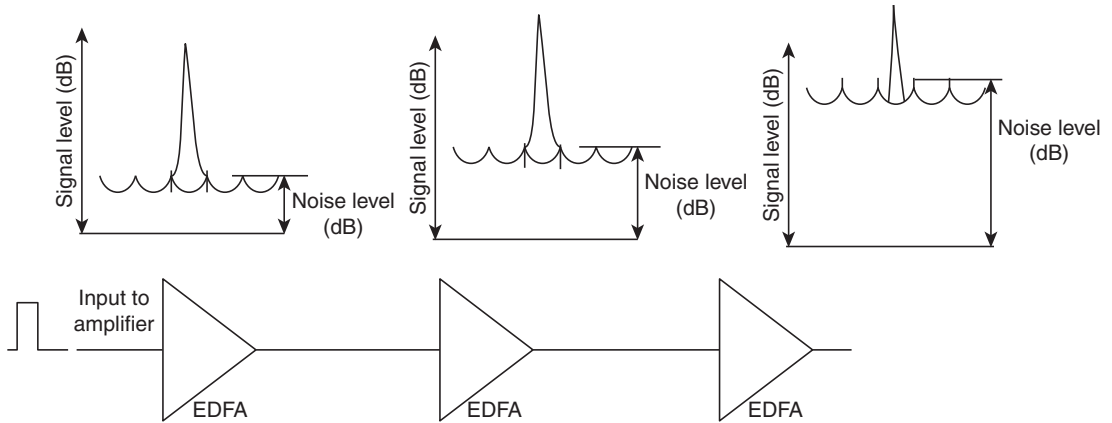


Figure 6.76 | ASE noise accumulations in multistage amplifier.

problem is overcome in optical amplifiers; they enhance the signal power level but at the same time add their own noise as shown in Fig. 6.75. This noise is ASE.

Amplified spontaneous noise is a severe problem in system design. As the number of amplifier stages increases, noise gets accumulated as shown in Fig. 6.76.

In a long-haul WDM link, the amplifiers are placed periodically at regular interval to boost the signal power. Each amplifier stage adds its own component of ASE noise and degrades the OSNR. Every amplifier amplifies the already-present noise and it is almost impossible to remove this noise. Therefore, OSNR is calculated at the end of an N -stage amplified system.

OSNR of each stage is given as

$$\text{OSNR} = \frac{P_{\text{signal}}}{\text{NF}_{\text{stage}} h \nu \nabla f} \quad (6.72)$$

where NF_{stage} is the noise figure of the stage, h is Planck's constant $= 6.6260 \times 10^{-34}$, ν is the optical frequency $= 193 \text{ THz}$, ∇f is the bandwidth that measures the $\text{NF} = 0.1 \text{ nm}$.

The total OSNR for the system is given as

$$\frac{1}{\text{OSNR}_{\text{final}}} = \frac{1}{\text{OSNR}_1} + \frac{1}{\text{OSNR}_2} + \frac{1}{\text{OSNR}_3} + \dots + \frac{1}{\text{OSNR}_N} \quad (6.73)$$

For N -stage system

$$\frac{1}{\text{OSNR}_{\text{final}}} = \sum_i \frac{1}{\text{OSNR}_i} \quad (6.74)$$

For a single amplifier with gain G , the OSNR is given by the following equation:

$$\text{OSNR} = \frac{P_{\text{signal}}}{P_{\text{ASE}}} = \frac{P_{\text{in}}}{2n_{\text{sp}}(G-1)h\nu\nabla f} \quad (6.75)$$

where G is amplifier gain, h is Planck's constant $= 6.6260 \times 10^{-34}$, ν is the optical frequency $= 193$ THz, ∇f is the bandwidth that measures the NF $= 0.1$ nm, $P_{\text{signal}} = P_{\text{in}}$ = input signal power and n_{sp} is the population inversion parameter given by

$$n_{\text{sp}} = \frac{N_2}{N_2 - N_1}$$

where N_1 is number of electrons in lower energy state and N_2 is number of electrons in higher energy state.

For an N -amplifier-stage system, each amplifier compensates for the loss of previous span. It is denoted as Γ and given in dB. So OSNR can be written as

$$\text{OSNR}_{\text{final}} = \frac{P_{\text{signal}}}{\text{NF} \Gamma h\nu\nabla f N} \quad (6.76)$$

where NF is noise figure. Taking log on both sides, we get

$$\text{OSNR}_{\text{dB}} = 158.93 + P_{\text{signal}} - \Gamma(\text{dB}) - \text{NF}_{\text{dB}} - 10 \log N - 10 \log \nabla f \quad (6.77)$$

Now $\nabla f = 0.1$ nm. Substituting this value we get

$$\text{OSNR}_{\text{dB}} = 58 + P_{\text{signal}}(\text{dBm}) - \Gamma(\text{dB}) - \text{NF}_{\text{dB}} - 10 \log N \quad (6.78)$$

6.6.5 Crosstalk

In WDM networks, several optical signals are multiplexed. Moreover, due to different fiber non-linear effects, there arise coupling and crosstalk between the multiplexed signals. These phenomena limit the ultimate WDM network performance. When a signal from one channel gets coupled with other channel, it becomes noise in the other channel. This severely affects the SNR and hence the error rate of the optical system.

Due to technology advancement in fiber fabrication technology and optical amplifier such as EDFAs, the distance limitation due to attenuation problem has been overcome to much extent. However, still chromatic dispersion (CD) and ASE noise are the major limitations. Along with these two major limitations, there are several types of fiber non-linearities that can further limit the performance of any fiber-optic transmission system. These non-linearities are divided into two broad groups:

1. Scattering phenomenon.
2. Refractive index phenomenon.

6.6.6 DWDM System Performance

As already seen in Section 6.6.1, we know that the BER performance of a DWDM channel is determined by the OSNR that is delivered to the photodetector. For a typical commercial system for achieving an acceptably low BER of 10^{-15} , the required OSNR is approximately 20 dB, measured in a 0.1 nm bandwidth.

For achieving the acceptable OSNR, a relatively sophisticated analysis of the following parameters is done:

1. Signal strength per channel.
2. Frequency spacing between adjacent channels.
3. Distance between successive amplifiers.

The signal strength per channel is linearly proportional to the number of amplifiers as well as the SNR and noise of each amplifier. It is exponentially proportional to the loss between amplifiers.

The present laser technology and fiber non-linearities (non-linearities increase with increase in power after some threshold value of power) have put some constraint on the signal strength per channel. Thus, we can change only the amplifier spacing.

To get a clear idea, one should look at the list of performance parameters of commercially used optical system which is as follows:

1. Fiber loss: 0.3 dB/km.
2. Receiver optical bandwidth: 0.1 nm.
3. Noise figure of optical amplifiers: 5 dB.
4. Transmission length: 600 km.
5. The optical SNR at the receiver: 20 dB measured in the 0.1 nm bandwidth.

6.6.7 Non-Linear Effects

In optical fibers, the non-linear scattering effects are due to the inelastic scattering of a photon to a lower energy photon. The energy difference is absorbed by the molecular vibrations or phonons in the medium.

Thus, the energy of a light wave is transferred to another wave at a higher wavelength and lower energy such that energy difference appears in the form of photons. The light-wave signal is called as the pump wave. The other wave at higher wavelength is known as the Stokes wave. If phonon of right energy and momentum is available, the high-energy photon at the anti-Stokes frequency can also be created.

In optical fibers there are two non-linear scattering phenomenon related to vibrational excitation modes of silica. These phenomenon are known as stimulated Raman scattering (SRS) and stimulated Brillouin scattering (SBS). The fundamental difference between the two phenomenon is that SRS is due to optical photons while SBS is due to acoustic phonons. As a result of this difference, SRS can occur in both directions – forward and backward, whereas SBS occurs only in one direction, that is, backward.

At high optical power levels, the non-linear scattering processes cause disproportionate attenuation. It also causes the transfer of optical power from one mode to other modes in forward or backward direction at different frequencies. The stimulated scattering mechanisms SBS and SRS also provide optical gain but with a shift in frequency.^[29,30]

Another category of non-linear effects due to refractive index phenomena can be controlled by increasing the channel spacing or through careful choice of channel power.

6.6.7.1 Scattering Phenomena

Stimulated Brillouin Scattering is one subtype of scattering phenomenon. It is caused by the interaction between the optical signal and acoustic waves in the fiber. The power from the optical signal is scattered back toward the transmitter due to this phenomenon. SBS is a narrowband process. Each channel in a DWDM system gets affected individually. This effect is more pronounced in systems with greater power levels.

A second subtype of scattering phenomenon is known as stimulated Raman scattering (SRS). It is caused by the interaction of the optical signal with silica molecules in the fiber. In this phenomenon, the power from shorter wavelength, higher photon energy channels, gets transferred to longer wavelength, lower photon energy channels. SRS is a wideband phenomenon. It affects the entire optical transmission spectrum. SRS causes tilting of a spectrum of equal amplitude channels as it moves through the fiber. With increase in power and total width of DWDM spectrum, this effect is more pronounced. This effect can be minimized by using moderate powers and densely packed channels to reduce the total width of DWDM spectrum.^[31–33]

6.6.7.1.1 Stimulated Brillouin Scattering

Due to narrowband nature SBS is really a single-channel phenomenon. But then also it must be taken into account in WDM systems. SBS depends on a number of things; some of them are as follows:

1. **Signal line width:** SBS becomes more pronounced for the narrow linewidth. If the channel bandwidth is greater than 100 MHz, SBS is almost negligible.
2. **Signal power:** Below a certain threshold value (typically 5 mW per channel), SBS will not cause problem.
3. **Fiber core size:** For the smaller core size, the power concentration is more. Thus, SBS effect is more pronounced.
4. **Wavelength:** SBS effect is more pronounced in the 1,550 nm band than in the 1,310 nm band.

6.6.7.1.2 Stimulated Raman Scattering

Due to wideband nature, SRS occurs between signal channels and between groups of channels. The product of total power and total optical bandwidth must be less than 500 GHz/W to keep the signal degradation due to SRS below the acceptable level. For example, if we have 10 channels with a spacing of 200 GHz, the bandwidth would be $200 \times 10 = 2,000$ GHz. So the total power of all 10 channels would be limited to 250 mW.

If we have 20 channels spaced over the same range (using 100 GHz spacing), then the total power available to all channels would be the same as in the previous example (250 mW). A watt is a very large amount of power. In most WDM systems, each channel power is well below 10 mW, in order to avoid non-linear effects (refer to Section 3.4 for more details). In addition, SRS is influenced by fiber dispersion.

6.6.7.2 Refractive Index Phenomena

There are three types of non-linearities based on refractive index phenomena. These are:

1. **Self-phase modulation (SPM):** The phase of the light pulse gets modulated by the refractive index; this effect is called as SPM. SPM acts on each DWDM channel independently. It causes the widening of signal's spectrum that can cause crosstalk or an unexpected dispersion penalty.
2. **Cross-phase modulation (CPM):** Intensity fluctuations in adjacent channels cause CPM. This effect is unique to DWDM systems.

3. **Four-wave mixing (FWM):** In FWM of two or more waves, optical signals are non-linearly combined to produce a new optical frequency. Four-wave mixing is generally negligible in conventional SMF. It is more pronounced and causes more problems in the dispersion-shifted fiber.

All these non-linearities are caused because the refractive index, and hence the speed of propagation of light in a fiber, is dependent on the intensity of light. This has a significant effect in long-haul applications.

In DWDM system, as the signal propagates through the channel, the tilt and ripple effects get accumulated along the chain of amplifiers. Each amplifier in the chain introduces the ASE noise into the system. This causes decrease in SNR and hence the degradation of signal. At the receiver further noise gets added. In an optically amplified system, the BER is determined differently than in a conventional regenerated system. In a properly designed optically amplified system, probability of error in the reception of a binary value of zero is determined by the ASE noise value alone, while the probability of error in the reception of a binary value of one is determined by the signal mixing with the ASE.

6.6.7.2.1 Four-Wave Mixing

When a high-power optical signal is launched into a fiber, the non-linear effects come in to picture. One such non-linear effect, which is due to the third-order electric susceptibility, is called the *optical Kerr effect*. Four-wave mixing (FWM) is a type of optical Kerr effect. It occurs when light of two or more wavelengths is launched into a fiber. When light of three different wavelengths is launched into a fiber, a new wave having different wavelength than the three wavelengths is generated.

The intensity dependence of the refractive index in a WDM system using the frequencies $f_1, \dots, f_n$ induces phase shifts within a channel, apart from giving rise to signals at new frequencies, such as $(2f_i - f_j)$ and $(f_i + f_j - f_k)$. This phenomenon is called four-wave mixing.

SPM and CPM effects are significant at high bit-rate systems, whereas FWM effect is independent of bit rate; however, it critically depends on channel spacing and fiber chromatic dispersion.^[34]

In the transmission of dense wavelength-division multiplexed (DWDM) signals, FWM is undesirable, but for certain applications, it is useful. It provides the basic technology for measuring the non-linearity and chromatic dispersion of optical fibers. It is an effective technological basis for fiber-optic devices.

6.6.7.2.2 Theory of FWM

Four-wave mixing is a type of harmonic generation process. Two or more frequencies combine to generate wave at a different frequency that is sum or difference of the signals that are mixed as shown in Fig. 6.77. Three waves combine to generate a fourth frequency.

If input frequencies are designated as f_1, f_2, f_3 , the new frequency f_4 is

$$f_4 = f_1 + f_2 - f_3 \quad (6.79)$$

In dense WDM systems, the optical channels are typically close and spaced on a frequency grid typically separated by 100 or 200 GHz. If f_1 is the starting frequency, then f_2 is at frequency 100 GHz higher and f_3 is another 100 GHz higher.

$$\begin{aligned} f_4 &= f_1 + (f_1 + 100) - (f_1 + 200) \\ f_4 &= f_1 - 100 \end{aligned}$$

FWM is a weak effect, but it can accumulate if the signals on the optical channels remain in-phase with each other over long distances and it occurs when dispersion is zero.

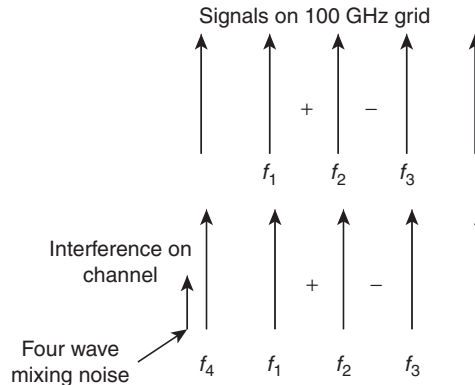


Figure 6.77 | Four-wave mixing.

FWM in WDM systems is influenced by the following two factors:

1. Channel spacing.
2. Fiber dispersion.

FWM increases rapidly with reduction in channel spacing. In practice, FWM occurs only between adjacent channels, even though it could affect multiple channels. For FWM to take effect, each channel must stay in-phase with its adjacent channel for some considerable distance. Thus, if fiber dispersion is high, then the effects of FWM are minimum. With DSF fiber having dispersion less than 1 ps/nm-km, the FWM effect is maximum.

FWM causes some degradation at channel spacings of less than 80 GHz for dispersion-shifted fibers (DSFs) with zero dispersion in the 1,550 nm band. FWM causes transfer of 80% of the energy in the two signals into sum or difference frequencies at a channel separation of 25 GHz. FWM effects disappear when channel separation is greater than 25 GHz for standard fiber in the 1,550 nm band having dispersion = 17 ps/nm/km. However, the effect on standard fiber also becomes severe at a channel separation of less than 15 GHz.

The impact of FWM can be minimized by placing WDM channels such that the generated signals do not fall within other WDM channels. Thus, they don't interfere with other channels too much. This does help, but it cannot overcome the problem of noise generated in the source WDM channels by power being transferred out of them. In addition, some WDM devices are difficult to construct if wavelength spacing is uneven.

Summary of Important Points

1. The FWM effect becomes greater as the channel spacing is reduced.
2. FWM is non-linear with signal power. As signal power increases, the effect increases exponentially.
3. If the signals at different wavelengths remain in-phase over long distance then this can cause FWM. The laser sources produce highly coherent light; and if there is no chromatic dispersion then the number of signals will stay in-phase over long distance. So FWM effect will be more. If dispersion effect is more, signals cannot stay in-phase over long distance and FWM effect will be reduced.
4. The FWM problem can be minimized by uneven distribution of channels in WDM.

6.6.7.2.3 Self-Phase Modulation

The refractive index of glass varies slightly with intensity of light passing through it. Hence changes in signal intensity cause the change of speed of light passing through the glass. This, in turn, modulates the phase of optical channel that creates it, which is called *self phase modulation* (SPM).

The phase shift is proportional to intensity of pulse. As the optical power rises and falls, these phase shifts also effectively shift the frequencies of some of the light. The shifts are in opposite directions at rising and falling parts of pulse. Thus, different parts of pulse undergo different phase shifts, which give rise to chirping of pulses.

Pulse chirping enhances the pulse-broadening effects of chromatic dispersion. Chirping effect is proportional to the transmitted signal power so that SPM effects are more pronounced in systems using high transmitted power.

6.6.7.2.4 Cross-Phase Modulation

In WDM systems, the intensity-dependent non-linear effects are enhanced because the combined signal from all channels can be quite high, even when individual channels are operated at moderate powers.

Intensity-dependent phase shift and consequent chirping induced by SPM are enhanced because of intensities of the signals in other channels. This effect is called as cross-phase modulation (CPM).

The effect can be significantly reduced by increasing the wavelength spacing between individual channels. Because of fiber chromatic dispersion, the propagation constant of individual channels is sufficiently different. Hence the pulses corresponding to different channels walk away rapidly from each other.

In general, all non-linear effects are weak and depend on long interaction lengths to build up to significant levels. So any mechanism that reduces the interaction length decreases the non-linear effect.

Solved Problems

PROBLEM 1

Make a power budget and calculate the maximum transmission distance for a 1,300 nm light-wave system operating at 100 Mb/s and using an LED for launching 0.1 mW of average power into the fiber. Assume 1 dB/km fiber loss, 0.2 dB splice loss at every 2 km, 1 dB connector loss at each end of fiber link, and 100 nW receiver sensitivity. Allow 6 dB system margin.

Solution: $P_i = 0.1 \text{ mW}$, $P_o = 100 \text{ nW}$, $\alpha = 1 \text{ dB/km}$. Connector loss is

$$N_c \times l_c = 2 \times 1 \text{ dB} = 2 \text{ dB}$$

Splice loss is

$$N_s \times l_s = \frac{L}{2} \times 0.2 \text{ dB}$$

Fiber attenuation is

$$\alpha \times L = 1 \text{ dB/km} \times L$$

$$\text{System margin} = 6 \text{ dB}$$

Also

$$P_i = 0.1 \text{ mW} = 10 \log(0.1 \times 10^{-3}) = -40 \text{ dBW}$$

$$P_o = 100 \text{ nW} = 10 \log(100 \times 10^{-9}) = -70 \text{ dBW}$$

Now

$$P_t = P_i - P_o$$

$$= \text{Connector loss} + \text{Splice loss} + \text{Fiber attenuation} + \text{System margin}$$

This implies

$$\begin{aligned} P_t &= -40 - (-70) \\ &= 2 \text{ dB} + \left(\frac{L}{2} \times 0.2 \text{ dB} \right) + 1 \text{ dB/km} \times L + 6 \text{ dB} \end{aligned}$$

After simplification we get

$$L = 20 \text{ km}$$

The maximum transmission distance for a 1,300 nm light-wave system operating at 100 Mbps is 20 km.

PROBLEM 2

Make a power budget for a short-haul system where the required data rate is 20 Mbps, BER is 1×10^{-9} errors/bit and it is operating at $\lambda = 850 \text{ nm}$. The Si PIN photodiode has a receiver sensitivity of about -42 dBm . GaAlAs LED can couple $50 \mu\text{W}$ into multimode fiber with a core diameter of $50 \mu\text{m}$. The connector loss is 1 dB per connector. Splices will be required at each kilometer with 0.5 dB per splicing loss, and fiber attenuation loss for the fiber is 3.5 dB/km .

Solution: $P_i = 50 \mu\text{W}$, $P_o = -42 \text{ dBm}$, $\alpha = 3.5 \text{ dB/km}$, length of short-haul system = 5 km . Connector loss is

$$N_c \times l_c = 2 \times 1 \text{ dB} = 2 \text{ dB}$$

Splice loss is

$$N_s \times l_s = \frac{L}{1} \times 0.5 \text{ dB} = 5 \times 0.5 = 2.5 \text{ dB}$$

Fiber attenuation is

$$\alpha \times L = 3.5 \text{ dB/km} \times 5 = 17.5 \text{ dB}$$

System margin is 6 dB . Also

$$P_i = 50 \mu\text{W} = 10 \log(50 \times 10^{-3}) = -13.01 \text{ dBm}$$

Now

$$\begin{aligned} P_t &= P_i - P_o \\ P_t &= -13 - (-42) = 29 \text{ dBm} \end{aligned}$$

Therefore

$$\begin{aligned} \text{Systemic loss} &= \text{Connector loss} + \text{Splice loss} + \text{Fiber attenuation} + \text{System margin} \\ &= 2 \text{ dB} + 2.5 \text{ dB} + 17.5 \text{ dB} + 6 \text{ dB} = 28 \text{ dB} \end{aligned}$$

Thus, we have an excess power margin of

$$29 \text{ dB} - 28 \text{ dB} = 1 \text{ dB}$$

Hence, the proposed system is feasible.

PROBLEM 3

Make a power budget and calculate the minimum source power required for a 40 km link length. Assume 0.5 dB/km fiber loss. There are 4 splices with 0.5 dB splice loss for each, 1 dB connector loss at each end of fiber link and -39 dBm receiver sensitivity. Allow 6 dB system margin.

Solution: $P_o = -39$ dB, $\alpha = 0.5$ dB/km. Connector loss is

$$N_c \times I_c = 2 \times 1 \text{ dB} = 2 \text{ dB}$$

Splice loss is

$$N_s \times I_s = 4 \times 0.5 \text{ dB} = 2 \text{ dB}$$

Fiber attenuation is

$$\alpha \times L = 0.5 \text{ dB/km} \times 40 \text{ km} = 20 \text{ dB}$$

System margin = 6 dB. Now

$$P_t = P_i - P_o = \text{Connector loss} + \text{Splice loss} + \text{Fiber attenuation} + \text{System margin}$$

$$P_t = P_i - (-39 \text{ dBm}) = 2 \text{ dB} + 2 \text{ dB} + 20 \text{ dB} + 6 \text{ dB}$$

After simplification we get

$$P_i = -9 \text{ dBm} = 0.13 \text{ mW}$$

The minimum source power required is 0.13 mW.

PROBLEM 4

Make a power budget and calculate the maximum transmission distance for a light-wave system operating at 20 Mbps and using a pigtailed LED source at 850 nm, launching 50 μW of average power into the fiber and a silicon PIN detector with sensitivity of -38.8 dBm. Allow 8 dB system margin. In the same system if laser source emitting 0 dBm power is used, what is the maximum permissible link length?

Solution: The power emitted by source is

$$P_i = 50 \mu\text{W} = 10 \log(50 \times 10^{-6}) = -13 \text{ dBm}$$

Also

$$\text{System margin} = 8 \text{ dB}$$

Assume connector loss = 2 dB (for two connectors). Then

$$\begin{aligned} P_t &= P_i - P_o \\ &= \text{Available transmission loss} \\ &= -13 - (-38.8) \\ &= 25.8 \text{ dB} \end{aligned}$$

If the fiber with a loss of 2.5 dB/km is used, then the maximum permissible link length is

$$P_t = P_i - P_o = 25.8 \text{ dB} = \text{Connector loss} + \text{Fiber attenuation} + \text{System margin}$$

or
$$25.8 \text{ dB} = 2 \text{ dB} + 2.5 \text{ dB/km} \times L + 8 \text{ dB}$$

Therefore

$$\text{Maximum permissible link length} = L \cong 6.3 \text{ km}$$

In the same system if laser source emitting 0 dBm power is used, then

$$P_t = P_i - P_o = 0 \text{ dB} - (-38.8) = 2 \text{ dB} + 2.5 \text{ dB/km} \times L + 8 \text{ dB}$$

So maximum permissible link length = $L \cong 11.5 \text{ km}$.

PROBLEM 5

Find the maximum permissible link length for a light-wave system with data rate = 20 Mbps, bit-error rate of 10^{-9} .

Transmitter: GaAlAs LED that can couple $60 \mu\text{W}$ average optical power level into the fiber

Receiver: Silicon PIN photodiode operating at 850 nm, require receiver input signal of -40 dBm

Fiber: $\alpha = 3.2 \text{ dB/km}$

Connector loss = 1 dB

Solution: The power emitted by source is given by

$$P_i = 60 \mu\text{W} = 10 \log(60 \times 10^{-6}) = -12.21 \text{ dBm}$$

Now

$$P_t = P_i - P_o = \text{Connector loss} + \text{Fiber attenuation} + \text{System margin}$$

$$P_t = -12.21 - (-40) = 2 \text{ dB} + 3.2 \text{ dB/km} \times L + 6 \text{ dB}$$

Therefore

$$\text{Maximum permissible link length} = L \cong 6.1 \text{ km}$$

PROBLEM 6

Make a rise time budget for a $0.85 \mu\text{m}$, 10 km fiber link designed to operate at 50 Mbps. The LED transmitter and the Si PIN receiver have rise times of 10 ns and 15 ns, respectively. The graded index fiber has a core index of 1.46, $\Delta = 0.01$ and $D = 80 \text{ ps/km-nm}$. The LED spectral width is 50 nm. Can the system be designed to operate with NRZ format?

Solution: We have $\lambda = 0.85 \mu\text{m}$, $\Delta\lambda = 50 \text{ nm}$, $L = 10 \text{ km}$, Bit rate = 50 Mbps, $D = 80 \text{ ps/km-nm}$, $n_{co} = 1.46$, $\Delta = 0.01$, $T_{TX} = 10 \text{ ns}$, $T_{RX} = 15 \text{ ns}$. Now

$$T_{\text{FIBER}} = D \times \Delta\lambda \times L$$

$$T_{\text{FIBER}} (\text{ns}) = 80 \times 10^{-3} \times 50 \times 10 = 40$$

Also

$$T_{\text{SYS}} = 1.1 \sqrt{T_{\text{TX}}^2 + T_{\text{RX}}^2 + T_{\text{FIBER}}^2}$$

$$T_{\text{SYS}} (\text{ns}) = 1.1 \sqrt{10^2 + 15^2 + 40^2} = 48.26$$

For good reception of signal

$$T_{\text{SYS}} \leq 0.7 \times \text{Pulse width}$$

Therefore

$$\text{Pulse width} = \frac{1}{\text{Bit rate}} = \frac{1}{50 \times 10^6} = 20 \text{ ns}$$

For NRZ data format,

$$T_{\text{SYS}} \leq 0.7 \times \text{Pulse width} \leq 14 \text{ ns}$$

Rise time budgeting criteria is not fulfilled, hence poor budgeting.

PROBLEM 7

Make a rise time budget for a 0.85 μm , 150 km fiber link designed to operate at 622 Mbps. The LED transmitter and the Si PIN receiver have rise times of 0.1 ns and 0.5 ns, respectively. The graded index fiber has $D = 18 \text{ ps/km-nm}$. The LED spectral width is 0.15 nm. Can the system be designed to operate with NRZ format?

Solution: We have $\lambda = 0.85 \mu\text{m}$, $\Delta\lambda = 0.15 \text{ nm}$, $L = 150 \text{ km}$, Bit rate = 622 Mbps, $D = 18 \text{ ps/km-nm}$, $T_{\text{TX}} = 0.1 \text{ ns}$, $T_{\text{RX}} = 0.5 \text{ ns}$. Now

$$T_{\text{FIBER}} = D \times \Delta\lambda \times L$$

$$T_{\text{FIBER}} (\text{ns}) = 18 \times 10^{-3} \times 0.15 \times 150 = 0.4$$

So

$$T_{\text{SYS}} = 1.1 \sqrt{T_{\text{TX}}^2 + T_{\text{RX}}^2 + T_{\text{FIBER}}^2}$$

$$T_{\text{SYS}} (\text{ns}) = 1.1 \sqrt{0.1^2 + 0.5^2 + 0.4^2} = 0.7$$

For good reception of signal

$$T_{\text{SYS}} \leq 0.7 \times \text{Pulse width}$$

$$\text{Pulse width} = \frac{1}{\text{Bit rate}} = \frac{1}{622 \times 10^6} = 1.6 \text{ ns}$$

For NRZ data format

$$T_{\text{SYS}} \leq 0.7 \times \text{Pulse width} \leq 1.1 \text{ ns}$$

Rise time budgeting criteria is fulfilled, hence good budgeting.

PROBLEM 8

Make a rise time budget for a 2 km fiber link designed to operate at 20 Mbps. The LED transmitter and the Si PIN receiver have rise times of 8 ns and 12 ns, respectively. The graded index fiber has material dispersion $D = 0.1$ ns/nm-km and $BW = 35$ MHz km. The LED spectral width is 40 nm. Can the system be designed to operate with NRZ format?

Solution: We have $\Delta\lambda = 40$ nm, $L = 2$ km, bit rate = 20 Mbps, $D = 0.1$ ns/km-nm, $T_{TX} = 8$ ns, $T_{RX} = 12$ ns, $BW = 35$ MHz km. Now

$$T_{FIBER} = D \times \Delta\lambda \times L$$

$$T_{FIBER} \text{ (ns)} = 0.1 \times 40 \times 2 = 8$$

So

$$T_{SYS} = 1.1 \sqrt{T_{TX}^2 + T_{RX}^2 + T_{FIBER}^2}$$

$$T_{SYS} \text{ (ns)} = 1.1 \sqrt{8^2 + 12^2 + 8^2} = 18.14$$

For good reception of signal

$$T_{SYS} \leq 0.7 \times \text{Pulse width}$$

Therefore

$$\text{Pulse width} = \frac{1}{\text{Bit rate}} = \frac{1}{20 \times 10^6} = 50 \text{ ns}$$

For NRZ data format

$$T_{SYS} \leq 0.7 \text{ Pulse width} \leq 35 \text{ ns}$$

Rise time budgeting criteria is fulfilled, hence good budgeting.

PROBLEM 9

Consider a parabolic index multimode fiber, $n_{co} = 1.46$, $\Delta = 0.01$, operating at 850 nm. If the source is an LD with spectral width of 1 nm, find the maximum bit rate that can be handled by fiber.

Solution: Multimode graded index fiber with parabolic index we have

$$T_{MODAL} = \frac{n_1}{2c} \Delta^2 L = \frac{1.46}{2 \times 3 \times 10^8} \times 0.01^2 \times 1 \cong 0.24 \text{ ns/km}$$

Total fiber rise time is given as

$$T_{FIBER} = \sqrt{T_{MODAL}^2 + T_{MATERIAL}^2} = \sqrt{0.24^2 + 0.085^2} \cong 0.25 \text{ ns/km}$$

For good reception of signal

$$T_{SYS} \leq 0.7 \times \text{Pulse width}$$

Therefore

$$\text{Pulse width} = \frac{0.25 \text{ ns/km}}{0.7} = 0.357 \text{ ns}$$

$$\text{Bit rate} = \frac{1}{\text{Pulse width}} = \frac{1}{0.357 \times 10^{-9}} = 2.8 \text{ Gb-km/s}$$

The maximum bit rate that can be handled by fiber is 2.8 Gb-km/s.

PROBLEM 10

A given system requires 500 GHz channel spacing over the wavelength interval of 1,536 to 1,556 nm. Center of this region is 1,546 nm. What is the number of possible channels in this wavelength interval?

Solution: First, find the wavelength spacing. We have

$$|\Delta\nu| = \frac{c}{\lambda^2} |\Delta\lambda|$$

$$\Delta\lambda = \left(\frac{\lambda^2}{c} \right) \Delta\nu = \frac{(1,546 \text{ nm})^2}{3 \times 10^8} (500 \times 10^9) = 3.98 \text{ nm}$$

The width of this wavelength interval is 20 nm. The number of channels is

$$N = \frac{20 \text{ nm}}{3.98 \text{ nm}} = 5.02$$

So at this frequency spacing, there can be 5 channels in this wavelength interval.

PROBLEM 11

A 2×2 biconical tapered fiber coupler has an input optical power level of $P_0 = 135 \mu\text{W}$. The coupler output powers are $P_1 = 60 \mu\text{W}$, $P_2 = 55 \mu\text{W}$, $P_3 = 4.3 \text{ nW}$. Find the four coupling parameters.

Solution: We have

$$\begin{aligned} \text{Splitting ratio} &= \left(\frac{P_2}{P_1 + P_2} \right) \times 100\% \\ &= \frac{55 \mu\text{W}}{(60 + 55) \mu\text{W}} \times 100\% \\ &= 47.8\% \end{aligned}$$

$$\text{Excess loss} = 10 \log \left(\frac{P_0}{P_1 + P_2} \right) = 10 \log \frac{135 \mu\text{W}}{(60 + 55) \mu\text{W}} = 0.696$$

$$\text{Insertion loss (port 0 to 1)} = 10 \log \left(\frac{P_0}{P_1} \right) = 10 \log \left(\frac{135 \mu\text{W}}{60 \mu\text{W}} \right) = 3.52 \text{ dB}$$

$$\text{Insertion loss (port 0 to 2)} = 10 \log \left(\frac{P_0}{P_2} \right) = 10 \log \left(\frac{135 \mu\text{W}}{55 \mu\text{W}} \right) = 3.89 \text{ dB}$$

$$\text{Return loss or crosstalk} = 10 \log \left(\frac{P_3}{P_0} \right) = 10 \log \left(\frac{4.3 \text{ nW}}{135 \mu\text{W}} \right) = -44.98 \text{ dB}$$

PROBLEM 12

Design a 4×25 span WDM link with an optical amplifier gain of 24 dB and NF equal to 5 dB. Calculate the final OSNR if the input signal power is 0 dB. Calculate the signal power at the receiver. Check whether the system will work if the receiver sensitivity is -26 dB? Also check the same if input signal power is 10 dB.

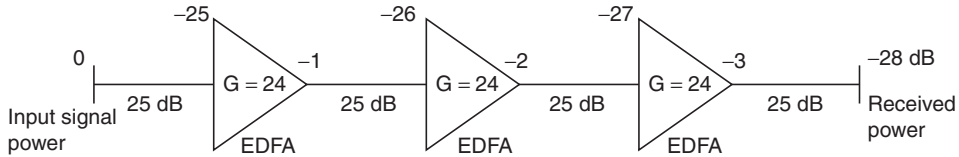


Figure 6.78 | Signal power level at various stages for input signal power of 0 dB.

Solution: See Fig. 6.78. We have

$$\text{Number of span} = N = 4$$

$$\text{Noise figure} = \text{NF} = 5 \text{ dB}$$

$$\text{Span loss} = \Gamma = 25 \text{ dB}$$

Now

$$\text{OSNR}_{\text{dB}} = 58 + P_{\text{signal}} (\text{dBm}) - \Gamma (\text{dB}) - \text{NF}_{\text{dB}} - 10 \log N$$

$$\text{OSNR}_{\text{dB}} = 58 + 0 - 25 (\text{dB}) - 5 - 10 \log 4$$

$$\text{OSNR}_{\text{dB}} = 22 \text{ dB}$$

Power at receiver = -28 dB. If the receiver sensitivity is -26 dB, the system will not work. The solution is to:

- (a) Increase the gain of the amplifier.
- (b) Increase the input power of the transmitter.

If the input signal power is 10 dB, refer Fig. 6.79, the power at receiver = -18 dB. But the receiver sensitivity is -26 dB, so the system will work.

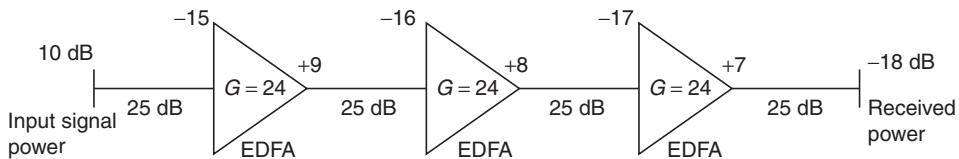


Figure 6.79 | Signal power level at various stages for input signal power of 10 dB.

PROBLEM 13

For a WDM link, if P_{signal} is 0 dB, $\text{OSNR}_{\text{final}} = 20$ dB, total length = 300 km, bit rate = 5 Gbps, NF = 5 dB, fiber type is SMF with $\alpha = 0.2$ dB/km, then calculate the number of spans in the link.

Solution: Total loss over the complete length = $300 \times 0.2 = 60$ dB

Attenuation per span = $60/N$, where N = Number of spans

Now

$$\text{OSNR}_{\text{dB}} = 58 + P_{\text{signal}} (\text{dBm}) - \Gamma (\text{dB}) - 10 \log N$$

$$20 = 58 + 0 - \frac{60}{N} - 5 - 10 \log N$$

$$-33 = -\frac{60}{N} - 10 \log N$$

Solving for N gives $N = 2$. Therefore, the total number of spans in this link are 2.

PROBLEM 14

Find the total length of a system having $\text{OSNR} = 20$ dB, dispersion of fiber is 17 ps/nm-km, span loss is 25 dB, NF of the amplifier is 4 dB, dispersion tolerance is given as 1,600 ps/nm, and $P_{\text{signal}} = 10$ dB.

Solution: We have

$$\text{OSNR}_{\text{dB}} = 58 + P_{\text{signal}} (\text{dBm}) - \Gamma (\text{dB}) - \text{NF}_{\text{dB}} - 10 \log N$$

$$20 = 58 + 10 - 25 - 4 - 10 \log N$$

$$19 = 10 \log N$$

$$N = 80$$

Now

$$\text{Total length} = 80 \times \frac{\text{Span loss}}{\alpha} = 80 \times \frac{25}{0.2} = 10,000 \text{ km (theoretical limit)}$$

But due to dispersion,

$$\text{Maximum length} = [1,600 \text{ ps-nm}] / [17 \text{ ps/nm-km}] = 94 \text{ km}$$

SUMMARY

1. Fiber-optic communication systems are classified into three broad categories as follows:
 - (a) Point-to-point links.
 - (b) Local area networks.
 - (c) Distribution networks.
2. To ensure the performance of an optical communication system, two types of system analysis are carried out. These are power budgeting and rise-time budgeting.
3. Wavelength division multiplexing (WDM) seems to be a promising solution for increasing demand of channel capacity. In WDM several different wavelengths are transmitted over one SMF at the same time.

4. If the spacing between the channels is below 200 GHz, then it is called as dense WDM (DWDM).
5. The WDM or DWDM system involves a number of passive and active components.
6. A coupler combines light into or splits light out of a fiber.
7. An optical isolator is an optical component that allows light transmission in only one direction.
8. An optical circulator is an isolator having three or more ports.
9. Optical add/drop multiplexers are specifically designed for WDM systems. The main function of an optical multiplexer is to couple two or more wavelengths into the same fiber. On the other hand, the demultiplexer separates a multiplicity of wavelengths in a fiber and directs them to many fibers.
10. Arrayed waveguide gratings (AWGs) are essential components for the realization of DWDM and optical networks. They are optical wavelength (de)multiplexers used in DWDM.
11. An acousto-optic tunable filter is a solid-state electro-optical device. It is an electronically tunable spectral band-pass filter.
12. Fiber gratings are passive optical components, which selectively reflect and transmit light of certain wavelengths.
13. *Bit error rate* is the most common figure of merit for digital links. It is defined as the number of bit errors (N_{error}) occurring over a specific time interval, divided by the total number of bits (N_{total}) sent during that interval^[28] and is commonly abbreviated as BER.
14. The eye diagram is a simple and powerful measurement technique for assessing the data-handling ability of a digital transmission system.
15. OSNR is useful in the design and installation of optical networks. It is also used to check the health and status of individual optical channels in WDM system.
16. At high transmitted power levels or at high bit rates, non-linear effects place limitations on system performance. There are two types of non-linear effects as follows:
 - (a) Non-linear scattering, for example SRS, SBS.
 - (b) Non-linear effects based on refractive index, for example SPM, CPM, FWM.
17. Non-linear effects due to refractive index phenomena can be controlled by increasing the channel spacing or through careful choice of channel power.

MULTIPLE-CHOICE QUESTIONS

1. According to ITU G.692 standard, the maximum link distance for a system without amplifier is
 - (a) 200 km
 - (b) 160 km
 - (c) 640 km
 - (d) 100 km
2. What does an add/drop multiplexer do?
 - (a) Optical-to-electrical conversion
 - (b) Amplifies the signal
 - (c) Diverts one or more optical channels and may add new signals in their place
 - (d) Switches signal between different wavelengths
3. Fiber Bragg grating
 - (a) Reflects the selected wavelength and absorbs other wavelengths
 - (b) Reflects the selected wavelength and transmits the other light
 - (c) Transmits the selected wavelength and reflects the other wavelength
 - (d) Transmits the selected wavelength and absorbs other light
4. LPFG typically have grating periods of the order of
 - (a) 0.1 micrometer to 100 micrometer
 - (b) 0.1 millimeter to 100 millimeter
 - (c) 100 micrometers to a millimeter
 - (d) 100 millimeter to few centimeter

5. FWM in WDM systems is influenced by two factors:
 - (a) Channel spacing and fiber dispersion
 - (b) Channel spacing and fiber attenuation
 - (c) Fiber attenuation and optical power
 - (d) Fiber dispersion and optical power
6. *Manchester* coding in digital optical communication is also called
 - (a) Unipolar line coding
 - (b) Bipolar line coding
 - (c) Nonpolar line coding
 - (d) Bi-phase line coding
7. How many 2×2 MZIs will be required to implement four-channel wavelength multiplexer?
 - (a) Three
 - (b) Four
 - (c) Six
 - (d) Eight
8. A 4×4 crossbar switch is realized using how many 2×2 switches?
 - (a) 8
 - (b) 16
 - (c) 32
 - (d) 4
9. What material is used in electro-optic modulators and switches?
 - (a) Gallium arsenide
 - (b) Lithium niobate
 - (c) Silica or silicon
 - (d) Indium phosphide
10. What type of WDM system requires an optical circulator?
 - (a) Mach–Zehnder interferometer
 - (b) Tunable optical filter
 - (c) Michelson interferometer
 - (d) Fiber Bragg grating

REVIEW QUESTIONS

1. What is meant by line coding? Explain NRZ and RZ codes in brief.
2. What are the two analyses usually carried out to ensure the desired performance of optical fiber transmission link? Explain any one of them.
3. Explain the convenient budget analysis for determining the dispersion limitation of an optical fiber link.
4. Explain the point-to-point link design with reference to the choice of components and their associated characteristics.
5. Explain concept of WDM and hence explain key system features of WDM.
6. What is DWDM? Compare WDM and DWDM.
7. Write short note on 2×2 fiber coupler.
8. Write a short note on photonic switching.
9. What is the main function of an optical multiplexer?
10. Could a passive optical demultiplexer be used as a multiplexer? If so, why?
11. Explain the following terms related to optical coupler.
 - (a) Splitting ratio
 - (b) Crosstalk
 - (c) Insertion loss
 - (d) Excess loss
12. With neat sketches explain the operation of optical isolator.
13. Describe the operation of OADM.
14. Write a note on:
 - (a) MZI modulator
 - (b) MZI as demultiplexer
15. With neat sketch explain the mode coupling phenomenon in FBG and LPFG.
16. Briefly describe the communication and sensing applications of optical fiber gratings.
17. Describe the applications of optical switches.
18. Explain the following system performance parameters:
 - (a) Bit error rate
 - (b) Eye diagram
 - (c) OSNR
 - (d) ASE noise
 - (e) Crosstalk

EXERCISES

1. A digital optical fiber link of overall length 7 km is operating at 20 Mbit/s using an RZ code. An LED emitting at $0.85\text{ }\mu\text{m}$ with graded index fiber to a PIN photodiode is a suitable choice for the system components, giving no dispersion equalization penalty. An LED which is capable of launching an average of $100\text{ }\mu\text{W}$ of optical power (including connector loss into a $50\text{ }\mu\text{m}$ core diameter graded index fiber) is chosen. The proposed fiber cable has an attenuation of 2.6 dB/km and requires splicing every km with a loss of 0.5 dB per splice. There is also a connector loss at the receiver of 1.5 dB . The receiver requires mean incident optical power of -41 dBm in order to give the necessary BER of 10^{-10} , and it is predicted that a safety margin of 6 dB will be required. Write down the optical power budget for the system and hence determine its viability.
2. A single-mode long-haul digital link operating at $1.2\text{ }\mu\text{m}$ has a laser source, which launches power of -3 dBm , fiber cable loss is 0.3 dB/km , splice loss is 0.2 dB , mean power required is -54 dBm at 35 Mbps ($\text{BER} = 10^{-9}$), and -45 dBm at 400 Mbps ($\text{BER} = 10^{-9}$). Calculate the maximum possible link length without repeaters at 35 Mbps and 400 Mbps with no dispersion equalization penalty.
3. A system is designed to transmit 600 Mbit/s through 50 km of fiber with attenuation of 0.3 dB/km . The system contains two connector pairs with 1.5 dB loss, a laser source couples 0 dBm into the fiber and the receiver is having sensitivity of -34 dBm . How many splices with average loss of 0.15 dB can the system contain if the system margin must be at least 8 dB .
4. For an optical fiber link of 15 km and BW of 100 Mbps , receiver sensitivity is -50 dBm , fiber loss is 2 dB/km , power launch is 0 dBm into the fiber by transmitter, detector coupling loss is 1 dB , and splicing loss of 0.4 dB/splice for 10 splices. Determine the feasibility of system.
5. Design a 4×25 span WDM link with optical amplifier gain = 20 dB and $\text{NF} = 6\text{ dB}$. Calculate final OSNR if the input power is 0 dB . Calculate signal power at the receiver. The receiver sensitivity is -25 dB at $\text{BER } 10^{-15}$. Does the system work?
6. An optical fiber link of length 2 km employs an LED which launches mean optical power of -10 dBm into a multimode optical fiber. The fiber cable exhibits a loss of 3.5 dB/km with splice losses calculated at 0.7 dB/km . In addition there is a connector loss at receiver of 1.6 dB . The PIN photodiode receiver has a sensitivity of -25 dBm for an SNR of 50 dB and with a modulation index of 0.5 . It is estimated that a safety margin of 4 dB is required. Assuming there is no dispersion equalization penalty:
 - (a) Perform an optical power budget for the system operating under the above conditions and check the validity.
 - (b) Calculate the possible increase in length which may be achieved using injection laser source, which launches mean optical power of 0 dBm into fiber. Assume safety margin of 7 dB .
7. A 100 Mbit/s signal is to be sent through a 100 m length of fiber with eight connector pairs to a receiver with sensitivity of -30 dBm . The fiber loss is 4 dB/km and the average connector loss is 1.0 dB . If the system margin is 5 dB , what is the minimum power that the light source must launch into fiber?
8. What is the response time of a system having transmitter response time of 2 ns , receiver response time of 1 ns and 100 m of multimode fiber with dispersion of 20 ns/km including both modal and chromatic dispersion?
9. The 10–90% rise times for possible components to be used in D-IM analog optical fiber link are specified as.
 LED source: 8 ns .
 Fiber cable: Intermodal: 9 ns/km .
 Intramodal: 2 ns/km .
 APD Detector: 2 ns .
 The desired link length without repeaters is 5 km and the required BW is 6 MHz . Determine the above combination of components to give an adequate response.

10. A 2×2 biconical tapered fiber, coupler has an input optical power $P_0 = 300 \mu\text{W}$. The output powers at the three other ports are $P_1 = 150 \mu\text{W}$, $P_2 = 65 \mu\text{W}$, and $P_3 = 8.3 \text{ nW}$. Calculate the splitting ratio, excess loss, insertion loss, and crosstalk for this coupler.
11. If one takes a spectral band of 0.8 nm (or, equivalently, a mean frequency spacing of 100 GHz) within which narrow-line width lasers are transmitting, how many channels can be sent in the 1525–1565 nm band on a single fiber?
12. Make a rise time budget for a 0.85 μm , 10 km fiber link designed to operate at 15 Mbps. The LED transmitter and the Si PIN receiver have rise times of 10 ns and 12 ns, respectively. The graded index fiber has a core index of 1.46, $\Delta = 0.01$, and $D = 70 \text{ ps/km-nm}$. The LED spectral width is 50 nm. Can the system be designed to operate with NRZ format?
13. Make a rise time budget for a 2 km fiber link designed to operate at 20 Mbps. The LED transmitter and the Si PIN receiver have rise times of 8 ns and 12 ns, respectively. The graded index fiber has material dispersion $D = 0.1 \text{ ns/nm-km}$ and $\text{BW} = 35 \text{ MHz km}$. The LED spectral width is 40 nm. Can the system be designed to operate with NRZ format?
14. Consider a parabolic index multimode fiber, $n_{\text{co}} = 1.45$, $\Delta = 0.01$, operating at 850 nm. If the source is an LD with spectral width of 1.2 nm, find the maximum bit rate that can be handled by fiber.
15. A given system requires 400 GHz channel spacing over the wavelength interval of 1,540 to 1,560 nm. Center of this region is 1,550 nm. What is the number of possible channels in this wavelength interval?
16. For a WDM link, if P_{signal} is 0 dB, $\text{OSNR}_{\text{final}} = 20 \text{ dB}$, total length = 600 km, bit rate = 5 Gbps, $\text{NF} = 5 \text{ dB}$, fiber type is SMF with $\alpha = 0.2 \text{ dB/km}$, then calculate the number of spans in the link.
17. Find the total length of a system having $\text{OSNR} = 30 \text{ dB}$, dispersion of fiber is 17 ps/nm-km, span loss is 25 dB, NF of the amplifier is 4 dB, dispersion tolerance is given as 1,600 ps/nm, and $P_{\text{signal}} = 12 \text{ dB}$.

ANSWERS

Multiple-Choice Questions

- | | |
|--------|---------|
| 1. (b) | 6. (d) |
| 2. (c) | 7. (a) |
| 3. (b) | 8. (b) |
| 4. (c) | 9. (b) |
| 5. (a) | 10. (d) |

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7 | Optical Amplifiers

Learning Objectives

After completing this chapter, you will be able to understand:

- The necessity of optical amplifiers.
 - The different types of optical amplifiers, their comparison.
 - Semiconductor optical amplifiers, their types, and cavity gain calculations.
 - Rare-earth-doped fiber amplifiers, such as erbium-doped fiber amplifier (EDFA) and thulium-doped fiber amplifier.
 - Applications of optical amplifiers.
-

In long distance communication systems, the attenuation in fibers and various losses due to optical components (such as multiplexers and couplers) are compensated by inserting regenerative repeaters at equidistant points. Due to the cumulative loss of signal strength, the signal becomes too weak to be detected; the signal strength has to be restored before the signal gets buried under background noise. This task is accomplished by a **regenerative repeater**.

7.1 Basic Concept

A regenerative repeater converts the optical signal to an electrical signal and restores the signal, that is, it compensates for signal loss and dispersion (as shown in Fig. 7.1), and converts the signal back into an optical signal for further transmission.

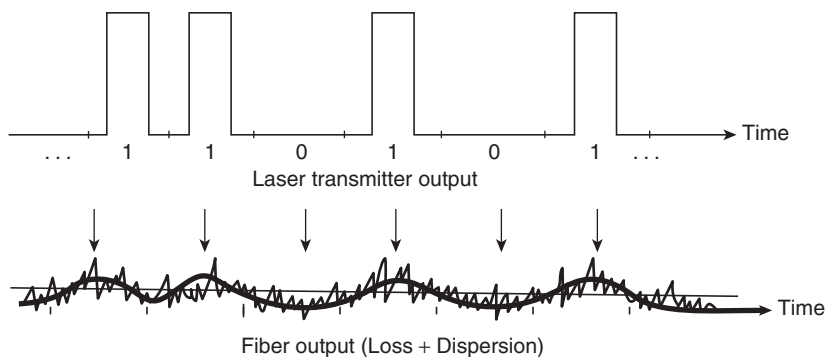


Figure 7.1 | Regeneration in digital communication system.

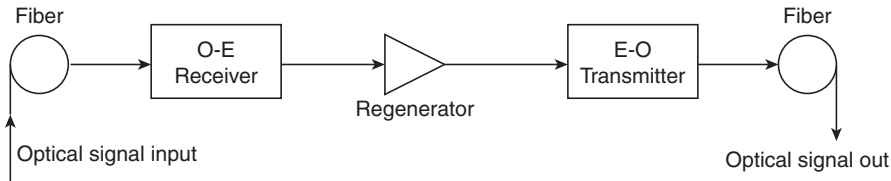


Figure 7.2 | Simplified block diagram of a repeater.

The various functional blocks involved in regenerative repeaters are shown in Fig. 7.2. If the performance of an optical communication link would be limited only by the attenuation of the signal, a simple amplification of the signal would be sufficient to regenerate the original signal. However, it is obvious that the regeneration of a signal due to a repeater is not only amplification. In general, as a signal travels along a wire, a cable, or an optical fiber (it does not matter whether it is transmitted electrically or optically), the signal is distorted.

As shown in Fig. 7.3, the job of repeater is different than being only an amplifier. A repeater receives the signal, converts it to an electrical signal, re-clocks and re-shapes it, amplifies it, and converts it back to an optical signal before coupling the signal back in the optical fiber. It is important to mention that this process is code- and timing-sensitive. The repeaters have to be designed to handle the transmission code and the timing scheme. This process works well for moderate-speed, single-wavelength operation; it is very complex and expensive for high-speed, multiple-wavelength systems. Many researchers have worked hard to find an amplifying device which works in the entire optical domain.

Optical amplifiers have many advantages over repeaters, such as:

1. Optical amplifiers are insensitive to the bit rate or signal formats. Thus, a system using optical amplifiers can be more easily upgraded to a higher bit rate, without replacing the amplifier.
2. Optical amplifiers have fairly large gain bandwidths and as a consequence, a single amplifier can simultaneously amplify several wavelength division multiplexing (WDM) signals, in contrast to a separate regenerator for each wavelength.

Thus, optical amplifiers have become an essential component in high-performance optical communication systems. During the 1980s, most optical communication systems operated at a wavelength of 1310 nm (this was the 2nd generation of optical communication systems). This changed at the beginning of 1990s, when operating wavelengths were increased to 1550 nm (3rd generation). In the mid-1990s, erbium-doped fiber amplifiers (EDFAs) were introduced, and the electro-optical repeaters were replaced by all-optical repeaters.

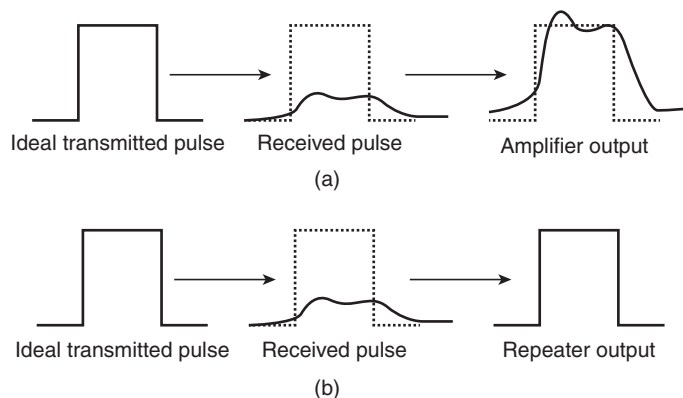


Figure 7.3 | (a) Amplifier function. (b) Repeater function.

7.2 Basic Applications of Optical Amplifiers

Optical amplifiers can be used in both linear and non-linear modes of operation. The common applications are shown in Fig. 7.4.^[1]

1. **In-line optical amplifiers:** Optical amplifiers are used as optical linear repeaters in a long-haul transmission system. In optical transmission systems that use single longitudinal mode lasers, the effects of fiber dispersion may be small, and the main limitation on repeater spacing is the signal attenuation as a result of fiber loss. Such systems do not require a complete regeneration of the signal at each repeater, and linear amplification of the signal is sufficient. Thus, linear optical amplifiers can be used as repeaters.

The gain of each amplifier is chosen to compensate exactly for the signal loss incurred in the preceding fiber section of length L :

$$G = \exp(+\alpha L)$$

2. **Pre-amplifier:** Optical amplifiers are used as pre-amplifiers to boost weak optical signals before detection. Use of a semiconductor laser amplifier before a photodetector to linearly amplify the optical signal can increase the detection sensitivity. The improvement can be particularly significant for bit rates in excess of 1 Gbit/s.

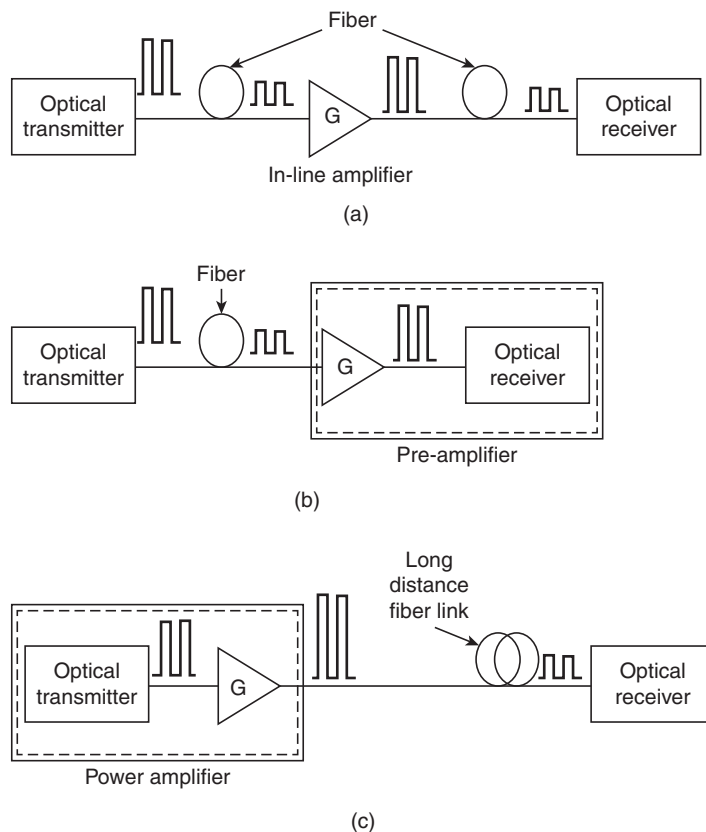


Figure 7.4 | Applications of optical amplifiers: (a) In-line amplifier, (b) pre-amplifier, (c) power amplifier.

3. **Power amplifier:** Power or booster amplifiers are placed immediately after an optical transmitter to boost the transmitted power. The amplifier inputs are generally -8 dBm or greater, and the power amplifier gain must be greater than 5 dB in order for it to be more advantageous than using a pre-amplifier at the receiver. For example, this boosting technique, when used along with an optical pre-amplifier at the receiving end, can enable repeater-less undersea transmission up to distances of 200–250 km. It can also be used as an optical gain block to compensate for losses in splitting networks, for example, to provide fan-out capability for future optical networks.

7.3 Types of Optical Amplifiers

An optical amplifier is a single in-line component as shown in Fig. 7.5. It does not require electrical-to-optical and optical-to-electrical conversions, in contrast to a repeater. All optical fiber amplifiers increase the power level of incident light through stimulated emission or an optical power transfer process. Here, the device absorbs the energy supplied from an external source, called a **pump**. The pump supplies energy to electrons in an active medium and raises them to higher energy levels to produce population inversion. The incoming signal photon triggers these excited electrons to drop to lower levels through a stimulated emission process. One incoming trigger photon stimulates many excited electrons to emit photons of equal energy, thus producing an amplified optical signal.

Two main categories of optical amplifiers are as follows:

1. **Cavity amplifier/semiconductor optical amplifiers (SOAs):** Here amplification is done by simulated emission from injected carriers.
2. **Fiber amplifiers (FA):** Here amplification is provided by stimulated Raman scattering or doping with rare-earth materials, such as erbium (Er) or thulium.

7.4 Semiconductor Optical Amplifier (SOA)

The SOA is basically a pn junction^[4, 6] as shown in Fig. 7.6. Light is amplified through stimulated emission when it propagates through the active region. The depletion region formed at the junction acts as the active region. The two ends of the active region are given an anti-reflection coating to eliminate ripples in the amplifier gain.

A semiconductor consists of two bands of electron energy levels: valence band and conduction band. These bands are separated by an energy difference called as **band gap energy** (E_g). In a p-type semiconductor material, shown in Fig. 7.7, at thermal equilibrium there is a small concentration of electrons in the conduction band. However, in the population inversion condition, the electron concentration is much higher. Population inversion is achieved by forward-biasing a pn junction.^[2]

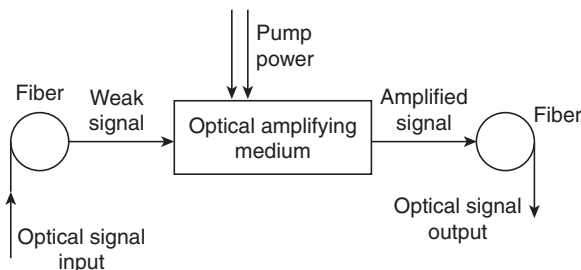


Figure 7.5 | Simplified block diagram of optical amplifier.

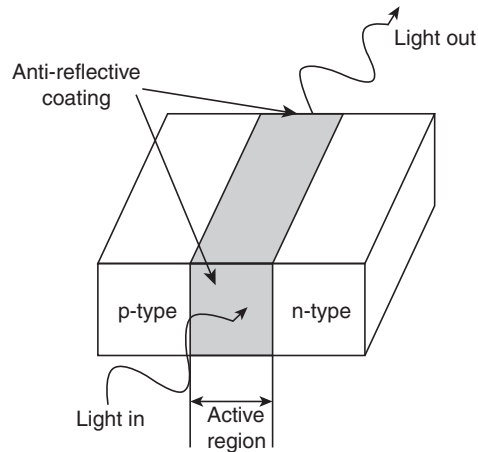


Figure 7.6 | Block diagram of a semiconductor optical amplifier.

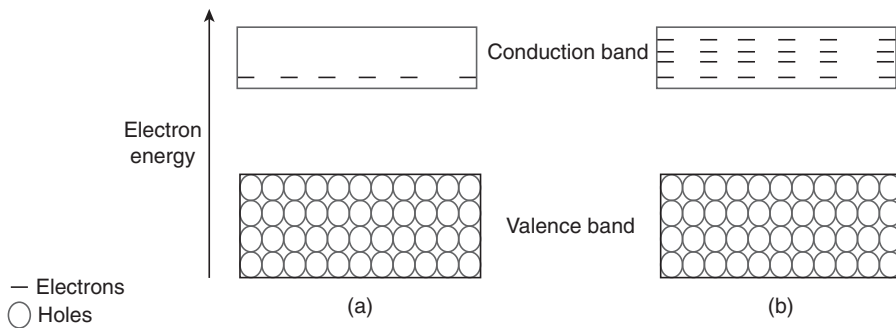


Figure 7.7 | Energy bands and electron concentration in a p-type semiconductor at (a) thermal equilibrium and (b) population inversion.

In a pn junction, the holes diffuse from the p-type semiconductor to the n-type semiconductor, and electrons diffuse from the n-type semiconductor to the p-type semiconductor and form a depletion region. When the junction is forward-biased, the width of depletion region reduces and there is a drift of electrons from the n-type region to the p-type region, which increases the electron concentration in the conduction band of the p-type region. This results in population inversion with a sufficiently high forward-bias. In this case, the pn junction acts as an optical amplifier.

The following are the characteristics of a SOA^[5]:

1. Polarization dependent – requires a polarization maintaining fiber.
2. Relatively high gain ~20 dB.
3. Output saturation power 5–10 dBm.
4. Large bandwidth.
5. Can operate in the 800, 1,300, and 1,500 nm wavelength regions.
6. Compact and can be easily integrated with other devices.
7. Can be integrated into arrays.
8. High noise figure and crosstalk levels due to non-linear phenomena such as 4-wave mixing.

7.4.1 Types of SOA

SOAs are classified into two main groups:

1. Fabry–Perot amplifiers (FPAs).
2. Traveling wave amplifiers (TWAs).

The main difference between the two types of amplifiers is the facet reflectivity, as shown in Fig. 7.8.

In FPAs, the facet reflectivities are of the order of 0.01 to 0.3 and a highly resonant amplifier is formed. The transmission characteristic comprises very narrow pass bands as shown in Fig. 7.9.^[3] The mode zero corresponds to peak gain wavelength and the mode spacing $\delta\lambda$ can be obtained as follows:

$$\delta\lambda = \frac{\lambda^2}{2nL} \quad (7.1)$$

FPAs are normally biased below the normal lasing threshold current, and light entering one facet appears amplified at the other facet, together with inherent noise. These devices are very sensitive to fluctuations in bias current, temperature, and signal polarization. Because of their resonant nature and high internal field, they are used in non-linear applications such as pulse shaping and bistable elements.

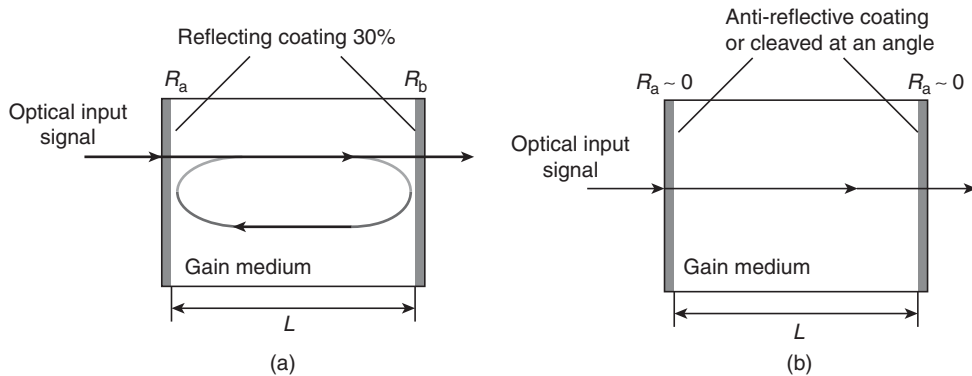


Figure 7.8 | (a) Fabry–Perot amplifier and (b) traveling wave amplifier.

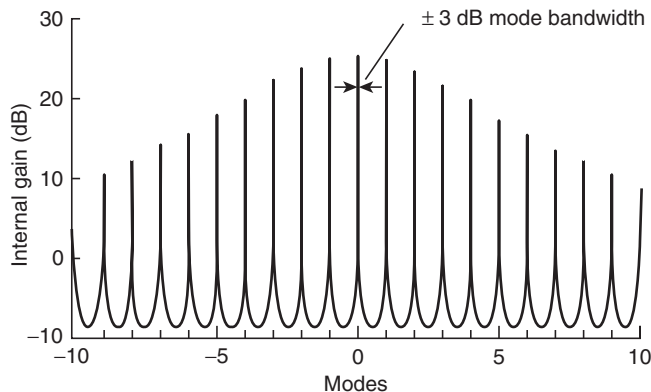


Figure 7.9 | Fabry–Perot amplifier pass band.

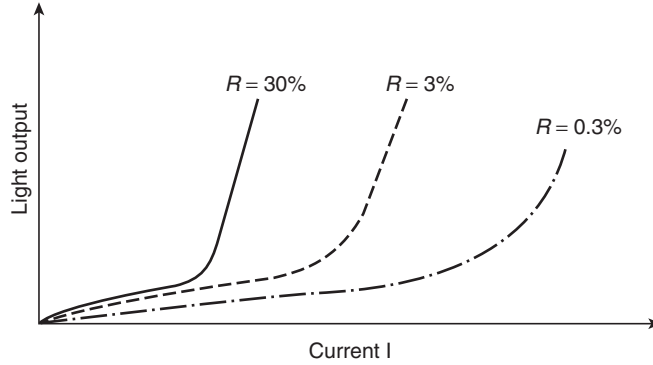


Figure 7.10 | Output light versus current characteristics for SOA.

In traveling wave SOAs, to eliminate or reduce the end reflectivities, a thin layer of silicon oxide or silicon nitride is applied on the end facets. The reflectivities are reduced to 1×10^{-3} or less; this operates the traveling wave amplifier (TWA) in the single-pass amplification mode. The effects of this are: suppressed resonance, increased spectral bandwidth, and increased lasing current threshold. This also makes the transmission characteristics less dependent upon fluctuations in bias current, temperature, and input signal polarization. Hence, TWAs are superior to FPAs, particularly for linear applications.

When compared to an FPA, a TWA requires significantly higher bias currents for operation. The narrow spectral bandwidth of FPAs provides inherent noise filtering, which is not obtained with TWAs. Therefore, they are subject to increased level of noise.

Figure 7.10 shows the output light versus current characteristics for SOA with different facet reflectivities.^[3] The lasing current threshold increases with decrease in facet reflectivity.

7.4.2 Cavity Gain of SOA

The cavity gain of SOA as a function of signal frequency f is given as^[3]

$$G(f) = \frac{(1 - R_1)(1 - R_2)G_s}{(1 - \sqrt{R_1 R_2} G_s)^2 + 4\sqrt{R_1 R_2} G_s \sin^2 \phi} \quad (7.2)$$

where R_1 and R_2 are the input and output facet reflectivities; G_s is the single-pass gain; and ϕ is the single-pass phase shift through the amplifier, given by

$$\phi = \frac{\pi(f - f_o)}{\delta f} \quad (7.3)$$

Here f_o is the Fabry–Perot resonant frequency and δf is the free spectral range of the SOA. The 3-dB spectral bandwidth of SOA is expressed as

$$B_{\text{FPA}} = 2(f - f_o) = \frac{2\delta f}{\pi} \sin^{-1} \left[\frac{1 - \sqrt{R_1 R_2} G_s}{2(\sqrt{R_1 R_2} G_s)^{1/2}} \right]$$

This implies

$$B_{\text{FPA}} = \frac{c}{\pi n L} \sin^{-1} \left[\frac{1 - \sqrt{R_1 R_2 G_s}}{2(\sqrt{R_1 R_2 G_s})^{1/2}} \right] \quad (7.4)$$

7.4.3 Crosstalk in SOAs

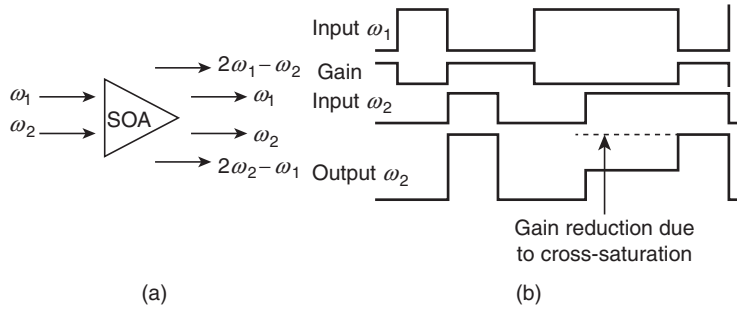


Figure 7.11 | Crosstalk in SOA: (a) Interchannel crosstalk and (b) cross-saturation.

Crosstalk is any distortion of a channel caused by the presence of another channel. There are two types of crosstalk in SOAs, as indicated in Fig. 7.11:

1. **Interchannel crosstalk:** When two wavelengths (channels) enter a SOA, their non-linear interference produces new signals at the **beat** (combination of sums and differences) **frequencies**.
2. **Cross-saturation:** For a SOA working in the saturated mode ($I > I_s$), the gain undergoes an opposite change when one channel changes from ON to OFF. This gain change results in variations in the amplification of another signal because all signals share the same gain produced by one active medium.

7.5 Raman Amplifier

Stimulated Raman scattering is a very important non-linear effect because it affects the signal-to-noise ratio (SNR) in a WDM system and it can also be used for amplification of the optical signals in a long-haul optical communication link. Spontaneous Raman scattering was discovered by Sir C. V. Raman. In spontaneous Raman scattering, a small portion of the incident light is transformed into a new wave with a lower or higher frequency. This transformation is because of the interaction of the photon with the vibrational modes of the material.

The transformation efficiency of stimulated Raman scattering is very high as compared to that of spontaneous Raman scattering. Stimulated Raman scattering can transform a large fraction of the incident light into a new frequency-shifted wave.

7.5.1 Raman Scattering

Spontaneous Raman scattering can be observed in any material. If the scattered light is studied by irradiating a medium by an intense monochromatic beam of light with a spectrometer, the scattered light will show new wavelengths in addition to the original wavelength. The wave which irradiates the material is called the **pump wave**, waves with lower frequencies are called **Stokes waves**, and waves with higher frequencies are called the **anti-Stokes waves**. The intensity of the Stokes waves is many orders of magnitude higher than the intensity of the anti-Stokes waves, as shown in Fig. 7.12.

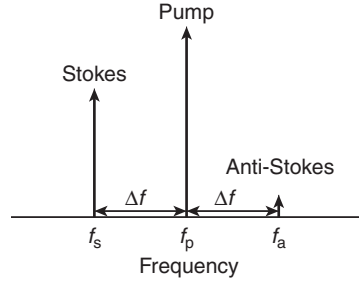


Figure 7.12 | Frequency distribution of pump, Stokes, and anti-Stokes waves.

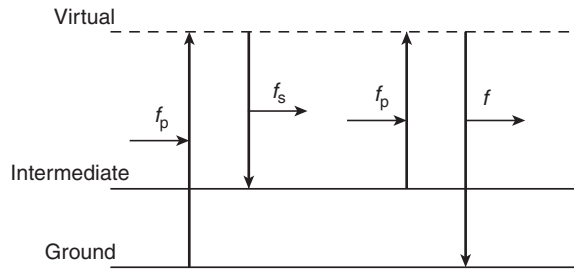


Figure 7.13 | The quantum-mechanical energy diagram for Raman scattering.

The origin of the generation of these new frequencies lies in the energy exchange between the photons and the material molecules.^[7] In the Stokes generation process, the incoming photon of frequency f_p excites a molecule from the ground state to the virtual state, as shown in Fig. 7.13. The molecule returns to the intermediate state releasing a Stokes photon of frequency f_s . Since the energy difference between the virtual and intermediate levels is smaller than the energy difference between the ground and the virtual states, the frequency of the Stokes photon is lower than that of the incident photon ($f_s < f_p$). In the anti-Stokes generation process, the incoming photon of frequency f_p excites a molecule from the intermediate state to the virtual state. The molecule returns to the ground state, releasing an anti-Stokes photon of frequency f_a . Since the energy difference between the virtual and intermediate levels is smaller than the energy difference between the ground and the virtual states, the frequency of the anti-Stokes photon is higher than the incident photon ($f_a > f_p$).

If energy difference between the intermediate level and the ground level is

$$\Delta E = h\Delta f$$

where h is the Planck's constant, then we get

$$f_s = f_p - \Delta f = f_p - \frac{\Delta E}{h} \quad (7.5)$$

Energy ΔE is the vibrational energy and in the Stokes generation process, it is delivered to the molecule. In anti-Stokes generation process, the molecule supplies vibrational energy to the photon, thereby increasing the total energy of the incident photon. In thermal equilibrium, the density of molecules in the intermediate state is much smaller than the density in the ground state, and therefore the number of anti-Stokes photons is much smaller compared to that of Stokes photons.

The vibrational energy depends upon the molecular resonances. The Raman scattering depends upon particular material resonances. In crystalline material, these resonances show a very narrow bandwidth. Therefore, the frequencies of the Stokes and the anti-Stokes waves reflect material property.

7.5.2 Raman Scattering in Optical Fibers

The basic material used in optical fibers is glass, which is not crystalline, but amorphous in nature. The molecular resonance frequencies of the vibrational modes in glass overlap with each other to give a rather broad frequency band. Optical fibers, therefore, show Raman scattering over a large frequency range. The energy conversion process between the pump and the Stokes waves is characterized by a parameter called the Raman gain, g_R .

The Raman gain depends mainly on the material composition of the fiber core and the contained dopants. A typical Raman gain spectrum of silica glass fiber is shown in Fig. 7.14.

The Raman gain for silica glass has one main maximum and many small side maxima. If minor peaks in the spectrum are included, the Raman gain has a wide bandwidth of about 40 THz. The maximum gain is practically constant over a bandwidth from 9 THz to 16 THz. The mean of the high gain region is around 13 THz (which at 1550 nm wavelength corresponds to $\Delta\lambda = 112$ nm). The maximum Raman gain in pure glass is

$$g_{R(\max)} = 1.9 \times 10^{-13} \text{ m/W}$$

Raman gain is inversely proportional to the wavelength of the pump and depends on the polarization of the wave. It is much higher for parallel polarization (when the pump and the Stokes waves have same polarization) compared to perpendicular polarization (i.e., when the pump and the Stokes waves have orthogonal polarization).

In optical fibers however, due to birefringence, the polarization states of the pump and the Stokes wave change continuously, and if the fiber is long enough, the effective Raman gain is the mean of the parallel and perpendicular gains. So, in a fiber we may say that the Raman gain is polarization-independent.

Raman scattering can be seen in the forward as well as the backward directions. In optical communication, forward Raman scattering is of importance because it causes crosstalk in a multichannel WDM system. Backward scattering can be exploited for signal amplification in fiber Raman amplifier, which we will discuss next.

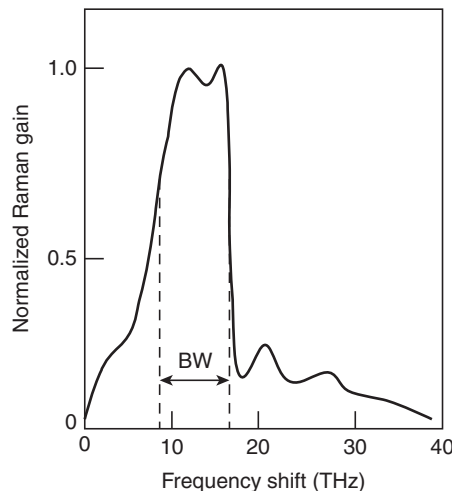


Figure 7.14 | Raman gain spectrum of silica glass.

7.5.3 Fiber Raman Amplifier

For optical amplification, we need stimulated Raman scattering. The basic difference between spontaneous and stimulated Raman scattering is shown in Fig. 7.15.

If the intensity of the incident field is below a threshold, spontaneous scattering occurs. The important thing to note is that the stimulated process can be used for light amplification. If a small signal at Stokes frequency is present along with the pump, the signal gets amplified while retaining all the characteristics of the input signal.

The differential equations governing the intensities of the pump and the Stokes waves are

$$\frac{dI_s}{dz} = g_R I_p I_s - \alpha_s I_s \quad (7.6)$$

$$\frac{dI_p}{dz} = -\frac{\omega_p}{\omega_s} g_R I_p I_s - \alpha_p I_p \quad (7.7)$$

where I_p and I_s are the intensities of the pump and the signal, respectively; ω_p and ω_s are the frequencies; and α_p and α_s are the attenuation constants of the pump and the Stokes waves. The equations show that the Stokes waves grow exponentially with distance whereas the pump waves decay exponentially with distance.

In the first approximation, it is generally assumed that the pump depletion due to Raman scattering is small. The first term on the RHS in Eq. (7.7) can be neglected. The pump then decays along the fiber due to fiber attenuation, and its intensity is given by

$$I_p = I_p(0)e^{-\alpha_p z} \quad (7.8)$$

Introducing I_p in Eq. (7.6), we get the intensity of the Stokes wave as

$$I_s = I_s(0) \exp [g_R I_p(0) L_{\text{eff}} - \alpha_s z] \quad (7.9)$$

where L_{eff} is the effective interaction length of the pump, and is given as

$$L_{\text{eff}} = \frac{1 - \exp(-\alpha_p z)}{\alpha_p} \quad (7.10)$$

For long fibers,

$$L_{\text{eff}} = \frac{1}{\alpha_p} \quad (7.11)$$

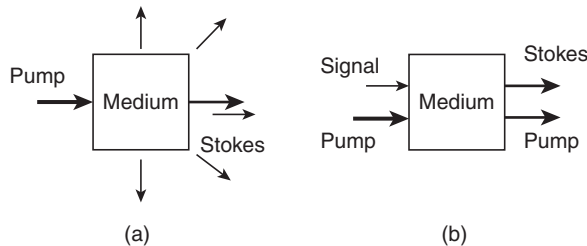


Figure 7.15 | (a) Spontaneous Raman scattering. (b) Stimulated Raman scattering.

The threshold for stimulated scattering is defined as the input intensity I_{pth} value of the pump for which Stokes wave shows gain in the fiber. The Raman gain process then must exceed the fiber loss to give

$$g_R I_{\text{pth}} L_{\text{eff}} \approx \alpha_s z \quad (7.12)$$

Assuming that the length of the fiber is approximately L_{eff} we get the threshold intensity for stimulated Raman scattering (SRS) as

$$I_{\text{pth}} \approx \frac{\alpha_s}{g_R} \quad (7.13)$$

For a single mode fiber, typically the core effective area is $80 \mu\text{m}^2$, $\alpha \approx 0.2 \text{ dB/km}$, Raman gain is $7 \times 10^{-14} \text{ m/W}$, and the Raman threshold power is

$$P_{\text{th}} = I_{\text{pth}} A_{\text{eff}} \approx 53 \text{ mW}$$

where I_{pth} is the threshold value of input intensity used for pumping and A_{eff} is the core effective area. Above this threshold pump power, there will be gain for any signal above the noise. However, the Stokes power is still orders of magnitude smaller than the pump power.

If we define the threshold as the input power for which the output powers of pump and Stokes are equal, then its value is approximately

$$I_{\text{th}} = \frac{16}{g_R L_{\text{eff}}} \quad (7.14)$$

Typical power to achieve this threshold intensity in an SM fiber is about 1 W.

There has been renewed interest in Raman amplification in the recent years due to several reasons:

1. Laser diodes which deliver the required power at several wavelengths are commercially available.
2. The whole transmission bandwidth of the optical fiber can be utilized using Raman amplification. (The EDFA makes use of a very small fraction of the transmission window.)
3. In an optical fiber, multiple Raman processes can go on simultaneously, and so ultra wideband amplifiers can be realized using multiple pumps.
4. As compared to the EDFA or the SOA, it is a distributed amplifier.

The signals levels on a long-haul optical link with EDFA and with Raman amplification are shown in Fig. 7.16. The arrows indicate the location of EDFAs or pumps for Raman amplification.

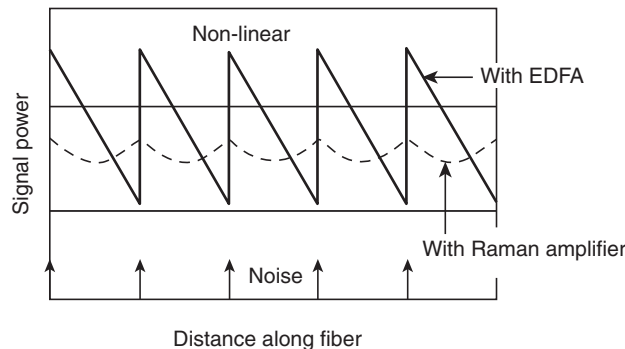


Figure 7.16 | The signal levels on a long-haul optical link with EDFA and with Raman amplification.

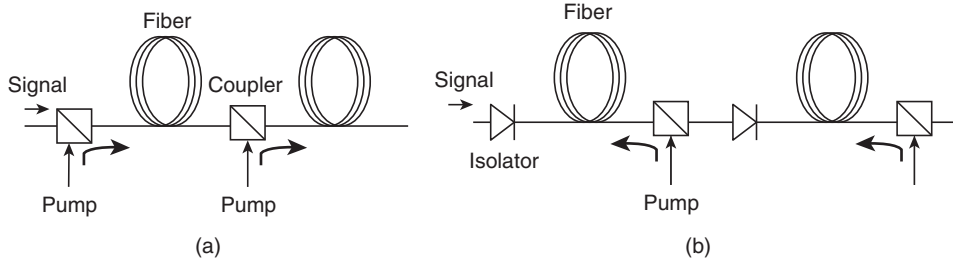


Figure 7.17 | (a) Forward pumping and (b) backward pumping.

It can be seen from Fig. 7.16 that the signal level for Raman amplification remains more or less constant at a level much higher than the noise floor in the system. In the case of EDFA, the signal level reaches close to the noise floor just before it goes through the amplifier, and just after passing through the amplifier it increases to such a high value that the non-linear effects such as SPM, XPM, FWM become effective. In the presence of non-linear effects, the signal gets distorted. Lumped amplification systems such as EDFAs have worse noise levels and distortion performances.

7.5.4 Pumping Schemes for Raman Amplifier

The Raman amplifier can be pumped with forward as well as backward pump, since Raman gain is independent of the pump direction. The two schemes are shown in Fig. 7.17.

The forward pump scheme has two main drawbacks:

1. The pump level is higher where signal is stronger, thereby increasing the signal power at the beginning of the section.
2. The residual co-propagating pump is to be filtered.

The maximum length or gain beyond which the Raman amplifiers no longer improve the system performance is determined by the double Rayleigh back-scattering (DRBS) multipath interference (MPI) noise. This can be explained as follows.

Suppose a signal is amplified by a co-propagating pump. Due to Rayleigh scattering, a part of the signal is scattered back. The SRS, being direction-independent, amplifies the backward scattered signal. The scattered light becomes strong enough to generate a double back-scattered signal which travels with the pump. Superposition between the signal and the time-delayed double back-scattered light leads to time-dependent noise.

7.5.5 Broadbanding of Raman Amplifier

The Raman amplifier with single pump gives a bandwidth of about 7 THz which is approximately 60 nm. The transmission window of the fiber is about 400 nm (1200 nm to 1600 nm). A broadband amplifier, therefore, is very desirable. Using multiple pumps, wide-band amplifiers with a very small gain ripple can be designed. It should be kept in mind, however, that in a multiple-pump scheme, there is exchange of power between the pumps themselves, due to Raman process.

7.5.6 Impact of Raman Scattering on WDM Systems

If only one channel propagates in an optical fiber, the Raman effect is observable only if the input power of the channel is of the order of the threshold for the forward Raman scattering. Since the threshold is very high, SRS can be neglected in a single channel system.

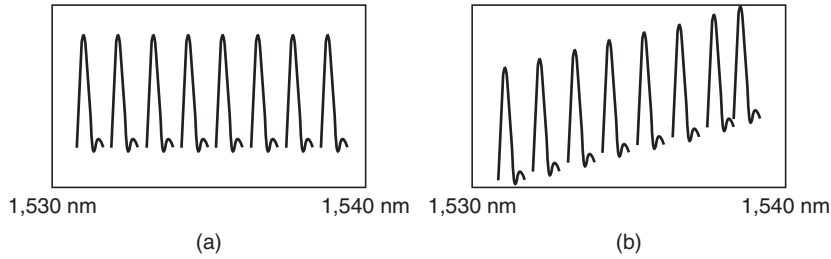


Figure 7.18 | WDM channels (a) without Raman scattering and (b) with Raman scattering.

A completely different behavior can be seen in WDM systems. Here the initial Stokes wave is not generated by a spontaneous process because the waves of the right frequency shifts are already present in the system. Further, the input power of the Stokes wave is more or less same as the input power of the pump (channels of shorter wavelengths act as pump for channels of longer wavelengths).

Since the Raman gain peaks around a 13 THz frequency shift, channels separated by about 100 nm influence each other maximally. In other words, channels from different bands, such as L and C, influence each other more. Nevertheless, channels within a band also influence each other though to a lesser extent.

For dense wavelength division multiplexing (DWDM) systems, the Raman interaction is very complicated. Every wavelength acts as a pump for wavelengths longer than itself and as a Stokes wave for wavelengths shorter than itself. Hence, due to Raman scattering, every channel receives power and every channel loses power. There is a systematic flow of power from higher frequency channels to the lower frequency channels. So if all channels had equal power to start with, at the end the spectrum will be as shown in Fig. 7.18.

Decrease in channel power is a problem in WDM systems, as it reduces SNR and increases the bit error rate (BER). The decrease in power of a channel can be estimated analytically assuming a triangular Raman gain profile from 0 to 15 THz. If the acceptable reduction in the channel power is 1 dB, the power-bandwidth product is

$$[nP](N-1)\Delta f < 500 \text{ W} \cdot \text{GHz}$$

where n is the number of DWDM channels, P is the power per channel, and Δf is the channel separation.

If the number of channels is less, the maximum power per channel decreases as $1/n$, but if the number of channels is more, the power decreases as $1/n^2$. A plot of maximum power per channel as a function of number of channels is shown in Fig. 7.19.

In a long-haul system with periodic amplification, SNR performance in the presence of Raman scattering has been estimated. If the acceptable decrease in SNR in the channel with the smallest wavelength is 0.5 dB, the number of transmittable channels n can be calculated using

$$n(n-1) = \frac{8.7 \times 10^{15}}{P\Delta f L_{\text{eff}}} \quad (7.15)$$

where L_{eff} is the effective interaction length of the system with amplifiers. The noise power in the system due to the amplifiers is

$$N = 2mh\nu n_{\text{sp}} B_o (G-1) \quad (7.16)$$

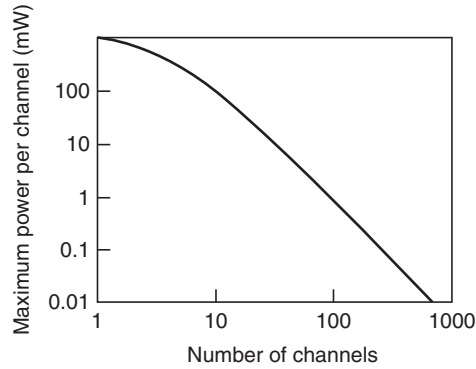


Figure 7.19 | Maximum power per channel versus number of channels.

where m is number of amplifiers $m = L/L_A$ (L being the link length and L_A the amplifier spacing), $h\nu$ is the photon energy, n_{sp} is excess noise factor of the amplifier, B_o is the optical bandwidth of the receiver, and G is the gain of each amplifier. If the required SNR is R , the average input power in each channel is

$$P = 2 \left(\frac{L}{L_A} \right) h\nu n_{sp} B_o (G-1) R \quad (7.17)$$

If the number of channels is large, the channel capacity can be obtained as

$$C < \sqrt{\frac{1.8 \times 10^{14}}{2 \left(\frac{L}{L_A} \right) h\nu n_{sp} L_{eff} (G-1) R}} \quad (7.18)$$

Here it is assumed that channel spacing is 6 times the data rate and receiver filter bandwidth is 4 times the data rate.

7.6 Rare-Earth-Doped Fiber Amplifiers

Optical long-haul data transmission systems today usually operate in the C-band, from 1530 to 1570 nm. The standard amplifier for these systems is the erbium-doped fiber amplifier (EDFA), which provides gain more or less in the whole C-band. For high data rate links, dense wavelength division multiplexing (DWDM) systems with increasing bit-rates per channel and decreasing channel spacing are used. Of course, this process cannot be continued without limitations. Another approach to reach very high bit-rates without violating the 1 bit/s/Hz limit is the usage of transmission bands besides the C- and L-bands. The usable range of the low-loss window of transmission fibers is usually considered to be 1200–1700 nm. Figure 7.20 indicates the spectral position of a couple of rare-earth-doped fiber amplifiers in this window. Amplifiers in the S- and S+-bands are of the most practical importance, on the one hand due to their spectral location close to the C-band and on the other hand because the dispersion compensation is more easy at shorter wavelengths than at the long wavelength edge of the transmission window. These S-band amplifiers use thulium for the active core instead of erbium and are therefore called thulium-doped fiber amplifiers (TDFAs).

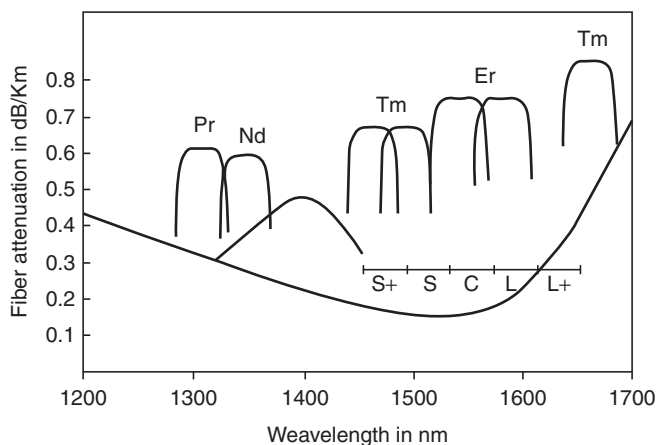


Figure 7.20 | Rare-earth-doped fiber amplifiers in the low-loss window of transmission fibers.

7.6.1 Erbium-Doped Fiber Amplifier (EDFA)

An EDFA is shown in Fig. 7.21. It consists of a 10–30 m length of silica fiber whose core is lightly doped with ionized atoms of rare-earth elements such as erbium (Er^{3+}), ytterbium (Yb), thulium (Tm), or praseodymium (Pr), while the host fiber material can be standard silica, fluoride glass, or telluride glass. The operating region is decided by the host material and the doping element. For long-haul telecommunication applications, silica fiber doped with erbium is the most popular choice, which is known as erbium-doped fiber amplifier or EDFA.

Instead of using external current injection to excite electrons to higher energy levels, this amplifier uses optical pumping. The fiber is pumped using a pump signal from a laser, typically at a wavelength of 980 nm or 1480 nm.

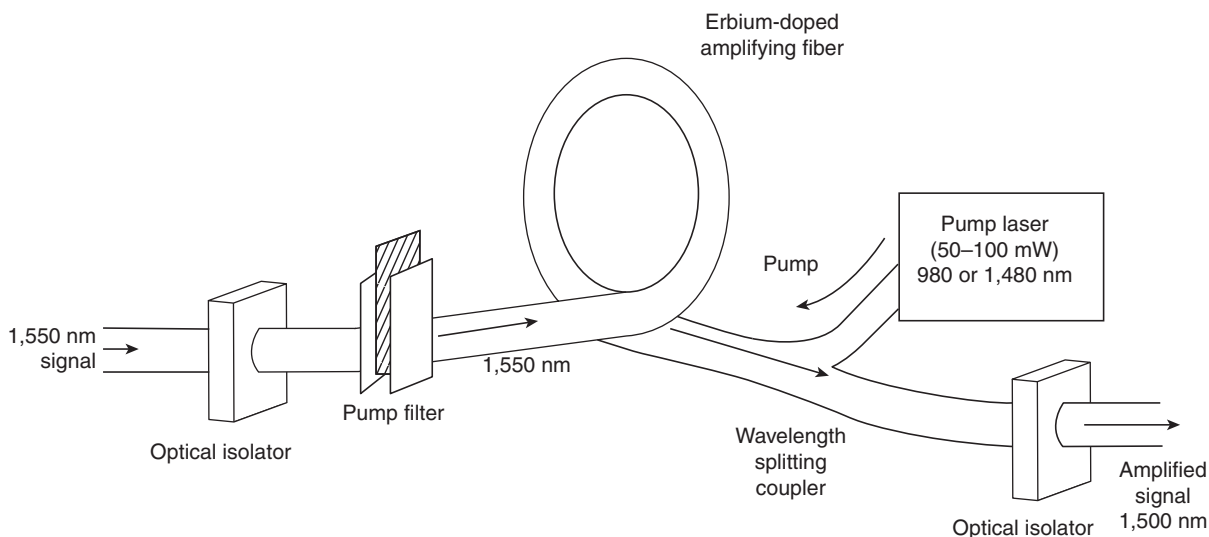


Figure 7.21 | An erbium-doped fiber amplifier.

To combine the output of the pump laser with the input signal, the doped fiber is preceded by a wavelength-selective coupler. Usually an optical isolator is used at the input and/or output of any amplifier to prevent back reflections. EDFAs are more preferred in today's telecommunication systems, because of a combination of several factors as listed below:

1. High power transfer efficiency from pump to signal (>50%).
2. Wide spectral band amplification with a relative flat gain (>20 dB) – useful for WDM applications.
3. Saturation output > 1 mW (10–25 dBm).
4. Long gain-time constant (>100 ms) to overcome patterning effects and intermodulation distortions (low noise).
5. Large dynamic range.
6. Low noise figure.
7. Polarization independence.
8. Suitable for long-haul applications.

However, this amplifier has some drawbacks also, which are as follows:

1. Relatively large devices (10–30 m lengths of fiber) – not easily integrated with other devices.
2. Amplified spontaneous emission (ASE). There is always some output even with no signal input due to some excitation of ions in the fiber (spontaneous noise).
3. Crosstalk effects.
4. Gain saturation effects.

7.6.1.1 Principle of Operation

Three-level energy diagram of Er^{3+} is shown in Fig. 7.22. The energy levels are labeled as E_1 , E_2 , and E_3 in order of increasing energy. These levels are actually bands of closely spaced energy levels that form a manifold due to the effect known as **Stark splitting**.

1. E_3 is the pump band, with an energy difference of 1.27 eV from ground level. This energy corresponds to a wavelength of 980 nm.

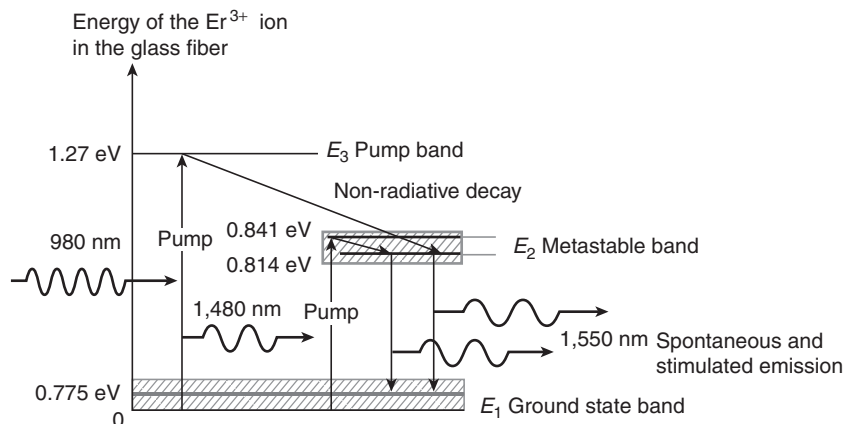


Figure 7.22 | Three energy levels of Er^{3+} ions in silica glass.

2. E_2 is the metastable band. The term “metastable” means that the lifetimes for transitions from this state to the ground state are very long when compared to the lifetimes of the states that led to this level. The top of the metastable band has an energy difference of 0.841 eV from ground level, corresponding to a wavelength of 1480 nm. The bottom of the metastable band has energy difference of 0.814 eV from ground level, corresponding to a wavelength of 1530 nm.
3. E_1 is the ground state band.

The upward arrow indicates wavelengths at which the amplifier can be pumped to excite the ions into the higher energy level. The 980 nm transition corresponds to the band gap between the E_1 and E_3 levels. The 1480 nm transition corresponds to the gap between the bottom of E_1 and top of E_2 band. The downward transition represents the wavelength of photons emitted due to spontaneous and stimulated emission.

A pump laser emitting photons of wavelength 980 nm is used to excite ions from the ground state to the pump level. These excited ions decay very quickly (in about 1 μ s) from pump band to the metastable band. During this decay, the excess energy is released as mechanical vibrations in the fiber. In the metastable state, the electrons of excited ions tend to populate the lower end of band and have a very long fluorescence time of about 10 ms.

Another wavelength (1480 nm) can also be used for pumping. Absorption of a 1480 nm pump photon excites an electron from E_1 to top of E_2 , which is lightly populated. These electrons then tend to move down to the more populated end of E_2 . Some of the ions can decay back to ground level in the absence of external stimulating photon flux. This phenomenon is known as **spontaneous emission** and adds to the amplifier noise. In the stimulated emission process, a signal photon triggers an excited ion to drop to the ground state, thereby emitting a new photon having the same energy, wave vector, and polarization as the incoming signal photon.

In a two-energy-level atomic system, only an optical signal at the frequency f_c satisfying $hf_c = E_2 - E_1$ is amplified. If these levels are spread into bands, all the frequencies that correspond to the energy difference between some energy in the E_2 band and some energy in the E_1 band can be amplified. In an EDFA, the set of frequencies that can be amplified by stimulated emission from E_2 band to the E_1 band corresponds to the wavelength range 1525–1570 nm, a bandwidth of 50 nm.

7.6.1.2 EDFA Architecture

The complete setup of an optical fiber amplifier in different configurations is shown in Fig. 7.23. It consists of a doped fiber, one or more pump lasers, a passive wavelength coupler, an optical isolator, and tap couplers at the input and output sides to monitor the input and output. The dichroic coupler couples the pump and signal optical powers into the fiber amplifier. The optical isolator prevents the amplified signal from reflecting back into the device.

The three possible configurations of an EDFA are as follows:

1. Co-directional pumping.
2. Counter-directional pumping.
3. Dual pumping.

In co-directional pumping, the pump light is injected from the same direction as the signal flow. However, in counter-directional pumping, the pump power is injected in the opposite direction to the signal flow. This arrangement allows higher gain, but co-directional pumping gives better noise performance. The dual pumping scheme can be employed for more gain of around +35 dB, which is almost double that of single pumping scheme (+17 dB).

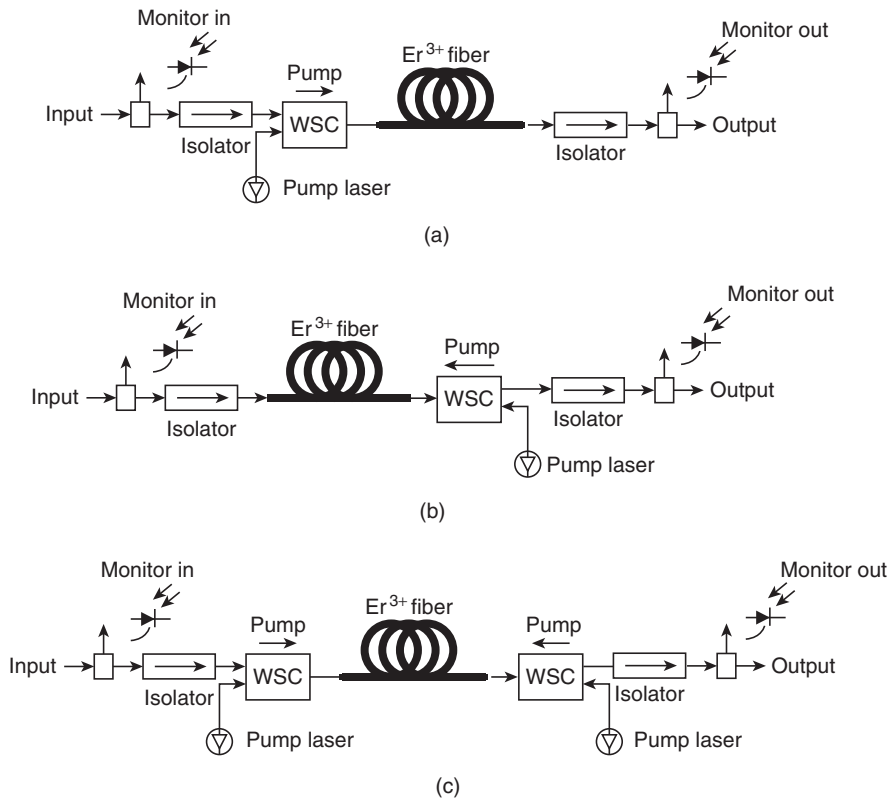


Figure 7.23 | Different configurations of EDFA. Here WSC is wavelength selective coupler.

7.6.1.3 Amplifier Gain and Efficiency

The populations at the various levels within a band are different; therefore, EDFA gain becomes a function of the wavelength. The gain flattening of EDFAs has been a research issue in recent years, with the development of high capacity WDM optical communication systems. For single-channel systems, the gain variation is not a problem. However, as the number of channels increases, transmission problems arise, because a conventional EDFA has intrinsic non-uniform gain. They typically present gain peaking at about 1530 nm and the useful gain bandwidth may be reduced to less than 10 nm as shown in Fig. 7.24.

The gain of EDFAs depends on a large number of device parameters such as erbium-ion concentration, amplifier length, core radius, and pump power. To increase the gain bandwidth of an amplified light wave system, several methods can be used, but equalizing optical filters operating as spectrally selective loss elements appear to be the best solution.

The factors controlling the degree of gain uniformity are as follows:

1. Concentrations of the active ion (erbium).
2. Optical gain flattening filter.

An additional (second) pump laser can also be used at each end of the fiber – one pump beam propagates with signal beam, while the other propagates against it. This ensures that population inversion and gain remain constant along the fiber.

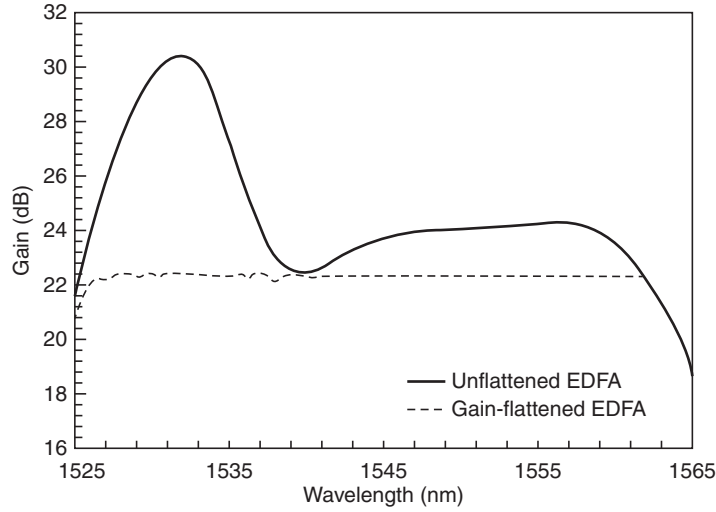


Figure 7.24 | Gain band of EDFA.

Gain equalization can be accomplished in several ways:

1. Long period fiber gratings.^[8]
2. Chirped fiber Bragg gratings.^[9]
3. Thin film filters.^[10]

The input and output powers of an EDFA can be expressed in terms of the principle of energy conservation as

$$P_{\text{signal out}} \leq P_{\text{signal in}} + \frac{\lambda_p}{\lambda_s} P_{\text{pump in}} \quad (7.19)$$

where $P_{\text{pump in}}$ is the input pump power, and λ_p and λ_s are the pump and signal wavelengths, respectively. We can rewrite the above equation as

$$G = \frac{P_{\text{signal out}}}{P_{\text{signal in}}} \leq 1 + \frac{\lambda_p P_{\text{pump in}}}{\lambda_s P_{\text{signal in}}} \quad (7.20)$$

Thus

$$P_{\text{signal in}} = \left(\frac{(\lambda_p / \lambda_s) P_{\text{pump input}}}{G - 1} \right) \quad (7.21)$$

Maximum output signal power depends on the ratio λ_p / λ_s . Thus the power conversion efficiency (PCE) is defined as

$$\text{PCE} = \frac{P_{\text{signal out}} - P_{\text{signal in}}}{P_{\text{pump in}}} \quad (7.22)$$

7.6.1.4 Noise in EDFA

The dominant noise generated in an EDFA is amplified spontaneous emission (ASE). This is generated due to the spontaneous recombination of electrons and holes in the amplifier medium. This recombination gives rise to

a broad spectral background of photons that get amplified along with the optical signal. The output spectrum for an EDFA amplifying a signal at 1540 nm with a pumping signal at 1480 nm is shown in Fig. 7.25 along with ASE.

The total optical field is sum of signal field E_s and the spontaneous emission field E_n . The total photodetector current i_{total} is proportional to the square of the electric field of the optical signal. Mathematically,

$$i_{\text{total}} \propto (E_s + E_n)^2 = E_s^2 + E_n^2 + 2E_s \cdot E_n \quad (7.23)$$

Here the first term arises due to the signal, the second term due to noise, and the third term is a beat signal (mixing component) between signal and noise. Signal degradation comes from beat signals generated at noise–noise and noise–signal interferences. Each “slice” of noise can interfere with another “slice” to generate a beat signal at frequencies that are combinations of the sum and difference of the input frequencies as indicated in Fig. 7.26. Noise–noise beating can be easily removed by a narrowband filter. Noise–signal beating, however, cannot be filtered because it is within a signal’s bandwidth. The noise figure based on signal–noise beating is F_n , which varies from 3.5 to 9 dB for EDFAs.

Spontaneous emission in the amplifier degrades the SNR by adding to the noise during the amplification process. SNR degradation is quantified through the amplifier noise figure NF_{EDFA} :

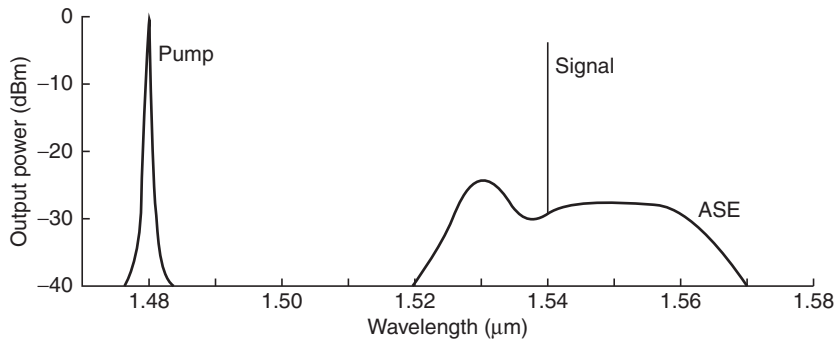


Figure 7.25 | Output spectrum of an EDFA amplifying a signal at 1540 nm with a pumping signal at 1480 nm.

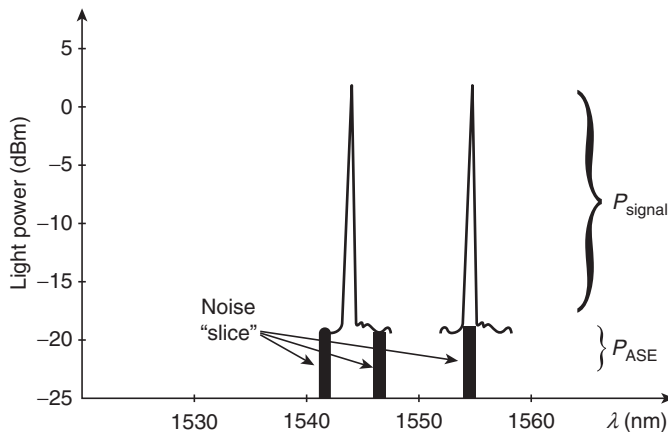


Figure 7.26 | Noise in EDFA.

$$NF_{EDFA} = \frac{SNR_{in}}{SNR_{out}} \quad (7.24)$$

The amplifier has an amplification factor G given by

$$G = \frac{P_{\text{signal out}}}{P_{\text{signal in}}} \quad (7.25)$$

The SNR of the input signal is

$$SNR_{in} = \frac{\langle I \rangle^2}{\sigma_s^2} = \frac{(RP_{\text{signal in}})^2}{2q(RP_{\text{signal in}})B_c} = \frac{P_{\text{signal in}}}{2hvB_c} \quad (7.26)$$

$$\sigma_s^2 = 2q(RP_{\text{signal in}})B_c \quad (7.27)$$

Spontaneous emission population inversion factor n_{sp} is given by

$$n_{sp} = \frac{N_2}{N_2 - N_1} \quad (7.28)$$

where N_1 and N_2 are the population densities for the excited and ground states of the amplifying medium. The SNR of the amplified signal becomes

$$SNR_{out} = \frac{(RGP_{\text{signal in}})^2}{\sigma^2} \quad (7.29)$$

and the amplifier noise figure is

$$NF_{EDFA} = \frac{2n_{sp}(G-1)}{G} \approx 2n_{sp} \quad (7.30)$$

For most amplifiers, $NF_{EDFA} > 3$ dB and can be 6–8 dB.

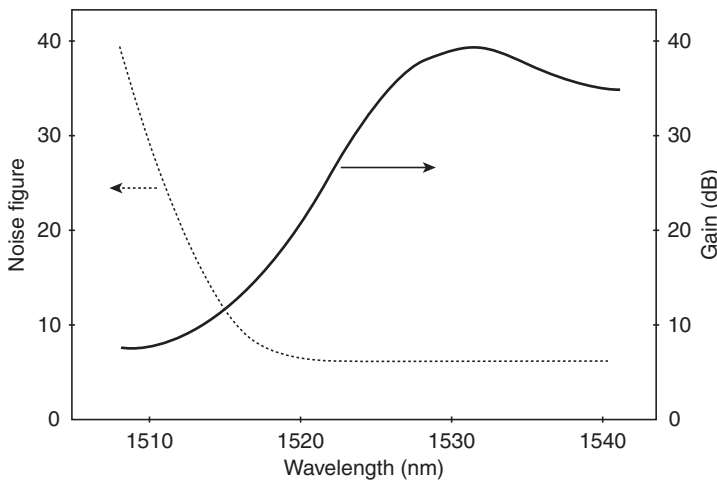


Figure 7.27 | Characteristic plot of gain and noise figure for an erbium-doped fiber amplifier pumped ~30 mW at 980 nm.

7.6.1.5 Optical OSNR

When a number of optical amplifiers are cascaded in a series, ASE noise of each stage gets added, and the signal entering the optical receiver may contain a significant level of ASE noise. In this case, we have to evaluate the optical signal-to-noise ratio (OSNR). This is defined as the ratio of the EDFA optical signal output power P_{out} to the unpolarized ASE optical noise power P_{ASE} . Mathematically

$$\text{OSNR(dB)} = 10 \log \frac{P_{\text{out}}}{P_{\text{ASE}}} \quad (7.31)$$

OSNR does not depend on factors such as data format, optical filter bandwidth, or pulse shape, but only on average optical signal power P_{out} and the average optical noise power.

So far we have concentrated on EDFAs operating in the C-band. Several different wavelength bands have been designated for WDM and EDFAs have been designed to operate in these bands.

The wavelength divisions have been designated as

S-band: 1,480–1,520 nm
 C-band: 1,521–1,560 nm
 L-band: 1,561–1,620 nm

L-band EDFAs operate on the same principle as C-band EDFAs. The gain spectrum of erbium is much flatter intrinsically in the L-band than in the C-band. Therefore, the design of gain-flattening filter for L-band is easier, but the gain coefficient in the L-band is about three times smaller than in the C-band, so a much longer doped fiber length or a higher erbium doping concentration is required.

7.6.2 Thulium-Doped Fiber Amplifier (TDFA)

S-band amplifiers use thulium for the active core instead of erbium and are therefore called thulium-doped fiber amplifiers (TDFAs).^[11–14] The transition used for S-band amplification is 3→1 with the central wavelength of the fluorescence being 1460 nm as shown in Fig. 7.28. The lower level of the amplifier transition is not the ground state of the thulium ion, and the TDFA thus is basically a four-level amplifier. In contrast to the EDFA, an unpumped TDFA shows no signal attenuation. The first pump transition uses 790 nm and excites thulium ions from the ground state to level 3. The next step is a stimulated emission process, which ends up in level 1. At this point the system would terminate, since the lifetime of level 1 in fluoride glasses is about 11 ms, due to the low phonon energy of these glasses. Thus, a second pump transition is needed to depopulate level 1. One possibility is the excited state absorption (ESA) 1→5 at 1050 nm. Thulium ions in level 5 will undergo a fast non-radiative decay ending in the upper amplifier level 3, so the energy loop is closed. In practice, the large lifetime of level 1 leads to the fact that the second pump is much more important than the first one and the TDFA is even working when the 790 nm pump is omitted, although with lower power conversion efficiency.

In a pumped TDFA, level 1, due to its large lifetime, plays the role of the ground state, and in fact, in many respects the TDFA behaves as a quasi-three-level amplifier. When compared to the EDFA, the 1050 nm pump in the TDFA corresponds to a 980 nm pump in an EDFA, and the EDFA pumped at 1480 nm corresponds to a TDFA with its second pump at 1400 nm. In analogy to the EDFA, the TDFA pumped at 1050 nm is best for pre-amplifiers, as it has the lower PCE, but a population inversion level of about 100%, resulting in a low noise

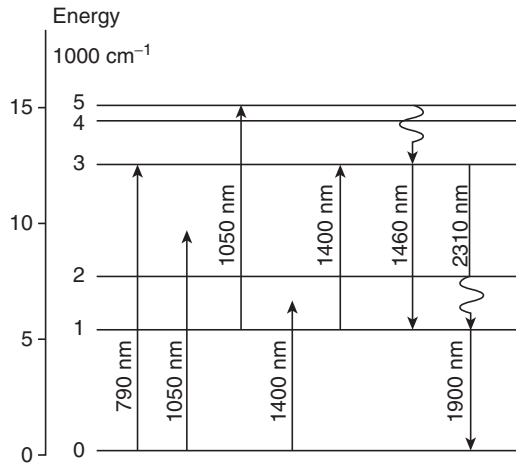


Figure 7.28 | Energy level diagram for thulium.

figure, whereas the TDFA pumped at 1400 nm is good for power amplifiers, as it has the higher PCE, but a lower inversion level, thus a higher noise figure.

7.7 System Applications

With increasing demand for systems with greater capacity and transparent routing, advances must be made in signal amplification and processing. Many of the important advances in optical fiber networks have been made possible by optical amplifiers. Since light is scattered over long distances while traveling through an optical fiber, optical amplifiers are primarily used to boost signals, though they also are used in other applications such as optical switching and wavelength conversion.^[15, 16]

Until recent developments in the fabrication and design of SOAs, fiber amplifiers had dominated conventional optical communication system applications such as in-line amplification used to compensate for fiber losses.

7.7.1 Wavelength Conversion

Due to carrier density changes, SOAs exhibit non-linear properties. These non-linear properties create problems for the use of SOAs as simple linear gain elements; however, these can be exploited to perform signal processing applications. Currently, all signal processing in WDM network nodes is accomplished by optical-to-electrical conversion and then carrying out the process electronically and again using electrical-to-optical conversion. By using SOA, O-E-O conversion is not required. Signal processing directly in the optical domain is possible; this reduces the cost drastically.

Integrated SOA-based devices are an important component of future all-optical systems because of their compact size, low optical and electrical power consumption, polarization independence, and high speed. Wavelength conversion is one of the important functionalities in the optical domain.

In broadband optical networks, these devices are used to avoid wavelength blocking in cross-connects in the WDM systems. In packet switching networks, they can be used to change the wavelength of certain signals so as to avoid packet contention and to reduce the need for optical buffering.

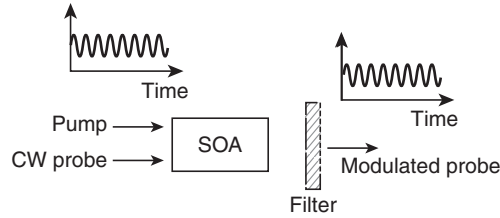


Figure 7.29 | Simple wavelength converter using SOA using cross-gain modulation.

The non-linear properties of SOAs can be exploited in the following ways for wavelength conversion:

1. Cross-gain modulation (XGM).
2. Cross-phase modulation (XPM).
3. Four-wave mixing (FWM).

The non-linear properties of an SOA can affect all the input signal levels. A strong signal at one wavelength can affect the gain of a weak signal at another wavelength. By using this property, wavelength conversion can be obtained by injecting a strong signal (pump) with a harmonic modulation at a certain angular frequency along with a weaker data signal (CW probe) into an SOA as shown in Fig. 7.29. Due to XGM, the stronger signal will force the weaker signal to its modulation. The result is a scheme by which the wavelength of a signal (CW probe) can be converted to that of another input signal (pump) with a single wavelength. This type of conversion is extremely efficient, at high bandwidth of the order of 10 GHz and bit rates up to 100 Gbit/s.

7.7.2 Optical Gates

Future high-speed WDM and time-division multiplexed (TDM) optical networks will require high speed all-optical logic gates that can be either optically or electronically controlled. These optical gates can be implemented using SOAs. The functionality of the gate can be controlled by turning ON or OFF the current to the SOA. SOAs can be used to form gate arrays due to their compact size and fitness for integration. A switching time of the order of 100 ps can be obtained from an SOA. All-optical logic gates can be used for signal processing applications in high-speed networks.

Figure 7.30 shows the NOR gate created by sending three signals into an SOA: two data signals A and B, and another probe signal (λ_s) to modulate the incoming signal. When both the data signals are data-low and the modulating signal (probe) is at the frequency of the signal constructed between the two data signals, the output signal will be at the wavelength of the probe signal. For all other data combinations, the output signal will not be at the selected wavelength.



Figure 7.30 | SOA NOR gate.

7.7.3 Multiplexers

In optical time-division multiplexed (OTDM) network nodes, components such as optical time-division demultiplexers (OTDDMs) and add/drop multiplexers (ADM) are required. These components work by dropping one channel from an incoming time-division data stream, while the other channels remain unchanged. In WDM networks, this ability to add and drop wavelength channels is useful for wavelength routing. A wavelength ADM separates a particular wavelength channel without interference from adjacent channels. A wavelength demultiplexer or an integrated SOA with a tunable filter can separate a particular channel. The filter can be tuned to reflect a particular wavelength channel. A new wavelength channel is added to replace the filtered or dropped channel without affecting the remaining channels.

7.7.4 Wavelength Selection and Spectral Inversion

A wavelength selector can be created using SOA gates positioned between two phased array wavelength demultiplexers. Using this scheme, nanosecond-scale wavelength selection can be achieved. The spectral inversion effect is used as a dispersion compensation technique. It can be implemented using optical phase conjugation (OPC) in a SOA structure in high-bit-rate transmission systems using standard single-mode fibers.

7.7.5 SOAs in CWDM Systems

The International Telecommunication Union (ITU) has defined a CWDM standard (G.694.2). CWDM is the best technology for the metro and enterprise layers of the network at low cost, giving connection flexibility and increased throughput. It covers the entire low-loss transmission window of a standard single-mode optical fiber (1260 nm to 1620 nm). CWDM systems (>100 km) require a low-cost optical amplifier operating across the entire optical bandwidth to extend the capacity and distance. CWDM EDFAs can operate in the C-band and L-band and therefore can cover not more than two channels of CWDM at a time. EDFAs cannot amplify channels in 1300 nm band. Therefore, SOA is the only viable technology that can be used to meet these expanding applications between 1260 nm and 1620 nm.

Solved Problems

PROBLEM 1

An FPA has 32% facet reflectivities and a single-pass gain of 4.8 dB. The amplifier has a 280 μm long active region, a mode spacing of 1 nm, and a peak gain of 1500 nm. Determine the refractive index of active medium.

Solution: The refractive index of the active medium of the FPA at peak gain wavelength can be obtained by using Eq. (7.1). It is given by

$$\delta\lambda = \frac{\lambda^2}{2nL}$$

$$\Rightarrow n = \frac{\lambda^2}{2\delta\lambda L} = \frac{(1500 \times 10^{-9})^2}{2 \times 1 \times 10^{-9} \times 280 \times 10^{-6}} = 4.0$$

PROBLEM 2

An FPA has facet reflectivities of 30% and a single-pass gain of 4.8 dB. The device has an active region with a refractive index of 3.75, a peak gain wavelength of 1500 nm with a spectral bandwidth of 240 GHz. Determine the length of active region and mode spacing.

Solution: The 3-dB spectral bandwidth of SOA is expressed as

$$B_{\text{FPA}} = \frac{c}{\pi n L} \sin^{-1} \left[\frac{1 - \sqrt{R_1 R_2 G_s}}{2(\sqrt{R_1 R_2 G_s})^{1/2}} \right]$$

$$\Rightarrow L = \frac{c}{\pi n B_{\text{FPA}}} \sin^{-1} \left[\frac{1 - \sqrt{R_1 R_2 G_s}}{2(\sqrt{R_1 R_2 G_s})^{1/2}} \right]$$

Now gain at 4.8 dB = 3.02. Therefore

$$B_{\text{FPA}} = \frac{3 \times 10^8}{\pi \times 3.75 \times 240 \times 10^9} \sin^{-1} \left[\frac{1 - \sqrt{0.3 \times 0.3 \times 3.02}}{2(\sqrt{0.3 \times 0.3 \times 3.02})^{1/2}} \right] = 300 \text{ } \mu\text{m}$$

Mode spacing can be calculated as follows:

$$\delta\lambda = \frac{\lambda^2}{2nL} = \frac{(1500 \times 10^{-9})^2}{2 \times 3.75 \times 300 \times 10^{-6}} = 1 \text{ nm}$$

PROBLEM 3

For an EDFA to be used as a power amplifier, what should be the minimum pump power required to obtain 12 dBm output at 1540 nm, if the amplifier input is 0 dBm and the pump wavelength is 980 nm.

Solution: We have

$$P_{\text{signal out}} \leq P_{\text{signal in}} + \frac{\lambda_p}{\lambda_s} P_{\text{pump in}}$$

where $P_{\text{pump in}}$ is the input pump power and λ_p and λ_s are the pump and signal wavelengths, respectively. Then

$$P_{\text{pump in}} \geq \frac{\lambda_s}{\lambda_p} (P_{\text{signal out}} - P_{\text{signal in}})$$

$$12 \text{ dBm} = 15.84 \text{ mW}$$

$$\Rightarrow P_{\text{pump in}} = \frac{1540}{980} (15.84 \text{ mW} - 1 \text{ mW}) = 23.32 \text{ mW}$$

PROBLEM 4

An EDFA is pumped at 980 nm with a 25 mW pump power. If the gain at 1550 nm is 20 dB, then find the maximum input and output power.

Solution: Minimum input power

$$P_{\text{signal in}} \leq \frac{(\lambda_p/\lambda_s)P_{\text{pump input}}}{G-1}$$

$$\Rightarrow P_{\text{signal in}} \leq \frac{(980/1550)(25 \text{ mW})}{100-1} = 160 \text{ } \mu\text{W}$$

Maximum output power is

$$P_{\text{signal out}} = P_{\text{signal in}} + \frac{\lambda_p}{\lambda_s} P_{\text{pump in}}$$

$$P_{\text{signal out}} = 160 \text{ } \mu\text{W} + \frac{980}{1550}(25 \text{ mW}) = 15.96 \text{ mW}$$

PROBLEM 5

A semiconductor optical amplifier has facet reflectivities of 30% and a single-pass gain of 5 dB. The device has an active region of length 350 μm , a mode spacing of 1 nm, peak gain wavelength is 1550 nm. Calculate the refractive index of active region.

Solution: The refractive index of the active medium of SOA at peak gain wavelength can be obtained by using Eq. (7.1). We have

$$\delta\lambda = \frac{\lambda^2}{2nL}$$

$$\Rightarrow n = \frac{\lambda^2}{2\delta\lambda L} = \frac{(1500 \times 10^{-9})^2}{2 \times 1 \times 10^{-9} \times 350 \times 10^{-6}} = 3.21$$

PROBLEM 6

A semiconductor optical amplifier has facet reflectivities of 30% and a single-pass gain of 5 dB. The device has an active region of length 350 μm , a mode spacing of 1 nm, peak gain wavelength is 1550 nm, and the refractive index of active region is 3.75. Calculate the spectral bandwidth of amplifier.

Solution: The 3-dB spectral bandwidth of the SOA is expressed as

$$B_{\text{FPA}} = \frac{c}{\pi n L} \sin^{-1} \left[\frac{1 - \sqrt{R_1 R_2 G_s}}{2(\sqrt{R_1 R_2 G_s})^{1/2}} \right]$$

Now

$$\text{Gain 5 dB} = 3.16$$

Therefore

$$B_{\text{FPA}} = \frac{3 \times 10^8}{\pi \times 3.75 \times 350 \times 10^{-6}} \sin^{-1} \left[\frac{1 - \sqrt{0.30 \times 0.30 \times 3.16}}{2 \times (\sqrt{0.30 \times 0.30 \times 3.16})^{1/2}} \right]$$

$$\Rightarrow B_{\text{FPA}} = 1.11 \times 10^{11} \text{ Hz}$$

SUMMARY

- The repeater in optical communication link
 - Receives the signal.
 - Converts the received optical signal to electrical signal.
 - Re-clocks it.
 - Re-shapes it.
 - Amplifies it.
 - Converts it back to an optical signal.
- An optical amplifier is more advantageous as compared to repeater because
 - It is insensitive to bit rate or signal format, hence system upgradation is easily possible.
 - It has fairly large gain bandwidth, thus it can amplify several WDM signals simultaneously.
- Optical amplifiers can be used as
 - In-line amplifiers.
 - Pre-amplifiers.
 - Power amplifiers.
- There are two main categories of optical amplifier:
 - Cavity amplifier/semiconductor optical amplifier (SOA).
 - Fiber amplifier.
- EDFAs are preferred in today's telecommunication systems.
- The dominant noise generated in an EDFA is amplified spontaneous emission (ASE).
- The optical signal-to-noise ratio (OSNR) is defined as the ratio of the EDFA's optical signal output power P_{out} to the unpolarized ASE optical noise power P_{ASE} .

MULTIPLE-CHOICE QUESTIONS

- How is the job of a repeater different from that of amplifier?
 - They are both the same.
 - Repeater receives the signal, converts the signal in an electrical signal, re-clocks and re-shapes it, amplifies it, converts it back to an optical signal before coupling the signal back in the optical fiber.
 - Repeater only re-clocks and reshapes the signal, while amplifier amplifies the signal.
 - None of these.
- What is the main difference between Fabry–Perot amplifiers (FPAs) and traveling wave amplifiers (TWAs)?
 - Facet reflectivity.
 - Cavity length.
 - Gain medium.
 - No difference.
- Which is the operating window of an erbium-doped fiber amplifier?
 - Low dispersion window around 1300 nm.
 - Low attenuation window around 1550 nm.
 - Both windows.
 - None of these.
- Which wavelength is most suitable for pumping an erbium-doped fiber amplifier?
 - 850 nm
 - 980 nm
 - 1300 nm
 - 1550 nm
- What is the similarity between an erbium-doped fiber amplifier and a fiber Raman amplifier?
 - Both of them operate in the all-optical domain.
 - Both of them operate around 1550 nm.
 - Both of them can be used for multichannel operation.
 - All of the above.
- What is the difference between erbium-doped fiber amplifier and fiber Raman amplifier?
 - EDFA requires population inversion, while FRA does not require it.
 - EDFA operates on the principle of stimulated emission.
 - FRA operates on the principle of stimulated Raman scattering.
 - There is no difference.
- Optical amplifiers can be used as
 - In-line amplifiers to compensate for loss.

- (b) Power amplifiers to follow the transmitter.
 - (c) Pre-amplifiers to precede the receiver.
 - (d) All of the above.
8. Which of the following optical amplifiers is most suited for multichannel bidirectional operation?
 - (a) Semiconductor optical amplifier.
 - (b) Erbium-doped fiber amplifier.
 - (c) Fiber Raman amplifier.
 - (d) None of these
 9. EDFA gain depends on which of the following factors?
 - (a) Pump power
 - (b) Doping concentration
 - (c) Length of fiber
 - (d) All of these
 10. Thulium-doped fiber amplifiers work in which band?
 - (a) C-band
 - (b) S-band
 - (c) L-band
 - (d) U-band

REVIEW QUESTIONS

1. With the help of neat sketch explain how the job of repeater in an optical communication system is different from that of an amplifier.
2. Explain the different possible applications of optical amplifier in a practical fiber optic communication system.
3. Which are the different types of optical semiconductor amplifiers? Compare them.
4. Explain the basic principle of operation of semiconductor optical amplifier.
5. What is Raman scattering? How is it useful in optical signal amplification?
6. Explain the pumping schemes for Raman amplifier.
7. With the help of energy level diagram, explain the operating principle of optical fiber amplifier. Enlist the benefits of fiber amplifier technology over semiconductor amplifier technology.
8. With the help of neat sketch describe EDFA architectures.
9. Write a note on S-band amplifier.
10. Distinguish between the amplification processes in
 - (a) An erbium-doped fiber amplifier.
 - (b) A fiber Raman amplifier.

EXERCISES

1. An FPA has facet reflectivities of 30% and a single-pass gain of 5.0 dB. The amplifier has a 300 μm long active region, a mode spacing of 1 nm and a peak gain wavelength of 1500 nm. Determine the 3 dB spectral bandwidth of device.
2. An EDFA power amplifier produces $P_{\text{signal out}} = 27 \text{ dBm}$ for an input signal of 2 dBm at 1540 nm. Find the amplifier gain and minimum pump power required.
3. Determine the refractive index of the active medium in an FPA that has 30% facet reflectivities and a single-pass gain of 5.1 dB. The amplifier has a 340 μm long active region, a mode spacing of 1.2 nm, and a peak gain of 1350 nm.
4. Calculate the minimum pump power required in order to obtain a 15 dBm output at 1550 nm for an EDFA power amplifier, if the amplifier input is 0 dBm and the pump wavelength is 980 nm.
5. An EDFA is pumped at 980 nm with a 30 mW pump power. If the gain at 1550 nm is 25 dB, then find the maximum input and output power.
6. Calculate the refractive index of active region of a semiconductor optical amplifier having facet reflectivities of 30% and a single-pass gain of 5 dB. The device has an active region of length 325 m, a mode spacing of 1.3 nm, and a peak gain wavelength of 1550 nm.

7. Calculate the spectral bandwidth of a semiconductor optical amplifier that has facet reflectivities of 35% and a single pass gain of 4 dB. The device has an active region of length 400 μm , a mode spacing of 1 nm, peak gain wavelength is 1550 nm, and the refractive index of active region is 3.75.

ANSWERS

Multiple-Choice Questions

- | | |
|--------|---------|
| 1. (b) | 6. (a) |
| 2. (a) | 7. (d) |
| 3. (b) | 8. (b) |
| 4. (b) | 9. (d) |
| 5. (d) | 10. (b) |

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8

Basic Instrumentation for Optical Measurements

Learning Objectives

After completing this chapter, you will be able to:

- Study measurement setups for attenuation, dispersion, optical power, numerical aperture (NA), and bandwidth in optical communication systems.
 - Study the important instruments involved in various optical measurements, such as optical spectrum analyzer (OSA), optical power meter, and optical wavelength meter.
 - Study the troubleshooting instrument called optical time-domain reflectometer (OTDR).
-

The basic measurements for any optical fiber communication system are optical power, attenuation, wavelength, numerical aperture (NA), bandwidth, data rate, and dispersion. The measurements may be performed by design engineers at laboratory level for testing the performance of the system, or during installation of the system in the field or in troubleshooting the already installed fiber optic system. The important instruments used for these measurements are optical power meter, optical time-domain reflectometer (OTDR), optical spectrum analyzer (OSA), wave meters, etc. It is preferable to make measurements without interrupting the communication signal.

8.1 Attenuation Measurement

Typically, the following three methods are used for attenuation measurement in optical fiber:

1. Cutback method.
2. Insertion loss method.
3. Optical time-domain reflectometer method.

8.1.1 Cutback Method

This is the most common technique used for the measurement of total attenuation per unit length in the optical fiber. The measurement setup is shown in Fig. 8.1. It consists of the following:

1. A white light source (Xenon arc or tungsten halogen lamp).
2. A monochromator (prism or diffraction grating) to select a particular wavelength.
3. A mode stripper to remove the light launched in the fiber cladding.
4. A test fiber.
5. A photodetector.

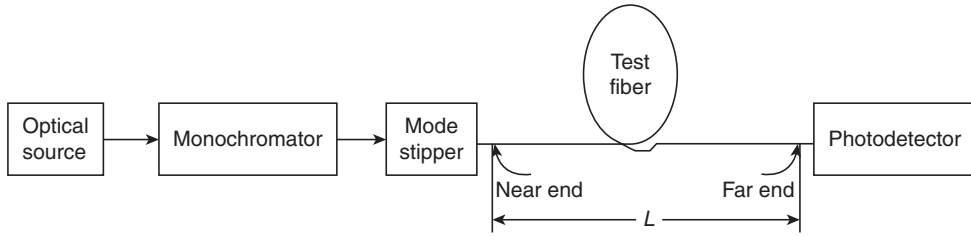


Figure 8.1 | Experimental setup for fiber attenuation measurement using cutback method.

8.1.1.1 Measurement of Attenuation Constant

The optical power (P_{farend}) and the corresponding voltage (V_{farend}) are measured at the output or the far end of the fiber of length L_1 . After this, the long fiber is cut to a short length ($L_2 = \text{a few meters}$) from the source and the output power is measured as P_{nearend} . The corresponding voltage is V_{nearend} .^[1] Here, the loss per unit length is given as

$$\alpha(\text{dB/km}) = \frac{10 \log P_{\text{nearend}} - 10 \log P_{\text{farend}}}{L} = \frac{10}{L_1 - L_2} \log_{10} \left(\frac{P_{\text{nearend}}}{P_{\text{farend}}} \right) \quad (8.1)$$

Also,

$$\alpha(\text{dB/km}) = \frac{20 \log V_{\text{nearend}} - 20 \log V_{\text{farend}}}{L} = \frac{20}{L_1 - L_2} \log_{10} \left(\frac{V_{\text{nearend}}}{V_{\text{farend}}} \right) \quad (8.2)$$

Since the fiber needs to be cut for measurement, this method is also called the *destructive method*.

8.1.2 Insertion Loss Method

The cutback method cannot be used for cables with connectors. In this case, the insertion loss technique is used. The basic setup for the measurement is shown in Fig. 8.2(a). The wavelength tunable optical source and photodetectors are coupled to a short-length fiber through an optical connector as shown in Fig. 8.2(b). The launching system includes the following:

1. A *mode scrambler* is used to ensure an equilibrium-mode distribution in a multimode fiber.
2. A *cladding mode stripper* is used to allow fundamental-mode propagation in single-mode fiber.
3. A *filter* is used for selection of a particular source wavelength.

8.1.2.1 Measurement of Attenuation Constant

Connector *A* of the short-length launching fiber is attached to connector *B* of the photodetector as shown in Fig. 8.2(a), and power $P_1(\lambda)$ is measured. The optical fiber to be tested is then connected between the power transmitting and receiving systems. The received power is recorded as $P_2(\lambda)$. The attenuation of the fiber is calculated as follows:

$$\alpha = 10 \log \frac{P_1(\lambda)}{P_2(\lambda)} \quad (8.3)$$

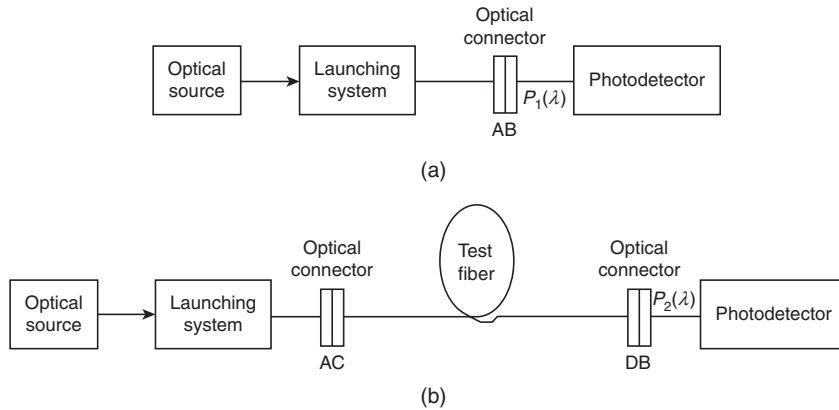


Figure 8.2 | Experimental setup for fiber attenuation measurement using insertion loss methods: (a) Reference measurement and (b) fiber attenuation measurement.

It gives the total attenuation, that is, fiber attenuation plus connector attenuation.

8.1.3 Optical Time-Domain Reflectometer (OTDR) Method

This technique is very useful for the field testing of an installed optical fiber link, because it requires access at only one end of the fiber, as shown in Fig. 8.3. In the laboratory, both ends of the fiber are easily accessible, and so cutback or insertion loss technique can be employed. However, in a field test, only one end of the link is easily available. Another advantage of this system is that the fiber need not be cut; therefore, it is also known as the *non-destructive method*. It works on the principle of measurement of backscattered light rather than transmitted light.

OTDR works on the principle of Rayleigh scattering and Fresnel reflection to measure the attenuation in the fiber. It gives a characteristic trace or profile of length of fiber versus back-reflected signal level on its display screen by launching a sharp optical pulse into the fiber and measuring the travel time and the back-reflected

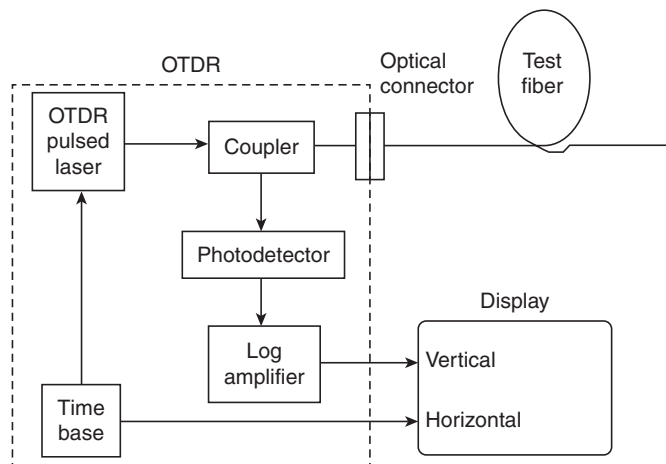


Figure 8.3 | Experimental setup for fiber attenuation measurement using OTDR.

signal strength. We have already studied Rayleigh scattering in detail in Section 3.2.3.1. When a light pulse is sent through the fiber, it strikes the dopant particles in glass and gets scattered in all directions. Very small part of light (approximately 0.0001%) gets scattered in backward direction. It is given as ^[2, 3]

$$P_{\text{backscattered}}(t) = \frac{1}{2} P_{\text{incident}} S \Upsilon_{\text{R}} \tau v_g e^{(-\alpha v_g t)} \quad (8.4)$$

where P_{incident} is a optical power launched into the fiber, S is the fraction of captured optical power, Υ_{R} is the Rayleigh scattering coefficient, τ is the incident optical pulse width, v_g is the group velocity in the fiber, and α is the attenuation coefficient per unit length of fiber. The fraction of captured power S is

$$S = \frac{(\text{NA})^2}{4n_1^2} \quad (8.5)$$

where n_1 is the refractive index of core.

When a light enters from one medium to another medium of a different refractive index, Fresnel reflection (refer to Section 2.5) occurs. When light of power P_{incident} is perpendicular to the glass–air interface, the reflected power $P_{\text{reflected}}$ is given by

$$P_{\text{reflected}} = P_{\text{incident}} \left(\frac{n_{\text{fiber}} - n_{\text{air}}}{n_{\text{fiber}} + n_{\text{air}}} \right)^2 \quad (8.6)$$

where n_{fiber} is the refractive index of fiber core and n_{air} is the refractive index of air.

8.1.3.1 Working of OTDR

An OTDR consists of the following:

1. Laser light source.
2. Coupler or splitter.
3. Photodetector.
4. Display section.
5. Log amplifier.

The laser diode launches sharp optical pulses into the test fiber through a coupler. Due to Rayleigh backscattering, which occurs along the whole length of fiber, some part of light pulses is reflected back from the other end of the fiber. Due to fiber losses, the returned radiation declines with constant slope.

The coupler has three ports. It connects the laser output to the test fiber and the backscattered signal from test fiber to the photodetector. Light cannot go directly from source to the detector. The photodetector converts the backscattered optical power into corresponding electrical level. This section is designed such that it can detect extremely small levels of backscattered light. The logarithmic amplifier further boosts the electrical signal level.^[4]

Figure 8.4 shows the typical OTDR display screen. The vertical axis measures the back-reflected light in dB, and its scale is logarithmic. The horizontal axis gives the distance between the instrument and the measurement point in the fiber. The time base unit gives a synchronizing signal to the laser as well as the horizontal section of the display unit. The loss per unit length is determined by measuring the slope of the pattern on the screen.

The fault location, that is, fiber length L from front end of fiber, is calculated from the time axis on the display screen. Time t on display screen represents the time taken by the pulse to reach the fault location and then return to the front end of the fiber. Therefore, L can be calculated as follows:

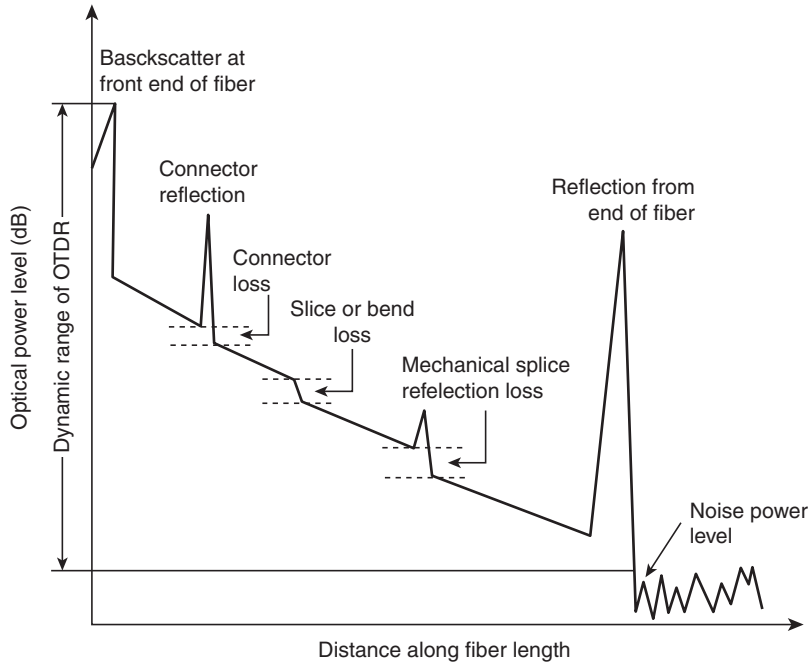


Figure 8.4 | OTDR plot.

$$L = \frac{ct}{2n_1} \quad (8.7)$$

where n_1 is the refractive index of core and c is velocity of light in free space.

8.1.3.2 Uses of OTDR

OTDR is used in the following manners:

1. To measure:
 - (a) Length of fiber.
 - (b) Overall attenuation in fiber.
 - (c) Connector and splicing losses.
 - (d) Optical return loss.
2. To locate faults such as breaks.
3. To get evidence of macrobending or microbending.

8.1.3.3 Performance Parameters of OTDR

8.1.3.3.1 Dynamic Range

This range decides the maximum length of fiber over which various measurements can be taken. The dynamic range of the instrument is decided by the total pulse power and the sensitivity of the photodetector. A powerful

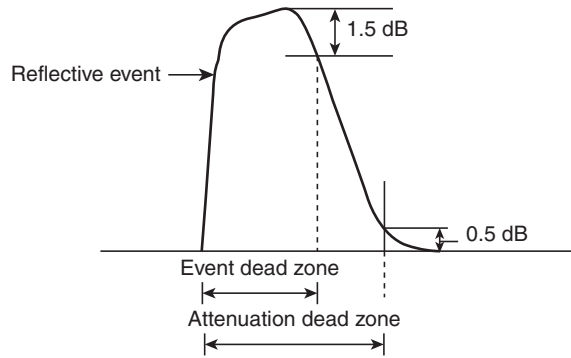


Figure 8.5 | OTDR dead zone.

source and a sensitive sensor give large dynamic range, whereas a weak source and a low-sensitivity sensor give smaller dynamic range. The pulse power can be increased by increasing the amount of light emitted or increasing the pulse width. The output power cannot be increased beyond the maximum output level. If the pulse width is increased, the dead zone also increases.

As shown in Fig. 8.5, the dynamic range is calculated by taking the difference between backscatter at the front end of fiber and noise floor power.

8.1.3.3.2 Dead Zone

It is the distance over which the high reflection covers up the lower level of backscatter. When the OTDR measures the strong reflection (Fresnel reflection), it becomes momentarily blind. There are two categories of dead zone as shown in Fig. 8.5.

1. **Event dead zone:** It is the distance following the reflective event that is needed by the photodetector to be able to detect a next reflective event (-1.5 dB).
2. **Attenuation dead zone:** It is the distance following the reflective event that is needed by the photodetector to be able to detect a splice (0.5 dB).

The typical specifications and performance parameters of commercial OTDR are given in Table 8.1 as follows.

Table 8.1 | Typical specifications of OTDR

Item	Specifications
Wavelength range	(1550 ± 25) nm
Applicable fiber	SM (ITU-T G.652)
Distance range	500 m, 1 km, 2 km, 10 km, 50 km, 100 km, 400 km
Pulse width	3 ns, 10 ns, 50 ns, 100 ns, 500 ns, 1 μ s, 5 μ s, 20 μ s
Dynamic range	30 dB
Event dead zone	0.8 m
Attenuation dead zone (Typ)	10 m

8.2 Optical Power Measurement

The energy associated with a photon is given by Planck's Law, stated mathematically as follows:

$$E = \frac{hc}{\lambda}$$

where h is Planck's constant, c is the velocity of light in free space, and λ is the operating wavelength.

Optical power is a measure of the rate of arrival of a photon at a photodetector. In other words, it is a measure of energy transfer rate. Thus, *optical power* is a function of time as shown in Fig. 8.6; therefore, it is measured as average power or peak power and is expressed in watts or joules per second. *Peak power* is the maximum power level in a pulse, which varies with time. Average power is an average over a longer period of time. For the same data, the average power in return to zero (RZ) data format is half that of non-return to zero (NRZ) format.

Optical power meter consists of the following blocks (refer to Fig. 8.7):

1. **Photodiode:** Optical power meter measures the average power over a selected wavelength band. Therefore, the user should know the wavelengths of light to be measured. Different photodiode materials are suitable for different wavelength ranges, for example:^[5]
 - (a) Silicon photodiode: 280–1,050 nm.
 - (b) Germanium or indium–gallium arsenide photodiode: 800–1,700 nm.
2. **Transimpedance amplifier:** It is a precision amplifier with autogain ranging. It converts input photocurrent to amplified output voltage. A digital display with controller and A to D converter gives the power

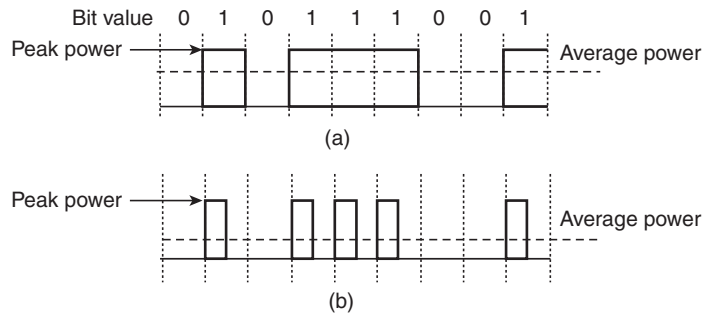


Figure 8.6 | Peak and average power in: (a) NRZ optical pulses and (b) RZ optical pulses as a function of time.

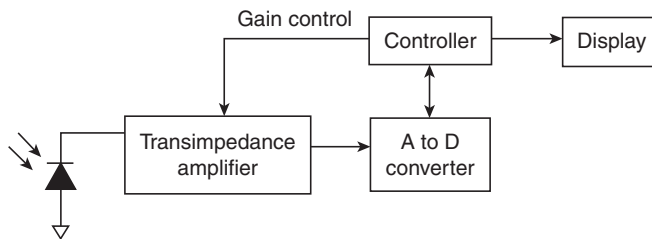


Figure 8.7 | Block diagram of optical power meter.

in linear units, such as milliwatts or nanowatts, or in logarithmic units such as dB or dBm. The relation between the linear power in watts (P) and logarithmic unit power in dBm (P_{dBm}) is given by

$$P_{\text{dBm}} = 10 \log_{10} \frac{P \text{ (W)}}{1 \text{ mW}} \quad (8.8)$$

The typical specifications of optical power meter are given in Table 8.2.

Table 8.2 | Typical specifications of optical power meter

Item	Specifications
Wavelength range	400–850 nm
Light-receiving element	Si-PIN Diode
Received light power range	1 μW to 100 mW
Maximum light-receiving level	100 mW
Maximum power density	5 mW/mm ²
Input type	Spatial light

8.3 Dispersion Measurement

In multimode fibers, each mode in an optical pulse travels at slightly different speed and thus reaches the fiber end at slightly offset times. This gives rise to the pulse broadening of light wave signal.^[6] There are many ways to measure dispersion effect. In this section, we will study intermodal dispersion measurement in time domain and frequency domain.

8.3.1 Pulse Dispersion Measurement in Time Domain

The test setup for pulse dispersion measurement in time domain is shown in Fig. 8.8. A narrow optical pulse (100–400 ps) from a suitable source such as AlGaAs injection laser is injected into one end of fiber through a mode scrambler. A mode scrambler is used to ensure an equilibrium-mode distribution in multimode fiber. The pulse travels through a test fiber of 1 km length. The pulse gets broadened due to various dispersion phenomena. A high-speed avalanche photodiode receives the pulses and gives them to sampling oscilloscope. The electrical pulse through the delay circuit is used for triggering the oscilloscope and for input pulse measurement. The output pulse width is measured and noted. Then the test fiber is cut to a short length of around 1% of its original length, and the same procedure is repeated to measure the input pulse width. The variable delay block is used to offset the delay difference between test fiber and short length reference fiber.

8.3.2 Pulse Dispersion Measurement in Frequency Domain

A narrow band continuous-wave optical signal from injection laser is sinusoidally modulated using RF source or a microwave signal. The modulated signal is launched into the test fiber through a mode scrambler to ensure the equilibrium-mode distribution as shown in Fig. 8.9. At the other end of test fiber, avalanche photodiode measures the output power $P_{\text{out}}(f)$ as a function of modulation frequency f . Then the test fiber is cut to a short length of around 1% of its original length, and the same procedure is repeated to measure the input power $P_{\text{in}}(f)$.^[7] The baseband frequency response of test fiber is given as

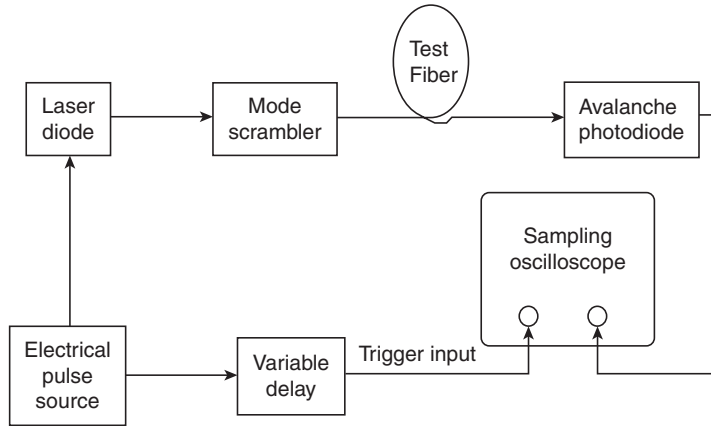


Figure 8.8 | Test setup for pulse dispersion measurement in time domain.

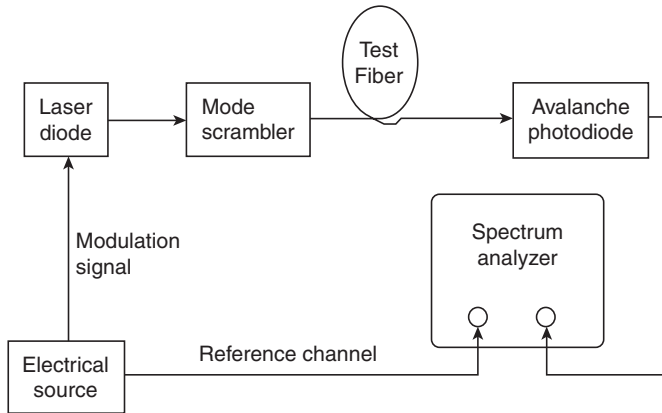


Figure 8.9 | Test setup for pulse dispersion measurement in frequency domain.

$$H(f) = \frac{P_{\text{out}}(f)}{P_{\text{in}}(f)} \quad (8.9)$$

In this method, we get an amplitude versus frequency curve, which is more useful than time-domain measurement from system design point of view.

8.4 Fiber Numerical Aperture Measurement

Numerical aperture (NA) is a measurement of the ability of an optical fiber to capture light. It specifies the angular range over which an optical fiber can accept light. The characteristics of a fiber – such as V number, number of modes propagating through fiber, etc. – depend on NA. In this section, we will study the different methods available for the measurement of NA, which are as follows:

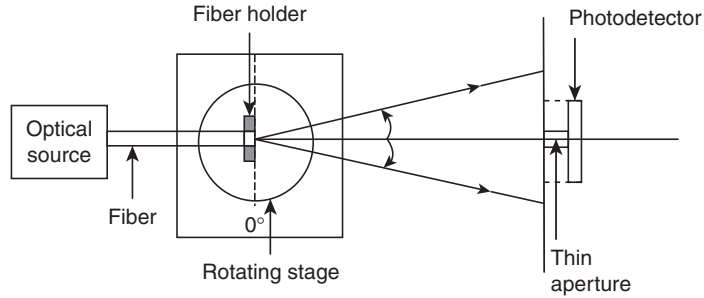


Figure 8.10 | Test setup for NA measurement using scanning photodetector and rotating stage method.

1. Scanning photodetector and rotating stage method.
2. Trigonometric method.

8.4.1 Scanning Photodetector and Rotating Stage Method

The test setup for NA measurement using scanning photodetector and rotating stage method is shown in Fig. 8.10. Optical source having emission wavelength of 850 nm is used to launch the light in an optical fiber of approximately 2 m length. The fiber end faces are properly prepared to ensure square smooth termination. A rotating arrangement is made to rotate the fiber in both directions from the center (0°). The fiber output end is positioned with fiber holder. A photodetector is placed at 10–20 cm from the fiber output end. The output powers at different angles of rotation are measured as a function of angle of rotation in both directions from 0° reference. A graph is plotted between sine of angle of rotation and the output optical power. The nature of such plot is shown in Fig. 8.11. The acceptance angle is maximum when the output power drops to 5% of its maximum value.^[8] The values corresponding to this power are marked as x_1 and x_2 on the graph to calculate the NA. Then

$$NA = \frac{x_1 + x_2}{2} \quad (8.10)$$

NA measurement using far field scanning measurement can also be performed by keeping the photodetector on rotating stage and the fiber end fixed to 0° reference.

8.4.2 Trigonometric Method

Trigonometric method is used for rough measurement of an NA. The setup for trigonometric NA measurement is shown in Fig. 8.12. The optical source launches the light in test fiber over the full range of NA. The fiber ends are prepared to ensure square smooth termination. A screen is positioned at distance B from fiber output end such that the far field pattern is displayed on it. The test fiber is then aligned to obtain maximum intensity of far field pattern. As shown in Fig. 8.12, the pattern size on the screen is measured as A with the help of vernier caliper. By using the simple trigonometric relation we can write

$$NA = \sin \theta = \frac{A/2}{\sqrt{(A/2)^2 + B^2}}$$

$$NA = \sin \theta = \frac{A}{\sqrt{(A)^2 + 4B^2}} \quad (8.11)$$

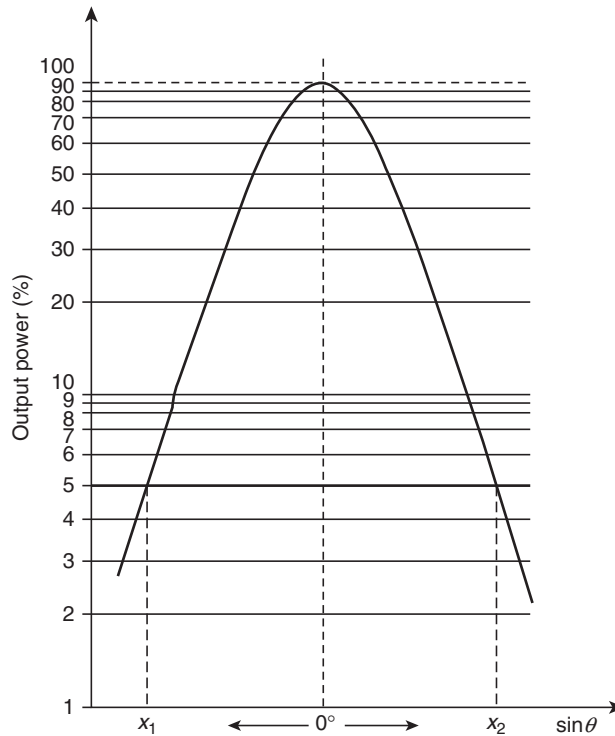


Figure 8.11 | Graph for numerical aperture measurement.

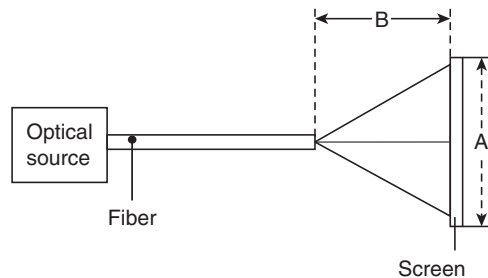


Figure 8.12 | Setup for trigonometric numerical aperture measurement.

8.5 Bandwidth and Data Rate

Bandwidth is a measure of the highest frequency the system can transmit. Figure 8.13 shows the optical loss of a fiber as a function of modulating frequency for analog fiber optic system. We can see that the optical loss is more at high frequencies. Bandwidth can be calculated from this curve. It is the point on a curve where the response has been dropped by 3 dB or half-power, that is, 5 GHz point. The system can transmit higher frequencies, but in this case the attenuation caused will be more.

Data rate gives the maximum speed of digital system in bits per second. Bit error rate measures the quality of digital signal.

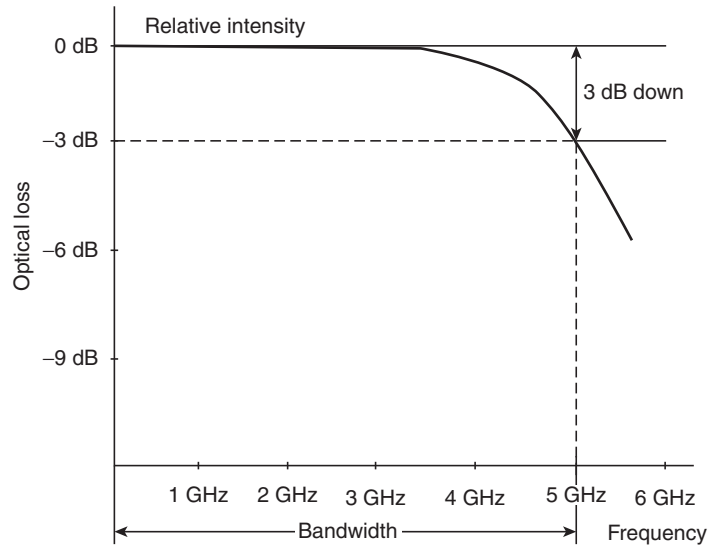


Figure 8.13 | Optical loss of a fiber as a function of modulating frequency for analog fiber optic system.

8.6 Optical Spectrum Analyzer

Optical spectrum analyzer (OSA) displays the optical power as a function of wavelength. OSAs can be divided into the following two categories:

1. Diffraction grating-based OSAs.
2. Interferometer-based architectures:
 - (a) Fabry-Perot interferometer-based OSAs.
 - (b) Michelson interferometer-based OSAs.

Refer Fig. 8.14 for the simplified block diagram of OSA in order to understand the basic operation. The input light from an optical fiber is passed through wavelength tunable optical filter. It may be a monochromator

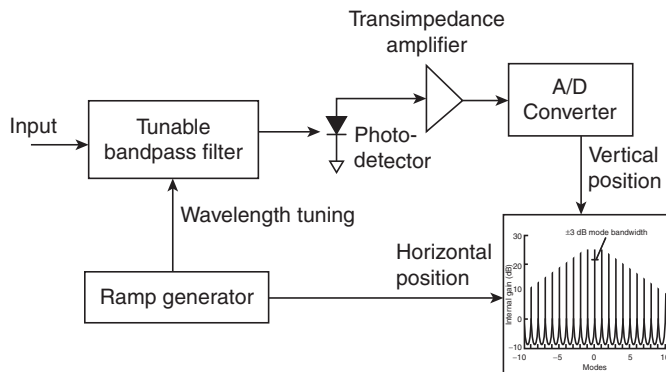


Figure 8.14 | Simplified block diagram of optical spectrum analyzer.

(diffraction grating) or an interferometer. The individual spectral components are resolved by this filter. The photodetector converts the optical signal to equivalent electrical current. The transimpedance amplifier converts the photocurrent to equivalent voltage. This voltage is converted to digital form by A to D converter. This digitized amplitude signal is applied to vertical section of display. The ramp generator gives the horizontal sweep signal to horizontal section of display and the tuning signal to the filter. The display shows optical power versus wavelength trace.^[9]

8.6.1 Diffraction Grating-Based OSA

The basic idea of diffraction-based tunable filtering is shown in Fig. 8.16. The input light from an optical fiber is focused onto a diffraction grating as shown in Fig. 8.15. The diffraction grating spreads out the wavelength spectrum. A diffraction grating is a mirror having extremely narrow grooves on its surface. The spacing between the grooves is equal to the wavelength of interest. When the light beam strikes the grating, it gets reflected in a number of directions.

1. The first reflection is $m = 0$, that is, zero-order beam. The direction of this beam is same as the reflection from plane mirror. It is not separated into different wavelengths.
2. $m = 1$ beam results from constructive interference of reflections having path-length difference equal to one wavelength from adjacent grooves. The reflection angle of each wavelength is different, and thus the light beam is separated into different wavelengths.
3. $m = 2$ beam results from constructive interference of reflections having path-length difference equal to two wavelengths from adjacent grooves.
4. $m = 3$ beam results from constructive interference of reflections having path-length difference equal to three wavelengths from adjacent grooves, and so on.

To cover the entire wavelength range, OSA utilizes multiple-order beams.

The diffracted spectrum is focused onto a flat surface having a narrow aperture. It transmits the narrow band of selected wavelengths toward a photodetector to measure power at that particular wavelength. To measure power at different wavelengths the diffraction grating is tilted, so different wavelengths pass through the aperture. Thus the entire wavelength spectrum can be scanned by slow and continuous tilting of grating. The resolution of this OSA is variable, typically ranging from 0.1 nm to 5–10 nm.

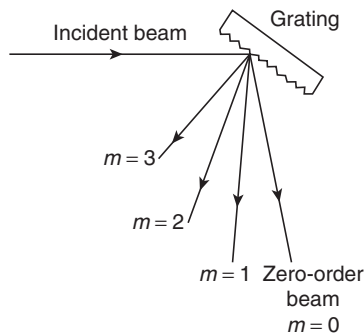


Figure 8.15 | Schematic representation of grating operation (wavelength separation).

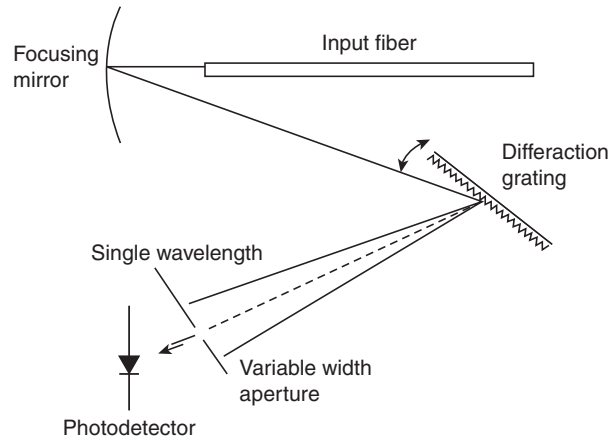


Figure 8.16 | Schematic representation of tunable wavelength filtering in diffraction grating-based OSA.

8.6.2 Interferometer-Based Architectures

8.6.2.1 Fabry–Perot Interferometer-Based OSA

In this OSA, the incoming light is filtered by Fabry–Perot resonant cavity. The cavity consists of two highly reflective parallel mirrors as shown in Fig. 8.17. The reflection coefficient and spacing between the mirrors decide the resolution of OSA. By varying the distance between the mirrors, wavelength can be varied. Thus it has a fixed, narrow resolution, typically 100 MHz–10 GHz. Due to narrow resolution, measurement span is much more limited than the diffraction grating-based OSAs.

8.6.2.2 Michelson Interferometer-Based OSA

The Michelson interferometer shown in Fig. 8.18 consists of a beam splitter and two mirrors. The beam splitter splits the incoming beam into two equal portions. The first half is transmitted through the beam splitter and the other half is reflected toward the other arm of interferometer. The mirror on horizontal arm is movable and hence its length is variable. The other arm on lower side is fixed. The mirror fixed to lower arm reflects the beam toward beam splitter. One-half of this will travel toward the detector and other half gets reflected toward the source.

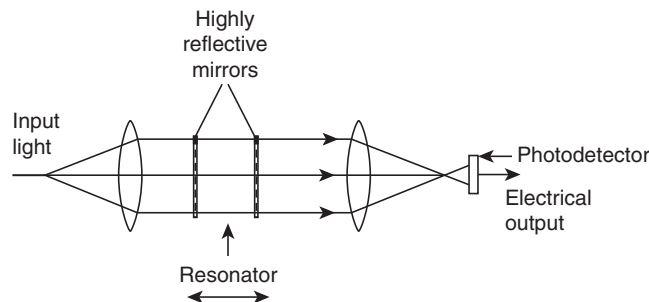


Figure 8.17 | Schematic representation of tunable wavelength filtering in Fabry–Perot interferometer-based OSA.

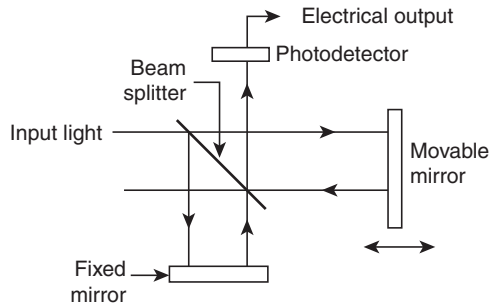


Figure 8.18 | Schematic representation of tunable wavelength filtering in Michelson interferometer-based OSA.

While the mirror fixed to horizontal arm also reflects the beam toward beam splitter, one-half will travel toward the source and other half will get reflected toward the detector.

The amplitudes of optical beams reaching the detector are the same. Thus, the two beams can interfere. The interference pattern depends on the path-length difference which in turn depends on the wavelength. Thus, by changing the path length by moving the mirror, different wavelengths can be selected for measurement of optical power.

Michelson interferometer-based OSAs are used for direct coherence-length measurements. The spectrum is displayed by calculating the Fourier transform of a measured interference pattern. The typical specifications of commercial OSA are given in Table 8.3.

Table 8.3 | Typical important specifications of OSA

Item	Specifications
Wavelength range	600–1,700 nm
Span	0.2–1,100 nm
Wavelength accuracy	±0.1 nm
Wavelength resolution	0.02 nm
Dynamic range	80 dB
Applicable to	Single-mode and multimode fiber

8.7 Wavelength Meter

Michelson interferometer is used for wavelength measurement. The construction is as shown in Fig. 8.18. The light beams reflected from the mirrors and reaching the detector have the same amplitude. Thus, the two beams can interfere constructively or destructively. Initially, the moving mirror is adjusted such that the two interfering beams cancel out at the detector. Then the mirror is moved slowly by known distance D . Thereafter, the signal amplitude at detector is monitored for dark–light–dark transaction cycles N (dark: cancel out the beam, light: addition of beam amplitude). The wavelength can be calculated by using following equation:

$$\lambda = \frac{2D}{N} \quad (8.12)$$

Note that the distance of the arm is multiplied by 2 because the beam makes a round trip through the arm.

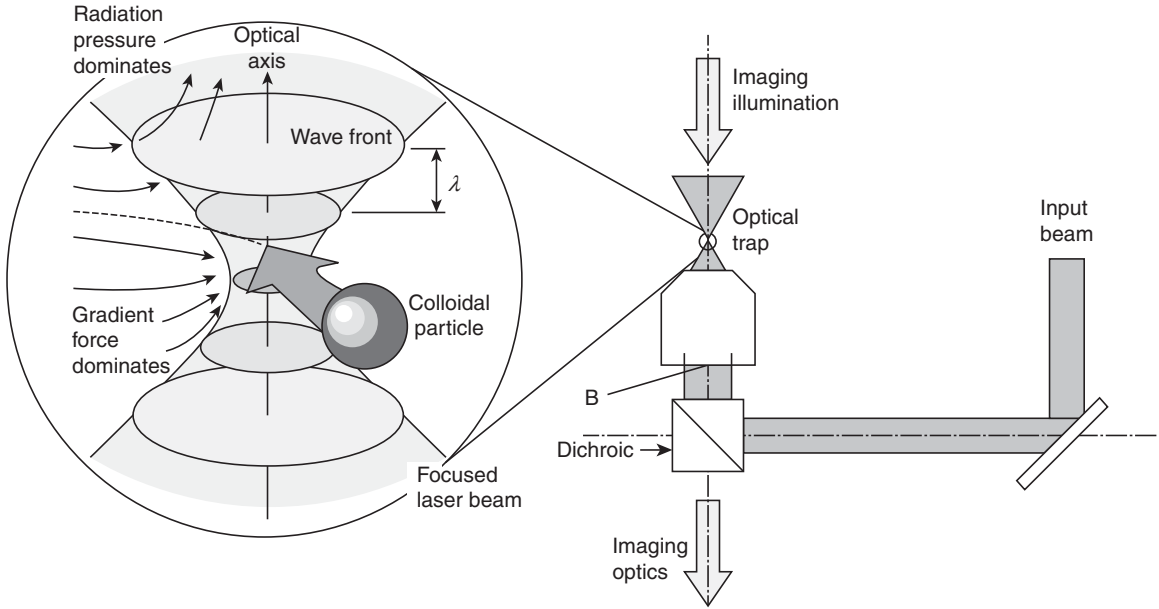


Figure 8.19 | Optical tweezers principle.

8.8 Optical Tweezers

Optical tweezers are scientific instruments that use laser light pressure to trap the microscopic objects as small as single atom. First optical tweezers were demonstrated by A. Ashkin et al. in 1986.^[10]

The laser beam (typically Nd-YAG) is focused at a spot by a high-quality microscope objective lens. This small laser spot acts as an optical trap as shown in Fig. 8.19. The necessary condition for trapping of object is: the refractive index of object to be trapped should be more than the refractive index of surrounding. Commonly

$$\frac{n_{\text{object}}}{n_{\text{surrounding}}} \approx 1.1 \text{ to } 1.2$$

The optical trap holds the object at its center. The light at the center of the beam is brighter than the light at the edges. The light rays are bent according the laws of reflection and refraction. Two types of forces are acting on the object due to laser light (see Fig. 8.20):

1. **Scattering force:** It is due to reflection in the direction of laser propagation.
2. **Gradient force:** It is due to refraction directed toward laser focal point.

The net force holds the object at the narrowest point of beam that is called *beam waist*.

8.8.1 Applications

1. Tracking of bacteria movement.
2. Cell organization.
3. To trap viruses, bacteria, living cells, small metal particles, and DNA strands.
4. Measurement of forces.

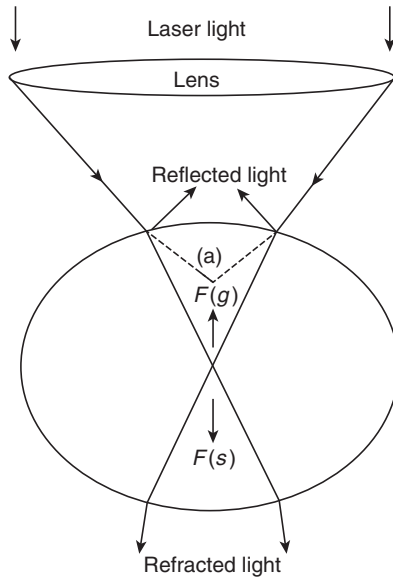


Figure 8.20 | Forces on dielectric sphere due to reflection and refraction of light.

Solved Problems

PROBLEM 1

In a fiber attenuation measurement using cutback method following data is obtained: length of fiber is 2 km and the corresponding output power is $60 \mu\text{W}$; length of cutback fiber is 2 m and the corresponding output power is $400 \mu\text{W}$. Determine the attenuation per kilometer at $1,100 \text{ nm}$.

Solution: $L_1 = 2 \text{ km}$, $L_2 = 2 \text{ m}$, $P_{\text{nearend}} = 400 \mu\text{W}$, $P_{\text{farend}} = 60 \mu\text{W}$. Then the attenuation per kilometer is given by

$$\begin{aligned}\alpha(\text{dB/km}) &= \frac{10 \log P_{\text{nearend}} - 10 \log P_{\text{farend}}}{L} \\ &= \frac{10}{L_1 - L_2} \log_{10} \left(\frac{P_{\text{nearend}}}{P_{\text{farend}}} \right) \\ \Rightarrow \alpha(\text{dB/km}) &= \frac{10}{2 - 0.002} \log_{10} \left(\frac{400}{60} \right) \\ &= 4.12 \text{ dB/km}\end{aligned}$$

PROBLEM 2

In a trigonometric method of NA measurement, the screen is placed 10 cm from the fiber end face. The fiber illuminated from visible source gives the output pattern of size 9 cm . Calculate the NA of fiber.

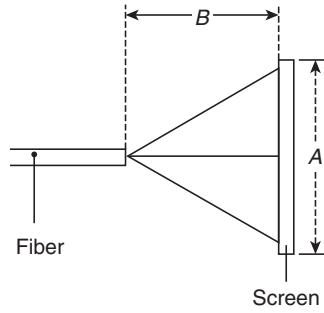


Figure 8.21 | Setup for trigonometric method of NA measurement.

Solution: See Fig. 8.21. We have

$$\begin{aligned} \text{NA} &= \frac{A}{\sqrt{(A)^2 + 4B^2}} \\ &= \frac{9}{\sqrt{(9)^2 + 4 \times (10)^2}} = 0.41 \end{aligned}$$

PROBLEM 3

In a fiber-attenuation measurement using cutback method, calculate the attenuation per kilometer for a multimode fiber of 1.5 km length operating at 850 nm. The photodetector gives 1.2 V output for 1.5 km fiber. If the fiber is cut back to 2 m length, the output voltage increases to 12 V.

Solution: $L_1 = 1.5$ km, $L_2 = 2$ m, $V_{\text{nearend}} = 12$ V, $V_{\text{farend}} = 1.2$ V. Then the attenuation per kilometer is given by

$$\begin{aligned} \alpha (\text{dB/km}) &= \frac{20 \log V_{\text{nearend}} - 20 \log V_{\text{farend}}}{L} = \frac{20}{L_1 - L_2} \log_{10} \left(\frac{V_{\text{nearend}}}{V_{\text{farend}}} \right) \\ &= \frac{20}{1.5 - 0.002} \log_{10} \left(\frac{12}{1.2} \right) = 13.35 \end{aligned}$$

Therefore

$$\alpha = 13.35 \text{ dB/km}$$

PROBLEM 4

In scanning photodetector and rotating stage NA measurement method, the angular limit of the far field pattern is found to be 25° , and when the same is displayed on the screen, its size is 20 cm. Calculate the NA of fiber and distance of fiber output end face from the screen.

Solution: Given that $\theta = 25^\circ$, pattern size on screen = 20 cm. See Fig. 8.22. We have,

$$\text{NA} = \sin \theta = \sin 25^\circ = 0.534$$

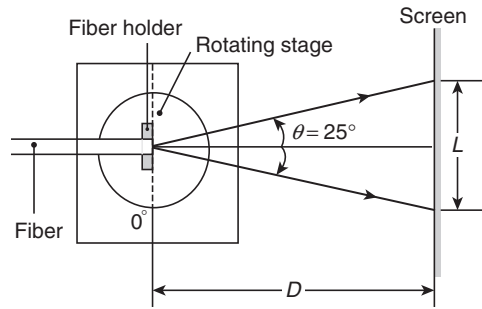


Figure 8.22 | Setup for NA measurement.

Referring to Fig. 8.22, we have

$$\tan \theta = \frac{L/2}{D}$$

Therefore,

$$D = \frac{L/2}{\tan \theta} = \frac{20/2}{\tan 25} = 21.44 \text{ cm}$$

SUMMARY

- Optical attenuation is measured using the following methods:
 - Cutback method.
 - Insertion loss method.
 - OTDR method.
- Cutback is a destructive method requiring access to both ends of fiber.
- OTDR works on the principle of Rayleigh scattering and Fresnel reflection to measure attenuation in fiber.
- Optical power meters are calibrated for 850, 1,300, and 1,550 nm.
- Numerical aperture (NA) is a measurement of the ability of an optical fiber to capture light. It specifies the angular range over which an optical fiber can accept light. Different methods of measurement of NA are:
 - Scanning photodetector and rotating stage method.
 - Trigonometric method.
- Optical spectrum analyzer displays the optical power as a function of wavelength. OSAs are divided into the following two categories:
 - Diffraction grating-based OSAs.
 - Interferometer-based architectures.
- A wavelength meter uses a Michelson interferometer to measure the wavelength of light.
- Optical tweezers are scientific instruments that use laser light pressure to trap the microscopic objects as small as single atom.

MULTIPLE-CHOICE QUESTIONS

- The test setup for optical loss measurement includes
 - Optical spectrum analyzer and power meter
 - Power meter and a length of fiber calibrated to work together
 - Wavelength meter and power meter
 - Light source and power meter

2. A wavelength meter uses which optical system for wavelength measurement?
 - (a) Fabry–Perot interferometer
 - (b) Michelson interferometer
 - (c) Mach–Zehnder interferometer
 - (d) Optical spectrum analyzer
3. Optical spectrum analyzer records
 - (a) Optical power as a function of wavelength
 - (b) Fiber losses as a function of length
 - (c) Optical power as a function of length
 - (d) Fiber losses as a function of wavelength
4. Optical time-domain reflectometer plots
 - (a) Intensity of reflected light as a function of wavelength
 - (b) Intensity of reflected light as a function of time
 - (c) Attenuation of reflected light as a function of time
 - (d) Attenuation of reflected light as a function of wavelength

REVIEW QUESTIONS

1. Explain the following attenuation measurement techniques with neat sketches.
 - (a) Cutback method
 - (b) Insertion loss method
 - (c) Optical time-domain reflectometer method
2. Describe the dispersion measurement in multi-mode fiber using time-domain method and frequency-domain method.
3. Explain scanning photodetector and rotating stage method and trigonometric method of numerical aperture measurement.
4. Write a short note on OTDR.
5. Explain the working of optical power meter with block schematic.
6. With a neat sketch, explain the operation of wave meter.
7. With block schematic, explain the working of OSA.
8. With a neat sketch, explain the operating principle of optical tweezers.

EXERCISES

1. Determine the attenuation in a 3 km multi-mode fiber attached in spectral loss measurement setup. The measured output voltage for 3 km fiber length is 1.8 V. Then the fiber is cut to 3 m length, and output voltage is again measured. Its value is 9 V.
2. In scanning photodetector and rotating stage NA measurement method, the angular limit of the far field pattern is found to be 23° , and the same is displayed on the screen at a distance of 20 cm from fiber output end face. Calculate the NA of fiber and its display size on screen.
3. Determine the attenuation per kilometer in a fiber using cutback method. During measurement the following data is obtained: length of fiber is 5 km and the corresponding output power is $100 \mu\text{W}$, length of cut back fiber is 5 m and the corresponding output power is $700 \mu\text{W}$ at 1100 nm.
4. Calculate the NA of fiber using the trigonometric method, where the screen is placed 15 cm from the fiber end face. The fiber is illuminated from visible source and gives an output pattern of size 7 cm.

ANSWERS

Multiple-Choice Questions

1. (d)
2. (b)
3. (a)
4. (b)

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9 | Optical Sensors

Learning Objectives

After completing this chapter, you will be able to understand:

- Classification of fiber-optic sensors.
 - Intensity-modulated sensors.
 - Phase-modulated sensors.
 - Wavelength-modulated sensors.
 - Polarization-modulated sensors.
 - Applications of sensors.
-

Fiber-optics technology was primarily developed for telecommunication applications. In previous chapters, we have studied how optical fibers can be used for telecommunication applications. However, due to the advances in fabrication technology, high quality and competitive price optoelectronic components and fibers are available; this has contributed to the expansion of guided wave technology for sensing applications.

A fiber-optic sensor detects changes in the light guided through an optical fiber when it is affected by external physical, chemical, biomedical, or any other parameters.

The basic fiber-optic system consists of optical source, optical fiber, modulating element, optical detector, signal processor, and readout. The light output of the source is coupled to the optical fiber, which is then given to the modulating element, where light is modulated by the measurand (e.g., temperature, pressure). Modulated light is detected by photodetector, and processed and calibrated to get the direct readout of measured quantity.

The attractive features of fiber-optic sensing are as follows:

1. High sensitivity.
2. Large bandwidth.
3. Operation at very high temperature, pressure, or voltage.
4. Multiplexed or distributed measurement.
5. Long range operation.
6. Explosion-proof and immune to radio frequency and electromagnetic interference due to non-electrical (passive) method of operation.
7. Small size, light weight, and great flexibility, allowing access to restricted areas.
8. Resistant to chemically aggressive and ionizing environments.
9. Easy interface with optical data communication systems and secure data transmission.

9.1 Classification of the Fiber-Optic Sensors

Fiber-optic sensors are classified as follows:

1. **Based on sensing site or location:** Waveguide sensors are subdivided as:

- (a) Intrinsic.
- (b) Extrinsic.

In telecomm applications, the fiber cables are designed to be isolated from the outside environment. In sensing applications, on the other hand, the fiber is designed such that the light wave propagation is strongly affected by the external parameters. By observing the light propagating through the fiber, the external parameter is measured in terms of changes in intensity of light, phase, wavelength, and polarization.

2. **Based on operating principle:** Based on the optical parameter modulated by the measurand, or in other words based on operating principle, waveguide sensors can be divided into four basic categories:
 - (a) *Intensity modulated:* The intensity of light launched into the fiber is changed intrinsically or extrinsically.
 - (b) *Phase modulated:* The phase of monochromatic light propagating through the fiber is changed intrinsically by the measurand.
 - (c) *Wavelength modulated:* The wavelength of light is changed extrinsically by the measurand.
 - (d) *Polarization modulated:* The direction of the *E*-field vector with respect to the direction of the light's propagation is changed by measurand.
3. **Based on application:** In this case, the fiber-optic sensors are classified as:
 - (a) *Physical sensors:* Used to measure physical quantities like temperature, pressure, etc.
 - (b) *Chemical sensors:* Used for gas analysis, pH measurement, etc.
 - (c) *Biomedical sensors:* Used in measurement of blood flow, glucose content, etc.

9.2 Intrinsic Sensors

Intrinsic sensors are true waveguide sensors in which the sensing element is the waveguide itself and sensing takes place within the fiber, as shown in Fig. 9.1. The light need not come outside the fiber.

9.3 Extrinsic Sensors

Extrinsic sensors make use of an optical transducer or modulator coupled to the waveguide, the optical constants of which are modulated by the measurand. Sensing takes place in a region outside the fiber, as shown in Fig. 9.2. The light propagating through input fiber comes out, reaches the measurand site, where the information is impressed upon the light in response to measurand, and then modulated light comes back to the output fiber. The information may be impressed in the form of intensity, frequency, polarization phase, or spectral content.

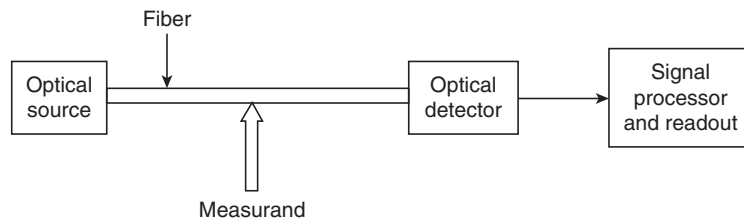


Figure 9.1 | Intrinsic fiber-optic sensor.

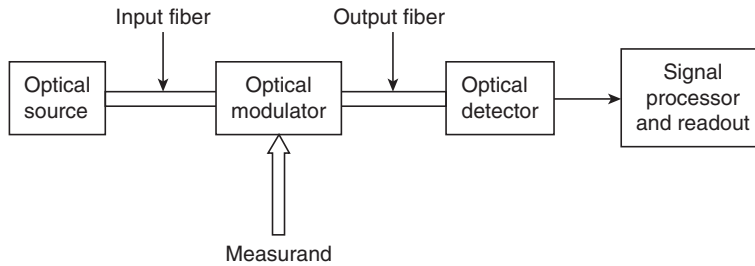


Figure 9.2 | Extrinsic fiber-optic sensor.

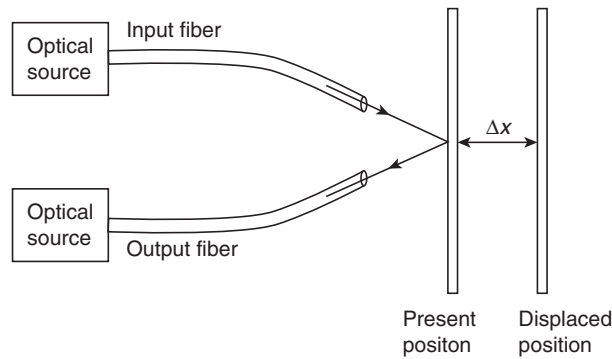


Figure 9.3 | Intensity-modulated reflection-type sensor.

9.4 Intensity-Modulated Sensors

These sensors convert a physical parameter into a change in the intensity of light launched into a fiber intrinsically or extrinsically. There are a number of schemes for intensity modulation.

9.4.1 Reflection-Type Sensors

The intensity of light arriving at the detector depends upon the position of the reflecting surface. The displacement of the reflecting surface causes a change in the intensity of light detected. Typically, a broadband source and multimode fiber are used for sensing. The measurement setup is as shown in Fig. 9.3. This sensor arrangement is similar to a movable reflector. Output power is proportional to displacement Δx ; therefore, it is used as distance or pressure sensor.

9.4.2 Transmission-Type Sensors

A fixed light is launched into one of the fibers. The second fiber undergoes displacement due to measurand. The displacement may be

1. Longitudinal or axial, as shown in Fig. 9.4(a).

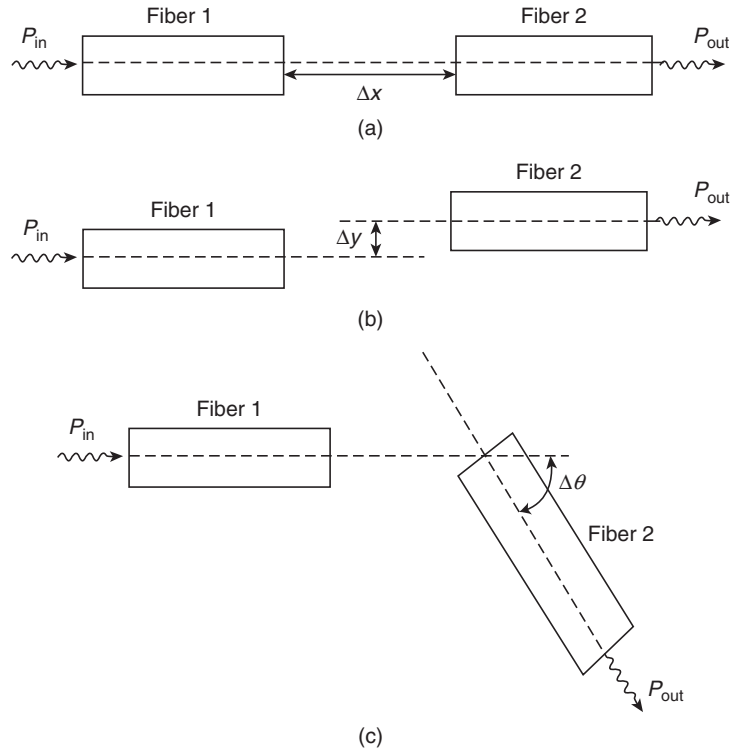


Figure 9.4 (a) Longitudinal, (b) lateral, and (c) angular displacement of fiber 2 with respect to fiber 1 due to measurand, such as displacement or strain.

2. Lateral or transverse, as shown in Fig. 9.4(b).
3. Angular, as shown in Fig. 9.4(c).

Any parameter that can cause either longitudinal or lateral or angular displacement can be measured with this sensor. However, as there is no reference signal for measurement. This measurement scheme suffers from light source intensity fluctuation, and the fiber loss varies. This can add error in the measurement. Figure 9.5 shows the sensitivity curve for different types of transmission-type sensors. Lateral displacement is most sensitive but not linear throughout the operation range, while longitudinal displacement is less sensitive but gives linear output.

9.4.3 Differential Intensity Sensors

To overcome the disadvantage of transmission-type sensors, the modified sensor arrangement shown in Fig. 9.6 is used. In order to have a reference for the measurement, a differential arrangement is made. Light from an optical source is launched into the transmitting fiber. There are two receiving fibers, placed such that in equilibrium condition, the lateral offsets between transmitting fiber axis and receiving fiber axis (Δy_1 for fiber 1 and Δy_2 for fiber 2) are equal, so both fibers will receive equal amount of power. The output of both fibers is given to similar detectors. In equilibrium condition, outputs of both detectors are equal ($V_1 = V_2$). Displacement due to the measurand is applied to transmitting fiber as shown in Fig. 9.6. Due to this, lateral offset of fiber 1 (Δy_1) decreases and lateral offset of fiber 2 (Δy_2) increases. Consequently, the received power variation is proportional to the displacement of the transmitting fiber.

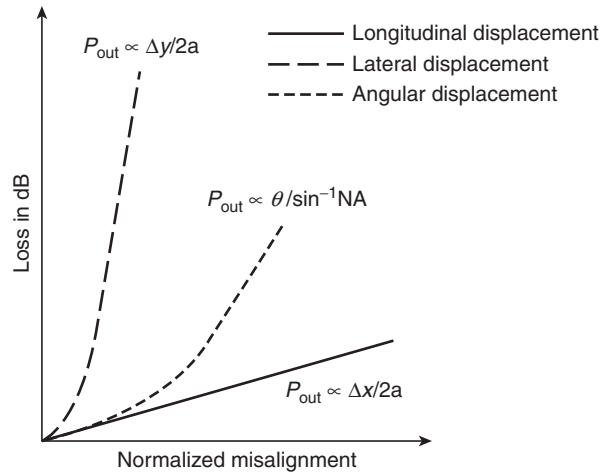


Figure 9.5 | Sensitivity curve for different transmission type sensors.

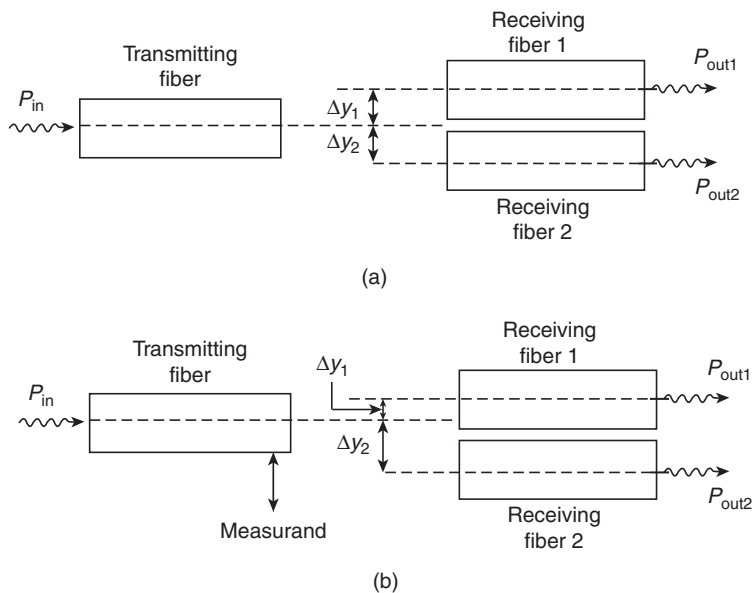


Figure 9.6 | Differential intensity sensors: (a) Equilibrium condition and (b) transmitting fiber displaced by measurand.

9.4.4 Microbending Sensor

Microbending sensors are intrinsic sensors. Figure 9.7 shows the simple setup of a microbending sensor. The light is launched into a fiber and its output is connected to a photodetector. In between, a microbend transducer is placed to sense the pressure. A microbend transducer has tapered teeth that produce microbends on the fiber when pressure is applied. Microbending results in the loss of higher order modes. With increase in pressure,

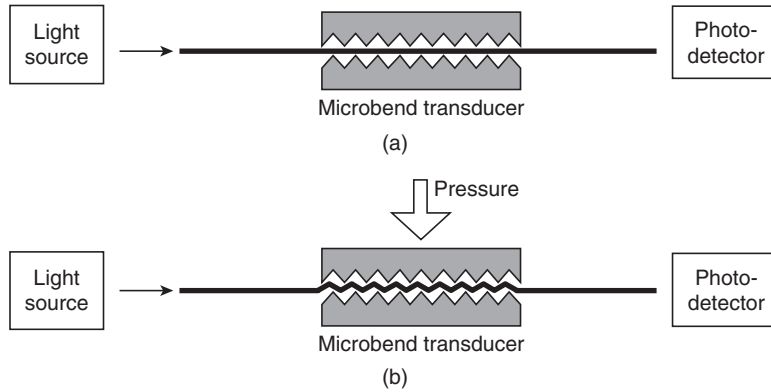


Figure 9.7 | Microbending sensors: (a) No pressure, straight fiber and (b) with pressure, microbending of fiber.

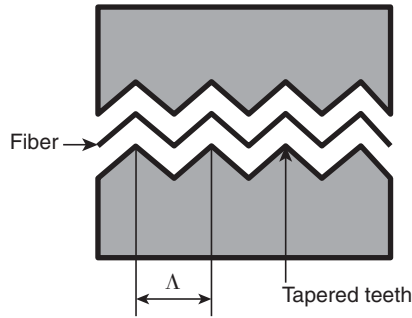


Figure 9.8 | Details of microbending transducer.

microbending increases and the optical power received by photodetector decreases. Thus, the physical quantity pressure is converted into a change in intensity.

The loss of signal in a sensor is maximum for periodic microbending with a bend pitch Λ (refer Fig. 9.8), given by

$$\Lambda = \frac{\alpha \pi a n}{NA} \quad (9.1)$$

where α is the core refractive index profile parameter, a is the radius of the core, n is the refractive index of the core, and NA is the numerical aperture of the fiber.

All the intensity-modulated sensors mentioned above have the following features:

1. Versatility.
2. Simple design and easy signal interpretation.
3. Low cost.
4. Reliability.

However, they usually suffer from intensity fluctuations and low sensitivity. These sensors need some referencing and are suitable for digital applications, where an ON–OFF signal is required.

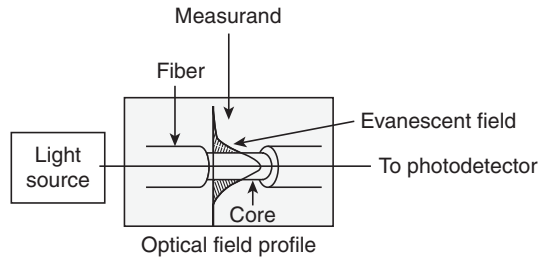


Figure 9.9 | Evanescent wave fiber-optic chemical sensor.

9.4.5 Evanescent Wave Fiber Sensors

The light energy profile of optical fiber is not confined totally to the core only; some energy penetrates the cladding and it is called an *evanescent wave*. This leaked optical energy is utilized for chemical sensing. If the cladding of the fiber is removed, then this evanescent wave can interact with the outside environment as shown in Fig. 9.9. The chemical absorbs the light, and the change in light intensity is calibrated as a change in the concentration of the chemical.

Evanescent wave fiber sensors can be used in biomedical applications such as blood component meters for detection of cholesterol and uric acid.

9.5 Phase-Modulated or Interferometric Sensors

These sensors can modulate the phase of light that in turn modulates the intensity. The modulation is detected interferometrically or using a fiber optic gyroscope. The most commonly used interferometers are Mach–Zehnder and Fabry–Perot.

9.5.1 Fiber Mach–Zehnder Interferometer Sensor

An interferometric sensor configuration consisting of a Mach–Zehnder interferometer is shown in Fig. 9.10. The light of intensity I_0 from the laser source is launched in one fiber that is split by a 2×2 coupler having a 50:50 coupling ratio. The equally divided light is sent through the sensing arm and the reference arm. The reference arm is shielded while the sensing arm is subject to the measurand directly. The measurand changes the phase of the light by changing its length, or refractive index, or both. This creates a phase difference $\Delta\phi$ between the light wave with intensity I_1 in the sensing arm and a light wave with intensity I_2 in the reference arm.

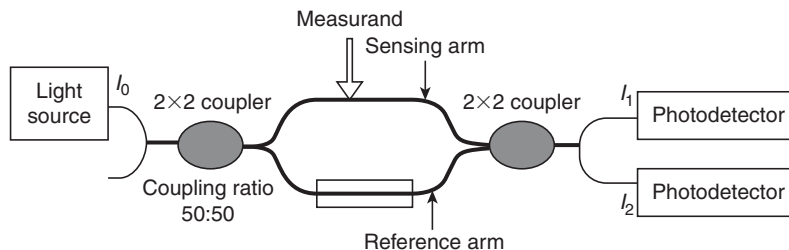


Figure 9.10 | A fiber-optic Mach–Zehnder interferometric sensor.

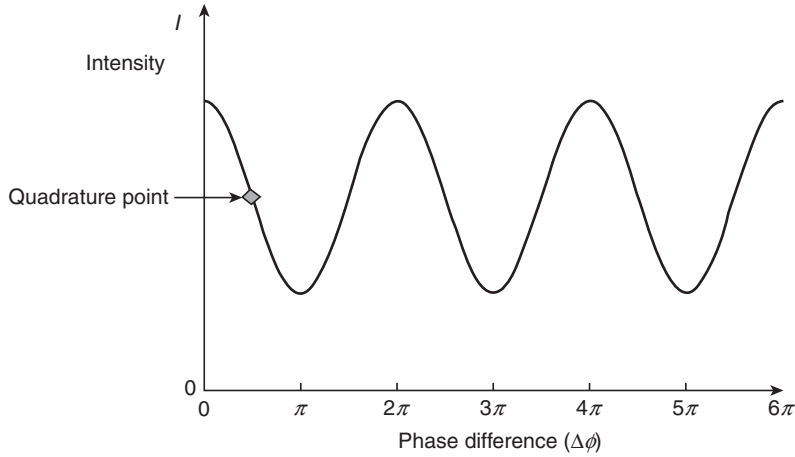


Figure 9.11 | Sensor operation point.

The couplers on the input and output sides have a coupling ratio of 50:50, so the coupling coefficient $k_1 = k_2 = k = 0.5$. The fiber arms consist of single-mode fibers with negligible loss. Therefore,

$$I_1 = \frac{I_0}{2}(1 + \cos \Delta\phi) = I_0 \cos^2 \left(\frac{\Delta\phi}{2} \right) \quad (9.2)$$

$$I_2 = \frac{I_0}{2}(1 - \cos \Delta\phi) = I_0 \sin^2 \left(\frac{\Delta\phi}{2} \right) \quad (9.3)$$

where $I_0 = I_1 + I_2$ is the input intensity. Now

1. If $\Delta\phi = 0$, then $I_1 = I_0$ and $I_2 = 0$, and all power launched will be available at port 1.
2. If $\Delta\phi = \pi$, then $I_1 = 0$ and $I_2 = I_0$, and all power launched will be available at port 2.
3. When $\Delta\phi = (2k+1)\pi/2$, the sensor has maximum sensitivity. This point is called the **quadrature point** as shown in Fig. 9.11. Thus, a fixed bias of $\pi/2$ is induced and if θ is the phase change due to the measurand, then

$$\Delta\phi = \frac{\pi}{2} + \theta \quad (9.4)$$

So

$$\begin{aligned} I_2 &= \frac{I_0}{2} \left[1 - \cos \left(\frac{\pi}{2} + \theta \right) \right] \\ \Rightarrow I_2 &= \frac{I_0}{2} [1 + \sin \theta] \\ \Rightarrow I_2 &\approx \frac{I_0}{2} [1 + \theta] \end{aligned} \quad (9.5)$$

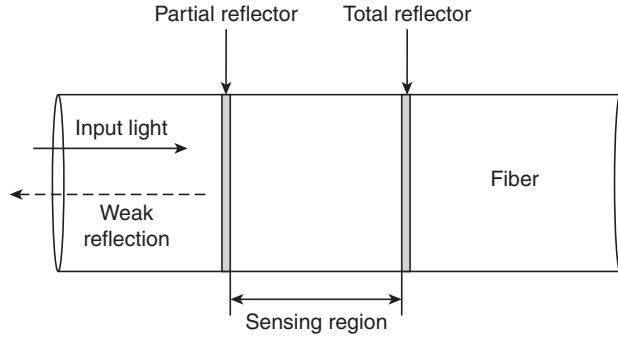


Figure 9.12 | Fiber Fabry–Perot sensor.

Similarly,

$$I_1 \approx \frac{I_0}{2} [1 - \theta] \quad (9.6)$$

An interferometric sensor is highly sensitive to the measurand, but it is also sensitive to wavelength fluctuations. Hence, it requires a highly coherent laser light source.

9.5.2 Fiber Fabry–Perot Interferometer Sensors

These are intrinsic sensors. They detect a phase shift within a resonant cavity. These sensors consist of a piece of fiber that has reflective layers on each end as shown in Fig. 9.12. The two reflective layers form a cavity that acts as a sensing element. One layer is partial reflector and the other is a total reflector. The sensor internal reflectors can be made by thin film deposition on the cleaved fiber end followed by fusion splicing.

The Fabry–Perot interferometer has a series of resonances at characteristic wavelength defined by cavity length and refractive index, when an exact number of half-wavelengths of the light fit within the separation distance between the mirrors.

The resonance condition along the axis of cavity is given as

$$L = \frac{\lambda q}{2n} \quad (9.7)$$

where λ is the emission wavelength, n is the refractive index of amplifying medium, and q is an integer. Change in phase shift causes change in refractive index, and that is reflected as change in resonance wavelength according to Eq. (9.7).

9.5.2.1 Uses

This sensor can be used to sense temperature. Any change in temperature causes variation in the optical length of the Fabry–Perot cavity and accordingly the resonance wavelength changes.

9.5.3 Fiber-Optic Gyroscope

It is used to sense angular velocity. It consists of a loop of single-mode fiber (refer to Fig. 9.13). A coherent light source launches light into an input fiber coupler. The input beam passes through a polarizer that is used to make

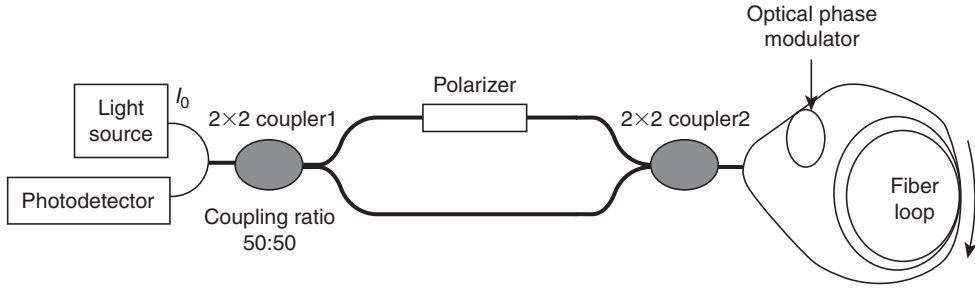


Figure 9.13 | Fiber-optic gyroscope.

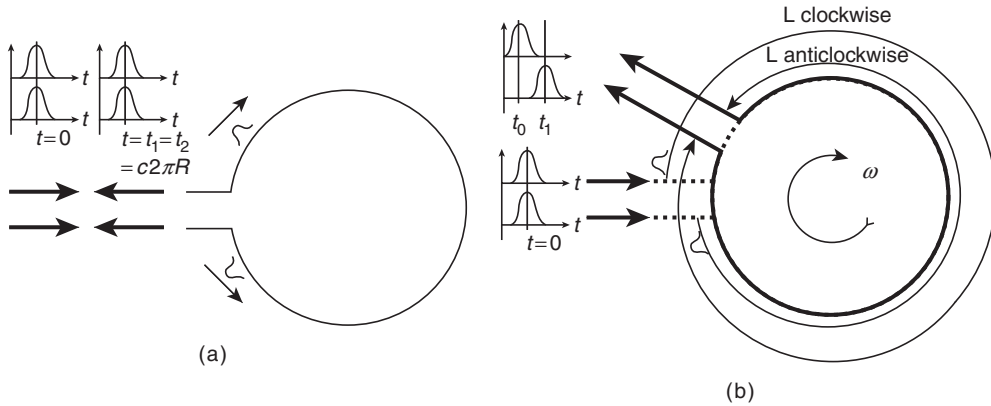


Figure 9.14 | Sagnac effect: (a) Fiber loop in stationary condition and (b) fiber loop rotating in clockwise direction.

certain the mutuality of light beams traveling in the opposite direction through the fiber coil. The second or output directional coupler splits the light beam into two equal halves and sends them into two ends of fiber coil, where the light beam passes through a modulator.

It is used to produce a time-altering output signal indicative of angular rotation. The light beam emerging from respective ends recombines at output coupler, then passes through the polarizer, and finally reaches the detector through the input coupler.

When the fiber loop is stationary, the counter-propagating beams through it will take the same time to emerge, as shown in Fig. 9.14(a).

If the fiber loop is rotating in clockwise direction with angular velocity ω , the clockwise traveling beam has to travel more distance before emerging, as shown in Fig. 9.14(b), whereas the anticlockwise beam has to travel less. As there is a relative phase shift between the two beams, hence there is a corresponding shift in the interference pattern at the detector. This phenomenon is known as **Sagnac effect**. The phase shift is proportional to the rotational speed. The transit time of the beam is

$$t = \left(\frac{2\pi a}{c} \right) \quad (9.8)$$

where a is the radius of loop. The longer path length for a clockwise beam is

$$L_{\text{clockwise}} = 2\pi a + a\omega t \quad (9.9)$$

The shorter path length for an anticlockwise beam is

$$L_{\text{anticlockwise}} = 2\pi a - a\omega t \quad (9.10)$$

The difference between the propagation times of clockwise and anticlockwise beams is

$$\begin{aligned} \Delta t &= \frac{L_{\text{clockwise}}}{c} - \frac{L_{\text{anticlockwise}}}{c} \\ &= \frac{2\pi a + a\omega t}{c} - \frac{2\pi a - a\omega t}{c} \end{aligned} \quad (9.11)$$

So

$$\Delta t = \frac{2a\omega t}{c} = \frac{4\pi a^2 \omega}{c^2} \quad (9.12)$$

If the light waves are continuous and the frequency is f , the phase difference corresponding to Δt is

$$d\phi = 2\pi f \Delta t = \frac{8\pi a^2 \omega}{c^2} = \frac{8\pi A \omega}{c^2} \quad (9.13)$$

where A is the fiber loop area.

Uses

Gyroscopes are used for rotation sensing in aircraft and missiles.

9.6 Wavelength-Modulated Sensors

In these sensors, the wavelength of light is modulated by the measurand. There are several schemes available for this, as follows:

1. Fiber Bragg grating sensors.
2. Fluorescent sensors.
3. Blackbody sensors.

9.6.1 Fiber Bragg Grating Sensors

In Section 6.6.10, we have studied the details of fabrication techniques and applications of fiber Bragg gratings. Fiber Bragg gratings are in-fiber components, generally written in Ge-doped fiber using UV light. If Λ is the grating period and n_{eff} is the effective refractive index of grating in the fiber core, then the reflected wavelength is called **Bragg wavelength**, which is defined as

$$\lambda_B = 2n_{\text{eff}} \Lambda \quad (9.14)$$

The Bragg grating resonance, which is the center wavelength of light back reflected from a Bragg grating (λ_B), depends on the effective index of refraction of the core and the periodicity of the grating. Changes in strain and

temperature will affect the effective index of refraction and the periodic spacing between the grating planes. The shift in the Bragg grating center wavelength $\Delta\lambda_B$ due to strain and temperature changes is given by

$$\Delta\lambda_B = 2\left(\Lambda \frac{\partial n_{\text{eff}}}{\partial L} + n_{\text{eff}} \frac{\partial \Lambda}{\partial T}\right)\Delta L + 2\left(\Lambda \frac{\partial n_{\text{eff}}}{\partial T} + n_{\text{eff}} \frac{\partial \Lambda}{\partial T}\right)\Delta T \quad (9.15)$$

The first term in the equation represents the strain effect on an optical fiber. This corresponds to a change in the grating period and a strain-induced change in the refractive index due to the photoelastic effect. To make our calculations easier, the strain affected may be expressed as

$$\Delta\lambda_B = \lambda_B (1 - P_e) \varepsilon \quad (9.16)$$

where λ_B is the applied strain and P_e is an effective strain-optic constant defined as

$$P_e = \frac{n_{\text{eff}}^2}{2} [P_{12} - \nu(P_{11} + P_{12})] \quad (9.17)$$

P_{11} and P_{12} are components of the strain-optic tensor and ν is Poisson's ratio. The second term in the equation represents the effect of temperature on an optical fiber. A shift in the Bragg wavelength due to the thermal expansion changes the grating period and the refractive index. The wavelength shift due to the temperature change may be expressed as

$$\Delta\lambda_B = \lambda_B (\alpha_\Lambda + \alpha_n) \Delta T \quad (9.18)$$

where

$$\alpha_\Lambda = \left(\frac{1}{\Lambda}\right) \left(\frac{\partial \Lambda}{\partial T}\right)$$

is the thermal expansion coefficient and

$$\alpha_n = \left(\frac{1}{n_{\text{eff}}}\right) \left(\frac{\partial n_{\text{eff}}}{\partial T}\right)$$

represents the thermo-optic coefficient. Thus, the fiber Bragg grating (FBG) sensor can be used for temperature and strain measurements. Fiber grating-based sensors have many advantages over conventional electric and alternative fiber-optic sensor configurations. Some of them are as follows:

1. Wavelength-encoded output.
2. Self-referencing feature.
3. Linear output.
4. Small and lightweight.
5. Wavelength division multiplexing (WDM) and time division multiplexing (TDM).
6. Mass producible.
7. Durable.
8. Single- and multi-point sensing.

The basic principle of FBG sensors is the measurement of an induced shift in the wavelength of an optical source due to a measurand, such as strain or temperature. A basic reflective FBG sensor system is shown in Fig. 9.15. A broadband light source is used to interrogate the grating, from which a narrowband slice is reflected. The peak wavelength of the reflected spectrum can be compared to λ_B .

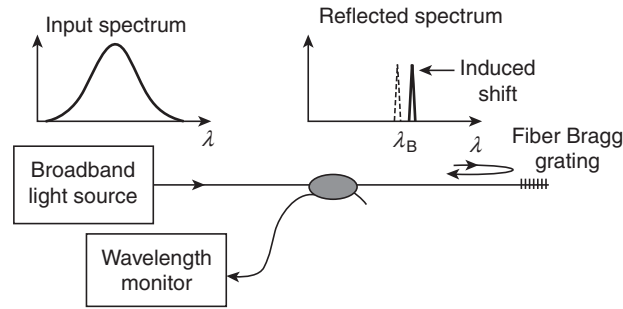


Figure 9.15 | Basic reflective FBG sensor system.

9.6.2 Fluorescent Sensors

Luminescence-based fiber sensors are usually based on fluorescence or amplified spontaneous emission occurring in rare earth materials. Fluorescence is the phenomenon of emission of light by a material upon absorption of electromagnetic radiation, such as ultraviolet, visible, or infrared. These sensors can be used for sensing temperature, humidity, chemicals, etc. The sensor probe is prepared by using a crystal of fluorescent material connected at the tip of fiber, or the fiber itself can be doped with fluorescent material.

The fiber doped with luminescent material is used for a chemical concentration detection application. The setup for the same is shown in Fig. 9.16.

A laser pulse is launched into the fiber doped with fluorescent material. The luminescence occurs at longer wavelength than the laser wavelength. The luminous intensity $I(t)$ at the detector decays exponentially. Therefore, mathematically we can write

$$I(t) = I_0 e^{-(k_1 + k_2 C)t} \quad (9.19)$$

where I_0 is the initial luminescence intensity, t is the time, k_1 , k_2 are the constants, and C is the chemical concentration. From the luminous intensity curve, the time constants can be computed and concentration C of chemical can be obtained by the mathematical formula given in Eq. (9.19).

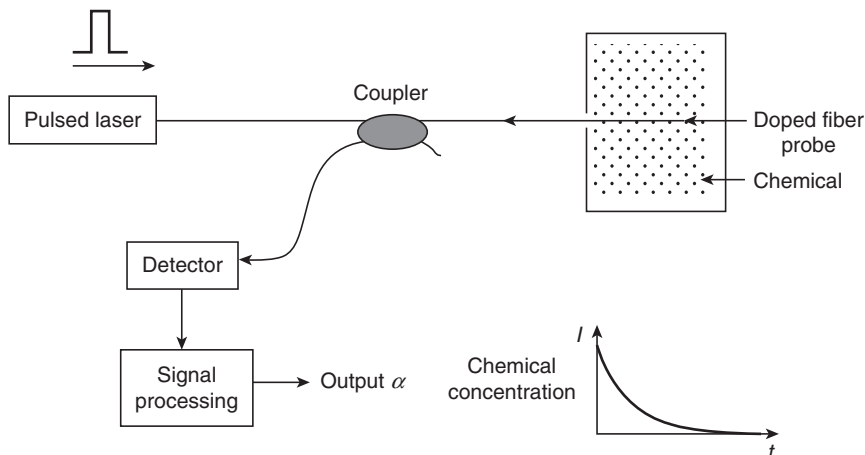


Figure 9.16 | Fiber-optic fluorescence chemical sensor.

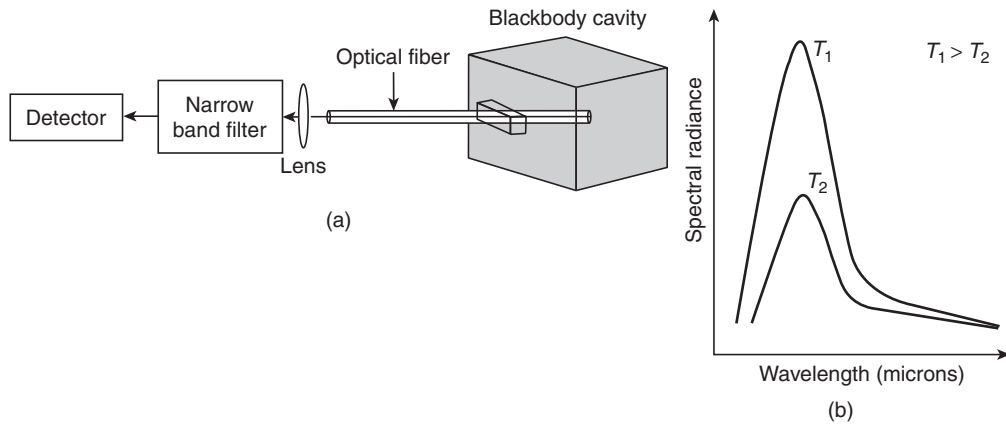


Figure 9.17 | Blackbody fiber-optic sensor: (a) Measurement setup and (b) blackbody radiation curve.

9.6.3 Blackbody Sensor

The operating principle of this sensor is very simple. The basic measurement set consists of a blackbody cavity, optical fiber, lens, narrowband filter, and detector as shown in Fig. 9.17(a). When the temperature of a blackbody cavity starts increasing, it will glow and act as a light source. This light is picked up by the fiber probe tip kept inside the blackbody cavity. The detector and narrow band filters are used to determine the blackbody radiation curve, as shown in Fig. 9.17(b).

9.7 Polarization Fiber Sensor

The sensor can modulate the phase of light. The original signal and phase-shifted signal interfere to modulate the light intensity, so that phase shift is converted in terms of light intensity.

A pressure sensor setup is shown in Fig. 9.18. A light signal having two polarizations is passed through the sensing fiber. The pressure signal changes the refractive index of the sensing fiber, and this in turn changes the polarization of light. Hence, there is delay difference between the two polarizations. For measurement purposes, the two polarizations are separated by a beam splitter. One polarization is rotated by 90° and the path length of other is equalized, after which both are combined. The output is high if there is no delay between the two and it is low if one is 180° behind the other as shown in Fig. 9.19.

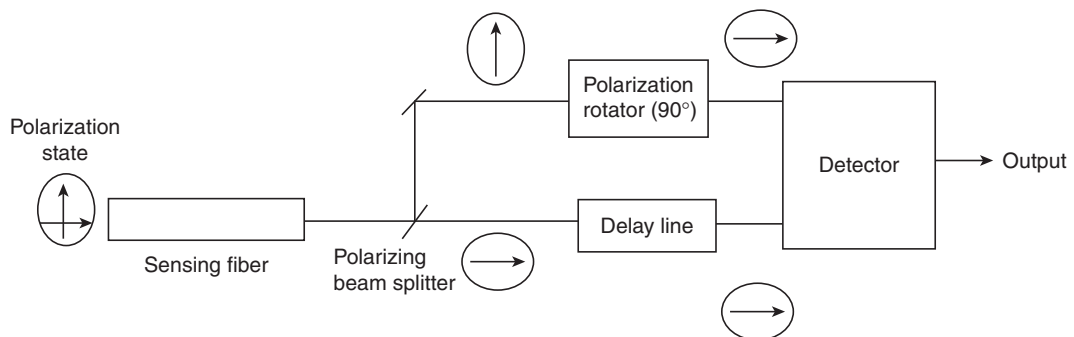


Figure 9.18 | Polarization fiber sensor setup.

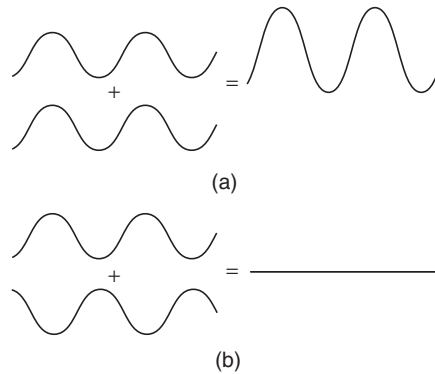


Figure 9.19 | Sensor outputs: (a) In-phase polarization: high output and (b) out-of-phase polarization: no output.

9.8 Fiber-Optic Biomedical Sensor

Fiber-optic sensors are more suitable for biomedical applications due to their miniature size and flexibility. Fibers can be easily inserted in catheters and needles that allow localized measurements inside blood and tissues. The fiber materials – such as glass and plastic – are non-toxic and biocompatible and can therefore be used for continuous monitoring.

The different biomedical applications include the following:

1. Blood pressure.
2. Body temperature.
3. Blood flow.
4. Blood pH.
5. Blood proteins.
6. Lipids.
7. Cholesterol.

9.8.1 Blood Flow Measurement

With the help of fiber optics, invasive and contact measurement can be easily done. The basic scheme of fiber-optic laser Doppler flowmetry is shown in Fig. 9.20. The He–Ne laser is launched into the optical fiber probe,

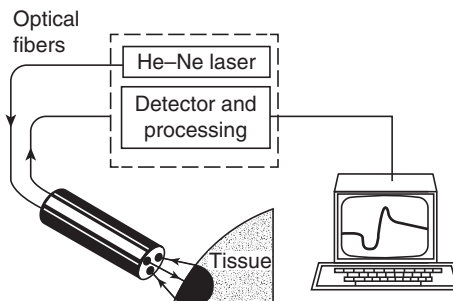


Figure 9.20 | Fiber-optic laser Doppler flowmetry setup.

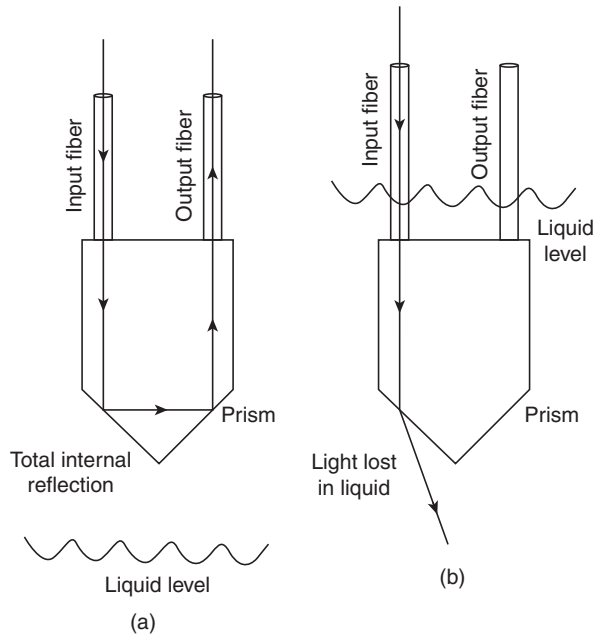


Figure 9.21 | Fiber-optic liquid level sensors: (a) Liquid level below sensor and (b) liquid level above sensor.

which is guided toward the body tissues or vascular system to be studied. The light gets scattered, diffused, and partly absorbed. The light experiences a Doppler shift upon hitting the moving blood cells. The shift in the wavelength is proportional to the blood flow rate. By analyzing the received spectrum of light through detection and processing, the blood flow rate is obtained.

9.9 Fiber-Optic Liquid Level Sensor

The schematic drawing in Fig. 9.21 shows the principle of operation of a fiber-optic liquid level sensor. An input fiber delivers light to the prism. If the liquid is below the sensor level, the light from the fiber experiences total internal reflection at the base of prism and is directed back into the output fiber. If the prism is covered with liquid, total internal reflection cannot occur at the angle at which the light strikes the prism. Thus, no light is reflected back in the output fiber. Many such sensors can be fixed at different levels to detect a particular level of liquid, or one sensor at the top can be fixed to prevent overfilling of the tank. The output signal is continuously monitored, and when the light signal stops, it indicates that the tank is full.

9.10 Fiber-Optic Probe

The fiber-optic probe is used for checking the presence of parts on an assembly line. The input fiber delivers light from source and the output fiber collects the emerging light if no part is present on the assembly line as shown in Fig. 9.22(a). If a part is present on assembly line, the light is blocked by the part as shown in Fig. 9.22(b) and no light is collected by the output fiber.

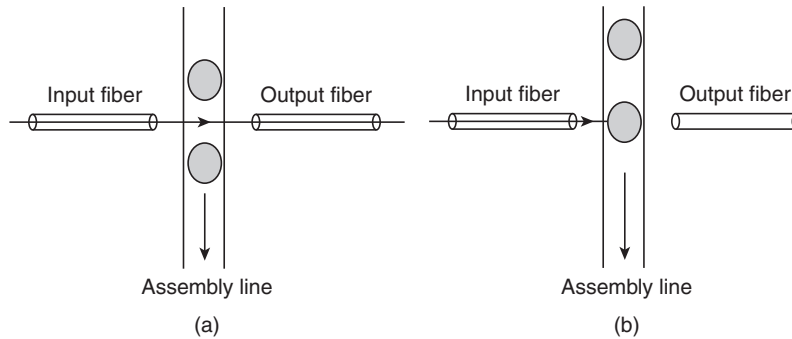


Figure 9.22 Fiber-optic probe to check presence of parts on assembly line: (a) Light passed due to absence of part and (b) light blocked due to presence of part.

9.11 Distributed Sensor

The sensors which provide a single measurand value are called as **point sensors**. However, when it is required to monitor the measurand at a number of points, then an array of point sensors is used, and is called a **distributed sensor**. The following are some areas where distributed sensors are used:

1. Structural health monitoring of bridges, buildings, dams, ships, aircrafts, spacecrafts, etc.
2. Temperature profile monitoring of furnaces, generators, boilers, power transformers, etc.

In Section 9.10, we saw that an FBG is suitable for single-point as well as multi-point sensing. The wavelength-encoded nature of the output of an FBG allows the formation of an inline sensor array and multipoint sensing. The number of sensors can be multiplexed by using WDM or TDM.

9.11.1 Wavelength Division Multiplexing Interrogation

Several FBGs in reflection mode are arranged inline in a single fiber as indicated in Fig. 9.23. Each FBG has a different resonance wavelength. The array is designed such that the resonance wavelengths are equally spaced and distributed over the available spectral width. With the help of a broadband light source, such as an edge-emitting

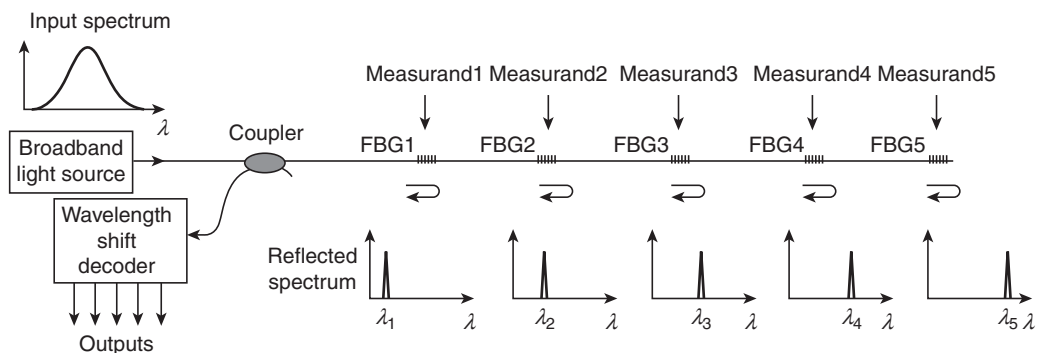


Figure 9.23 WDM interrogation of an FBG array.

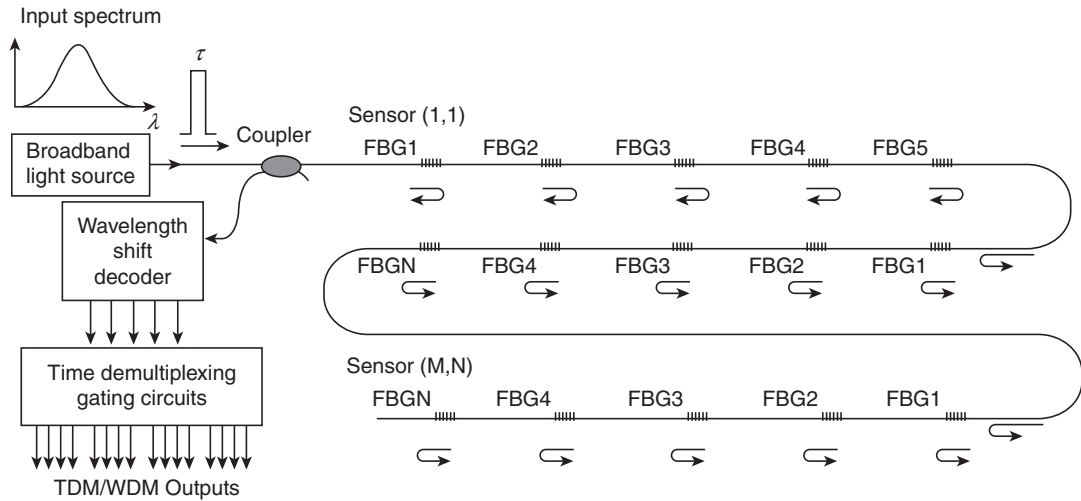


Figure 9.24 | Time-multiplexed WDM interrogation of FBG array.

LED or a superluminescent solid state or fiber source, light is launched into the fiber with the help of coupler. FBG1 reflects λ_1 ; FBG2 reflects λ_2 , and so on.

Due to the measurand there is a specific shift in the resonance wavelength of each FBG, which can be determined. The back-reflected signal can be analyzed by using an optical spectrum analyzer, a Fabry–Perot tunable filter, or a color glass filter. With this technique, dozens of sensors can be simultaneously interrogated. To further increase the number of sensors per fiber, TDM in combination with WDM can be used.

9.11.2 Time Division Multiplexing Interrogation

A large number of FBG sensors are arranged in groups as shown in Fig. 9.24. Each group is scanned separately one at a time. A short pulse of light is launched, by which the reflections from every FBG returns to the detector at different times, depending upon the location of FBG with respect to source. The sensor (1, 1) output will be available first at the detector as it is located nearest to the source; then the output from sensor (1, 2), sensor (1, 3), and so on.

SUMMARY

1. A fiber-optic sensor detects changes when the light guided through an optical fiber is affected by external physical, chemical, biomedical, or any other parameter.
2. Fiber optic sensor systems are classified into three broad categories – based on sensing site or location, based on operating principle, and based on application.
3. Based on sensing site or location, waveguide sensors are further divided as intrinsic and extrinsic.
4. Intrinsic sensors are true waveguide sensors in which the sensing element is the waveguide itself and sensing takes place within the fiber. The light need not come outside the fiber.
5. Extrinsic sensors make use of an optical transducer or modulator coupled to the waveguide,

the optical constants of which are modulated by the measurand. Sensing takes place in a region outside the fiber.

6. Based on operating principle, waveguide sensors can be divided into four basic categories – intensity modulated, phase modulated, wavelength modulated, and polarization modulated.
7. Fiber Fabry–Perot interferometer sensors can be used to sense temperature. Any change in temperature causes variation in the optical length of the Fabry–Perot cavity, and the resonance wavelength changes accordingly.
8. Fiber-optic gyroscope is used to sense angular velocity.
9. The basic principle of fiber Bragg grating (FBG) sensors is the measurement of an induced shift in the wavelength of an optical source due to a measurand, such as strain or temperature.
10. Fiber-optic sensors are widely used in biomedical applications due to their miniature size and flexibility.

MULTIPLE-CHOICE QUESTIONS

1. What is the operating principle of the FBG sensor?
 - (a) It induces a shift in wavelength
 - (b) It induces a change in polarization
 - (c) It induces a change in intensity
 - (d) It induces a change in refractive index
2. Which of the following is an intensity-modulated sensor?
 - (a) Fabry–Perot sensor
 - (b) Differential fiber-optic sensor
 - (c) Fiber Bragg grating sensor
 - (d) Fiber-optic gyroscope
3. Interferometry can be used to sense
 - (a) Change in wavelength of input light
 - (b) Change in intensity of light
 - (c) Change in refractive index due to pressure
 - (d) Change in optical absorption
4. The refractive index of a fiber can change due to
 - (a) Temperature change
 - (b) Sound waves
 - (c) Pressure change
 - (d) All of these
 - (e) None of these
5. Which of the following is a wavelength-modulated sensor?
 - (a) Fabry–Perot sensor
 - (b) Differential fiber-optic sensor
 - (c) Fiber-optic fluorescence temperature sensor
 - (d) Fiber-optic gyroscope
6. Microbending sensor is used to sense
 - (a) Pressure
 - (b) Temperature
 - (c) Both of these
 - (d) None of these
7. Which of the following is a phase-modulated sensor?
 - (a) Differential fiber-optic sensor
 - (b) Fiber Bragg grating sensor
 - (c) Fiber-optic gyroscope
 - (d) Fiber-optic fluorescence temperature sensor
8. Fiber-optic distributed sensing can be used for
 - (a) Structural health monitoring of bridges
 - (b) Monitoring of aircraft wings
 - (c) Temperature profile monitoring of furnaces
 - (d) All of these
9. FBG sensor is not suitable for which measurement?
 - (a) Strain
 - (b) Temperature
 - (c) Liquid level
 - (d) Pressure
10. All optical fiber interferometers can measure
 - (a) Intensity
 - (b) Phase
 - (c) Wavelength shift
 - (d) None of these

REVIEW QUESTIONS

1. With neat sketches explain the intrinsic and extrinsic fiber-optic sensors.
2. Give the classification of fiber-optic sensors.
3. Which are the different configurations of transmissive-type optical fiber sensor? Explain with sketch and sensitivity curve.
4. What is the disadvantage of transmissive-type sensor measurement? How it is overcome in differential intensity sensor?
5. Write a note on microbending sensor.
6. With neat sketch and mathematical equation describe the Mach–Zehnder interferometric sensor.
7. Describe the fiber Fabry–Perot interferometer sensor.
8. What is a distributed sensor?
9. Explain WDM and TDM for a distributed sensor.
10. Explain the fiber-optic gyroscope with a neat sketch.
11. What are the advantages of optical fiber in biomedical application?
12. Explain the fiber-optic laser Doppler blood flow meter.

ANSWERS

Multiple-Choice Questions

- | | |
|--------|---------|
| 1. (a) | 6. (c) |
| 2. (b) | 7. (c) |
| 3. (c) | 8. (d) |
| 4. (d) | 9. (c) |
| 5. (c) | 10. (b) |

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Index

A

- absorption, 63–64
- acceptance angle, 19–20
- acousto-optic tunable filter, 237–238
- active glass fibers, 37
- add/drop multiplexers (ADM), 324
- amplified spontaneous emission (ASE)
 - noise, 276–278, 315
- analog system, 211
- anti-Stokes photon, 307
- anti-Stokes waves, 306–307
- apodized grating, 247
- arrayed waveguide gratings (AWGs),
 - 234–237
- attenuation, 24–25, 78
- attenuation due to Rayleigh scattering, 64–65
- attenuation measurement
 - constant, 332–333
 - cutback method, 331–332
 - insertion loss method, 332
 - optical time-domain reflectometer (OTDR) method, 333–336
- avalanche effect, 183
- avalanche multiplication, 184
- avalanche photodiode (APD), 183–185
- axial ray, 18

B

- band gap energy, 114
- bandgap energy of photodiode material,
 - 172–175, 178
- bandgap energy of semiconductor, 114
- bandwidth, 341–342
- bandwidth-distance product, 73
- beam waist, 346
- bending losses, 65–66
- biasing, 113–114
- biomedical sensor, 367–368
- bi-phase coding, 153
- bipolar (RZ) code, 153
- birefringence, 72–73
 - of the TeO_2 crystal, 238
- bit error rate, 274–275, 312, 341
- blackbody sensors, 366

- Boltzmann constant, 133
- Boltzmann statistics, 133
- Bragg diffracter, 238
- Bragg grating of a mode, 244
- Bragg-type scattering, 84
- Bragg wavelength, 243, 363
- breakout cables, 44
- Brillouin fiber amplifiers, 84
- Brillouin scattering, 78, 280
- built-in voltage, 114

C

- cabling losses, 66
- cavity amplifier, 302
- C-band, 321
- chain rule for waveguide dispersion, 76–77
- chalcogenide glass fiber, 37
- chirped fiber grating, 247
- chromatic dispersion, 68–69
- circular polarization of light, 13
- circulators, 224–225
- cladding mode stripper, 332
- cleaved coupled cavity (CCC) laser, 145
- cleaving, 159
- core-cladding boundary conditions,
 - 243–245
- core-cladding refractive index difference, 65
- Corning glass works, 2
- critical angle, 15–16
 - for crystal-air interface, 121
- critical radius, 67
 - of curvature for single-mode fiber, 68
- cross-phase modulation (CPM), 280, 283
- cross-phase modulation (XPM or CPM),
 - 78, 90–91
- cross-saturation, 306
- crosstalk, 222, 278–279
- cut-in voltage, 113
- cutoff wavelength, 34–35

D

- dark current, 186
- dark-light-dark transaction cycles, 345
- data rate, 341–342
- demultiplexer (DEMUX), 218

- dense wavelength division multiplexing (DWDM), 61, 219–221
- dense wavelength division multiplexing (DWDM) systems, 313
- depletion region, 113
- detectivity, 186–187
- differential intensity sensors, 356–357
- diffraction grating, 145
- diffraction of light, 13–14
- diffusion length, 124–125
- digital system, 211
- direct modulation, 154
- direct recombination, 116–117
- dispersion, 61, 68–74
 - intermodal or modal, 72
 - intramodal or chromatic, 68–69
 - material, 69–71
 - maximum bit rate and, 77–78
 - measurement in frequency domain,
 - 338–339
 - measurement in time domain, 338
 - polarization-mode, 72–74
 - waveguide, 71–72
- dispersion compensated fiber (DCF),
 - 47–48
- dispersion-compensated fiber (DCF), 47
- dispersive media, 68
- distributed Bragg reflection (DBR) laser, 109
- distributed Bragg reflector (DBR) laser,
 - 144–145
- distributed feedback (DFB) laser, 109
- distributed feedback laser (DFB), 145
- distributed sensor, 369
- distribution networks, 213
- dome light-emitting diode (LED), 126
- double crucible method, 41
- double heterojunction (DH) laser, 145–146
- double heterostructure (DH), 125
- downward transition rate, 132
- drift time, 172, 181

E

- edge-emitting device, 131, 369–370
- edge-emitting LED (EELED), 127–128

Einstein coefficient, 132
 electromagnetic radiation, 11
 electron-hole pairs, 110–111
 electro-optic effects, 84
 electrostriction effect, 83
 elliptically polarized light, 13
 erbium-doped fiber amplifier (EDFA), 313–321
 erbium-doped fiber amplifiers (EDFAs), 300
 error probability, 274
 evanescent field, 65, 67
 evanescent wave, 359
 evanescent wave fiber sensors, 359
 excess loss, 223
 excimer-pumped dye laser, 256
 excited state absorption (ESA), 321
 external modulation, 154
 external quantum efficiency of LED, 121
 extrinsic absorption, 63–64
 extrinsic semiconductor, 112
 extrinsic sensors, 354–355
 eye diagram, 275–276

F

fabrication methods, 25, 35
 double-crucible technique, 41
 drawing and coating, 40
 MCVD process, 39
 plasma enhanced MCVD (PMCD), 40
 preform generation techniques, 37–39
 Fabry–Perot amplifiers (FPA), 304–305
 Fabry–Perot cavity, 81
 Fabry–Perot filters, 239
 Fabry–Perot interferometer, 230–231
 based OSA, 344
 sensors, 361
 Fabry–Perot resonant frequency, 305
 Faraday effect, 223
 FBG-based optical add/drop multiplexer, 227
 fiber beat length, 30
 fiber birefringence, 30
 fiber Bragg grating (reflection grating), 246
 applications of, 260–265
 characteristics, 248–253
 fabrication techniques, 253–260
 phase mask method, 257–258
 point-by-point technique for, 258–259
 sensors, 264–265
 transverse holographic method, 256–257
 fiber Bragg grating sensors, 363–365
 fiber grating, 242–243
 fiber materials, 35–37
 fiber-optic cable construction, 41–45
 fiber-optic cables, 41
 fiber optic connector, 35, 43

fiber-optic point-to-point link, 210
 fiber-optic systems
 advantages, 4–5
 applications of, 7–8
 disadvantages, 5
 evolution, 2–4
 fiber Raman amplifier, 309–311
 fiber sensors, 84
 flicker noise, 186
 fluorescent sensors, 365–366
 forward bias, 113–114
 forward current, 114
 four-wave mixing (FWM), 78, 85–87, 281–282
 applications of, 87–88
 reduction in, 87
 four-wave-mixing phenomenon, 85–88
 applications of, 87–88
 effect on optical transmission in single-mode fiber, 86
 factors influencing efficiency, 87
 non-linear refractive index, 86
 reduction in, 87
 free spectral range (FSR), 231
 full width at half maximum (FWHM), 26
 fused biconical tapered fiber coupler, 221
 fusion splicing, 158

G

Gaussian apodization, 248
 Gaussian beam, 32–33
 Gaussian pulse, 68, 88
 glass fibers, 36–37
 group delay per unit length, 72, 74
 group velocity, 26
 group velocity dispersion (GVD), 76
 guided modes, 20–23
 guiding layer (grating), 145
 gyroscope, fiber-optic, 361–363

H

halide glass fibers, 37
 heterojunctions, 124
 Hill gratings, 253
 hollow core fiber, 47
 hollow core fibers, 47
 homojunction, 119, 124

I

impact ionization, 183
 indirect recombination, 117–118
 inhomogeneities, 65
 injection electroluminescence, 114
 in-line optical amplifiers, 301
 insertion loss, 222

integrated SOA-based devices, 322
 intensity-modulated sensors, 355–359
 interchannel crosstalk, 306
 intermodal dispersion, 26
 internal quantum efficiency of LED, 121, 131
 intramodal dispersion, 26–27
 intrinsic absorption, 63
 intrinsic semiconductor, 110–112
 intrinsic sensors, 354
 irradiance, 12
 isolators, 223–224

J

Johnson noise, 186

K

Kerr effect, 84–85
 vs scattering effects, 91
 Kramers–Kronig relation, 254

L

L-band, 321
 light amplification by stimulated emission of radiation (LASER)
 characteristics, 147–149
 diode rate equations, 132–135
 emission process of photons, 131
 properties and applications, 149–151
 light-emitting diode (LED), 114–131
 characteristic of, 130–131
 direct and indirect recombination, 116–118
 dome, 126
 double heterojunction, 124–125
 edge-emitting LED (EELED), 127–128
 GaAs/Ga1-xAlxAs based, 125
 materials, 118–119
 planar, 125–126
 polymer (PLED), 129–130
 quantum efficiency of, 119–123
 spectral responses of, 131
 structure, 125–130
 super luminescent LEDs (SLED), 128–129
 surface-emitting light diodes (SELED), 127
 linear effects in optical fiber
 absorption, 63–64
 attenuation, 61–63
 bending losses, 65–68
 dispersion, 68–74
 distortions, 76–77

group delay, 74–76
 linear scattering, 64–65
 maximum bit rate, 77–78
 linear electro-optic effect, 84
 linearly polarized light, 13
 linear scattering, 64
 line coding, 152–153
 linewidth, 109
 liquid level sensor, fiber-optic, 368
 local area network (LAN), 211–213
 long-period fiber gratings (LPFGs), 246–247
 loose tube cable, 44–45
 losses in fiber joints, 160

M

Mach–Zehnder interferometer (MZI), 227, 231–233
 sensor, 359–361
 macrobending losses, 67–68
 macrobends, 65–66
 Maiman, Theodore, 2
 material dispersion, 69–71
 maximum bit rate, 77–78
 mechanical protection to the fiber, 17
 mechanical splice, 159
 mechanical waves, 11
 meridional ray, 18
 Michelson interferometer, 227–228
 OSA, 344–345
 microbending losses, 66–67
 microbending sensors, 357–359
 microbends, 65–66
 micro-resonator geometries, 241–242
 microring, 241
 microwaves, 11
 Mie scattering, 65
 minority carrier injection, 114
 modal dispersion, 72
 mode coupling, 244
 mode field diameter (MFD), 33–34
 mode partitioning noise (MPN), 109
 mode scrambler, 332
 modes of the waveguide, 17, 21
 modified chemical vapor deposition (MCVD), 39
 monochromatic light wave, 74
 multimode fibers, 17, 20, 25–26, 29, 31, 33, 35, 66
 multiple quantum wells (MQW), 147
 multiplexers, 324
 MZ intensity modulator, 233–234

N

network provisioning, 273–274
 noise equivalent power (NEP), 186
 noise-noise beating, 319
 non-linear effects in optical fiber, 78, 279

non-linear refractive index, 78
 non-return-to-zero (NRZ) code, 152
 normalized frequency, 20, 23, 28–30, 32
 normalized propagation constant, 71
 n-type GaAs, 124–125
 n-type semiconductor, 112
 numerical aperture, 30
 numerical aperture (NA), 339–341

O

O-E-O conversion, 322
 OH harmonics, 64
 on-off (RZ) code, 152
 optical add/drop multiplexers (OADMs), 226–227, 262–263, 273
 optical amplifiers
 applications of, 301–302
 basic concept, 299–300
 types of, 302
 optical connectors, 156–157
 optical cross-connects (OXC), 272–273
 optical Kerr effects, 78, 85
 optical multiplexers, 226–227
 optical power, 337–338
 optical receiver, 190–192
 equivalent circuit of, 192
 receiver amplifier configurations, 192–193
 optical signal monitoring (OSM), 273
 optical signal-to-noise ratio (OSNR), 276–278
 optical spectrum analyzer (OSA), 342–345
 optical splices, 157–160
 optical-switching technologies
 acousto-optic, 270
 applications of, 272–274
 electro-optic, 267
 large, 271–272
 liquid-crystal, 269–270
 microelectromechanical systems (MEMS), 266–267
 optomechanical, 266
 parameters for evaluating an, 272
 semiconductor optical amplifiers (SOAs), 270–271
 thermo-optic, 268–269
 optical time-division demultiplexers (OTDDMs), 324
 optical transmitter, 155–156
 optical tweezers, 346–347
 orthogonal polarizations, 73

P

packaging losses, 66
 Passive Star Coupler (PSC), 222

phase encoded (PE) code, 152–153
 phase matching condition, 238
 phase shift, 155
 photodetector materials, 174–176
 photodiode, 171–178, 337
 absorption coefficient, 173
 avalanche (APD), 183–185
 circuits, 187–188
 depletion region, 172
 direct and indirect absorption, 174
 electron-hole pairs (EHPs), 172
 long wavelength cutoff, 178
 photodetector materials, 174–176
 PIN, 180–183
 pn, 178–179
 quantum efficiency, 176
 responsivity, 176–177
 rise time, 183
 photonic crystal fiber (PCF), 46–47
 photons, energy of, 79
 photosensitive fiber, 48
 photosensitivity of a fiber, 253–254
 photosensitivity of a glass fiber, 48
 phototransistors
 applications, 189–190
 definition of, 188
 equivalent circuit of, 188–189
 features of, 188
 PIN photodiode, 180–183
 planar light-emitting diode (LED), 125–126
 Planck's blackbody radiation distribution law, 133
 Planck's constant, 79, 115, 277, 307
 plasma enhanced MCVD (PMCD), 40
 plastic fibers, 37
 pn Junction, 112–114
 pn photodiode, 178–179
 Pockels effect, 85
 point sensors, 369
 point-to-point fiber-optic communication network, 209–215
 analog transmission, 211
 block diagram, 210
 digital system, 211
 distribution networks, 213
 evolution of, 210
 local area network (LAN), 211–213
 system design considerations, 213–215
 system design steps, 215
 polarization, 13
 polarization fiber sensor, 366–367
 polarization maintaining fiber, 45–46
 polarization-maintaining fibers, 45–46

polarization-mode dispersion, 72–74
 polarization mode dispersion (PMD), 26
 polar (RZ) code, 153
 polymer LED (PLED), 129–130
 population inversion, 133–136, 138, 142, 146
 power amplifier, 302
 power budgeting, 215–216
 pre-amplifiers, 301
 preform generation techniques, 37–39
 probe, fiber-optic, 368–369
 profile parameter, 28–30
 protection switching, 273
 p-type GaAs, 124
 p-type semiconductor, 112
 pulse delaying and advancement, 84
 pulse dispersion, 68
 pumping, 131

Q

quadratic electro-optic effect (QEO effect), 85
 quantum efficiency, 176
 quantum well devices, 145–147

R

radiation modes, 23
 Raman amplifier, 306–313
 broadbanding of, 311
 impact of Raman Scattering on WDM systems, 311–313
 Raman fiber amplifier, 81
 Raman laser, fiber-based, 81
 Raman scattering, 78
 Raman threshold power, 78
 rare-earth-doped fiber amplifiers, 313–322
 Rayleigh scattering, 25, 64–65
 reach-through APD, 184
 reflection, 14–15
 reflection-type sensors, 355
 refraction, 14–15
 refractive index phenomena, 15, 71, 229–230, 280–283
 refractive index profile, 26–29, 31, 37–38
 regenerative repeater, 299
 relative intensity noise (RIN), 109
 return loss, 222
 return-to-zero (RZ) code, 152
 reverse bias, 113–114
 ring resonator-based wavelength filters, 239–241
 rise time budgeting, 216–217
 root mean square (rms) noise values, 275

S

sampled fiber grating, 247
 S-band amplifiers, 321

second order electro-optic effect, 84
 self-phase modulation, 88–89
 applications of, 90
 self-phase modulation (SPM), 78, 280, 282–283
 semiconductor optical amplifier (SOA), 302–306
 cavity gain of, 305–306
 crosstalk in, 306
 in CWDM systems, 324
 non-linear properties of, 323
 optical gates of, 323
 type of, 304–305
 sensors, fiber-optic
 biomedical, 367–368
 classification, 353–354
 distributed, 369–370
 extrinsic, 354–355
 intensity-modulated, 355–359
 intrinsic, 354
 liquid level, 368
 phase-modulated or interferometric, 359–363
 probe, 368
 wave-length modulated, 363–367
 shot noise, 186
 shunt resistance, 186–187
 simple fiber-optic patch cable, 41–43
 single-mode fiber, 26–27, 31, 33–35, 45–46
 microbending loss in, 66
 single-mode fibers, 63
 single-mode graded-index fibers, 35
 skew ray, 18
 Snell's law of refraction, 16
 solitons, 90, 92
 space charge region, 113
 specialty optical fibers, 45–48
 specific detectivity, 187
 splicing, 34–35
 splitting ratio, 221
 spontaneous Brillouin scattering, 81
 spontaneous emission, 114, 116, 119, 131–133, 139, 151, 316
 Stark splitting, 315
 stimulated Brillouin scattering (SBS), 81–83, 283
 applications of, 84–85
 schemes to reduce the power penalty, 83–84
 vs SRS, 85
 stimulated emission, 119, 128–129, 131–136, 139, 151

stimulated Raman scattering (SRS), 78–81, 280, 306
 stimulated scattering, 78
 Stokes waves, 78, 306, 308
 super luminescent LEDs (SLED), 128–129
 surface-emitting light diodes (SELED), 127

T

tellurium dioxide (TeO₂) crystal-based
 acousto-optic tunable filter, 238–239
 thermal equilibrium condition, 132
 thermal erasing of grating, 259
 threshold current, 109
 thulium-doped fiber amplifiers (TDFAs), 313, 321–322
 tight-buffered internal/external cables, 44
 tilted fiber grating, 247
 time division multiplexing
 interrogation, 370
 total internal reflection (TIR), 15–16
 transimpedance amplifier, 337–338
 transmission-type sensors, 355–356
 transmissivity, 64
 transverse wave, 13
 traveling wave amplifiers (TWAs), 304–305
 tunable filter, 238–239

U

unguided modes, 23
 unpolarized light, 13

V

vertical cavity surface-emitting laser (VCSEL), 147
 V number, 20–21, 66

W

waveguide dispersion, 71–72
 wavelength conversion, 88
 wavelength division multiplexing (WDM), 217–219
 components, 221–274
 interrogation of an FBG array, 369–370
 standards, 221
 wavelength measurement, 345
 wavelength-stabilized lasers, 263–264
 Western electric, 2

Z

zero dispersion, 63, 71, 86–87
 wavelength, 71
 zip duplex fiber-optic patch cable, 43–44
 zirconium tungstate (ZrW₂O₈), 265

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SYSTEMS AND COMPONENTS

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